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AUTHOR
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CLASS 10

CBSE

SAMPLE QUESTION PAPERS

CBSE · CLASS 10 · 2026-27
15
SAMPLE PAPERS

WITH CRISP NOTES, MIND MAPS & TOPPER'S ANSWER SHEETS

SCIENCE

WITH CBSE BOARD PAPERS 2026, 2025 & 2024 SOLVED

- 3 Official CBSE Papers Fully Solved
- 7 Solved + 5 Practice Papers
- 50% Competency Based Questions
- CBSE Topper's Answer Sheets
- Step-wise Marking Scheme
- Chapter-wise Crisp Notes & Mind Maps

AS PER LATEST
CBSE PATTERN 2026-27

BOARD EXAM 2027

What's Inside

1. Pattern, Crisp Notes and Mind Maps

Exam pattern, mind maps, CBQ bank	Page
Latest CBSE pattern 2026-27 and how to use this book	3
Mind maps: all 13 chapters	44
Competency Based Questions bank (chapter-wise)	57

Chemistry: Crisp Notes	Page
1. Chemical Reactions and Equations	5
2. Acids, Bases and Salts	8
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4. Carbon and its Compounds	14

Biology: Crisp Notes	Page
5. Life Processes	17
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Physics: Crisp Notes	Page
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10. The Human Eye and the Colourful World	32
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Environment: Crisp Notes	Page
13. Our Environment	41

2. Official CBSE Board Papers

Paper (with official marking scheme)	Page
SP 1: CBSE Board Exam 2026 (Set 31/1/1)	79
SP 2: CBSE Board Exam 2025 (Set 31/1/3) + Topper Answers	101
SP 3: CBSE Board Exam 2024 (Set 31/5/1) + Topper Answers	140

3. Solved Sample Papers

Paper	Paper	Solutions	Toppers
Sample Paper 4	173	184	191
Sample Paper 5	194	205	214
Sample Paper 6	217	228	236
Sample Paper 7	239	249	257
Sample Paper 8	260	272	281
Sample Paper 9	284	296	304
Sample Paper 10	307	317	325

4. Unsolved Practice Papers

Paper	Page
Sample Paper 11	329
Sample Paper 12	341
Sample Paper 13	352
Sample Paper 14	364
Sample Paper 15	374
Answer key (MCQ and A-R), Papers 11-15 + score tracker	385

How to use

Read the Crisp Notes and mind map of a chapter, solve its competency questions, then write one full paper every week in 3 hours. Check with the solutions and step marks.

LATEST CBSE PATTERN

Class X Science (086) · 2026-27

80 marks · 3 hours · 39 questions

EduSpark

1. Paper design: three subject-wise sections

Section	Q. No.	Question type	Marks each	Total
A: Biology	1 to 7	MCQ	1	7
(incl. Our Environment)	8, 9	Assertion-Reason	1	2
	10 to 12	Short answer (SA-I)	2	6
	13, 14	Short answer (SA-II)	3	6
	15	Case / source based	4	4
	16	Long answer	5	5
Section A total				**30**
B: Chemistry	17 to 23	MCQ	1	7
	24	Assertion-Reason	1	1
	25	Short answer (SA-I)	2	2
	26, 27	Short answer (SA-II)	3	6
	28	Case / source based	4	4
	29	Long answer	5	5
Section B total				**25**
C: Physics	30, 31	MCQ	1	2
	32	Assertion-Reason	1	1
	33, 34	Short answer (SA-I)	2	4
	35 to 37	Short answer (SA-II)	3	9
	38	Case / source based	4	4
	39	Long answer	5	5
Section C total · Grand total				**25 · 80**

2. Unit-wise marks

Unit	Marks
I. Chemical Substances: Nature and Behaviour (Ch 1–4)	25
II. World of Living (Ch 5–8)	25
III. Natural Phenomena (Ch 9–10)	12
IV. Effects of Current (Ch 11–12)	13
V. Natural Resources (Ch 13)	5
Theory total	**80**

3. Competency mix

Competency	Weight
Knowledge and understanding	50%
Application of knowledge	30%
Analyse, evaluate, create	20%

Internal choice: about 33% of questions.
Internal assessment (20): periodic assessment 5, multiple assessment 5, portfolio 5, practical work 5.

#Board ka scene: Not in the board paper this year: Periodic Classification, Evolution topics, electric motor, electromagnetic induction and generator (periodic assessment only). Boxed information in NCERT is also not examined.

How to use this book

Step 1: Crisp notes (13 chapters)

Revise one chapter in three pages: definitions, equations, diagrams, the points boards repeat ("Board ka favourite") and the mistakes that cost marks.

Step 2: Official CBSE papers 1 to 3

Real board papers of 2026, 2025 and 2024 with CBSE's own marking scheme. Papers 2 and 3 also carry the real answer books of CBSE toppers, so you see exactly how a full-marks answer looks.

Step 3: Solved papers 4 to 10

Written exactly on the 2026-27 CBSE pattern and format. Attempt first in 3 hours, then check with the marking-scheme style solution and give yourself marks honestly.

Step 4: Practice papers 11 to 15

Final mocks in exam conditions. Answer key for all objective questions is at the end of the book.

3-hour time plan

Time	What to do
First 15 min	Read the whole paper; mark internal choices you will attempt
Next 30 min	All 20 objective questions (MCQ + Assertion-Reason) of all three sections
Next 75 min	2-mark and 3-mark questions, section by section
Next 45 min	Case-based (4) and long answers (5) with neat diagrams
Last 15 min	Revise: units, balanced equations, labels, arrows on rays

Answer like a topper

Diagrams: pencil, labelled, arrows on every ray and field line; title below.

Equations: balanced, with state symbols and conditions (heat, catalyst, sunlight).

Numericals: formula, substitution with units, final answer boxed with unit.

Differences: two columns, same basis in each row.

Long answers: sub-parts labelled (a), (b), (c); keywords underlined.

Case based: answer only what is asked, use data from the passage.

#Nitesh Bhaiya ki tip: Har solved paper pehle khud 3 ghante me likho, phir marking scheme se check karo. Jo galti ho, use notes ke "Common mistakes" box me mark kar do. 15 paper ke baad board exam bas ek aur paper lagega.

Chemical Reactions and Equations

Most asked: balancing with state symbols; types of reactions with examples; oxidation-reduction; corrosion and rancidity

Every chemical change forms new substances: the whole chapter is about writing, balancing and classifying them.

1. Chemical Reactions and Equations

Chemical reaction: a process in which reactants change into products with new properties; atoms are only rearranged.

Signs of a reaction: change in **state**, change in **colour**, **evolution of gas**, change in **temperature**, formation of a **precipitate**.

$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$ dazzling white flame, white ash

Skeletal equation: unbalanced ($Mg + O_2 \longrightarrow MgO$). **Balanced equation:** equal number of atoms of each element on both sides, as required by the **Law of Conservation of Mass** (mass is neither created nor destroyed).

Balancing by hit-and-trial (NCERT example)

$$Fe + H_2O \longrightarrow Fe_3O_4 + H_2 \text{ (skeletal)}$$

1. Start with the compound having the **maximum atoms** (Fe_3O_4): 4 O on RHS, so write $4H_2O$.
2. Balance H: $4H_2$. Balance Fe: 3Fe.
3. Check every element again; add state symbols.

$$3Fe(s) + 4H_2O(g) \longrightarrow Fe_3O_4(s) + 4H_2(g)$$

Making equations informative: states (s), (l), (g), (aq); conditions such as heat, pressure, catalyst, sunlight above/below the arrow.

$$CO(g) + 2H_2(g) \xrightarrow{340\ atm} CH_3OH(l)$$

$$6CO_2(aq) + 12H_2O(l) \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6(aq) + 6O_2(aq) + 6H_2O(l)$$

2. Types of Chemical Reactions

(a) Combination: A + B ⟶ AB

$CaO(s) + H_2O(l) \longrightarrow Ca(OH)_2(aq) + Heat$ quick lime to slaked lime

$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$ whitewash: shiny finish

$$C(s) + O_2(g) \longrightarrow CO_2(g) \text{ burning of coal}$$

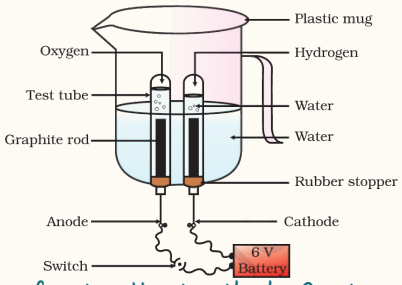
$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l) \text{ formation of water}$$

Exothermic: heat released (burning of natural gas $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$ + energy; respiration $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$ + energy; decomposition of vegetable matter into compost). **Endothermic:** heat absorbed (most decomposition reactions, photosynthesis).

(b) Decomposition: AB ⟶ A + B

Type	Energy	Example
Thermal	heat	$2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$ (green crystals turn brown, smell of burning sulphur)
Thermal	heat	$CaCO_3(s) \xrightarrow{Heat} CaO(s) + CO_2(g)$ (manufacture of quick lime, used for cement)
Thermal	heat	$2Pb(NO_3)_2(s) \xrightarrow{Heat} 2PbO(s) + 4NO_2(g) + O_2(g)$ (brown fumes)

Electrolytic	electricity	$2H_2O(l) \xrightarrow{\text{Electricity}} 2H_2(g) + O_2(g)$
Photolytic	light	$2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$ (white to grey); $2AgBr(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Br_2(g)$: used in black and white photography



Electrolysis of water: H_2 at cathode, O_2 at anode, volume ratio 2 : 1

(c) Displacement: A + BC ⟶ AC + B

A **more reactive** metal displaces a less reactive metal from its salt solution.

$Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$

$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$

$Pb(s) + CuCl_2(aq) \longrightarrow PbCl_2(aq) + Cu(s)$

Reverse reactions do not occur: $Cu + FeSO_4 \longrightarrow$ no reaction, because copper is less reactive than iron.

(d) Double displacement: AB + CD ⟶ AD + CB

Exchange of ions between two compounds. A reaction that gives an insoluble solid is a **precipitation reaction.**

$Na_2SO_4(aq) + BaCl_2(aq) \longrightarrow BaSO_4(s) + 2NaCl(aq)$ white ppt

$Pb(NO_3)_2(aq) + 2KI(aq) \longrightarrow PbI_2(s) + 2KNO_3(aq)$ yellow ppt

(e) Oxidation and reduction (redox)

Oxidation	*Reduction*
gain of oxygen	loss of oxygen
loss of hydrogen	gain of hydrogen
substance oxidised = reducing agent	substance reduced = oxidising agent

$2Cu(s) + O_2(g) \xrightarrow{\text{Heat}} 2CuO(s)$ brown to black

$CuO(s) + H_2(g) \xrightarrow{\text{Heat}} Cu(s) + H_2O(l)$ black to brown

Here CuO is **reduced** (oxidising agent) and H_2 is **oxidised** (reducing agent). Other examples: $ZnO + C \longrightarrow Zn + CO$ (ZnO reduced, C oxidised); $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$ (MnO_2 reduced, HCl oxidised).

3. Oxidation in Everyday Life

	Corrosion	*Rancidity*
What	metal attacked by moisture, air, acids	fats and oils oxidised
Sign	rust (iron, reddish-brown), black coat (silver, Ag_2S), green coat (copper, basic carbonate)	bad smell and taste
Prevention	painting, oiling, greasing, galvanising, alloying	antioxidants, N_2 flushing in chip packets, airtight containers, refrigeration

Rusting of iron

Iron rusts only when **both air (oxygen) and water (moisture)** are present: $4Fe + 3O_2 + 2xH_2O \longrightarrow 2Fe_2O_3.xH_2O$ (rust, hydrated iron(III) oxide). **Galvanisation** = coating iron/steel with a thin layer of **zinc**; the zinc protects iron even if the coating is scratched. Corrosion damages car bodies, bridges, iron railings and ships every year.

4. Key Observations at a Glance

Reaction	What you observe
Mg ribbon burnt in air	dazzling white flame, white powder (MgO)
$Pb(NO_3)_2(aq)$ + $KI(aq)$	yellow precipitate of PbI_2
Zn granules + dil. H_2SO_4	bubbles of H_2, flask becomes warm
CaO + water	hissing sound, beaker becomes very hot
Heating $FeSO_4.7H_2O$	green crystals lose water, turn brown/red; smell of burning sulphur
Heating $Pb(NO_3)_2$	brown fumes of NO_2, yellow residue of PbO
AgCl kept in sunlight	white turns grey (silver)
Iron nail in $CuSO_4(aq)$	brown coating on nail; blue solution turns pale green
$BaCl_2(aq)$ + $Na_2SO_4(aq)$	white precipitate of $BaSO_4$
Cu powder heated in air	brown surface turns black (CuO)

Classify the reaction (frequent MCQ)

Reaction	Type
$2KClO_3 \xrightarrow{Heat} 2KCl + 3O_2$	thermal decomposition
$NaOH + HCl \longrightarrow NaCl + H_2O$	double displacement (neutralisation)
$CuSO_4 + Zn \longrightarrow ZnSO_4 + Cu$	displacement (also redox)
$2H_2O_2 \xrightarrow{light} 2H_2O + O_2$	decomposition
$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$	combustion, exothermic, redox
$Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe$	displacement + redox (thermite)

Board ka favourite

→Balance with state symbols: Fe + H_2O (steam), $BaCl_2$ + Na_2SO_4, $Pb(NO_3)_2$ + KI.

→Heating $FeSO_4$ and $Pb(NO_3)_2$: observations + equations (3 marks).

→Electrolysis of water: why is the volume of H_2 double? why add a few drops of dilute H_2SO_4?

→Identify substance oxidised / reduced in CuO + H_2 and ZnO + C.

→Why is silver chloride stored in dark bottles? Why are chips packets flushed with N_2?

→Why does the colour of $CuSO_4$ solution change when an iron nail is dipped in it?

→Assertion-Reason on displacement (Fe + $CuSO_4$) and on decomposition being endothermic.

Common mistakes

→Changing subscripts to balance (H_2O to H_2O_2): only coefficients may change.

→Missing state symbols when the question asks for them, or writing $H_2O(l)$ for steam instead of (g).

→Calling every precipitation reaction "displacement": it is **double displacement**.

→Writing the reducing agent as the substance reduced: the reducing agent itself gets **oxidised**.

→Forgetting that Cu cannot displace Fe/Zn from their salts (Cu is less reactive).

→Writing rust as Fe_2O_3 only: rust is hydrated iron(III) oxide, $Fe_2O_3.xH_2O$.

Yaad karo

"O mila, H gaya": Oxygen mila ya Hydrogen gaya = **oxidation**; ulta = **reduction**. Dono saath hote hain, isliye **redox**.

Acids, Bases and Salts

Most asked: pH and its importance in daily life; reactions of acids with metals and carbonates; chlor-alkali process; preparation and uses of bleaching powder, baking soda, washing soda and Plaster of Paris

Sour, bitter or salty: every substance here is sorted by the ions it gives in water.

1. Acids and Bases

Property	Acid	Base
Taste	sour	bitter, soapy to touch
In water gives	H^+ ions (as H_3O^+)	OH^- ions
Blue litmus	turns red	no change
Red litmus	no change	turns blue
Phenolphthalein	colourless	pink
Methyl orange	red	yellow
Examples	HCl, H_2SO_4, HNO_3, CH_3COOH	NaOH, KOH, $Ca(OH)_2$, NH_4OH

Indicators: natural (litmus from **lichen**, turmeric, red cabbage, hydrangea petals), synthetic (phenolphthalein, methyl orange) and **olfactory** (smell changes): onion and vanilla lose their smell in a base; clove oil also works.

Alkali: a base that is **soluble in water** (NaOH, KOH). All alkalis are bases, but not all bases are alkalis ($Cu(OH)_2$, $Al(OH)_3$ are insoluble).

Strong vs weak: a strong acid/base ionises completely (HCl, H_2SO_4, NaOH); a weak one ionises partly (CH_3COOH, NH_4OH). **Dilution:** always add **acid to water** slowly with stirring, never water to acid: the process is highly **exothermic** and the mixture may splash.

2. Chemical Properties

(a) Acid + metal ⟶ salt + hydrogen

$$Zn(s) + 2HCl(aq) \longrightarrow ZnCl_2(aq) + H_2(g)$$

Test for H_2: a burning candle near the gas burns with a **pop sound**. Some bases also react with metals: $2NaOH + Zn \longrightarrow Na_2ZnO_2$ (sodium zincate) + H_2.

(b) Acid + metal carbonate / hydrogencarbonate

$$Na_2CO_3(s) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$$

$$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$$

Test for CO_2: passed through **lime water**, it turns milky ($CaCO_3$); excess CO_2 removes the milkiness (soluble $Ca(HCO_3)_2$).

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

$$CaCO_3(s) + H_2O(l) + CO_2(g) \longrightarrow Ca(HCO_3)_2(aq)$$

(c) Neutralisation: acid + base ⟶ salt + water

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

Neutralisation in daily life: antacid for acidity, lime for acidic soil, baking soda on a bee sting, factory waste neutralised before it is released.

(d) Metal oxide + acid; non-metal oxide + base

$$CuO(s) + 2HCl(aq) \longrightarrow CuCl_2(aq) + H_2O(l)$$ blue-green solution

$$Ca(OH)_2(aq) + CO_2(g) \longrightarrow CaCO_3(s) + H_2O(l)$$

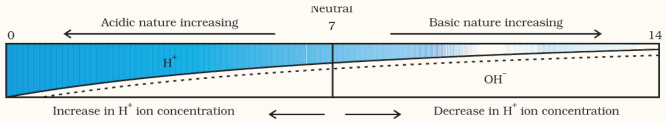
So **metal oxides are basic** and **non-metal oxides are acidic**.

Acids and bases in water

Acids show acidic behaviour only in water: HCl(g) does not change dry blue litmus; in water $HCl + H_2O \longrightarrow H_3O^+ + Cl^-$. Glucose and alcohol solutions contain H but give no H^+ ions, so they do **not conduct** and are not acids. Distilled water does not conduct; rain water does (dissolved acids/ions).

3. pH Scale

pH runs from **0 to 14** (p = potenz, power). pH 7 = neutral, **below 7 = acidic**, above 7 = basic. Higher H^+ concentration means **lower pH**. Measured with **universal indicator** (shows different colours at different pH).



Variation of pH with $H^+(aq)$ and $OH^-(aq)$ concentration

Substance	pH (approx.)
Gastric juice	~1.2
Lemon juice	~2.2
Pure water / blood	7 / 7.4
Milk of magnesia	~10
Sodium hydroxide solution	~14

Importance of pH in everyday life

- →**Our body** works within pH 7.0 to 7.8. Rain with pH below **5.6** is **acid rain**; it makes rivers acidic and harms aquatic life.
- →**Soil:** plants need a specific pH range; acidic soil is treated with quick lime/slaked lime, basic soil with organic matter.
- →**Digestion:** the stomach makes HCl; excess acid (indigestion) is relieved by an **antacid** such as milk of magnesia, $Mg(OH)_2$.
- →**Tooth decay** starts when mouth pH falls below **5.5**; bacteria acting on sugar make acid that corrodes enamel (calcium hydroxyapatite). Basic toothpaste neutralises it.
- →**Self-defence:** bee sting and nettle sting inject **methanoic acid**; rub baking soda (bee) or dock leaf (nettle) on the spot.

Natural source	Acid
Vinegar	acetic (ethanoic) acid
Orange, lemon	citric acid
Tamarind	tartaric acid
Tomato	oxalic acid
Sour milk (curd)	lactic acid
Ant / nettle sting	methanoic acid

4. Salts

Salt: formed by neutralisation. Salts with the same positive or negative radical form a **family** (NaCl, Na_2SO_4 = sodium salts; NaCl, KCl = chloride salts).

Salt from	pH	Example
strong acid + strong base	7 (neutral)	NaCl, K_2SO_4
strong acid + weak base	< 7 (acidic)	NH_4Cl, $CuSO_4$
weak acid + strong base	> 7 (basic)	Na_2CO_3, CH_3COONa

Common salt (NaCl): obtained by evaporating sea water; large crystals of brown **rock salt** are mined like coal. It is the raw material for NaOH, bleaching powder, baking soda and washing soda.

Chlor-alkali process: sodium hydroxide

$$2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{electricity}} 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Electricity through **brine**: Cl_2 at the **anode**, H_2 at the **cathode**, NaOH near the cathode. Named after the products, chlor (chlorine) and alkali (NaOH).

Product	Uses
NaOH	soaps and detergents, paper making, artificial fibres, de-greasing metals
Cl_2	water treatment, PVC, disinfectants, CFCs, pesticides, bleaching powder
H_2	fuels, margarine (hydrogenation of oils), ammonia for fertilisers
HCl (H_2 + Cl_2)	cleaning steel, ammonium chloride, medicines, cosmetics

Four important salts

Compound	Preparation	Uses
Bleaching powder $CaOCl_2$	$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$ (Cl_2 on dry slaked lime)	bleaching cotton, linen, wood pulp; disinfecting drinking water; oxidising agent

Baking soda $NaHCO_3$ (sodium hydrogen-carbonate)	$NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$	baking powder (with tartaric acid), antacid, soda-acid fire extinguisher. On heating: $2NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$
Washing soda $Na_2CO_3.10H_2O$	heat baking soda to get Na_2CO_3, then recrystallise: $Na_2CO_3 + 10H_2O \longrightarrow Na_2CO_3.10H_2O$	glass, soap, paper industries; borax; cleaning agent; removes **permanent hardness** of water
Plaster of Paris $CaSO_4.{}^1/_2H_2O$	heat gypsum at **373 K**: $CaSO_4.2H_2O \longrightarrow CaSO_4.{}^1/_2H_2O + 1{}^1/_2H_2O$	plaster for fractured bones, toys, decoration, smooth surfaces. With water it sets back to gypsum, a hard mass

Water of crystallisation: fixed number of water molecules in one formula unit of a salt: $CuSO_4.5H_2O$ (blue), gypsum $CaSO_4.2H_2O$, washing soda (10). Heating blue copper sulphate crystals makes them **white** (anhydrous); adding water restores the blue colour.

Quick revision

1. Acids turn blue litmus red; bases turn red litmus blue. Phenolphthalein turns pink in bases.

Board ka favourite

→Chlor-alkali process: equation, products at each electrode, one use of each product.
→Baking soda: preparation, why cake becomes soft and spongy (CO_2), why tartaric acid is added (to neutralise bitter Na_2CO_3).
→Plaster of Paris: preparation, why stored in moisture-proof containers, setting reaction.
→pH questions: tooth decay at pH 5.5, antacid, acid rain, ranking solutions by pH or by H^+ concentration.
→Metal carbonate + acid, test for the gas evolved (lime water), and what happens with excess CO_2.
→Why does dry HCl gas not change the colour of dry blue litmus paper? (No H^+ ions without water.)
→Milkman adds a little baking soda to fresh milk: why? (pH becomes slightly alkaline, so milk takes longer to set as curd.)
→Why curd/sour substances should not be kept in brass and copper vessels.

Common mistakes

→Writing the formula of bleaching powder as $CaCl_2$: it is $CaOCl_2$.
→Writing water molecules of POP as 2: POP has ${}^1/_2\ H_2O$; gypsum has 2.
→Saying "higher pH = stronger acid": lower pH means more H^+, a stronger acid.
→Adding water to concentrated acid in the answer of a dilution question.
→Calling every base an alkali: only water-soluble bases are alkalis.
→Writing NaOH at the anode in the chlor-alkali process: Cl_2 forms at the anode, H_2 and NaOH at the cathode.

Yaad karo

POP = Pura aadha: Plaster of Paris me sirf **aadha** H_2O, gypsum me **do**. Washing soda me **das** (10). **Acid in water**: "AW" = Acid pehle Water me. **BRA**: Blue litmus Red = Acid.

METALS AND NON-METALS

Most asked: reactivity series and displacement; formation of ionic compounds (electron-dot); extraction of metals (roasting, calcination, reduction, refining); corrosion and its prevention

Why gold shines for centuries while iron rusts in a monsoon: the answer is reactivity.

1. Physical Properties

Property	Metals	Non-metals
Lustre	lustrous (shiny)	dull (except iodine, graphite)
Hardness	hard (except Na, K: cut with knife)	soft (except diamond, the hardest natural substance)
Malleable / ductile	yes (gold and silver are the most malleable and ductile)	no, brittle
Conduction	good conductors of heat and electricity (Ag, Cu best; Pb, Hg poor)	poor (except graphite)
State	solid (Hg liquid)	solid, liquid (Br_2), gas
Sonorous	yes (used in bells)	no
m.p.	high (except Ga, Cs: melt on the palm)	low (except diamond, graphite)

2. Chemical Properties of Metals

(a) With oxygen: metal oxide (basic)

$2Cu + O_2 \longrightarrow 2CuO$ (black) ; $4Al + 3O_2 \longrightarrow 2Al_2O_3$

Amphoteric oxides react with both acids and bases: Al_2O_3, ZnO.

$$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$$

$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$ sodium aluminate

Na and K catch fire in air, so they are **kept under kerosene**. Mg, Al, Zn, Pb get a protective oxide layer. Fe burns only as filings; Cu does not burn but forms black CuO; Ag and Au do not react. **Anodising:** thickening the protective oxide layer on aluminium by electrolysis.

(b) With water: metal hydroxide/oxide + H_2

$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$ + heat violent, H_2 catches fire

$Ca(s) + 2H_2O(l) \longrightarrow Ca(OH)_2(aq) + H_2(g)$ Ca floats (H_2 bubbles stick)

$Mg(s) + 2H_2O(l) \xrightarrow{\text{hot}} Mg(OH)_2(aq) + H_2(g)$ Mg also floats

$2Al + 3H_2O(g) \longrightarrow Al_2O_3 + 3H_2$; $3Fe + 4H_2O(g) \longrightarrow Fe_3O_4 + 4H_2$ steam

Pb, Cu, Ag, Au do not react with water at all.

(c) With dilute acids: salt + H_2

$Fe + 2HCl \longrightarrow FeCl_2 + H_2$; $Mg + 2HCl \longrightarrow MgCl_2 + H_2$

H_2 is **not evolved with HNO_3** (strong oxidising agent: oxidises H_2 to water) except with Mg and Mn and very dilute HNO_3. **Aqua regia:** 3 parts conc. HCl + 1 part conc. HNO_3; dissolves gold and platinum.

(d) With salt solutions: displacement

A more reactive metal displaces a less reactive one: $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$.

Chemical properties of non-metals

Metals	Non-metals
form **basic** oxides (Na_2O, MgO); Al_2O_3, ZnO amphoteric	form **acidic** oxides (SO_2, CO_2, P_2O_5) or **neutral** oxides (CO, H_2O, N_2O)
displace H_2 from dilute acids (those above H)	do not displace H_2 from acids (cannot give electrons to H^+)

lose electrons: **electropositive**, reducing agents	gain electrons: **electronegative**, oxidising agents
form ionic chlorides (NaCl)	form covalent chlorides (CCl_4, PCl_3) and hydrides (H_2S, NH_3, CH_4)

Reactivity series (most to least reactive)

K → Na → Ca → Mg → Al → Zn → Fe → Pb → [H] → Cu → Hg → Ag → Au

Yaad karo

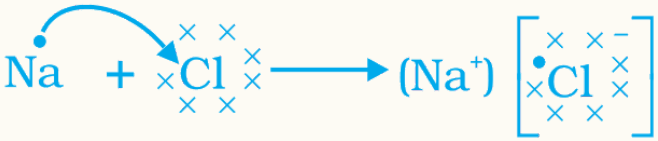
"Kya Naya Car Maine Abhi Zindagi Faltu Pe Hi Chalai, Mujhe Aaj Gussa": K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au. H se upar wale metals acids se H_2 dete hain.

3. Ionic Compounds

Metals **lose** electrons (form cations), non-metals **gain** electrons (form anions) to reach a noble-gas octet. The electrostatic attraction between them is an **ionic (electrovalent) bond**.

$$Na \rightarrow Na^+ + e^-$$
$$2,8,1 \quad 2,8$$
(Sodium cation)

$$Cl + e^- \rightarrow Cl^-$$
$$2,8,7 \quad 2,8,8$$
(Chloride anion)



Formation of NaCl: Na (2,8,1) gives 1 e^- to Cl (2,8,7)

MgO: Mg (2,8,2) gives two electrons to O (2,6): Mg^{2+} O^{2-}. Na_2O: two Na atoms give one electron each to O.

$MgCl_2$: Mg (2,8,2) gives one electron to each of two Cl atoms: $Mg^{2+} + 2Cl^-$.

Property of ionic compounds	*Reason*
solid and hard, brittle	strong force of attraction between ions
high m.p. and b.p.	a lot of energy needed to break strong inter-ionic attraction
soluble in water, insoluble in kerosene and petrol	water is polar; solvent molecules interact with ions
conduct electricity in molten state or in solution, not as solid	ions become free to move; in solid they are fixed

4. Occurrence and Extraction of Metals

Mineral: naturally occurring compound of a metal. **Ore:** mineral from which a metal is extracted profitably. **Gangue:** unwanted soil, sand in the ore, removed first (concentration).

Steps: **ore** ⟶ concentration (remove gangue) ⟶ conversion to metal (by method below) ⟶ **refining**. Metals at the bottom of the series (Au, Ag, Pt, Cu) occur **free (native)**; the most reactive (K, Na, Ca, Mg, Al) are never found free.

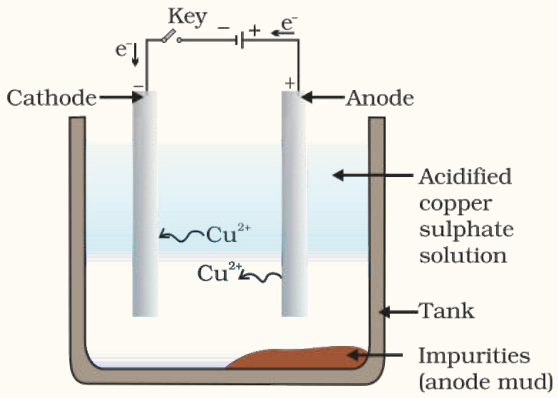
Activity	*Method*	*Example*
low (Cu, Hg)	heating the ore alone	$2HgS + 3O_2 \xrightarrow{heat} 2HgO + 2SO_2$; $2HgO \xrightarrow{heat} 2Hg + O_2$; $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$; $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
middle (Zn, Fe, Pb)	convert to oxide, then reduce with carbon	**Roasting** (sulphide ore, excess air): $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$. **Calcination** (carbonate ore, limited air): $ZnCO_3 \rightarrow ZnO + CO_2$. Reduction: $ZnO + C \rightarrow Zn + CO$
high (K, Na, Ca, Mg, Al)	electrolysis of molten chloride/oxide	Na at cathode: $Na^+ + e^- \rightarrow Na$; Cl_2 at anode: $2Cl^- \rightarrow Cl_2 + 2e^-$

Yaad karo

C-C, R-S: Calcination = Carbonate ore (limited air); Roasting = Sulphide ore (excess air). Dono ke baad oxide ko carbon se reduce karte hain.

Thermit reaction: $Fe_2O_3(s) + 2Al(s) \longrightarrow 2Fe(l) + Al_2O_3(s)$ + heat; highly exothermic, molten iron used to **join railway tracks** and cracked machine parts.

Electrolytic refining (copper, zinc, silver, gold)



Electrolytic refining of copper

Anode: impure metal; **cathode:** thin strip of pure metal; **electrolyte:** acidified solution of a salt of the metal ($CuSO_4$). Pure metal deposits on the cathode; soluble impurities go into solution, insoluble ones settle below the anode as **anode mud**.

5. Corrosion and its Prevention

Silver turns black (Ag_2S, reacts with sulphur in air); copper gets a green coat of **basic copper carbonate**; iron gets reddish-brown **rust** (needs both air and moisture).

Rusting experiment: iron nails in three test tubes: (A) tap water (air + water): **rusts**; (B) boiled distilled water with a layer of oil (no air): no rust; (C) dry air over anhydrous $CaCl_2$ (no moisture): no rust. So rusting needs **both air and water**.

Prevention: painting, oiling, greasing, **galvanising** (zinc coating), chrome plating, anodising, making **alloys**.

Alloy	Composition	Use / property
Stainless steel	Fe + Ni + Cr	does not rust; utensils
Brass	Cu + Zn	lower conductivity than Cu
Bronze	Cu + Sn	statues, medals
Solder	Pb + Sn	low m.p.; welding electrical wires
Amalgam	Hg + any metal	dental fillings

Pure gold is 24 carat and very soft; ornaments are made of **22 carat** gold (22 parts gold + 2 parts Cu or Ag).

Board ka favourite

- → Electron-dot formation of NaCl, $MgCl_2$, MgO, Na_2O with the ions formed.
- → Why ionic compounds have high m.p. and conduct only in molten/aqueous state.
- → Roasting vs calcination with equations; reduction of ZnO with carbon; thermit reaction.
- → Electrolytic refining of copper: labelled diagram, what is anode mud.
- → Metals X, Y, Z with water/acid observations: arrange in order of reactivity (case-based).
- → Rusting experiment with three test tubes: purpose of boiled water, oil layer and anhydrous $CaCl_2$.
- → Why Na is kept in kerosene; why H_2 is not evolved with HNO_3; amphoteric oxides with equations.

Common mistakes

- → Writing "$Na + H_2O \longrightarrow NaOH + H_2$" unbalanced: correct is $2Na + 2H_2O \longrightarrow 2NaOH + H_2$.
- → Mixing roasting (sulphide ore, excess air) with calcination (carbonate ore, limited air).
- → In refining, putting impure copper at the cathode: impure metal is always the **anode**.
- → Drawing electron transfer without showing the charges on the ions formed.
- → Saying all metals are solids or all non-metals are non-conductors: Hg is liquid, graphite conducts.
- → Writing that aluminium does not corrode because it is unreactive: Al is **reactive**; its oxide layer protects it.

Carbon and its Compounds

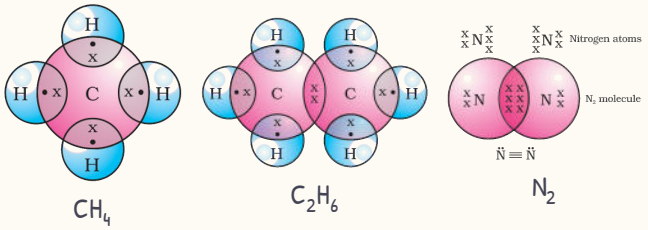
Most asked: electron-dot structures (CH_4, C_2H_6, CO_2, N_2); homologous series and IUPAC names; addition vs substitution; ethanol and ethanoic acid reactions; soap micelle and hard water

One element, millions of compounds: carbon does it with covalent bonds, catenation and tetravalency.

1. Covalent Bonding

Carbon (2,4) can neither lose 4 nor gain 4 electrons easily (C^{4+}/C^{4-} would be unstable), so it **shares** electrons. A bond formed by sharing a pair of electrons is a **covalent bond**.

Molecule	Bond	Shared pairs
H_2, Cl_2, CH_4	single (H–H)	1 per bond
O_2, CO_2 (O=C=O), C_2H_4	double	2
N_2 (N≡N), C_2H_2	triple	3



Electron-dot structures

In H_2O oxygen shares one pair with each H; in NH_3 nitrogen shares three pairs; CO_2 has two C=O double bonds (O::C::O), each O keeping two lone pairs; O_2 has O=O.

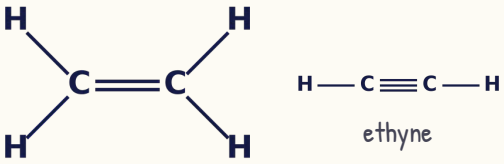
Properties of covalent compounds: low melting and boiling points (weak intermolecular forces), generally **poor conductors** of electricity (no ions formed).

2. Versatile Nature of Carbon

→ **Catenation:** carbon links with other carbon atoms to form long chains, branched chains and rings (strong, stable C–C bonds). Silicon shows it only up to 7–8 atoms.

→ **Tetravalency:** valency 4, bonds with C, H, O, N, S, Cl; small size makes these bonds strong.

Saturated compounds have only single bonds (alkanes, C_nH_{2n+2}); **unsaturated** have double (alkenes, C_nH_{2n}) or triple bonds (alkynes, C_nH_{2n-2}). Carbon compounds also form **chains, branches and rings** (cyclohexane C_6H_{12}, benzene C_6H_6). Same molecular formula, different structure = **structural isomers** (n-butane and iso-butane, C_4H_{10}).



ethene ethyne

Unsaturated hydrocarbons

Functional groups and homologous series

Group	Formula	Suffix / prefix	Example
Halo	–Cl, –Br	chloro-, bromo-	CH_3Cl chloromethane
Alcohol	–OH	-ol	CH_3CH_2OH ethanol
Aldehyde	–CHO	-al	HCHO methanal
Ketone	>C=O	-one	CH_3COCH_3 propanone
Carboxylic acid	–COOH	-oic acid	CH_3COOH ethanoic acid
Alkene / alkyne	C=C / C≡C	-ene / -yne	C_3H_6 propene, C_2H_2 ethyne

Homologous series: compounds with the same functional group, successive members differing by **$–CH_2–$ (14 u)**; same chemical properties, gradual change in physical properties (m.p., b.p., solubility). E.g. CH_4, C_2H_6, C_3H_8, C_4H_{10}.

C	Alkane C_nH_{2n+2}	Alkene C_nH_{2n}	Alkyne C_nH_{2n-2}
1	CH_4 methane	–	–
2	C_2H_6 ethane	C_2H_4 ethene	C_2H_2 ethyne

3	C_3H_8 propane	C_3H_6 propene	C_3H_4 propyne
4	C_4H_{10} butane	C_4H_8 butene	C_4H_6 butyne

Naming: (i) count carbons: meth (1), eth (2), prop (3), but (4), pent (5), hex (6); (ii) add prefix/suffix of the functional group; (iii) if the suffix starts with a vowel drop the final "e" of the alkane name (propane ⟶ propanone). E.g. $CH_3CH_2CH_2OH$ propanol, CH_3CH_2CHO propanal, C_2H_5Br bromoethane.

3. Chemical Properties

(a) Combustion

$$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O + \text{heat} + \text{light}$$

$$CH_3CH_2OH + 3O_2 \longrightarrow 2CO_2 + 3H_2O + \text{heat} + \text{light}$$

Saturated hydrocarbons burn with a clean **blue flame**; unsaturated ones give a **yellow, sooty flame**. Limited air gives a sooty flame even for saturated fuels (blackened vessels = air holes blocked).

(b) Oxidation

$$CH_3CH_2OH \xrightarrow[\text{or acidified } K_2Cr_2O_7]{\text{alk. } KMnO_4 + \text{heat}} CH_3COOH$$

Alkaline $KMnO_4$ and acidified $K_2Cr_2O_7$ are **oxidising agents** (they add oxygen). Purple $KMnO_4$ is decolourised while ethanol is being oxidised.

(c) Addition (unsaturated only): hydrogenation

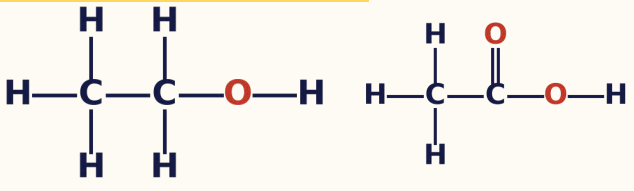
$$R_2C{=}CR_2 + H_2 \xrightarrow{\text{Ni / Pd catalyst}} R_2CH{-}CHR_2$$

Used to make **vegetable ghee** from vegetable oils. Animal fats contain saturated fatty acids (harmful); oils with unsaturated fatty acids are healthier.

(d) Substitution (saturated): in sunlight

$$CH_4 + Cl_2 \xrightarrow{\text{sunlight}} CH_3Cl + HCl$$

4. Ethanol and Ethanoic Acid



ethanol C_2H_5OH ethanoic acid CH_3COOH

Structural formulae

	Ethanol (C_2H_5OH)	Ethanoic acid (CH_3COOH)
Nature	liquid, soluble in water, neutral (no effect on litmus)	weak acid; pure acid freezes at 290 K in winter: **glacial acetic acid**; 5–8% solution is **vinegar**
Uses	solvent, tinctures, cough syrups, alcoholic drinks, fuel (with petrol)	vinegar for preserving pickles; making esters
With Na	$2Na + 2CH_3CH_2OH \longrightarrow 2CH_3CH_2ONa + H_2$	with NaOH: $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$
Special	conc. H_2SO_4, 443 K: $CH_3CH_2OH \longrightarrow CH_2{=}CH_2 + H_2O$ (dehydration)	with $NaHCO_3$: $CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + H_2O + CO_2$

Denatured alcohol: industrial ethanol made unfit for drinking by adding poisonous **methanol** (and a blue dye). Methanol is oxidised to methanal in the liver; even small amounts can cause **blindness and death**. Drinking ethanol slows body processes and impairs judgement.

Esterification and saponification

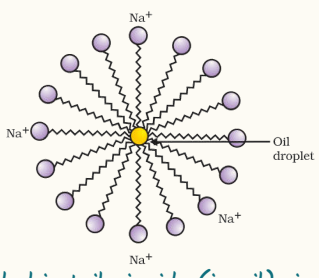
$$CH_3COOH + CH_3CH_2OH \xrightarrow{\text{acid}} CH_3COOCH_2CH_3 + H_2O$$

$$CH_3COOC_2H_5 \xrightarrow{NaOH} C_2H_5OH + CH_3COONa$$

The ester (ethyl ethanoate) is **sweet-smelling**; the reverse reaction with NaOH is **saponification**. Esters are used in perfumes and flavouring agents. Alkaline hydrolysis of esters (fats) gives **soap**.

5. Soaps and Detergents

Soap: sodium or potassium salt of a long-chain carboxylic acid ($C_{17}H_{35}COONa$). **Detergent:** sodium salt of a long-chain sulphonic acid or ammonium salt with chloride/bromide ions.



Micelle: hydrophobic tails inside (in oil), ionic heads outside (in water)

Cleansing action: the **ionic (hydrophilic) end** stays in water, the **hydrocarbon (hydrophobic) tail** dissolves in oil/dirt. Molecules form clusters called **micelles** with the oily dirt trapped in the centre; the micelles stay in solution as a colloid (they repel one another) and are rinsed away.

Soap	Detergent
forms **scum** (insoluble Ca/Mg salts) with hard water	works in hard water too: no scum
biodegradable	some are non-biodegradable
used for bathing, washing	shampoos, washing clothes

Why soap fails in hard water: Ca^{2+} and Mg^{2+} ions react with soap to form an insoluble, curdy precipitate called **scum**, so a lot of soap is wasted: $2C_{17}H_{35}COONa + Ca^{2+} \longrightarrow (C_{17}H_{35}COO)_2Ca\downarrow + 2Na^+$. Detergents' charged ends do not form insoluble salts with these ions.

Board ka favourite

→Electron-dot structures: CH_4, C_2H_6, C_2H_4, CO_2, N_2, O_2, H_2O, NH_3, cyclohexane.

→IUPAC names and the next homologue; why homologues show similar chemical properties.

→Addition vs substitution with equations; hydrogenation of oils and its use.

→Ethanol to ethene (conc. H_2SO_4, 443 K) and ethanol to ethanoic acid; esterification.

→Test to distinguish ethanol and ethanoic acid: add $NaHCO_3$, brisk effervescence (CO_2) with the acid only.

→Why are carbon compounds poor conductors? Why does carbon form covalent and not ionic bonds?

→Micelle formation with diagram; why soap does not work in hard water; soap vs detergent.

Common mistakes

→Showing 8 electrons on H in electron-dot structures: hydrogen needs only **2**.

→Writing ethanoic acid as $C_2H_4O_2$ in equations where the structure is asked: write CH_3COOH.

→Forgetting the catalyst (Ni/Pd) in hydrogenation, sunlight in substitution, conc. H_2SO_4 in esterification.

→Counting the C of –COOH/–CHO wrongly: CH_3COOH has 2 C, so **eth**anoic acid.

→Saying covalent compounds conduct electricity in solution: they generally do not (no ions).

→Calling $KMnO_4$ a reducing agent in the oxidation of ethanol: it is the **oxidising agent**.

Yaad karo

"Meri Eklauti Pyari Bahan": Meth 1, **Eth** 2, **Prop** 3, **But** 4. **Blue flame = saturated**, peeli dhuaan-daar flame = unsaturated.

Quick test: ethanol or ethanoic acid?

Test	Ethanol	Ethanoic acid
Blue litmus	no change	turns red
$NaHCO_3$ / Na_2CO_3	no reaction	brisk effervescence, CO_2 turns lime water milky
Smell	pleasant, spirit-like	smell of vinegar

Quick revision

1. Carbon forms covalent bonds (shares electrons) due to tetravalency and catenation.
2. Alkanes C_nH_{2n+2}, alkenes C_nH_{2n}, alkynes C_nH_{2n-2}.
3. Homologous series: successive members differ by CH_2 (14 u); same chemical properties.
4. Saturated compounds burn with a blue flame; unsaturated give a sooty yellow flame.
5. Addition of hydrogen (Ni catalyst) turns vegetable oils into vanaspati ghee.
6. Ethanol + Na → sodium ethoxide + H_2; hot conc. H_2SO_4 dehydrates ethanol to ethene.
7. Esterification: ethanoic acid + ethanol → ester (sweet smell); saponification makes soap.
8. Soap micelle: hydrophobic tail in oil, ionic head in water; does not lather in hard water.

LIFE PROCESSES

Most asked: photosynthesis, human digestion (role of enzymes, HCl, bile), breakdown of glucose, double circulation and the heart, nephron and urine formation

Nutrition, respiration, transportation and excretion keep the body maintained even when it is resting.

1. Life Processes and Nutrition

Life processes: the processes that together perform the maintenance job of the body: **nutrition, respiration, transportation, excretion.** Unicellular organisms manage by simple diffusion; multicellular organisms need specialised organs.

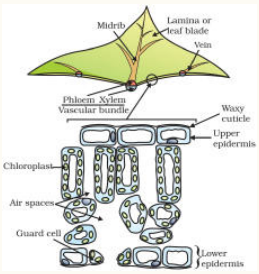
Autotrophic nutrition	Heterotrophic nutrition
makes own food from CO_2 and water using sunlight	takes ready-made complex food and breaks it with enzymes
green plants, some bacteria	animals, fungi; saprophytic (fungi), parasitic (Cuscuta, ticks, lice, tapeworm), holozoic (Amoeba, humans)

Photosynthesis: conversion of CO_2 and water into carbohydrates in the presence of sunlight and chlorophyll; extra stored as **starch** (plants) and **glycogen** (animals).

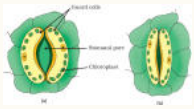
$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

Three events: (i) absorption of light energy by chlorophyll; (ii) conversion of light energy to chemical energy and **splitting of water** into hydrogen and oxygen; (iii) **reduction of CO_2** to carbohydrates. Desert plants take in CO_2 at night and store it as an acid, used by day.

Stomata: pores on leaves for gas exchange and water loss (transpiration). **Guard cells** swell when water flows in, so the pore **opens**; they shrink when water flows out, so it closes. Plants take nitrogen from soil as nitrates or nitrites.



Cross-section of a leaf



Open and closed stoma

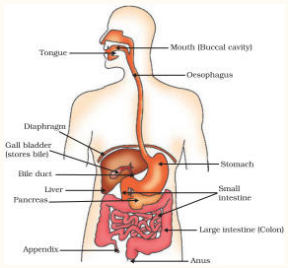
Amoeba: engulfs food with **pseudopodia**, forming a **food vacuole**; digested food diffuses into the cytoplasm; undigested food is thrown out.

Human digestive system

Mouth → Oesophagus → Stomach → Small intestine → Large intestine → Anus

Part	Secretion / enzyme	Action
Mouth	saliva: **salivary amylase**	starch → simple sugar; tongue mixes food
Stomach	gastric glands: **HCl**, **pepsin**, mucus	HCl makes medium acidic (pepsin works), kills germs; pepsin digests proteins; mucus protects stomach lining
Small intestine	**bile** (liver): no enzyme	makes food alkaline; **emulsifies fats** into small globules
	pancreatic juice: **trypsin**, **lipase** (and amylase)	trypsin: proteins; lipase: emulsified fats
	intestinal juice	proteins → amino acids; carbohydrates → glucose; fats → fatty acids + glycerol
Large intestine	none	absorbs water; rest leaves via anus (anal sphincter)

Villi on the inner wall of the small intestine increase surface area for absorption; their blood vessels take food to every cell. **Peristalsis** (rhythmic contraction of the gut muscles) pushes food along.



Human alimentary canal

2. Respiration

Food (glucose) is broken down in cells to release energy, stored as **ATP** (energy currency of the cell). Step 1 (glycolysis) occurs in the **cytoplasm** in all cases.

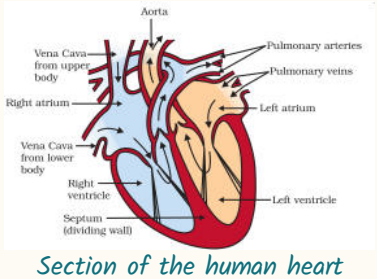
Pathway	Where	Products
Anaerobic (no O_2)	yeast (fermentation)	glucose → pyruvate → **ethanol** + CO_2 + little energy
Lack of O_2	our muscle cells	pyruvate → **lactic acid** + energy (causes cramps)
Aerobic (O_2)	mitochondria	pyruvate → CO_2 + H_2O + much more energy

Human respiratory system: nostrils (hairs and mucus filter air) → pharynx → larynx → **trachea** (rings of cartilage stop it collapsing) → bronchi → bronchioles → **alveoli**. Alveoli give a huge surface (about 80 m^2) with thin walls and blood capillaries. Breathing in: ribs lift, **diaphragm flattens**, chest cavity enlarges.

Haemoglobin in RBCs carries O_2; CO_2 is carried mostly dissolved in plasma (more soluble in water). Aquatic animals breathe faster: dissolved O_2 in water is low. Plants exchange gases by diffusion through stomata, lenticels and root surface; at night only CO_2 is released, by day respiration CO_2 is used in photosynthesis.

3. Transportation

Blood: plasma (carries food, CO_2, nitrogenous waste, salts), RBCs (O_2), WBCs (defence), **platelets** (clotting at injury). **Lymph** (tissue fluid): carries digested fats from intestine and returns extra fluid to blood.



Section of the human heart

Heart: four chambers. Ventricles have **thick muscular walls** as they pump blood to organs (lungs and the whole body). **Valves** stop backflow.

Body → Vena cava → Right atrium → Right ventricle → Pulmonary artery → Lungs → Pulmonary vein → Left atrium → Left ventricle → Aorta → Body

Double circulation: blood passes through the heart **twice** per cycle. Separation of oxygenated and deoxygenated blood gives a highly efficient O_2 supply, needed by birds and mammals (high energy, constant body temperature). Fish: 2 chambers, single circulation. Amphibians, most reptiles: 3 chambers, some mixing.

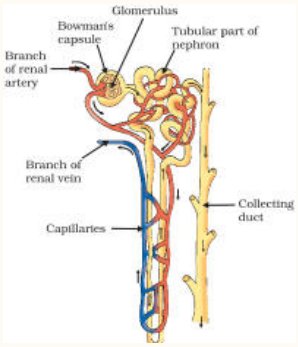
Arteries	Veins	Capillaries
carry blood **away** from heart	carry blood **to** heart	link arteries and veins
thick elastic walls (high pressure)	thin walls, **valves**	one-cell-thick walls: exchange with cells
pulmonary artery: deoxygenated blood	pulmonary vein: oxygenated blood	

Xylem	Phloem
water and minerals, from roots upward	food (sucrose), amino acids: **translocation**, both ways
vessels, tracheids; dead cells	sieve tubes, companion cells; living
root pressure (night), **transpiration pull** (day): no energy	uses **ATP**; osmotic pressure pushes food to where needed

Transpiration: loss of water vapour from aerial parts; creates suction that pulls water up, helps absorb minerals and regulates temperature.

4. Excretion

Human excretory system: a pair of **kidneys**, ureters, urinary bladder, urethra. The basic filtering unit is the nephron (about a million per kidney).



Structure of a nephron

Glomerulus filters blood in Bowman's capsule → Tubule reabsorbs glucose, amino acids, salts, water → Collecting duct → Urine

About **180 L** of filtrate forms daily but only 1–2 L is excreted as urine; the rest is reabsorbed. Urine is stored in the bladder under nervous control. **Haemodialysis** (artificial kidney) removes nitrogenous waste by diffusion through a semi-permeable tube, but there is no reabsorption.

Plants: extra O_2 and water vapour leave by diffusion and transpiration; wastes are stored in vacuoles, in leaves that fall off, as **resins and gums** in old xylem, or excreted into soil.

Board ka favourite (last 6 years)

- →Photosynthesis equation and its 3 events; role of stomata and guard cells.
- →Functions of HCl, pepsin, bile, trypsin, lipase; role of villi.
- →Flow chart of breakdown of glucose by 3 pathways.
- →Double circulation; why ventricles are thick; separation of blood in birds and mammals.
- →Nephron structure and urine formation; dialysis.
- →Diagrams: alimentary canal, heart, nephron, stomata, leaf section.

Common mistakes

- →Writing that bile contains enzymes (it has **none**; it emulsifies fats).
- →Saying all arteries carry oxygenated blood: **pulmonary artery** carries deoxygenated blood.
- →Mixing up ethanol (yeast) and lactic acid (muscles) in anaerobic respiration.
- →Writing that plants respire only at night: they respire **all the time**.
- →Writing that filtration happens in the tubule: it happens in the **glomerulus**; the tubule does **selective reabsorption**.

Yaad karo

Digestion ka order: **"Mere Saath Sab Log Aao"**: Mouth, Stomach, Small intestine, Large intestine, Anus. Artery = Away from heart.

Quick revision

1. Photosynthesis: $6CO_2 + 12H_2O \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$ (chlorophyll, sunlight).
2. Stomata open and close by guard cells; they also lose water by transpiration.
3. Digestion: saliva (amylase), stomach (pepsin, HCl, mucus), small intestine (bile, pancreatic juice, villi absorb).
4. Aerobic respiration gives 38 ATP; anaerobic in yeast gives ethanol + CO_2; in muscles, lactic acid (cramps).

Control and Coordination

Most asked: neuron and synapse, reflex arc, parts of the brain and their functions, plant hormones and tropisms, endocrine glands and hormones

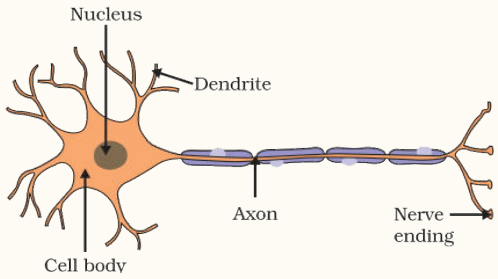
Animals use nerves and hormones to control and coordinate the body; plants use only hormones.

1. Nervous System in Animals

Body activities must be controlled and body parts must work together (**coordination**). Animals use two systems: the **nervous system** (fast, electrical) and the **endocrine system** (slower, chemical hormones).

Receptors: specialised tips of nerve cells that detect information: **gustatory** (taste, tongue), **olfactory** (smell, nose), photoreceptors (eye), phonoreceptors (ear), thermoreceptors (skin).

Neuron: the structural and functional unit of the nervous system. **Dendrites** pick up information → a chemical reaction creates an **electrical impulse** → impulse travels from dendrite to **cell body** and along the **axon** to the nerve ending → nerve ending releases **chemicals** that cross the synapse (gap) and start a new impulse in the next neuron (or a muscle/gland cell).



Structure of a neuron

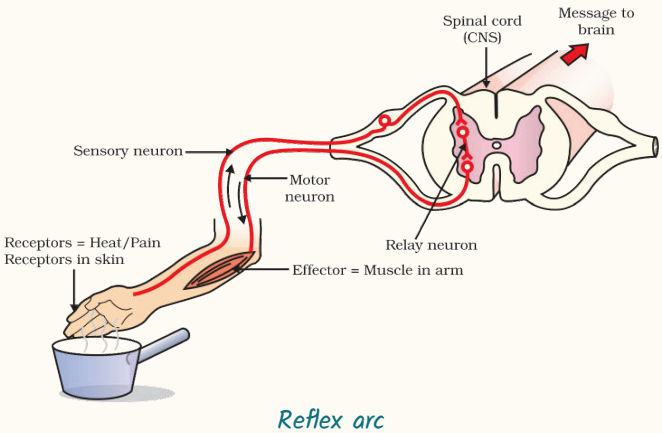
Dendrite → Cell body → Axon → Nerve ending → Synapse → Next neuron

Neuron type	Carries impulse
Sensory (afferent)	from receptor to spinal cord / brain
Motor (efferent)	from brain / spinal cord to effector (muscle, gland)
Relay (inter-neuron)	between sensory and motor neurons in the CNS

Reflex action and reflex arc

Reflex action: a sudden, automatic, involuntary response to a stimulus without conscious thinking, e.g. pulling the hand away from a flame, knee-jerk, blinking, sneezing. The pathway is the **reflex arc**; it is formed in the **spinal cord**, so the response is quick, and the information also reaches the brain.

Receptor (skin) → Sensory neuron → Relay neuron in spinal cord → Motor neuron → Effector (muscle)

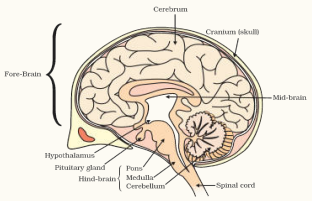


Reflex arc

Part	Made of	Job
Central (CNS)	brain, spinal cord	receives information, decides, sends orders
Peripheral (PNS)	cranial nerves (brain), spinal nerves (spinal cord)	link CNS with all body parts

Human brain

Part	Region	Functions
Fore-brain	cerebrum (largest)	main thinking part; sensory areas (sight, hearing, smell), motor areas for voluntary muscles; association areas; **hunger** centre
Mid-brain		involuntary actions such as reflex movement of head, neck and eyes
Hind-brain	**cerebellum**	precision of voluntary actions, **posture and balance** (walking, riding, picking a pencil)
	pons	regulates breathing
	medulla	blood pressure, salivation, vomiting (involuntary actions)



Human brain

Protection: brain sits in a bony box (**cranium**) cushioned by a fluid (cerebrospinal fluid) that absorbs shocks; the spinal cord is protected by the **vertebral column.**

Voluntary	Involuntary	Reflex
under our will; fore-brain	not under our will; mid- and hind-brain	automatic; spinal cord
writing, talking, walking	heartbeat, breathing, salivation	knee-jerk, blinking, sneezing

How muscles move: a nerve impulse reaches the muscle; muscle cells have special **proteins** that change their shape and arrangement, so the muscle cell becomes shorter.

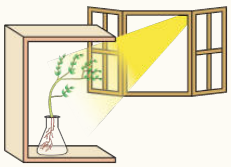
2. Coordination in Plants

Plants have **no nervous system** and no muscles. They respond by (i) **immediate movement** not linked to growth and (ii) slow **growth movements** (tropisms).

Touch-me-not (Mimosa pudica): leaves fold on touch. Cells pass the message using electrical-chemical means; cells then **change shape by changing the amount of water** in them (swelling or shrinking). No growth is involved.

Immediate (nastic) movement	Growth (tropic) movement
not directional, not due to growth	directional; due to unequal growth
fast; cells gain or lose water	slow; seen after hours or days
folding of Mimosa leaves	shoot bending towards light

Tropism	Stimulus	Example
Phototropism	light	shoot bends towards light (+), root away (–)
Geotropism	gravity	root grows down (+), shoot up (–)
Hydrotropism	water	roots grow towards water
Chemotropism	chemicals	pollen tube grows towards ovule
Thigmotropism	touch	tendrils coil around a support (less growth on the side in contact)


Phototropism

Geotropism

Plant hormone	Main function
Auxin	made at shoot tip; cell elongation; on the shady side it makes cells grow longer, so the shoot bends **towards light**
Gibberellins	growth of the stem
Cytokinins	cell division; high in fruits and seeds
Abscisic acid	**inhibits growth**; wilting of leaves (stress hormone)

3. Hormones in Animals

Hormones: chemical messengers secreted by **endocrine (ductless) glands** directly into the blood; they act on target cells. Electrical impulses reach only cells connected to nerves and cells need time to reset, so hormones are needed for long-lasting control.

Gland	Hormone	Function / disorder
Hypothalamus	releasing hormones	stimulates pituitary to release hormones (e.g. growth hormone releasing factor)
Pituitary	growth hormone	growth and development; less: **dwarfism**, more: gigantism
Thyroid	**thyroxin**	carbohydrate, protein, fat metabolism; needs **iodine**; lack of iodine: **goitre**
Pancreas	**insulin**	regulates blood sugar; less insulin: **diabetes**
Adrenal	**adrenaline**	fight or flight: faster heartbeat, more breathing, blood diverted to skeletal muscles
Testes	testosterone	male changes at puberty
Ovaries	oestrogen	female changes at puberty, menstrual cycle

Feedback mechanism: timing and amount of a hormone are regulated by feedback. Example: blood sugar rises → pancreas cells detect it → more insulin; sugar falls → insulin secretion reduces. **Iodised salt** is advised to prevent goitre.

Nervous control	Hormonal control
electrical impulses, chemicals at synapses	chemicals (hormones) through blood
very fast, short-lived	slower, long-lasting
reaches only connected cells	reaches all cells; acts on target cells

Board ka favourite (last 6 years)

- →Draw and label a neuron; how an impulse crosses a synapse.
- →Reflex arc flow chart and why reflexes are formed in the spinal cord.
- →Functions of cerebrum, cerebellum, medulla, pons.
- →Movement in Mimosa vs phototropism; role of auxin in bending.
- →Thyroxin and iodine (goitre), insulin (diabetes), adrenaline (emergency); feedback mechanism.

Common mistakes

- →Writing that reflex actions are controlled by the brain: the reflex arc is in the **spinal cord**.
- →Calling Mimosa folding a tropism: it is **not directional and not growth**.
- →Saying auxin collects on the side facing light: it diffuses to the **shady side**.
- →Mixing cerebellum (balance) with cerebrum (thinking).

Yaad karo

Plant hormones: **"Aaj Ghar Chalo, Aaram"**: Auxin, Gibberellin, Cytokinin grow karate hain, Abscisic acid aaram (growth band).

Quick revision

1. Neuron: dendrite → cell body → axon → nerve ending; gap is the synapse (chemical signal).
2. Reflex arc: receptor → sensory neuron → spinal cord → motor neuron → effector.
3. Cerebrum: thinking and voluntary actions; cerebellum: balance and posture; medulla: involuntary actions.

How do Organisms Reproduce?

Most asked: modes of asexual reproduction, flower LS and fertilisation, human reproductive systems and placenta, contraception and STDs

Reproduction makes copies of the organism; DNA copying with small variations keeps species alive.

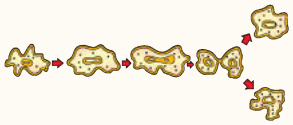
1. Why Reproduce? DNA Copying and Variation

Reproduction is not essential for an individual to survive, but it is essential for the **species**. The DNA in the nucleus carries information for making proteins, so the basic event in reproduction is the **creation of a DNA copy** by chemical reactions, along with extra cellular apparatus. Copying is never 100% accurate, so **variations** arise.

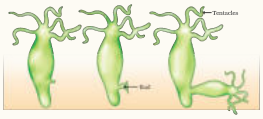
Importance of variation: maintains stability of populations in their **niche**; helps the species **survive** drastic changes (e.g. a rise in water temperature may kill most bacteria, but heat-resistant variants survive). Variation is the basis of evolution.

2. Asexual Reproduction (single parent)

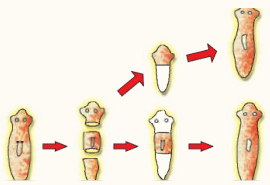
Mode	How it happens	Examples
Binary fission	cell divides into two equal halves (any plane in Amoeba; definite plane in Leishmania)	Amoeba, Leishmania (kala-azar), bacteria
Multiple fission	one cell divides into many daughter cells at once	**Plasmodium** (malaria)
Fragmentation	mature filament breaks into pieces; each grows into a new individual	Spirogyra
Regeneration	a cut body part grows into a complete organism by specialised cells (not same as reproduction)	Planaria, Hydra
Budding	a bud grows out due to repeated cell division at one site, detaches when mature	Hydra, yeast
Spore formation	sporangia release spores with thick walls that survive harsh conditions	Rhizopus (bread mould)
Vegetative propagation	new plants from root, stem or leaf	Bryophyllum (leaf-notch buds), potato, banana, rose, sugarcane



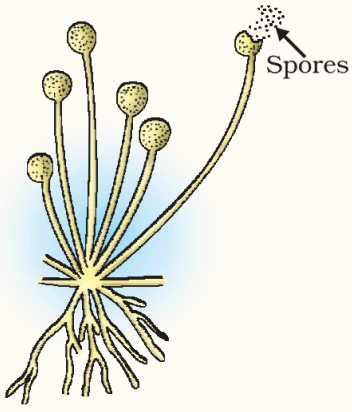
Binary fission in Amoeba



Budding in Hydra



Regeneration in Planaria



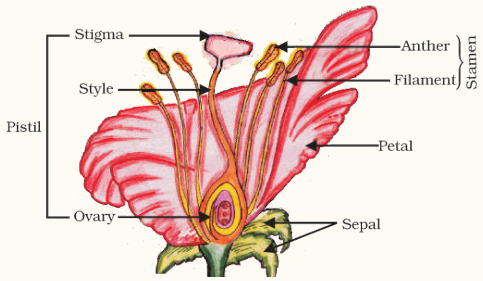
Spore formation in Rhizopus

Vegetative propagation: artificial methods are **layering, grafting, cutting** and **tissue culture** (new plants from cells of the growing tip, in an artificial medium). Advantages: plants that have lost the capacity to produce seeds (banana, orange, rose, jasmine) can be grown; plants bear flowers and fruits **earlier**; all plants are **genetically similar** to the parent, so good characters are kept.

3. Sexual Reproduction in Flowering Plants

Two parents contribute **germ cells (gametes)** that fuse. Germ cells carry **half the number of chromosomes** (made by a special division, meiosis), so the zygote restores the original number and the DNA content stays constant across generations. Mixing of DNA from two parents gives much more variation.

Asexual reproduction	*Sexual reproduction*
one parent; no gametes	two parents; male and female gametes fuse
offspring nearly identical (clones)	offspring show variation
fast, simple; common in lower organisms	slower; helps species adapt and survive



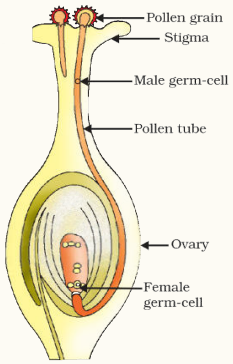
Longitudinal section of a flower

Part	*Role*
Sepals, petals	protect bud; attract insects
Stamen (male): anther, filament	anther makes **pollen grains** (male germ cells)
Pistil / carpel (female): stigma, style, ovary	ovary has **ovules**; each ovule has an **egg cell**

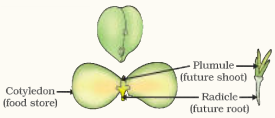
Bisexual flowers have both stamens and pistil (Hibiscus, mustard); **unisexual** have either (papaya, watermelon).

Pollination: transfer of pollen from anther to stigma; **self-pollination** (same flower) or **cross-pollination** (another flower, by wind, water, insects, animals). A **pollen tube** grows from the pollen through the style to reach the ovule; the male germ cell fuses with the egg: fertilisation.

Zygote → embryo → Ovule → seed → Ovary → fruit → Petals, sepals, stamens wither and fall



Germination of pollen on stigma

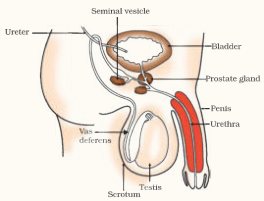


Germination of a seed

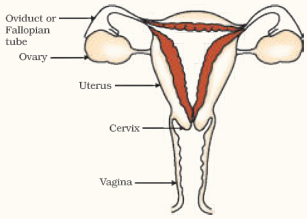
Seed contains the embryo (future plant): **plumule** (future shoot), **radicle** (future root) and **cotyledons** (stored food). It germinates under suitable conditions.

4. Reproduction in Human Beings

Puberty: the period of sexual maturation. Common changes: thick hair in armpits and genital area, oily skin, pimples. Boys: facial hair, voice cracks, penis enlarges. Girls: breasts enlarge, menstruation begins. Changes are slow and vary between individuals.



Male reproductive system



Female reproductive system

Organ	*Function*
Testes (in scrotum)	make sperms and **testosterone**; scrotum keeps them about 2–3 °C below body temperature, needed for sperm formation
Vas deferens	carries sperms; unites with the tube from the bladder (urethra: common passage)
Seminal vesicles, prostate	add fluid: nutrition and easy transport of sperms (semen)
Ovaries	make eggs (one egg each month by one ovary) and hormones **oestrogen, progesterone**
Oviduct (fallopian tube)	carries egg; site of **fertilisation**
Uterus	embryo implants and develops; opens into vagina through cervix

The zygote divides and the embryo **implants** in the thick, blood-rich uterine lining. Placenta: a disc embedded in the uterine wall with **villi** on the embryo side that give a large surface for **glucose and oxygen** to pass from mother to embryo and for wastes to pass back. Development takes about **nine months**; birth by rhythmic contraction of uterine muscles.

Menstruation: if the egg is not fertilised, it lives about one day; the thickened uterine lining breaks down and comes out as blood and mucus. Cycle: about **28–30 days**, lasting 2–8 days.

5. Reproductive Health

Sexually transmitted diseases (STDs): bacterial: gonorrhoea, syphilis; **viral**: warts, **HIV-AIDS**. Using a condom (covering for penis) during sex helps prevent transmission.

Method of contraception	*How it works*	*Example*
Barrier (mechanical)	stops sperm reaching egg; also protects from STDs	condom, diaphragm
Hormonal	changes hormonal balance so eggs are not released; side effects possible	oral pills
IUCD	placed in uterus; may cause irritation	loop, **copper-T**
Surgical	vas deferens blocked (**vasectomy**) or fallopian tube blocked (**tubectomy**)	permanent

Pre-natal sex determination is **prohibited by law**; reckless female foeticide is reducing the child sex ratio. Family planning protects women's health (child bearing, repeated pregnancies) and controls population.

Board ka favourite (last 6 years)

→Binary vs multiple fission; regeneration in Planaria; budding in Hydra; spore formation in Rhizopus (with diagrams).

→Advantages of vegetative propagation and tissue culture.

→LS of flower, pollen tube growth; post-fertilisation changes.

→Functions of testes, scrotum, seminal vesicles, oviduct, uterus; role of placenta.

→Contraceptive methods; why condoms prevent STDs; why pre-natal sex determination is banned.

Common mistakes

→Writing that ovary becomes seed: **ovule → seed, ovary → fruit.**

→Saying fertilisation takes place in the uterus: it occurs in the **oviduct.**

→Calling regeneration and fragmentation the same.

→Writing that germ cells have the full chromosome number (they have **half**).

Yaad karo

Asexual modes: **"Fauji Fauji Rahe, Bina Sone Vaale"**: Fission, Fragmentation, Regeneration, Budding, Spore formation, Vegetative propagation. Ovule = seed, Ovary = fruit (O-V-A-R-Y bada, fruit bhi bada).

Quick revision

1. Asexual modes: fission, fragmentation, regeneration, budding, spore formation, vegetative propagation.
2. DNA copying gives variation, which helps species survive.
3. Flower: stamen (anther, filament) is male; pistil (stigma, style, ovary) is female.
4. After fertilisation: ovule becomes seed, ovary becomes fruit.
5. Testis makes sperm and testosterone; ovary makes egg, oestrogen and progesterone.
6. Fertilisation happens in the oviduct; embryo implants in the uterus.

HEREDITY

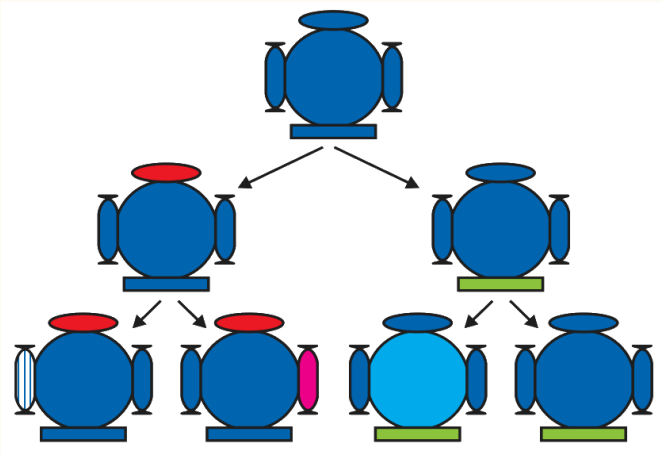
Most asked: monohybrid cross with Punnett square (3:1 and 1:2:1), dihybrid cross 9:3:3:1, Mendel's laws, how traits are expressed, sex determination in humans

Traits pass from parents to offspring through genes; small variations appear in every generation.

1. Heredity and Variation

Heredity: the transmission of characters (traits) from parents to offspring. **Variation:** differences between individuals of the same species. **Gene:** a segment of **DNA** on a chromosome that carries information for one protein, and so controls one trait.

Accumulation of variation: DNA copying in reproduction is never perfect, so small variations appear. **Asexual** reproduction gives very little variation; **sexual** reproduction gives much greater diversity because DNA from two parents mixes. Variations pass on to the next generation and accumulate. Not all variations survive equally; nature selects the ones useful in an environment (e.g. heat-resistant bacteria survive a heat wave).



Creation of diversity over succeeding generations (sexual reproduction)

Term	Meaning	Example
Allele	alternative forms of a gene for one trait	T (tall), t (short)
Dominant	trait that shows itself in F1 (in Tt)	tall (T)
Recessive	trait hidden in F1; shows only when two copies (tt)	short (t)
Genotype	genetic make-up	TT, Tt, tt
Phenotype	visible trait	tall, short
Homozygous / pure	both alleles same	TT, tt
Heterozygous / hybrid	two different alleles	Tt
F1, F2	first and second filial generations	offspring, then their offspring

Rules of inheritance: both parents contribute **equal genetic material**; so each trait is controlled by **two copies** of a gene, one from each parent. A child's trait depends on which copy is dominant.

2. Mendel's Contributions

Gregor Johann Mendel (1822–1884), the "Father of Genetics", worked on the **garden pea** (*Pisum sativum*). He used **mathematics** and kept count of individuals with each trait in each generation.

Why pea? clear **contrasting traits**; naturally **self-pollinating** but easy to cross-pollinate; short life cycle; many seeds per generation.

Character	Dominant	Recessive
Seed shape	round	wrinkled
Seed colour	yellow	green
Flower colour	violet	white
Pod shape	full (inflated)	constricted
Pod colour	green	yellow
Flower position	axial	terminal
Plant height	tall	short (dwarf)

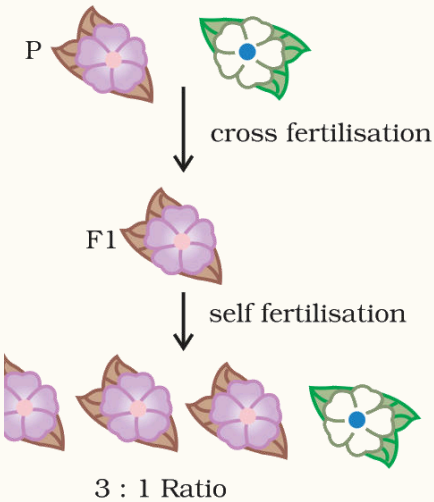
Monohybrid cross (one pair of traits): tall × short

P: TT (tall) × tt (short) → Gametes: T and t → F1: all Tt (tall) → F1 selfed: Tt × Tt

Gametes	T	t
T	TT (tall)	Tt (tall)
t	Tt (tall)	tt (short)

F2 phenotypic ratio = 3 tall : 1 short; **genotypic ratio** = 1 TT : 2 Tt : 1 tt

Conclusions: (i) the shortness trait was **not lost** in F1, it was only hidden: it reappeared in F2, so traits are inherited as separate units (genes). (ii) Of the two traits, the one expressed in F1 is **dominant**. (iii) Each gamete carries **only one** copy of each gene (law of **segregation**). A single copy of T is enough to make the plant tall.



Inheritance of a trait over two generations (3 : 1)

Cross	Offspring	Tells us
TT × tt	all Tt (tall)	parents pure
Tt × tt	1 tall : 1 short	tall parent is hybrid (Tt)
Tt × Tt	3 tall : 1 short	both parents hybrid

Dihybrid cross (two pairs of traits)

P: RRYY (round yellow) × rryy (wrinkled green) → F1: all RrYy (round yellow) → F1 gametes: RY, Ry, rY, ry

♀ / ♂	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F2 phenotypic ratio = 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green

Mendel's actual F2 count (556 seeds): 315 round yellow, 108 round green, 101 wrinkled yellow, 32 wrinkled green. Two **new combinations** (round green, wrinkled yellow) appear, so the two traits are inherited **independently** of each other (law of **independent assortment**).

Point	Monohybrid	Dihybrid
traits studied	one pair	two pairs
F2 phenotypic ratio	3 : 1	9 : 3 : 3 : 1
F2 genotypic ratio	1 : 2 : 1	9 kinds (1:2:1:2:4:2:1:2:1)
law shown	dominance, segregation	independent assortment

3. How do Traits get Expressed?

DNA → **protein**. A section of DNA giving information for one protein is a **gene**. Example: plant height depends on the amount of a **plant hormone** (growth hormone), which is made by an **enzyme**.

Gene → Enzyme (protein) → Hormone amount → Height

If the gene makes an **efficient enzyme**, much hormone is made and the plant is **tall**; if the gene

is altered, the enzyme is less efficient, less hormone is made and the plant is **short**. Each germ cell has **one set** of chromosomes; each chromosome pair separates independently during gamete formation, so both parents contribute one copy of each gene and the original number is restored in the zygote.

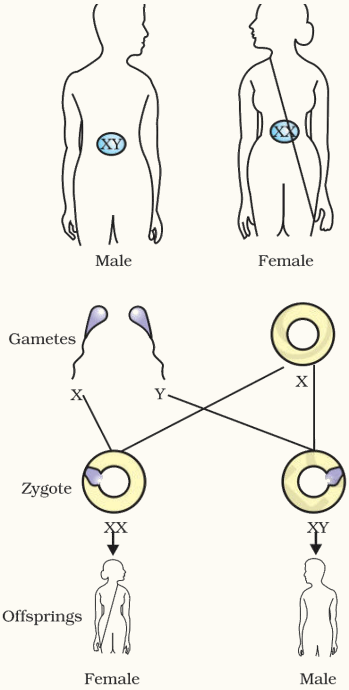
4. Sex Determination

In some animals sex is decided by **environment**: the **temperature** at which fertilised eggs are kept (some reptiles, e.g. turtles), or individuals can change sex (snails). In **humans** it is **genetic**.

Humans have **23 pairs** of chromosomes: **22 pairs autosomes + 1 pair sex chromosomes**. Female: **XX** (a perfect pair); male: **XY** (Y is short). All eggs carry X; half the sperms carry X and half carry Y.

Gametes	X (egg)	X (egg)
X (sperm)	XX girl	XX girl
Y (sperm)	XY boy	XY boy

Chance of a boy or a girl = 50% each; the sex of the child depends on whether the egg is fertilised by an X-carrying or a Y-carrying sperm, so the **father** decides. The mother contributes only X.



Sex determination in human beings

Mendel's laws in one line

→**Law of dominance:** in a hybrid (Tt) only the dominant trait is expressed.
→**Law of segregation:** the two copies of a gene separate during gamete formation; each gamete gets one.
→**Law of independent assortment:** genes for different traits are inherited independently (dihybrid 9:3:3:1).

Board ka favourite (last 6 years)

→Monohybrid cross with Punnett square: give F1, F2, both ratios; why short trait disappears in F1.
→Dihybrid cross: F2 ratio 9:3:3:1 and what it proves (independent inheritance).
→Why variation is more in sexual reproduction.
→How a gene controls a trait (height example).
→Sex determination in humans; why the father decides; Mendel's reasons for choosing pea.

Common mistakes

→Writing 1:2:1 as the phenotypic ratio: it is the **genotypic** ratio; phenotypic is **3:1**.
→Writing gametes as TT or Tt: a gamete carries **one** allele (T or t).
→Saying the mother decides the sex: **all eggs carry X**; the sperm decides.
→Calling 9:3:3:1 a genotypic ratio: it is **phenotypic**.

Yaad karo

3:1 = ek trait (mono), **9:3:3:1** = do traits (di); total 16 boxes. Ladka ya ladki: **papa ka Y** decide karta hai.

Quick revision

1. Mendel's monohybrid cross: F2 ratio 3 : 1 (phenotype), 1 : 2 : 1 (genotype).
2. Dihybrid cross F2 ratio: 9 : 3 : 3 : 1, showing independent inheritance.

Light: Reflection and Refraction

Most asked: ray diagrams for mirrors and lenses, mirror/lens formula numericals, refractive index, power of a lens

Light travels in straight lines; it bounces off mirrors (reflection) and bends through glass, water and lenses (refraction).

1. Reflection and Spherical Mirrors

Laws of reflection: (i) $\angle i = \angle r$; (ii) the incident ray, the normal at the point of incidence and the reflected ray lie in the **same plane**. They hold for spherical mirrors too.

Plane mirror image: virtual, erect, same size (m = +1), as far behind as the object is in front, **laterally inverted**.

Real image: rays actually meet, can be taken on a screen, always inverted. **Virtual image:** rays only appear to meet, cannot be taken on a screen, always erect.

Concave mirror (reflecting surface curved inwards) is converging; **convex mirror** (curved outwards) is diverging.

Key terms

- →**Pole (P):** centre of the reflecting surface. **Centre of curvature (C):** centre of the sphere of which the mirror is a part.
- →**Radius of curvature (R):** PC. **Principal axis:** line through P and C.
- →**Principal focus (F):** point where rays parallel to the axis meet (concave) or appear to diverge from (convex) after reflection. **Focal length f** = PF. **Aperture:** diameter of the reflecting surface.

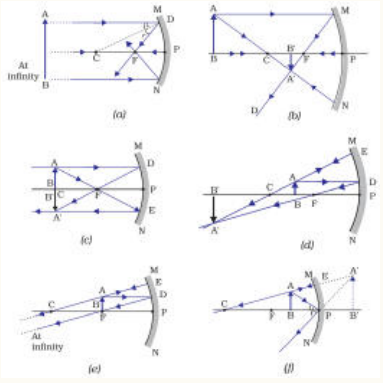
$R = 2f$ (F lies midway between P and C)

Rules for ray diagrams (mirrors)

1. Ray parallel to the axis passes **through** F (concave) or appears to come from F (convex).
2. Ray through F (or directed towards F) becomes **parallel** to the axis.
3. Ray through C (or towards C) retraces its path.
4. Ray striking P obliquely reflects at an **equal angle** with the axis.

Images by a concave mirror

Object at	*Image at*	*Size and nature*
Infinity	F	Point-sized; real, inverted
Beyond C	Between F and C	Diminished; real, inverted
C	C	Same size; real, inverted
Between C and F	Beyond C	Enlarged; real, inverted
F	Infinity	Highly enlarged; real, inverted
Between P and F	Behind mirror	Enlarged; virtual, erect



Concave mirror: object at six positions

Convex mirror: image always behind the mirror between P and F (at F for object at infinity):

virtual, erect, diminished.

Uses of concave mirror: torches, headlights, searchlights (bulb at F gives a parallel beam); shaving and make-up mirrors (enlarged, erect image); dentist's mirror; solar furnaces. **Uses of convex mirror:** rear-view mirrors of vehicles (always erect image, **wider field of view**), blind turns, shop security mirrors.

2. Sign Convention, Mirror Formula, Magnification

1. Object always on the **left**; all distances from the pole P (optical centre O for lens).
2. Distances along incident light (right) are **+**; against it (left) **−**.
3. Heights above the axis **+**, below **−**.

	u	v real	v virtual	f
Concave mirror	−	−	+	−
Convex mirror	−	never	+	+
Convex lens	−	+	−	+
Concave lens	−	never	−	−

> Mirror: $1/v + 1/u = 1/f$ | $m = h'/h = -v/u$

m negative: real, inverted; m positive: virtual, erect. $|m| > 1$ enlarged, $|m| < 1$ diminished.

Solved. A 4.0 cm object is 25.0 cm from a concave mirror, f = 15.0 cm. u = −25, f = −15: $1/v = -1/15 + 1/25 = -2/75$, **v = −37.5 cm** (screen 37.5 cm in front). $h' = -vh/u$ = **−6.0 cm**: real, inverted, enlarged.

Solved. Convex rear-view mirror, R = 3.00 m, bus at 5.00 m. u = −5.00 m, f = +1.50 m: $1/v = 1/1.50 + 1/5.00$, **v = +1.15 m** (behind). $m = -v/u$ = **+0.23**: virtual, erect, diminished.

3. Refraction of Light

Refraction: bending of light when it passes obliquely from one transparent medium to another, because its **speed changes**. Examples: pencil looks bent in water, a pond looks shallower, letters look raised under a glass slab.

Glass slab: ray bends **towards** the normal entering glass (rarer to denser) and **away** on leaving; emergent ray is **parallel** to the incident ray but laterally displaced. A ray along the normal goes undeviated.

Laws of refraction

1. Incident ray, refracted ray and normal lie in the **same plane**.
2. **Snell's law:** for a given colour and pair of media, sin i / sin r = constant (refractive index of medium 2 w.r.t. medium 1).

> $n_{21} = v_1/v_2$ | $n_m = c/v$ | $n_{21} = 1/n_{12}$

$c = 3 \times 10^8$ m/s. Absolute refractive index: air 1.0003, water 1.33, crown glass 1.52, diamond 2.42 (highest). **Optically denser** = higher n, slower light; not the same as mass density (kerosene, n = 1.44, is optically denser than water but lighter).

Solved. n of glass = 1.50. Speed of light in glass $v = c/n = 3 \times 10^8/1.50$ = **2×10^8 m/s.** If n_{water} = 1.33, $n_{glass\ w.r.t.\ water} = 1.50/1.33$ = **1.13**.

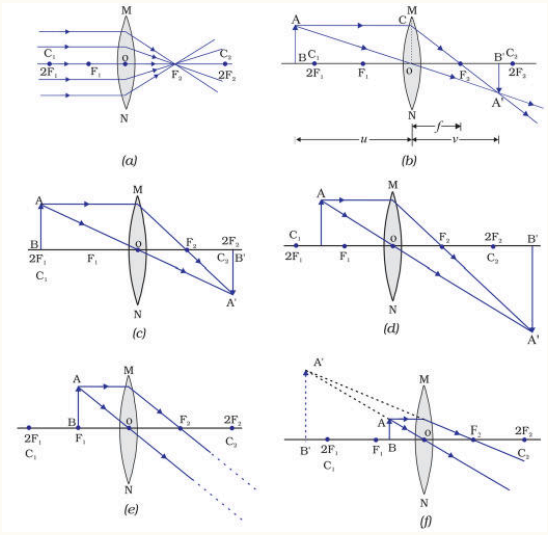
4. Spherical Lenses

Convex lens (thicker at the middle) is converging; **concave lens** (thinner at the middle) is diverging. Optical centre O: a ray through it goes **undeviated**. A lens has two foci F_1, F_2; $2F_1$, $2F_2$ are at twice the focal length.

Rules for ray diagrams (lenses)

1. Ray parallel to the axis passes through F_2 (convex) or appears to diverge from F_1 (concave).
2. Ray through F_1 (or towards F_2 for concave) emerges **parallel** to the axis.
3. Ray through O goes straight, undeviated.

Object at	Image at	Size and nature
Infinity	F_2	Point-sized; real, inverted
Beyond $2F_1$	F_2 to $2F_2$	Diminished; real, inverted
$2F_1$	$2F_2$	Same size; real, inverted
F_1 to $2F_1$	Beyond $2F_2$	Enlarged; real, inverted
F_1	Infinity	Highly enlarged; real, inverted
F_1 to O	Object side	Enlarged; virtual, erect



Convex lens: object at six positions

Concave lens: image always between F_1 and O on the object side: **virtual, erect, diminished** (at F_1 for object at infinity).

Lens formula, magnification, power

Lens: $1/v - 1/u = 1/f$ | $m = h'/h = v/u$

$P = 1/f$ (f in metres) | SI unit: dioptre (D) | $P = P_1 + P_2 + \ldots$

1 D = power of a lens of focal length 1 m. Convex lens: **P positive**; concave lens: **P negative**. An optician's prescription +2.0 D means a convex lens of f = +0.5 m. Uses: convex lens in magnifying glass (object within F), camera, spectacles for hypermetropia; concave lens in spectacles for myopia.

Solved. A 2.0 cm object is 10 cm from a concave lens, f = 15 cm. u = −10, f = −15: $1/v = 1/f + 1/u = -1/15 - 1/10 = -1/6$, **v = −6 cm**. $m = v/u = +0.6$, h′ = **+1.2 cm**: virtual, erect, diminished, same side.

Board ka favourite

→Ray diagram: concave mirror object between F and C / between P and F; convex lens object between F_1 and $2F_1$ / at $2F_1$.

→Mirror/lens formula numerical: find v, m, nature (3 marks).

→Refractive index from speed of light; which medium is optically denser.

→Power of a lens and combination; "convex lens of f = 50 cm, P = ?" (+2 D).

→Why convex mirror as rear-view mirror; concave mirror in headlights and by dentists.

→Case-based: magnifying glass, glass slab and lateral displacement.

Common mistakes

→Writing u as positive. Object distance is always negative.

→Mixing formulas: mirror has 1/v + 1/u and m = −v/u; lens has 1/v − 1/u and m = v/u.

→Power with f in cm. Convert to metres first (f = 25 cm gives P = 4 D).

→Ray diagrams without arrowheads, or virtual rays drawn solid (use dotted lines).

→Forgetting the nature of the image (position, size, nature all three).

1 minute recap

→Mirror: 1/v + 1/u = 1/f, m = −v/u; lens: 1/v − 1/u = 1/f, m = v/u; R = 2f; P = 1/f (m).

→Concave mirror, convex lens: real image except object inside F. Convex mirror, concave lens: always virtual, erect, diminished.

→n = c/v; higher n = optically denser = slower light; denser to rarer bends away from normal.

Yaad karo

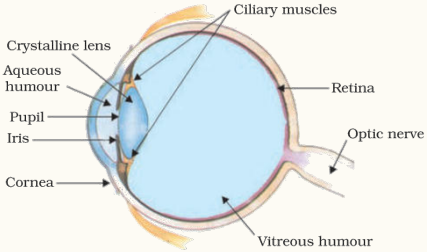
Mirror mein plus, lens mein minus (1/v + 1/u vs 1/v − 1/u). Concave mirror aur convex lens: object jitna F ke paas, image utni badi aur door. Convex mirror aur concave lens hamesha chhota, seedha, virtual.

The Human Eye and the Colourful World

Most asked: myopia and hypermetropia with correction, dispersion by a prism, twinkling of stars, blue sky and Tyndall effect

The eye focuses light with a natural convex lens; prisms, raindrops and air molecules bend and scatter light to give colours.

1. The Human Eye



The human eye

Part	Function
Cornea	Thin transparent bulge; **most refraction** happens here
Iris	Dark muscular diaphragm; controls the size of the pupil
Pupil	Opening that regulates light entering (small in bright light)
Eye lens	Convex, flexible; **fine adjustment** of focal length
Ciliary muscles	Change the curvature, hence focal length, of the lens
Retina	Light-sensitive screen (rods and cones); image is **real and inverted**
Optic nerve	Carries electrical signals to the brain

Cornea → Aqueous humour → Pupil → Eye lens → Vitreous humour → Retina → Optic nerve → Brain

Rods respond to the intensity of light (dim-light vision); **cones** respond to colour. The brain interprets the inverted image as erect.

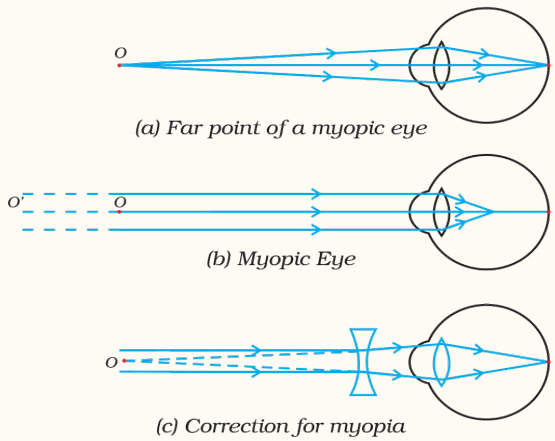
Power of accommodation: ability of the eye lens to adjust its focal length. Far object: ciliary muscles relax, lens thin, f longer. Near object: muscles contract, lens thick, f shorter.

Near point (least distance of distinct vision) = **25 cm** for a normal young eye; **far point = infinity**. Range of vision: 25 cm to infinity.

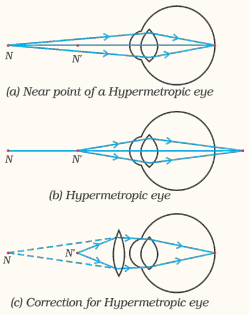
Cataract: the lens becomes milky and cloudy in old age, causing partial or complete loss of vision; restored by surgery.

2. Defects of Vision and their Correction

	Myopia (near-sightedness)	Hypermetropia (far-sightedness)
Sees clearly	Near objects only	Distant objects only
Point shifted	Far point < infinity	Near point > 25 cm
Image forms	**In front of** the retina	**Behind** the retina
Causes	Excessive curvature of eye lens; eyeball elongated	Focal length too long; eyeball too small
Correction	**Concave** lens (P negative)	**Convex** lens (P positive)



Myopic eye and its correction with a concave lens



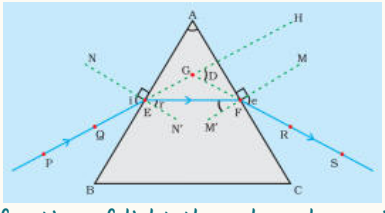
Hypermetropic eye and its correction with a convex lens

Presbyopia: with age the ciliary muscles weaken and the lens loses flexibility, so the near point recedes. Often both defects occur together: corrected by **bifocal lenses** (upper part concave for distance, lower part convex for reading). Today contact lenses and surgery also correct defects.

Solved. (i) Far point of a myopic eye = 80 cm. Lens must form the image of a distant object at 80 cm: f = **−80 cm**, P = 1/(−0.8) = **−1.25 D** (concave). (ii) Near point of a hypermetropic eye = 1 m. u = −25 cm, v = −100 cm: 1/f = 1/v − 1/u = −1/100 + 1/25 = 3/100, f = +33.3 cm, **P = +3.0 D** (convex).

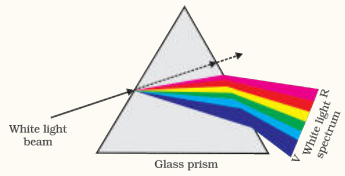
3. Refraction through a Prism and Dispersion

Prism: a transparent body with two triangular bases and three rectangular faces. **Angle of prism (A):** angle between two refracting faces. The emergent ray bends towards the base; the angle between the incident ray (produced) and emergent ray is the **angle of deviation (D)**.



Refraction of light through a glass prism

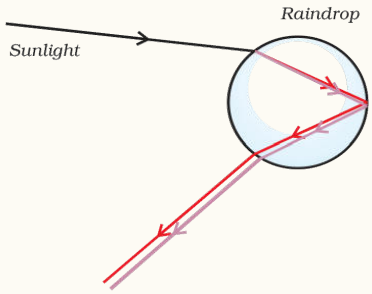
Dispersion: splitting of white light into its seven colours, VIBGYOR. Each colour bends through a different angle: **red bends least, violet most** (different speeds in glass). The band of colours is the **spectrum**.



Dispersion of white light by a glass prism

Newton: a second identical prism placed **inverted** recombines the colours into white light, so sunlight is made of seven colours. Any light giving a similar spectrum is called white light.

Rainbow: a natural spectrum seen after rain, always **opposite to the Sun**. Tiny water droplets act like prisms: **refraction and dispersion** on entering, **internal reflection** inside, and refraction on leaving the drop.



Rainbow formation in a water droplet

4. Atmospheric Refraction

The atmosphere has layers of varying optical density (hot air is rarer than cold air), so light bends as it passes through them. Objects seen through hot air above a fire appear to **flicker**.

Effect	Reason
Stars **twinkle**	Stars are distant **point sources**; starlight is refracted continuously by changing atmosphere, so its apparent position and brightness fluctuate
Stars appear higher than they are	Starlight bends towards the normal on entering denser lower layers
Planets do not twinkle	Planets are nearer, so they are **extended sources**; fluctuations from many points cancel out

5. Scattering of Light

Tyndall effect: scattering of light by colloidal particles makes the path of a beam visible, e.g. sunlight entering a smoky room or through the canopy of a dense forest. The colour scattered depends on particle size: **very fine particles scatter blue** (short wavelength) most; large particles scatter all colours equally (white).

Observation	Explanation
Sky is **blue**	Air molecules are smaller than the wavelength of visible light and scatter blue far more than red
Sky looks dark to astronauts at great height	No atmosphere there, so no scattering
Danger signals are **red**	Red is scattered least (longest wavelength), so it is seen from far, even in fog or smoke
Clouds are white	Water droplets are large and scatter all colours equally

Board ka favourite

→Myopia/hypermetropia: causes, ray diagram of defect and correction (3–5 marks), and power of the corrective lens.

→Draw dispersion of white light by a prism; which colour deviates most and least.

→Why stars twinkle but planets do not.

→Why the sky is blue; why danger signals are red; Tyndall effect examples.

→Function of ciliary muscles, iris, pupil; near point and far point values.

→Formation of a rainbow (labelled raindrop diagram).

Common mistakes

→Mixing lenses: myopia needs a **concave** (diverging) lens, hypermetropia a **convex** lens.

→Writing that the cornea only protects; it does most of the refraction.

→Saying red bends most in a prism. Violet deviates most.

→Rainbow: forgetting the **internal reflection** step.

→Calling the blue sky an effect of refraction; it is scattering.

→Hypermetropia numerical: u = −25 cm (normal near point) and v = −(defective near point), not the other way round.

1 minute recap

→Near point 25 cm, far point infinity; accommodation by ciliary muscles.

→Myopia: image in front of retina, concave lens, f = −(far point). Hypermetropia: image behind retina, convex lens.

→Dispersion: different colours, different speeds in glass; violet deviates most.

→Twinkling: atmospheric refraction. Blue sky and Tyndall effect: scattering.

Yaad karo

"Myopia = Minus lens" (concave, P negative), **"Hyper = Plus"** (convex). Prism mein **Violet Very much** mudta hai, Red Rarely.

Quick revision

1. Power of accommodation: ciliary muscles change the focal length of the eye lens.
2. Near point 25 cm, far point infinity for a normal eye.
3. Myopia: concave lens; hypermetropia: convex lens; presbyopia: bifocal lens.
4. Prism: dispersion into VIBGYOR; violet deviates most, red least.

ELECTRICITY

Most asked: Ohm's law and V-I graph, series/parallel numericals, R = ρl/A, Joule's heating, power and kW h

What makes charges flow, what controls the current, how resistors combine, and how current produces heat and power.

1. Electric Current and Potential Difference

Electric current (I): rate of flow of charge through a cross-section of a conductor. In metals the charge carriers are **electrons**; **conventional current** flows opposite to electron flow (from + to – outside the cell).

$$I = Q/t \quad | \quad 1\ A = 1\ C/1\ s \quad | \quad Q = ne$$

Charge on an electron $e = 1.6 \times 10^{-19}$ C, so 1 C ≈ 6×10^{18} electrons. 1 mA = 10^{-3} A, 1 μA = 10^{-6} A. Current is measured by an **ammeter**, always connected **in series** (low resistance).

Potential difference (V): work done to move a unit positive charge from one point to another in a circuit. A cell maintains a potential difference by its chemical action. Measured by a **voltmeter**, connected **in parallel** (high resistance).

$$V = W/Q \quad | \quad 1\ V = 1\ J/1\ C$$

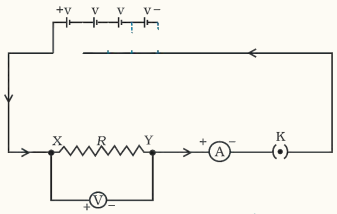
Circuit symbols

Component	Symbol	Component	Symbol
Cell		Battery	
Key open		Key closed	
Wire joint		Wires crossing	
Bulb		Resistor	
Rheostat		Ammeter	
Voltmeter			

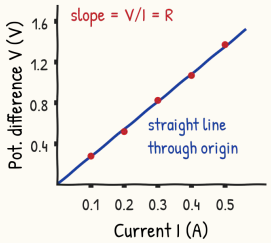
2. Ohm's Law and Resistance

Ohm's law: the current through a metallic conductor is directly proportional to the potential difference across its ends, **provided its temperature remains the same.**

$$V \propto I \quad | \quad V = IR \quad | \quad 1\ \Omega = 1\ V/1\ A$$



Circuit to verify Ohm's law



V-I graph: a straight line through the origin; slope = R

Resistance (R): property of a conductor to oppose the flow of charge. A **rheostat** (variable resistance) changes the current without changing the source voltage. Good conductor: low R; insulator: very high R.

Factors affecting resistance

$$R = \rho\, l/A \quad | \quad \text{SI unit of } \rho\text{: } \Omega\ m$$

→R ∝ length l; R ∝ 1/area A (thicker wire, less resistance).

→R depends on the **material** and **temperature**. ρ (resistivity) depends only on material and temperature, not on l or A.

→Metals: $\rho \approx 10^{-8}$ to 10^{-6} Ω m (alloys higher); insulators: 10^{12} to 10^{17} Ω m. Copper and aluminium (low ρ) for transmission wires.

→Alloys (nichrome) have higher ρ and do not oxidise (burn) readily at high temperature: used in heaters, irons, toasters.

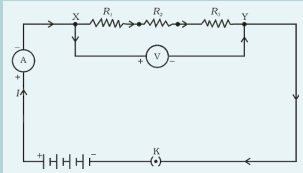
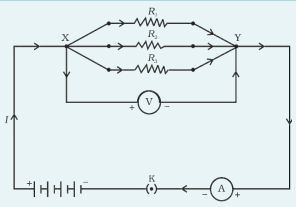
Material	ρ (Ω m)	Material	ρ (Ω m)
Silver	1.60×10^{-8}	Tungsten	5.20×10^{-8}
Copper	1.62×10^{-8}	Nichrome	100×10^{-6}
Aluminium	2.63×10^{-8}	Glass	10^{10} to 10^{14}

Wire stretched to double length: volume constant, so A halves and R becomes **4R**. Diameter halved: A becomes A/4, R becomes 4R.

Solved. A wire 1 m long, diameter 0.3 mm, has R = 26 Ω. $\rho = RA/l = 26 \times \pi(0.15 \times 10^{-3})^2/1$ = **1.84×10^{-6} Ω m** (manganese). If the diameter is doubled, R becomes 26/4 = 6.5 Ω.

Solved. A heater draws 4 A from a 60 V line. R = V/I = **15 Ω**; on 120 V it draws I = 120/15 = **8 A**.

3. Resistors in Series and Parallel

Series	Parallel
	
Same **current** through each resistor	Same **potential difference** across each
$V = V_1 + V_2 + V_3$	$I = I_1 + I_2 + I_3$
$R_s = R_1 + R_2 + R_3$	$1/R_p = 1/R_1 + 1/R_2 + 1/R_3$
R_s greater than the largest resistance	R_p less than the smallest resistance

Why parallel for homes: in series, if one appliance fails the whole circuit breaks, and all get the same current though each needs a different current. In parallel each gadget gets the full voltage, has its own switch and draws the current it needs; total resistance is lower.

Solved. 5 Ω, 10 Ω and 30 Ω in parallel across 12 V: $1/R_p = 1/5 + 1/10 + 1/30 = 10/30$, **$R_p$ = 3 Ω**, I = 12/3 = **4 A**. Currents: 2.4 A, 1.2 A, 0.4 A. In series the same three give R_s = 45 Ω, I = 12/45 = 0.27 A.

Solved (mixed). R_1 = 10 Ω and R_2 = 40 Ω in parallel, joined in series with R_3 = 2 Ω, across 6 V. $R_p = (10 \times 40)/50$ = 8 Ω; total R = 8 + 2 = **10 Ω**; I = 6/10 = **0.6 A**. V across R_3 = 1.2 V, across the parallel pair = 4.8 V, so I_1 = 0.48 A, I_2 = 0.12 A.

Two resistors only: $R_p = R_1R_2/(R_1 + R_2)$; n equal resistors R: $R_s = nR$, $R_p = R/n$, so $R_s/R_p = n^2$.

4. Heating Effect of Electric Current

In a purely resistive circuit the energy from the source is entirely dissipated as heat.

Joule's law of heating: heat produced is proportional to the square of the current, the resistance, and the time.

$$H = I^2Rt = VIt = (V^2/R)\,t$$

→**Electric iron, toaster, heater, geyser:** high-ρ alloy elements.

→**Bulb:** filament of **tungsten** (melting point 3380 °C), bulb filled with inert **nitrogen and argon** to prolong filament life; most energy is given out as heat, some as light.

→**Fuse:** wire of a metal or alloy of suitable low melting point (Al, Cu, Fe, Pb), connected **in series** with the device; melts when current exceeds the safe limit. Ratings 1 A, 2 A, 3 A, 5 A, 10 A. A 1 kW iron at 220 V draws 4.5 A, so it needs a 5 A fuse.

Solved. 100 J of heat is produced each second in a 4 Ω resistor. $P = I^2R$ gives $100 = I^2 \times 4$, $I = 5$ A, so $V = IR =$ **20 V.**

5. Electric Power

Electric power: rate at which electrical energy is consumed or dissipated. SI unit: watt (W); 1 W = 1 V × 1 A.

$$P = VI = I^2R = V^2/R \quad | \quad E = Pt$$

$$1 \text{ kW h} = 3.6 \times 10^6 \text{ J} = 1 \text{ unit}$$

Solved. A 400 W refrigerator runs 8 h a day. Energy in 30 days = 0.4 kW × 8 h × 30 = **96 kW h**; at Rs 3.00 per unit, cost = **Rs 288.** A bulb rated 100 W, 220 V: $R = V^2/P = 484$ Ω, $I = P/V = 0.45$ A.

Which bulb glows brighter? In **parallel** (same V) the higher-wattage bulb (lower R) is brighter, as $P = V^2/R$. In **series** (same I) the lower-wattage bulb (higher R) is brighter, as $P = I^2R$.

Board ka favourite

→State Ohm's law; draw the circuit to verify it and the V-I graph.

→Equivalent resistance of mixed series-parallel networks and current through each resistor (3–5 marks).

→$R = \rho l/A$: effect of doubling length, halving radius, stretching the wire.

→Why alloys in heating elements; why tungsten in bulbs; fuse and its rating.

→Power and energy: units consumed, cost, which bulb glows brighter (series vs parallel).

→Disadvantages of series connection for domestic circuits.

Common mistakes

→Connecting the voltmeter in series or ammeter in parallel in diagrams.

→Adding 1/R values and forgetting to take the reciprocal at the end.

→Thinking resistivity changes with length or thickness; only resistance does.

→Using time in hours with power in watts for joules; use seconds for J, hours with kW for kW h.

→Writing Ohm's law without the condition "temperature remains constant".

1 minute recap

→$I = Q/t$, $V = W/Q$, $V = IR$ (temperature constant), $R = \rho l/A$.

→Series: $R_s = \Sigma R$, same I. Parallel: $1/R_p = \Sigma 1/R$, same V.

→$H = I^2Rt$; $P = VI = I^2R = V^2/R$; 1 kW h = 3.6×10^6 J.

Yaad karo

"V = IR, Virat India Ruled". Series mein **current same**, parallel mein **voltage same.** kW h ko J mein badalna ho toh × 3.6×10^6.

Magnetic Effects of Electric Current

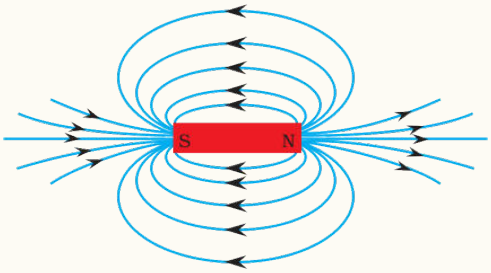
Most asked: field lines and their properties, right-hand thumb rule, solenoid, Fleming's left-hand rule, domestic circuits and safety

A current-carrying wire behaves like a magnet: its field, the force it feels in a magnetic field, and safe wiring at home.

1. Magnetic Field and Field Lines

Oersted (1820): a compass needle placed near a current-carrying wire is deflected, so **electricity and magnetism are linked**. Reversing the current reverses the deflection. The unit of magnetic field strength, the oersted (CGS), is named after him.

Magnetic field: the region around a magnet in which its force can be detected. It has both **magnitude and direction**. Direction at a point = the direction in which the north pole of a compass needle points there.



Field lines around a bar magnet

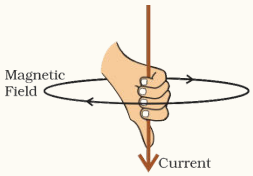
Properties of magnetic field lines

1. Outside the magnet they go from **N to S**; inside, from S to N. They are **closed curves**.
2. Their **closeness** shows strength: crowded near the poles, where the field is strongest.
3. **No two field lines ever intersect**: at the crossing point a compass would point in two directions at once, which is impossible.

2. Field due to a Current-carrying Conductor

(a) Straight wire

Field lines are **concentric circles** centred on the wire, in a plane perpendicular to it. Field strength ∝ current I and ∝ 1/(distance from the wire). Reversing the current reverses the field direction.

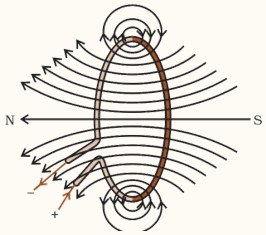


Right-hand thumb rule

Right-hand thumb rule: hold the current-carrying conductor in the right hand with the **thumb along the current**; the curled **fingers show the direction of the field lines**. (Also called Maxwell's corkscrew rule.)

(b) Circular loop

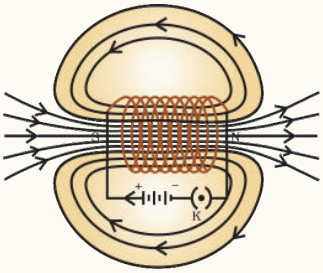
Every small part of the loop acts like a straight wire. Near the wire the lines are circles; at the **centre** they are nearly **straight**, and all segments add their fields in the same direction. For a coil of n turns the field is **n times** that of one turn; it increases with current.



Field lines of a current-carrying circular loop

(c) Solenoid

Solenoid: a coil of many circular turns of insulated copper wire wrapped closely in the shape of a cylinder. Its field pattern is **like that of a bar magnet**: one end acts as N, the other as S.



Field lines of a current-carrying solenoid

→Inside the solenoid the field lines are **parallel straight lines**: the field is **uniform** (same at all points).

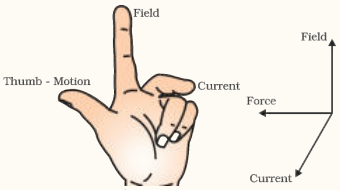
→Strength increases with number of turns, current, and a soft iron core.

Electromagnet: a soft iron core placed inside a solenoid gets strongly magnetised while current flows and loses magnetism when it stops. Used in cranes, electric bells, relays.

Electromagnet	*Permanent (bar) magnet*
Temporary: magnetic only while current flows	Retains magnetism
Strength can be changed (turns, current)	Fixed strength
Polarity reverses with current	Fixed polarity
Soft iron core	Steel, alnico, cobalt alloys

3. Force on a Current-carrying Conductor

A conductor carrying current in a magnetic field experiences a force (Ampère). The displacement of the rod **reverses** when either the current or the field direction is reversed. The force is **maximum when the current is perpendicular to the field** and zero when parallel.



Fleming's left-hand rule

Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the **left hand** mutually perpendicular. Forefinger = Field, middle finger = current, thumb = motion (force). Devices using this: electric motors, loudspeakers, microphones, measuring instruments.

A moving charge (an electron or alpha particle beam) is a current, so it is also deflected in a magnetic field; for electrons take the current opposite to their motion.

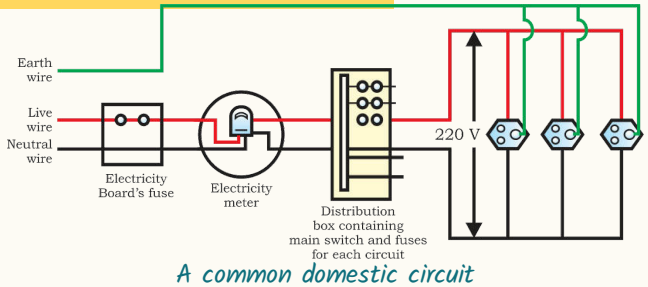
Magnetism in medicine: weak ionic currents in the heart and brain produce magnetic fields inside the body; **MRI** (magnetic resonance imaging) uses this to image internal organs.

4. Direct and Alternating Current

Direct current (DC)	*Alternating current (AC)*
Flows in one direction only	Reverses direction periodically
Cells, batteries, solar cells	Power stations, household supply
Cannot be stepped up or down easily	Transmitted over long distances without much energy loss

In India AC has a **frequency of 50 Hz**: it changes direction every **1/100 s**. The **advantage of AC** over DC: electric power can be transmitted over long distances with much less loss.

5. Domestic Electric Circuits



A common domestic circuit

Wire	*Insulation*	*Role*
Live (positive)	Red	At 220 V
Neutral (negative)	Black	At 0 V; potential difference between live and neutral is 220 V
Earth	Green	Connected to a metal plate deep in the earth; joined to metal bodies of appliances

→Supply enters through the main fuse and the electricity meter, then the distribution box.

→Two circuits: **15 A** for high-power appliances (geyser, air cooler, iron), **5 A** for bulbs and fans.

→All appliances are connected **in parallel**, each with its own switch, so each gets 220 V and works independently.

→**Earth wire:** gives a low-resistance path to current leaking to the metal body; keeps the body at earth potential, so the user does not get a severe shock.

→**Fuse / MCB:** protects circuits and appliances from **short-circuiting** and **overloading** by cutting off excess current.

Short-circuit: live and neutral wires come in direct contact (damaged insulation or faulty appliance); resistance becomes very small, so current rises abruptly. **Overloading:** too many appliances on one socket, or an accidental hike in supply voltage, draws excess current.

Board ka favourite

→Draw field lines of a bar magnet; why two field lines never cross.

→Field pattern of a straight wire, loop and solenoid; right-hand thumb rule.

→Solenoid vs bar magnet; how to make an electromagnet; uniform field inside a solenoid.

→Fleming's left-hand rule: find direction of force (often for an electron or alpha beam; case-based).

→Earth wire, fuse, short-circuit vs overloading; why appliances are in parallel; 15 A vs 5 A circuits.

→Frequency of AC in India; advantage of AC over DC.

Quick revision

1. Field lines go from N to S outside a magnet; they never cross.
2. Right-hand thumb rule: direction of field around a straight conductor.

Common mistakes

→Field lines drawn without arrows, or crossing each other.

→Using the right hand for the force rule. Force is **Fleming's left-hand** rule; the right-hand thumb rule is for the field direction.

→For an electron beam, taking current along the motion; current is opposite.

→Saying field lines start at N and end at S; they are closed loops (S to N inside).

→Mixing up short-circuit (live touches neutral) with overloading (too much load).

1 minute recap

→Field lines: N to S outside, closed, never cross; crowded = strong.

→Straight wire: concentric circles (right-hand thumb rule). Solenoid: like a bar magnet, uniform field inside.

→Force: maximum when I is perpendicular to B; direction by Fleming's left-hand rule.

→AC in India: 50 Hz. Live red, neutral black, earth green; fuse in series in the live wire.

Yaad karo

FBI for the left hand: **F**orce = thumb, **B** (field) = forefinger, **I** (current) = middle finger. Wire colours: **Red live, Black neutral, Green earth.**

Frequently asked in Boards

1. Draw the field lines around a bar magnet. (2 marks)
2. State Fleming's left-hand rule. (2 marks)
3. What is the function of a fuse? (1 mark)
4. Draw the field pattern of a current-carrying solenoid. (3 marks)

Our Environment

Most asked: food chain and 10 percent law; biological magnification; ozone depletion and CFCs; biodegradable vs non-biodegradable waste

Every bite of food links us to a chain of energy that starts with the Sun, and every bit of waste comes back to us.

1. Ecosystem

Environment: everything that surrounds us: living (biotic) and non-living (abiotic) components. **Ecosystem:** all the organisms of an area together with the non-living factors, interacting with each other (biotic + abiotic).

	Natural ecosystem	Artificial (man-made) ecosystem
Examples	forest, pond, lake, grassland, ocean	garden, crop field, aquarium, park
Self-sustaining	yes	no, needs human care

Component	Examples
Biotic	plants, animals, micro-organisms
Abiotic	temperature, rainfall, wind, soil, minerals, water, sunlight

Producers, consumers and decomposers

→**Producers (autotrophs):** green plants and blue-green algae; make food (sugar, starch) by photosynthesis using sunlight and chlorophyll.

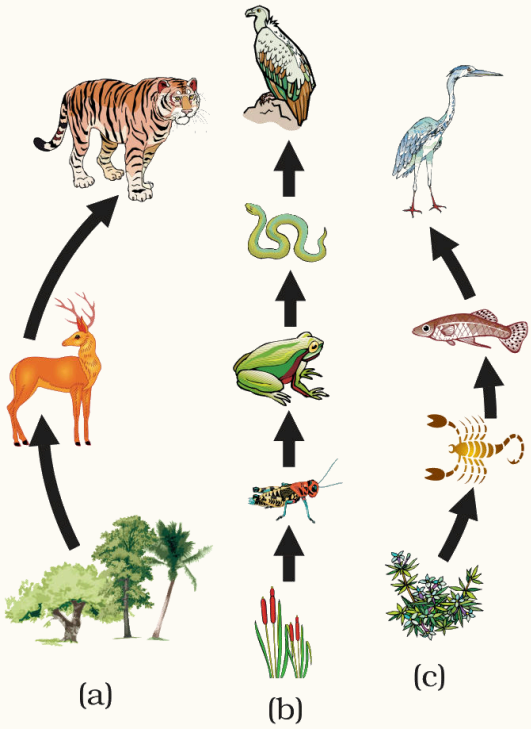
→**Consumers (heterotrophs):** depend on producers directly or indirectly. Herbivores (cow, deer), carnivores (lion, tiger), omnivores (human, crow), parasites (lice, Cuscuta).

→**Decomposers:** bacteria and fungi; break down dead remains and waste into simple substances that go back into the soil, and so **recycle nutrients** and keep the environment clean.

2. Food Chain, Trophic Levels and Energy Flow

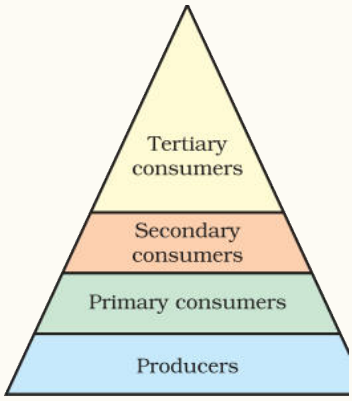
Food chain: a series of organisms in which each one feeds on the one before it. Each step is a **trophic level.**

Grass → Grasshopper → Frog → Snake → Eagle



Food chains: (a) forest (b) grassland (c) pond

Trophic level	Organisms
T1	producers (plants)
T2	primary consumers (herbivores)
T3	secondary consumers (small carnivores)
T4	tertiary consumers (large carnivores)



Trophic levels

Energy flow and the 10 percent law

→Green plants capture about **1% of the sunlight** falling on leaves and convert it into food energy.

→Only about **10%** of the energy of one trophic level is passed on to the next; the rest is used in life processes and lost as heat (**10 percent law**).

→Energy flow is **unidirectional**: energy that passes to herbivores does not come back to autotrophs.

→Food chains usually have only **3 or 4 steps**: very little energy is left after the fourth level. Numbers decrease at higher trophic levels.

Example: if producers trap 10,000 J, herbivores get 1,000 J, primary carnivores 100 J and secondary carnivores 10 J.

Why an aquarium needs cleaning but a pond does not: a pond is a natural, self-sustaining ecosystem with decomposers that recycle wastes; an aquarium (artificial) lacks enough decomposers, so wastes accumulate.

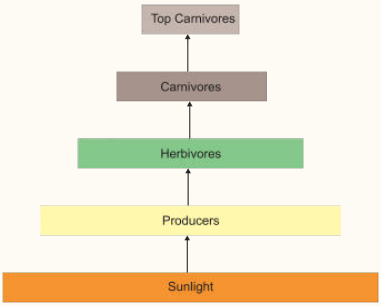
Pyramid of numbers: the number of organisms and the energy available both **decrease** from producers to top carnivores, so top carnivores are fewest; the loss of energy at each step is why the pyramid narrows upward.

Food web: many interlinked food chains, because one organism may be eaten by several others. A food web is more stable than a single chain.

A food web

Food chain	Food web
a single straight sequence of who eats whom	a network of many interconnected food chains
removal of one organism breaks the chain	alternative food paths keep it stable
energy flows along one path	energy flows along many paths

If one trophic level is removed: e.g. if all herbivores die, carnivores starve and plants grow unchecked; if decomposers are absent, dead bodies and wastes pile up and nutrients are not returned to the soil.



Energy flow in an ecosystem

3. Biological Magnification

Non-biodegradable pesticides and chemicals (e.g. **DDT**) enter the food chain through soil and water. They are not broken down or excreted, so their **concentration increases at each higher trophic level.** Humans, at the top of many chains,

get the **maximum** concentration (found in grains, fruits, vegetables, meat).

4. Ozone Layer and its Depletion

$$O_2 \xrightarrow{UV} O + O \; ; \; O + O_2 \longrightarrow O_3 \text{ (ozone)}$$

Ozone (O_3) is a deadly poison at ground level, but in the **upper atmosphere (stratosphere)** it shields the Earth from harmful **ultraviolet (UV) radiation** of the Sun, which causes skin cancer, cataract and damage to the immune system.

The amount of ozone fell sharply in the 1980s, mainly due to **chlorofluorocarbons (CFCs)** used as refrigerants and in fire extinguishers. In **1987 UNEP** got an agreement to freeze CFC production at 1986 levels; CFC-free refrigerators are now mandatory worldwide.

5. Managing the Garbage we Produce

Biodegradable	*Non-biodegradable*
broken down by bacteria and saprophytes	not broken down by biological processes
vegetable peels, paper, cotton, wood, leather, dung	plastics, polythene, glass, DDT, metal cans
cause little lasting harm; give manure	persist for a long time; harm various members of the ecosystem

Problems: changing lifestyle produces huge amounts of waste; disposable cups, plastic packaging and e-waste keep increasing.
Solutions: segregate waste at source (wet and dry bins), composting of kitchen waste, recycling of paper, plastic and metals, reuse, use cloth and jute bags, and the **3 Rs**: reduce, reuse, recycle.

Harm by non-biodegradable waste: plastic bags choke drains and cause flooding, cattle swallow polythene and die, plastics and chemicals pollute soil and water and enter food chains.

Kulhads (disposable clay cups) were suggested to replace plastic cups on trains; using them on a large scale would remove fertile **top soil**, so they are not a perfect solution. Disposable paper cups are better as paper is biodegradable.

Board ka favourite

→Calculate energy at a trophic level using the 10 percent law (e.g. 20,000 J at producers: tertiary consumer gets 20 J).
→Biological magnification: what it is, why humans are most affected.
→Ozone formation equation, its role, cause of depletion (CFCs), the 1987 UNEP step.
→Why decomposers are important; what will happen if all organisms of one trophic level are killed.
→Why food chains have only 3-4 trophic levels; why energy flow is unidirectional.
→Classify given wastes as biodegradable or not; two ways to reduce non-biodegradable waste.

Common mistakes

→Starting a food chain with an animal or with the Sun: a food chain begins with a **producer**.
→Drawing the arrows the wrong way: arrows point from food to eater (grass ⟶ deer).
→Saying 10% energy is lost at each level: 10% is **passed on**, about 90% is lost.
→Confusing ozone depletion with the greenhouse effect: CFCs deplete stratospheric ozone.
→Calling decomposers consumers of living organisms: they feed on **dead** remains.

Yaad karo

10% ka rule: har level pe energy ka sirf ek zero kat jaata hai (10000 ⟶ 1000 ⟶ 100 ⟶ 10). **Poison ulta chalta hai**: energy ghatti hai, DDT ka concentration badhta hai.

Chemical Reactions and Equations

1 Signs of a Reaction

- **Colour change:** $CuSO_4$ blue fades
- **Gas evolved:** Zn + dil. H_2SO_4 gives H_2
- **Heat change:** CaO + water gets hot
- $Pb(NO_3)_2$ + KI: yellow PbI_2
- **Change of state:** solid, liquid or gas
- Mg burns with dazzling white flame, leaves white MgO ash
- Rub Mg ribbon with sandpaper to remove its oxide layer

2 Writing Equations

- **Skeletal:** formulae, not balanced
- **Balanced:** same atoms on both sides
- Balancing obeys conservation of mass
- Change coefficients, not subscripts
- **State symbols:** (s), (l), (g), (aq)
- $3Fe + 4H_2O(g) \rightarrow Fe_3O_4 + 4H_2$
- Heat, light, catalyst or pressure written above the arrow

3 Combine & Decompose

- **Combination:** A + B → AB
- $CaO + H_2O \rightarrow Ca(OH)_2$ + heat
- Whitewash: shiny $CaCO_3$ layer
- $2FeSO_4 \rightarrow Fe_2O_3 + SO_2 + SO_3$
- $Pb(NO_3)_2$ heated: brown NO_2
- $2H_2O \rightarrow 2H_2 + O_2$ (2 : 1)
- **Light:** $2AgCl \rightarrow 2Ag + Cl_2$

CHEMICAL REACTIONS

Chemistry

4 Displacement Reactions

- More reactive metal displaces less reactive one
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- Blue solution turns pale green; nail turns brown
- $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
- **Double displacement:** ions swap
- $BaCl_2 + Na_2SO_4$: white $BaSO_4$
- **Precipitation:** insoluble solid forms

5 Oxidation & Reduction

- **Oxidation:** gain O or loss of H
- **Reduction:** loss of O or gain of H
- **Redox:** both occur together
- $CuO + H_2 \rightarrow Cu + H_2O$: CuO reduced, H_2 oxidised
- $ZnO + C \rightarrow Zn + CO$: C is oxidised
- $2Cu + O_2 \rightarrow 2CuO$ (brown to black)
- **Exothermic:** heat given out

6 Daily Life & Exam Focus

- **Rust:** $Fe_2O_3.xH_2O$, needs air + water
- Prevent rust: paint, oil, galvanise
- **Rancidity:** oxidised fats smell bad
- Chips packets flushed with N_2; antioxidants added
- AgCl stored in dark bottles
- **Diagram:** electrolysis of water, H_2 at cathode
- Dil. H_2SO_4 makes water conduct

Quick Tip: Balance karte waqt subscript kabhi mat badlo, sirf aage ka number badlo, aur state symbols zaroor likho!

Acids, Bases and Salts

ACIDS, BASES & SALTS

Chemistry

1 Acids & Bases

- **Acid:** sour, gives H^+ ions in water
- **Base:** bitter, soapy, gives OH^- ions
- **Alkali:** water-soluble base, e.g. NaOH
- **Strong:** HCl, H_2SO_4; **weak:** acetic
- Dry HCl gas: no H^+ ions, dry litmus unchanged
- Always add acid to water slowly, never water to acid
- Conduct in water due to ions

2 Indicators

- **Litmus:** acid red, base blue
- **Phenolphthalein:** base turns it pink
- **Methyl orange:** acid red, base yellow
- **Turmeric:** reddish-brown with base
- **Olfactory:** onion, vanilla, clove oil lose smell in base
- Natural: litmus, red cabbage
- **Universal indicator:** pH colours

3 Reactions of Acids

- Acid + metal → salt + H_2 (pop test)
- $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
- $2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$
- $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$
- CO_2 turns lime water milky ($CaCO_3$); excess CO_2 clears it
- **Neutralisation:** salt + water
- Metal oxides basic, non-metal acidic

4 pH Scale

- **pH:** H^+ concentration, 0 to 14
- 7 neutral, below 7 acid, above 7 base
- More H^+ ions means lower pH
- Our body works at pH 7.0 to 7.8
- **Acid rain:** pH below 5.6
- Tooth decay below pH 5.5
- **Antacid:** milk of magnesia

5 Important Salts

- **Chlor-alkali:** brine → NaOH + Cl_2 + H_2
- Cl_2 at anode, H_2 at cathode
- **Bleaching powder:** $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$
- **Baking soda:** $NaHCO_3$, antacid
- Heat: $NaHCO_3$ gives CO_2, cake rises
- **Washing soda:** $Na_2CO_3.10H_2O$
- **PoP:** $CaSO_4.½H_2O$

6 Salts & Exam Focus

- Strong acid + strong base: pH 7
- Strong acid + weak base: acidic
- Weak acid + strong base: basic
- **Water of crystallisation:** $CuSO_4.5H_2O$ blue, white on heating
- PoP + water sets as gypsum
- Tartaric acid in baking powder removes bitter taste
- Curd not kept in copper vessels

Quick Tip: pH yaad rakho: 7 se kam acid, 7 se zyada base. Chaaron salts ke formula aur ek-ek use pakka karo!

Metals and Non-metals

1 Physical Properties

- **Metals:** lustrous, malleable, ductile
- Sonorous; good conductors of heat
- Ag and Cu are the best conductors
- Au, Ag: most malleable metals
- **Exceptions:** Na, K soft; Hg liquid at room temperature
- **Non-metals:** brittle and dull
- Iodine lustrous; graphite conducts; diamond hardest

2 Reactions of Metals

- $2Mg + O_2 \rightarrow 2MgO$ (basic oxide)
- **Amphoteric:** Al_2O_3, ZnO (acid + base)
- Na, K stored in kerosene
- $2Na + 2H_2O \rightarrow 2NaOH + H_2$ + heat
- Mg floats in hot water; Al, Fe, Zn need steam
- Metal + dil. HCl $\rightarrow$ salt + H_2
- HNO_3 gives no H_2 (oxidiser)

3 Reactivity Series

- K > Na > Ca > Mg > Al > Zn
- Fe > Pb > H > Cu > Hg > Ag > Au
- Metals above H give H_2 with acids
- $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
- Cu cannot displace Fe
- **Aqua regia:** conc. HCl : HNO_3 = 3 : 1, dissolves gold
- Non-metal oxides acidic: SO_2, CO_2

METALS & NON-METALS

Chemistry

4 Ionic Compounds

- Formed by transfer of electrons
- Na (2,8,1) loses $1e^- \rightarrow Na^+$
- Cl (2,8,7) gains $1e^- \rightarrow Cl^-$
- Mg gives $2e^-$ to two Cl atoms: $MgCl_2$
- Hard solids with high m.p. and b.p.
- Soluble in water, not in kerosene
- Conduct only when molten or dissolved (free ions)

5 Extraction of Metals

- **Ore:** mineral worth extracting
- **Gangue:** sand, soil impurities
- **Roasting:** sulphide ore, excess air
- **Calcination:** carbonate ore
- $ZnO + C \rightarrow Zn + CO$
- **Thermit:** $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3$, joins rails
- **Refining:** impure Cu anode, pure Cu cathode, anode mud

6 Corrosion & Exam Focus

- Ag turns black (Ag_2S); Cu turns green
- Rust needs both air and moisture
- Rust test: 3 test tubes
- **Galvanising:** zinc coat on iron
- **Alloys:** brass (Cu+Zn), bronze (Cu+Sn), solder (Pb+Sn)
- 22 carat: 22 parts gold in 24
- **Diagram:** refining of copper

Quick Tip: Reactivity series ki line ratt lo, displacement, extraction aur corrosion sab isi se nikalta hai!

Carbon and its Compounds

1 Covalent Bonding

- C (2,4) cannot lose or gain $4e^-$
- **Covalent bond:** shared electron pair
- Single: H_2, CH_4; double: O_2, CO_2
- **Triple:** N_2 (N≡N), ethyne C_2H_2
- Low m.p. and b.p.: weak forces between molecules
- Poor conductors: no ions
- **Diagram:** electron-dot CH_4, CO_2, N_2

2 Versatile Carbon

- **Catenation:** chains, branches, rings
- **Tetravalency:** 4 bonds; C, H, O, N, Cl
- **Alkane:** C_nH_{2n+2}, saturated
- **Alkene:** C_nH_{2n}; **alkyne:** C_nH_{2n-2}
- Rings: benzene C_6H_6
- **Isomers:** n-butane, iso-butane
- Silicon shows catenation only up to 7-8 atoms

3 Homologous Series

- Same functional group
- Members differ by $-CH_2-$ (14 u)
- Same chemical, gradually changing physical properties
- **-OH** alcohol (-ol); **-CHO** aldehyde (-al)
- **C=O** ketone (-one); **-COOH** acid
- Meth 1, eth 2, prop 3, but 4, pent 5
- Propane → propanone: final "e" dropped before vowel

4 Chemical Properties

- $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat
- Saturated: blue flame
- Alk. $KMnO_4$ oxidises ethanol
- **Addition:** H_2 + Ni/Pd catalyst
- Hydrogenation turns vegetable oil into vanaspati ghee
- $CH_4 + Cl_2 \rightarrow CH_3Cl + HCl$ in sunlight
- Substitution in saturated, addition in unsaturated compounds

5 Ethanol & Ethanoic Acid

- Na + ethanol → ethoxide + H_2
- Ethanol + hot conc. H_2SO_4 → ethene
- **Vinegar:** 5-8% ethanoic acid
- Glacial acetic acid: freezes in winter
- Ethanoic acid + $NaHCO_3$ gives CO_2 (lime water test)
- **Ester:** acid + alcohol, sweet smell
- **Saponification:** ester + NaOH → soap

6 Soaps & Exam Focus

- **Soap:** Na/K salt of fatty acid
- Ionic head in water, carbon tail in oil
- **Micelle:** traps oily dirt in its centre, rinsed away
- Hard water (Ca, Mg salts) forms scum
- **Detergents:** work in hard water
- **Diagram:** soap micelle
- Common error: calling chlorination an addition reaction

CARBON COMPOUNDS

Chemistry

Quick Tip: Har functional group ka suffix aur ek example likh ke rakho, naming ke 2 marks pakke!

Life Processes

1 Nutrition in Plants

- **Autotrophs:** make their own food
- $6CO_2 + 12H_2O \rightarrow$ glucose $+ 6O_2$
- Light absorbed, water split
- Extra glucose stored as starch
- **Stomata:** gas exchange; guard cells swell to open
- Desert plants take in CO_2 at night, use it by day
- Variegated leaf iodine test

2 Human Digestion

- **Heterotrophs:** Amoeba, fungi
- **Saliva:** amylase, starch → sugar
- **Stomach:** HCl, pepsin, mucus
- **Bile** (liver): emulsifies fats, makes medium alkaline
- **Pancreas:** trypsin, lipase
- **Villi:** large surface for absorption
- Large intestine absorbs water

3 Respiration

- Glucose → pyruvate in cytoplasm
- Aerobic in mitochondria → $CO_2 + H_2O$ + energy
- Yeast (no O_2) → ethanol + CO_2
- Muscles: lactic acid, cramps
- **ATP:** energy currency of the cell
- **Alveoli:** site of gas exchange
- Haemoglobin carries O_2 in blood

LIFE PROCESSES

Biology

6 Excretion & Exam Focus

- **Nephron:** unit of the kidney
- Glomerulus: filtration of blood
- Glucose, amino acids, salts, water reabsorbed
- Ureter → bladder → urethra
- **Dialysis:** artificial kidney
- Plant wastes stored in old leaves, bark, resins, gums
- **Diagrams:** digestive system, heart, nephron, stomata

5 Transport in Plants

- **Xylem:** water, minerals upward
- Root pressure works at night
- **Transpiration pull:** main force by day
- Transpiration also cools the leaf
- **Phloem:** sucrose moves both ways (translocation)
- Translocation uses energy from ATP
- Plants need less energy

4 Transport in Humans

- Heart: 4 chambers
- **Double circulation:** blood passes the heart twice per cycle
- Ventricles thick: pump blood to body
- Valves stop backflow of blood
- **Arteries:** thick, elastic walls
- **Capillaries:** one cell thick walls
- **Platelets:** clot; **lymph:** carries fats

Quick Tip: Chaar diagram roz practice karo: digestive system, heart, nephron aur stomata. Har saal aate hain!

Control and Coordination

1 Neuron & Impulse

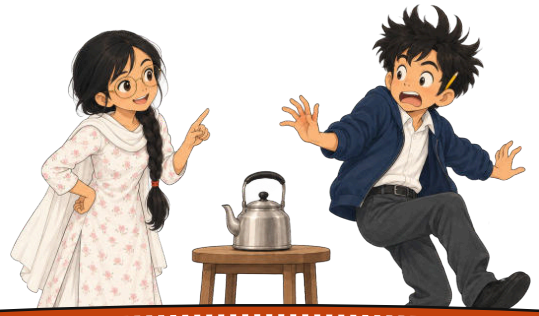
- **Receptors:** detect stimuli
- Dendrite → cell body → axon
- Impulse is an electrical signal
- **Synapse:** gap between two neurons
- Chemicals released at the axon end cross the synapse
- Impulse goes one way only
- **Diagram:** labelled neuron

2 Reflex Action

- Quick, automatic, no thinking
- E.g. hand pulled back from flame
- Receptor → sensory neuron
- Relay → motor neuron → muscle
- Reflex arcs are formed in the spinal cord, not brain
- Saves time; brain told later
- **Effector:** muscle or gland

3 Human Brain

- **CNS:** brain + spinal cord
- **Cerebrum:** thinking, memory
- **Hypothalamus:** hunger
- **Cerebellum:** balance, posture
- **Medulla:** BP, salivation, vomiting
- **Pons:** regulates breathing
- Skull and fluid-filled membranes protect the brain



CONTROL & COORDINATION

Biology

6 Animal Hormones

- **Adrenaline:** fight or flight
- **Thyroxin:** iodine needed
- **Pituitary:** growth hormone
- **Insulin:** lack causes diabetes
- Testosterone, oestrogen: changes at puberty
- **Feedback:** blood sugar controls insulin release
- Hormones travel in blood

5 Plant Hormones

- **Auxin:** shoot tip, cell elongation
- Auxin moves to the shaded side, which grows more
- **Gibberellins:** stem growth
- **Cytokinins:** cell division
- **Abscisic acid:** stops growth, wilting
- Abscisic acid is the stress hormone
- Mimosa folding is not a tropism

4 Plant Movements

- No nerves: chemical control
- **Mimosa:** cells lose water, leaves fold (no growth)
- **Tropism:** directional growth
- Shoot to light: phototropism
- Root down: geotropism; to water: hydrotropism
- Pollen tube to ovule: chemotropism
- Tendrils coil round support

Quick Tip: Cerebrum = soch, cerebellum = balance, ye do kabhi mix mat karna. Reflex arc spinal cord me banta hai!

How do Organisms Reproduce?

1 DNA & Variation

- DNA carries protein information
- DNA copying is never 100% exact
- Variations are the basis of evolution
- Variation helps species survive changes in environment
- E.g. heat-resistant bacteria survive a heat wave
- Stable copying keeps the body design
- Reproduction maintains population

2 Asexual Reproduction

- **Binary fission:** Amoeba, Leishmania
- **Multiple fission:** Plasmodium
- **Fragmentation:** Spirogyra
- **Regeneration:** Planaria, Hydra
- **Budding:** Hydra
- **Spores:** Rhizopus sporangia, thick-walled spores
- Regeneration is not fragmentation

3 Vegetative Propagation

- New plant from root, stem or leaf
- **Bryophyllum:** buds on leaf margins
- Sugarcane, rose, grapes: by layering or grafting
- Banana, orange, rose: lost the ability to form seeds
- Offspring identical to parent
- **Tissue culture:** callus to plantlets
- Used for orchids, ornamental plants

REPRODUCTION

Biology

4 Flowering Plants

- **Stamen:** anther makes pollen
- **Pistil:** stigma, style, ovary
- Papaya unisexual; Hibiscus bisexual
- **Pollination:** self or cross (wind, water, insects)
- Pollen tube reaches the ovule
- Ovule → seed; ovary → fruit
- **Diagram:** LS of flower, pollen tube

5 Human Reproduction

- **Puberty:** body hair, voice change
- **Testes:** sperms + testosterone
- Scrotum keeps testes cooler
- **Ovary:** one egg a month; oestrogen
- Fertilisation in the oviduct
- **Placenta:** villi give food and O_2, remove waste
- No fertilisation: uterus lining sheds (menstruation)

6 Reproductive Health

- **Bacterial STDs:** gonorrhoea, syphilis
- **Viral STDs:** warts, HIV-AIDS
- **Condoms:** also prevent STDs
- Oral pills change hormone balance
- **Copper-T** placed in the uterus
- **Surgery:** vasectomy, tubectomy
- Pre-natal sex determination is banned (female foeticide)
- A healthy sex ratio is needed

Quick Tip: Ovule se seed, ovary se fruit, aur fertilisation oviduct me hota hai. Ye teen line exam me bachayengi!

Heredity

1 Basic Terms

- **Heredity:** traits pass to offspring
- **Variation:** differences within one species
- **Gene:** DNA segment for one protein, one trait
- **Alleles:** forms of a gene, e.g. T, t
- **Genotype:** genetic make-up (TT, Tt, tt)
- **Phenotype:** visible trait (tall, short)
- **Pure** TT / tt (homozygous); **hybrid** Tt

2 Variation

- DNA copying is never perfect: variations arise
- Asexual reproduction: very little variation
- Sexual: DNA of two parents mixes, more diversity
- Variations pass on and accumulate
- Nature selects useful variations
- E.g. heat-resistant bacteria survive a heat wave

3 Mendel's Experiments

- Gregor Mendel: "Father of Genetics"
- Worked on garden pea, *Pisum sativum*
- **Why pea:** contrasting traits, short life
- **Dominant:** round, yellow seed; violet flower; tall
- **Recessive:** wrinkled, green seed; white flower; dwarf
- Counted each trait in every generation (maths)

HEREDITY

Biology

6 Expression and Sex

- **Gene** → enzyme → hormone → height
- More hormone made: tall plant
- 22 autosome pairs + 1 sex pair
- **Female** XX; **male** XY (Y is short)
- Eggs: X only; sperms: X or Y
- Father decides sex; 50 : 50 chance
- Turtles: egg temperature decides sex
- Snails can change their sex

5 Dihybrid Cross and Laws

- **P:** RRYY (round yellow) x rryy
- **F1:** all RrYy; gametes RY, Ry, rY, ry
- **F2** = 9 : 3 : 3 : 1 (16 boxes)
- New combos: round green, wrinkled yellow
- **Dominance:** dominant shows in Tt
- **Segregation:** one allele per gamete
- **Independent assortment:** traits separate
- Mendel's F2: 315, 108, 101, 32 seeds

4 Monohybrid Cross

- **P:** TT (tall) x tt (short)
- **F1:** all Tt tall; short trait only hidden
- **F2 phenotype** = 3 tall : 1 short
- **F2 genotype** = 1 TT : 2 Tt : 1 tt
- **Tt x tt** gives 1 tall : 1 short
- A gamete carries one allele: T or t
- **Conclusion:** traits pass as separate units (genes)

Quick Tip: 3:1 = ek trait, 9:3:3:1 = do traits. 1:2:1 genotypic hai, phenotypic nahi. Ladka ya ladki: papa ka Y decide karta hai.

Light: Reflection and Refraction

1 Reflection Basics

- **Laws:** ∠i = ∠r; i, r, normal in one plane
- **Plane mirror:** virtual, erect, same size
- Image laterally inverted, as far behind
- **Real image:** on screen, inverted
- **Virtual:** rays appear to meet, erect
- **Concave** converges; **convex** diverges
- **Terms:** pole P, centre C, focus F
- **R = 2f**: F midway between P and C

2 Mirror Images and Uses

- **Object at C:** image at C, same size, real
- **C to F:** beyond C, enlarged, real, inverted
- **P to F:** behind mirror, enlarged, virtual
- **Convex:** always virtual, erect, diminished
- **Concave uses:** headlights, shaving, dentist
- **Convex use:** rear-view, wider field of view
- **Via C:** ray retraces its path

3 Formulas and Signs

- **Mirror:** 1/v + 1/u = 1/f; m = −v/u
- **Lens:** 1/v − 1/u = 1/f; m = v/u
- **Power:** P = 1/f (f in m); unit dioptre D
- Distances from P (or O); object on left
- Along light +, against −; u always −
- m − : real, inverted; m + : virtual, erect
- |m| > 1 enlarged; |m| < 1 diminished

LIGHT

Physics

4 Refraction

- Light bends as its speed changes
- **Snell's law:** sin i / sin r = constant
- $\mathbf{n_m}$ = c/v; c = 3 x 10^8 m/s
- $\mathbf{n_{21}} = v_1/v_2$; $n_{21} = 1/n_{12}$
- Rarer → denser: towards normal
- **Glass slab:** lateral displacement
- **n:** water 1.33, diamond 2.42
- Kerosene optically denser than water

5 Spherical Lenses

- **Convex** converges; **concave** diverges
- Ray via optical centre: undeviated
- **At $2F_1$:** image at $2F_2$, same size
- **F_1-$2F_1$:** beyond $2F_2$, enlarged, real
- **Within F_1:** virtual, erect, enlarged
- **Concave lens:** virtual, erect, diminished
- Convex P +, concave P −; $P = P_1 + P_2$
- **Uses:** magnifier, camera (convex)

6 Exam Focus

- Ray diagrams: all six positions
- Numericals: find v, m, nature (3 marks)
- u = −25, f = −15 cm: v = −37.5 cm
- Convex lens f = 50 cm: P = +2 D
- f in metres: 25 cm gives 4 D
- Virtual rays dotted; use arrows
- Always state position, size, nature
- Don't mix: mirror +, lens − formula

Quick Tip: Mirror mein plus, lens mein minus (1/v + 1/u vs 1/v − 1/u). Convex mirror aur concave lens hamesha chhota, seedha, virtual.

The Human Eye and the Colourful World

1 Parts of the Eye

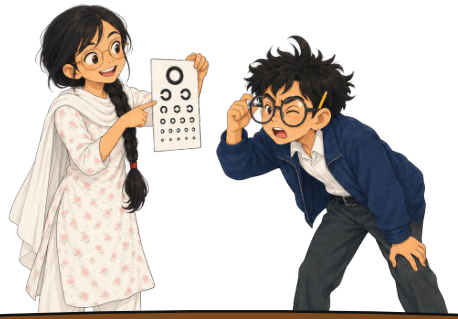
- **Cornea:** most refraction occurs here
- **Iris:** controls size of pupil
- **Pupil:** small in bright light
- **Eye lens:** convex, flexible; fine focusing
- **Ciliary muscles:** change lens curvature
- **Retina:** image real, inverted
- **Rods:** light intensity; **cones:** colour
- **Optic nerve:** signals to the brain

2 Accommodation

- **Accommodation:** lens changes f
- **Far:** muscles relax, lens thin, f long
- **Near:** muscles contract, lens thick
- **Near point** = 25 cm (young eye)
- **Far point** = infinity
- Range of vision: 25 cm to infinity
- **Cataract:** lens milky; surgery
- Brain sees inverted image as erect

3 Defects of Vision

- **Myopia:** near-sighted; image before retina
- Cause: lens too curved, eyeball long
- Fix: **concave lens**, P negative
- **Hypermetropia:** image behind retina
- Cause: f too long, eyeball too small
- Fix: **convex lens**, P positive
- **Presbyopia:** old age; bifocal lens
- Far point 80 cm: f = −80 cm, P = −1.25 D



HUMAN EYE & COLOURS

Physics

6 Scattering of Light

- **Tyndall effect:** colloid scatters light
- E.g. sunbeam in a smoky room
- Fine particles scatter blue light
- **Blue sky:** air scatters blue most
- **Red signals:** least scattered
- **White clouds:** big drops scatter all
- Astronauts see a dark sky
- Colour depends on particle size

5 Atmospheric Refraction

- Air layers differ in optical density
- Hot air rarer: objects flicker above fire
- **Stars twinkle:** point sources
- Starlight path keeps changing
- Stars look higher than actual
- **Planets** don't twinkle: extended source
- Many-point light fluctuations cancel
- Light bends towards normal in denser layers

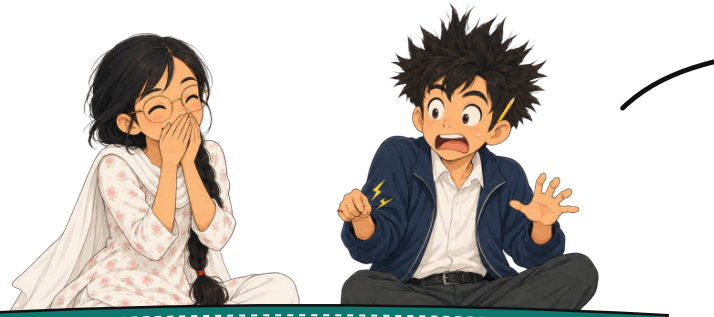
4 Prism and Dispersion

- Prism: 2 triangular, 3 rectangular faces
- Emergent ray bends towards base
- **Angle of deviation:** between rays
- **Dispersion:** white light → VIBGYOR
- Red bends least, violet most
- **Newton:** 2nd inverted prism: white
- **Rainbow:** dispersion + internal reflection
- Rainbow always opposite the Sun

Quick Tip: Myopia = Minus lens (concave), Hyper = Plus (convex). Prism mein Violet sabse zyada mudta hai, Red sabse kam. Blue sky = scattering, refraction nahi.

ELECTRICITY

Physics



1 Current and Voltage

- **Current:** I = Q/t; 1 A = 1 C/1 s
- **Q = ne**; e = 1.6 x 10^{-19} C
- 1 C ≈ 6 x 10^{18} electrons
- Current flows + to −, against e^-
- **p.d.:** V = W/Q; 1 V = 1 J/C
- **Ammeter** in series (low resistance)
- **Voltmeter** in parallel (high resistance)

2 Ohm's Law

- V ∝ I at constant temperature
- **V = IR**; 1 Ω = 1 V/1 A
- V-I graph: straight line through origin
- Slope of V-I graph = R
- **Rheostat:** varies current, same source
- **Solved:** 4 A at 60 V: R = 15 Ω
- Same heater on 120 V draws 8 A
- Always state "temperature constant"

3 Resistance and ρ

- **R = ρl/A**; ρ in Ω m
- R ∝ length; R ∝ 1/area
- ρ depends only on material, temperature
- **Metals:** ρ 10^{-8} to 10^{-6} Ω m
- **Cu, Al:** low ρ, used for transmission
- **Nichrome:** high ρ, no oxidation: heaters
- Stretched to double length: R becomes 4R
- Diameter halved: R becomes 4R

4 Series vs Parallel

- **Series:** $R_s = R_1 + R_2 + R_3$; same I
- **Parallel:** $1/R_p = 1/R_1 + 1/R_2 + ...$; same V
- R_s > largest R; R_p < smallest R
- Two resistors: $R_p = R_1R_2/(R_1 + R_2)$
- n equal: $R_s = nR$, $R_p = R/n$
- 5, 10, 30 Ω parallel: R_p = 3 Ω
- **Homes use parallel:** full V, own switch

5 Heating Effect

- **Joule's law:** H = I^2Rt = VIt
- Heat ∝ I^2, R and time
- **Uses:** iron, toaster, heater, geyser
- **Bulb:** tungsten filament (melts 3380 °C)
- Bulb filled with nitrogen and argon
- **Fuse:** low melting wire, in series
- 1 kW iron at 220 V: 4.5 A, use 5 A fuse
- Fuse ratings: 1, 2, 3, 5, 10 A

6 Electric Power

- **P = VI = I^2R = V^2/R**; unit watt
- **E = Pt**; 1 kW h = 3.6 x 10^6 J
- 1 kW h = 1 unit of electricity
- 400 W fridge, 8 h x 30 days = 96 units
- 100 W, 220 V bulb: R = 484 Ω
- Parallel: higher-watt bulb glows brighter
- Series: lower-watt bulb glows brighter
- Cost at Rs 3 per unit: Rs 288

Quick Tip: V = IR yaad rakho. Series mein current same, parallel mein voltage same. kW h ko J mein badalna ho to 3.6 x 10^6 se guna karo.

Magnetic Effects of Electric Current

1 Magnetic Field

- **Oersted:** current deflects compass
- **Field:** region of magnetic force
- Has magnitude and direction
- Direction: compass N-pole points
- Field lines: N to S outside, S to N inside
- Closed curves; never intersect
- Two lines never cross: compass can't point two ways
- Crowded lines = stronger field

2 Current-carrying Wire

- **Straight wire:** concentric circles around it
- Field ∝ current; ∝ 1/distance
- **Right-hand thumb rule:** thumb = current
- Curled fingers give field direction
- Also called Maxwell's corkscrew rule
- Reverse current: field reverses
- **Circular loop:** lines straight at centre
- n turns: field n times stronger

3 Solenoid and Electromagnet

- **Solenoid:** closely wound coil
- Field like a bar magnet: N and S ends
- Inside: uniform, parallel lines
- **Electromagnet:** soft iron core
- Soft iron core strengthens the field
- Temporary; set by turns, current
- Permanent magnet: steel, alnico
- **Uses:** cranes, electric bells, relays

MAGNETIC EFFECTS

Physics

4 Force on Conductor

- Conductor in a field feels a force
- Reverse current or field: force reverses
- Max when I ⊥ field; zero if parallel
- **Fleming's left-hand rule**
- Forefinger field, middle current, thumb force
- Electron beam: I opposite to motion
- **MRI:** images organs via body's fields
- **Uses:** loudspeakers, microphones

5 AC and DC

- **DC:** one direction; cells, batteries
- **AC:** reverses direction periodically
- AC: power stations, home supply
- India: AC at **50 Hz**
- AC changes direction every 1/100 s
- **AC advantage:** long distance, less loss
- DC cannot be stepped up or down easily

6 Domestic Circuits

- **Live** red 220 V; **neutral** black; **earth** green
- Main fuse → meter → distribution box
- **15 A:** geyser, iron; **5 A:** bulbs, fans
- Appliances in parallel: each gets 220 V
- **Earth wire:** leak path, prevents shock
- **Short circuit:** live touches neutral
- **Overloading:** too many appliances
- **Fuse / MCB** cuts off excess current

Quick Tip: Field ke liye right-hand thumb rule, force ke liye Fleming ka LEFT hand. Wire colours: Red live, Black neutral, Green earth.

Our Environment

1 Ecosystem

- **Ecosystem:** biotic + abiotic, interacting
- **Biotic:** plants, animals, microbes
- **Abiotic:** temperature, rainfall, soil
- **Natural:** forest, pond; self-sustaining
- **Artificial:** garden, crop field, aquarium
- Aquarium needs cleaning by us
- Pond: decomposers recycle the wastes

2 Components

- **Producers:** green plants, blue-green algae
- Make food by photosynthesis
- **Herbivores:** cow, deer; **carnivores:** lion
- **Omnivores:** human, crow
- **Parasites:** lice, Cuscuta
- **Decomposers:** bacteria and fungi
- Decomposers recycle nutrients to soil

3 Food Chain and Web

- Each organism feeds on the one before
- Grass → grasshopper → frog → snake → eagle
- **T1** producers; **T2** herbivores
- **T3, T4:** small, then large carnivores
- Chain begins with a producer, not Sun
- Arrows point from food to eater
- **Food web:** interlinked chains, more stable
- Herbivores die: carnivores starve

4 Energy Flow

- Plants trap about 1% of sunlight
- **10% law:** only 10% passes to next level
- Rest used in life processes, lost as heat
- 10,000 J → 1,000 J → 100 J → 10 J
- Energy flow is unidirectional
- Chains have only 3-4 steps
- **Pyramid:** numbers, energy fall upward

5 Bio-magnification and Ozone

- **Biomagnification:** DDT rises up the chain
- Not broken down or excreted
- Humans at top get the most
- $O_2 + UV \rightarrow O + O$; $O + O_2 \rightarrow O_3$
- Stratospheric ozone blocks UV rays
- UV causes skin cancer, cataract
- **CFCs** (refrigerants) deplete ozone
- **1987 UNEP:** CFC output frozen at 1986 level

6 Waste Management

- **Biodegradable:** peels, paper, cotton, wood
- **Non-biodegradable:** plastic, glass, DDT, cans
- Plastic chokes drains, kills cattle
- Segregate wet and dry waste at source
- Compost, recycle; use jute bags
- **3 Rs:** reduce, reuse, recycle
- Kulhads remove fertile top soil

OUR ENVIRONMENT

Biology

Quick Tip: 10% ka rule: har level pe ek zero kat jaata hai (10000, 1000, 100, 10). Energy ghatti hai, DDT ka concentration badhta hai.

SECTION

COMPETENCY BASED QUESTIONS

Chapter-wise

About 50% of the 2026-27 paper tests competencies: applying concepts to real-life situations, data, experiments and cases. Each of the 13 chapters has eight such questions (MCQ, assertion-reason, case and data based, source based with sub-parts, and application or higher-order thinking), followed by concise model answers with marks.

As per the latest CBSE pattern: 50% competency-based questions

CHAPTER 1 Chemical Reactions and Equations

Q1. MCQ **[1 mark]**

A student places a clean iron nail in a test tube of blue copper sulphate solution and leaves it for 30 minutes. Which pair of observations correctly describes the result?

(a) Nail turns silvery white; solution turns colourless

(b) Nail gets a reddish-brown coating; solution turns pale green

(c) Nail gets a black coating; solution turns deep blue

(d) Nail dissolves completely; solution turns yellow

Q2. MCQ **[1 mark]**

In the reaction $CuO + H_2 \rightarrow Cu + H_2O$, carried out by passing hydrogen over heated black copper oxide, which statement is correct?

(a) Copper oxide is oxidised and hydrogen is reduced

(b) Copper oxide is reduced and hydrogen is oxidised

(c) Both are oxidised

(d) It is a decomposition reaction

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Packets of potato chips are flushed with nitrogen gas before sealing.
Reason (R): Nitrogen is an inert gas that prevents the oils and fats in the chips from getting oxidised.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

A sample of 5 g of white silver chloride kept in a watch glass near a sunny window turned grey within an hour, while an identical sample kept in a closed cupboard stayed white.
(i) Write the balanced chemical equation for the change.
(ii) Name the type of reaction and state one practical use of this reaction.

Q5. CASE / DATA BASED **[3 marks]**

A student recorded the following when three reactions were carried out in the lab:

Reaction	Observation
P: Quicklime + water	Hissing sound, beaker becomes hot
Q: Lead nitrate solution + potassium iodide solution	Yellow solid settles
R: Lead nitrate crystals heated strongly	Brown fumes evolve

(i) Classify P, Q and R by type of reaction. (ii) Name the yellow solid in Q and the brown gas in R.

Q6. CASE / DATA BASED **[3 marks]**

A student heated 2 g of pale green ferrous sulphate crystals in a dry boiling tube and noted the following:

Stage	Observation
Gentle heating	Water droplets on the cooler part of the tube; crystals turn white
Strong heating	Reddish-brown solid left; gas with smell of burning sulphur

(i) Write the balanced equation for the change on strong heating. (ii) Why do the crystals lose their green colour on gentle heating? (iii) Identify the type of reaction.

Q7. SOURCE BASED **[4 marks]**

During electrolysis of water, a student passed current through acidified water using two carbon electrodes covered with inverted test tubes. After 15 minutes the tube over the cathode held about 20 mL of gas and the tube over the anode about 10 mL. A burning splinter brought near the first tube produced a pop sound, while a glowing splinter rekindled in the gas of the second tube.

(i) Identify the gas collected at the cathode. **[1]**

(ii) Why is a few drops of dilute sulphuric acid added to the water? **[1]**

(iii) Explain the 2 : 1 ratio of the gas volumes with the help of a balanced equation, and name the type of reaction. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A shopkeeper notices that copper vessels left in his store develop a green layer and silver ornaments in a display case turn black. (i) Name the substances formed in each case. (ii) Name the process. (iii) Suggest two practical methods to slow this process down.

ANSWER KEY

Answer Q1 (b) Iron displaces copper from copper sulphate: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. Copper deposits on the nail (reddish-brown) and iron(II) sulphate makes the solution pale green.

Answer Q2 (b) Copper oxide loses oxygen (reduced) while hydrogen gains oxygen (oxidised); this is a redox reaction, and the black solid turns reddish-brown.

Answer Q3 (a) Both statements are true. Removing oxygen with unreactive nitrogen prevents oxidation of fats, which is what causes rancidity, so R correctly explains A.

Answer Q4 (i) $2AgCl(s) \rightarrow 2Ag(s) + Cl_2(g)$ in the presence of sunlight. (ii) Photolytic decomposition (decomposition by light); it is used in black and white photography.

Answer Q5 (i) P: combination reaction (exothermic); Q: double displacement (precipitation) reaction; R: thermal decomposition reaction. (ii) Yellow solid: lead iodide (PbI_2); brown gas: nitrogen dioxide (NO_2).

Answer Q6 (i) $2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$. (ii) The crystals lose their water of crystallisation ($FeSO_4.7H_2O$), which gives them their colour. (iii) Thermal decomposition reaction.

Answer Q7 **(i)** Hydrogen gas (it burns with a pop sound). **(ii)** Pure water is a poor conductor; the acid supplies ions so that electric current can flow through it. **(iii)** $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. Two molecules of hydrogen form for every one molecule of oxygen, so hydrogen occupies twice the volume. It is an electrolytic decomposition reaction.

Answer Q8 (i) The green layer on copper is basic copper carbonate [$CuCO_3.Cu(OH)_2$]; the black layer on silver is silver sulphide (Ag_2S). (ii) Corrosion. (iii) Keep articles in airtight boxes or with moisture-absorbing silica gel; apply a protective coating such as lacquer or wax, or polish them regularly.

CHAPTER 2 Acids, Bases and Salts

Q1. MCQ **[1 mark]**

Four colourless solutions were tested with universal indicator and gave the following pH values: P = 2, Q = 6.5, R = 9, S = 13. Which solution would react most vigorously with a piece of magnesium ribbon, and which is most likely to be a solution of sodium hydroxide?

(a) Q and R (b) P and S

(c) S and P (d) R and Q

Q2. MCQ **[1 mark]**

A person stung by a honeybee feels a burning pain caused by methanoic acid. Which household substance, applied to the sting, would give the most suitable relief?

(a) Lemon juice (b) Vinegar

(c) Baking soda paste (d) Tomato juice

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Dry hydrogen chloride gas does not change the colour of dry blue litmus paper.

Reason (R): Hydrogen ions are produced by HCl only in the presence of water.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

A farmer finds that the soil pH of his field has dropped to 5.2 after years of heavy fertiliser use, and his crop yield has fallen.

(i) Is the soil acidic or basic? (ii) Name one substance he can add to correct it and explain why it works.

Q5. CASE / DATA BASED **[3 marks]**

A student adds dilute hydrochloric acid to four test tubes containing the substances listed below and records whether a gas is produced.

Tube	Substance
1	Zinc granules
2	Sodium carbonate
3	Sodium hydrogencarbonate
4	Copper turnings

(i) In which tubes does the gas turn lime water milky? (ii) Name the gas in tube 1 and give a test for it. (iii) Why is no gas seen in tube 4?

Q6. CASE / DATA BASED **[3 marks]**

Equal volumes of hydrochloric acid and acetic acid, both of the same concentration, were taken in two test tubes, and an equal length of magnesium ribbon was added to each.

Acid	pH	Time for ribbon to dissolve
Hydrochloric acid	1	45 s
Acetic acid	3	6 min

(i) Which acid is stronger and how does the data show it? (ii) Explain the difference in terms of ions. (iii) Name the gas released in both tubes.

Q7. SOURCE BASED **[4 marks]**

Bleaching powder is made by passing chlorine gas over dry slaked lime. Chlorine itself is obtained when electricity is passed through brine, a process that also gives hydrogen and a strong alkali. Bleaching powder is used to disinfect drinking water, to bleach cotton and linen in textile mills, and as an oxidising agent in industries.

(i) Name the process by which brine is electrolysed and the alkali formed. **[1]**

(ii) Give the chemical formula of bleaching powder. **[1]**

(iii) Write the balanced equation for the preparation of bleaching powder and state one precaution in storing it. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A baker replaces baking soda with baking powder in a cake recipe and finds that the cake no longer tastes bitter. (i) Explain the chemistry behind the bitterness when only baking soda is used. (ii) What does baking powder contain that removes this problem? (iii) Write the reaction that makes the cake soft and spongy.

ANSWER KEY

Answer Q1 (b) P (pH 2) is the strongest acid and releases hydrogen most vigorously with Mg; S (pH 13) is a strong base such as NaOH.

Answer Q2 (c) Baking soda is a mild base and neutralises the methanoic acid; the other three are acidic.

Answer Q3 (a) Both are true. Acidic behaviour depends on H^+ (H_3O^+) ions, which form only when HCl dissolves in water, so R explains A.

Answer Q4 (i) Acidic (pH less than 7). (ii) Quicklime (calcium oxide), slaked lime (calcium hydroxide) or chalk (calcium carbonate); being basic, it neutralises the excess acid in the soil and raises the pH towards neutral.

Answer Q5 (i) Tubes 2 and 3 (carbon dioxide is released). (ii) Hydrogen; it burns with a pop sound when a burning splinter is brought near. (iii) Copper is less reactive than hydrogen, so it cannot displace hydrogen from dilute HCl.

Answer Q6 (i) Hydrochloric acid: it has the lower pH and reacts much faster. (ii) HCl ionises almost completely in water, giving a high concentration of H^+ ions; acetic acid ionises only partially. (iii) Hydrogen gas.

Answer Q7 **(i)** Chlor-alkali process; the alkali is sodium hydroxide (NaOH). **(ii)** $CaOCl_2$ (calcium oxychloride). **(iii)** $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$. It should be stored in an airtight container in a cool, dry place, as it loses chlorine on exposure to air and moisture.

Answer Q8 (i) On heating, $NaHCO_3$ forms sodium carbonate, which is bitter: $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. (ii) Baking powder is baking soda mixed with a mild edible acid such as tartaric acid, which neutralises the sodium carbonate. (iii) $NaHCO_3 + H^+$ (from tartaric acid) $\rightarrow CO_2 + H_2O$ + sodium salt of the acid; the CO_2 released makes the cake rise.

CHAPTER 3 Metals and Non-metals

Q1. MCQ **[1 mark]**

A jeweller wants a metal that can be beaten into the thinnest sheets for decorating sweets and drawn into very fine wires. Which metal best meets both requirements?

(a) Sodium (b) Mercury
(c) Gold (d) Zinc

Q2. MCQ **[1 mark]**

An iron gate in a coastal town rusts much faster than an identical gate in a dry inland town. The most likely reason is that the coastal air contains more:

(a) nitrogen (b) moisture and salt
(c) carbon dioxide only (d) dust particles

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Aluminium vessels do not corrode easily even though aluminium is a reactive metal.
Reason (R): Aluminium reacts with oxygen to form a thin, tough layer of aluminium oxide that protects the metal from further reaction.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

Strips of four metals P, Q, R and S were placed in solutions of different salts. The table shows whether a displacement reaction took place.

Metal	$FeSO_4$	$CuSO_4$	$ZnSO_4$
P	No	Yes	No
Q	Yes	Yes	Yes
R	No	No	No
S	Yes	Yes	No

(i) Arrange P, Q, R and S in decreasing order of reactivity. (ii) Which of these metals could be silver?

Q5. CASE / DATA BASED **[3 marks]**

A student burns a piece of magnesium ribbon and a small amount of sulphur powder separately, then dissolves each product in water and tests with litmus.
(i) Write the equation for the burning of magnesium. (ii) Predict the colour change in litmus for each solution. (iii) What general conclusion about oxides of metals and non-metals do the results support?

Q6. CASE / DATA BASED **[3 marks]**

Three samples were tested by a student and the results recorded:

Sample	Lustre	On hammering	Melting point	Conducts electricity
A	Shiny	Flattens	1085 °C	Yes
B	Dull	Breaks into powder	115 °C	No
C	Shiny, grey	Breaks	Above 3500 °C	Yes

(i) Classify A and B as metal or non-metal. (ii) Sample C is a non-metal. Identify it and state the property that makes it an exception. (iii) Which sample can be used for making electrical wires?

Q7. SOURCE BASED **[4 marks]**

Ores of metals in the middle of the reactivity series, such as zinc, are often found as carbonates or sulphides. In a smelting unit, zinc blende (ZnS) is first heated strongly in excess air. The oxide obtained is then heated with coke in a furnace. Since it is easier to obtain a metal from its oxide, the first step is essential before reduction.

(i) Name the process of heating a sulphide ore in excess air. **[1]**
(ii) What is the role of coke in the second step? **[1]**
(iii) Write balanced equations for both steps. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

Sodium chloride does not conduct electricity in its solid form but conducts when molten or dissolved in water. A student also observes that it has a very high melting point (801 °C). (i) Explain these properties on the basis of the type of bond present. (ii) Show the formation of NaCl by the transfer of electrons (Na: 2,8,1; Cl: 2,8,7).

ANSWER KEY

Answer Q1 (c) Gold (along with silver) is the most malleable and ductile metal; sodium is too soft and reactive, mercury is a liquid and zinc is comparatively brittle.

Answer Q2 (b) Rusting needs both air and moisture; dissolved salts in humid sea air speed up the process.

Answer Q3 (a) Both are true and R explains A: the protective oxide layer stops further oxidation. This property is enhanced industrially by anodising.

Answer Q4 (i) Q > S > P > R. (ii) R, because it displaces none of the metals and is therefore less reactive than copper, iron and zinc.

Answer Q5 (i) $2Mg + O_2 \rightarrow 2MgO$. (ii) Magnesium oxide solution turns red litmus blue (basic); sulphur dioxide solution turns blue litmus red (acidic). (iii) Metal oxides are generally basic and non-metal oxides are generally acidic.

Answer Q6 (i) A is a metal (lustrous, malleable, conducts); B is a non-metal (dull, brittle, non-conductor; it could be sulphur). (ii) C is graphite, a form of carbon; it is a non-metal that conducts electricity. (iii) A, as it is ductile and a good conductor.

Answer Q7 **(i)** Roasting. **(ii)** Coke (carbon) acts as the reducing agent, removing oxygen from zinc oxide. **(iii)** $2ZnS + 3O_2 \rightarrow 2ZnO + 2SO_2$ (roasting); $ZnO + C \rightarrow Zn + CO$ (reduction).

Answer Q8 (i) NaCl is an ionic compound. Strong electrostatic forces between oppositely charged ions require much energy to break, giving a high melting point. In the solid the ions are fixed, so there is no conduction; in the molten state or in solution the ions move freely and carry current. (ii) Na (2,8,1) $\rightarrow Na^+$ (2,8) + e^-; Cl (2,8,7) + $e^- \rightarrow Cl^-$ (2,8,8); $Na^+ + Cl^- \rightarrow NaCl$.

CHAPTER 4 Carbon and its Compounds

Q1. MCQ **[1 mark]**

The formulae of four hydrocarbons are C_3H_6, C_4H_{10}, C_5H_8 and C_6H_{14}. Which two are saturated and belong to the same homologous series?

(a) C_3H_6 and C_5H_8

(b) C_4H_{10} and C_6H_{14}

(c) C_3H_6 and C_4H_{10}

(d) C_5H_8 and C_6H_{14}

Q2. MCQ **[1 mark]**

A student wants to name the compound $CH_3CH_2CH_2CHO$. Its correct IUPAC name is:

(a) Propanal (b) Butanal

(c) Butanone (d) Butanoic acid

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Carbon forms a very large number of compounds.

Reason (R): Carbon has the property of catenation and a valency of four.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

A cook notices that the gas burner gives a yellow sooty flame and blackens the bottom of the vessel, although it gave a clean blue flame earlier.

(i) What does the yellow flame indicate? (ii) Suggest what has gone wrong and how it can be corrected.

Q5. CASE / DATA BASED **[3 marks]**

Two test tubes contain samples labelled X and Y. X is a liquid with a smell of vinegar; Y is a liquid used in cough syrups and as a fuel additive. When a pinch of sodium hydrogencarbonate is added to each, only X gives brisk effervescence.

(i) Identify X and Y. (ii) Write the equation for the reaction of X with sodium hydrogencarbonate. (iii) Name the sweet-smelling product formed when X and Y are warmed with a few drops of concentrated sulphuric acid.

Q6. CASE / DATA BASED **[3 marks]**

The boiling points of the first members of the alcohol series are given below.

Alcohol	Formula	Boiling point (°C)
Methanol	CH_3OH	65
Ethanol	C_2H_5OH	78
Propanol	C_3H_7OH	97

(i) What trend is seen and why? (ii) Write the formula of the next member and state the difference in molecular mass between successive members. (iii) Why do all three show similar chemical properties?

Q7. SOURCE BASED **[4 marks]**

Soaps are sodium or potassium salts of long-chain carboxylic acids. A soap molecule has an ionic end that dissolves in water and a long carbon chain that dissolves in oil. When soap is added to water containing oily dirt, the molecules arrange themselves into clusters that trap the oil in the centre, which can then be rinsed away. In hard water, however, a scummy white solid forms and little lather is produced.

(i) What name is given to the clusters formed by soap molecules? **[1]**

(ii) Which end of the soap molecule faces the oil droplet? **[1]**

(iii) Why does soap form scum in hard water, and what is used instead to clean clothes in such water? **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

Vegetable oils are converted into vanaspati ghee in industry by heating them with hydrogen in the presence of a nickel catalyst. (i) Name this reaction and state the structural change in the oil molecules. (ii) Health experts advise using vegetable oils rather than animal fats for cooking. Give the reason. (iii) Suggest a simple test to distinguish an unsaturated compound from a saturated one.

ANSWER KEY

Answer Q1 (b) Both fit the alkane formula C_nH_{2n+2} and differ by a CH_2 unit ($2 \times 14 = 28$ u).

Answer Q2 (b) It has four carbon atoms (but-) and an aldehyde group (-al) at the end of the chain.

Answer Q3 (a) Both are true. Catenation (forming long chains, branches and rings) together with tetravalency lets carbon bond with many atoms, so R explains A.

Answer Q4 (i) Incomplete combustion of the fuel, which releases unburnt carbon (soot). (ii) The air holes of the burner are probably blocked, so less oxygen reaches the fuel; cleaning the air inlets restores a supply of air and the clean blue flame.

Answer Q5 (i) X: ethanoic acid (CH_3COOH); Y: ethanol (C_2H_5OH). (ii) $CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$. (iii) Ethyl ethanoate, an ester (esterification).

Answer Q6 (i) Boiling point rises with increasing molecular size and mass in a homologous series. (ii) C_4H_9OH (butanol); successive members differ by CH_2, that is 14 u. (iii) They have the same functional group (-OH), which decides chemical properties.

Answer Q7 **(i)** Micelles. **(ii)** The hydrophobic (hydrocarbon tail) end faces the oil droplet. **(iii)** Hard water contains calcium and magnesium salts, which react with soap to form insoluble precipitates (scum), wasting the soap. Detergents are used instead, as their calcium and magnesium salts are soluble and they lather well in hard water.

Answer Q8 (i) Hydrogenation (an addition reaction); carbon-carbon double bonds of the unsaturated oil take up hydrogen and become single bonds, producing a saturated fat. (ii) Animal fats contain saturated fatty acids, which are considered harmful to health, while vegetable oils are rich in healthier unsaturated fatty acids. (iii) Add bromine water: an unsaturated compound decolourises it; a saturated compound does not.

CHAPTER 5 Life Processes

Q1. MCQ **[1 mark]**

A student keeps a potted plant in a dark room for 48 hours. A strip of black paper is then clipped across the middle of one leaf on both surfaces and the plant is kept in bright sunlight for 6 hours. The leaf is boiled in alcohol and tested with iodine solution. Which observation is expected?

(a) The whole leaf turns blue-black

(b) Only the covered strip turns blue-black

(c) Only the uncovered parts turn blue-black

(d) No part of the leaf turns blue-black

Q2. MCQ **[1 mark]**

During a 400 m sprint, an athlete's leg muscles begin to ache and cramp. Which process in the muscle cells is mainly responsible for this?

(a) Aerobic respiration producing carbon dioxide and water

(b) Anaerobic respiration producing lactic acid

(c) Fermentation producing ethanol and carbon dioxide

(d) Breakdown of glucose into pyruvate in the mitochondria

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): The walls of the ventricles of the human heart are thicker than those of the atria.
Reason (R): Ventricles have to pump blood under high pressure to the lungs and to all parts of the body.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

A student recorded her breathing rate at rest and after running for 5 minutes.

Condition	Breaths per minute
At rest	16
After running	34

(i) By what percentage did the breathing rate increase? Why does the body need this increase? (2)
(ii) After the run she felt cramps in her leg muscles. Name the substance responsible and the process that produced it. (1)

Q5. CASE / DATA BASED **[2 marks]**

In a healthy adult, the kidneys filter about 180 L of fluid from the blood every day, but only about 1.5 L of urine is passed out.
(i) Calculate the percentage of the filtrate that is reabsorbed. (1)
(ii) Name the part of the nephron where filtration occurs and state one substance that is selectively reabsorbed. (1)

Q6. CASE / DATA BASED **[3 marks]**

A student took four similar leafy shoots and placed each in a measuring cylinder containing 50 mL of water, covering the water surface with a thin layer of oil. Petroleum jelly was applied to different leaf surfaces as shown. The water left after 24 hours was recorded.

Shoot	Treatment	Water left (mL)
A	No jelly	38
B	Jelly on upper surface only	40
C	Jelly on lower surface only	48
D	Jelly on both surfaces	49

(i) Why was a layer of oil placed on the water? (1)
(ii) Which leaf surface has more stomata? Justify using the data. (1)
(iii) State one way in which this water loss helps the plant. (1)

Q7. SOURCE BASED **[4 marks]**

Two test tubes, A and B, each received 5 mL of 1% starch solution. To tube A, 1 mL of fresh saliva was added; to tube B, 1 mL of saliva that had been boiled for 5 minutes. Both tubes were kept in a water bath at 37 °C. Every 2 minutes, a drop from each tube was placed on a tile with dilute iodine solution. After 10 minutes, the drop from A gave a brown-yellow colour while the drop from B was still blue-black.

(i) Name the enzyme in saliva that acted on starch in tube A. **[1]**

(ii) Why was the water bath kept at 37 °C? **[1]**

(iii) Explain the different results of tubes A and B. Name the product formed in tube A. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A patient's bile duct is blocked by a gallstone, so bile cannot reach the small intestine. Predict how digestion of a fried meal would be affected and explain why. Also state one other function of bile that is lost.

ANSWER KEY

Answer Q1 (c) Only the uncovered parts received light and made starch by photosynthesis; the covered strip had no light, so no starch formed there (the dark period had removed old starch).

Answer Q2 (b) When oxygen supply falls short during intense activity, pyruvate is converted into lactic acid in the muscle cells; its build-up causes cramps.

Answer Q3 (a) Both are true. Ventricles send blood over long distances under high pressure, so they need thick muscular walls, whereas atria only push blood into the ventricles.

Answer Q4 (i) Increase = (34 − 16)/16 × 100 = 112.5%. Exercising muscles need more energy, so more O_2 must be taken in and more CO_2 removed for faster aerobic respiration.
(ii) Lactic acid, formed by anaerobic respiration in muscle cells when oxygen supply falls short: glucose → pyruvate → lactic acid + energy.

Answer Q5 (i) Reabsorbed = (180 − 1.5)/180 × 100 = 99.2% (approx.).
(ii) Filtration occurs in the glomerulus inside Bowman's capsule; glucose (also amino acids, salts and most of the water) is reabsorbed in the tubule.

Answer Q6 (i) To stop direct evaporation from the water surface, so that the loss measured is due only to the plant.
(ii) The lower surface: blocking it (C) cut the loss from 12 mL to 2 mL, while blocking the upper surface (B) reduced it only to 10 mL.
(iii) Transpiration creates a suction pull that draws water and dissolved minerals up through the xylem from the roots; it also helps keep the leaves cool.

Answer Q7 **(i)** Salivary amylase. **(ii)** 37 °C is body temperature, the optimum temperature at which the enzyme works best. **(iii)** In A the active amylase digested starch into a simple sugar (maltose), so iodine no longer turned blue-black. In B boiling denatured (destroyed) the enzyme, so starch remained and gave the blue-black colour.

Answer Q8 Bile salts emulsify fats, breaking large fat globules into tiny droplets that give lipase a larger surface to act on; without bile, fat digestion and absorption become poor and undigested fat passes out. Bile also makes the food coming from the stomach alkaline, which pancreatic enzymes need; this neutralisation of acid would be reduced.

CHAPTER 6 Control and Coordination

Q1. MCQ **[1 mark]**

A gymnast walks along a narrow balance beam without falling and makes precise turns. Which part of the brain mainly coordinates this balance and posture?

(a) Cerebrum (b) Cerebellum
(c) Medulla (d) Hypothalamus

Q2. MCQ **[1 mark]**

When the leaves of a touch-me-not (*Mimosa pudica*) plant are touched, they fold within seconds. Which statement correctly explains this movement?

(a) Cells at the touched spot grow faster on one side
(b) Auxin moves rapidly to the shaded side of the leaf
(c) Cells change shape by changing the amount of water in them
(d) Nerve impulses travel along special nerve fibres in the stem

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): The tendril of a pea plant coils around a support it touches.
Reason (R): The part of the tendril in contact with the support grows faster than the part away from it.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

While cooking, a person accidentally touches a hot pan and pulls the hand away in about 0.1 s, even before feeling pain.
(i) Write, in the correct order, the path the impulse follows in this action. (2)
(ii) Why is it useful that this action does not wait for the brain? (1)

Q5. CASE / DATA BASED **[2 marks]**

A lab technician recorded fasting blood glucose levels of three people (normal range 70–100 mg/dL).

Person	Fasting glucose (mg/dL)
P	88
Q	172
R	95

(i) Which person is likely diabetic? Name the hormone whose deficiency causes this. (1)
(ii) Name the gland that secretes this hormone and state its role. (1)

Q6. CASE / DATA BASED **[3 marks]**

In an experiment, identical groups of dwarf pea seedlings were sprayed with different plant hormones, and the average stem length was measured after three weeks.

Group	Spray	Average stem length (cm)
P	Water only (control)	12
Q	Hormone X	31
R	Hormone Y	7

(i) Identify hormone X, which promotes stem growth. (1)
(ii) Hormone Y also caused some leaves to wilt. Name it and state its general role. (1)
(iii) Why was group P included in the experiment? (1)

Q7. SOURCE BASED **[4 marks]**

A student grew bean seedlings in a closed cardboard box that had a single small window on the right side. After four days, all the shoots were bent towards the window. When the box was rotated so that the window faced left, the newly growing tips bent towards the left within two days. The roots in the pot did not bend towards the light.

(i) Name the movement shown by the shoots. **[1]**

(ii) Name the plant hormone responsible for this bending. **[1]**

(iii) Explain how this hormone makes the shoot bend towards light. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A potted seedling is laid on its side for two days. Its shoot tip turns upwards while its root tip turns downwards. Explain this observation. Why is this response useful to the plant?

ANSWER KEY

Answer Q1 (b) The cerebellum, part of the hindbrain, maintains posture and balance and makes voluntary movements precise.

Answer Q2 (c) The movement is independent of growth; plant cells lose or gain water, change shape, and the leaflets fold. Plants have no nerves; the information passes electrically and chemically from cell to cell.

Answer Q3 (c) A is true but R is false. The part of the tendril away from the support grows faster than the part in contact, so the tendril curls around the support.

Answer Q4 (i) Receptor in skin (heat) → sensory neuron → relay neuron in the spinal cord → motor neuron → effector muscle of the arm (contracts).
(ii) The reflex arc is completed in the spinal cord, so the response is very fast and prevents injury; thinking through the brain would take longer.

Answer Q5 (i) Person Q; deficiency of insulin.
(ii) Pancreas; insulin lowers blood sugar by helping cells take up glucose (and its storage), keeping the level normal.

Answer Q6 (i) Gibberellin, which promotes growth of the stem.
(ii) Abscisic acid; it inhibits growth and causes wilting of leaves.
(iii) Group P is the control: it shows normal growth without any hormone, so the effect of each hormone can be judged by comparison.

Answer Q7 **(i)** Positive phototropism. **(ii)** Auxin. **(iii)** Auxin is made at the shoot tip and, when light comes from one side, diffuses towards the shaded side. There it makes the cells elongate more, so the shaded side grows longer and the shoot bends towards the light.

Answer Q8 The plant responds to gravity (geotropism): the shoot is negatively geotropic and grows away from gravity, while the root is positively geotropic and grows towards it; unequal distribution of growth hormone causes unequal growth on the two sides. This keeps the shoot growing up towards light for photosynthesis and the root growing down into soil for anchorage, water and minerals.

Exam tip: case and data based questions

1. Read the question parts first, then the passage, so you know what to look for.
2. Quote numbers from the table or graph in your answer; examiners look for them.
3. Keep the unit with every value.

CHAPTER 7 How do Organisms Reproduce?

Q1. MCQ **[1 mark]**

A gardener places a Bryophyllum leaf on moist soil. After some days, small plantlets grow from the notches along its margin. This is an example of:

(a) Spore formation
(b) Fragmentation
(c) Vegetative propagation
(d) Regeneration

Q2. MCQ **[1 mark]**

A health worker explains family-planning methods to a community. Which method both prevents pregnancy and protects against sexually transmitted infections such as HIV-AIDS?

(a) Oral contraceptive pills
(b) Copper-T placed in the uterus
(c) Use of condoms
(d) Surgical blocking of the vas deferens

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Offspring produced by sexual reproduction show more variation than those produced by asexual reproduction.
Reason (R): Sexual reproduction combines DNA from two different individuals.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

In a culture, a lab technician placed 5 Amoeba cells. Each cell divides into two every 2 hours under ideal conditions.
(i) How many Amoeba will be present after 8 hours? (1)
(ii) Name this mode of reproduction and state how it differs in Leishmania. (1)

Q5. CASE / DATA BASED **[3 marks]**

A student examined two flowers and recorded:

Flower	Stamens	Pistil
X (hibiscus)	Present	Present
Y (papaya)	Present	Absent

(i) Classify X and Y as bisexual or unisexual. (1)
(ii) Which flower can show self-pollination? Give a reason. (1)
(iii) After fertilisation, what do the ovary and ovules develop into? (1)

Q6. CASE / DATA BASED **[3 marks]**

A student dusted pollen grains of a flowering plant onto glass slides containing sugar solutions of different strengths and counted the grains that had produced a tube after 3 hours (100 grains per slide).

Sugar solution (%)	0	5	10	20
Grains with a tube	2	35	68	21

(i) At which concentration did pollen germinate best? Suggest why sugar is needed. (1)
(ii) In a flower, through which part does the pollen tube grow, and what does it carry to the ovule? (1)
(iii) After fertilisation, what do the ovule and the ovary develop into? (1)

Q7. SOURCE BASED **[4 marks]**

In a human female, one ovary releases an egg roughly every month. At the same time the inner lining of the uterus becomes thick and spongy and is richly supplied with blood. If the egg meets a sperm, the zygote implants in this lining and develops into an embryo, which stays connected to the mother through a special tissue. If the egg is not fertilised, the lining slowly breaks down and comes out through the vagina as blood and mucus.

(i) Where in the female reproductive system does fertilisation take place? **[1]**
(ii) What is the process of monthly breakdown of the uterine lining called? **[1]**
(iii) Name the special tissue that connects the embryo to the mother and state two of its functions. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A farmer owns a mango tree that gives exceptionally sweet fruit and wants 50 new trees with exactly the same fruit quality, bearing fruit as early as possible. Should the farmer raise them from seeds? Suggest a suitable method and give two reasons for your choice.

ANSWER KEY

Answer Q1 (c) Buds in the leaf notches develop into new plants, which is vegetative propagation from a leaf.

Answer Q2 (c) A condom is a barrier method: it prevents sperm from reaching the egg and also blocks the transfer of infection-causing microbes. The other methods do not prevent infection.

Answer Q3 (a) Both are true. Fusion of gametes from two parents brings new combinations of DNA, so variation is greater than in asexual reproduction where only copying errors cause variation.

Answer Q4 (i) 8 h = 4 divisions; number = $5 \times 2^4 = 80$.
(ii) Binary fission. In Amoeba the cell can split in any plane, while in Leishmania it splits in a definite plane (lengthwise) with respect to its whip-like flagellum.

Answer Q5 (i) X is bisexual; Y is unisexual (male).
(ii) X, because it has both stamens and pistil, so its own pollen can reach its stigma; Y has no pistil.
(iii) The ovary develops into the fruit and the ovules into seeds.

Answer Q6 (i) 10% sugar (68 grains); sugar acts like the sugary fluid on the stigma and provides nourishment for tube growth.
(ii) The pollen tube grows through the style to reach the ovule in the ovary, carrying the male germ-cell to fuse with the female germ-cell.
(iii) The ovule develops into a seed and the ovary develops into a fruit.

Answer Q7 **(i)** In the oviduct (fallopian tube). **(ii)** Menstruation. **(iii)** Placenta. It passes glucose and oxygen from the mother's blood to the embryo, and removes wastes produced by the embryo into the mother's blood.

Answer Q8 Seeds should not be used, because seeds come from sexual reproduction and the new plants may differ from the parent. Grafting (vegetative propagation) of a branch of the sweet tree onto a rootstock is suitable: the new plants are genetically identical to the parent, so the fruit quality is preserved, and they bear fruit much earlier than seed-grown plants.

CHAPTER 8 Heredity

Q1. MCQ **[1 mark]**

A pure tall pea plant (TT) is crossed with a pure dwarf plant (tt). The F_1 plants are self-pollinated and 800 F_2 plants are obtained. About how many F_2 plants are expected to be dwarf?

(a) 0 (b) 200
(c) 400 (d) 600

Q2. MCQ **[1 mark]**

In a pea plant, height depends on the amount of a growth hormone, whose production needs a particular enzyme. A change in the gene for this enzyme makes the enzyme work less efficiently. The plant will most likely be:

(a) tall, because the gene is still present
(b) short, because less growth hormone is made
(c) tall, because the hormone is made in other parts
(d) short, because the plant loses its chromosomes

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): A mouse whose tail has been cut off still produces offspring with normal tails.
Reason (R): Changes in non-reproductive tissues do not change the DNA of the germ cells.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

Pea plants with round, yellow seeds (RRYY) were crossed with plants having wrinkled, green seeds (rryy). The F_1 plants were self-pollinated, and 640 F_2 seeds were counted:

Round yellow	Round green	Wrinkled yellow	Wrinkled green
360	120	120	40

(i) What will be the appearance of the F_1 seeds? (1)
(ii) Write the F_2 ratio. (1)
(iii) What do the new combinations (round green and wrinkled yellow) show? (1)

Q5. CASE / DATA BASED **[2 marks]**

A couple already has three daughters. Some relatives say the fourth child is now certain to be a son and that the mother decides the sex of the child.
(i) What is the probability that the fourth child will be a boy? (1)
(ii) Which parent's gamete actually decides the sex of the child? Explain briefly. (1)

Q6. CASE / DATA BASED **[3 marks]**

The table shows the number of chromosomes in the body cells of three organisms.

Organism	Body cell	Gamete
Human	46	?
Pea	14	?
Dog	78	?

(i) Write the number of chromosomes in the gamete of each organism. (1)
(ii) Explain why gametes must carry only half the number of chromosomes. (1)
(iii) A pea zygote is formed by fertilisation. How many chromosomes does it have, and how many come from each parent? (1)

Q7. SOURCE BASED **[4 marks]**

A researcher crossed pure-breeding pea plants having violet flowers with pure-breeding plants having white flowers. All the F_1 plants had violet flowers. When F_1 plants were allowed to self-pollinate, the F_2 generation had 912 plants with violet flowers and 304 plants with white flowers. No plant had pale or mixed-coloured flowers.

(i) Which flower colour is dominant? **[1]**

(ii) Calculate the ratio of violet to white plants in F_2. **[1]**

(iii) Using V and v, write the genotypes of the F_1 plants and of the three types of F_2 plants, with their genotypic ratio. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

Two tall pea plants were crossed and, out of 120 offspring, about 30 were dwarf. A student argues that dwarf plants cannot arise from two tall parents. Explain the result with the genotypes of the parents and a cross, and state the number of offspring that would be pure tall.

ANSWER KEY

Answer Q1 (b) F_2 ratio tall : dwarf = 3 : 1, so dwarf = 800 × 1/4 = 200.

Answer Q2 (b) A gene carries the information to make a protein (here the enzyme); a less efficient enzyme produces less hormone, so the plant grows short.

Answer Q3 (a) Both are true. Cutting the tail is an acquired trait; it does not alter the DNA in the germ cells, so it is not inherited.

Answer Q4 (i) All round and yellow (Rr Yy), as round and yellow are dominant.
(ii) 360 : 120 : 120 : 40 = 9 : 3 : 3 : 1.
(iii) The traits for seed shape and seed colour are inherited independently of each other.

Answer Q5 (i) 1/2 (50%); each birth is independent of earlier births.
(ii) The father's. All eggs carry an X chromosome, while half the sperms carry X and half carry Y; an X sperm gives a girl (XX) and a Y sperm a boy (XY).

Answer Q6 (i) Human 23, pea 7, dog 39.
(ii) Gametes are made by a special division that halves the chromosome number, so that when two gametes fuse, the original number is restored and stays constant from generation to generation.
(iii) 14 chromosomes: 7 from the male gamete and 7 from the female gamete.

Answer Q7 **(i)** Violet. **(ii)** 912 : 304 = 3 : 1. **(iii)** F_1: Vv. F_2: VV, Vv and vv in the ratio 1 : 2 : 1 (VV and Vv violet, vv white).

Answer Q8 Both parents must be heterozygous (Tt), each carrying the recessive dwarf allele. Tt × Tt → TT : Tt : tt = 1 : 2 : 1, so 1/4 are dwarf (tt) = 30 of 120. Pure tall (TT) = 1/4 × 120 = 30 plants.

CHAPTER 9 Light: Reflection and Refraction

Q1. MCQ **[1 mark]**

A dentist holds a concave mirror 10 cm from a tooth and sees a real image that is three times the size of the tooth. Where is the image formed?

(a) 30 cm in front of the mirror

(b) 30 cm behind the mirror

(c) 15 cm in front of the mirror

(d) 15 cm behind the mirror

Q2. MCQ **[1 mark]**

A ray of light falls on a plane mirror fixed on a wall at an angle of incidence of 35°. What is the angle between the incident ray and the reflected ray?

(a) 35° (b) 55°

(c) 70° (d) 110°

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): A ray of light passing through the optical centre of a thin convex lens emerges without any deviation.

Reason (R): Near the optical centre, the two surfaces of a thin lens are almost parallel, so the lens behaves like a thin glass slab there.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

A student notes the speed of light in three media:

Medium	Speed of light (m/s)
Water	2.25×10^8
Crown glass	2.0×10^8
Diamond	1.24×10^8

(i) Find the absolute refractive index of water ($c = 3 \times 10^8$ m/s).

(ii) A ray passes obliquely from water into crown glass. Will it bend towards or away from the normal? Why?

Q5. CASE / DATA BASED **[3 marks]**

In a lab, a student uses a convex lens of focal length 20 cm and gets a sharp, inverted image of a candle flame on a screen kept 60 cm from the lens.

(i) How far is the candle from the lens?

(ii) Find the magnification and the power of the lens.

Q6. CASE / DATA BASED **[3 marks]**

In the glass-slab experiment, a student measured the angle of incidence (i) and the angle of refraction (r) at the first surface:

i	sin i	r	sin r
30°	0.500	19.5°	0.334
45°	0.707	28.1°	0.471
60°	0.866	35.3°	0.578

(i) Calculate sin i / sin r for each reading and name the law it verifies. (ii) Find the speed of light in this glass ($c = 3 \times 10^8$ m/s). (iii) How is the emergent ray related to the incident ray?

Q7. SOURCE BASED **[4 marks]**

A city bus is fitted with a convex rear-view mirror whose radius of curvature is 2.0 m. The driver notices that a car 4.0 m behind the bus looks much smaller in the mirror, yet the mirror shows a wide stretch of the road, including both lanes and the pavement. Such mirrors are preferred on vehicles even though they make objects appear farther than they really are.

(i) State two reasons why a convex mirror is preferred as a rear-view mirror. **[1]**

(ii) Find the focal length of the mirror with the correct sign. **[1]**

(iii) Find the position of the image of the car and the magnification produced. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A watchmaker holds a convex lens of focal length 15 cm at 10 cm from a tiny screw. Find the position and nature of the image and the magnification. Why must the screw be kept within 15 cm of the lens for this use?

ANSWER KEY

Answer Q1 (a) Real image, so $m = -3 = -v/u$; with $u = -10$ cm, $v = 3u = -30$ cm, i.e. 30 cm in front of the mirror.

Answer Q2 (c) Angle of reflection = angle of incidence = 35°, so the angle between the two rays is 35° + 35° = 70°.

Answer Q3 (a) Both are true and R explains A: a thin parallel slab produces no deviation (only a negligible lateral shift).

Answer Q4 (i) $n = c/v = (3 \times 10^8)/(2.25 \times 10^8) = 1.33$. (ii) Towards the normal, because light slows down ($2.25 \rightarrow 2.0 \times 10^8$ m/s) on entering the optically denser glass.

Answer Q5 (i) $1/v - 1/u = 1/f \rightarrow 1/60 - 1/u = 1/20 \rightarrow 1/u = (1 - 3)/60 = -1/30$, so u = −30 cm (30 cm in front of the lens). (ii) $m = v/u = 60/(-30) = -2$ (real, inverted, twice the size). $P = 1/f(m) = 1/0.20 = +5$ D.

Answer Q6 (i) The ratio is 1.50 in every case (0.500/0.334, 0.707/0.471, 0.866/0.578); this verifies Snell's law, and 1.5 is the refractive index of the glass. (ii) $v = c/n = (3 \times 10^8)/1.5 = 2 \times 10^8$ m/s. (iii) It is parallel to the incident ray but laterally displaced.

Answer Q7 **(i)** It always forms an erect, diminished image, and it gives a much wider field of view. **(ii)** $f = R/2 = +2.0/2 = +1.0$ m (positive, as the focus is behind a convex mirror). **(iii)** u = −4.0 m; $1/v = 1/f - 1/u = 1/1 + 1/4 = 1.25$, so v = +0.8 m (virtual, behind the mirror). $m = -v/u = -0.8/(-4.0) = +0.2$ (erect, one-fifth size).

Answer Q8 $1/v = 1/f + 1/u = 1/15 + 1/(-10) = (2 - 3)/30 = -1/30$, so v = −30 cm (same side as the screw). $m = v/u = (-30)/(-10) = +3$: virtual, erect and three times enlarged. Only when the object lies between the optical centre and F does a convex lens give an erect, magnified virtual image, so the screw must be within 15 cm.

CHAPTER 10 The Human Eye and the Colourful World

Q1. MCQ **[1 mark]**

A student can read a book clearly but cannot read the blackboard; his far point is 2 m. Which lens will correct his vision?

(a) Convex lens of power +0.5 D

(b) Concave lens of power −0.5 D

(c) Concave lens of power −2 D

(d) Convex lens of power +2 D

Q2. MCQ **[1 mark]**

A 60-year-old reader cannot see distant road signs clearly and also needs to hold a newspaper far away to read it. The most suitable spectacles have:

(a) a concave lens only

(b) a convex lens only

(c) bifocal lenses: concave in the upper part and convex in the lower part

(d) bifocal lenses: convex in the upper part and concave in the lower part

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): To an astronaut in space, far above the Earth's atmosphere, the sky appears dark instead of blue. Reason (R): Scattering of sunlight does not occur in the absence of an atmosphere.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

An eye test shows that a person's near point has shifted to 75 cm, so a newspaper has to be held at arm's length.
(i) Name the defect.
(ii) Calculate the focal length and power of the lens needed to read comfortably at 25 cm.

Exam tip

1. Draw a quick labelled diagram whenever the question allows it; it earns marks even if the words fall short.
2. Use the exact scientific term from the chapter, not an everyday word.

Q5. CASE / DATA BASED **[2 marks]**

A traffic engineer compares wavelengths of light used in signals:

Colour	Approx. wavelength (nm)
Red	700
Green	530
Blue	450

(i) Which colour is scattered least by fog and smoke particles?
(ii) Use this to explain why danger signals are red.

Q6. CASE / DATA BASED **[3 marks]**

A student photographs the Sun at three times on a clear day and records its colour:

Time	Position of the Sun	Apparent colour
6:05 a.m.	Just above the horizon	Reddish-orange
12:15 p.m.	Overhead	White
6:20 p.m.	Near the horizon	Reddish

(i) Why does the Sun look reddish near the horizon? (ii) Why does it look white at noon? (iii) The Sun was visible a little before the time of actual sunrise given in the almanac. Name the phenomenon responsible.

Q7. SOURCE BASED **[4 marks]**

A student passes a narrow beam of white light through a glass prism (apex pointing up) and sees a band of seven colours on a screen. She then places a second identical prism in an inverted position close to the first. The colours disappear and a patch of white light is seen on the screen. The same splitting of sunlight by water droplets after a shower produces a rainbow.

(i) Name the phenomenon observed with the first prism. **[1]**

(ii) Which colour appears at the bottom of the band? Give a reason. **[1]**

(iii) What does the result with the second prism show about white light? Explain briefly. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

During a night-sky observation, a student notices that stars twinkle while planets shine steadily. The next morning she learns that the Sun is visible about 2 minutes before it actually crosses the horizon.
(i) Explain both observations.
(ii) How would the length of a day change if the Earth had no atmosphere?

ANSWER KEY

Answer Q1 (b) Myopia is corrected by a concave lens with f = −(far point) = −2 m, so P = 1/f = −0.5 D.

Answer Q2 (c) The upper concave part corrects distant (myopic) vision; the lower convex part corrects near vision (presbyopia).

Answer Q3 (a) Both are true and R explains A: the blue colour of the sky is due to scattering by air molecules; with no atmosphere, there is no scattering and the sky looks dark.

Answer Q4 (i) Hypermetropia (long-sightedness). (ii) Object at u = −25 cm must give a virtual image at v = −75 cm. 1/f = 1/v − 1/u = −1/75 + 1/25 = 2/75, so f = +37.5 cm (convex lens). P = 100/37.5 = +2.67 D.

Answer Q5 (i) Red, as it has the longest wavelength and fine particles scatter longer wavelengths the least. (ii) Red light is scattered least, so it travels farthest through fog or smoke and remains visible from a large distance.

Answer Q6 (i) Near the horizon, sunlight travels a longer path through the atmosphere; most blue light of shorter wavelength is scattered away and mainly red light reaches the eye. (ii) At noon the path is shorter, so little scattering takes place and all colours reach us together. (iii) Atmospheric refraction, which advances sunrise (and delays sunset) by about 2 minutes.

Answer Q7 **(i)** Dispersion of white light. **(ii)** Violet: it has the shortest wavelength and is deviated (bent) the most towards the base of the prism. **(iii)** White light is made of seven colours. The inverted prism bends each colour equally in the opposite direction, so all colours recombine and emerge as white light.

Answer Q8 (i) Starlight undergoes continuous atmospheric refraction in layers of changing density; a star is a point source, so the light entering the eye flickers (twinkling). A planet is an extended source, a collection of point sources, so the variations average out. The same refraction bends sunlight over the horizon, showing the Sun early. (ii) Without an atmosphere there would be no advance sunrise or delayed sunset, so the day would be shorter by about 4 minutes.

CHAPTER 11 Electricity

Q1. MCQ **[1 mark]**

An electrician has three 6 Ω resistors and needs an equivalent resistance of 9 Ω. How should they be connected?

(a) All three in series

(b) All three in parallel

(c) Two in parallel, and this pair in series with the third

(d) Two in series, and this pair in parallel with the third

Q2. MCQ **[1 mark]**

A manufacturer must choose a material for the heating element of an electric toaster. Which property makes nichrome the best choice?

(a) Low resistivity and low melting point

(b) High resistivity and resistance to oxidation at high temperature

(c) High conductivity and low cost

(d) Low resistivity and high density

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): When the length of a wire is doubled keeping its area of cross-section the same, its resistance doubles.

Reason (R): The resistivity of a material depends on the length of the wire.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

A family uses a 1500 W room heater on 220 V mains for 2 hours every day in December (31 days). Electricity costs Rs 7 per unit.

(i) Find the current drawn by the heater.

(ii) Find the energy used in the month and the cost.

Q5. CASE / DATA BASED **[2 marks]**

A student records the following for a nichrome wire at constant temperature:

V (volt)	1.0	2.0	3.0	4.0
I (ampere)	0.25	0.50	0.75	1.00

(i) Find the resistance of the wire and name the law shown.

(ii) Predict the current at 5.0 V.

Q6. CASE / DATA BASED **[3 marks]**

A heater coil is made from a nichrome wire 2.0 m long with a cross-sectional area of 1.0×10^{-7} m^2. The resistivity of nichrome is 1.1×10^{-6} Ω m.

(i) Calculate the resistance of the coil. (ii) Find the current it draws from 220 V mains. (iii) How much heat does it produce in 5 minutes?

Q7. SOURCE BASED **[4 marks]**

A household circuit runs at 220 V. A fuse wire, made of a material with a low melting point, is connected in the circuit to protect appliances. When the current exceeds the rated value, the heat produced ($H = I^2Rt$) melts the wire and breaks the circuit. On one line, a 1.1 kW electric iron and a 660 W microwave oven are often used together.

(i) Why is the fuse always connected in series with the appliances? **[1]**

(ii) Calculate the current drawn by the iron alone. **[1]**

(iii) Which fuse rating, 5 A or 10 A, should be used on this line when both appliances run together? Justify. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

Two bulbs rated 100 W, 220 V and 60 W, 220 V are first connected in series and then in parallel across 220 V mains. Which bulb glows brighter in each case? Support your answer with calculation.

ANSWER KEY

Answer Q1 (c) Two 6 Ω in parallel give 3 Ω; adding 6 Ω in series gives 3 + 6 = 9 Ω.

Answer Q2 (b) An alloy like nichrome has high resistivity, so it produces more heat, and it does not oxidise (burn) readily when red hot.

Answer Q3 (c) A is true since R = ρl/A is proportional to l. R is false: resistivity depends only on the material and temperature, not on the dimensions.

Answer Q4 (i) I = P/V = 1500/220 = 6.8 A (approx.). (ii) E = P × t = 1.5 kW × 2 h × 31 = 93 kWh (93 units). Cost = 93 × 7 = Rs 651.

Answer Q5 (i) R = V/I = 1.0/0.25 = 4 Ω (constant for every pair); V is proportional to I, i.e. Ohm's law. (ii) I = V/R = 5.0/4 = 1.25 A.

Answer Q6 (i) $R = \rho l/A = (1.1 \times 10^{-6} \times 2.0)/(1.0 \times 10^{-7}) = 22\ \Omega$. (ii) $I = V/R = 220/22 = 10$ A. (iii) $H = VIt = 220 \times 10 \times 300 = 6.6 \times 10^5$ J.

Answer Q7 **(i)** So that the whole current passes through it; when it melts, the entire circuit is broken. **(ii)** $I = P/V = 1100/220 = 5$ A. **(iii)** Total power = 1100 + 660 = 1760 W, so $I = 1760/220 = 8$ A. A 5 A fuse would blow during normal use, so a 10 A fuse is suitable.

Answer Q8 $R = V^2/P$: $R_{100} = 48400/100 = 484\ \Omega$; $R_{60} = 48400/60 = 807\ \Omega$. In series the current is the same, so $P = I^2R$ is larger for the larger resistance: the 60 W bulb glows brighter. In parallel each gets 220 V, so $P = V^2/R$ is larger for the smaller resistance: the 100 W bulb glows brighter.

CHAPTER 12 Magnetic Effects of Electric Current

Q1. MCQ **[1 mark]**

A long straight vertical wire carries current upwards. What is the direction of the magnetic field at a point due east of the wire?

(a) Towards north (b) Towards south

(c) Towards east (d) Vertically upwards

Q2. MCQ **[1 mark]**

A student wants to increase the strength of the magnetic field at the centre of a circular coil. Which change will NOT help?

(a) Increasing the current

(b) Increasing the number of turns

(c) Reducing the radius of the coil

(d) Reversing the direction of the current

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Two magnetic field lines never intersect each other.

Reason (R): If they intersected, the compass needle at the point of intersection would have to point in two directions at once, which is impossible.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Q4. CASE / DATA BASED **[2 marks]**

A user gets a mild shock on touching the metal body of a washing machine. An electrician finds that the green wire from the socket is loose.

(i) Name this wire and state its function.

(ii) Why is it especially important for appliances with a metal body?

Q5. CASE / DATA BASED **[3 marks]**

A student winds a solenoid on a cardboard tube and counts the iron pins it lifts:

Turns	Current (A)	Pins lifted
50	1	4
50	2	8
100	2	16

(i) What two conclusions can be drawn from the data? (ii) What should be placed inside the tube to make a strong electromagnet, and why?

Q6. CASE / DATA BASED **[3 marks]**

The following appliances in a room are connected to one 15 A circuit on 220 V mains and are often switched on together:

Appliance	Power
Air conditioner	2000 W
Room heater	1500 W
4 bulbs	100 W each

(i) Calculate the total current drawn. (ii) What will happen in the circuit, and what is this condition called? (iii) Why are household appliances connected in parallel?

Q7. SOURCE BASED **[4 marks]**

In an experiment, a light aluminium rod is hung horizontally by two thin wires so that it lies between the poles of a strong horseshoe magnet, with the field perpendicular to the rod. When a current is passed through the rod, it is pushed out of the gap. On reversing the current, the rod moves into the gap. When the rod is turned to lie along the field, it does not move at all.

(i) Name the rule used to find the direction of the force on the rod. **[1]**

(ii) What happens to the rod if both the current and the magnetic field are reversed? **[1]**

(iii) The field is vertically upwards and the current in the rod flows from west to east. Find the direction of the force. When is this force maximum? **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

A beam of electrons moving horizontally from south to north enters a region where the magnetic field acts vertically downwards.
(i) In which direction will the beam be deflected? Explain.
(ii) Would an electron at rest in this field experience any force? Why?

ANSWER KEY

Answer Q1 (a) By the right-hand thumb rule (thumb up), the field lines are anticlockwise when seen from above; at a point to the east the field points north.

Answer Q2 (d) Reversing the current reverses the direction of the field but does not change its strength.

Answer Q3 (a) Both are true and R explains A: the field at any point has only one direction.

Answer Q4 (i) Earth wire; it connects the metal body to a metal plate buried deep in the earth. (ii) If live wire insulation fails and touches the body, the leakage current flows to the earth through this low-resistance path, keeping the body at earth potential, so the user does not get a severe shock.

Answer Q5 (i) The magnetic field strength increases with current (rows 1 and 2) and with the number of turns (rows 2 and 3). (ii) A soft iron core: it gets strongly magnetised by the field of the solenoid and loses its magnetism when the current is switched off.

Answer Q6 (i) Total power = 2000 + 1500 + 400 = 3900 W; I = P/V = 3900/220 = 17.7 A (approx.). (ii) The current exceeds 15 A, so the fuse melts (or the MCB trips) and the circuit is cut off; this is overloading. (iii) In parallel each appliance gets the full 220 V, can be switched on or off independently, and draws its own required current.

Answer Q7 **(i)** Fleming's left-hand rule. **(ii)** It is displaced in the same direction as before, because reversing both quantities leaves the direction of force unchanged. **(iii)** Force acts towards the south (forefinger up, middle finger east, thumb points south). The force is maximum when the current is perpendicular to the magnetic field.

Answer Q8 (i) The conventional current is opposite to electron flow, i.e. from north to south. Applying Fleming's left-hand rule with the field downwards and current towards the south, the force acts towards the east, so the beam bends east. (ii) No; a magnetic field exerts force only on a moving charge (a current), not on a charge at rest.

CHAPTER 13 Our Environment

Q1. MCQ **[1 mark]**

In the food chain grass → grasshopper → frog → snake → eagle, the grass traps 50,000 J of energy. How much energy is available to the snake?

(a) 500 J (b) 50 J
(c) 5 J (d) 5000 J

Q2. MCQ **[1 mark]**

Which of the following groups contains only biodegradable substances?

(a) Polythene bag, cow dung, paper
(b) Leather, glass bottle, cotton cloth
(c) Paper, fruit peels, cotton cloth
(d) Aluminium foil, wood, leaves

Q3. ASSERTION-REASON **[1 mark]**

Assertion (A): Ozone present in the upper atmosphere is harmful to living organisms.
Reason (R): At ground level, ozone is a deadly poison.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q4. CASE / DATA BASED **[3 marks]**

A survey of a lake near farmland gives the concentration of a pesticide:

Level	Pesticide (ppm)
Lake water	0.02
Phytoplankton	0.5
Small fish	2
Large fish	8
Fish-eating bird	25

(i) Name the phenomenon shown and calculate how many times more concentrated the pesticide is in the bird than in the water.
(ii) Why does the concentration increase at each level?

Q5. CASE / DATA BASED **[2 marks]**

A school eco-club weighs one day's waste:

Waste	Mass (kg)
Fruit and vegetable peels	12
Paper	8
Plastic wrappers	5
Glass bottles	3
Aluminium cans	2

(i) What percentage of the waste is biodegradable?
(ii) Suggest one proper way to handle the non-biodegradable part.

Q6. CASE / DATA BASED **[3 marks]**

A field survey of a grassland counted organisms at each level of a food chain:

Trophic level	Organism	Number
T1	Grass plants	6000
T2	Grasshoppers	450
T3	Frogs	35
T4	Snakes	3

(i) Why do the numbers fall sharply at each higher level? (ii) Why do food chains rarely have more than four or five trophic levels? (iii) What would happen to this grassland if all decomposers were removed?

Q7. SOURCE BASED **[4 marks]**

High in the atmosphere, ultraviolet radiation from the Sun splits oxygen molecules into free oxygen atoms, which then combine with oxygen molecules to form ozone. This ozone layer shields the Earth from harmful UV rays. Its thickness began to fall sharply in the 1980s, mainly due to synthetic chemicals such as chlorofluorocarbons (CFCs) used as refrigerants and in fire extinguishers. In 1987, an international agreement was reached to freeze CFC production at 1986 levels.

(i) State one harmful effect of UV radiation on human beings. **[1]**

(ii) Write the reactions showing the formation of ozone. **[1]**

(iii) Why is ozone depletion a global concern? Suggest two steps an individual can take to reduce it. **[2]**

Q8. APPLICATION / HOTS **[3 marks]**

In a pond ecosystem, insects are eaten by frogs and frogs are eaten by snakes. Suppose all the frogs are removed.
(i) Predict what will happen to the populations of insects and snakes.
(ii) If, instead, all decomposers in the pond disappeared, what would be the long-term effect?

ANSWER KEY

Answer Q1 (b) By the 10% law: grasshopper 5000 J → frog 500 J → snake 50 J.

Answer Q2 (c) Paper, fruit peels and cotton are broken down by bacteria and fungi; polythene, glass and aluminium are non-biodegradable.

Answer Q3 (d) A is false: ozone in the upper atmosphere is protective, as it absorbs harmful UV radiation from the Sun. R is true.

Answer Q4 (i) Biological magnification; 25/0.02 = 1250 times. (ii) The pesticide is non-biodegradable and is not excreted or broken down, so it accumulates in the body and passes up the chain; each organism eats many from the lower level, so it gets concentrated at every trophic level.

Answer Q5 (i) Biodegradable = peels + paper = 20 kg out of 30 kg = 66.7%. (ii) Segregate plastic, glass and aluminium and send them for recycling instead of burning or dumping them.

Answer Q6 (i) Only about 10% of the energy at one level passes to the next; the rest is lost as heat and used in life processes, so fewer organisms can be supported at higher levels. (ii) After four or five levels, too little energy remains to support another population. (iii) Dead bodies and wastes would pile up, nutrients would not return to the soil, and plant growth would decline.

Answer Q7 **(i)** It can cause skin cancer (also cataract and damage to the immune system). **(ii)** $O_2 \rightarrow O + O$ (in presence of UV); $O + O_2 \rightarrow O_3$. **(iii)** The ozone layer protects all life on Earth and CFCs released anywhere spread through the atmosphere, so depletion affects everyone. Steps: use CFC-free refrigerators and air conditioners, and get old appliances serviced or disposed of properly so the gases do not leak.

Answer Q8 (i) Insects will increase rapidly since their predator is gone, and may damage plants; snakes will decline due to lack of food. This shows the balance of a food chain is disturbed when one level is removed. (ii) Dead plants and animals would pile up and nutrients would not be returned to the water and soil, so producers would starve of nutrients and the whole ecosystem would collapse.

How to attack competency-based questions

1. Read the questions before the passage. Underline what each part asks.
2. Find the concept behind the story: every case is built on one or two textbook ideas.
3. Match the data: read table headings, units and graph axes before answering.
4. Answer in the words of the chapter. Name the law, rule or process.
5. For assertion-reason, decide if A and R are each true first, then check if R explains A.
6. Show the formula, the substitution and the unit in every numerical.
7. Give a reason with every choice: 'because' turns a 1-mark guess into a full answer.
8. Do not spend more than 6 to 7 minutes on one case-based set of 4 marks.
9. Leave a line after each part so you can add a point later.

Nitesh Bhaiya ki tip: passage lamba dikhe to ghabrao mat, answer usually 2-3 lines me hi hota hai.

SECTION

OFFICIAL CBSE BOARD PAPERS

Sample Papers 1 to 3

Paper 1: CBSE Board Exam 2026 (Set 31/1/1), on the current section-wise pattern (Biology, Chemistry, Physics), with the official marking scheme.
Paper 2: CBSE Board Exam 2025 (Set 31/1/3) with the marking scheme and the complete Topper's Answer Sheet.
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Series : MKL1N

SET ~ 1

रोल नं.
Roll No.

प्रश्न–पत्र कोड
Q.P. Code **31/1/1**

परीक्षार्थी प्रश्न–पत्र कोड को उत्तर–पुस्तिका के मुख–पृष्ठ पर अवश्य लिखें।
Candidates must write the Q.P. Code on the title page of the answer-book.

विज्ञान
SCIENCE

निर्धारित समय : 3 घण्टे *अधिकतम अंक : 80*
Time allowed : 3 hours *Maximum Marks : 80*

नोट/ NOTE ( )

(I) कृपया जाँच कर लें कि इस प्रश्न–पत्र में मुद्रित पृष्ठ **23** हैं।
Please check that this question paper contains **23** printed pages.

(II) कृपया जाँच कर लें कि इस प्रश्न–पत्र में **39** प्रश्न हैं।
Please check that this question paper contains **39** questions.

(III) प्रश्न–पत्र में दाहिने हाथ की ओर दिए गए प्रश्न–पत्र कोड को परीक्षार्थी उत्तर–पुस्तिका के मुख–पृष्ठ पर लिखें।
Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.

(IV) **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर–पुस्तिका में यथास्थान पर प्रश्न का क्रमांक अवश्य लिखें।**
Please write down the Serial Number of the question in the answer-book at the given place before attempting it.

(V) इस प्रश्न–पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न–पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न–पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर–पुस्तिका पर कोई उत्तर नहीं लिखेंगे।
15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

GENERAL INSTRUCTIONS :

Read the following instructions carefully and follow them:

(i) *This question paper contains **39** questions. **All** questions are **compulsory.***

(ii) *Question paper is divided into **THREE** sections – **A, B** and **C**.*
Section A : Biology (30 Marks)
Section B : Chemistry (25 Marks)
Section C : Physics (25 Marks)

(iii) *The question paper has MCQs, VSAs, SAs, LAs and C/S-BQs. Marks are given against each question.*

(iv) *There are **case based** questions (CBQs) with three sub-questions and are of **4** marks each.*

(v) ***Divide your answer sheet into three sections as per question paper – Section A (Biology), Section B (Chemistry) and Section C (Physics). It is compulsory to answer each question in its respective section. Do not mix answers of one section into the other section.***

(vi) *Instructions are given with each section and question, wherever necessary.*

(vii) ***Kindly note that a separate question paper has been provided for visually impaired candidates.***

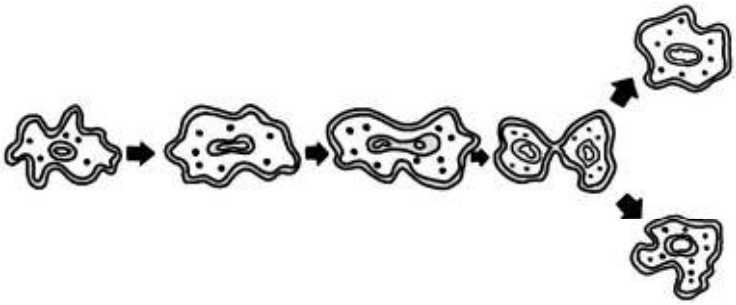
(viii) *There is no overall choice in the question paper. However, an internal choice has been provided in few questions. **Only one** of the choices in such questions must be attempted.*

SECTION – A

1. Which structure in a leaf is mainly responsible for gaseous exchange ? **1**

(A) Xylem (B) Stomata
(C) Phloem (D) Cuticle

2. Identify the type of reproduction shown in the diagram given below : **1**



(A) Budding (B) Fragmentation
(C) Spore Formation (D) Binary Fission

3. Identify the correct statement for *spirogyra*, *leishmania* and *hydra* : **1**

(A) they reproduce sexually. (B) they are unicellular.
(C) they are multicellular. (D) they reproduce asexually.

4. Human brain has various parts or regions that help in different actions, responses and coordination. From the following, identify the part responsible for precision of voluntary actions : **1**

(A) Cerebrum (B) Cerebellum
(C) Medulla (D) Pons

5. Pancreas secretes pancreatic juice which contain certain enzyme that helps in digestion of food. **1**

Choose the correct option from the following :

(A) Trypsin digests emulsified fats and lipase digests proteins.
(B) Trypsin digests proteins and lipase digests emulsified fats.
(C) Trypsin and lipase both digests fats.
(D) Trypsin digests proteins and lipase digests carbohydrates.

6. Sex is determined by different factors in various species. However, in human beings, it is determined genetically.
Which amongst the following option(s) is/are correct for human beings ? **1**

(i) Gamete carrying X chromosome from female parent.
(ii) Gamete carrying X chromosome from male parent.
(iii) Gamete carrying Y chromosome from male parent.

Options :

(A) (ii) and (iii)
(B) (i) only
(C) (i) and (iii)
(D) (iii) only

7. Which of the following group is not 'biodegradable' ? **1**

(A) Vegetable peels, dead leaves, paper
(B) Cow dung, leather bag, water
(C) Polythene bag, rubber band, ball pen
(D) Paper, fruits, bones

( )

Directions : Question numbers 8 and 9 are Assertion and Reason based questions. Two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below :

(A) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

8. **Assertion (A) :** Reflex actions do not involve thinking.

Reason (R) : Most reflex actions are controlled by the spinal cord. **1**

9. **Assertion (A) :** In human beings, the respiratory pigment is haemoglobin present in red blood cells.

Reason (R) : Haemoglobin has a very high affinity for carbon dioxide. **1**

10. What is the function of diaphragm in human respiratory system ? Where is it present in human body ? **2**

11. (a) State two differences between the act of chewing food and salivation on sight of food. **2**

OR

(b) State two differences between pollination and fertilization. **2**

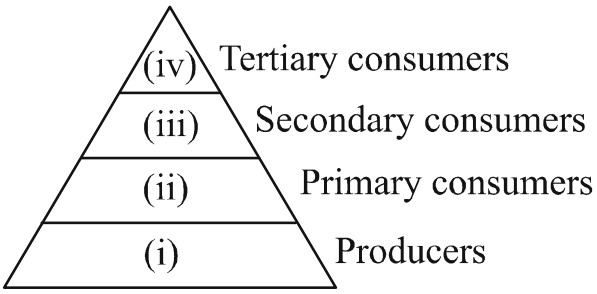
12. Draw a neat diagram to show germination of pollen on the female reproductive part of the flower. Name and label only the following parts : **2**

(a) The part that receives the pollen grain.

(b) The structure that carries the male germ cell to reach the female germ cell.

( )

13. Given below is a pyramid showing various trophic levels in an ecosystem : **3**



(a) From the organisms listed below, identify which one is to be placed at which trophic level ?
Deer, Grass, Lion, Snake, Rabbit

(b) Discuss the reason why primary consumers will have more energy as compared to secondary consumers ?

(c) Why is the base of the pyramid broad ?

14. Give differences between the following : **3**

(a) Nephron and neuron

(b) Sensory nerve and motor nerve

(c) Consumers and decomposers

15. Mendel took garden pea plants with different characteristics, such as height to study the inheritance pattern of factors (genes). He crossed tall pea plant with short pea plant and obtained all the tall plants in the F_1 generation.

Answer the following questions :

(a) Why only tall pea plants were observed in F_1 progeny ? **1**

(b) By which method did Mendel obtain F_2 progeny ? **1**

(c) (i) Write one difference between dominant and recessive trait. **2**

OR

(c) (ii) Write two observations made by Mendel about F_1 progeny. **2**

( )

16. (a) Given below are certain situations. Analyse each and describe its possible impact : **5**

(i) A population of bacteria living in temperate waters whose temperature increased by global warming.

(ii) The sperm encounters the egg when it reaches the oviduct in human females.

(iii) Self pollination does not occur in a flower that contains only pistil.

(iv) Egg does not get fertilised in a human female.

(v) When the seed is placed under appropriate condition of water and air in the soil ?

OR

(b) Given below are certain situations. Analyse and describe what would happen when : **5**

(i) Spores are liberated from blob-like structures of the bread mould?

(ii) Leaves of bryophyllum fall on wet soil ?

(iii) A pollen from different species land on the stigma of totally unrelated species ?

(iv) Copper-T is placed in the uterus of a human female ?

(v) *Spirogyra* breaks into smaller fragments upon maturation ?

SECTION – B

17. (i) $AgNO_3 + NaCl \longrightarrow NaNO_3 + AgCl$

(ii) $K_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2KCl$

Which of the following options clearly describes both the reactions ? **1**

(A) (i) is double displacement, (ii) is displacement reaction.

(B) Both, (i) and (ii) are displacement reactions and precipitation reactions.

(C) Both, (i) and (ii) are double displacement reactions and precipitation reactions.

(D) (i) is displacement, (ii) is double displacement reaction.

18. Which one of the following can be used as an acid-base indicator by a visually impaired (blind) student ? **1**

(A) Turmeric (B) Vanilla essence

(C) Methyl orange (D) Litmus

( )

19. The gases evolved on heating lead (II) nitrate crystals are : **1**

(A) NO and O_2 (B) N_2 and NO_2

(C) NO_2 and H_2 (D) NO_2 and O_2

20. The formula of functional group for aldehyde is : **1**

(A) –COOH (B) –CHO

(C) $-\overset{\overset{O}{\|}}{C}-$ (D) –OH

21. Which of the following is a poor conductor of electricity ? **1**

(A) Pb (B) Cu

(C) Ag (D) Al

22. The natural sources of oxalic acid, lactic acid and methanoic acid respectively are : **1**

(A) tomato, curd, ant-sting

(B) tomato, orange, nettle-sting

(C) orange, milk, ant-sting

(D) orange, sour milk, nettle-sting

23. When an element 'X' reacts with water, it starts floating. Identify the element 'X' : **1**

(A) Potassium (B) Calcium

(C) Sodium (D) Iron

Directions : For question number 24, two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below :

(A) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

24. **Assertion (A) :** Carbon shares its valence electrons with other atoms of carbon or with atoms of other elements.

Reason (R) : The shared electrons belong to the outermost shells of both the atoms and lead to both atoms attaining the noble gas configuration. **1**

25. (a) What are amphoteric oxides ? **1**

(b) Categorise the following based on their nature :
ZnO, Na_2O, CO_2 **1**

26. (a) Name the substance oxidised and reduced in the following reaction :
$ZnO + C \longrightarrow Zn + CO$ **1**

(b) Balance the following chemical reaction :
$Pb(NO_3)_2 + KI \longrightarrow PbI_2 + KNO_3$ **1**

(c) Give one example each of electrolytic decomposition and decomposition by sunlight. **1**

27. (a) Explain chlor-alkali process with chemical equation. Name the products formed at anode and cathode. **3**

OR

(b) Write the preparation of following compounds with balanced chemical equation : **3**

(i) Baking soda

(ii) Bleaching powder

(iii) Plaster of Paris

28. Read the following passage and answer the questions given below :

Most of metals occur in combined state in form of ores. Carbonate ores are converted into oxides by calcination and sulphide ores by roasting. Oxides are reduced with suitable reducing agent like carbon to get free metal. Highly reactive metals like – Al, Mg are also used as reducing agents to obtain metal from their oxides. Most reactive metals are obtained by electrolytic reduction of their molten ores. Alloying is a very good method of improving the properties of a metal. We can get desired properties by this method. The electrical conductivity and melting point of an alloy is less than that of pure metals.

(a) Why carbonate or sulphide ores are converted to oxides before extraction of metal from it ? **1**

(b) Write a reaction in which Aluminium is used as a reducing agent to obtain metal from its oxide. **1**

(c) (i) How is copper obtained from its ore (Cu_2S) ? Give equations of the reactions. **2**

OR

(c) (ii) (I) Why highly reactive metals cannot be obtained from their oxides by using carbon as a reducing agent ? **1**

(II) Why solder, an alloy of lead and tin, is used for welding electrical wires together ? **1**

29. (a) (i) Give reasons for the following : **3**

(I) Covalent compounds are poor conductor of electricity.

(II) Soap does not form lather in hard water.

(III) Carbon shows catenation but silicon does not.

(ii) Write chemical equations for the following : **2**

(I) Oxidation of ethanol by acidified $K_2Cr_2O_7$.

(II) Hydrogenation of ethene.

OR

(b) Mohan heated ethanol with a compound 'X' in the presence of a few drops of conc. H_2SO_4 and observed a sweet smelling compound 'Y' is formed. When 'Y' is treated with sodium hydroxide it gives back ethanol and a compound 'Z'. **5**

(i) Identify 'X', 'Y' and 'Z'.

(ii) Write the role of conc. H_2SO_4 in the reaction.

(iii) Write the chemical equations involved and name the reactions.

SECTION – C

30. A convex lens of focal length 15 cm, is forming a real image. If the size of image is same as the size of object, then position of object and position of image will be, respectively : **1**

(A) – 15 cm and – 15 cm from lens

(B) – 15 cm and + 15 cm from lens

(C) – 30 cm and + 30 cm from lens

(D) – 30 cm and – 30 cm from lens

( )

31. When you look at an object very close to your eyes, the : **1**

(A) Ciliary muscles of your eye contract and the eye lens becomes thick.

(B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick.

(C) Ciliary muscles of your eye contract and the eye lens becomes thin.

(D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

Directions : For question number 32, two statements are given, one labelled as Assertion (A) and the other is labelled as Reason (R). Select the correct answer to this question from the codes (A), (B), (C) and (D) as given below :

(A) Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both, Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

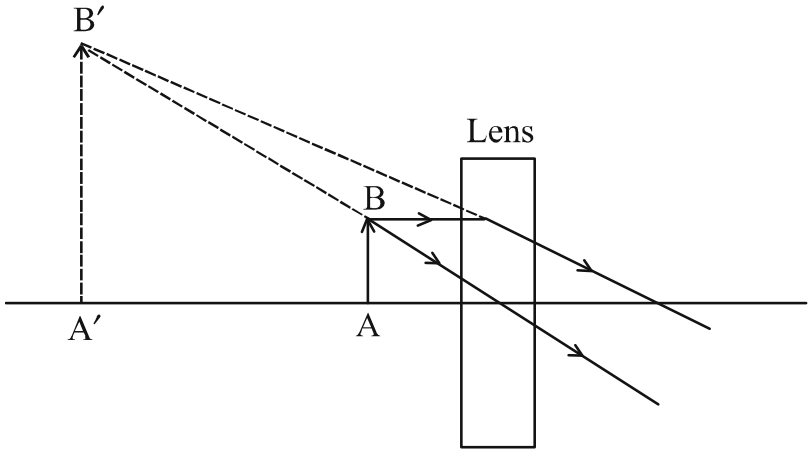
(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

32. **Assertion (A) :** When rays of white light pass through a prism, on emerging they give spectrum of seven colours.

Reason (R) : It is due to the scattering of light that red light bends minimum and violet light bends the maximum. **1**

33. The given figure shows image formation by a lens. Analyse the figure and answer the following questions : **2**



(a) What is the type of lens used for image formation in the given ray diagram ?

(b) If the real image is formed at a distance of 30 cm from the lens and the size of image is twice the size of the object, then where was the object placed ?

34. (a) The resistance of a wire of 0.01 cm radius and 1.0 cm length is 7 Ω. Calculate its resistivity. **2**

OR

(b) An electric heater is rated 220 V; 11 A. Calculate the power consumed if the heater is operated at 200 V. **2**

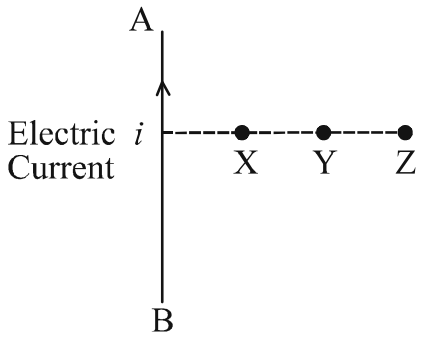
35. A person is unable to read a book placed closer than 1 meter from his eyes. Identify the defect of vision in his eyes. Draw the ray diagrams to show the defect of vision and its correction. **3**

36. (a) Describe an activity to show that a current carrying conductor, placed in an external magnetic field experiences a force. **3**

(b) Imagine that you are sitting in a chamber with your back to one wall. An electron beam, moving horizontally towards the front wall from the back wall, is deflected by a strong magnetic field to your right side. Find the direction of the magnetic field.

37. (a) The pattern of magnetic field due to a current carrying wire depends upon the shape made by that wire. Justify. **3**

(b) A current carrying straight wire AB is shown in the given diagram. Out of X, Y and Z on which point will the strength of magnetic field be maximum and why ?



38. Read the following passage and answer the questions given :

Lenses can form different types of images depending upon their focal length and position of object. A convex lens can create real, inverted or virtual, erect images, while a concave lens forms only virtual and diminished images. The focal length determines the power of lens. Convex lenses have positive focal length while concave lenses have negative focal length by convention. When lenses are placed together, their combined power is determined by the sum of their individual powers. Ray diagrams help to visualize how light converges or diverges through lens to form an image.

(a) A convex lens of focal length 20 cm is used to form an image. If an object is placed at 40 cm from the lens, what will be the position and nature of image ? **1**

(b) Illustrate the formation of image with the help of ray diagram, when the object is placed between the optical centre and principal focus of concave lens. **1**

(c) (i) A lens combination consists of a convex lens of focal length 30 cm and a concave lens of focal length 15 cm placed together. Find the equivalent focal length and power of this lens combination. **2**

OR

(c) (ii) Two lenses are placed in contact. One is a concave lens with focal length 2 m and the other is a convex lens with focal length 1.5 m. What type of lens will the combination behave as (convex or concave) ? Give reason. **2**

39. (a) (i) Due to change in length and area of cross-section of a conductor, resistance of conductor changes while resistivity does not change. Why ? **5**

(ii) Conductors of electric toasters and electric iron are made of an alloy rather than a pure metal. Why ?

(iii) Define the S.I. unit of electric current.

OR

(b) (i) How many bulbs of resistance 8 Ω each should be connected in parallel combination to draw a current of 2 A from a battery of 4 V ? **5**

(ii) Name the device used for measuring electric current. How is it connected in a circuit ?

(iii) State Joule's law of heating.

MARKING SCHEME
SCIENCE (Subject Code-086)
(PAPER CODE: 31/1/1) (10-01-86K)

<table>
<tr><th>Q.No.</th><th>EXPECTED OUTCOMES/VALUE POINTS</th><th>Marks</th><th>Total Marks</th></tr>
<tr><td colspan="4">SECTION – A
(Biology)</td></tr>
<tr><td>1.</td><td>(B) / Stomata</td><td>1</td><td>1</td></tr>
<tr><td>2.</td><td>(D) / Binary Fission</td><td>1</td><td>1</td></tr>
<tr><td>3.</td><td>(D) / they reproduce asexually</td><td>1</td><td>1</td></tr>
<tr><td>4.</td><td>(B) / Cerebellum</td><td>1</td><td>1</td></tr>
<tr><td>5.</td><td>(B) / Trypsin digests proteins and lipase digests emulsified fats.</td><td>1</td><td>1</td></tr>
<tr><td>6.</td><td>(A) / (ii) and (iii)</td><td>1</td><td>1</td></tr>
<tr><td>7.</td><td>(C) / Polythene bag, rubber band, ball pen</td><td>1</td><td>1</td></tr>
<tr><td>8.</td><td>(A) / Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).</td><td>1</td><td>1</td></tr>
<tr><td>9.</td><td>(C) / Assertion (A) is true, but Reason (R) is false.</td><td>1</td><td>1</td></tr>
<tr><td>10.</td><td>• The main function of the diaphragm is to flatten during inhalation, which expands the chest cavity and draws the air into the lungs / helps in breathing.
• It is located at the base of chest cavity.</td><td>1
1</td><td>2</td></tr>
<tr><td>11.</td><td>(a)
<table>
<tr><th>Chewing of food</th><th>Salivation on sight of food</th></tr>
<tr><td>i) It is a voluntary action.
ii) It is controlled by forebrain.</td><td>i) It is a reflex/involuntary action.
ii) It is controlled by medulla in the hind brain.</td></tr>
</table>
(any other suitable difference)
OR
(b)
<table>
<tr><th>Pollination</th><th>Fertilization</th></tr>
<tr><td>i) It is the transfer of pollen grains from anther to the suitable stigma.
ii) It occurs in plants.</td><td>i) It is the fusion of male gamete with the female gamete.
ii) It occurs in both plants and animals.</td></tr>
</table>
(any other suitable difference)</td><td>1
1
1
1</td><td>2</td></tr>
</table>

12.

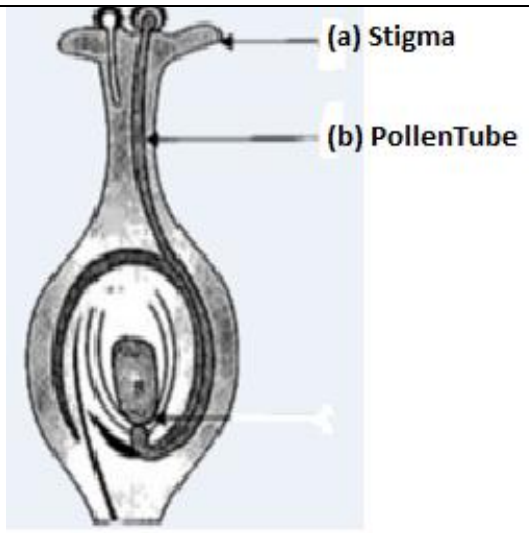


Diagram — **1**

(a) Stigma — ½
(b) Pollen Tube — ½ — **2**

13.

(a) (i) Grass
(ii) Deer, Rabbit — **1**
(iii) Snake, Lion
(iv) Lion

(b) Primary consumers feed on green plants which have large amount of energy. Only 10% of its energy is available/passed for the next secondary consumer /trophic level. — **1**

(c) The base is broad as the number/energy/mass of producers is usually the highest in comparison to other trophic levels of the pyramid. — **1** — **3**

14.

(a)

Nephron	Neuron
(i) Filtration/Structural/ Functional unit of the kidney.	(i) Structural/ Functional unit of the nervous system.
(ii) Filters nitrogenous wastes from the blood.	(ii) Transmits information from one part of the body to another.

1

(any one, any other suitable difference)

(b)

Sensory Nerve	Motor Nerve
(i) Carries impulse from receptors to CNS/ Brain and Spinal cord.	(i) Carries impulse from CNS/Brain and Spinal cord to the motor area/ effector organ.

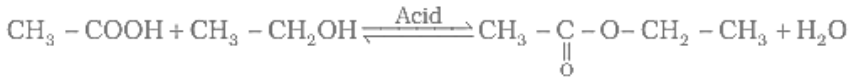
1

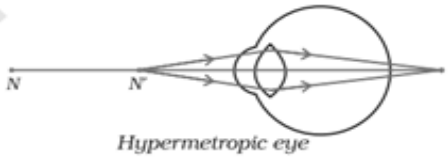
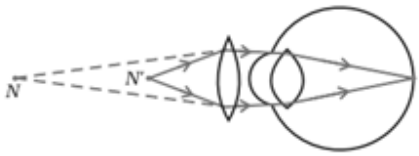
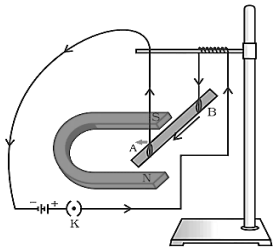
(any other suitable difference)

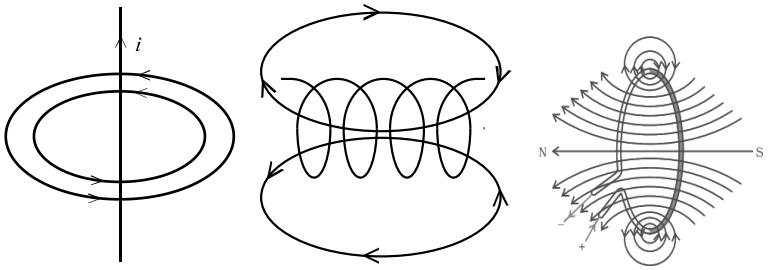
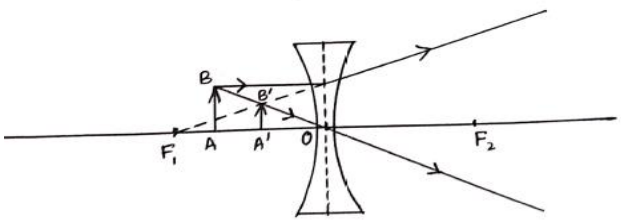
	(c) Consumers: (i) Organisms that feed on producers and other consumers. (ii) Transfer energy through the food chain. Decomposers: (i) Organisms that breakdown dead organic matter into simpler inorganic substances. (ii) Recycle nutrients back into the environment. (any one, any other suitable difference)	1	3
15.	(a) In F_1 progeny, pea plants have 'Tt' where 'T' is dominant over 't' so all the plants of F_1 progeny were tall. / Tall height is dominant trait over short height. (b) Self-pollination (c) (i) Dominant trait: i) Expresses itself over recessive trait. ii) Expresses in both conditions-TT and Tt. Recessive trait: i) Unable to express itself in presence of a dominant trait. ii) Expresses itself only when it is 'tt' or in pure condition. (any one, any other suitable difference) **OR** (c) (ii) Mendel's observations: • All plants of F_1 progeny were tall. • No medium/ no short height plants observed in F_1 progeny. • F_1 progeny resembled one parent only. (any two observations)	1 1 2 1+1	4
16.	(a) (i) Most of these bacteria would die, but the few variants resistant to heat would survive and grow further. (ii) Fertilization occurs to form a zygote. (iii) Cross pollination may occur leading to fruit formation. / No fertilization. / No fruit formation. (iv) If the egg is not fertilised, the thick and spongy lining of the uterus breaks and comes out through vagina as blood and mucus, known as menstruation. / Menstruation will take place. (v) The seed will develop into a seedling. / Germination will take place. **OR** (b) (i) When spores land on a substance and get adequate moisture and temperature, it will develop into new *Rhizopus*.	1 1 1 1 1 1	

	(ii) New plants grow from the buds located in the notches of the leaf. (iii) The pollen tube will not be formed. / No fertilisation will take place. (iv) Fertilization /Pregnancy will be prevented. (v) Each fragment or piece grows into a new individual organism.	**1** **1** **1** **1**	**5**
	SECTION – B **(Chemistry)**		
17.	(C) / Both, (i) and (ii) are double displacement reactions and precipitation reactions.	**1**	**1**
18.	(B) / Vanilla essence	**1**	**1**
19.	(D) / NO_2 and O_2	**1**	**1**
20.	(B) / -CHO	**1**	**1**
21.	(A) / Pb	**1**	**1**
22.	(A) / tomato, curd, ant-sting	**1**	**1**
23.	(B) / Calcium	**1**	**1**
24.	(A) / Both, Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).	**1**	**1**
25.	(a) Metal oxides which can react with both acids as well as bases to produce salt and water. (b) ZnO – Amphoteric oxide Na_2O – Basic oxide CO_2 – Acidic oxide	**1** **1**	**2**
26.	(a) • Substance oxidised - C • Substance reduced - ZnO (b) $Pb(NO_3)_2 + \mathbf{2}\ KI \rightarrow PbI_2 + \mathbf{2}\ KNO_3$ (c) • $2H_2O \xrightarrow{\text{electricity}} 2H_2 + O_2$ • $2AgCl \xrightarrow{\text{sunlight}} 2Ag + Cl_2$ (any other example in each case)	 ½ ½ **1** ½ ½	**3**
27.	(a) • When electricity is passed through brine, it decomposes to form sodium hydroxide (alkali) and chlorine, hence this process is called chlor-alkali process. $2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{Electricity}} 2NaOH(aq) + H_2(g) + Cl_2(g)$ • At anode: Cl_2 • At cathode: H_2 **OR**	**1** **1** ½ ½	

	(b) (i) $NaCl + H_2O + NH_3 + CO_2 \rightarrow NaHCO_3 + NH_4Cl$ (ii) $Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$ / $2Ca(OH)_2 + 2Cl_2 \rightarrow Ca(ClO)_2 + CaCl_2 + 2H_2O$ (iii) $CaSO_4.2H_2O \xrightarrow{373K} CaSO_4.\frac{1}{2}H_2O + 1\frac{1}{2}H_2O$ (deduct ½ mark for no / incorrect balancing)	**1** **1** **1**	**3**
28.	(a) Because it is easier to obtain metal from its oxide. / Because it is easier to reduce metal oxide to metal (b) $Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$ $3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + Heat$ (balancing is optional) (any one equation) (c) (i) $2Cu_2S + 3O_2(g) \xrightarrow{Heat} 2Cu_2O(s) + 2SO_2(g)$ $2Cu_2O + Cu_2S \xrightarrow{Heat} 6Cu(s) + SO_2(g)$ **OR** (c)(ii) (I) Because highly reactive metals have more affinity for oxygen than carbon. (II) Because of its low melting point.	**1** **1** **1** **1** **1** **1**	**4**
29.	(a) (i) (I) They do not give rise to charged particles/ ions. (II) Soap reacts with calcium and magnesium salts present in hard water and forms insoluble substances called Scum. (III) C-C bonds are strong and stable whereas Si-Si bonds are relatively weak. (ii) (I) $CH_3CH_2OH \xrightarrow{\text{acidified } K_2Cr_2O_7 + heat} CH_3COOH$ (II) $CH_2{=}CH_2 + H_2 \xrightarrow{Ni} CH_3\text{-}CH_3$ **OR** (b) (i) X - CH_3COOH/ethanoic acid /acetic acid Y - $CH_3COOC_2H_5$/$CH_3COOCH_2CH_3$/ ester/ ethyl ethanoate Z - CH_3COONa/sodium ethanoate/sodium acetate (ii) Catalyst/ dehydrating agent	**1** **1** **1** **1** **1** ½ ½ ½ ½	

	(iii) •  • Esterification reaction • $CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$ • Saponification reaction/ De-esterification reaction	1 ½ 1 ½	5
	SECTION – C **(Physics)**		
30.	(C) / - 30 cm and + 30 cm from lens	**1**	**1**
31.	(A) / Ciliary muscles of your eye contract and the eye lens become thick	**1**	**1**
32.	(C) / Assertion (A) is true, but Reason (R) is false.	**1**	**1**
33.	(a) Convex lens / Converging lens (b) m = - 2, v= 30 cm $m = \frac{v}{u}$ $u = \frac{v}{m}$ $u = \frac{30}{-2}$ u = -15cm / The object was placed at 15 cm in front of the lens.	**1** ½ ½	**2**
34.	(a) r = 0.01 cm=1 x 10^{-4} m l = 1 cm = 0.01 m $R = \rho \frac{l}{A}$ $\rho = \frac{RA}{l} = \frac{R \times \pi r^2}{l}$ $\rho = \frac{7 \times 22 \times 10^{-8}}{7 \times 0.01}$ ρ = 22 x 10^{-8} x 10^2 ρ = 22 x 10^{-6} Ωm =2.2 x 10^{-5} Ωm **OR** (b) Resistance of electric heater $R = \frac{V}{I}$ $R = \frac{220}{11}$ R = 20 Ω $P = \frac{V^2}{R}$ $P = \frac{200 \times 200}{20}$ P = 2000W/ 2 kW	½ 1 ½ ½ ½ ½ ½	**2**

35.	• Hypermetropia/ Far sightedness •  *Hypermetropic eye* •  *Correction for Hypermetropic eye* (deduct ½ mark for not showing the direction of ray of light)	**1** **1** **1**	**3**
36.	(a) **Procedure** Take a small aluminium rod AB and using two connecting wires suspend it horizontally from a stand. Place a strong horse-shoe magnet in such a way that the rod lies between the two poles perpendicularly. Connect the aluminium rod in series with a battery and a key. Now pass a current through the aluminium rod from one end to another. / **(Procedure can also be explained with a <u>Diagram)</u>**  **<u>Observation</u>** It is observed that the **rod is displaced** on passing current though it. (b) Magnetic field will be vertically downwards.	**1** **1** **1**	**3**

37.	(a)  (any two cases) / • Magnetic field pattern for a current carrying straight conductor- concentric circles. • Magnetic field pattern for a current carrying solenoid- magnetic field lines similar to that of a bar magnet. • Magnetic field pattern for a current carrying circular loop- a pair of concentric circles with parallel straight lines at the centre. (any two cases) (b) • At X. • Magnetic field decreases as the distance from the conducting wire increases.	**1+1** **½** **½**	**3**
38.	(a) • Position - Image will form at 40 cm / 2F /C • Nature – Real and inverted / **Alternate answer** f = +20 cm, u = -40 cm Using Lens Formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ $\frac{1}{v} - \frac{1}{-40} = \frac{1}{20}$ $\frac{1}{v} = \frac{1}{20} + \frac{1}{-40}$ v = + 40 cm Position - 40 cm on the other side of the lens Nature - Real and inverted (b)  	**½** **½** **1**	

	(c) (i) f_1 = 30 cm = 0.3 m, f_2 = -15cm = -0.15 m $P = \frac{1}{f}$ $P_1 = \frac{+1}{0.3}$ D ; $P_2 = \frac{-1}{0.15}$ D Equivalent power, $P = P_1 + P_2$ P = - 3.33D Equivalent focal length, $f = \frac{1}{P}$ $f = \frac{-1}{3.33}$ = - 0.3 m = - 30 cm **OR** (c) (ii) • Combination Lens will behave like convex lens • f_1 = - 2 m, f_2 = 1.5 m $P = \frac{1}{f}$ $P_1 = \frac{-1}{2}$ D , $P_2 = \frac{+1}{1.5}$ D $P = P_1 + P_2$ $P = \frac{1}{6}$ f = + 6m The focal length of combination is positive. / **Alternate answer** • Combination Lens will behave like convex lens. • Convex lens - Less 'f', More 'P_1' Concave lens - More 'f', Less 'P_2' Combined Power = P_1+P_2, which will be positive.	½ ½ ½ ½ **1** **1**	**4**
39.	(a) (i) As Resistance, $R = \rho \frac{l}{A}$, it changes with change in length and area of cross section of conductor. But resistivity of conductor is the characteristic property of material and hence it does not change. (ii) The resistivity of an alloy is generally higher than that of its constituent metals. / Alloys do not oxidise (burn) readily at high temperatures. (iii) 1 Ampere is constituted by the flow of 1 Coulomb of charge per second. / 1A= 1C/1s **OR**	**1** **1** **2** **1**	

	(b) (i) V= 4V, I= 2A Resistance of circuit $R = \frac{V}{I}$ $R = \frac{4}{2}$ $R = 2\Omega$ Let 'n' be the number of bulbs $\frac{1}{R} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + ---n$ $\frac{1}{R} = \frac{n}{8}$ $\frac{1}{2} = \frac{n}{8}$ $n = 4$ Therefore, 4 bulbs of resistance 8 Ω should be connected in parallel. (ii) • Ammeter • In series (iii) Heat generated through a current carrying conductor is directly proportional to square of current, resistance of conductor and time for which current flows in conductor. / $H = I^2 R t$	½ ½ ½ ½ 1 1 1	5

What this marking scheme teaches

1. A correct formula earns marks on its own, even before the calculation.
2. The final answer with its unit carries a separate mark.
3. Keywords carry the marks: use the exact NCERT term.
4. In 5-mark questions, marks are split across sub-parts; attempt every part.

Series : 1GEFH **SET~3**

प्रश्न-पत्र कोड
Q.P. Code **31/1/3**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें।
Candidates must write the Q.P. Code on the title page of the answer-book.

	नोट		NOTE
(I)	कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **27** हैं।	(I)	Please check that this question paper contains **27** printed pages.
(II)	प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें।	(II)	Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(III)	कृपया जाँच कर लें कि इस प्रश्न-पत्र में **39** प्रश्न हैं।	(III)	Please check that this question paper contains **39** questions.
(IV)	**कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में यथा स्थान पर प्रश्न का क्रमांक अवश्य लिखें।**	(IV)	**Please write down the Serial Number of the question in the answer-book at the given place before attempting it.**
(V)	इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है। प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा। 10.15 बजे से 10.30 बजे तक परीक्षार्थी केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे। #	(V)	15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the candidates will read the question paper only and will not write any answer on the answer-book during this period.

विज्ञान
SCIENCE

*निर्धारित समय : **3** घण्टे*	*अधिकतम अंक : **80***
*Time allowed : **3** hours*	*Maximum Marks : **80***

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper comprises **39** questions. **All** questions are **compulsory**.*

(ii) *This question paper is divided into **five** sections – **A**, **B**, **C**, **D** and **E**.*

(iii) ***Section A** – Questions No. **1** to **20** are Multiple Choice Questions. Each question carries **1** mark.*

(iv) ***Section B** – Questions No. **21** to **26** are Very Short Answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of **30** to **50** words.*

(v) ***Section C** – Questions No. **27** to **33** are Short Answer type questions. Each question carries **3** marks. Answer to these questions should in the range of **50** to **80** words.*

(vi) ***Section D** – Questions No. **34** to **36** are Long Answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of **80** to **120** words.*

(vii) ***Section E** – Questions No. **37** to **39** are of **3** source-based/case-based units of assessment carrying **4** marks each with sub-parts.*

(viii) *There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.*

SECTION A

*Select and write the most appropriate option out of the four options given for each of the questions no. **1** to **20**. There is no negative marking for incorrect response.* *20×1=20*

1. The metals obtained from their molten chlorides by the process of electrolytic reduction are :

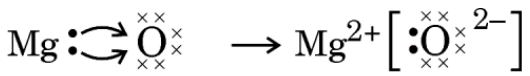
(A) Gold and silver

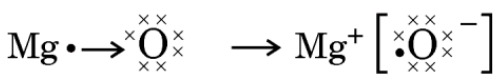
(B) Calcium and magnesium

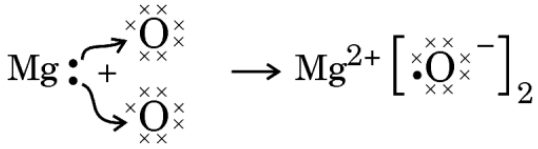
(C) Aluminium and silver

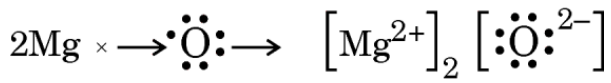
(D) Sodium and iron

2. The formation of magnesium oxide is correctly shown in option :

(A) 

(B) 

(C) 

(D) 

3. Reaction between two elements A and B, forms a compound C. A loses electrons and B gains electrons. Which one of the following properties will ***not*** be shown by compound C ?

(A) It has high melting point.

(B) It is highly soluble in water.

(C) It has weak electrostatic forces of attraction between its oppositely charged ions.

(D) It conducts electricity in its molten state or aqueous solution.

4. Consider the following reactions :

(i) Dilute hydrochloric acid reacts with sodium hydroxide.

(ii) Magnesium oxide reacts with dilute hydrochloric acid.

(iii) Carbon dioxide reacts with sodium hydroxide.

It is found that in each case :

(A) Salt and water is formed.

(B) Neutral salts are formed.

(C) Hydrogen gas is formed.

(D) Acidic salts are formed.

5. Tooth enamel is made up of calcium hydroxyapatite (a crystalline form of calcium phosphate). This chemical starts corroding in the mouth when the pH is :

(A) 7 (B) 5

(C) 10 (D) 14

6. The products formed when Aluminium and Magnesium are burnt in the presence of air respectively are :

(A) Al_3O_4 and MgO_2 (B) Al_2O_3 and MgO

(C) Al_3O_4 and MgO (D) Al_2O_3 and MgO_2

7. Electrolysis of water is a decomposition reaction. The mass ratio (M_H : M_O) of hydrogen and oxygen gases liberated at the electrodes during electrolysis of water is :

(A) 8 : 1 (B) 2 : 1

(C) 1 : 2 (D) 1 : 8

8. The breakdown of glucose has taken the following pathway :

Glucose $\xrightarrow{(a)}$ Pyruvate + Energy $\xrightarrow{(b)}$ Lactic acid + Energy

The sites 'a' and 'b' respectively are :

(A) Mitochondria and Oxygen deficient muscle cells

(B) Cytoplasm and Oxygen rich muscle cells

(C) Cytoplasm and Yeast cells

(D) Cytoplasm and Oxygen deficient muscle cells

9. If pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY), the seeds developed by the plants of F_1 generation will be :

(A) 50% round and green (B) 75% wrinkled and green

(C) 100% round and yellow (D) 75% wrinkled and yellow

10. The correct/true statement(s) for a bisexual flower is/are :

(i) They possess both stamen and pistil.

(ii) They possess either stamen or pistil.

(iii) They exhibit either self-pollination or cross-pollination.

(iv) They cannot produce fruits on their own.

(A) (i) only (B) (iv) only

(C) (i) and (iii) (D) (i) and (iv)

11. The plant hormone whose concentration stimulates the cells to grow longer on the side of the shoot which is away from light is :

(A) Cytokinins

(B) Gibberellins

(C) Adrenaline

(D) Auxins

12. Secretion of less saliva in mouth will effect the conversion of :

(A) proteins into amino acids

(B) fats into fatty acids and glycerol

(C) starch into simple sugars

(D) sugars into alcohol

13. The percentage of solar energy which is ***not*** converted into food energy by the leaves of green plants in a terrestrial ecosystem is about :

(A) 1% (B) 10%

(C) 90% (D) 99%

14. Which of the following groups do ***not*** constitute a food chain ?

(i) Wolf, rabbit, grass, lion

(ii) Plankton, man, grasshopper, fish

(iii) Hawk, grass, snake, grasshopper, frog

(iv) Grass, snake, wolf, tiger

(A) (i) and (iv) (B) (i) and (iii)

(C) (ii) and (iii) (D) (ii) and (iv)

15. The phenomenon responsible for making the smoke particles visible when a beam of sunlight enters a smoke filled room through a narrow hole is :

(A) scattering of light

(B) dispersion of light

(C) reflection of light

(D) internal reflection of light

16. Mirror 'X' is used to concentrate sunlight in solar furnace and Mirror 'Y' is fitted on the side of the vehicle to see the traffic behind the driver. Which of the following statements are true for the two mirrors ?

(i) The image formed by mirror 'X' is real, diminished and at its focus.

(ii) The image formed by mirror 'Y' is virtual, diminished and erect.

(iii) The image formed by mirror 'X' is virtual, diminished and erect.

(iv) The image formed by mirror 'Y' is real, diminished and at its focus.

(A) (i) and (ii) (B) (ii) and (iii)

(C) (iii) and (iv) (D) (i) and (iv)

For Questions number 17 to 20, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

17. *Assertion (A) :* The amount of ozone in the atmosphere began to drop sharply in the 1980s.

Reason (R) : The oxygen atoms combine with molecular oxygen to form ozone.

18. *Assertion (A) :* No two magnetic field lines are found to cross each other.

Reason (R) : The compass needle cannot point towards two directions at the point of intersection of two magnetic field lines.

19. *Assertion (A) :* A human child bears all the basic features of human beings.

Reason (R) : It looks exactly like its parents, showing very little variations.

20. *Assertion (A) :* Decomposition reactions are generally endothermic reactions.

Reason (R) : Decomposition of organic matter into compost is an exothermic process.

SECTION B

Questions no. ***21*** *to* ***26*** *are Very Short Answer Type questions.*

21. An object is placed at a distance of 10 cm in front of a concave mirror of focal length 15 cm. Use mirror formula to determine the position of the image formed by this mirror. *2*

22. (a) Consider two lamps A and B of rating 50 W; 220 V and 25 W; 220 V respectively. Find the ratio of the resistances of the two lamps (i.e. $R_A : R_B$). *2*

OR

(b) Heat produced per second due to a current in a resistor of 4 Ω is 400 joules. Calculate the potential difference across the resistor. *2*

23. Draw labelled diagrams to show different stages of budding in *Hydra*. *2*

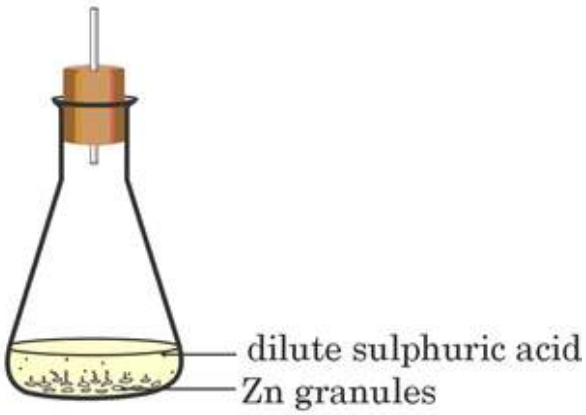
24. (a) Besides minimising the loss of blood, why is it essential to plug any leak in a blood vessel ? Name the component of blood which helps in this process and state how this component perform this function. *2*

OR

(b) (i) The transport system in plants is relatively slower than in animals. Give reasons.

(ii) State the role of phloem in the transport of materials in plants. *2*

25. A student performs the following experiment in his school laboratory.



List two observations to justify that in this experiment a chemical change has taken place. *2*

26. Translate the following statements into chemical equations and then balance them : *2*

(a) Nitric acid reacts with calcium hydroxide to form calcium nitrate and water.

(b) Sodium chloride reacts with silver nitrate to form silver chloride and sodium nitrate.

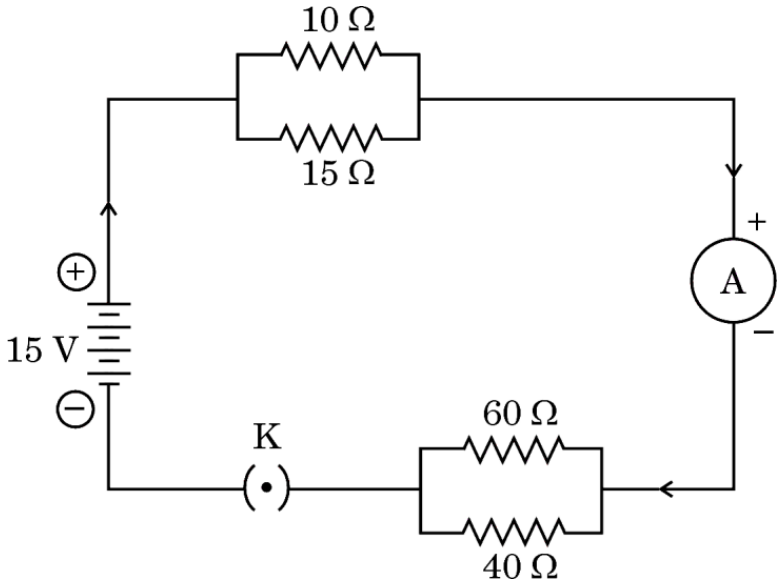
SECTION C

Questions no. ***27*** *to* ***33*** *are Short Answer Type questions.*

27. (a) Define one volt potential difference between two points in an electric field.

(b) Draw a schematic diagram of an electric circuit of a cell of 1·5 V, 5 Ω and 10 Ω resistor and a plug key, all connected in series. Calculate the current drawn from the cell when the key is closed. *3*

28. Consider the following electric circuit :



Calculate the values of the following : *3*

(a) The total resistance of the circuit

(b) The total current drawn from the source

(c) Potential difference across the parallel combination of 10 Ω and 15 Ω resistors

29. Draw ray diagrams to show the nature, position and relative size of the image formed by a convex mirror when the object is placed (i) at infinity and (ii) between infinity and pole P of the mirror. *3*

30. (a) How many chromosomes are present in human beings ? Out of these how many are sex chromosomes ?

(b) Explain how, in sexually reproducing organisms, the number of chromosomes in the progeny is maintained. *3*

31. A hormone 'X' is secreted in blood when a person is under scary situation.

(a) Identify the hormone 'X' and the gland that secretes it.

(b) Explain its role in dealing with scary or emergency situations. *3*

32. (a) With the help of an activity, explain the conditions under which iron articles get rusted. *3*

OR

(b) (i) Name two metals which react violently with cold water. List any three observations which a student notes when these metal are dropped in a beaker containing water.

(ii) Write a test to identify the gas evolved (if any) during the reaction of these metals with water. *3*

33. (a) "Displacement reactions also play a key role in extracting metals in the middle of the reactivity series." Justify this statement with two examples.

(b) Why can metals high up in the reactivity series not be obtained by reduction of their oxides by carbon ? *3*

SECTION D

Questions no. ***34*** *to* ***36*** *are Long Answer Type questions.*

34. (a) (i) The power of a lens 'X' is – 2·5 D. Name the lens and determine its focal length in cm. For which eye defect of vision will an optician prescribe this type of lens as a corrective lens ?

(ii) "The value of magnification 'm' for a lens is – 2." Using new Cartesian Sign Convention and considering that an object is placed at a distance of 20 cm from the optical centre of this lens, state :

(I) the nature of the image formed;

(II) size of the image compared to the size of the object;

(III) position of the image, and

(IV) sign of the height of the image.

(iii) The numerical values of the focal lengths of two lenses A and B are 10 cm and 20 cm respectively. Which one of the two will show higher degree of convergence/divergence ? Give reason to justify your answer. *5*

OR

(b) (i) Draw a ray diagram to show the refraction of a ray of light through a rectangular glass slab when it falls obliquely from air into glass.

(ii) State Snell's law of refraction of light.

(iii) Differentiate between the virtual images formed by a convex lens and a concave lens on the basis of :

(I) object distance, and

(II) magnification. *5*

35. (a) (i) Write the functions of the following parts of human female reproductive system :

(I) Ovary

(II) Fallopian tube

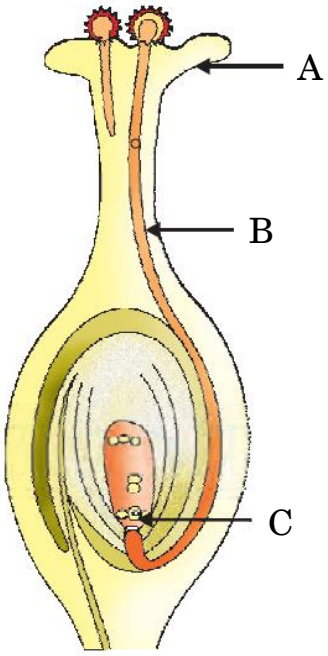
(III) Uterus

(ii) State briefly two contraceptive methods used by human males. *5*

OR

(b) (i) Differentiate between self-pollination and cross-pollination.

(ii) Identify A, B and C in the diagram given below and write one function of each. *5*



36. (a) (i) Draw electron dot structure of chlorine molecule.
(Atomic Number of Chlorine = 17)

(ii) What happens when chlorine reacts with methane in the presence of sunlight ? Write the name of the reaction.

(iii) Name the two oxidising agents used for the conversion of alcohols to acids.

(iv) List four differences in properties between covalent compounds and ionic compounds. *5*

OR

(b) (i) Give reason why carbon forms compounds mainly by covalent bonding.

(ii) Why do covalent compounds have low melting and boiling points.

(iii) Give reason for the following :

I. Covalent compounds are bad conductors of electricity.

II. Carbon shows catenation. *5*

SECTION E

The following questions are Source-based / Case-based questions. Read the case carefully and answer the questions that follow.

37. In our homes, we receive the supply of electric power through a main supply also called mains, either supported through overhead electric poles or by underground cables. In our country the potential difference between the two wires (live wire and neutral wire) of this supply is 220 V.

(a) Write the colours of the insulation covers of the line wires through which supply comes to our homes. *1*

(b) What should be the current rating of the electric circuit (220 V) so that an electric iron of 1 kW power rating can be operated ? *1*

(c) (i) What is the function of the earth wire ? State the advantage of the earth wire in domestic electric appliances such as electric iron. *2*

OR

(c) (ii) List two precautions to be taken to avoid electrical accidents. State how these precautions prevent possible damage to the circuit/appliance. *2*

38. Seawater contains many salts dissolved in it. Common salt is separated from these salts. Deposits of solid salt are also found in several parts of the world. These large crystals are often brown due to impurities. This is called rock salt and is mined like coal. The common salt is an important raw material for chemicals of daily use.

(a) Write balanced chemical equations to show the products formed during electrolysis of brine. *1*

(b) List two uses of any one product obtained during electrolysis of brine. *1*

(c) (i) A mild non-corrosive basic salt 'A', used for faster cooking, is strongly heated to produce a compound 'B', that is used for removing permanent hardness of water. Identify A and B and also write the equation for the reaction that occurs when A is heated. *2*

OR

(c) (ii) Define water of crystallisation. Give two examples of salts that have water of crystallisation. *2*

39. The maintenance functions of all living organisms must go on even when they are not doing anything particular. Even when we are just sitting in a class or even asleep, this maintenance job has to go on. These maintenance processes require energy to prevent damage and break-down of cells and tissues, which is obtained by the individual organism from the food prepared by the autotrophs, called producers.

(a) Name and define the process by which green plants prepare food. *1*

(b) Write chemical equation involved in the above process. *1*

(c) (i) State in proper sequence the events that occur in synthesis of food by desert plants. *2*

OR

(c) (ii) Explain giving reasons what happens to the rate at which the green plants will prepare food *2*

(I) during cloudy weather, and

(II) when stomata get blocked due to dust.

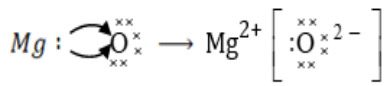
SECONDARY SCHOOL EXAMINATION, 2025

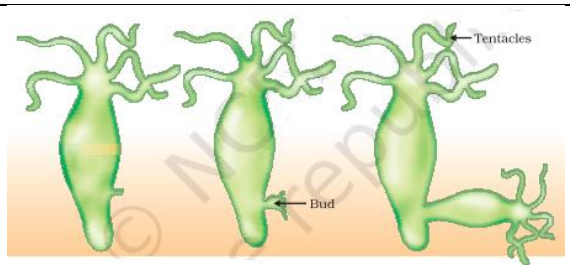
MARKING SCHEME

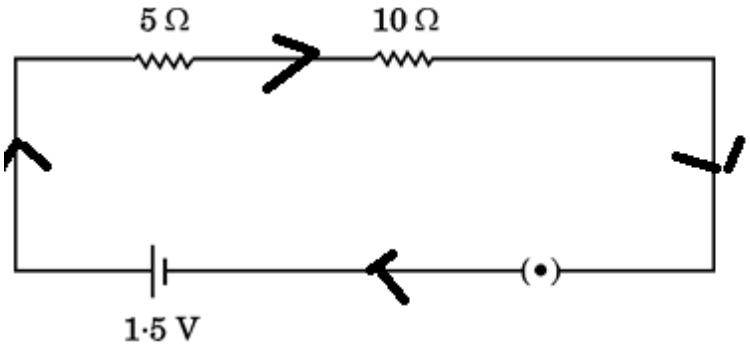
CLASS: X [SCIENCE (Subject Code–086)]

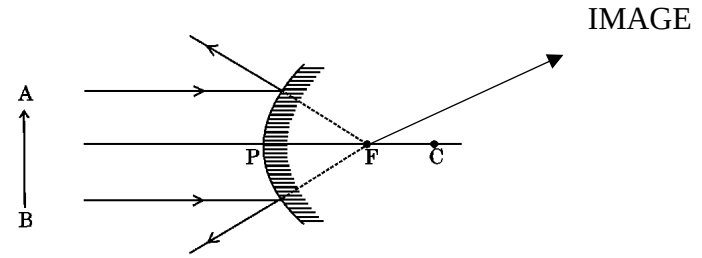
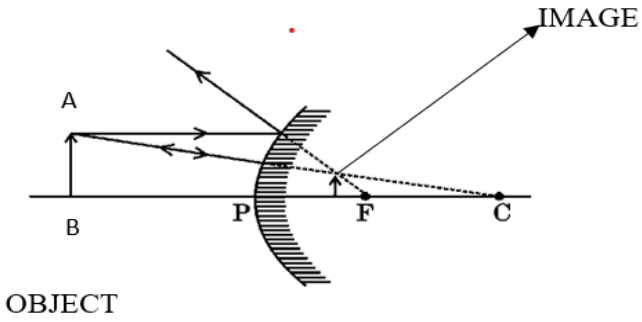
[Paper Code: 31/1/3]

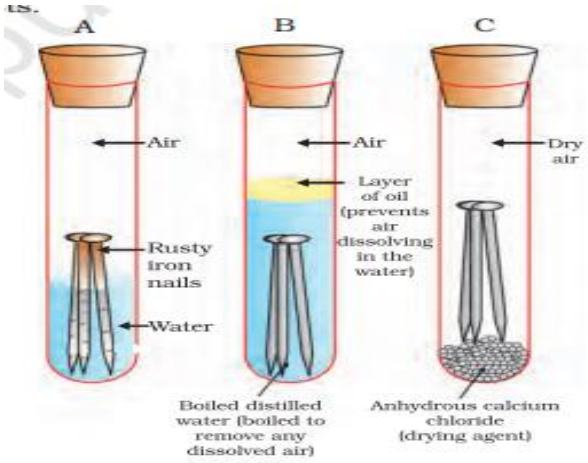
Maximum Marks: 80

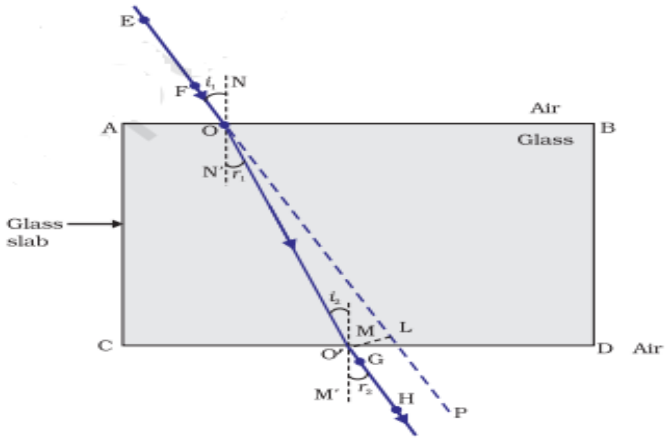
Q. No.	EXPECTED ANSWERS / VALUE POINTS	Marks	Total Marks
	SECTION A	**1**	**1**
1.	B / Calcium and Magnesium	1	1
2.	A / 	1	1
3.	C / It has weak electrostatic forces of attraction between its oppositely charged ions.	1	1
4.	A / Salt and water is formed	1	1
5.	B /5	1	1
6.	B / Al_2O_3 and MgO	1	1
7.	D / 1 : 8	1	1
8.	D / Cytoplasm and Oxygen deficient muscle cells.	1	1
9.	C / 100% round and yellow	1	1
10.	C / (i) and (iii)	1	1
11.	D / Auxins	1	1
12.	C / starch into simple sugars	1	1
13.	D / 99%	1	1
14.	D / (ii) and (iv)	1	1
15.	A / Scattering of light	1	1
16.	A / (i) and (ii)	1	1
17.	B / Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of Assertion (A).	1	1
18.	A / Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).	1	1
19.	C / Assertion (A) is true, but Reason (R) is false.	1	1
20.	B / Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of Assertion (A).	1	1
	SECTION B		
21.	Here u = – 10 cm; f = – 15 cm ; $\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$ $\frac{1}{-15}=\frac{1}{v}+\frac{1}{-10}$ $\frac{1}{v}=\frac{1}{-15}+\frac{1}{10}$ v =+30 cm Position of image is 30 cm from the mirror	½ ½ 1	2
22.	(a) $P=\frac{V^2}{R}$	½	

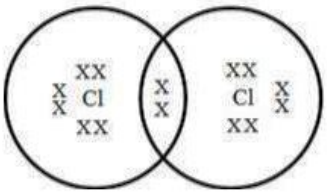
	$R=\frac{V^2}{P}$ $R_A=\frac{(220)^2}{50}$ $R_B=\frac{(220)^2}{25}$ $\frac{R_A}{R_B}=\frac{1}{2}$ / $P=\frac{V^2}{R}$ Since V is same means P is inversely proportional to R. Power of lamp A is double that of B hence Resistance of A is half that of B. $\frac{R_A}{R_B}=\frac{1}{2}$ **OR** (b) $H=\frac{V^2}{R}\times t$ $400=\frac{V^2}{4}\times 1$ $V^2=1600$ V=40V	½ 1 1 1 ½ ½ 1	2
23.	 diagram labelling	1 1	2
24.	(a) • Plugging of the leak in blood vessels prevents lowering of the blood pressure / maintains the efficiency of the pumping system. • Platelets • Help to clot the blood at the site of injury. **OR** (b) (i) Plants have low energy needs because they have a large proportion of dead cells in many tissues / Plants have low energy needs as they do not move (ii) Translocation of soluble products of photosynthesis from leaves to other parts of the plant / It transports amino acids and other substances to storage organs of roots, fruits and seeds and to growing organs.	1 ½ ½ 1 1	2

25.	• Evolution of gas • Change / Rise in temperature	1 1	2
26.	(a) $2\,HNO_3 + Ca(OH)_2 \longrightarrow Ca(NO_3)_2 + 2\,H_2O$ (b) $NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$ (**Deduct half mark if equation is not balance**d)	1 1	 2
	SECTION C		
27.	(a) It is the amount of work done to bring one coulomb charge from one point to another in the field is 1 joule. / 1V=$\frac{1J}{1C}$ (b)  **Current** = $\frac{\textbf{Potential difference}}{\textbf{Resistance}} = \frac{\textbf{1·5 V}}{(\textbf{5 Ω + 10 Ω})}$ $= \frac{1{\cdot}5\,V}{15\,\Omega} = 0{\cdot}1$ A	1 1 1	3
28.	(a) $\frac{1}{R_1} = \frac{1}{10\,\Omega} + \frac{1}{15\,\Omega} = \frac{1}{6\,\Omega} \Rightarrow R_1 = 6\ \Omega$ $\frac{1}{R_2} = \frac{1}{60\,\Omega} + \frac{1}{40\,\Omega} = \frac{100}{2400\,\Omega} \Rightarrow R_2 = 24\ \Omega$ ∵ R_1 and R_2 are in series ∴ $R_{total} = R_1 + R_2 = (6 + 24) = 30\ \Omega$ (b) $V = IR \Rightarrow I = \frac{V}{R} = \frac{15\,V}{30\,\Omega} = 0{\cdot}5$ A (c) $V = IR_1 = 0{\cdot}5\ A \times 6\ \Omega = 3{\cdot}0$ V	 1 1 1	 3

29.	(i)  OBJECT (ii)  OBJECT **(Note: Deduct ½ mark if arrows are not drawn.)**	1 ½ 1 ½	3
30.	(a) • 23 pairs or 46 chromosomes • 1 Pair or 2 chromosomes **(**b) In sexually reproducing organisms chromosomes occur in pairs. The members of the pair separate at the time of gamete formation. The two germ cells fuse and form zygote thus restore the same number of chromosomes as their parents**.** **(Award marks if explained by any other method.)**	½ ½ 2	3
31.	(a) Adrenaline Adrenal Gland **(b**) It acts on heart. Heart beats faster, resulting in more supply of oxygen to skeletal muscles. The blood to the digestive system and skin is reduced due to contraction of muscles around small arteries in these organs. Breathing rate increases because of contraction of diaphragm and the rib muscles**.**	½ ½ 2	3
32.	(a) ➢ Take three test tubes and place clean iron nails in each of them. ➢ Label these test tubes A, B and C.		

	➢ Pour some water in test tube A and cork it. ➢ Pour boiled distilled water in test tube B, add about 1 mL of oil and cork it. The oil will float on water and prevent the air from dissolving in the water. ➢ Put some anhydrous calcium chloride in test tube C and cork it. Anhydrous calcium chloride will absorb the moisture, if any, from the air. Iron nails rust in test tube A, but they do not rust in test tubes B and C. Rusting of iron takes place when exposed to both air and water. /  Iron nails rust in test tube A, but they do not rust in test tubes B and C. Rusting of iron takes place when exposed to both air and water. **(Note: if a student explains activity through description or through labelled diagram, give full marks.)** **OR** (b) (i) • Sodium, Potassium, Lithium **(any two)** • Observations: ➢ A violent reaction occurs. ➢ Large amount of heat is evolved. ➢ Evolved gas catches fire. (ii) The gas (bubbles) burns with a pop sound	3 ½,½ 1 ½ ½	3
33.	(a) $3\ MnO_2(s)\ +\ 4\ Al(s) \longrightarrow 3\ Mn(l)\ +\ 2\ Al_2O_3(s)\ +\ heat$ $Fe_2O_3(s)\ +\ 2\ Al(s) \longrightarrow 2\ Fe(l)\ +\ Al_2O_3(s)\ +\ heat$ **(Award marks if explained through statement)** (b) Metals towards the top of the reactivity series (Na, Mg, Ca) have more affinity for oxygen than carbon.	1 1 1	3
	SECTION D		
34.	(a) (i) • Concave lens	½	

	• $P = \frac{1}{f(m)}$ $-2{\cdot}5 = \frac{1}{f}$ $f = \frac{10}{-2{\cdot}5} = -0{\cdot}4$ m $= -40$ cm • Myopia (ii) (I) Real and inverted (II) magnified image /size of image is double the size of object (III) beyond 2F/ on the other side as that of object (IV) Negative (iii) • The lens with focal length 10 cm • less focal length, more converging/diverging power **OR** (b) (i)  **(if arrows not marked , deduct half mark)** (ii) The ratio of sine of angle of incidence to the sine of angle of refraction is a constant, for the light of a given colour and for a given pair of media. / $\frac{\sin i}{\sin r} = constant$ (iii) Convex Lens | Concave Lens (I) Object to be placed between O and F | Object can be placed anywhere in front of the lens (II) Magnified image | Diminished image	½ ½ ½ ½×4 ½ ½ 2 1 1 1	5
35.	(a) (i) (I) Ovary: Produces female gamete (egg) and female hormones(oestrogen). (II) Fallopian tube: Site of Fertilization (III) Uterus: Site of implantation and embryonic development. (ii) Methods of contraception used by males: • Mechanical barrier - Condoms	½,½ 1 1 1	

	• Surgical method – blocking the vas deferens in males (Vasectomy) **OR** (b) (i) Self-pollination: Pollen grains are transferred from stamen to the stigma of the same flower. Cross-pollination: Transfer of pollen grains from stamen of one flower to the stigma of another flower of same species. (ii) A – Stigma : Receives pollen and provides suitable environment for its germination. B –Pollen tube : Carries males germ cells (gametes) to the female gamete situated in the ovary. C – Egg Cell (Female germ cell) : Fuses with male gamete and forms Zygote.	1 1+1 1 1 1	5
36.	**(a) (i)**  (ii) • $CH_4 + Cl_2 \xrightarrow{Sunlight} CH_3Cl + HCl$ / Chloromethane is formed; • substitution reaction **(iii)** Alkaline $KMnO_4$/ Acidified $KMnO_4$ /Acidified $K_2Cr_2O_7$ **(any two)** **(iv)** Covalent compounds: Low melting and boiling points; Poor conductors of electricity; Generally soft if solids; Generally soluble in oil or non-polar solvents Ionic Compounds: High melting and boiling points; Conduct electricity in aqueous solution and molten state.; Hard crystalline solids; Generally soluble in soluble in water/polar solvents **OR** **(b)** (i) Carbon could gain four electrons forming C^{4-} anion. But it would be difficult for the nucleus with six protons to hold on to ten electrons. Carbon could also	1 ½ ½ ½, ½ ½×4	

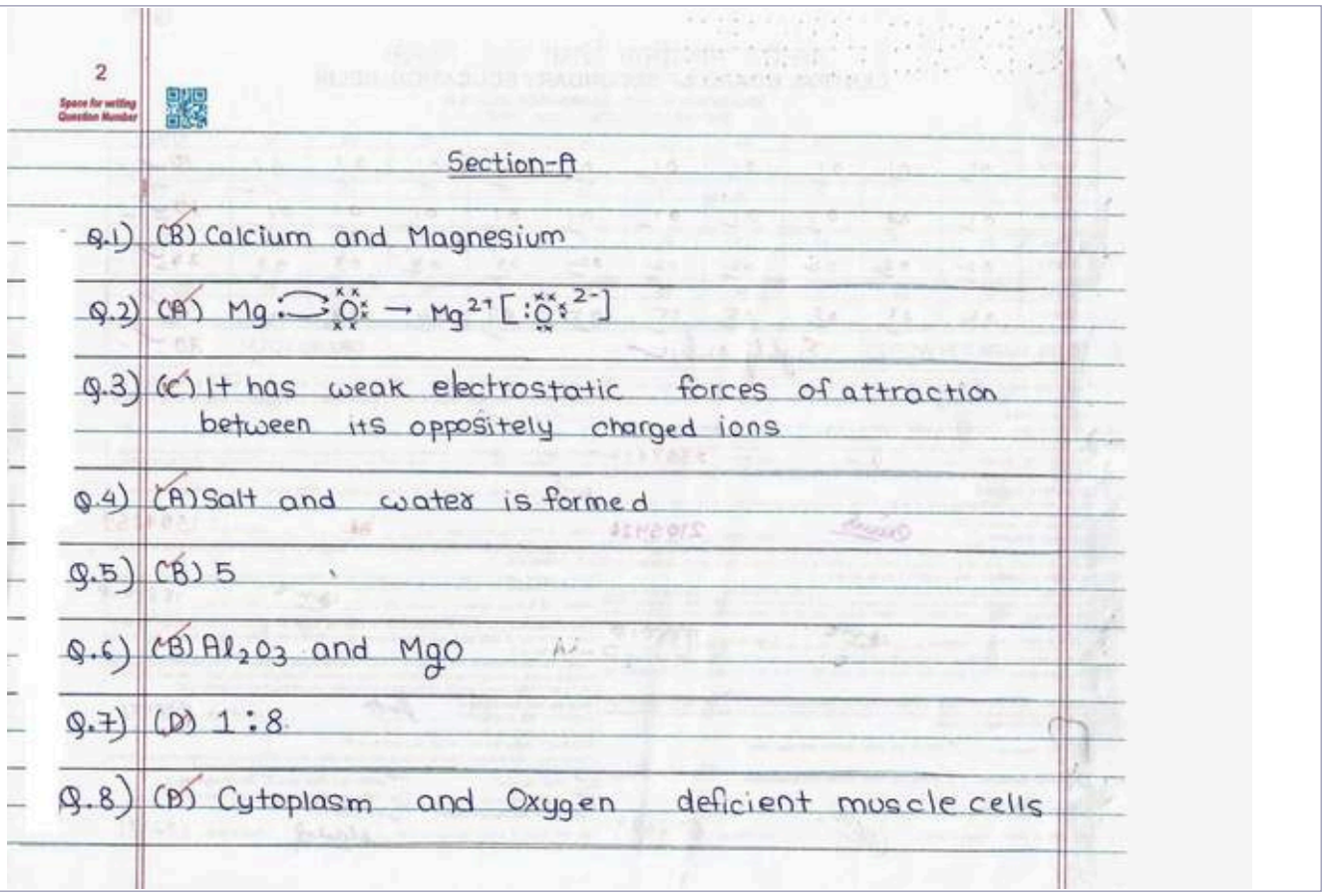
	lose four electrons forming C^{4+} cation. But it would require a large amount of energy to remove four electrons leaving behind a carbon cation with six protons in its nucleus. $\therefore$ It shares four electrons of the outermost shell to form covalent bond/covalent compound.	2	
	(ii) There are weak forces of attraction between the molecules of covalent compound.	1	
	(iii) (I) Covalent compounds do not form ions hence are poor conductors.	1	
	(II) Carbon-carbon single bond is very strong and stable.	1	5
	SECTION E		
37.	(a) Live wire- Red Neutral wire- Black	½ ½	
	(b) Power, P = 1 kW = 1 × 1000 W = 1000 W Voltage, V = 220 V Current drawn I =? $P = V \times I$ $I = \frac{1000\ W}{220\ V} = 4{\cdot}54$ A Current rating should be of 5A.	½ ½	
	(c) (i) • The earth wire provides a low resistance conducting path for the current which ensures that any leakage of current to flow to the metallic body of the appliances, keeps its potential to that of the earth. • The user will not get an electric shock. **OR**	1 1	
	(c) (ii) • Fuse wire • Earth wire • A fuse in a circuit prevents damage to the circuit due to overloading. • Earth wire prevents electric shock due to leakage of current.	½ ½ ½ ½	4
38.	(a) $2\,NaCl + 2\,H_2O \xrightarrow{\text{electricity}} 2\,NaOH + H_2 + Cl_2$ (b) Uses of NaOH : Degreasing metals/ Soaps and Detergents/ paper making/ artificial fibres/ preparation of bleach Uses of H_2: As fuel/ Margarine/ In preparation of ammonia for fertilizers/preparation of HCl Uses of Cl_2 : Disinfectant/ PVC/ water treatment/ in swimming pools/ CFC's/ preparation of bleach/ preparation of HCl/ pesticides **(Any two uses of anyone product)**	1 ½,½	

	(c) (i) A – $NaHCO_3$/ Sodium Hydrogen Carbonate/Baking soda B – Na_2CO_3 / Sodium Carbonate $2\,NaHCO_3 \xrightarrow{heat} Na_2CO_3 + H_2O + CO_2$ **OR** (c) (ii) • The fixed number of water molecules present in one formula unit of a salt. • $CuSO_4.5H_2O$/Copper Sulphate pentahydrate/Blue vitriol • $CaSO_4.2H_2O$/Gypsum/Calcium sulphate dihydrate • $Na_2CO_3.10H_2O$/Washing Soda/Sodium carbonate decahydrate • $FeSO_4.7H_2O$/ Green Vitriol/Ferrous sulphate heptahydrate • $CaSO_4$. ½ H_2O /Calcium Sulphate hemihydrate/ POP **(Any other two examples)**	½ ½ 1 1 ½,½	4
39.	(a) Photosynthesis A process by which green plants capture sunlight and convert it to chemical energy with the help of chlorophyll / Process by which carbon dioxide and water is converted into carbohydrates in the presence of sunlight chlorophyll and water. (b) $6CO_2 + 12H_2O \xrightarrow[\text{Sunlight}]{\text{Chlorophyll}} C_6H_{12}O_6 + 6O_2 + 6H_2O$ (c) (i) • Absorption of light energy by chlorophyll • Conversion of light energy to chemical energy. • Reduction of carbon dioxide to carbohydrates. • Desert plants take up CO_2 at night and prepare intermediate, which is acted upon by the energy absorbed by the chlorophyll during the day. **OR** (c) (ii) (I) Decrease the rate of photosynthesis due to low amount of sunlight. (II) Decreases the rate of photosynthesis due to reduced gaseous exchange. .	½ ½ 1 2 1 1	4

Topper's Answer Sheet

CBSE Board Exam 2025, Set 31/1/3: answer book of a CBSE topper, as published by CBSE. Pages in question order.

Answer book page 1: Q.1 to Q.8



2

Section-A

Q.1) (B) Calcium and Magnesium

Q.2) (A) $Mg : \ddot{\underset{\cdot\cdot}{O}} : \rightarrow Mg^{2+}[:\ddot{\underset{\cdot\cdot}{O}}:^{2-}]$

Q.3) (C) It has weak electrostatic forces of attraction between its oppositely charged ions

Q.4) (A) Salt and water is formed

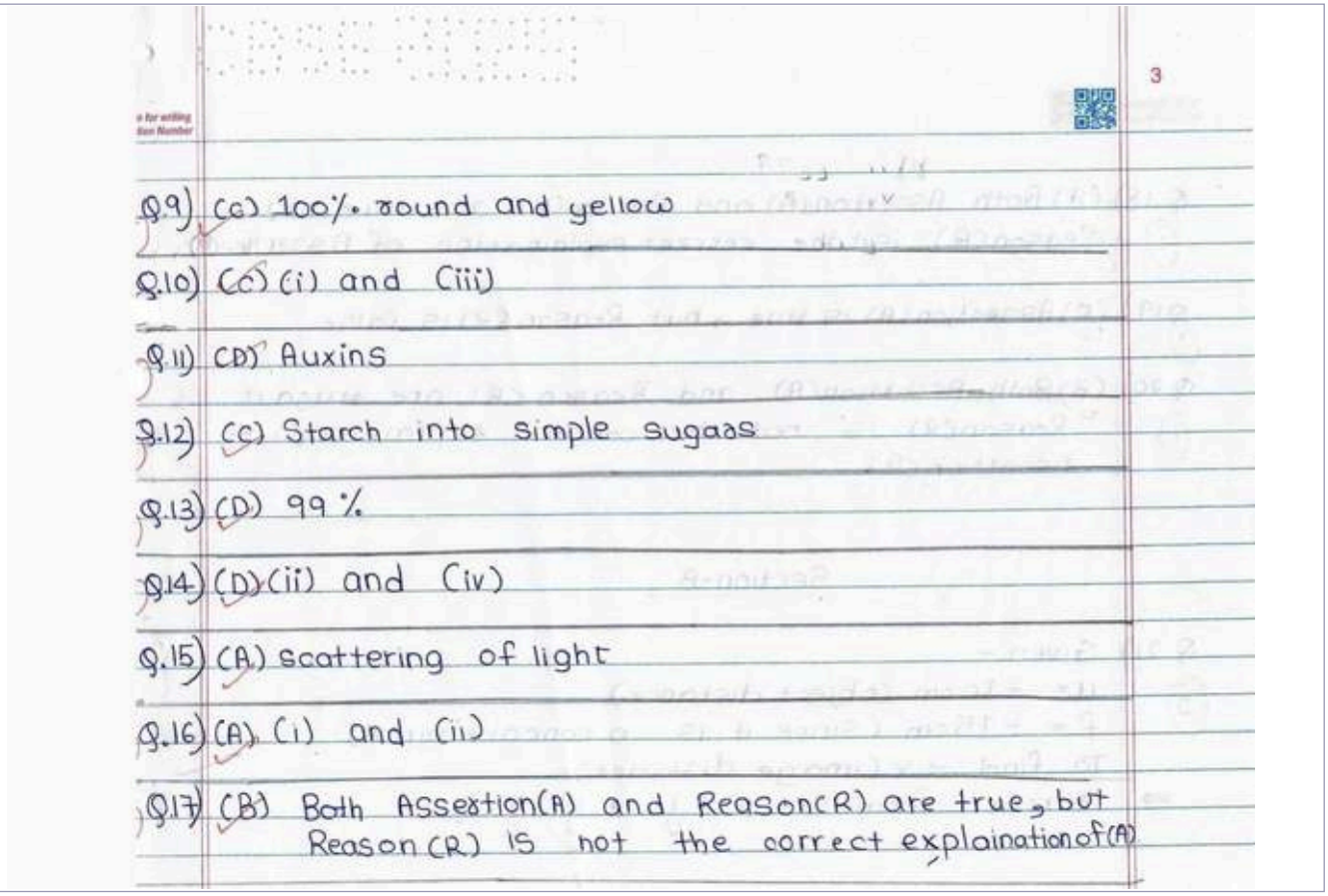
Q.5) (B) 5

Q.6) (B) Al_2O_3 and MgO

Q.7) (D) 1 : 8

Q.8) (D) Cytoplasm and Oxygen deficient muscle cells

Answer book page 2: Q.9 to Q.17



3

Q.9) (C) 100% round and yellow

Q.10) (C) (i) and (iii)

Q.11) (D) Auxins

Q.12) (C) Starch into simple sugars

Q.13) (D) 99 %

Q.14) (D) (ii) and (iv)

Q.15) (A) scattering of light

Q.16) (A) (i) and (ii)

Q.17) (B) Both Assertion(A) and Reason(R) are true, but Reason (R) is not the correct explaination of (A)

Answer book page 3: Q.18 to Q.21

4

Q.18) (A) Both Assertion(A) and Reason(R) are true and Reason(R) is the correct explaination of Assertion(A).

Q.19) (C) Assertion(A) is true, but Reason(R) is false.

Q.20) (B) Both Assertion(A) and Reason(R) are true, but Reason(R) is not the correct explaination of Assertion(A)

Section-B

Q.21) Given -

$u = -10\,cm$ (object distance)

$f = -15\,cm$ (since it is a concave mirror)

To find → v (image distance)

⇒ Mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Answer book page 4: Q.21 (contd.) and Q.22

5

⇒ $\frac{1}{v} - \frac{1}{10} = \frac{-1}{15}$

⇒ $\frac{1}{v} = \frac{-1}{15} + \frac{1}{10}$

⇒ $\frac{1}{v} = \frac{-10+15}{10\times15}$

⇒ $\frac{1}{v} = \frac{5}{\frac{10\times15}{2}}$

$v = +30\,cm$

Position of image formed by mirror = +30cm on the other side of mirror.

Q.22)

(a) Lamp A →

Power = 50 w

Volt = 220V

Answer book page 5: Q.22 (contd.)

6

Space for writing Question Number

We know,

$P = \frac{V^2}{R}$

$R = \frac{V^2}{P}$

$R_A = \frac{220 \times 220}{50}$

$= 44 \times 22$

$= 968\ \Omega$ ✓

Lamp B →

Power = 25 ω

Volt = 220 V

$P = \frac{V^2}{R}$

$R_B = \frac{220 \times 220}{25}$

$= 1936\ \Omega$ ✓

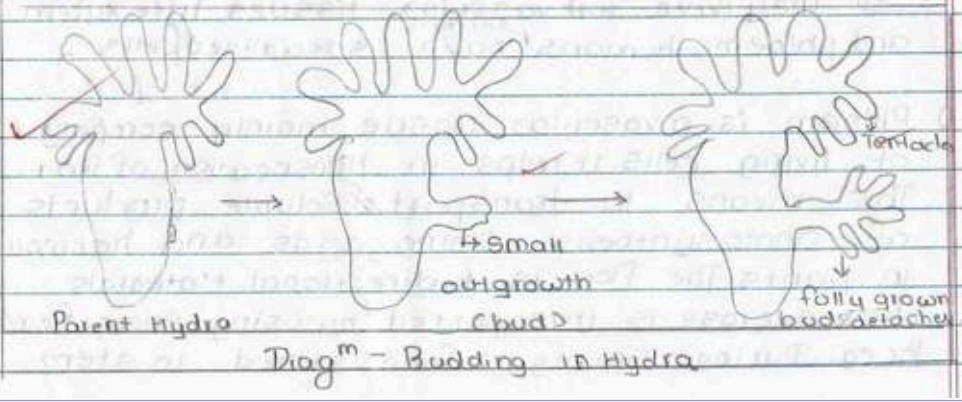
Answer book page 6: Q.22 (contd.) and Q.23

7

$\therefore \frac{R_A}{R_B} = \frac{968}{1936}$

$R_A : R_B = \boxed{1 : 2}$ ✓

Q.23) Hydra reproduces by the process of budding. With help of regenerative cells, there is outgrowth called bud which develops.



Diagᵐ - Budding in Hydra

Answer book page 7: Q.24

8

Q.24)

(b)(i) The transport system in plants is relatively slower than in animals because plants are mainly composed of dead cells. Plants are stationary and don't require energy to perform activities when compared to animals. Moreover, simple processes like diffusion and osmosis help in transport of materials from soil. They have vascular tissues like xylem and phloem, animals have specialised cells

(ii) Phloem is a vascular tissue mainly composed of living cells. It helps in transpcation of food. This means to transport of soluble products of photosynthesis, amino acids and hormones in plants. The flow is bidirectional. Materials like sucrose is transported by using energy from ATP eg. During spring, food stored in stem

Answer book page 8: Q.24 (contd.) and Q.25

9

and roots is transported to buds

Q.25) For a chemical change to occur there has to be-

i) evolution of gas
ii) change in temperature
iii) change in odour
iv) formation of precipitate

Here,

$$Zn + H_2SO_4 \longrightarrow ZnSO_4 + H_2\uparrow$$

Here 2 observations are seen-

i) Evolution of gas (H_2)
ii) Change in temperature since this is an exothermic reaction

Answer book page 9: Q.26 and Q.27

10

Space for writing Question Number

Q.26)

(a) $2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$

(b) $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

Section-C

Q.27)

(a) When 1 Joule of work is done to move 1 Coloumb of charge from one point to another in an electric field, it is said to be 1 Volt

$$V = \frac{W}{Q}$$

$$1V = \frac{1J}{1C}$$

$$1V = 1JC^{-1}$$

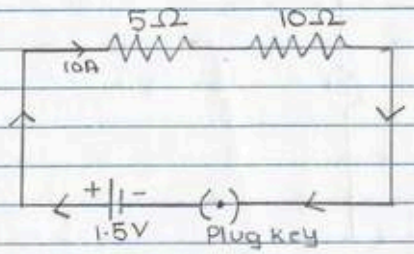
Answer book page 10: Q.27 (contd.)

11

Space for writing Question Number

Q.27)

(b)



Let $R_1 = 5\Omega$

$R_2 = 10\Omega$

To find → I (current)

Total resistance (R_s) $= R_1 + R_2$

$= 5 + 10$

$= 15\Omega$

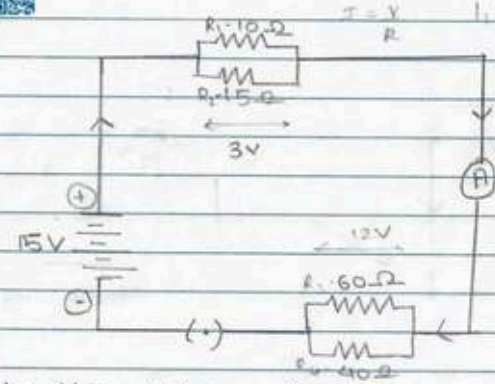
$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1.5}{15} = 0.1A$

Answer book page 11: Q.28

12

Q.28)



In the given circuit,

Let $R_1 = 10\,\Omega$

$R_2 = 15\,\Omega$

$R_3 = 60\,\Omega$

$R_4 = 40\,\Omega$

(a) Since R_1 and R_2 are in parallel.

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_5}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{15} = \frac{1}{R_5}$$

Answer book page 12: Q.28 (contd.)

13

$$\Rightarrow \frac{15+10}{150} = \frac{1}{R_5}$$

$$\Rightarrow \frac{25}{150}^{5} = \frac{1}{R_5}$$

$$30$$

$$R_5 = 6\,\Omega$$

Now R_3 and R_4 are in parallel.

$$\frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{R_6}$$

$$\Rightarrow \frac{1}{60} + \frac{1}{40} = \frac{1}{R_6}$$

$$\Rightarrow \frac{100}{60 \times 40} = \frac{1}{R_6}$$

$$\Rightarrow \therefore R_6 = 24\,\Omega$$

Now total $R_S = R_5 + R_6$

$= 30\,\Omega$ → Total resistance

Answer book page 13: Q.28 (contd.)

14

(b) Total current →

$R = 30\Omega$

$V = 15V$

$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1}{2} A$

(c) Total Resistance(R_5) of parallel combination of 10Ω and $15\Omega = 6\Omega$

$\therefore I = \frac{1}{2} A$

$\therefore V = IR$ (Ohm's law)

$= 3V$

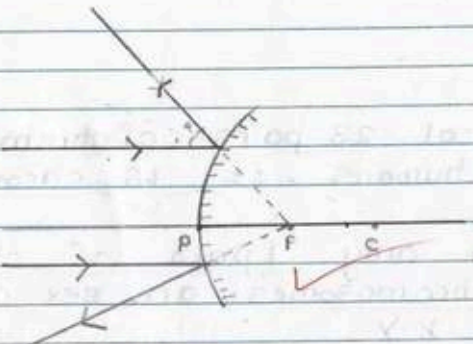
Voltage across them is 3V

Answer book page 14: Q.29

15

Q.29)

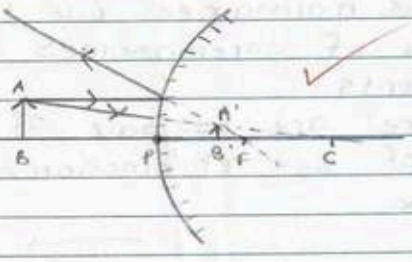
(i)



Position of object → At infinity

Nature of image → virtual, erect, diminished, At focus on other side

(ii)



Position of object → Between infinity and Pole

Nature of image - Virtual, Erect, diminished, between f and P on other side

Answer book page 15: Q.30

16

Space for writing Question Number

Q.30)

(a) There are total 23 pairs of chromosomes present in humans or 46 chromosomes are present.
Out of them only 1 pair of chromosomes or only 2 chromosomes are sex chromosomes.
In male → XY
In female → XX

(b)(i) In sexually reproducing organisms, the number of chromosomes is maintained due to inheritance of equal number of chromosomes from both the parents.

(ii) In humans, there are 22 pair of autosomes and 1 pair of sex chromosome. In male-XY and female XX.

→

Answer book page 16: Q.30 (contd.) and Q.31

17

Space for writing Question Number

(iii) The gametes ~~half~~ have only half the amount of DNA/~~eh~~ and chromosomes present in them.

(iv) Hence, when ~~these too~~ the ~~two~~ male and female gamete combine during fertilisation, original number of chromosomes is attained

(v) A diploid zygote is formed (2n) from haploid gametes (n).

Q.31)

(a) The hormone 'x' is adrenaline and the gland that secretes them is adrenal glands

(b) Adrenaline hormone prepares the body for fight or flight. There are 3 main process that take place →

i) It increases the heart rate

ii) It decreases blood supply to skin and digestive system due to contraction of

Answer book page 17: Q.31 (contd.) and Q.32

18

muscles around small arteries in these organs and diverts blood to skeletal muscles.

(iii) It increases breathing rate due to contraction of rib cage and diaphragm.

Q.32)

(a) Aim – To show iron articles get rusted only under specific conditions of moisture and air.

Materials required – Iron nails, three testube, calcium chloride, water, oil

Procedure –

(a) Take three test tubes A, B and C

(b) In test tube A, pour some water and put iron nails in it and cork it.

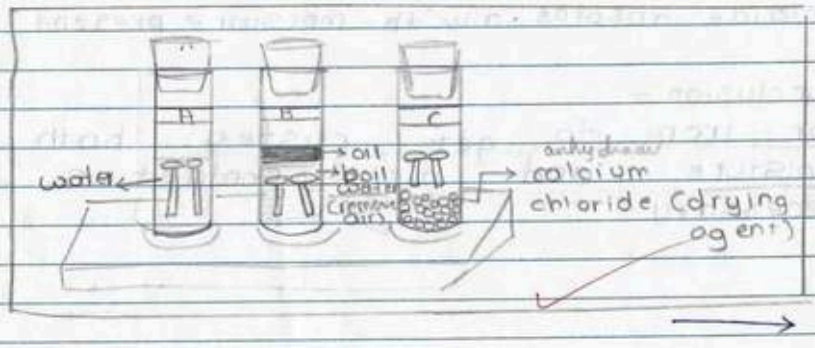
Answer book page 18: Q.32 (contd.)

19

(c) In test tube B, put iron nails in boiled distilled water and on top a layer of oil. and cork it

(d) In test tube C, put calcium chloride along with iron nails and cork it

(e) Do the experimental set up as shown below →



Answer book page 19: Q.32 (contd.)

20

Observation –

i) In test tube A, Iron nails develop red-brown flaky substance (rust)

ii) In test tube B and C no rust is observed as in test tube B boiled distilled water and layer of oil prevent air to interact with iron nail. In test tube C, calcium chloride absorbs all the moisture present.

Conclusion –

For iron to get rusted both moisture and air contact is necessary.

Answer book page 20: Q.33

21

Q.33)

(a) ~~(a)~~ Displacement reactions in which a high reactive metal displaces less reactive metal from its salt soln can be used to obtain two metals from middle of reactivity series → Manganese and Iron

(i) Reacⁿ of Aluminium with manganese oxide – Here Al being more reactive displaces Mn

$4Al + 3MnO_2 \longrightarrow 2Al_2O_3 + 3Mn + heat$

The heat evolved is so large that Mn is obtained in molten form

(ii) Reacⁿ of ~~Fe~~ ferric oxide with aluminium to obtain Iron →

Here Al replaces iron from its oxide

$Fe_2O_3 + Al \longrightarrow Fe + Al_2O_3 + heat$

→ This is also known as thermit reacⁿ.

Answer book page 21: Q.33 (contd.) and Q.34

22

(b) Metals high up in the reactivity series cannot be obtained by reduction of ~~metal~~ oxides by carbon as –

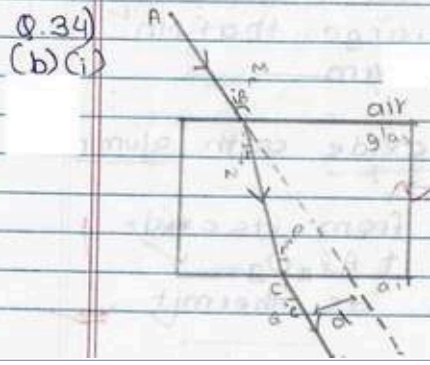
(i) These metals (K, Na, Ca) are highly reactive

(ii) They have more affinity for oxygen rather than carbon. ✓

Section - D

Q.34)

(b)(i)



Here,

AB – ~~re~~ incident ray

∠ABM → angle of incidence

∠NBC → angle of refraction

BC → refracted ray

∠QCP → angle of emergence

CP → emergent ray

d → lateral displacement

Answer book page 22: Q.34 (contd.)

23

(ii) Snell's law of refraction of light → The ratio of sine of angle of incidence to the sine of angle of refraction is constant for a given colour and given pair of media. ($i \geq 0 < 90°$)

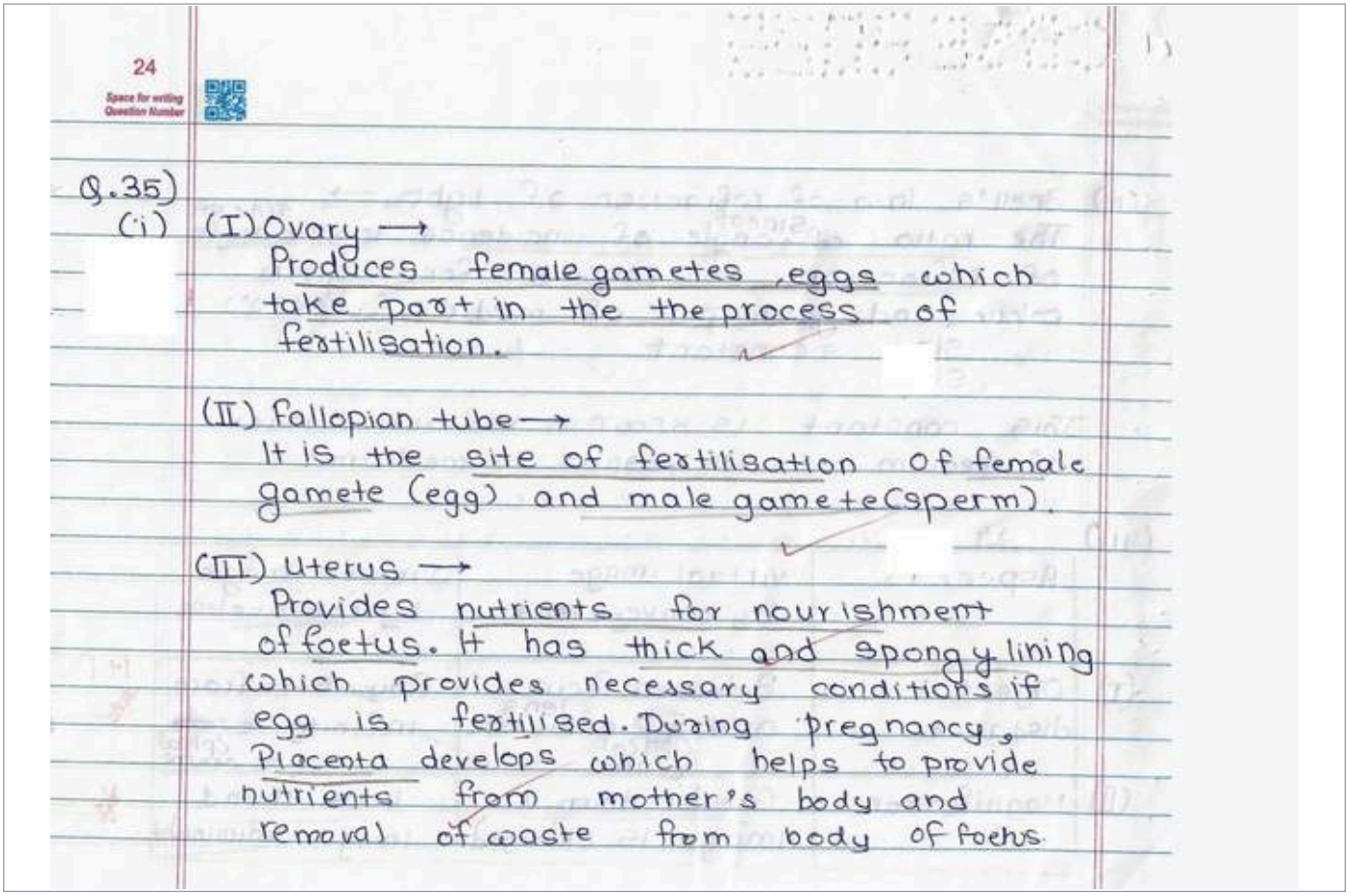
$$\frac{\sin i}{\sin r} = \text{constant}$$ ✓

This constant is known as refractive index of medium 2 with respect to medium 1.

(iii)

	Aspect	Virtual image by convex lens	Virtual image by concave lens
(I)	Object distance	Between focus and ~~optical~~ lens centre of ~~mirror~~	Anywhere from infinity to ~~optical~~ optical centre
(II)	Magnification	Greater than 1 Image is enlarged	Less than 1 Image is diminished

Answer book page 23: Q.35



24

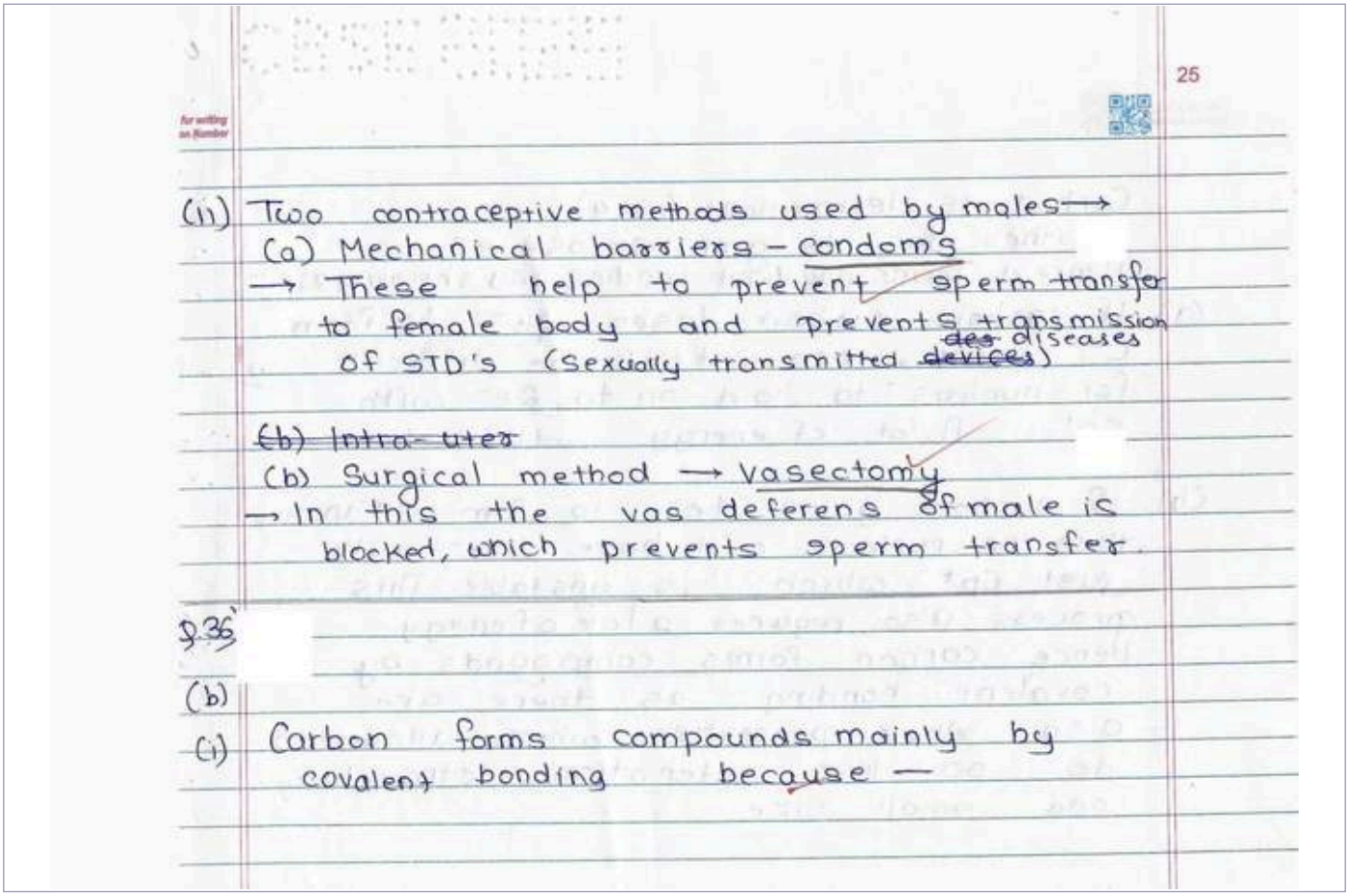
Q.35)

(i) (I) Ovary → Produces female gametes, eggs which take part in the the process of fertilisation.

(II) fallopian tube → It is the site of fertilisation of female gamete (egg) and male gamete (sperm).

(III) Uterus → Provides nutrients for nourishment of foetus. It has thick and spongy lining which provides necessary conditions if egg is fertilised. During pregnancy, Placenta develops which helps to provide nutrients from mother's body and removal of waste from body of foetus.

Answer book page 24: Q.35 (contd.) and Q.36



25

(ii) Two contraceptive methods used by males →

(a) Mechanical barriers – condoms → These help to prevent sperm transfer to female body and prevents transmission of STD's (Sexually transmitted ~~des~~ diseases ~~devices~~)

~~(b) Intra-uter~~

(b) Surgical method → Vasectomy → In this the vas deferens of male is blocked, which prevents sperm transfer.

~~Q.36~~

(b)

(i) Carbon forms compounds mainly by covalent bonding because —

Answer book page 25: Q.36 (contd.)

26

Carbon is tetravalent (2,4)
It either has to gain or lose e^-.
Hence it forms covalent bonding by sharing of e^-

(a) If ~~carbon~~ carbon loses $4e^-$ to form C^{4+} cation, then it will be difficult for nucleus to hold on to $2e^-$ with $6p^+$. A lot of energy will be lost

(b) If carbon gains $4e^-$ to form C^{4-} anion, then the nucleus will have $10e^-$ with just $6p^+$ which is unstable. This process also requires a lot of energy.
Hence carbon forms compounds by covalent bonding as there are also other properties which allow it to do so like catenation, tetravalency and small size.

Answer book page 26: Q.36 (contd.)

27

(II) Covalent compounds have low melting and boiling point as they have weak intermolecular forces of attraction which can easily be overcome due to heat

(III)

(I) Covalent compounds don't have any free electrons, they only constitute of shared pair of electrons. Hence they are bad conductors of electricity.

(II) Carbon shows self-linkage of property of forming long chain of carbon atoms.
It forms saturated, unsatured compounds.
It can even form branched, cyclic chains.
Due to its tetravalency, it forms bonds with other carbon atoms.

Answer book page 27: Q.37

28

Space for writing Question Number

Section-E

Q.37)

(a)

Wire	Insulation cover
Live wire	Red
Neutral wire	Black
Earth wire	Green

(b) Given →

$P = 1000\ \omega$

$V = 220\ V$

To find → I

$\Rightarrow P = VI$

$\Rightarrow I = \frac{P}{V}$

$\Rightarrow I = \frac{1000}{220}$

$I = 4.5A$

Hence current rating should be atleast 5A.

Answer book page 28: Q.37 (contd.) and Q.38

29

(c)

(i) Earth wire is connected to the metallic body of appliances to prevent overflow of current in a circuit and to prevent fatal accidents and damaging of appliance. The earth wire is connected to a metal plate buried inside earth. Whenever there is an excess of current flowing through circuit, it offers low resistance path for current to flow in earth therefore saving appliances such as electric iron from damage and prevent overloading / breaking of circuit.

Q.38)

(a) $2NaCl + 2H_2O \xrightarrow{\text{electrolysis}} 2NaOH + H_2 + Cl_2$

(Caustic Soda) (Hydrogen) (Chlorine)

Answer book page 29: Q.38 (contd.)

30

(b)

	Products obtained	uses
i)	H_2 (hydrogen)	fuels, ammonia for fertilisers, margarine
ii)	NaOH (caustic soda)	De-greasing metal, soap and detergents, artificial paper making
iii)	Cl_2 (Chlorine)	PVCs, disinfectant, swimming pool

(c)(ii) Water of crystallisation -
Fixed no. of water molecules present in one formula unit of salt.
e.g. (i) Washing soda → $Na_2CO_3.10H_2O$
Sodium carbonate decahydrate

Answer book page 30: Q.38 (contd.) and Q.39

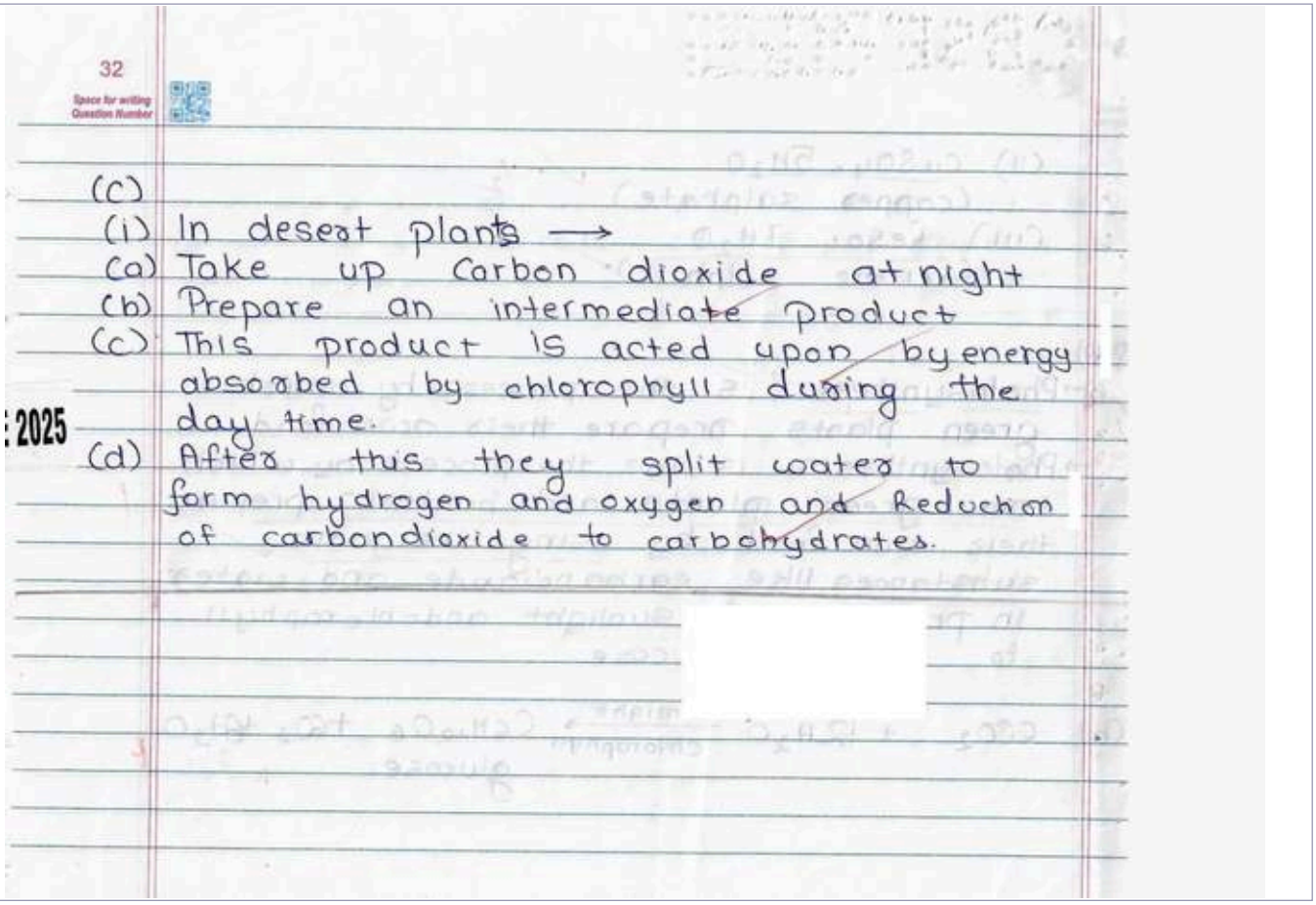
31

(ii) $CuSO_4.5H_2O$
(copper sulphate)
(iii) $FeSO_4.7H_2O$
(ferrous sulphate)

Q39
(a) → Photosynthesis is the process by which green plants prepare their own food.
→ Photosynthesis is the the process by which some green plants and bacteria prepare their own food by using inorganic substances like carbondioxide and water in presence of sunlight and chlorophyll to form glucose.

(b) $6CO_2 + 12H_2O \xrightarrow[\text{chlorophyll}]{\text{sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$
glucose

Answer book page 31: Q.39 (contd.)



What to learn from the topper's answer book

- Every MCQ answer is written with the option letter AND the option text, so there is no doubt for the examiner.
- Numericals: given data, formula, substitution with units and a boxed final answer, exactly like the marking scheme.
- Diagrams are drawn neatly in pencil with every part labelled and arrows shown on rays.
- Long answers are split into labelled sub-parts; key terms are underlined.
- No over-writing: each answer starts on a fresh line with the question number in the margin.

Series CABD5/5

Set-1

प्रश्न-पत्र कोड
Q.P. Code **31/5/1**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

विज्ञान
SCIENCE

निर्धारित समय : 3 घण्टे *अधिकतम अंक : 80*

Time allowed : 3 hours *Maximum Marks : 80*

	नोट		**NOTE**
(I)	कृपया जाँच कर लें कि इस प्रश्न–पत्र में मुद्रित पृष्ठ **27** हैं ।	(I)	Please check that this question paper contains **27** printed pages.
(II)	कृपया जाँच कर लें कि इस प्रश्न–पत्र में **39** प्रश्न हैं ।	(II)	Please check that this question paper contains **39** questions.
(III)	प्रश्न–पत्र में दाहिने हाथ की ओर दिए गए प्रश्न–पत्र कोड को परीक्षार्थी उत्तर–पुस्तिका के मुख–पृष्ठ पर लिखें ।	(III)	Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
(IV)	**कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर–पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**	(IV)	**Please write down the serial number of the question in the answer-book before attempting it.**
(V)	इस प्रश्न–पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न–पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न–पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर–पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।	(V)	15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper comprises **39** questions. **All** questions are compulsory.*

(ii) *This question paper is divided into **five** sections – **A**, **B**, **C**, **D** and **E**.*

(iii) ***Section A** – Questions No. **1** to **20** are multiple choice questions. Each question carries **1** mark.*

(iv) ***Section B** – Questions No. **21** to **26** are very short answer type questions. Each question carries **2** marks. Answer to these questions should be in the range of 30 to 50 words.*

(v) ***Section C** – Questions No. **27** to **33** are short answer type questions. Each question carries **3** marks. Answer to these questions should in the range of 50 to 80 words.*

(vi) ***Section D** – Questions No. **34** to **36** are long answer type questions. Each question carries **5** marks. Answer to these questions should be in the range of 80 to 120 words.*

(vii) ***Section E** – Questions No. **37** to **39** are of 3 source-based/case-based units of assessment carrying **4** marks each with sub-parts.*

(viii) *There is no overall choice. However, an internal choice has been provided in some sections. Only one of the alternatives has to be attempted in such questions.*

SECTION A

*Select and write the most appropriate option out of the four options given for each of the questions no. **1** to **20**.* *20×1=20*

1. To balance the following chemical equation, the values of the coefficients x, y and z must be respectively :

$$x\ Zn(NO_3)_2 \xrightarrow{\Delta} y\ ZnO + z\ NO_2 + O_2$$

(A) 4, 2, 2 (B) 4, 4, 2

(C) 2, 2, 4 (D) 2, 4, 2

2. Which of the following is a redox reaction, but ***not*** a combination reaction ?

(A) $C + O_2 \rightarrow CO_2$ (B) $2\ H_2 + O_2 \rightarrow 2\ H_2O$

(C) $2\ Mg + O_2 \rightarrow 2\ MgO$ (D) $Fe_2O_3 + 3\ CO \rightarrow 2\ Fe + 3\ CO_2$

3. The salt present in tooth enamel is :

(A) Calcium phosphate (B) Magnesium phosphate

(C) Sodium phosphate (D) Aluminium phosphate

4. An aqueous solution of sodium chloride is prepared in distilled water. The pH of this solution is :

(A) 6 (B) 8

(C) 7 (D) 3

5. A metal 'X' is used in thermit process. When 'X' is heated with oxygen, it gives an oxide 'Y', which is amphoteric in nature. 'X' and 'Y' respectively are :

(A) Mn, MnO_2 (B) Al, Al_2O_3

(C) Fe, Fe_2O_3 (D) Mg, MgO

6. The process in which transport of soluble products of photosynthesis takes place in plants is known as :

(A) Transpiration (B) Evaporation

(C) Conduction (D) Translocation

7. The correct sequence of events when someone's hand touches a hot object unconsciously :

(A) Receptors in skin → Motor neuron → Relay neuron → Sensory neuron → Effector muscle in arm

(B) Receptors in skin → Relay neuron → Sensory neuron → Motor neuron → Effector muscle in arm

(C) Receptors in skin → Sensory neuron → Relay neuron → Motor neuron → Effector muscle in arm

(D) Receptors in skin → Sensory neuron → Effector muscle in arm → Motor neuron → Relay neuron

8. Sense organ in which olfactory receptors are present is :

(A) Nose

(B) Skin

(C) Tongue

(D) Inner ear

9. The ***incorrect*** statement about placenta is :

(A) It is a disc embedded in the uterine wall.

(B) It contains villi on the embryo's side of the tissue.

(C) It has a very small surface area for glucose and oxygen to pass from mother to the embryo.

(D) The embryo gets nutrition from the mother's blood through it.

10. Select from the following the conditions responsible for the rapid spread of bread mould on a slice of bread :

(i) Formation of large number of spores

(ii) Presence of moisture and nutrients in bread

(iii) Low temperature

(iv) Presence of hyphae

(A) (i) and (ii)

(B) (ii) and (iv)

(C) (ii) and (iii)

(D) (iii) and (iv)

11. How will the image formed by a convex lens be affected, if the upper half of the lens is wrapped with a black paper ?

(A) The size of the image formed will be one-half of the size of the image due to complete lens.

(B) The image of upper half of the object will not be formed.

(C) The brightness of the image will reduce.

(D) The lower half of the inverted image will not be formed.

12. The phenomena of light involved in the formation of rainbow are :

(A) Refraction, reflection and dispersion

(B) Refraction, dispersion and internal reflection

(C) Reflection, dispersion and internal reflection

(D) Refraction, dispersion, scattering and total internal reflection

13. The colour of light for which the refractive index of glass is minimum, is :

(A) Red (B) Yellow

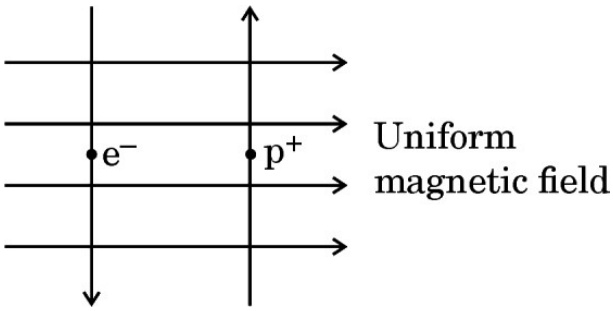
(C) Green (D) Violet

14. The current carrying device which produces a magnetic field similar to that of a bar magnet is :

(A) A straight conductor (B) A circular loop

(C) A solenoid (D) A circular coil

15.



A uniform magnetic field exists in the plane of paper as shown in the diagram. In this field, an electron (e^-) and a positron (p^+) enter as shown. The electron and positron experience forces :

(A) both pointing into the plane of the paper.

(B) both pointing out of the plane of the paper.

(C) pointing into the plane of the paper and out of the plane of the paper respectively.

(D) pointing out of the plane of the paper and into the plane of the paper respectively.

16. Which one of the following is ***not*** a natural ecosystem ?

(A) Pond ecosystem (B) Grassland ecosystem

(C) Forest ecosystem (D) Cropland ecosystem

For Questions number 17 to 20, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (A), (B), (C) and (D) as given below.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

17. *Assertion (A) :* Some vegetable oils are healthy.

Reason (R) : Vegetable oils generally have long unsaturated carbon chains.

18. *Assertion (A) :* Sex of the children will be determined by what they inherit from their mother.

Reason (R) : Women have XX sex chromosomes.

19. *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

20. *Assertion (A) :* Ozone layer protects the surface of the Earth from harmful UV radiations.

Reason (R) : Chlorofluorocarbons (CFCs) are responsible for depletion of ozone layer.

SECTION B

Questions no. ***21*** *to* ***26*** *are very short answer type questions.*

21. (a) Copper powder is taken in a china dish and heated over a burner. Name the product formed and state its colour. Write the chemical equation for the reaction involved. *2*

OR

(b) Write chemical equation for the chemical reaction which occurs when the aqueous solutions of barium chloride and sodium sulphate react together. Write the symbols of the ions present in the compound precipitated in the reaction. *2*

22. The melting and boiling points of carbon compounds are generally low and they are largely non-conductors of electricity. State two conclusions based on these two properties. *2*

23. (a) Sometimes while running, the athletes suffer from muscle cramps. Why ? How is the respiration in this case different from aerobic respiration ? *2*

OR

(b) Write the other name given to lymph. State its two functions. *2*

24. Some unicellular organisms such as Plasmodium and Leishmania differ in the manner in which they reproduce. Name and explain the reproductive process taking place in them. *2*

25. The heat produced at a point due to concentration of sunlight by a convex lens burns a paper.

(a) Explain why it happens.

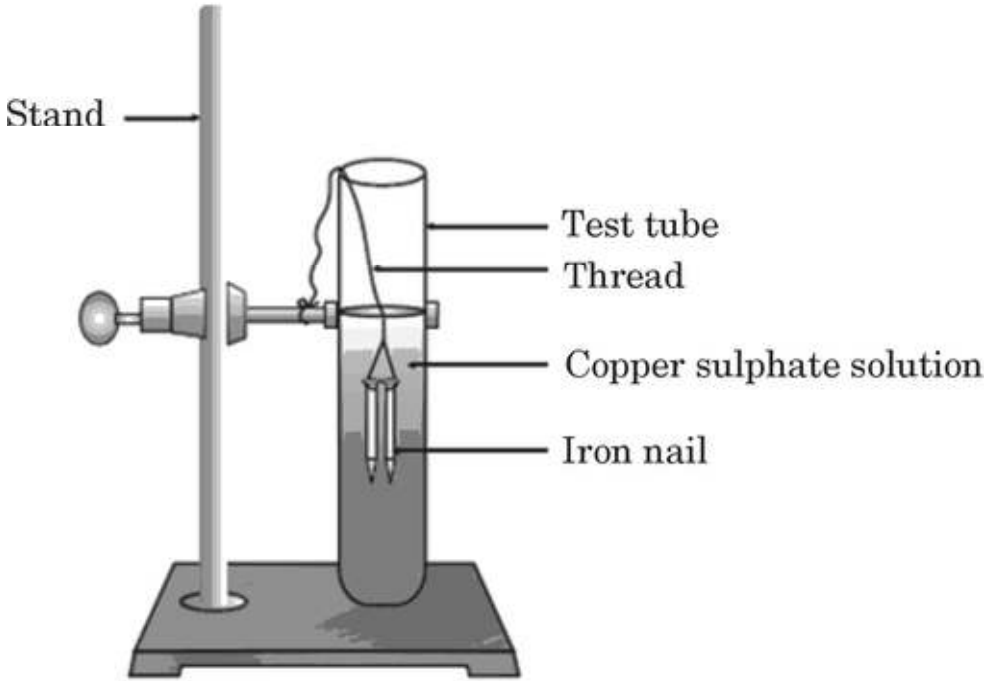
(b) Name the term (in the context of the lens used) given to the point at which the paper starts burning. What does the bright spot formed on the paper represent ? *2*

26. An electric source can supply a charge of 500 coulomb. If the current drawn by a device is 25 mA, find the time in which the electric source will be discharged completely. *2*

SECTION C

Questions no. ***27*** *to* ***33*** *are short answer type questions.*

27.



Study the experimental set-up shown in the diagram and write chemical equation for the chemical reaction involved. Name and define the type of reaction. List two other metals which can be used in place of iron to show the same type of reaction with copper sulphate solution. *3*

28. Name the ore of mercury and state the form in which it is found in nature. Write the chemical equations along with the condition required for the reactions involved in the extraction of mercury from its ore. *3*

29. Taking the example of any two animal hormones along with their gland of secretion, explain how these hormones help (i) in growth and development and (ii) regulate metabolism, in the body. *3*

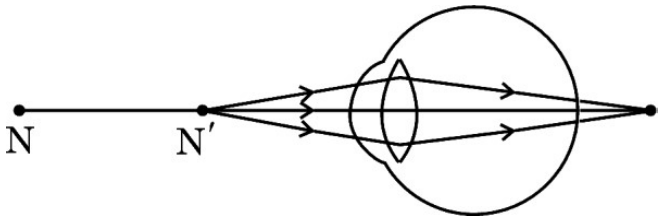
30. Mendel crossed pure tall pea plants (TT) with pure short pea plants (tt) and obtained F_1 progeny. When the plants of F_1 progeny were self-pollinated, plants of F_2 progeny were obtained.

(a) What did the plants of F_1 progeny look like ? Give their gene combination.

(b) Why could the gene for shortness not be expressed in plants of F_1 progeny ?

(c) Write the ratio of the plants obtained in F_2 progeny and state the conclusion that can be drawn from this experiment. *3*

31. (a) Study the diagram given below and answer the questions that follow :



(i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.

(ii) List two causes of this defect.

(iii) Name the type of lens used to correct this defect and state its role in this case. *3*

OR

(b) What is dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism. *3*

32. (a) What happens when a bundle of wires of soft iron is placed inside the coil of a solenoid carrying a steady current ? Name the device obtained. Why is it called so ?

(b) Draw the magnetic field lines inside a current carrying solenoid. What does this pattern of magnetic field lines indicate ? *3*

33. Differentiate between food chain and food web. In a food chain consisting of deer, grass and tiger, if the population of deer decreases, what will happen to the population of organisms belonging to the first and third trophic levels ? *3*

SECTION D

Questions no. ***34*** *to* ***36*** *are long answer type questions.*

34. (a) A few crystals of ferrous sulphate were taken in a dry boiling tube and heated. Tiny water droplets were observed in the tube after some time.

(i) From where did these water droplets appear ? Explain.

(ii) What colour change will be observed during heating ?

(iii) How many molecules of water are attached per molecule of $FeSO_4$ crystal ? Write the molecular formula of crystalline forms of (I) Copper sulphate, and (II) Sodium carbonate.

(iv) State how is Plaster of Paris obtained from gypsum. Write two uses of Plaster of Paris. *5*

OR

(b) An acid 'X' present in tamarind when mixed with 'Y', produces a mixture 'Z'. 'Z' on addition to a dough when heated makes cakes soft and spongy. 'Y' is prepared from common salt and helps in faster cooking.

(i) Write the common names of 'X', 'Y' and 'Z', and the chemical formula of 'Y'.

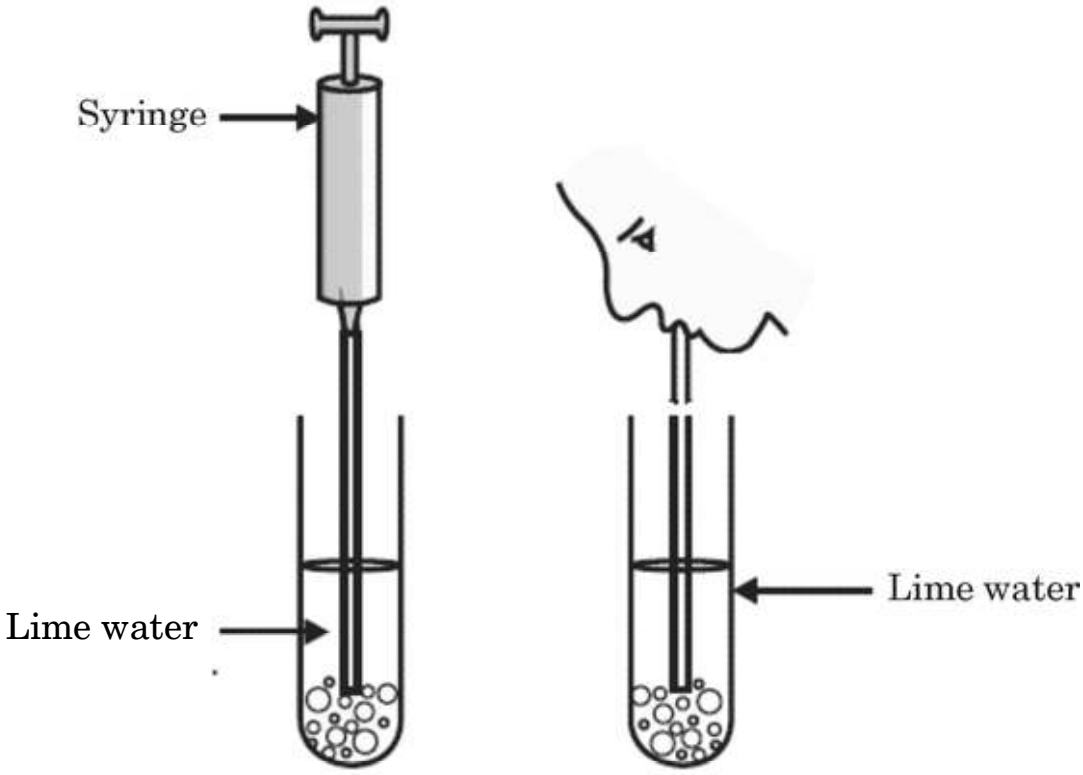
(ii) How is 'Y' prepared and how does it help in making cakes soft and spongy ? Illustrate the reaction with suitable chemical equation.

(iii) Write the name and chemical formula of a mild base other than 'Y' used as an antacid. *5*

35. (a) Design an experiment to demonstrate that carbon dioxide is essential for photosynthesis. Write the observation and conclusion of the experiment. *5*

OR

(b) (i)



In the experimental set-up shown above in diagram (I) atmospheric air is being passed into lime water with a syringe while in diagram (II) air is being exhaled into lime water. The time taken for the lime water to turn milky in both the test tubes is different. Give reason.

(ii) Draw the diagram of an open stomatal pore and label (I) Guard cells, and (II) Chloroplast on it. Mention two functions performed by stomata. *5*

36. (a) (i) The potential difference across the two ends of a circuit component is decreased to one-third of its initial value, while its resistance remains constant. What change will be observed in the current flowing through it ? Name and state the law which helps us to answer this question.

(ii) Draw a schematic diagram of a circuit consisting of a battery of four 1·5 V cells, a 5 Ω resistor, a 10 Ω resistor and a 15 Ω resistor and a plug key, all connected in series. Now find (I) the electric current passing through the circuit, and (II) potential difference across the 10 Ω resistor when the plug key is closed. *5*

OR

(b) (i) When is the potential difference between two points said to be 1 volt ?

(ii) A copper wire has a diameter of 0·2 mm and resistivity of $1{\cdot}6 \times 10^{-8}$ Ω m. What will be the length of this wire to make its resistance 14 Ω ? How much does the resistance change, if the diameter of the wire is doubled ? *5*

SECTION E

Questions no. ***37*** *to* ***39*** *are case-based/data-based questions with 3 short sub-parts. Internal choice is provided in one of these sub-parts.*

37. Carbon is a versatile element that forms the basis of all living organisms and many of the things we use. A large variety of compounds is formed because of its tetravalency. Compounds of carbon are formed with oxygen, hydrogen, nitrogen, sulphur, chlorine and many other elements.

Answer the following questions :

(a) What are hydrocarbons ? *1*

(b) List two properties by virtue of which carbon can form a large number of compounds. *1*

(c) (i) Write the formula of the functional group present in (1) aldehydes, and (2) ketones. Write chemical equation for the reaction that occurs between ethanoic acid and ethanol in the presence of a catalyst. *2*

OR

(c) (ii) What are structural isomers ? Write the structures of two isomers of butane (C_4H_{10}). *2*

38. Pollination is an important process in sexual reproduction of plants. It is an essential process that facilitates fertilisation in plants. Pollinating agents can be wind, water, insects and birds. Several changes take place in the flower after the fertilization has taken place.

(a) Write the main difference between self-pollination and cross-pollination. *1*

(b) Name the part of the flower which attracts insects for pollination. What happens to this part after fertilisation ? *1*

(c) (i) Define fertilisation. What is the fate of ovules and the ovary in a flower after fertilisation ? *2*

OR

(c) (ii) In a germinating seed, which parts are known as future shoot and future root ? Mention the function of cotyledon. *2*

39. A highly polished surface such as a mirror reflects most of the light falling on it. In our daily life we use two types of mirrors — plane and spherical. The reflecting surface of a spherical mirrors may be curved inwards or outwards. In concave mirrors, reflection takes place from the inner surface, while in convex mirrors reflection takes place from the outer surface.

(a) Define the principal axis of a concave mirror. *1*

(b) A ray of light is incident on a concave mirror, parallel to its principal axis. If this ray after reflection from the mirror passes through the principal axis from a point at a distance of 10 cm from the pole of the mirror, find the radius of curvature of the mirror. *1*

(c) (i) An object is placed at a distance of 10 cm from the pole of a convex mirror of focal length 15 cm. Find the position of the image. *2*

OR

(c) (ii) A mirror forms a virtual, erect and diminished image of an object. Identify the type of this mirror. Draw a ray diagram to show the image formation in this case. *2*

MARKING SCHEME

Secondary School Examination, 2024

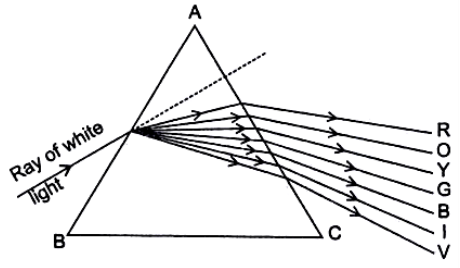
SCIENCE (Subject Code–086)

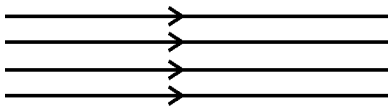
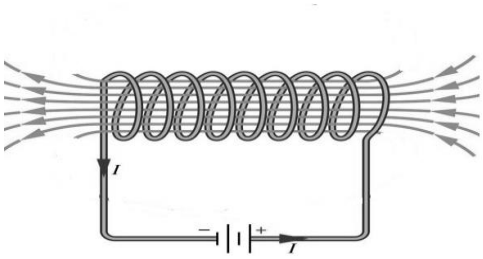
[Paper Code: 31/5/1]

Maximum Marks: 80

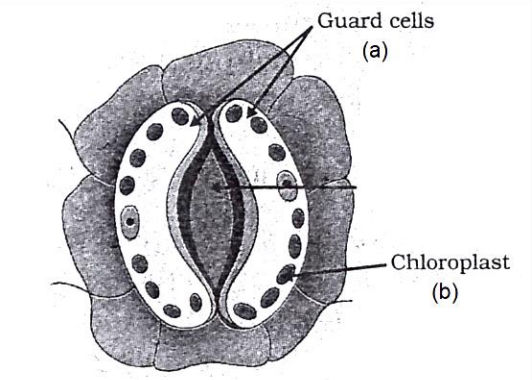
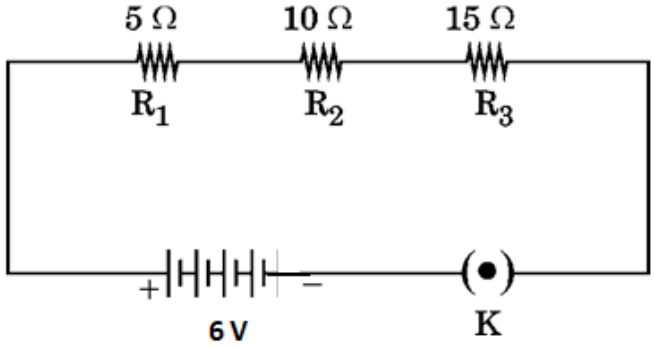
Q. No.	EXPECTED ANSWER / VALUE POINTS	Marks	Total Marks
	SECTION A		
1	(C) /2,2,4	1	1
2	(D) / Fe_2O_3 + 3 CO→ 2 Fe + 3 CO_2	1	1
3	(A) /Calcium Phosphate	1	1
4	(C)/ 7	1	1
5	(B) / Al, Al_2O_3	1	1
6	(D) / Translocation	1	1
7	(C)/ Receptors in skin →Sensory neuron →Relay neuron → Motor neuron→ Effector muscle in arm.	1	1
8	(A) / Nose	1	1
9	(C)/ It has a very small area for glucose and oxygen to pass from mother to the embryo	1	1
10	(A) / (i) and (ii)	1	1
11	(C) / The brightness of the image will reduce	1	1
12	(B) / Refraction, Dispersion and internal reflection	1	1
13	(A) / Red	1	1
14	(C) / A solenoid	1	1
15	(A) / both pointing into the plane of the paper.	1	1
16	(D)/ Crop land ecosystem	1	1
17	(A) / Both Assertion (A) and Reason (R) are the true and Reason (R) is a correct explanation of Assertion (A).	1	1
18	(D) / Assertion (A) is false, but Reason (R) is true.	1	1
19	(B) / Both Assertion (A) and Reason (R) are the true , but Reason (R) is not a correct explanation of Assertion (A).	1	1
20	(B) / Both Assertion (A) and Reason (R) are the true , but Reason (R) is not a correct explanation of Assertion (A).	1	1
	SECTION B		
21	(a) • Copper Oxide • Black $2Cu + O_2 \xrightarrow{Heat} 2CuO$ **OR** (b) $BaCl_2$ (aq)+ Na_2SO_4 (aq)→ $BaSO_4$ (s)+ 2NaCl (aq) Ba^{2+}, SO_4^{2-}	½ ½ 1 1 ½+½	2

22	• Low melting points and boiling points – Weak intermolecular forces of attraction. • Non-conductors of electricity – Bonding in these compounds does not give rise to any ions. / Covalent bonds or sharing of electrons do not form any charged particles.	1 1	2
23	(a) • Formation of lactic acid in muscles causes cramps. • Aerobic respiration takes place in the presence of oxygen whereas the respiration taking place above is due to lack of oxygen. / End products of aerobic respiration are $CO_2 + H_2O +$ Energy whereas in the above case, Lactic acid + Energy is formed. **OR** (b) • Tissue fluid / Extracellular fluid Functions : i. Carries digested and absorbed fats from the intestine. ii. Drains excess fluid from extracellular space back into the blood. iii. Fight against infections. **(Any two)**	1 1 1 ½ + ½	2
24	• Plasmodium: Multiple fission- A single cell divides into many daughter cells simultaneously. • Leishmania: Binary fission- Splitting of one cell into two daughter cells in definite orientation.	½ +½ ½+½	2
25	(a) The sun light is converged at a point by convex lens which generates heat causing the paper to burn. (b) •Principal Focus •Real image of the Sun.	1 ½ ½	2
26	$Q = I \times t$ $\therefore t = \frac{500\ C}{25 / 1000\ A}$ $= 20000\ s$	½ ½ 1	2
	SECTION C		
27	• $Fe(s) + CuSO_4\ (aq) \rightarrow FeSO_4(aq) + Cu(s)$ • Displacement reaction – A reaction in which a more reactive metal displaces a less reactive metal from its salt solution. • Zinc, Aluminium, Calcium, Magnesium **(Any two)**	1 ½ +½ ½ ,+½	3

28	• Cinnabar • Sulphide ore • $2HgS + 3O_2 \xrightarrow{Heat} 2HgO + 2SO_2$ • $2HgO \xrightarrow{Heat} 2Hg + O_2$	½ ½ 1 1	3
29	(i) •Growth hormone •Secreted by pituitary gland. •It stimulates growth in all organs. (ii) •Thyroxin •Secreted by thyroid gland. •It regulates carbohydrate, protein and fat metabolism for body growth.	½ x 3 ½ x 3	3
30	(a) •All Plants Tall •Gene combination: Tt (b) It is a recessive trait / it cannot be expressed in presence of dominant trait. (c) Tall : Short 3:1 Conclusion: Tall trait is dominant and short trait is recessive.	½ ½ 1 ½ ½	3
31	(a) (i) • Hypermetropia • Ciliary muscles/ eye lens (ii) • Focal length of the eye lens is too long. • Eyeball becomes too small. (iii) Converging lenses/ convex lens They provide the additional focussing power required for forming the image on the retina./ Helps to decrease the focal length of the eye lens. **OR** (b) The splitting of white light into its constituent colours is called dispersion. Cause: Different colours of white light bend through different angles with respect to incident ray.  	½ ½ ½ ½ ½ ½ 1 1 1	3

32	(a) • It gets magnetised • Electromagnet. • It behaves as a magnet only when current passes through the solenoid. (b)   **(Any one diagram)** • This pattern indicates that the magnetic field is uniform.	½ ½ ½ 1 ½	3
33	<table><tr><th>Food chain</th><th>Food web</th></tr><tr><td>It is a series of organisms feeding on one another at various levels</td><td>It is a network of interconnected food chains / series of branching lines which provides a number of feeding connections amongst different organisms.</td></tr></table> • Population of grass/ first trophic level will increase. • Population of tiger/ third trophic level will decrease.	1+1 ½ ½	3
	SECTION D		
34	(a) (i) The molecules of water of crystallisation in ferrous sulphate crystals get evaporated on heating. (ii) Green ⟶ White (iii) Seven / ($FeSO_4 \cdot 7H_2O$) (I) $CuSO_4 \cdot 5H_2O$ (II) $Na_2CO_3 \cdot 10H_2O$ (iv) • On heating gypsum ($CaSO_4 \cdot 2H_2O$) at 373 K it loses water molecules/ $CaSO_4 \cdot 2\,H_2O \xrightarrow[373\,K]{\Delta} CaSO_4 \cdot \frac{1}{2}\,H_2O + 1\frac{1}{2}\,H_2O$ Two uses of plaster of Paris: • Making toys / material for decoration • Supporting fractured bones **(or any other)**	1 ½ ½ ½ ½ 1 ½ + ½	

	OR (b) (i) X-Tartaric acid Y-Baking soda Z- Baking powder Y- $NaHCO_3$ (ii) $NaCl + H_2O + CO_2 + NH_3 \longrightarrow NH_4Cl + NaHCO_3$ $NaHCO_3 + H^+ \longrightarrow CO_2 + H_2O$ + Sodium salt of acid CO_2 released during heating makes the cake soft and spongy (iii) Magnesium hydroxide; $Mg(OH)_2$	½ ½ ½ ½ 1 ½ ½ 1	5
35	(a) • Take two healthy potted plants, A and B of nearly the same size. • Keep them in darkness for three days. (Destarch the plant) • Place a watch glass containing potassium hydroxide by the side of potted plant A but not in potted plant B. • Cover both the plants with separate bell jars and seal the bottom of the jars with Vaseline. • Keep both the plants in sunlight for two hours. • Pluck one leaf each from both the plants and test for the presence of starch with iodine solution. • <u>Observation</u>: The leaf of the potted plant A with KOH did not turn blue – black. The leaf of the potted plant B turns blue. • <u>Conclusion</u>: KOH absorbs CO_2 so photosynthesis did not occur in potted plant A. **OR** (b) (i) In set up (I) lime water turns milky in more time as compared to set up (II) because the air we exhaled contains high percentage of CO_2 as compared to atmospheric air.	½ x 6 1 1 1,1	

	(ii)  Open Stomatal Pore Two labellings : (I) Guard Cells (II) Chloroplast Two functions performed by stomata : • Gaseous exchange • Transpiration	1 ½ ½ ½ ½	5
36	(a) (i) • Current becomes one-third of its initial value. • Ohm's Law The potential difference across the ends of a conductor is directly proportional to the current flowing through it, provided its temperature remains the same. (ii)  Total Voltage = V = 4 × 1·5 V = 6 V Total resistance, R(s) = $R_1 + R_2 + R_3$ = 5 Ω + 10 Ω +15 Ω = 30 Ω (I) Current, $I = \frac{V}{R} = \frac{6\ V}{30\ \Omega} = 0{\cdot}2\ A$ (II) V = IR = 0·2 A × 10 Ω = 2 V	½ ½ 1 1 1 1	

OR

(b)

(i) When 1 joule of work is done to move a charge of 1 coulomb from one point to the other. — 1

(ii) d = 0·2 mm = 2×10^{-4} m; R = 14 Ω — ½

$\rho = 1{\cdot}6 \times 10^{-8}$ Ω m; A = $\pi d^2/4$

$R = \frac{\rho l}{A} = \frac{4\rho l}{\pi d^2}$ or $l = \frac{\pi d^2 R}{4\rho}$ — ½

$l = \frac{22}{7} \times \frac{(2 \times 10^{-4})^2}{4 \times 1{\cdot}6 \times 10^{-8}} \times 14$

= 27·5 m — 1

When the diameter is doubled, d' = 2d

A' = 4A — ½

$\frac{R'}{R} = \frac{A}{A'}$ or $R' = \frac{RA}{A'} = \frac{RA}{4A}$

$\frac{R^{l}}{14} = \frac{A}{4A}$

R' = 3·5 Ω — 1

Change (14·0 – 3·5) = 10·5 Ω — ½

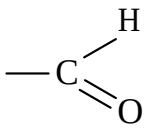
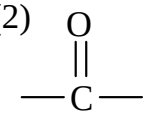
5

SECTION E

37

(a) Compounds formed by carbon and hydrogen only. — 1

(b) Tetravalency and Catenation — 1

(c) (i) (1)  (2)  — ½ + ½

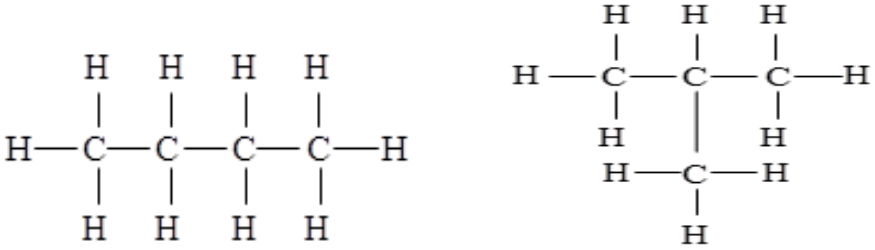
$CH_3COOH + C_2H_5OH \xrightarrow{Acid} CH_3COOC_2H_5 + H_2O$

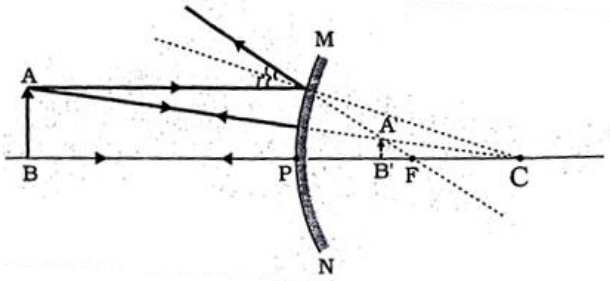
Ester — 1

OR

(c)

(ii) Compounds with identical molecular formula but different structures — 1

	Two isomers of butane C_4H_{10} 	½ + ½	4
38	(a) **Self-pollination**: Transfer of pollen grains from anther to the stigma of the same flower. **Cross-pollination**: Transfer of pollen grains from the anther of one flower to the stigma of another flower.	1	
	(b) Petals, they dry and fall off.	½ + ½	
	(c)(i) Fusion of male and female gametes to form a zygote Ovule – Seed, Ovary – fruit	1 ½ ½	
	OR		
	(c) (ii) Future shoot – Plumule, Future root – Radicle Cotyledon – Stores food.	½ ½ 1	4
39	(a) It is straight line passing through the pole and centre of curvature of a concave mirror.	1	
	(b) Radius of curvature ,R= 20 cm	1	
	(c) (i) u = –10 cm, f = +15 cm $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$ $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{15} - \frac{1}{-10}$ $\frac{1}{v} = \frac{1}{6}$ $\Rightarrow$ v = + 6 cm	½ ½ 1	
	OR		
	(c) (ii) Convex mirror / Diverging mirror	½	

	 **[Note: Deduct ½ mark if direction of rays is not shown]**	1 ½	4

*****

Lessons from the three board papers

1. Draw every ray diagram with arrows: this scheme deducts ½ mark when the direction of rays is not shown.

2. Write the formula first, then the substitution, then the answer with its unit; each step is marked.

3. Read both options of an internal choice before you start, and attempt only one.

4. Balance every chemical equation and add conditions (heat, catalyst) on the arrow.

5. In biology, label diagrams with straight lines and correct spellings.

6. Case-based questions come from NCERT activities and examples: revise them along with the theory.

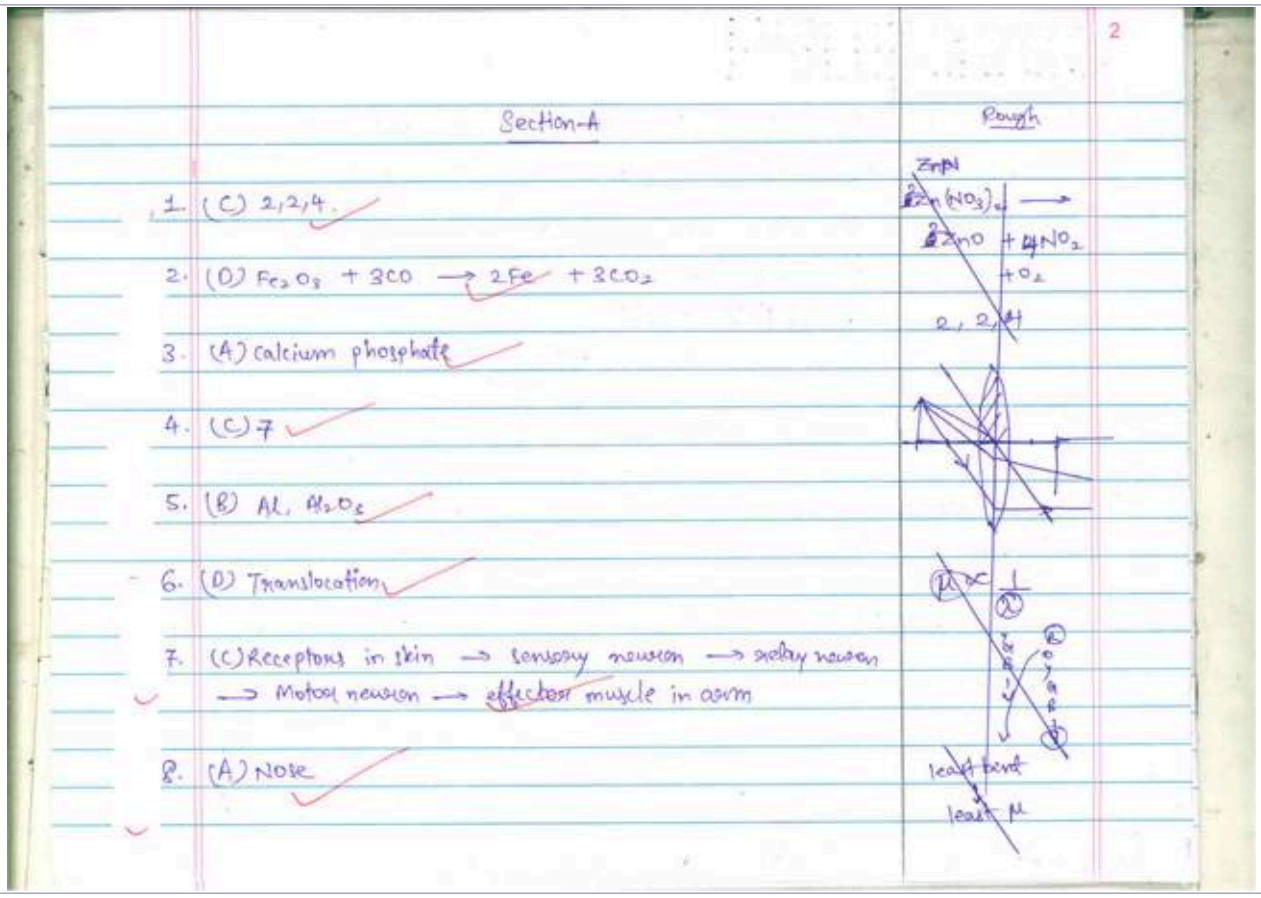
7. Practise these papers in 3 hours, then check against the marking scheme to see where marks are cut.

Nitesh Bhaiya ki tip: marking scheme ko answer key mat samjho, ye batata hai examiner kis point pe number deta hai.

Topper's Answer Sheet

CBSE Board Exam 2024, Set 31/5/1: answer book of a CBSE topper, as published by CBSE. Pages in question order.

Answer book page 1: Q.1 to Q.8



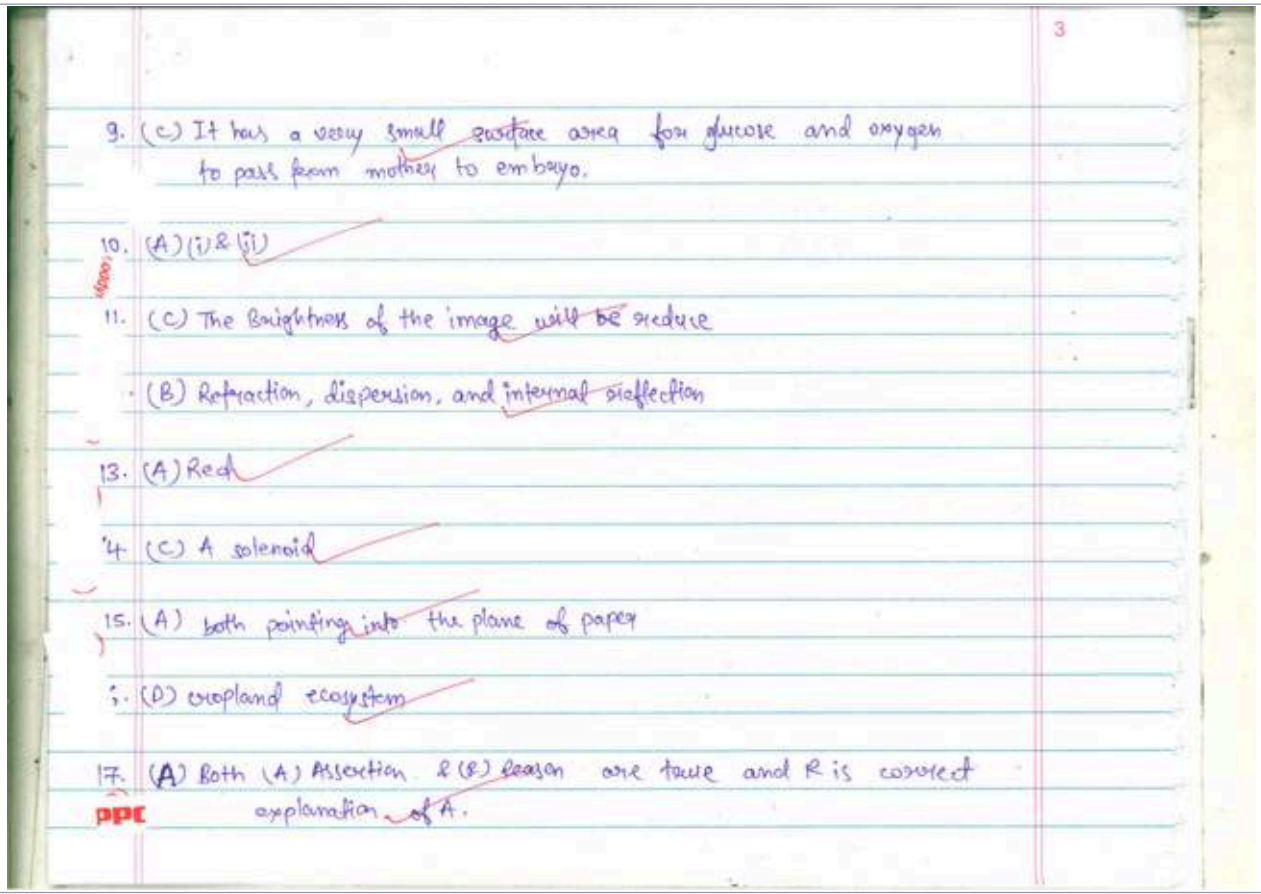
2

Section-A

1. (C) 2,2,4.
2. (D) $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
3. (A) calcium phosphate
4. (C) 7
5. (B) Al, Al_2O_3
6. (D) Translocation
7. (C) Receptors in skin → sensory neuron → relay neuron → Motor neuron → effector muscle in arm
8. (A) Nose

Rough

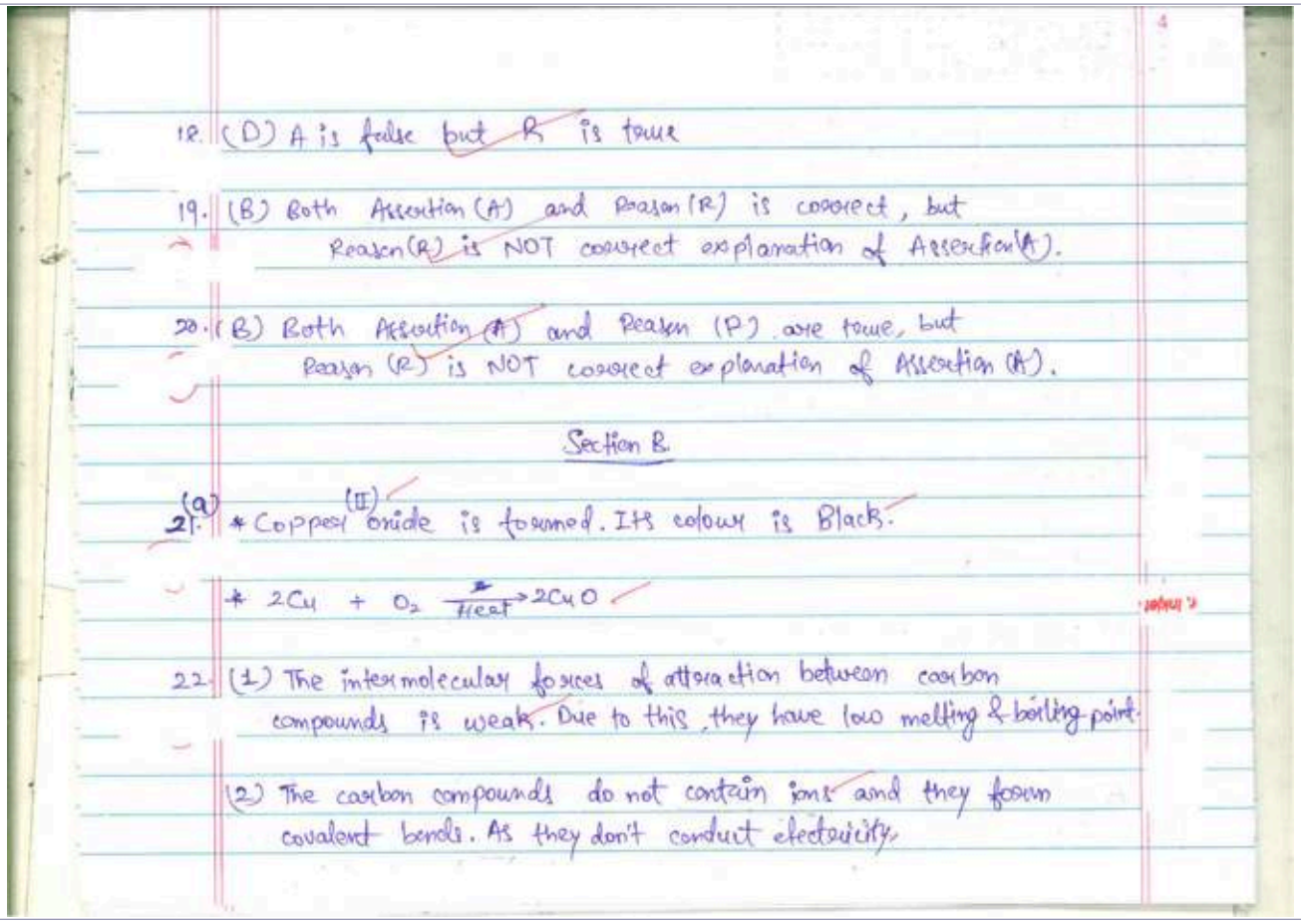
Answer book page 2: Q.9 to Q.17



3

9. (c) It has a very small surface area for glucose and oxygen to pass from mother to embryo.
10. (A) (i) & (ii)
11. (C) The brightness of the image will be reduce
12. (B) Refraction, dispersion, and internal reflection
13. (A) Red
14. (C) A solenoid
15. (A) both pointing into the plane of paper
16. (D) cropland ecosystem
17. (A) Both (A) Assertion & (R) Reason are true and R is correct explanation of A.

Answer book page 3: Q.18 to Q.22



18. (D) A is false but R is true

19. (B) Both Assertion (A) and Reason (R) is correct, but Reason (R) is NOT correct explanation of Assertion (A).

20. (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is NOT correct explanation of Assertion (A).

Section B

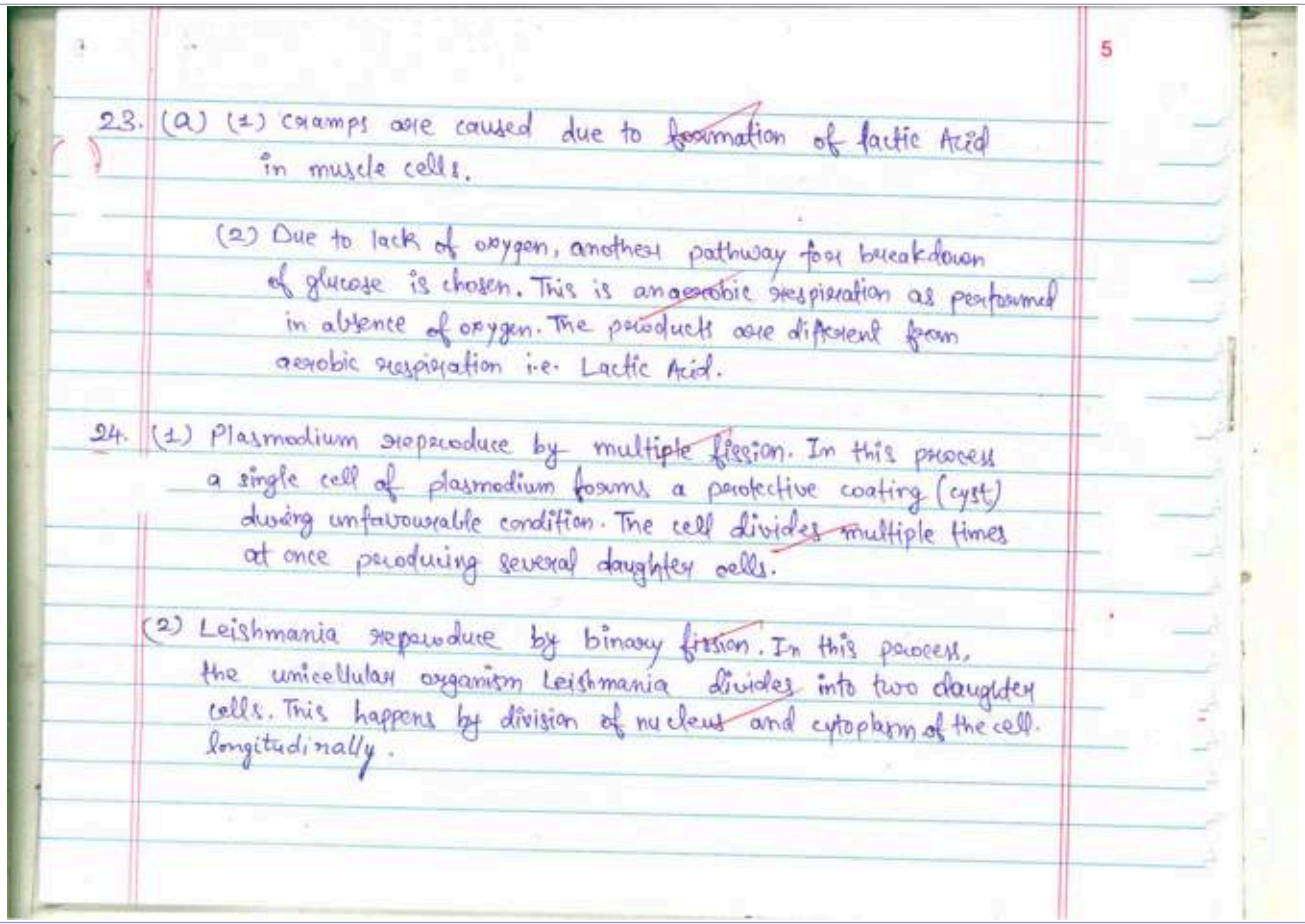
21. (a) * Copper (II) oxide is formed. Its colour is Black.

* $2Cu + O_2 \xrightarrow{Heat} 2CuO$

22. (1) The intermolecular forces of attraction between carbon compounds is weak. Due to this, they have low melting & boiling point.

(2) The carbon compounds do not contain ions and they form covalent bonds. As they don't conduct electricity.

Answer book page 4: Q.23 and Q.24



23. (a) (1) Cramps are caused due to formation of lactic Acid in muscle cells.

(2) Due to lack of oxygen, another pathway for breakdown of glucose is chosen. This is anaerobic respiration as performed in absence of oxygen. The products are different from aerobic respiration i.e. Lactic Acid.

24. (1) Plasmodium reproduce by multiple fission. In this process a single cell of plasmodium forms a protective coating (cyst) during unfavourable condition. The cell divides multiple times at once producing several daughter cells.

(2) Leishmania reproduce by binary fission. In this process, the unicellular organism Leishmania divides into two daughter cells. This happens by division of nucleus and cytoplasm of the cell. longitudinally.

Answer book page 5: Q.25 and Q.26

25. (a) The convex lens converges the sunlight by refracting (bending) the rays of sunlight & bringing it to one point.
Thus, the rays come together to produce heat & burn paper.

(b) Principal focus. Bright spot represents the image of sun.

26. $Q = 500C$
$I = 25mA = \frac{25}{1000} A$
$t = ?$

$I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} \Rightarrow t = \frac{500C}{\frac{25}{1000}A}$

$= \frac{500 \times 1000}{25} \quad \frac{20}{500 \times 1000}{25}$

$= 20000\,s$ or $333 m 20 s$

Answer book page 6: Q.27 to Q.29

27. (1) $Fe(s) + CuSO_4(aq) \longrightarrow FeSO_4(aq) + Cu(s)$
(2) Displacement reaction. In this reaction, a more reactive element displaces a less reactive element from its compound (solution)

(3) Zinc (Zn) & Aluminium (Al).

28. (1) Cinnabar. Mercury is found in its sulphide ore in nature.

(2) $2HgS + 3O_2 \xrightarrow{\Delta} 2HgO + 2SO_2$
$2HgO \xrightarrow{\Delta} 2Hg + O_2$
The condition required is heat. In first reaction presence of oxygen is also needed to oxidise mercury.

29. (i) Growth Hormone is secreted by pituitary gland. This helps in growth and development of bones and muscles in our body. Hence, it promotes the growth of height.

(ii) Thyroxine Hormone is secreted by Thyroid gland. It regulates the metabolism of carbohydrates, protein and fats in the body.

Answer book page 7: Q.30

30. (a) F1 progeny plants were all tall. Their gene combination is (Tt)

(b) Gene for shortness was suppressed in plants of F1 progeny because they were recessive gene. Recessive gene show appearance of a trait only in presence of another identical gene. In this case, a dominant gene for tallness was present. Hence, gene for shortness could not be expressed.

(c) 3:1 for tall plants : short plants.
The conclusion of experiment is the traits are controlled by two copies of gene, each acquired from one parent. But only one copy shows its appearance in the progeny, other is suppressed. The Hence, there are two types of genes, one dominant and other recessive. However, the suppressed copy of gene may show in its appearance in next generation, if another identical copy is present.

Answer book page 8: Q.31

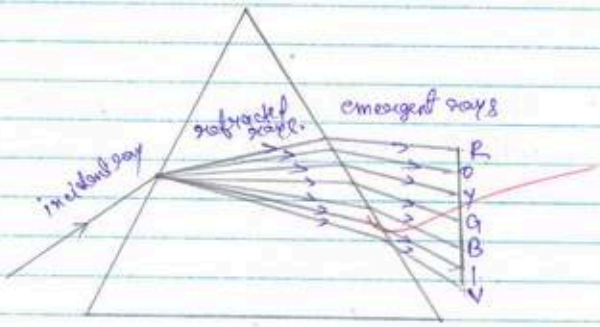
31. (b) (1) The splitting of white light into its component colours due to dependence of refractive index or wavelength of light, is called dispersion.

Hg_2S
HgS.

Hg_2O
HgO

(2) Dispersion of white light is caused by refraction. The white light is mixture of seven colours i.e VIBGYOR. Since refractive index of a medium is different for different wavelengths of light, the components of white light bends at different angles on entering another medium. As a result, they separate out and appear as distinct colours.

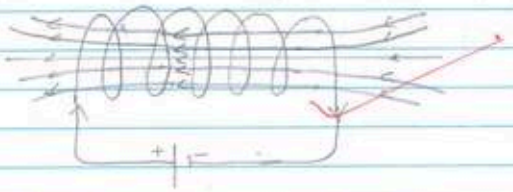
(3)



Answer book page 9: Q.32

32. (a) When bundle of soft iron is placed inside a coil of a solenoid carrying current, then the bundle of soft iron is magnetised due to magnetic field produced by the steady current. The device obtained is electromagnet. It is called so because the magnet is obtained by magnetic field due to an electric current.

(b)



The magnetic field lines are straight and parallel to each other indicating that there is a uniform magnetic field inside solenoid.

Answer book page 10: Q.33 and Q.34

33. (1) Food chain is a series of organisms that feeds on one another. It is generally depicted by a straight line. However, one organism is fed by several other kinds of organisms which in turn by are fed by many more organisms. So, A branched chain of food chain A food chain with several branches showing the flow of energy to different organisms from one is called food web.

(2) If population of deer decreases, then the population of organisms in 3rd trophic level (Tigers) will decrease as they'll have less food to eat and hence, they'll die. In contrast, the population of organisms in first trophic level / grass will increase dramatically as there will be less organisms to feed on them.

34. Section D

34. (a) (i) crystals of ferrous sulphate has 1 water molecules of crystallisation. These are present as an essential part of crystal in fixed numbers per unit formula of the compound. On heating

Answer book page 11: Q.34 (contd.)

12

ferrous sulphate crystals, these molecules of water comes out of the crystals and becomes free water which can be seen in the test tube.

$FeSO_4 \cdot 7H_2O \xrightarrow{\Delta} FeSO_4 + 7H_2O$,

(ii) Green colour of $FeSO_4$ crystal will fade and the crystal will become white in colour.

(iii) Seven molecules:
(I) $CuSO_4 \cdot 5H_2O$
(II) $Na_2CO_3 \cdot 10H_2O$

(iv) When gypsum is heated at 100°C (373K), then it loses 1½ molecule of water of crystallisation to become plaster of paris with only ½ molecule of water of crystallisation.

$CaSO_4 \cdot 2H_2O \xrightarrow{100°C} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$

Answer book page 12: Q.34 (contd.) and Q.35

13

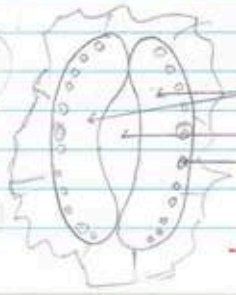
Uses of P.O.P:-
① for setting fractured bones in right position
② for making decoration items & toys.

35. (b) (i) Lime water turns milky due to formation of a calcium carbonate, while precipitate in it.

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

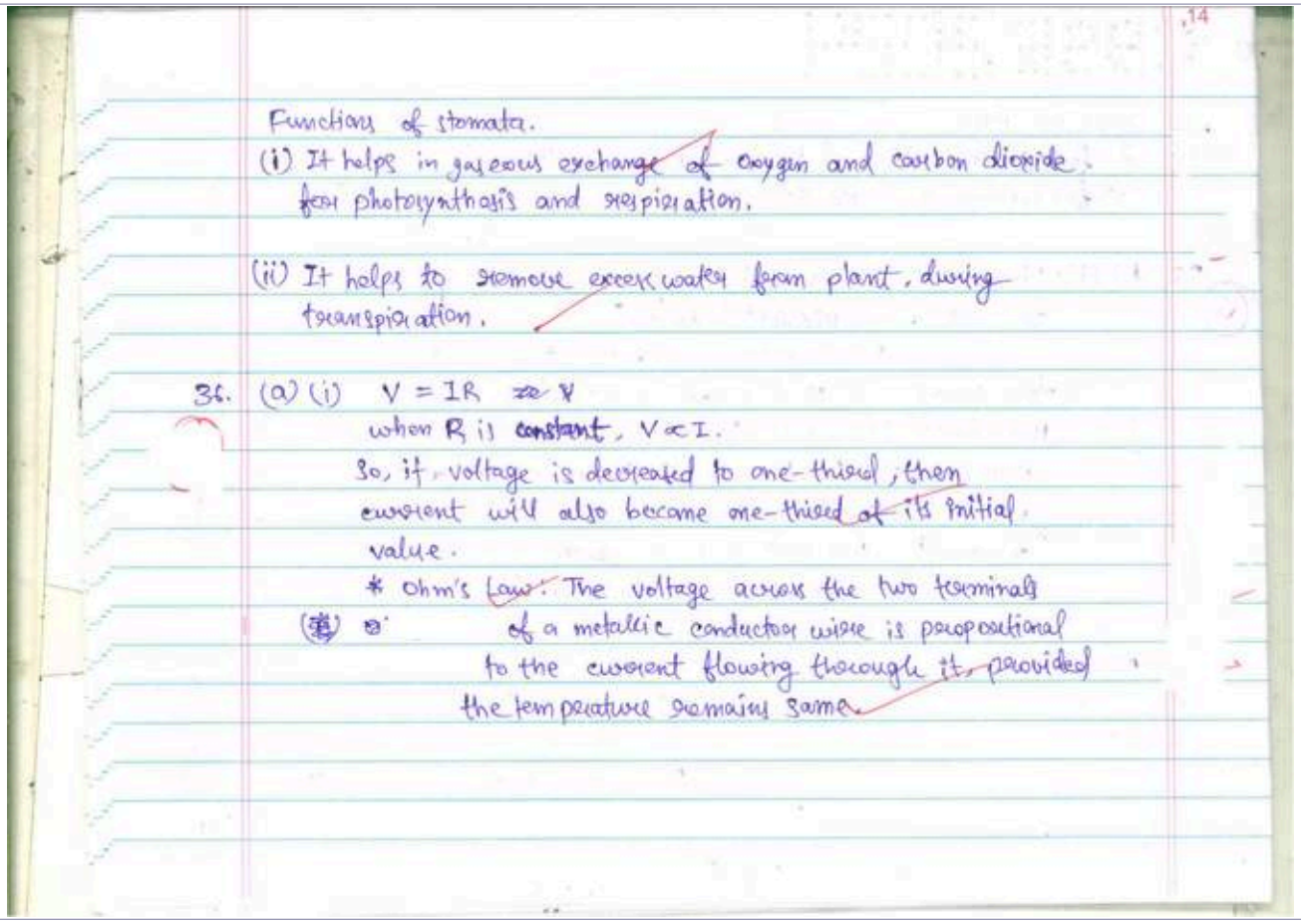
Now, in diagram (I), the atmospheric air is passed into the solution. Since, atmospheric air contains a very small percentage of CO_2, the solution turns milky slowly and takes a long time. While, air exhaled from mouth [diagram (II)] has a large percentage of CO_2 and hence it turns the lime water milky faster (comparatively).

(ii)



(I) guard cells
open stomatal pore.
(II) chloroplast

Answer book page 13: Q.35 (contd.) and Q.36



14

Functions of stomata.

(i) It helps in gaseous exchange of oxygen and carbon dioxide for photosynthesis and respiration.

(ii) It helps to remove excess water from plant, during transpiration.

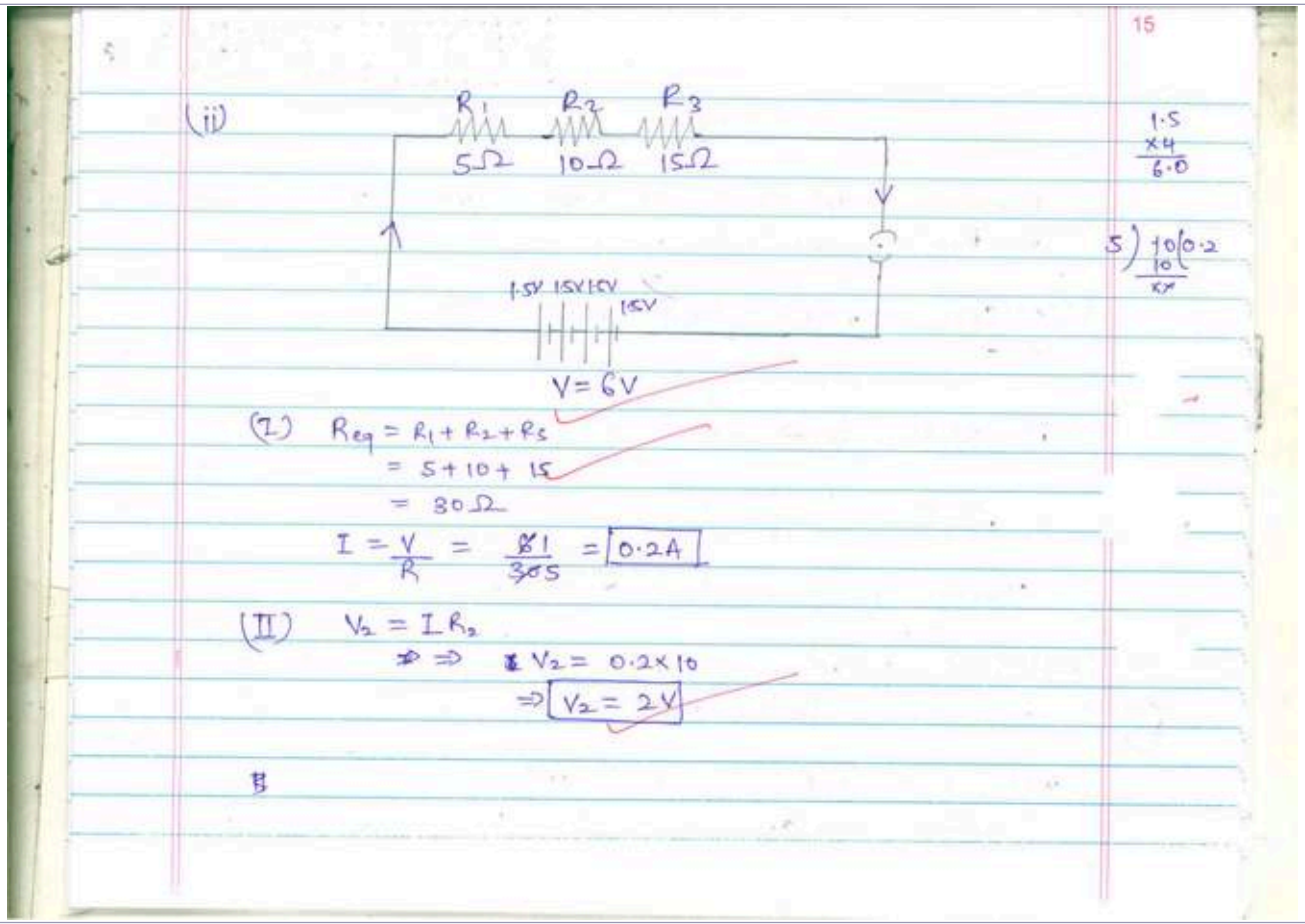
36. (a) (i) $V = IR$ ∝ V

when R is constant, $V \propto I$.

So, if voltage is decreased to one-third, then current will also became one-third of its initial value.

* Ohm's Law: The voltage across the two terminals of a metallic conductor wire is proportional to the current flowing through it, provided the temperature remains same.

Answer book page 14: Q.36 (contd.)



15

(ii)

$V = 6V$

(I) $R_{eq} = R_1 + R_2 + R_3$

$= 5 + 10 + 15$

$= 30\,\Omega$

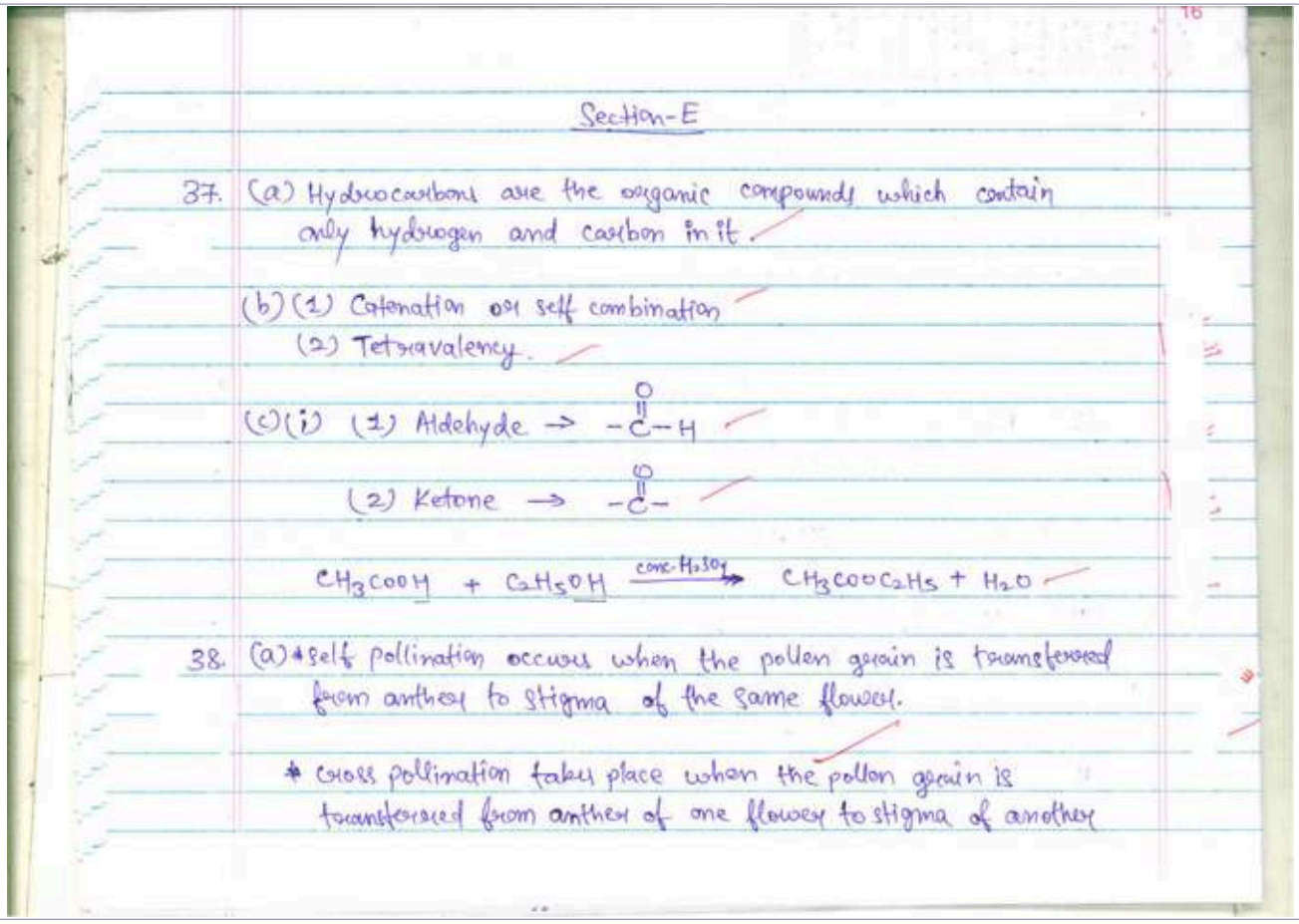
$I = \frac{V}{R} = \frac{6}{30} = 0.2A$

(II) $V_2 = IR_2$

$\Rightarrow V_2 = 0.2 \times 10$

$\Rightarrow V_2 = 2V$

Answer book page 15: Q.37 and Q.38



Section-E

37. (a) Hydrocarbons are the organic compounds which contain only hydrogen and carbon in it.

(b) (1) Catenation or self combination
(2) Tetravalency.

(c) (i) (1) Aldehyde → $-\overset{O}{\overset{\|}{C}}-H$

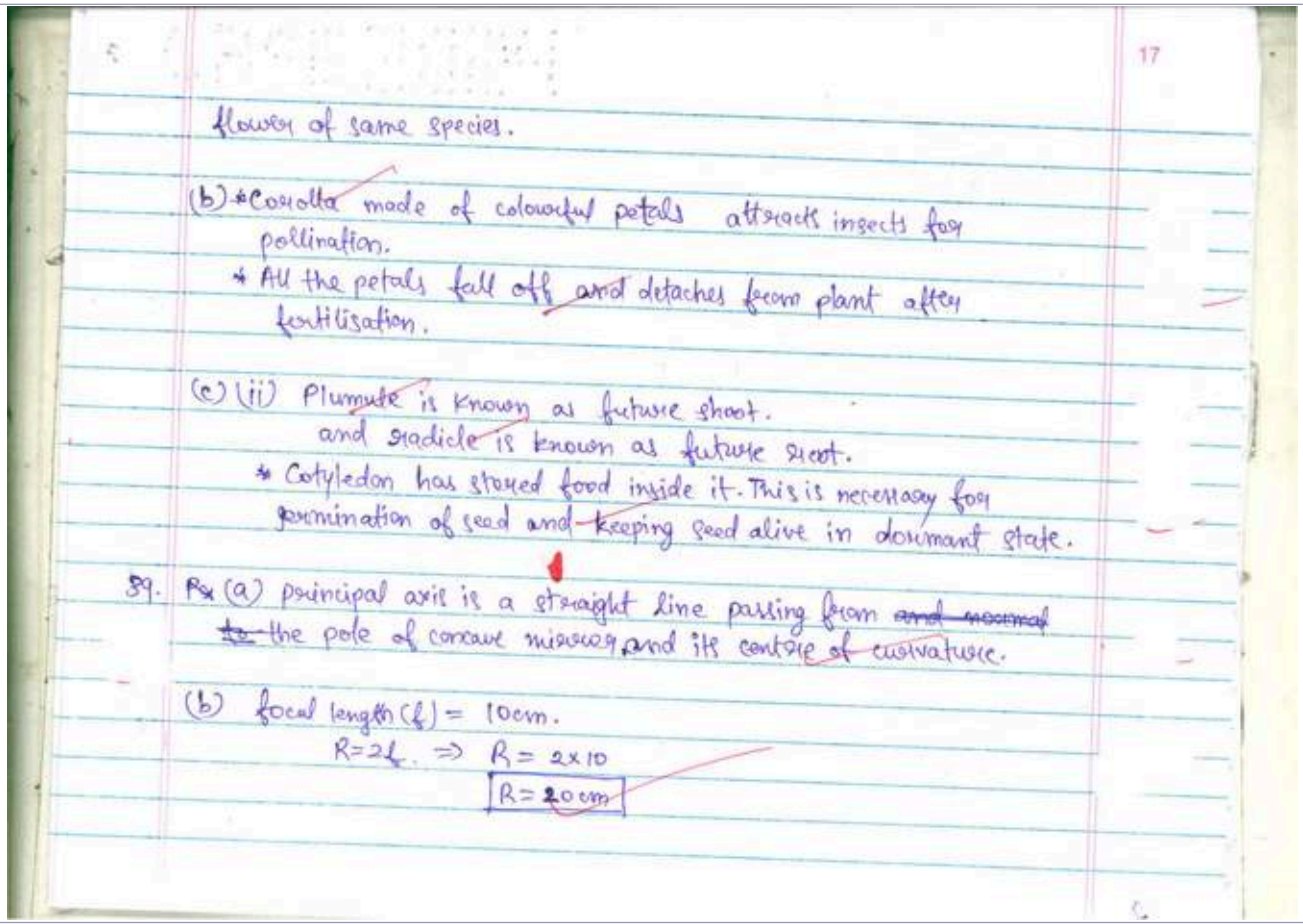
(2) Ketone → $-\overset{O}{\overset{\|}{C}}-$

$CH_3COOH + C_2H_5OH \xrightarrow{conc.\ H_2SO_4} CH_3COOC_2H_5 + H_2O$

38. (a) *Self pollination occurs when the pollen grain is transferred from anther to stigma of the same flower.

* Cross pollination take place when the pollen grain is transferred from anther of one flower to stigma of another

Answer book page 16: Q.38 (contd.) and Q.39



flower of same species.

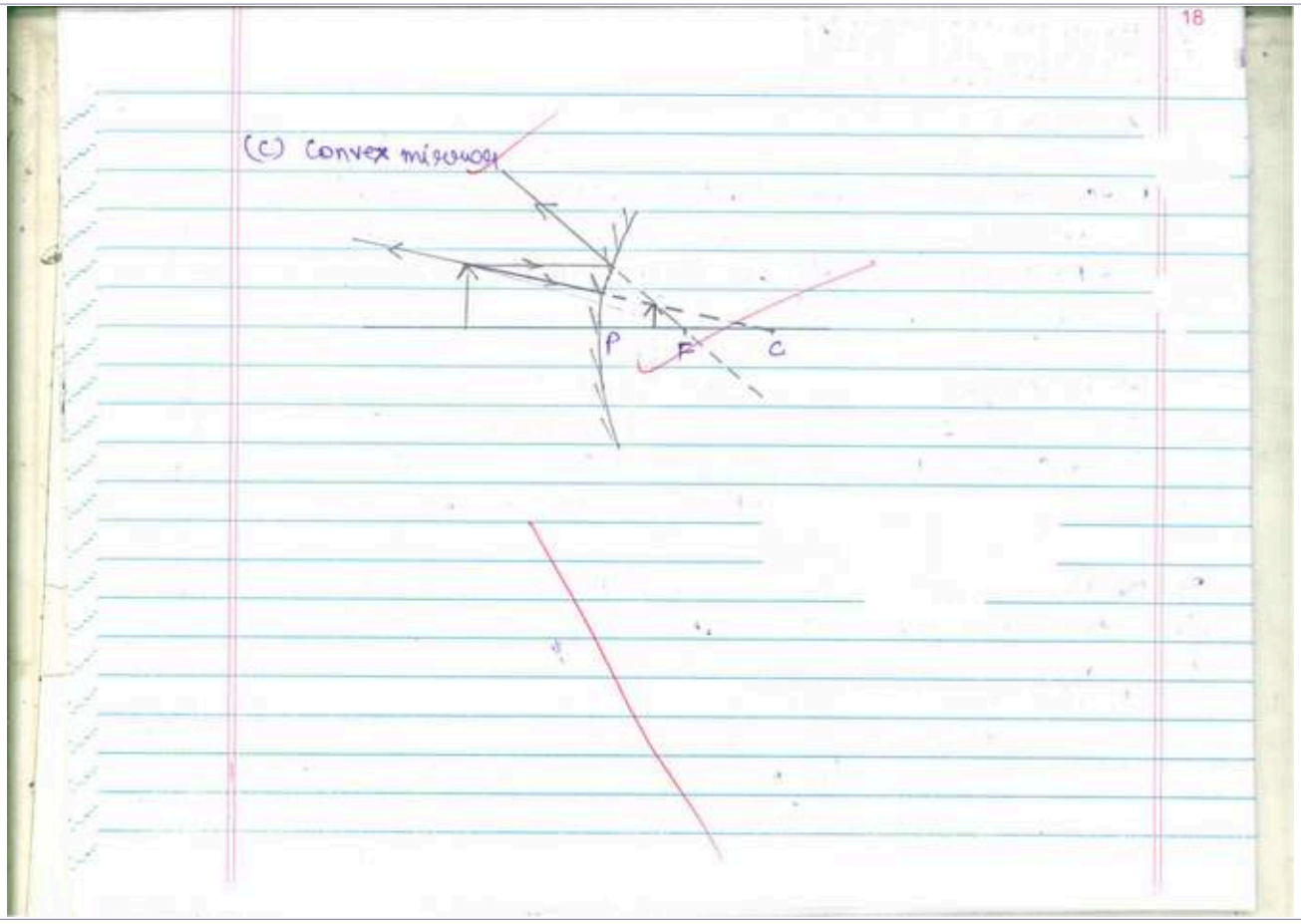
(b) *Corolla made of colourful petals attracts insects for pollination.
* All the petals fall off and detaches from plant after fertilisation.

(c) (ii) Plumule is known as future shoot.
and radicle is known as future root.
* Cotyledon has stored food inside it. This is necessary for germination of seed and keeping seed alive in dormant state.

39. Ex (a) principal axis is a straight line passing from ~~and meeting to~~ the pole of concave mirror and its centre of curvature.

(b) focal length (f) = 10cm.
$R = 2f \Rightarrow R = 2 \times 10$
$R = 20$ cm

Answer book page 17: Q.39 (contd.)



What to learn from the topper's answer book

- Each answer starts with the question number clearly written in the margin.
- Numericals are solved step by step with the formula first and units in every line.
- Reasons are given in one or two precise sentences using NCERT keywords.
- Diagrams and equations are placed exactly where they support the answer.
- Answers are complete but short: no extra story, only marking-scheme points.

CLASS 10 · SCIENCE

SECTION

SOLVED SAMPLE PAPERS

Sample Papers 4 to 10

Each paper is set exactly like the official 2026-27 Sample Question Paper: 39 questions, Section A Biology, Section B Chemistry, Section C Physics, internal choices, case-based questions and diagrams. The solution after every paper follows the CBSE marking-scheme style, with the marks for each step. After each solution, a Topper's Answer Sheet shows real CBSE topper answers to similar questions.

Latest CBSE 2026-27 pattern, fully solved with step marks

SAMPLE PAPER 04

SCIENCE · CODE NO. 086

Sample Question Paper 4

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

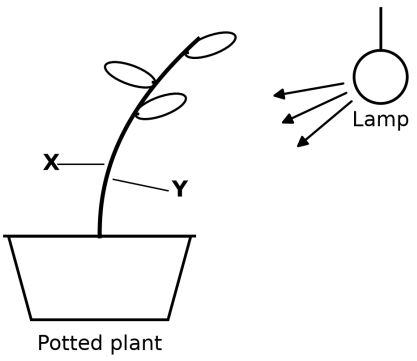
1 (1)

While tracing the path of blood through the human heart, Meera noted the blood vessel that brings oxygen-rich blood from the lungs back to the heart. Which vessel did she note?

A. Pulmonary artery
B. Pulmonary vein
C. Vena cava
D. Aorta

2 (1)

A potted plant was kept near a window so that light fell on it only from the right side, as shown below. After a few days the shoot bent towards the light. X and Y are the two sides of the shoot.



Which of the following correctly explains the bending of the shoot?

A. Auxin diffuses towards side X, so the cells on side X grow longer.
B. Auxin diffuses towards side Y, so the cells on side Y grow longer.
C. Gibberellin collects on side X and stops cell division on side X.
D. Abscisic acid collects on side Y and speeds up growth on side Y.

For visually impaired students

	A potted plant receives light only from its right side. After a few days the shoot bends towards the light. Which of the following correctly explains the bending of the shoot? **A.** Auxin diffuses towards the shaded side, so the cells on the shaded side grow longer. **B.** Auxin diffuses towards the lighted side, so the cells on the lighted side grow longer. **C.** Gibberellin collects on the shaded side and stops cell division there. **D.** Abscisic acid collects on the lighted side and speeds up growth there.	
3	A slice of moist bread left in a warm kitchen gets covered with a cottony, greyish growth within three days. The growth spreads quickly over the bread because the fungus reproduces by **A.** budding **B.** spore formation **C.** fragmentation **D.** regeneration	**1**
4	In a dihybrid cross, a pure pea plant with round yellow seeds (RRYY) is crossed with a pure plant with wrinkled green seeds (rryy). Which of the following can NOT be a gamete of an F_1 plant? **A.** RY **B.** rY **C.** ry **D.** Rr	**1**
5	The eco-club of a school places a green bin in the canteen for biodegradable waste only. Which group of items should go into the green bin? **A.** Vegetable peels, used tea leaves, paper bags **B.** Plastic bottles, leftover food, dry grass **C.** Glass bottles, cotton cloth, aluminium foil **D.** Polythene bags, fruit peels, wood shavings	**1**
6	During a 400 m sprint, an athlete feels cramps in the leg muscles. Which of the following pathways is responsible for the cramps? **A.** Glucose → Pyruvate → Ethanol + Carbon dioxide + Energy **B.** Glucose → Pyruvate → Lactic acid + Energy **C.** Glucose → Pyruvate → Carbon dioxide + Water + Energy **D.** Pyruvate → Glucose → Lactic acid + Energy	**1**
7	A doctor notices a swelling in the front of the neck of a patient and advises her to use iodised salt. The deficiency of which hormone, secreted by which gland, is the likely cause? **A.** Insulin – Pancreas **B.** Thyroxine – Thyroid **C.** Adrenaline – Adrenal gland **D.** Growth hormone – Pituitary	**1**

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** In human males, the testes lie outside the abdominal cavity, in the scrotum. **Reason (R):** Formation of sperms requires a temperature lower than the normal body temperature.	**1**
9	**Assertion (A):** A food chain generally has only three or four trophic levels. **Reason (R):** Only 1% of the energy of one trophic level is passed on to the next trophic level.	**1**

10 Ravi measured the amount of water absorbed by the roots of the same potted plant on two different days. The plant absorbed much more water on a hot, sunny and windy day than on a cool, humid and still day. Name the process responsible for this difference and explain how this process helps in the upward movement of water from the roots to the leaves of a tall tree. **2**

11 ***Students to attempt either option A or B.*** **2**

A. Ananya touches a hot pan and immediately pulls her hand back. Later, she walks to the sink to wash her hands. Which part of the nervous system controls each of these two actions? Why is the first action completed so quickly?

OR

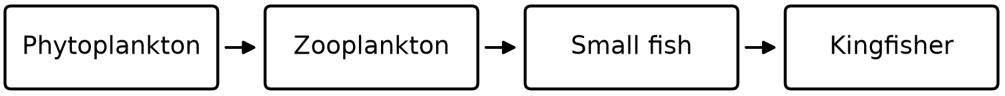
B. The tendrils of a pea plant coil around a bamboo stick placed near them. Name this type of movement. Explain how the tendril is able to coil around the support.

12 Mendel crossed pure tall pea plants (TT) with pure dwarf pea plants (tt). All F_1 plants were tall. When the F_1 plants were self-pollinated, 820 plants were obtained in the F_2 generation. **2**
(i) How many of the F_2 plants are expected to be dwarf? Show the working.
(ii) Why did the dwarf trait not appear in the F_1 generation?

13 Riya's grandfather has been told by his doctor that the sugar level in his blood is persistently high. He has to take regular injections of a hormone. **3**
(i) Name the hormone and the gland that secretes it.
(ii) Explain how the amount of this hormone in the blood is regulated by a feedback mechanism.
(iii) This gland does not have a duct. How then does the hormone reach the cells where it acts?

14 ***Students to attempt either option A or B.*** **3**

A. Study the following food chain found in a pond and answer the questions.



(i) At which trophic level is the small fish? What type of consumer is it?
(ii) Suppose all the kingfishers are removed from the pond. How will this affect the population of small fish and of zooplankton in the short term?
(iii) If the phytoplankton trap 20,000 J of energy, how much energy will be available to the kingfisher? Show the working.

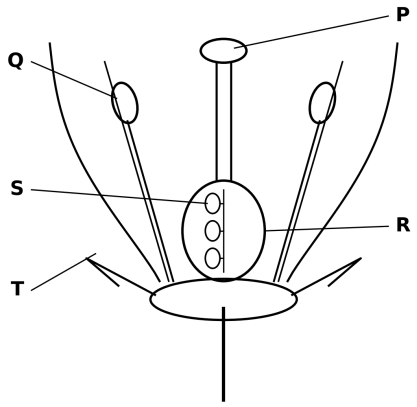
For visually impaired students

A pond food chain is: Phytoplankton → Zooplankton → Small fish → Kingfisher.
(i) At which trophic level is the small fish? What type of consumer is it?
(ii) Suppose all the kingfishers are removed from the pond. How will this affect the population of small fish and of zooplankton in the short term?
(iii) If the phytoplankton trap 20,000 J of energy, how much energy will be available to the kingfisher? Show the working.

OR

B. (i) Write the chemical equations that show how ozone is formed in the upper atmosphere.
(ii) Why is the ozone layer important for life on the Earth?
(iii) Name the group of chemicals mainly responsible for the depletion of the ozone layer and state one step taken at the international level to control it.

15 Kabir cut a flower lengthwise and observed it under a magnifying lens. He drew the diagram shown below and labelled some of its parts as P, Q, R, S and T. **4**



A. Which labelled part produces pollen grains, and which part receives them during pollination?

B. Which labelled part develops into the fruit, and which develops into the seed?

Attempt either sub-part C or D.

C. Describe the events that take place from the time a pollen grain lands on part P till a zygote is formed.

OR

D. A papaya plant bears flowers that have either only stamens or only a pistil. Can such a flower be self-pollinated? Justify your answer and distinguish between self-pollination and cross-pollination.

For visually impaired students

Kabir cut a flower lengthwise and observed its parts: sepal, stigma, anther, ovary and ovule.

A. Which part of a flower produces pollen grains, and which part receives them during pollination?

B. Which part of a flower develops into the fruit, and which develops into the seed?

Attempt either sub-part C or D.

C. Describe the events that take place from the time a pollen grain lands on the stigma till a zygote is formed.

OR

D. A papaya plant bears flowers that have either only stamens or only a pistil. Can such a flower be self-pollinated? Justify your answer and distinguish between self-pollination and cross-pollination.

16 ***Students to attempt either option A or B.*** **5**

A. I. Neha took three test tubes P, Q and R, each containing 5 mL of 1% starch solution, and kept them at 37 °C for 30 minutes after adding:

Test tube	Added to starch solution
P	1 mL of fresh saliva
Q	1 mL of boiled and cooled saliva
R	1 mL of water

She then added two drops of iodine solution to each test tube.

(a) In which test tube(s) will a blue-black colour appear?

(b) Explain the role of saliva in this experiment and why the result in test tube Q is different from that in P. (2)

II. Starch is only partly digested in the mouth. Explain how the digestion of food is completed and the digested food absorbed in the small intestine, giving the role of (a) bile, (b) pancreatic juice and (c) villi. (3)

OR

B. I. Manjeet placed a twig of an aquatic plant in a beaker of water and kept a lamp at different distances from it. She counted the number of gas bubbles given out by the twig per minute.

Distance of lamp (cm)	Bubbles per minute
10	48
20	30
30	16
40	6

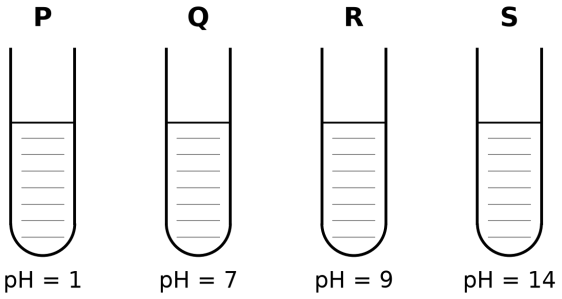
(a) Name the gas present in the bubbles.
(b) What conclusion can be drawn from the data about the rate of photosynthesis?
(c) Predict, with reason, the effect of dissolving a pinch of sodium hydrogencarbonate in the water on the number of bubbles. (3)
II. (a) How do guard cells control the opening and closing of stomata?
(b) Desert plants keep their stomata closed during the day. How do they carry out photosynthesis? (2)

SECTION B Chemistry

17 A clean copper strip is dipped in silver nitrate solution. After some time the solution turns blue and a shiny greyish-white layer forms on the strip. The reaction is an example of **1**

A. combination reaction
B. decomposition reaction
C. displacement reaction
D. double displacement reaction

18 Rohan's mother suffers from acidity after a heavy meal. Four solutions P, Q, R and S with the pH values shown are available. **1**



Which solution is most suitable for relieving her acidity?

A. P
B. Q
C. R
D. S

For visually impaired students

Rohan's mother suffers from acidity after a heavy meal. Four solutions P, Q, R and S have pH values 1, 7, 9 and 14 respectively. Which solution is most suitable for relieving her acidity?

A. P
B. Q
C. R
D. S

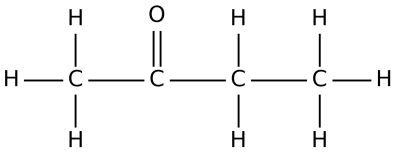
19 Zinc granules are added to four test tubes containing aqueous solutions of (I) iron(II) sulphate, (II) copper(II) sulphate, (III) aluminium sulphate and (IV) magnesium sulphate. In which test tubes will a chemical change be observed? **1**

A. I and II only **B.** II only
C. III and IV only **D.** I, II, III and IV

20 Which of the following hydrocarbons will decolourise bromine water? **1**

A. C_2H_6 **B.** C_3H_8
C. C_3H_6 **D.** CH_4

21 Study the structural formula of the compound given below. **1**



The correct IUPAC name of the compound is

A. butanal **B.** butanone
C. propanone **D.** butanoic acid

For visually impaired students

The correct IUPAC name of the compound $CH_3–CO–CH_2–CH_3$ is

A. butanal **B.** butanone
C. propanone **D.** butanoic acid

22 In a hospital, a white powder is mixed with water and applied over a bandage to hold a fractured bone in place. The powder sets into a hard mass. The chemical formula of the powder is **1**

A. $CaSO_4 \cdot 2H_2O$ **B.** $CaSO_4 \cdot ½H_2O$
C. $CaCO_3$ **D.** $Na_2CO_3 \cdot 10H_2O$

23 Cracked railway tracks are joined using the reaction **1**
$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$
Which of the following is correct for this reaction?

A. Al is oxidised and Fe_2O_3 is the oxidising agent.
B. Fe_2O_3 is oxidised and Al is the oxidising agent.
C. Al is reduced and Fe_2O_3 is the reducing agent.
D. Fe is oxidised and Al_2O_3 is the reducing agent.

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Solid sodium chloride does not conduct electricity, but molten sodium chloride does. **1**
Reason (R): In the solid state the ions are held in a rigid structure and cannot move, while in the molten state they are free to move.

25 In a laboratory, Sana passed dry hydrogen chloride gas over a dry blue litmus paper. The colour of the paper did not change. When she repeated the activity with a moist blue litmus paper, it turned red. Explain these observations with the help of a chemical equation. **2**

26 ***Students to attempt either option A or B.*** **3**

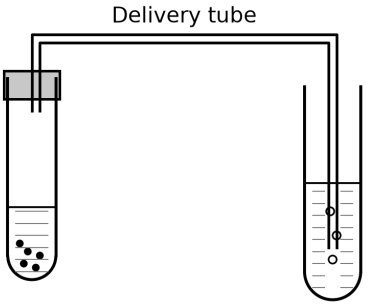
A. Lead nitrate powder is heated strongly in a dry boiling tube.
(i) What will be observed?
(ii) Write the balanced chemical equation for the reaction.
(iii) Name the type of reaction and give one reason for your answer.

OR

B. Zinc is found in nature as zinc blende (ZnS) and as calamine ($ZnCO_3$). Write balanced chemical equations for:
(i) the conversion of zinc blende into zinc oxide,
(ii) the conversion of calamine into zinc oxide, and
(iii) the reduction of zinc oxide to zinc metal using carbon.
Name the processes in (i) and (ii).

27 The atomic numbers of magnesium and chlorine are 12 and 17 respectively. **3**
(i) Write their electronic configurations.
(ii) Show the formation of magnesium chloride by the transfer of electrons.
(iii) Magnesium chloride has a high melting point and dissolves in water but not in kerosene. Give reasons.

28 Neha set up the apparatus shown below. She added dilute hydrochloric acid to sodium hydrogencarbonate in test tube A and passed the gas formed through lime water in test tube B. **4**



A. Name the gas evolved in test tube A and state what is observed in test tube B.

B. Write the balanced chemical equation for the reaction taking place in test tube A.

Attempt either sub-part C or D.

C. Neha kept passing the gas through test tube B for a long time. What change will she observe? Write chemical equations for the reactions that take place in test tube B, first in a short time and then over a long time.

OR

D. (i) Why is sodium hydrogencarbonate used as an ingredient of antacids?
(ii) Baking powder contains a mild edible acid along with sodium hydrogencarbonate. Why is the acid added?

For visually impaired students

Neha added dilute hydrochloric acid to sodium hydrogencarbonate in test tube A and passed the gas formed through lime water in test tube B with the help of a delivery tube.

A. Name the gas evolved in test tube A and state what is observed in test tube B.

B. Write the balanced chemical equation for the reaction taking place in test tube A.

Attempt either sub-part C or D.

C. Neha kept passing the gas through test tube B for a long time. What change will she observe? Write chemical equations for the reactions that take place in test tube B, first in a short time and then over a long time.

OR

D. (i) Why is sodium hydrogencarbonate used as an ingredient of antacids?
(ii) Baking powder contains a mild edible acid along with sodium hydrogencarbonate. Why is the acid added?

29 ***Students to attempt either option A or B.*** **5**

A. I. An organic compound X with molecular formula C_2H_6O reacts with sodium metal to give a gas that burns with a pop sound. When X is heated with excess concentrated sulphuric acid at 443 K, it forms a compound Y that decolourises bromine water. When X is heated with alkaline potassium permanganate, it forms a compound Z with molecular formula $C_2H_4O_2$.
(a) Identify X, Y and Z.
(b) Write the chemical equation for the conversion of X into Y and state the role of concentrated sulphuric acid. (3)
II. X and Z are warmed together in the presence of a few drops of concentrated sulphuric acid. A sweet-smelling substance is formed. Write the chemical equation for this reaction, name the product and give one use of such compounds. (2)

OR

B. I. Aman has two unlabelled bottles, one containing ethanol and the other ethanoic acid.
(a) Suggest a test using sodium hydrogencarbonate to identify the bottle containing ethanoic acid. State the observation and write the chemical equation.
(b) Why is pure ethanoic acid called glacial acetic acid? (3)
II. Ethyl ethanoate is heated with sodium hydroxide solution. Write the chemical equation for the reaction. Name this reaction and state where it is used. (2)

SECTION C Physics

30 Large shops fix a mirror in a corner near the ceiling so that the shopkeeper can watch a wide area of the shop. The image of every customer seen in this mirror is erect and smaller than the customer. The mirror used is **1**

A. a plane mirror
B. a concave mirror with the object between its pole and focus
C. a convex mirror
D. a concave mirror with the object beyond its centre of curvature

31 A copper wire of resistance 5 Ω is melted and drawn into a new wire of double the length. The resistance of the new wire will be **1**

A. 2.5 Ω **B.** 10 Ω
C. 20 Ω **D.** 1.25 Ω

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** A person suffering from myopia can see nearby objects clearly. **Reason (R):** In myopia, the image of a distant object is formed in front of the retina. **1**

33 Two electric bulbs are rated 40 W, 220 V and 100 W, 220 V.
(i) Calculate the resistance of each bulb.
(ii) If the two bulbs are connected in series across a 220 V supply, which bulb will glow brighter? Justify your answer. **2**

34 ***Students to attempt either option A or B.*** **2**

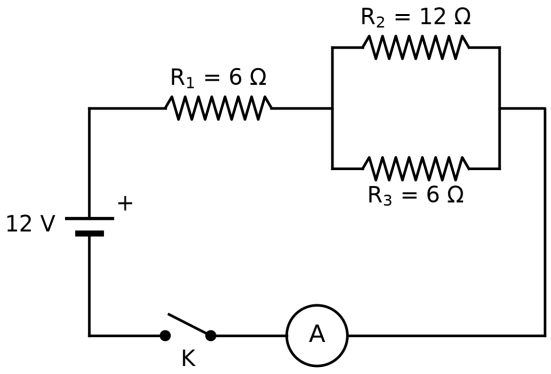
A. After a shower in the evening, Ishita saw a rainbow in the eastern sky. Explain how a rainbow is formed. Why is a rainbow seen only when the Sun is behind the observer?

OR

B. Why does the sky appear blue to a person on the Earth but dark to an astronaut in space?

35 Arjun uses a concave mirror of focal length 15 cm as a shaving mirror. He holds his face 10 cm from the mirror. **3**
(i) Find the position of the image.
(ii) Find the magnification.
(iii) State the nature of the image and explain why a concave mirror is preferred as a shaving mirror.

36 Study the circuit shown below. **3**



(i) Calculate the equivalent resistance of the circuit.
(ii) What will be the reading of the ammeter when the key K is closed?
(iii) Calculate the current through the resistor R_2.

For visually impaired students

A resistor R_1 = 6 Ω is connected in series with a parallel combination of R_2 = 12 Ω and R_3 = 6 Ω. The combination is connected to a 12 V battery through an ammeter and a key K.
(i) Calculate the equivalent resistance of the circuit.
(ii) What will be the reading of the ammeter when the key K is closed?
(iii) Calculate the current through the resistor R_2.

37 ***Students to attempt either option A or B.*** **3**

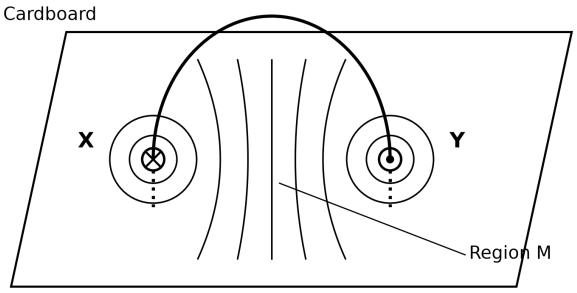
A. A straight horizontal conductor carrying current from west to east is placed in a uniform magnetic field directed vertically downwards.
(i) State the rule used to find the direction of the force acting on the conductor.

(ii) Find the direction of the force on the conductor.
(iii) What will happen to the direction of the force if the directions of both the current and the magnetic field are reversed?

OR

B. A 2 kW electric geyser is connected to a 220 V domestic line that has a 5 A fuse in the circuit.
(i) Calculate the current drawn by the geyser.
(ii) What will happen when the geyser is switched on? Suggest a suitable rating for the fuse.
(iii) Why is the metal body of the geyser connected to the earth wire?

38 Riya passed a thick copper wire, bent into a circular loop, through a horizontal cardboard at points X and Y. She sprinkled iron filings on the cardboard, passed a current through the loop and tapped the cardboard gently. The filings arranged themselves as shown below. **4**



A. What do the concentric circles of filings around points X and Y show about the magnetic field near the wire?

B. What does the pattern of filings in region M, at the centre of the loop, tell you about the magnetic field there?

Attempt either sub-part C or D.

C. Riya replaces the single loop with a coil of 50 turns of the same radius and doubles the current. How many times stronger will the magnetic field at the centre become? Justify your answer.

OR

D. A long coil of many circular turns is called a solenoid. Describe the magnetic field inside a current-carrying solenoid. How can this field be used to make an electromagnet?

For visually impaired students

Riya passed a thick copper wire, bent into a circular loop, through a horizontal cardboard at two points X and Y. She sprinkled iron filings on the cardboard and passed a current through the loop. Near X and Y the filings formed concentric circles, while at the centre of the loop they formed nearly straight lines. Attempt A and B given below in place of A and B above; C and D remain the same.

A. What do the concentric circles of filings around points X and Y show about the magnetic field near the wire?

B. What do the nearly straight lines of filings at the centre of the loop tell you about the magnetic field there?

39 ***Students to attempt either option A or B.*** **5**

A. I. An object 2 cm tall is placed 30 cm in front of a convex lens of focal length 20 cm. Using the lens formula, find the position, size and nature of the image. (3)
II. Draw a labelled ray diagram to show the formation of the image in part I. (2)

OR

B. I. An object 3 cm tall is placed 15 cm in front of a concave lens of focal length 10 cm. Find the position and size of the image, and the power of the lens. (3)
II. Draw a labelled ray diagram to show the formation of the image in part I and state the nature of the image. (2)

END OF QUESTION PAPER

Quick revision before the exam: formulas you must know

1. Mirror formula: $1/v + 1/u = 1/f$; magnification $m = -v/u = h'/h$.
2. Lens formula: $1/v - 1/u = 1/f$; magnification $m = v/u = h'/h$.
3. Power of a lens: $P = 1/f$ (f in metres); unit dioptre (D). Convex +, concave −.
4. Refractive index: $n = c/v = \sin i / \sin r$.
5. Ohm's law: $V = IR$. Resistance $R = \rho L/A$.
6. Series: $R = R_1 + R_2 + R_3$. Parallel: $1/R = 1/R_1 + 1/R_2 + 1/R_3$.
7. Electric power: $P = VI = I^2R = V^2/R$. Heat: $H = I^2Rt$.
8. Energy: 1 kWh = 3.6×10^6 J. Units used = power (kW) × time (h).
9. Sign convention: distances measured from the pole or optical centre; along the incident light positive.
10. Check every answer for units: cm, m, Ω, A, V, W, J, D.

Nitesh Bhaiya ki tip: paper khatam hone ke baad 10 minute bachake units aur diagrams ke labels check karo.

SOLUTION PAPER 04

SCIENCE · CODE NO. 086

Sample Paper 4: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

B. Pulmonary vein (1)

2 Ans

A. Auxin diffuses towards side X, so the cells on side X grow longer. (1)
(X is the shaded side; faster growth on the shaded side bends the shoot towards light.)

3 Ans

B. spore formation *(the fungus is Rhizopus, bread mould)* (1)

4 Ans

D. Rr *(a gamete carries only one allele of each gene)* (1)

5 Ans

A. Vegetable peels, used tea leaves, paper bags (1)

6 Ans

B. Glucose → Pyruvate → Lactic acid + Energy *(anaerobic respiration in muscle cells when oxygen is in short supply)* (1)

7 Ans

B. Thyroxine – Thyroid *(iodine deficiency causes goitre)* (1)

8 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

9 Ans

C. A is true but R is false. *(About 10% of the energy is passed on to the next trophic level.)* (1)

10 Ans

Transpiration (loss of water vapour through stomata of leaves); its rate is higher on a hot, sunny, windy day. (1)

Evaporation of water from leaf cells creates a suction (transpiration pull) that pulls water up through the continuous column of xylem vessels from roots to leaves. (1)

Total 2 marks

11 Ans

A.

Pulling the hand back: reflex action, controlled by the spinal cord (reflex arc). Walking to the sink: voluntary action, controlled by the brain (cerebrum). (½ + ½)

In a reflex arc the message goes from receptor to spinal cord and back to the muscle directly, without waiting for the brain to think, so the response is very quick and protects the body. (1)

OR

B.

Thigmotropism / directional growth movement in response to touch (contact). (1)

When the tendril touches the support, the cells on the side in contact grow less rapidly than the cells on the side away from the support; so the tendril bends and coils around the support. (1)

Total 2 marks

12 **Ans**

(i) F_2 ratio Tall : Dwarf = 3 : 1 (TT : Tt : tt = 1 : 2 : 1). Expected dwarf plants = ¼ × 820 = 205 **(1)**

(ii) F_1 plants are Tt; tallness (T) is dominant and dwarfness (t) is recessive, so the recessive trait is masked in the presence of the dominant allele. **(1)**

Total 2 marks

13 **Ans**

(i) Insulin; pancreas. **(½ + ½)**

(ii) When blood sugar rises, cells of the pancreas detect it and secrete more insulin; as the sugar level falls, insulin secretion is reduced. The timing and amount of hormone are thus controlled by the level of sugar itself (feedback mechanism). **(1)**

(iii) The pancreas (islets) acts as an endocrine (ductless) gland: it releases insulin directly into the blood, which carries it to the target cells. **(1)**

Total 3 marks

14 **Ans**

A.

(i) Third trophic level; secondary consumer (carnivore). **(½ + ½)**

(ii) Small fish will increase (no predator); zooplankton will decrease because more small fish eat them. **(½ + ½)**

(iii) 10% law: 20,000 J → 2,000 J (zooplankton) → 200 J (small fish) → 20 J available to the kingfisher. **(1)**

OR

B.

(i) $O_2 \xrightarrow{UV} O + O$; $O + O_2 \rightarrow O_3$ **(1)**

(ii) The ozone layer absorbs most of the harmful ultraviolet radiation from the Sun, which can cause skin cancer, cataract and damage to organisms. **(1)**

(iii) Chlorofluorocarbons (CFCs). In 1987 the UNEP agreement froze CFC production at 1986 levels; CFC-free refrigerants are now made. **(½ + ½)**

Total 3 marks

15 **Ans**

A. Q (anther) produces pollen grains; P (stigma) receives them. **(½ + ½)**

B. R (ovary) develops into the fruit; S (ovule) develops into the seed. **(½ + ½)**

C. The pollen grain on the stigma (P) germinates and a pollen tube grows out of it; the tube grows down through the style and reaches the ovule in the ovary; the male germ cell is released and fuses with the female germ cell (egg) in the ovule; this fusion (fertilisation) forms the zygote. **(½ × 4 = 2)**

OR

D. No. Self-pollination needs stamens and pistil in the same flower (or on the same plant); papaya flowers are unisexual, so pollen must come from another flower. **(1)**

Self-pollination: transfer of pollen from the anther to the stigma of the same flower (or same plant). Cross-pollination: transfer of pollen from the anther of one flower to the stigma of a flower on another plant. **(½ + ½)**

Total 4 marks

16 **Ans**

A. I.

(a) Test tubes Q and R. **(½)**

(b) Saliva contains the enzyme salivary amylase, which breaks down starch into sugar, so starch is absent in P and no blue-black colour appears. **(1)**

Boiling denatures (destroys) the enzyme, so starch in Q remains undigested. **(½)**

II.

(a) Bile (from liver) makes the food alkaline and emulsifies fats into small globules, increasing the surface for enzyme action. (1)

(b) Pancreatic juice has trypsin (proteins), lipase (fats) and amylase (starch); intestinal juice finally converts proteins to amino acids, carbohydrates to glucose and fats to fatty acids and glycerol. (1)

(c) Villi are finger-like projections of the inner lining of the small intestine that increase the surface area for absorption; they are richly supplied with blood vessels that take the absorbed food to every cell. (1)

OR

B. I.

(a) Oxygen. (1)

(b) As the lamp is moved away (light intensity decreases), the number of bubbles decreases, so the rate of photosynthesis decreases with decrease in light intensity. (1)

(c) The number of bubbles will increase, because sodium hydrogencarbonate supplies more carbon dioxide, a raw material for photosynthesis. (1)

II.

(a) When water flows into the guard cells they swell and the pore opens; when they lose water they shrink and the pore closes. (1)

(b) They open stomata at night, take in CO_2 and store it as an intermediate (acid); during the day this CO_2 is used for photosynthesis with the energy absorbed by chlorophyll. (1)

Total 5 marks

SECTION B Chemistry

17 Ans C. displacement reaction *($Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$)* (1)

18 Ans C. R *(a mild base, pH 9, neutralises excess acid; S, pH 14, is a strong alkali and harmful)* (1)

19 Ans A. I and II only *(zinc is more reactive than iron and copper, but less reactive than aluminium and magnesium)* (1)

20 Ans C. C_3H_6 *(propene, an unsaturated hydrocarbon with a C=C double bond)* (1)

21 Ans B. butanone (1)

22 Ans B. $CaSO_4 \cdot \frac{1}{2}H_2O$ *(Plaster of Paris)* (1)

23 Ans A. Al is oxidised and Fe_2O_3 is the oxidising agent. (1)

24 Ans A. Both A and R are true, and R is the correct explanation of A. (1)

25 Ans Acidic behaviour is due to H^+ ions, which are produced only in the presence of water. Dry HCl gas does not ionise, so it does not change the colour of dry litmus. (1)

With moisture, HCl ionises: $HCl + H_2O \rightarrow H_3O^+ + Cl^-$; the H^+ (H_3O^+) ions turn moist blue litmus red. (1)

Total 2 marks

26 **Ans**

A.

(i) Brown fumes (of nitrogen dioxide) are given off and a yellow residue (lead oxide) is left. **(1)**

(ii) $2Pb(NO_3)_2(s) \xrightarrow{heat} 2PbO(s) + 4NO_2(g) + O_2(g)$ **(1)**

(iii) Thermal decomposition reaction; a single reactant breaks down into simpler products on heating. **(½ + ½)**

OR

B.

(i) $2ZnS + 3O_2 \xrightarrow{heat} 2ZnO + 2SO_2$ (Roasting) **(½ + ½)**

(ii) $ZnCO_3 \xrightarrow{heat} ZnO + CO_2$ (Calcination) **(½ + ½)**

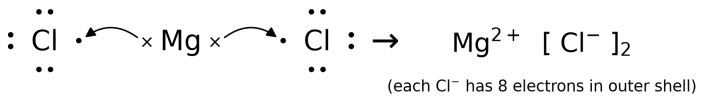
(iii) $ZnO + C \rightarrow Zn + CO$ **(1)**

Total 3 marks

27 **Ans**

(i) Mg (12): 2, 8, 2; Cl (17): 2, 8, 7 **(½ + ½)**

(ii) $Mg \rightarrow Mg^{2+} + 2e^-$; each Cl atom gains one electron: $2Cl + 2e^- \rightarrow 2Cl^-$; Mg^{2+} and $2Cl^-$ form $Mg^{2+}[Cl^-]_2$ ($MgCl_2$) **(1)**



(iii) Strong electrostatic forces between the oppositely charged ions need a lot of energy to break, so the melting point is high. Being ionic, it dissolves in water (a polar solvent) but not in kerosene (a non-polar/organic solvent). **(½ + ½)**

Total 3 marks

28 **Ans**

A. Carbon dioxide; lime water turns milky. **(½ + ½)**

B. $NaHCO_3 + HCl \rightarrow NaCl + H_2O + CO_2$ **(1)**

C. The milkiness first appears and then disappears on passing excess gas. **(½)**

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow$ (white, milky) + H_2O **(½)**

$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ (soluble) **(1)**

OR

D. (i) Being a mild (weak) base, it neutralises the excess acid in the stomach and gives relief. **(1)**

(ii) The acid reacts with sodium hydrogencarbonate to release CO_2, which makes the cake soft and spongy, and it neutralises the bitter-tasting sodium carbonate that would otherwise form on heating. **(1)**

Total 4 marks

29 **Ans**

A. I.

(a) X: ethanol (C_2H_5OH); Y: ethene ($CH_2=CH_2$); Z: ethanoic acid (CH_3COOH) **(½ × 3)**

(b) $CH_3–CH_2OH \xrightarrow{\text{hot conc. } H_2SO_4,\ 443\ K} CH_2=CH_2 + H_2O$ **(1)**

Concentrated sulphuric acid acts as a dehydrating agent (removes water from ethanol). **(½)**

II.

$CH_3COOH + CH_3CH_2OH \xrightarrow{acid} CH_3COOCH_2CH_3 + H_2O$ (esterification) **(1)**

Product: ethyl ethanoate (an ester). Use: in perfumes / as flavouring agents. **(½ + ½)**

OR

B. I.

(a) Add a pinch of sodium hydrogencarbonate to a little of each liquid. Brisk effervescence (CO_2 gas, turns lime water milky) is seen only with ethanoic acid; no reaction with ethanol. (1)

$CH_3COOH + NaHCO_3 \rightarrow CH_3COONa + H_2O + CO_2$ (1)

(b) The melting point of pure ethanoic acid is 290 K, so it often freezes during winter in cold climates and looks like ice (glacier). (1)

II.

$CH_3COOC_2H_5 + NaOH \rightarrow CH_3COONa + C_2H_5OH$ (1)

Saponification; used in the preparation of soap. (½ + ½)

Total 5 marks

SECTION C Physics

30 **Ans**

C. a convex mirror (1)

31 **Ans**

C. 20 Ω *(volume constant: length doubles, area halves; R = pl/A becomes 4 times)* (1)

32 **Ans**

B. Both A and R are true, and R is not the correct explanation of A. (1)

33 **Ans**

(i) $R = V^2/P$; $R_1 = (220)^2/40 = 1210\ \Omega$; $R_2 = (220)^2/100 = 484\ \Omega$ (½ + ½)

(ii) The 40 W bulb glows brighter. In series the same current flows through both, so $P = I^2R$ is greater for the bulb of higher resistance (40 W bulb). (½ + ½)

Total 2 marks

34 **Ans**

A.

Tiny water droplets in the air act like small prisms: sunlight is refracted and dispersed on entering a droplet, reflected internally, and refracted again on coming out, so the seven colours reach the eye separately. (1½)

These rays come back towards the side of the Sun, so the rainbow is seen only in the direction opposite to the Sun. (½)

OR

B.

Fine particles (molecules) of air scatter blue light (shorter wavelength) much more strongly than red light; the scattered blue light enters the eyes, so the sky looks blue. (1)

In space there is no atmosphere to scatter light, so the sky appears dark to an astronaut. (1)

Total 2 marks

35 **Ans**

(i) f = −15 cm, u = −10 cm; 1/v = 1/f − 1/u = −1/15 + 1/10 = 1/30; v = +30 cm (30 cm behind the mirror) (1)

(ii) m = −v/u = −(30)/(−10) = +3 (1)

(iii) Virtual, erect and magnified (3 times). Face held within the focus gives an enlarged, erect image, so fine details are seen clearly while shaving. (½ + ½)

Total 3 marks

36 **Ans**

(i) $R_2 \parallel R_3$: $1/R_p = 1/12 + 1/6 = 3/12$, $R_p = 4\ \Omega$; $R_{eq} = 6 + 4 = 10\ \Omega$ (1)

(ii) $I = V/R_{eq} = 12/10 = 1.2$ A (1)

(iii) p.d. across parallel part = 1.2 × 4 = 4.8 V; current through $R_2 = 4.8/12 = 0.4$ A (1)

Total 3 marks

37

A.

Ans

(i) Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular; if the forefinger points along the field and the middle finger along the current, the thumb gives the direction of the force. **(1)**

(ii) Towards the north (horizontal). **(1)**

(iii) The direction of the force remains the same (towards the north), as reversing both does not change it. **(1)**

OR

B.

(i) I = P/V = 2000/220 ≈ 9.1 A **(1)**

(ii) The current (9.1 A) exceeds the fuse rating, so the fuse wire melts and the circuit breaks; a fuse of 10 A rating should be used. **(½ + ½)**

(iii) If the live wire touches the metal body due to leakage, the current flows to the earth through the low-resistance earth wire, keeping the body at earth potential and preventing a severe electric shock. **(1)**

Total 3 marks

38

Ans

A. Near the wire, every part of the loop acts like a straight current-carrying conductor: the field lines are concentric circles around it, and the circles get larger (field weaker) farther from the wire. **(1)**

B. The lines are nearly straight and parallel, so the field at the centre is nearly uniform; it is perpendicular to the plane of the loop as field lines due to all parts of the loop act in the same direction there. **(1)**

C. $B \propto nI$ (for the same radius). New field = 50 × 2 = 100 times the original field. **(1 + 1)**

OR

D. Inside a solenoid the field lines are parallel straight lines, so the field is uniform; the solenoid behaves like a bar magnet with N and S poles at its ends. **(1)**

If a soft iron rod is placed inside the solenoid, the strong field magnetises it; it becomes an electromagnet that works only while current flows. **(1)**

Total 4 marks

39

Ans

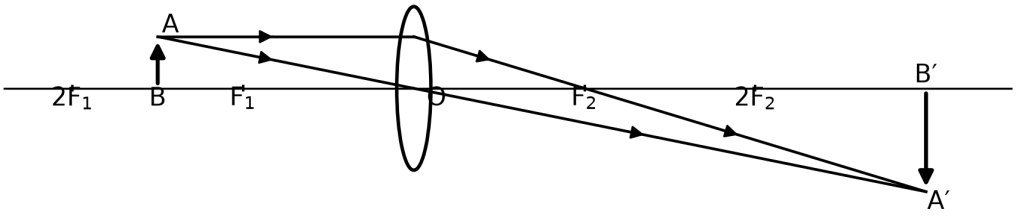
A. I.

f = +20 cm, u = −30 cm, h = +2 cm; 1/v = 1/f + 1/u = 1/20 − 1/30 = 1/60; v = +60 cm (on the other side of the lens) **(1)**

m = v/u = 60/(−30) = −2; h′ = m × h = −4 cm (image 4 cm tall) **(1)**

Real, inverted and magnified. **(1)**

II.



Correct rays (parallel ray through F_2, ray through O) with arrows **(1)**

Correct image position and labelling **(1)**

OR

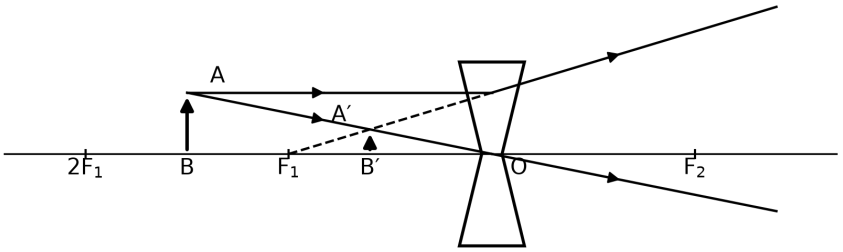
B. I.

f = −10 cm, u = −15 cm, h = +3 cm; 1/v = 1/f + 1/u = −1/10 − 1/15 = −1/6; v = −6 cm (6 cm in front of the lens, on the same side as the object) **(1)**

m = v/u = (−6)/(−15) = +0.4; h′ = 0.4 × 3 = +1.2 cm (1)

P = 1/f (in m) = 1/(−0.10) = −10 D (1)

II.



Correct ray diagram with arrows and labels (1)

Image is virtual, erect and diminished, between O and F_1. (1)

Total 5 marks

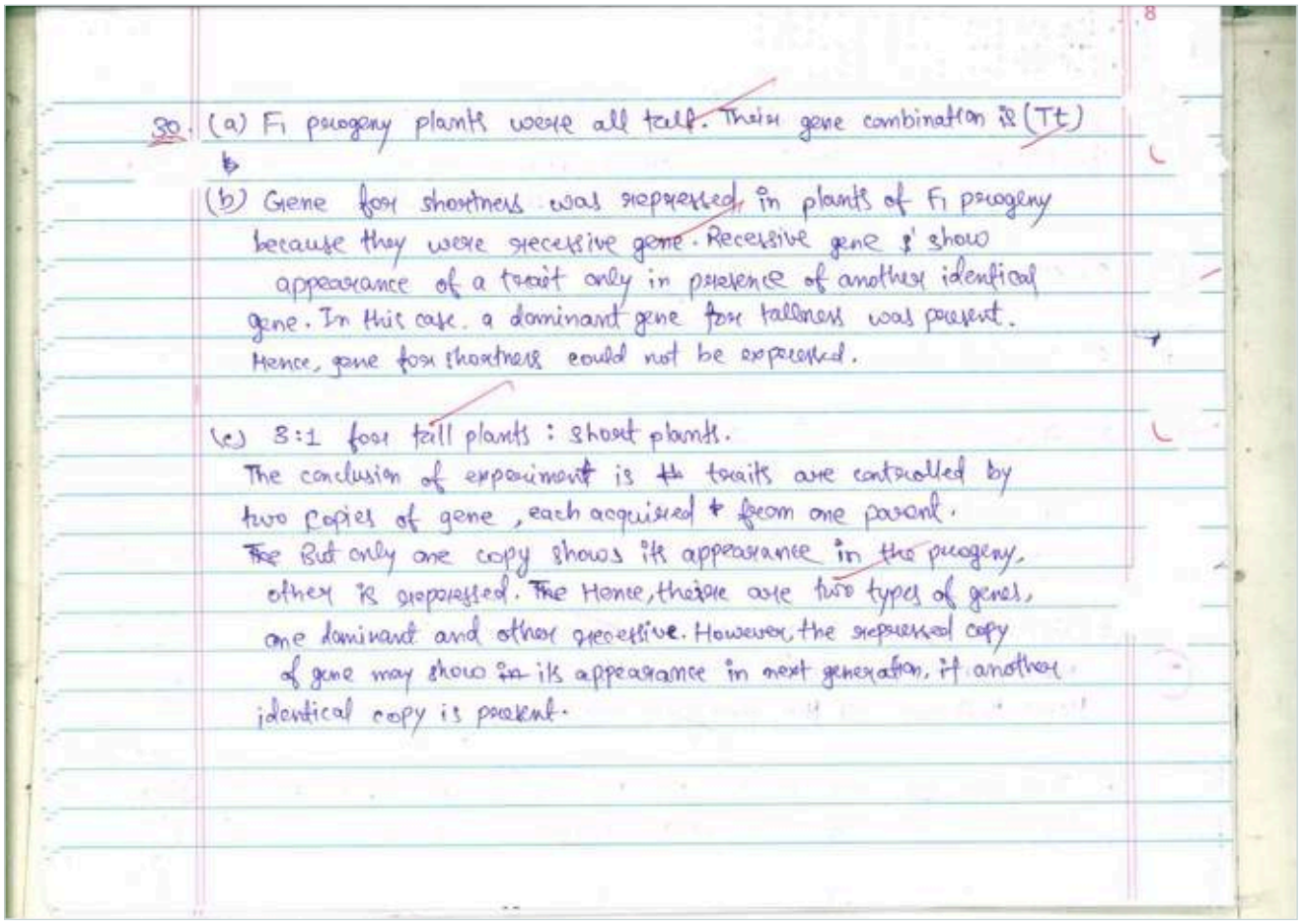
END OF SOLUTIONS

Common mistakes in this paper

1. Forgetting the minus sign for a concave lens or concave mirror focal length.
2. Writing magnification without its sign: + means virtual and erect, − means real and inverted.
3. Ray diagrams without arrows on the rays, or without labels F, 2F and O.
4. Unbalanced chemical equations, or missing state symbols when asked.
5. Writing 'energy' instead of 'ATP' or 'glucose' in biology answers that need the exact term.
6. Giving one point in a 2-mark answer: two marks usually means two clear points.
7. Leaving out units in the final answer of a numerical: half a mark is lost.
8. Copying the question instead of answering it in an assertion-reason item: write only the option letter.

Nitesh Bhaiya ki tip: solution padhte waqt step marks dekho, examiner wahi steps dhundhta hai.

Real CBSE topper's answer · Board Exam 2024 · similar to Q.12 of this paper (Mendel's monohybrid cross, F2 ratio)

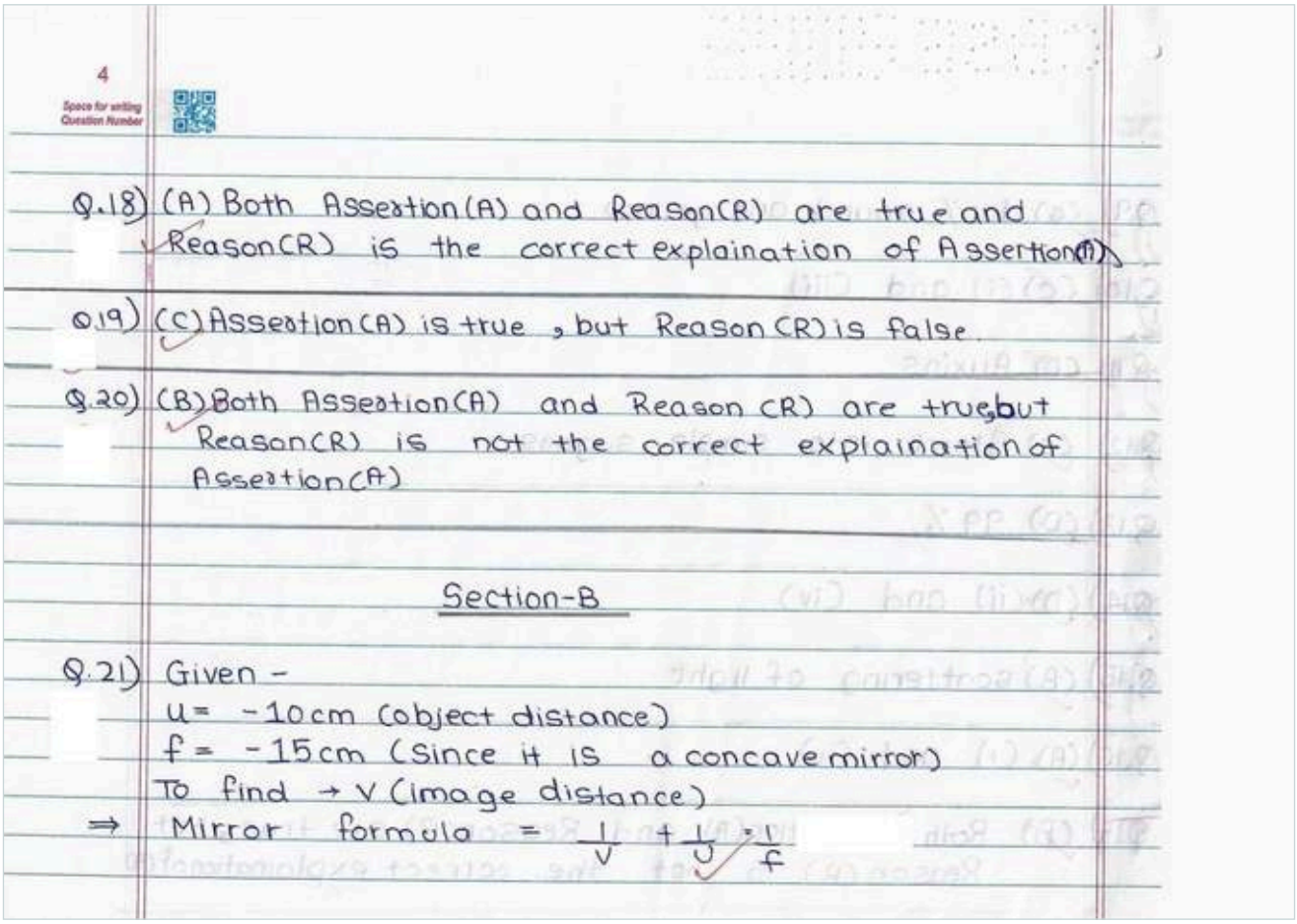


30. (a) F_1 progeny plants were all tall. Their gene combination is (Tt)

(b) Gene for shortness was suppressed in plants of F_1 progeny because they were recessive gene. Recessive gene is show appearance of a trait only in presence of another identical gene. In this case, a dominant gene for tallness was present. Hence, gene for shortness could not be expressed.

(c) 3:1 for tall plants : short plants.
The conclusion of experiment is the traits are controlled by two copies of gene, each acquired from one parent. But only one copy shows its appearance in the progeny, other is suppressed. Hence, there are two types of genes, one dominant and other recessive. However, the suppressed copy of gene may show its appearance in next generation, if another identical copy is present.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.35 of this paper (concave mirror, image by mirror formula)



4
Space for writing Question Number

Q.18) (A) Both Assertion(A) and Reason(R) are true and Reason(R) is the correct explaination of Assertion(A).

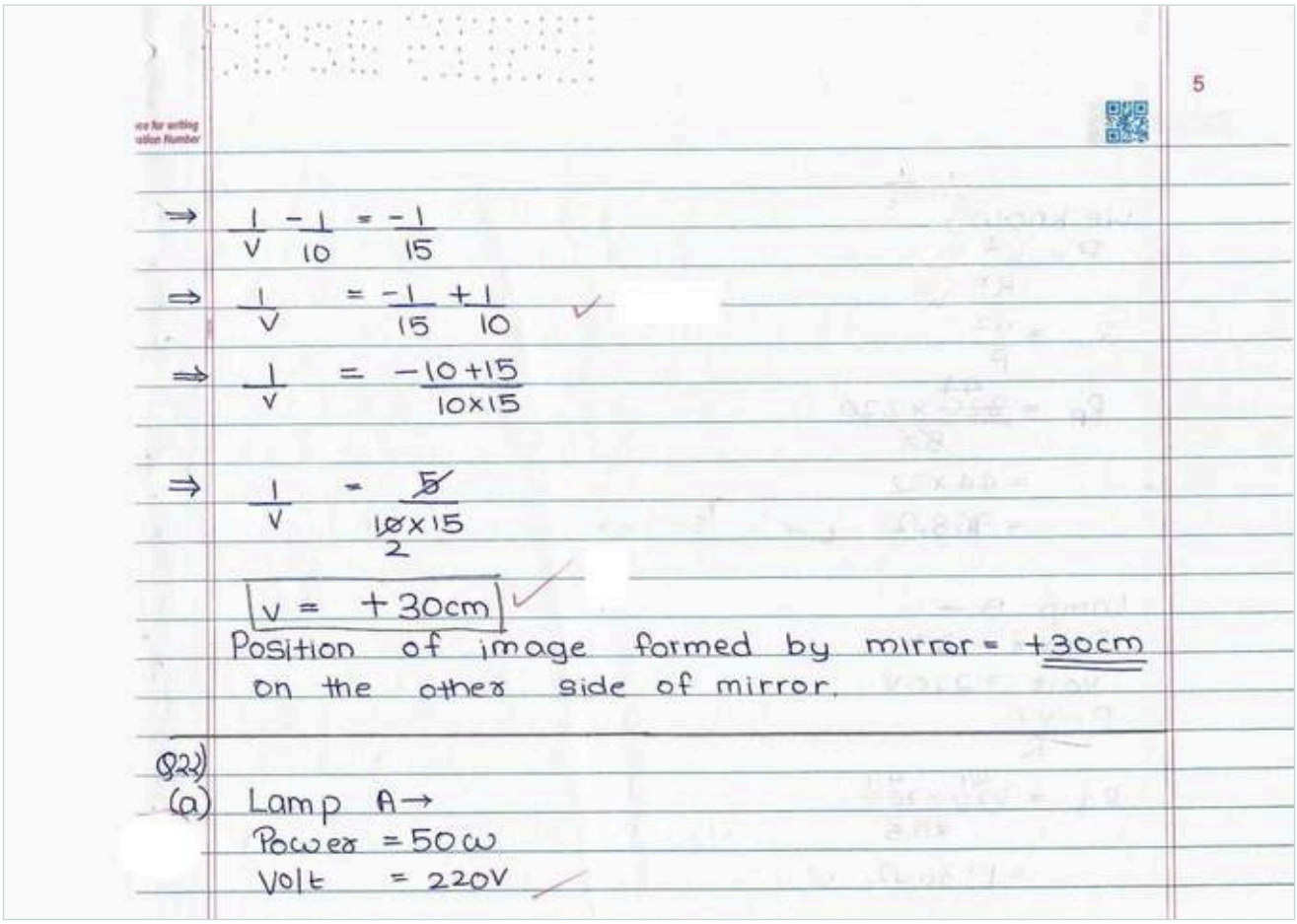
Q.19) (C) Assertion(A) is true, but Reason(R) is false.

Q.20) (B) Both Assertion(A) and Reason(R) are true, but Reason(R) is not the correct explaination of Assertion(A)

Section-B

Q.21) Given -
$u = -10$ cm (object distance)
$f = -15$ cm (Since it is a concave mirror)
To find → v (image distance)
⇒ Mirror formula = $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Real CBSE topper's answer · Board Exam 2025 · similar to Q.35 and Q.33 of this paper (magnification; resistance of rated bulbs)



5

$\Rightarrow \frac{1}{v} - \frac{1}{10} = \frac{-1}{15}$

$\Rightarrow \frac{1}{v} = \frac{-1}{15} + \frac{1}{10}$

$\Rightarrow \frac{1}{v} = \frac{-10+15}{10\times15}$

$\Rightarrow \frac{1}{v} = \frac{5}{\frac{10\times15}{2}}$

$v = +30\text{cm}$

Position of image formed by mirror = +30cm on the other side of mirror.

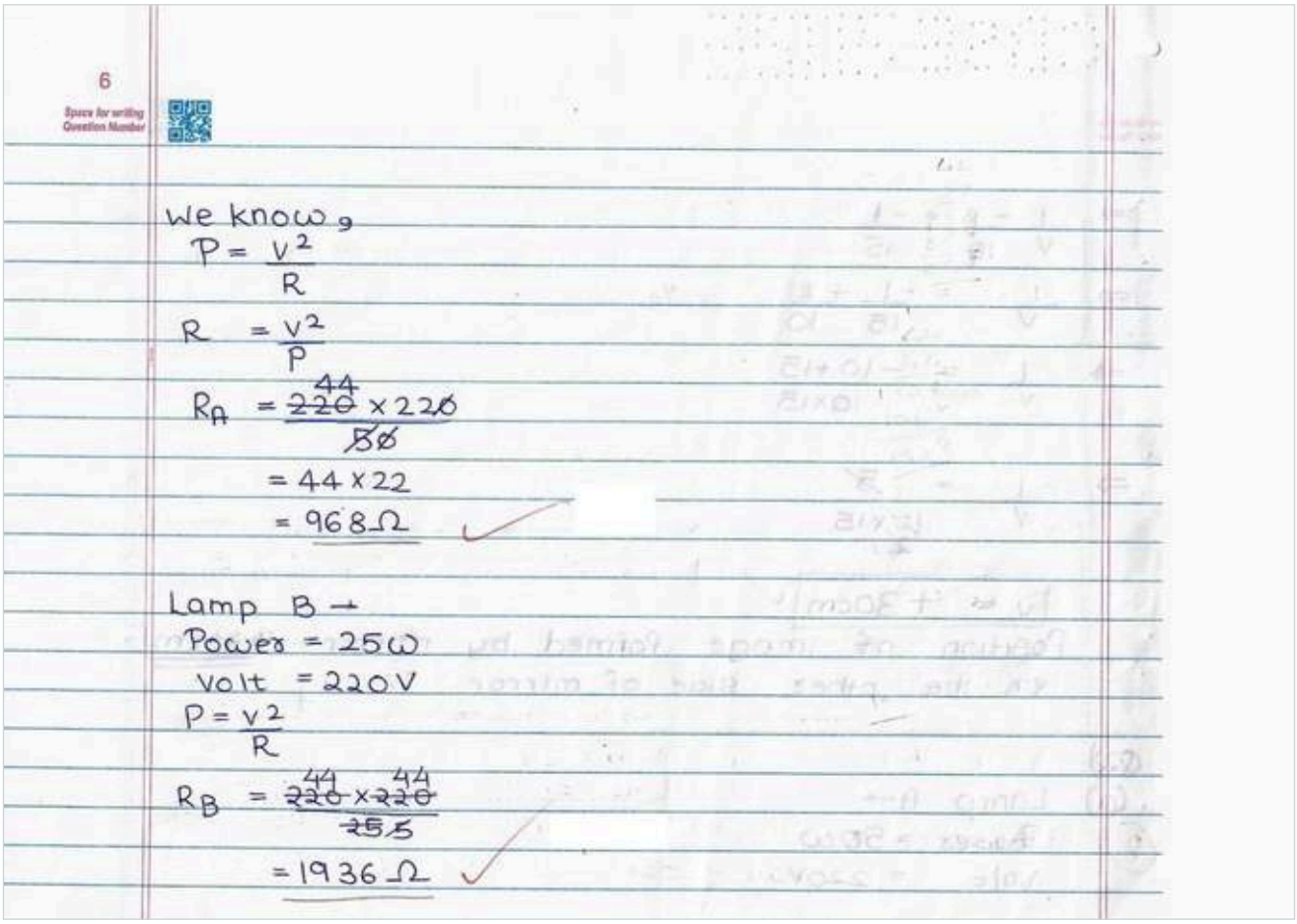
Q22)

(a) Lamp A →

Power = 50 ω

Volt = 220V

Real CBSE topper's answer · Board Exam 2025 · similar to Q.33 of this paper (bulbs rated in W and V: resistance and series use)



6

We know,

$P = \frac{V^2}{R}$

$R = \frac{V^2}{P}$

$R_A = \frac{220}{50} \times 220$

$= 44 \times 22$

$= 968\,\Omega$

Lamp B →

Power = 25 ω

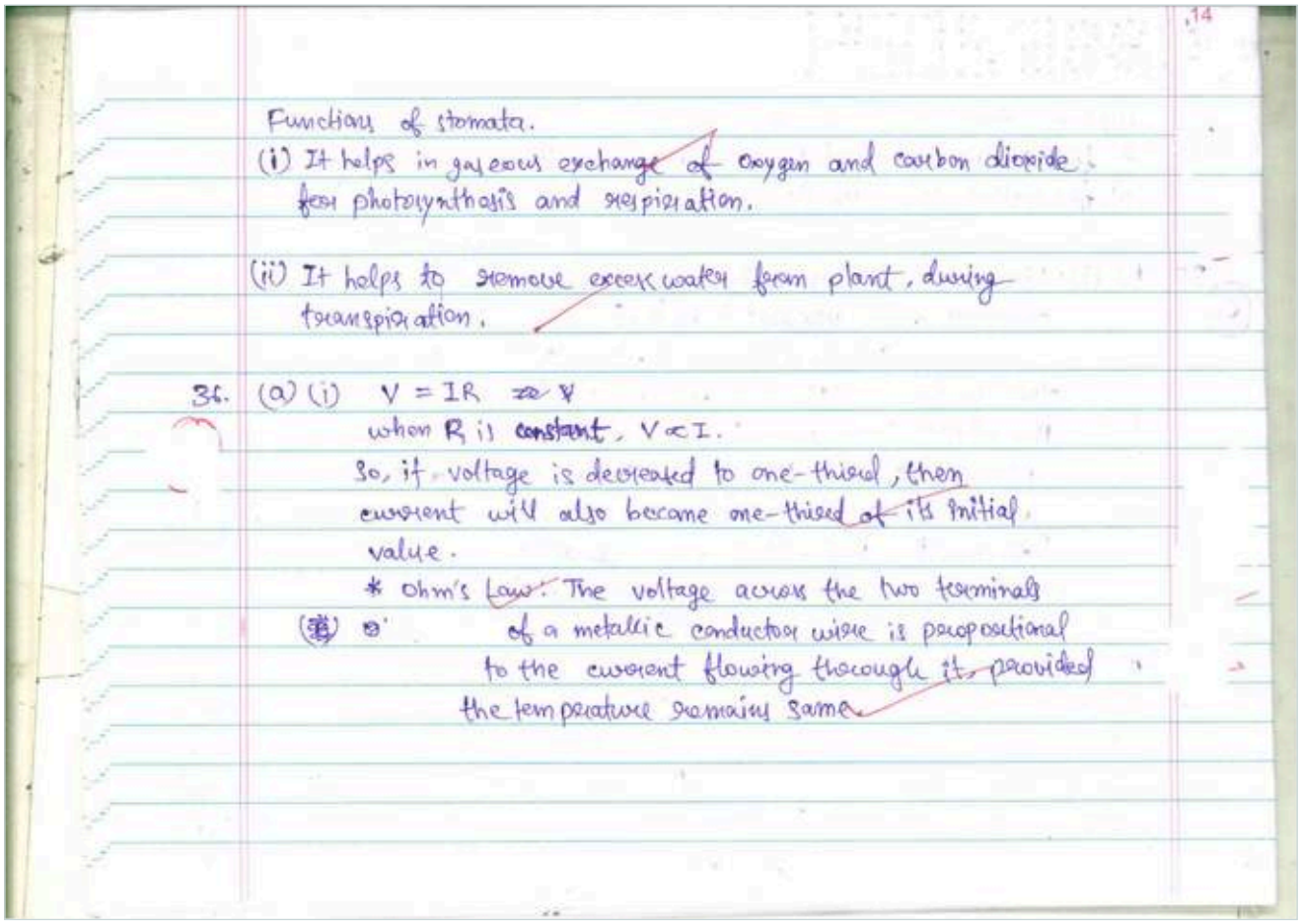
Volt = 220V

$P = \frac{V^2}{R}$

$R_B = \frac{220}{25} \times \frac{220}{5}$

$= 1936\,\Omega$

Real CBSE topper's answer · Board Exam 2024 · similar to Q.10 and Q.36 of this paper (functions of stomata; Ohm's law)



Functions of stomata.

(i) It helps in gaseous exchange of oxygen and carbon dioxide for photosynthesis and respiration.

(ii) It helps to remove excess water from plant, during transpiration.

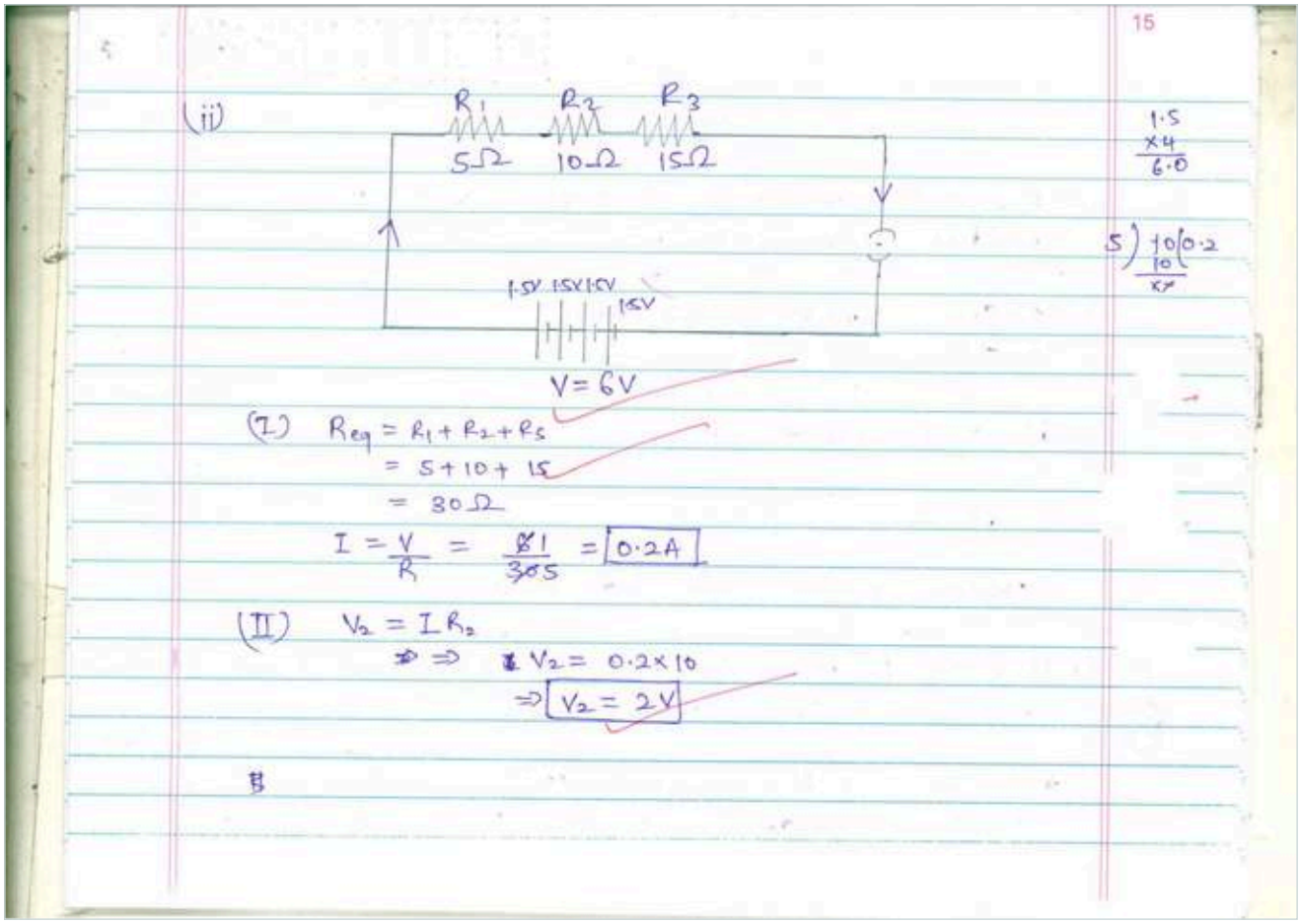
36. (a) (i) $V = IR$ ∝ V

when R is constant, $V \propto I$.

So, if voltage is decreased to one-third, then current will also became one-third of its initial value.

* Ohm's Law: The voltage across the two terminals of a metallic conductor wire is proportional to the current flowing through it, provided the temperature remains same.

Real CBSE topper's answer · Board Exam 2024 · similar to Q.36 of this paper (circuit: equivalent resistance and current)



(ii)

V = 6V

(I) $R_{eq} = R_1 + R_2 + R_3$

$= 5 + 10 + 15$

$= 30\,\Omega$

$I = \frac{V}{R} = \frac{6}{30} = 0.2A$

(II) $V_2 = IR_2$

$\Rightarrow V_2 = 0.2 \times 10$

$\Rightarrow V_2 = 2V$

SAMPLE PAPER 05

SCIENCE · CODE NO. 086

Sample Question Paper 5

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 (1)

Anita coated the lower surface of all the leaves of a potted hibiscus plant with petroleum jelly and left the upper surface uncoated. Which processes of the plant will be affected the most over the next two days?

A. Transpiration and exchange of gases through stomata

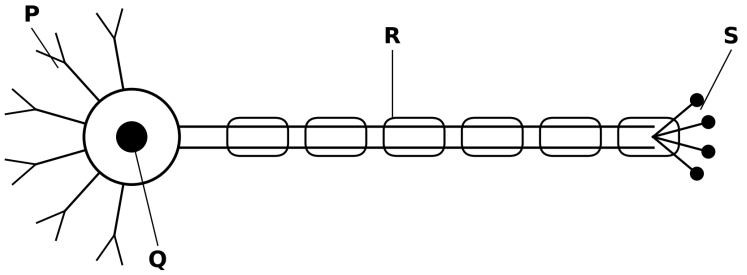
B. Absorption of water by root hairs

C. Transport of food through phloem in the stem

D. Conduction of water through the xylem of roots only

2 (1)

The diagram given below shows a nerve cell with four parts labelled P, Q, R and S.



In which labelled part is the electrical impulse converted into a chemical signal that is passed on to the next nerve cell?

A. P

B. Q

C. R

D. S

For visually impaired students

In which part of a neuron is information from the environment first acquired?

	A. Axon **B.** Dendrite **C.** Cell body **D.** Nerve ending	
3	A plant tissue-culture laboratory wants the cells of a callus to divide rapidly. Which plant hormone should be added to the culture medium for this purpose? **A.** Auxin **B.** Gibberellin **C.** Cytokinin **D.** Abscisic acid	**1**
4	A *Planaria* was cut into three pieces in a laboratory. After a few days, each piece had grown into a complete organism. Which statement best explains this observation? **A.** Each piece produced gametes which fused to form a new individual. **B.** Specialised cells in each piece divided and differentiated to form all the tissues of the body. **C.** Each piece formed a bud which detached from the parent body. **D.** The body of the organism breaks up naturally into many pieces at maturity.	**1**
5	A heterozygous tall pea plant (Tt) is crossed with a dwarf pea plant (tt). Out of 600 plants obtained in the next generation, how many are expected to be dwarf? **A.** 150 **B.** 200 **C.** 300 **D.** 450	**1**
6	Students of a school eco-club sorted the waste collected from the school canteen. Which of the following groups contains only non-biodegradable items? **A.** Polythene bags, aluminium foil, DDT **B.** Paper plates, cotton cloth, leather **C.** Fruit peels, wooden spoons, polythene bags **D.** Glass bottles, cow dung, woollen thread	**1**
7	In a grassland, the grass traps 20,000 J of solar energy. How much energy will be available to the snake in the food chain: Grass → Grasshopper → Frog → Snake? **A.** 2000 J **B.** 200 J **C.** 20 J **D.** 2 J	**1**

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** The opening and closing of stomata in a leaf is controlled by guard cells. **Reason (R):** Guard cells swell and the stomatal pore opens when water flows out of the guard cells.	**1**
9	**Assertion (A):** In human beings, the sex of the child is determined by the genetic contribution of the father. **Reason (R):** A male has one X and one Y chromosome, so half of the sperms carry an X chromosome and the other half carry a Y chromosome.	**1**

10 (2)

Ritu kept a potted seedling inside a closed cardboard box that had a small hole on one side only. After a week, she found that the shoot had bent towards the hole.

A. Name the movement shown by the shoot and the plant hormone responsible for it.

B. Explain how this hormone brings about the bending of the shoot towards the hole.

11 (2)

Students to attempt either option A or B.

A. Banana, rose and sugarcane are commonly grown by vegetative propagation. State two advantages of this method. Why is vegetative propagation the most practical method for growing banana?

OR

B. How is binary fission in *Amoeba* different from binary fission in *Leishmania*? Name the mode of reproduction in which a single cell divides into many daughter cells simultaneously and give one example of an organism showing it.

12 (2)

The ozone layer is often called the protective shield of the Earth.

A. How is ozone formed in the upper atmosphere? Write the chemical equations involved.

B. Name the group of chemicals mainly responsible for the depletion of the ozone layer, and state one step taken at the international level to check it.

13 (3)

To study the action of saliva on starch, Kabir set up four test tubes as shown in the table and tested each tube with iodine solution after 30 minutes.

Tube	Contents	Temperature	Colour with iodine
A	2 mL starch solution + 1 mL saliva	37 °C	No blue-black colour
B	2 mL starch solution + 1 mL boiled and cooled saliva	37 °C	Blue-black colour
C	2 mL starch solution + 1 mL saliva	Kept in ice	Blue-black colour
D	2 mL starch solution + 1 mL water	37 °C	Blue-black colour

A. Why does tube A not give a blue-black colour? Name the enzyme responsible and the product formed.

B. Explain why tubes B and C give a blue-black colour.

C. State the purpose of tube D. Name the enzyme that continues the digestion of the remaining starch in the small intestine.

14 (3)

In one of his experiments, Mendel crossed a pure-breeding pea plant having round seeds with a pure-breeding pea plant having wrinkled seeds. All the F_1 plants had round seeds. On self-pollination of F_1 plants, he obtained 5474 round seeds and 1850 wrinkled seeds in the F_2 generation.

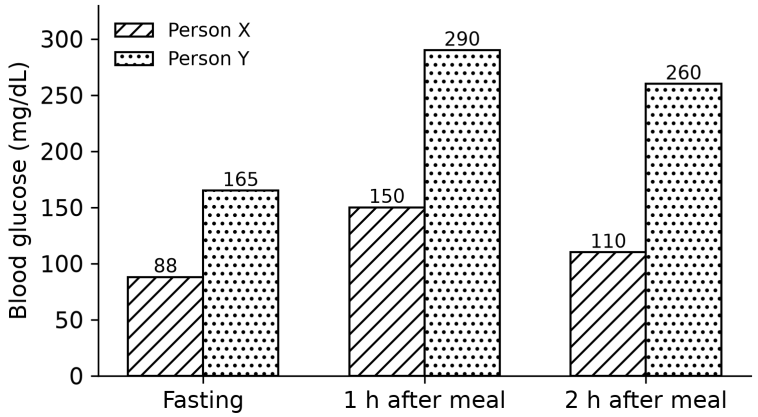
A. Which trait is dominant? Calculate the approximate ratio of round to wrinkled seeds in F_2.

B. Using the symbols R and r, write the genotypes of the two parents and the F_1 plants.

C. Show the F_2 generation with the help of a Punnett square and write the genotypic ratio.

15 (4)

Rohan accompanied his grandfather for a routine health check-up. The doctor measured the blood glucose level of two persons, X and Y, at three different times. The results are shown in the bar graph below. The doctor explained that a fasting blood glucose level above 126 mg/dL indicates a hormonal disorder, and that hormones are released in precise quantities controlled by feedback mechanisms.



A. Which person is most likely suffering from diabetes? Give one reason from the data.

B. Name the hormone and the gland whose deficiency causes this disorder.

Attempt either sub-part C or D.

C. Explain how the timing and amount of this hormone released in the body is regulated.

OR

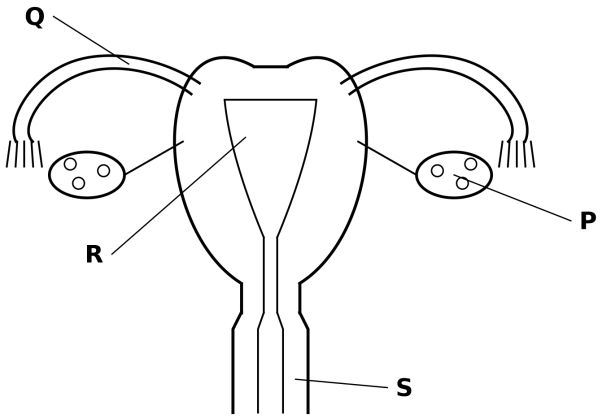
D. Rohan's grandmother has developed a swelling in the neck and the doctor advised her to use iodised salt. Name the disorder, the gland involved and the hormone it makes. State one function of this hormone.

For visually impaired students

Data for the graph: Person X: fasting 88 mg/dL, 1 h after meal 150 mg/dL, 2 h after meal 110 mg/dL. Person Y: fasting 165 mg/dL, 1 h after meal 290 mg/dL, 2 h after meal 260 mg/dL. Answer parts A to D using these data.

16 ***Attempt either option A or B.*** **5**

A. The diagram given below shows the human female reproductive system.



I. Identify the parts labelled P, Q, R and S. In which labelled part does (a) fertilisation take place, and (b) the embryo get implanted?

II. How does the developing embryo get nourishment inside the mother's body? What happens to the inner lining of the uterus if the egg is not fertilised?

OR

B. **I.** Why are the testes located outside the abdominal cavity in the scrotum? Name the hormone produced by the testes and state two of its functions. What is the role of the secretions of the seminal vesicles and the prostate gland?

II. A health worker explains different contraceptive methods to a group of couples. How do (a) condoms and (b) oral pills prevent pregnancy? Which of these two methods also protects against sexually transmitted diseases?

For visually impaired students

A. I. Name the parts of the human female reproductive system where (a) eggs are produced, (b) fertilisation takes place, (c) the embryo gets implanted, and (d) sperms are received during intercourse.

II. How does the developing embryo get nourishment inside the mother's body? What happens to the inner lining of the uterus if the egg is not fertilised?

OR

B. Same as option B above.

SECTION B Chemistry

17 Riya placed a clean iron nail in copper sulphate solution for 30 minutes. Which of the following correctly describes her observation and conclusion? **1**

A. The solution turns from blue to light green and a reddish-brown coating forms on the nail, as iron displaces copper.

B. The solution turns from light green to blue and the nail becomes shiny, as copper displaces iron.

C. There is no change, as copper is more reactive than iron.

D. The solution becomes colourless and a silvery coating forms on the nail.

18 Tooth enamel begins to corrode when the pH in the mouth falls below 5.5. Dentists advise brushing the teeth after eating sweets. This is because toothpaste is: **1**

A. acidic and kills the bacteria present on the teeth.

B. basic and neutralises the acid produced by bacteria from food particles.

C. neutral and simply washes away the food particles.

D. acidic and lowers the pH further to dissolve the food particles.

19 An element X forms an oxide X_2O_3 which reacts with dilute hydrochloric acid as well as with sodium hydroxide solution to form salt and water. The element X and the nature of its oxide are: **1**

A. sodium; basic oxide **B.** aluminium; amphoteric oxide

C. sulphur; acidic oxide **D.** copper; basic oxide

20 The IUPAC name and the functional group present in the compound CH_3COCH_3 are: **1**

A. propanal; aldehyde **B.** propanone; ketone

C. ethanoic acid; carboxylic acid **D.** propanol; alcohol

21 Steam reacts with red-hot iron according to the equation: a Fe + b H_2O → c Fe_3O_4 + d H_2. The values of a, b, c and d in the balanced equation are respectively: **1**

A. 3, 4, 1, 4

B. 3, 3, 1, 3

C. 1, 4, 3, 4

D. 2, 3, 1, 3

22 A hostel warden found that soap did not lather well in the borewell water even after boiling it. Which salt can be added to this water to remove its hardness? The same salt is also used in the glass, soap and paper industries. **1**

A. Baking soda **B.** Washing soda

C. Bleaching powder **D.** Plaster of Paris

23 Sameer noticed that the bottom of his cooking vessel was getting blackened while cooking on a gas stove. The most likely reason is that: **1**

A. the air holes of the burner are blocked and the fuel is not burning completely.
B. the fuel is burning completely with a blue flame.
C. the fuel contains only saturated hydrocarbons.
D. the food is being cooked at a very high temperature.

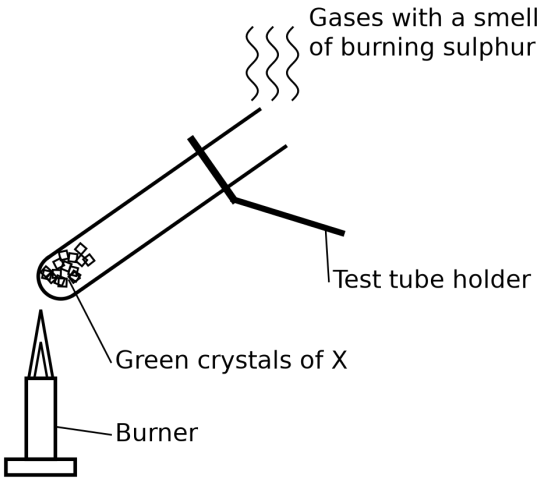
The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Silver articles become black after some time when exposed to air.
Reason (R): Silver reacts with oxygen present in air to form a black coating of silver oxide. **1**

25
A. Draw the electron dot structure of ethene (C_2H_4).
B. Why are most carbon compounds poor conductors of electricity and have low melting and boiling points? **2**

26 Pooja heated a few green crystals of a salt X in a dry boiling tube, as shown below. The crystals first turned white and then a reddish-brown residue was left behind. Gases with a characteristic smell of burning sulphur were given out. **3**



A. Identify the salt X and write the balanced chemical equation for its decomposition.
B. Why did the green crystals turn white before decomposing?
C. Name the type of reaction and give one example of a similar reaction used in industry, with its equation.

For visually impaired students

Pooja heated a few green crystals of a salt X in a dry boiling tube. The crystals first turned white and then a reddish-brown residue was left behind, with the evolution of gases having a smell of burning sulphur. Answer parts A, B and C as above.

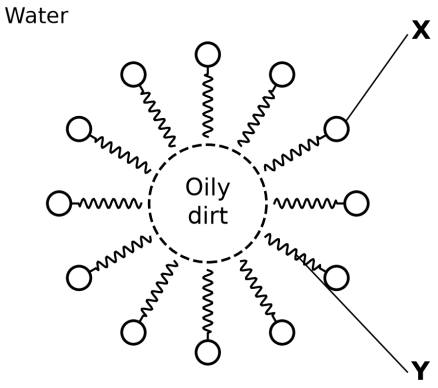
27 ***Students to attempt either option A or B.*** **3**

A. Baking powder is used to make cakes soft and spongy.
(i) What is baking powder made of?
(ii) Write the chemical equation for the reaction that takes place when baking soda is heated.
(iii) Why does a cake made with baking soda alone taste bitter, while one made with baking powder does not?

OR

B. A doctor uses a white powder, mixed with water, to set a fractured bone in the right position.
(i) Name the powder and write its chemical formula.
(ii) Write the chemical equation for its preparation from gypsum.
(iii) Why must this powder be stored in a moisture-proof container? Write the equation for the reaction involved.

28 Meera noticed that her grandmother washes oily kitchen cloths with soap. Her teacher explained that a soap molecule has two ends with different properties. In water, soap molecules arrange themselves into clusters called micelles, which trap the oily dirt, as shown below. **4**



A. In the micelle, identify the end labelled X and the part labelled Y, and state which of them is hydrophobic.

B. How does the formation of micelles help in removing oily dirt from cloth?

Attempt either sub-part C or D.

C. When Meera used the same soap with water drawn from a borewell, very little lather was formed and a white curdy precipitate appeared. Explain this observation. Why is a detergent more suitable for washing clothes in such water?

OR

D. State two differences between soaps and detergents. Mention one environmental problem caused by the use of detergents.

For visually impaired students

A. A soap molecule has an ionic end and a long hydrocarbon tail. Which of these is hydrophobic and which is hydrophilic? Parts B, C and D as above.

29 ***Attempt either option A or B.*** **5**

A. **I.** Pieces of three metals P, Q and R were added separately to solutions of zinc sulphate, copper sulphate and silver nitrate. The observations are recorded below.

Metal	$ZnSO_4$(aq)	$CuSO_4$(aq)	$AgNO_3$(aq)
P	No reaction	Displacement	Displacement
Q	No reaction	No reaction	Displacement
R	Displacement	Displacement	Displacement

(a) Arrange the metals P, Q and R in decreasing order of reactivity.
(b) Which of these metals will not liberate hydrogen gas from dilute hydrochloric acid? Give a reason.
(c) If P is iron, what colour change would be seen when it is added to copper sulphate solution?

II. Zinc is extracted from zinc blende (ZnS). Write the balanced chemical equations for the two main steps involved in obtaining zinc metal from this ore, and name the steps.

OR

B. **I.** Why can metals such as sodium and calcium not be obtained by heating their oxides with carbon? How are such metals extracted? Write the reactions taking place at the cathode and the anode during the extraction of sodium.

II. Draw a labelled diagram of the electrolytic refining of copper. What happens at the anode and at the cathode? What is anode mud?

For visually impaired students

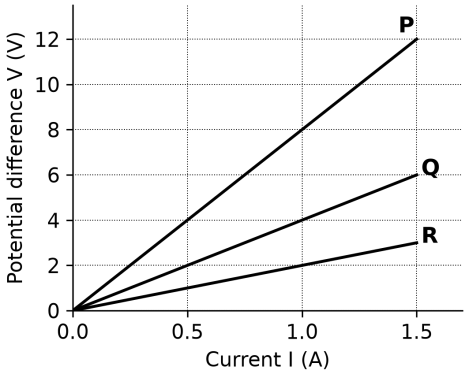
B. II. Describe the electrolytic refining of copper, naming the anode, the cathode and the electrolyte used. What happens at the anode and at the cathode? What is anode mud?

SECTION C Physics

30 The power of a lens used by Mr. Sharma is −2.5 D. The focal length and the nature of the lens are: **1**

A. +40 cm; convex
B. −40 cm; concave
C. −25 cm; concave
D. +25 cm; convex

31 The V–I graphs of three conductors P, Q and R are shown below. **1**



Which of the following statements is correct?

A. P has the highest resistance, equal to 8 Ω.
B. R has the highest resistance, equal to 2 Ω.
C. Q has a resistance of 2 Ω.
D. All three have equal resistance when the current is 1 A.

For visually impaired students

A potential difference of 4 V across a conductor drives a current of 0.5 A through it. The resistance of the conductor is:

A. 0.125 Ω
B. 2 Ω
C. 8 Ω
D. 20 Ω

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Stars appear to twinkle, but planets do not twinkle. **1**
Reason (R): Planets are much closer to the Earth and act as extended sources of light, so the variations in light from their many point sources average out to zero.

33 In a house, the metal body of an electric iron is connected to the earth wire through a three-pin plug. **2**

A. Why is this connection necessary for the safety of the user?

B. What is meant by overloading of an electric circuit? How does an electric fuse protect the circuit in such a case?

34 Name the type of spherical mirror used in each of the following and justify your answer: **2**

A. A dentist's mirror used to see a large image of the teeth.

B. The rear-view mirror of a scooter.

35 An electrician has a wire of length 10 m and area of cross-section 2×10^{-7} m^2, made of an alloy of resistivity 5×10^{-7} Ω m. **3**

A. Calculate the resistance of the wire.

B. The wire is stretched uniformly to double its length. Calculate its new resistance.

C. Will the resistivity of the wire change on stretching? Give a reason.

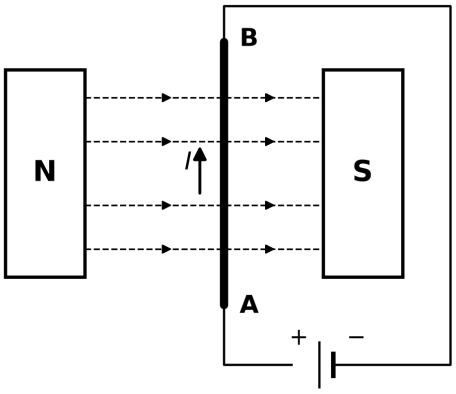
36 ***Students to attempt either option A or B.*** **3**

A. A student can see the blackboard clearly only when sitting in the first row; the far point of his eyes is 2 m.
(i) Name the defect of vision he is suffering from.
(ii) State two possible causes of this defect.
(iii) Find the nature and power of the lens required to correct the defect.

OR

B. (i) Why does the sky appear blue to an observer on the Earth, while it appears dark to an astronaut on the Moon?
(ii) Why are danger signal lights red in colour?
(iii) Why does the Sun appear reddish at sunrise?

37 A straight conductor AB is placed vertically between the poles of a strong magnet, as shown below. A current flows through the conductor from A to B. **3**



A. State Fleming's left-hand rule.

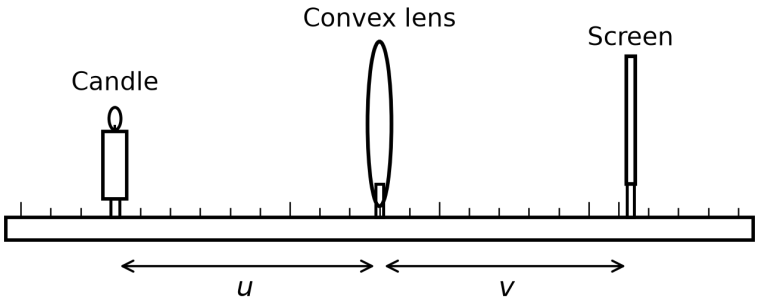
B. In which direction will the conductor AB experience a force?

C. How will the force on the conductor change if (i) the current is increased, and (ii) the directions of both the current and the magnetic field are reversed?

For visually impaired students

A straight conductor AB is held vertically between two magnetic poles, with the north pole on the left and the south pole on the right, so that the magnetic field is horizontal from left to right. A current flows upwards from A to B. Answer parts A, B and C as above.

38 In a school laboratory, Arjun placed a candle, a convex lens and a screen on an optical bench as shown below. For different positions of the candle, he adjusted the screen to get a sharp image and recorded the readings. **4**



Observation	Object distance *u* (cm)	Image distance *v* (cm)
1	−60	+30
2	−40	+40
3	−30	+60
4	−24	+120

A. Using observation 2, find the focal length of the lens. Give a reason.

B. Calculate the magnification in observation 3 and state the nature of the image.

Attempt either sub-part C or D.

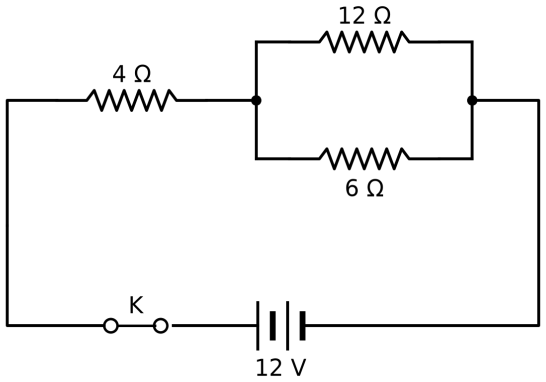
C. Arjun now places the candle 10 cm from the lens. Using the lens formula, find the position of the image. Will he get the image on the screen? Give a reason.

OR

D. Calculate the power of the lens used. In which observation is the image smaller than the object? State its nature.

39 ***Attempt either option A or B.*** **5**

A. **I.** Study the circuit shown below.



Calculate (a) the equivalent resistance of the circuit, (b) the current drawn from the battery, (c) the potential difference across the 4 Ω resistor, and (d) the current through the 12 Ω and the 6 Ω resistors.

II. Give two reasons why electrical appliances in a house are connected in parallel and not in series.

OR

B. **I.** Ishita has three resistors of 6 Ω each. Show, with the help of circuit diagrams, how she should connect them to get an equivalent resistance of (a) 9 Ω and (b) 4 Ω.
II. Draw a schematic circuit diagram to verify Ohm's law, using a battery, a key, a resistor, an ammeter, a voltmeter and a rheostat. In this experiment, the voltmeter reads 2.4 V when the ammeter reads 0.3 A. Find the resistance of the resistor and state the shape of the V–I graph obtained.

For visually impaired students

A. I. A 4 Ω resistor is connected in series with a parallel combination of a 12 Ω and a 6 Ω resistor, across a 12 V battery. Calculate (a) the equivalent resistance, (b) the current drawn from the battery, (c) the potential difference across the 4 Ω resistor, and (d) the current through the 12 Ω and the 6 Ω resistors.
II. Same as above.
OR
B. I. How should three resistors of 6 Ω each be connected to get an equivalent resistance of (a) 9 Ω and (b) 4 Ω? Show the calculation.
II. Name the instruments used to verify Ohm's law and state how the ammeter and voltmeter are connected. The voltmeter reads 2.4 V when the ammeter reads 0.3 A. Find the resistance and state the shape of the V–I graph.

END OF QUESTION PAPER

Quick revision: Electricity

1. Current I = Q/t (ampere); potential difference V = W/Q (volt).
2. Ammeter in series, voltmeter in parallel.
3. Resistance increases with length, decreases with area: R = ρL/A.
4. Series: same current, voltages add. Parallel: same voltage, currents add.
5. Household appliances are in parallel so each gets the full voltage and can be switched on alone.
6. Heating effect is used in the heater, iron, bulb filament and fuse.
7. Fuse wire: low melting point, high resistivity, connected in series.

Nitesh Bhaiya ki tip: circuit diagram me symbols NCERT wale hi banao, aur ammeter series me.

SOLUTION PAPER 05

SCIENCE · CODE NO. 086

Sample Paper 5: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

A. Transpiration and exchange of gases through stomata (1)

(In a dicot leaf like hibiscus, most stomata are on the lower surface; blocking them stops water vapour loss and gaseous exchange.)

2 Ans

D. S (nerve ending) (1)

For VI students: B. Dendrite

3 Ans

C. Cytokinin (1)

4 Ans

B. Specialised cells in each piece divided and differentiated to form all the tissues of the body. (Regeneration) (1)

5 Ans

C. 300 (1)

(Tt × tt gives Tt : tt = 1 : 1, so half of 600 = 300 plants are dwarf.)

6 Ans

A. Polythene bags, aluminium foil, DDT (1)

7 Ans

C. 20 J (1)

(10% law: grasshopper 2000 J → frog 200 J → snake 20 J)

8 Ans

C. A is true but R is false. (1)

(Guard cells swell when water flows into them, opening the pore.)

9 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

10 Ans

A.

Phototropism (positive phototropism of the shoot) (½)

Auxin (½)

B.

Auxin is synthesised at the shoot tip. When light falls on one side of the shoot, auxin diffuses towards the shaded side of the shoot. (½)

The higher concentration of auxin makes the cells on the shaded side elongate more, so the shoot bends towards light (the hole). (½)

Total 2 marks

11 Ans

A.

Advantages (any two): plants bear flowers and fruits earlier than those grown from seeds; all plants are genetically similar to the parent, so desirable characters are preserved; plants that have lost the capacity to produce seeds can be propagated. (1)

Banana (cultivated variety) does not produce viable seeds, so vegetative propagation is the only practical method of growing it. (1)

OR

B.

In *Amoeba*, the cell can split into two in any plane, whereas in *Leishmania* splitting takes place in a definite (longitudinal) plane with respect to the whip-like flagellum at one end. (1)

Multiple fission (½)

Example: *Plasmodium* (malarial parasite) (½)

Total 2 marks

12

Ans

A.

High-energy ultraviolet (UV) radiation splits oxygen molecules into free oxygen atoms: $O_2 \rightarrow O + O$ (in presence of UV) (½)

A free oxygen atom combines with an oxygen molecule to form ozone: $O + O_2 \rightarrow O_3$ (½)

B.

Chlorofluorocarbons (CFCs), used as refrigerants and in fire extinguishers (½)

In 1987, the United Nations Environment Programme (UNEP) reached an agreement to freeze CFC production at 1986 levels (Montreal Protocol); manufacture of CFC-free refrigerators. (any one) (½)

Total 2 marks

13

Ans

A.

Salivary amylase present in saliva has completely digested the starch into sugar, so no starch is left to give a blue-black colour with iodine. (½)

Enzyme: salivary amylase; product: maltose (sugar) (½)

B.

Tube B: boiling destroys (denatures) the enzyme, so starch remains undigested. (½)

Tube C: the enzyme becomes inactive at very low temperature, so starch is not digested. (½)

C.

Tube D is the control; it shows that starch is not broken down without the enzyme. (½)

Pancreatic amylase (present in pancreatic juice) (½)

Total 3 marks

14

Ans

A.

Round seed shape is dominant. (½)

Ratio = 5474 : 1850 ≈ 2.96 : 1 ≈ 3 : 1 (½)

B.

Parents: RR (round) and rr (wrinkled) (½)

F_1: Rr (round) (½)

C.

Punnett square ($F_1 \times F_1$: Rr × Rr): (½)

Gametes	R	r
R	RR (round)	Rr (round)
r	Rr (round)	rr (wrinkled)

Genotypic ratio RR : Rr : rr = 1 : 2 : 1 (½)

Total 3 marks

15

Ans

A.

Person Y. Fasting glucose level of Y (165 mg/dL) is above 126 mg/dL and remains very high (260 mg/dL) even 2 hours after the meal. (1)

B.

Hormone: insulin (½)

Gland: pancreas (½)

C.

The timing and amount of insulin released are regulated by a feedback mechanism. (½)

When the blood sugar level rises (e.g. after a meal), it is detected by the cells of the pancreas, which respond by producing more insulin. (1)

As the blood sugar level falls, insulin secretion is reduced. (½)

OR

D.

Disorder: goitre (swollen neck) (½)

Gland: thyroid gland; hormone: thyroxin (½)

Iodine is essential for the synthesis of thyroxin. Thyroxin regulates carbohydrate, protein and fat metabolism in the body so as to provide the best balance for growth. (1)

Total 4 marks

16

Ans

A. I.

P – Ovary; Q – Oviduct (fallopian tube); R – Uterus; S – Vagina (½ × 4) (2)

(a) Fertilisation: Q (oviduct); (b) Implantation: R (uterus) (1)

II.

The embryo gets nutrition from the mother's blood through the placenta, a disc-like special tissue embedded in the uterine wall. It has villi on the embryo's side which provide a large surface area for glucose and oxygen to pass from the mother to the embryo (and wastes to pass back). (1)

If the egg is not fertilised, the thick, spongy inner lining of the uterus slowly breaks and comes out through the vagina as blood and mucus; this is menstruation (lasts about 2 to 8 days, every month). (1)

VI students: (a) ovary (b) oviduct / fallopian tube (c) uterus (d) vagina; part II as above.

OR

B. I.

Sperm formation requires a temperature lower than the normal body temperature; the scrotum keeps the testes at this lower temperature. (1)

Hormone: testosterone. Functions: regulates the formation of sperms; brings about changes in appearance (secondary sexual characters) in boys at puberty. (1)

Secretions of the seminal vesicles and prostate gland are added to the sperms; they make transport of sperms easier and also provide nutrition to them. (1)

II.

(a) Condoms act as a mechanical barrier and prevent the sperm from reaching the egg. (½)

(b) Oral pills change the hormonal balance of the body so that eggs are not released. (½)

Condoms also protect against sexually transmitted diseases such as HIV-AIDS, gonorrhoea and syphilis, as they prevent the exchange of body fluids. (1)

Total 5 marks

SECTION B Chemistry

17

Ans

A. The solution turns from blue to light green and a reddish-brown coating forms on the nail, as iron displaces copper. (1)

($Fe + CuSO_4 \rightarrow FeSO_4 + Cu$)

18

Ans

B. basic and neutralises the acid produced by bacteria from food particles. (1)

19 **Ans**

B. aluminium; amphoteric oxide (1)

($Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$; $Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$)

20 **Ans**

B. propanone; ketone (1)

21 **Ans**

A. 3, 4, 1, 4 (1)

($3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$)

22 **Ans**

B. Washing soda ($Na_2CO_3 \cdot 10H_2O$) (1)

23 **Ans**

A. the air holes of the burner are blocked and the fuel is not burning completely. (1)

24 **Ans**

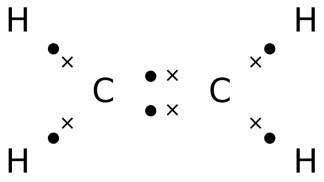
C. A is true but R is false. (1)

(Silver reacts with sulphur compounds in air to form a black coating of silver sulphide, Ag_2S.)

25 **Ans**

A.

Electron dot structure of ethene showing a double bond (two shared pairs) between the carbon atoms and one shared pair between each carbon and hydrogen atom: (1)



B.

Carbon compounds are formed by covalent bonding (sharing of electrons), so they do not form ions and hence are poor conductors of electricity. (½)

The intermolecular forces of attraction between their molecules are weak, so they have low melting and boiling points. (½)

Total 2 marks

26 **Ans**

A.

X is ferrous sulphate crystals ($FeSO_4 \cdot 7H_2O$). (½)

$2FeSO_4(s) \xrightarrow{Heat} Fe_2O_3(s) + SO_2(g) + SO_3(g)$ (½)

B.

On heating, the crystals lose their water of crystallisation ($FeSO_4 \cdot 7H_2O \rightarrow FeSO_4 + 7H_2O$), so the green colour changes to white anhydrous ferrous sulphate. (1)

C.

Thermal decomposition reaction (½)

Decomposition of limestone to quicklime used in the cement industry: $CaCO_3(s) \xrightarrow{Heat} CaO(s) + CO_2(g)$ (½)

Total 3 marks

27 **Ans**

A.

(i) Baking powder is a mixture of baking soda (sodium hydrogencarbonate) and a mild edible acid such as tartaric acid. (1)

(ii) $2NaHCO_3 \xrightarrow{Heat} Na_2CO_3 + H_2O + CO_2$ (1)

(iii) Baking soda alone forms sodium carbonate on heating, which is bitter. In baking powder, the tartaric acid neutralises it; $NaHCO_3 + H^+ \rightarrow CO_2 + H_2O$ + sodium salt of the acid. The CO_2 makes the cake soft and spongy. (1)

OR

B.

(i) Plaster of Paris; $CaSO_4 \cdot \frac{1}{2}H_2O$ (calcium sulphate hemihydrate) (1)

(ii) $CaSO_4 \cdot 2H_2O \xrightarrow{373\ K} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$ (1)

(iii) Plaster of Paris absorbs moisture and sets into a hard mass of gypsum, becoming useless: $CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O \rightarrow CaSO_4 \cdot 2H_2O$ (1)

Total 3 marks

28

Ans

A.

X is the ionic (carboxylate) end, which is hydrophilic; Y is the long hydrocarbon tail, which is hydrophobic. (1)

B.

The hydrocarbon tails of soap molecules dissolve in the oily dirt and the ionic ends face the water, so the oily dirt is collected in the centre of the micelle. The micelles stay in solution as an emulsion and the dirt is washed away with water. (1)

C.

Borewell water is hard; it contains calcium and magnesium salts. (½)

Soap reacts with these salts to form an insoluble substance called scum (curdy precipitate), so it does not form lather and a lot of soap is wasted. (½)

Detergents (ammonium or sulphonate salts of long-chain carboxylic acids) do not form insoluble precipitates with calcium and magnesium ions, so they remain effective in hard water. (1)

OR

D.

Soaps are sodium/potassium salts of long-chain carboxylic acids, whereas detergents are ammonium or sulphonate salts of long-chain carboxylic acids. (½)

Soaps form scum with hard water and are not effective in it, whereas detergents are effective in hard water. (½)

Some detergents are non-biodegradable; they cause foaming in rivers and water bodies and pollute water. (any one relevant problem) (1)

Total 4 marks

29

Ans

A. I.

(a) R > P > Q (1)

(b) Q; Q is less reactive than copper (it cannot displace copper), so it lies below hydrogen in the reactivity series and cannot displace hydrogen from dilute acids. (1)

(c) The blue colour of copper sulphate solution changes to pale green (and a reddish-brown deposit of copper forms on iron). (1)

II.

Roasting: $2ZnS(s) + 3O_2(g) \xrightarrow{Heat} 2ZnO(s) + 2SO_2(g)$ (1)

Reduction: $ZnO(s) + C(s) \rightarrow Zn(s) + CO(g)$ (1)

OR

B. I.

Sodium and calcium are very reactive and have more affinity for oxygen than carbon, so carbon cannot reduce their oxides. (½)

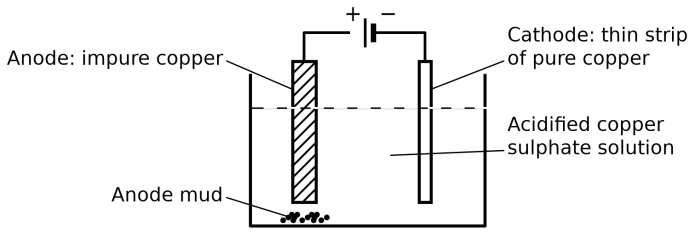
They are obtained by electrolytic reduction (electrolysis) of their molten chlorides. (½)

At cathode: $Na^+ + e^- \rightarrow Na$ (½)

At anode: $2Cl^- \rightarrow Cl_2 + 2e^-$ (½)

II.

Labelled diagram of electrolytic refining of copper: (1)



At the anode, impure copper dissolves into the electrolyte ($Cu \rightarrow Cu^{2+} + 2e^-$). At the cathode, an equivalent amount of pure copper from the electrolyte is deposited ($Cu^{2+} + 2e^- \rightarrow Cu$). (1)

Insoluble impurities settle down at the bottom of the anode; this is known as anode mud. (1)

VI students: anode – impure copper; cathode – thin strip of pure copper; electrolyte – acidified copper sulphate solution; rest as above.

Total 5 marks

SECTION C Physics

30 **Ans**

B. −40 cm; concave (1)

(f = 1/P = 1/(−2.5) m = −0.4 m = −40 cm)

31 **Ans**

A. P has the highest resistance, equal to 8 Ω. (1)

(Slope of V–I graph = R; P: 12 V / 1.5 A = 8 Ω)

For VI students: C. 8 Ω

32 **Ans**

A. Both A and R are true, and R is the correct explanation of A. (1)

33 **Ans**

A.

The earth wire provides a low-resistance conducting path for current. If there is leakage of current to the metal body, it flows to the earth and the potential of the body stays equal to that of the earth, so the user does not get a severe electric shock. (1)

B.

Overloading occurs when too many appliances are connected to a single socket, or when the supply voltage rises, so that the current drawn becomes too large. (½)

The large current heats the fuse wire, which melts (due to Joule heating) and breaks the circuit, preventing damage to the appliances and the wiring. (½)

Total 2 marks

34 **Ans**

A.

Concave mirror (½)

When the teeth are held between the pole and the focus, it forms an enlarged, erect (virtual) image. (½)

B.

Convex mirror (½)

It always forms an erect, diminished image and has a wider field of view, so the driver can see a large area of the traffic behind. (½)

Total 2 marks

35 Ans

A.

R = ρl/A (½)

$R = (5 \times 10^{-7}\ \Omega\ m \times 10\ m) / (2 \times 10^{-7}\ m^2) = 25\ \Omega$ (1)

B.

On stretching to double the length, the volume stays the same, so the area becomes half: $l' = 2l$, $A' = A/2$. (½)

$R' = \rho(2l)/(A/2) = 4R = 4 \times 25\ \Omega = 100\ \Omega$ (½)

C.

No. Resistivity depends only on the nature of the material (and temperature), not on its length or area of cross-section. (½)

Total 3 marks

36 Ans

A.

(i) Myopia (near-sightedness) (½)

(ii) Excessive curvature of the eye lens; elongation of the eyeball. (1)

(iii) Concave (diverging) lens. f = −(far point) = −2 m (½)

P = 1/f = 1/(−2 m) = −0.5 D (1)

OR

B.

(i) Fine particles in the Earth's atmosphere scatter shorter wavelengths (blue) much more strongly than red, and this scattered blue light enters our eyes. On the Moon there is no atmosphere, so there is no scattering and the sky appears dark. (1)

(ii) Red light has the longest wavelength and is scattered the least by fog or smoke, so it can be seen from a long distance. (1)

(iii) At sunrise, light from the Sun travels a longer distance through the atmosphere; most of the blue light and shorter wavelengths are scattered away, and mainly the red light reaches our eyes. (1)

Total 3 marks

37 Ans

A.

Stretch the thumb, forefinger and middle finger of the left hand so that they are mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of the current, then the thumb points in the direction of the force (motion) on the conductor. (1)

B.

The force on AB acts perpendicular to the plane of the paper, directed into the page (away from the reader). (1)

C.

(i) The force (displacement of the conductor) increases. (½)

(ii) The direction of the force remains the same (into the page), as reversing both the current and the field does not change it. (½)

Total 3 marks

38 Ans

A.

When u = v = 40 cm, the object is at 2F, so 2f = 40 cm and f = 20 cm. (1)

B.

m = v/u = (+60)/(−30) = −2 (½)

The image is real, inverted and magnified (twice the size of the object). (½)

C.

$1/v - 1/u = 1/f \Rightarrow 1/v = 1/20 + 1/(-10) = -1/20 \Rightarrow v = -20$ cm (1)

No. The image is virtual (formed on the same side as the candle, 20 cm from the lens), erect and magnified; a virtual image cannot be obtained on a screen. (1)

OR

D.

P = 1/f (in m) = 1/(0.20 m) = +5 D (1)

Observation 1 (u = −60 cm, v = +30 cm, m = −0.5) (½)

The image is real, inverted and diminished. (½)

Total 4 marks

39

Ans

A. I.

(a) Parallel combination: $1/R_p = 1/12 + 1/6 = 3/12 \Rightarrow R_p = 4\ \Omega$; equivalent resistance R = 4 Ω + 4 Ω = 8 Ω (1)

(b) I = V/R = 12 V / 8 Ω = 1.5 A (½)

(c) $V_{4\Omega} = IR = 1.5\ A \times 4\ \Omega = 6\ V$ (½)

(d) p.d. across the parallel combination = 12 V − 6 V = 6 V; $I_{12\Omega} = 6/12 = 0.5$ A; $I_{6\Omega} = 6/6 = 1$ A (1)

II.

In parallel, each appliance gets the full supply voltage (220 V) and can be switched on or off independently. (1)

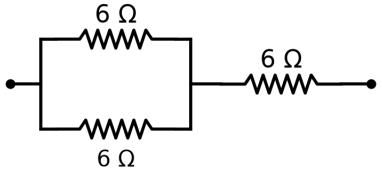
If one appliance fails, the others continue to work; also, each appliance draws the current it needs according to its own resistance, and the total resistance of the circuit is reduced. (1)

OR

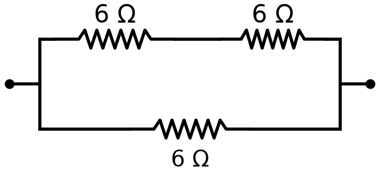
B. I.

(a) Two 6 Ω resistors in parallel (3 Ω) connected in series with the third: R = 3 Ω + 6 Ω = 9 Ω (1)

(b) Two 6 Ω resistors in series (12 Ω) connected in parallel with the third: R = (12 × 6)/(12 + 6) = 4 Ω (1)



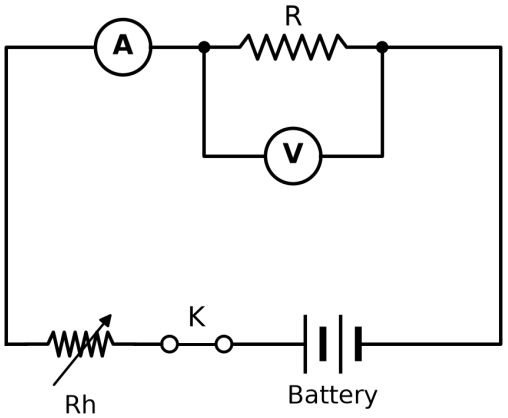
(a) 3 Ω + 6 Ω = 9 Ω



(b) (12 × 6)/(12 + 6) = 4 Ω

II.

Circuit diagram for verifying Ohm's law (ammeter in series, voltmeter in parallel with the resistor R): (1)



R = V/I = 2.4 V / 0.3 A = 8 Ω (1)

The V–I graph is a straight line passing through the origin. (1)

VI students (B. II): ammeter, voltmeter, battery, key, rheostat and resistor; ammeter in series and voltmeter in parallel with the resistor; R = 8 Ω; straight-line graph through the origin.

Total 5 marks

END OF SOLUTIONS

Common mistakes: Chemistry answers

1. Writing word equations when a balanced chemical equation is asked.
2. Mixing up oxidation (gain of oxygen or loss of hydrogen) and reduction.
3. Calling every reaction with heat a decomposition reaction; check what is formed.
4. pH: below 7 is acidic, above 7 is basic. A lower pH means a stronger acid.
5. Plaster of Paris is $CaSO_4 \cdot \frac{1}{2}H_2O$; gypsum is $CaSO_4 \cdot 2H_2O$.
6. Baking soda is $NaHCO_3$; washing soda is $Na_2CO_3 \cdot 10H_2O$.
7. Electron-dot structures with the wrong number of shared pairs.
8. Saying metals react with water to give acids: they give hydroxides (bases) and hydrogen.

Nitesh Bhaiya ki tip: formula ke saath naam bhi likho, examiner dono dekhta hai.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.10 of this paper (phototropism and plant movements)

22. a)

Movement of leaves of a sensitive plant	Movement of a shoot towards light
1. This is not a growth related movement and is due to a stimulus (Nastic movement)	1. This is a growth related movement (Tropic movement)
2. This is non-directional.	2. This is directional movement.
3. Movement is carried out by movement of water which is taken from the leaves to other part. This causes the leaf to shrink.	3. Movement is carried out due to hormone like Auxin which diffuses to area of shoot away from sunlight and regulates cell elongation in area of shoot present in the shade, which causes the plant to bend towards sunlight.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.13 of this paper (action of saliva on starch)

23. The enzyme salivary amylase is present in the fluid in our mouth cavity. Salivary gland produces the salivary amylase.
In the absence of this enzyme, digestion of complex starch to simple sugar (Maltose) would be difficult, thus affecting the digestive process.
Along with it, the ~~ing~~ swallowing of food would also be difficult as salivary amylase makes the food wet to easily flow down the oesophagus.

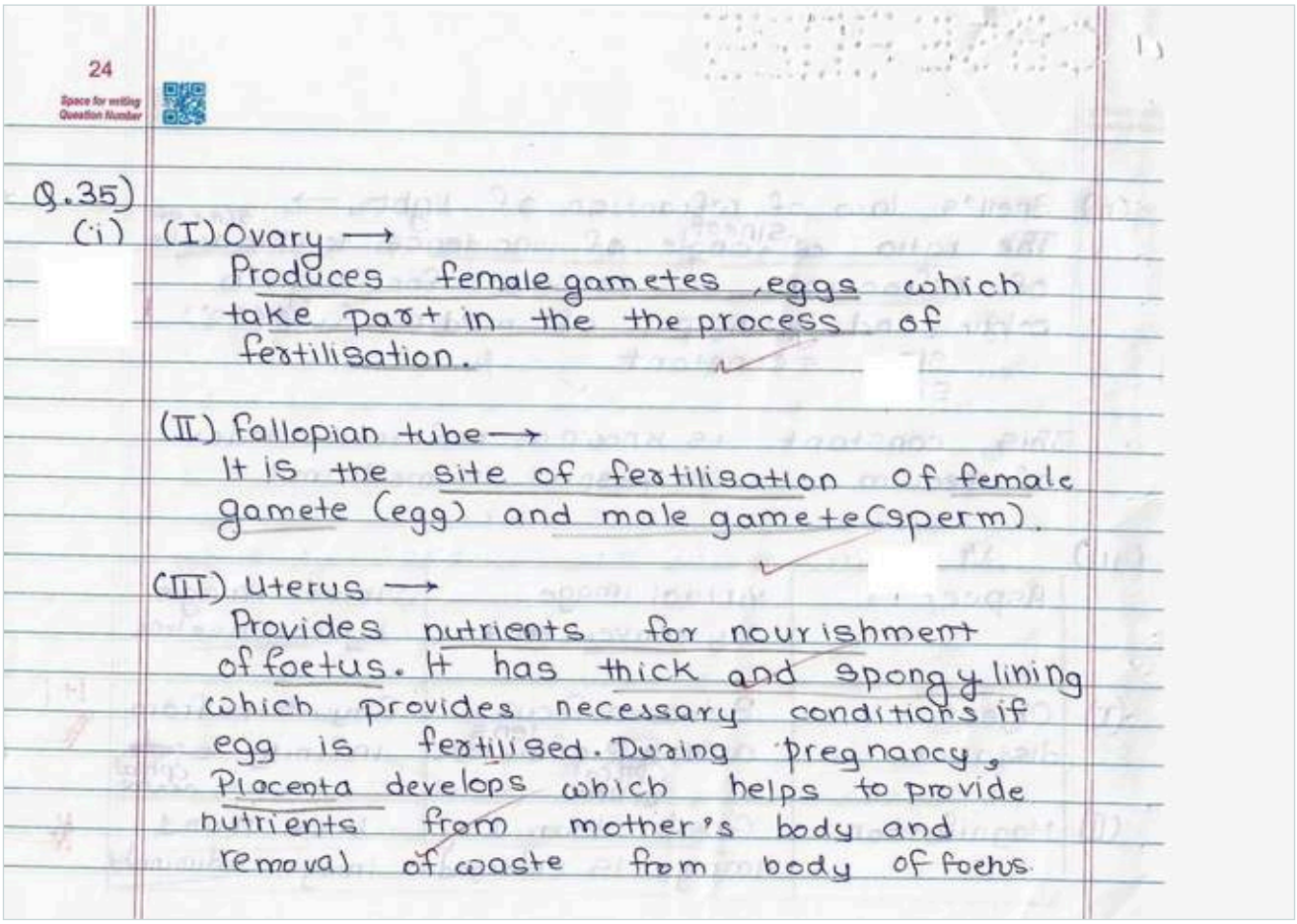
5

24. The above stated question can be solved by ~~Eh~~ Ohm's Law, which indicates that potential difference (V) across the conductor is directly proportional to the current (I) flowing through it.

$V \propto I$ or $V = IR$

where R is the constant and is the characteristic

Real CBSE topper's answer · Board Exam 2025 · similar to Q.16 of this paper (human female reproductive system)



24

Space for writing Question Number

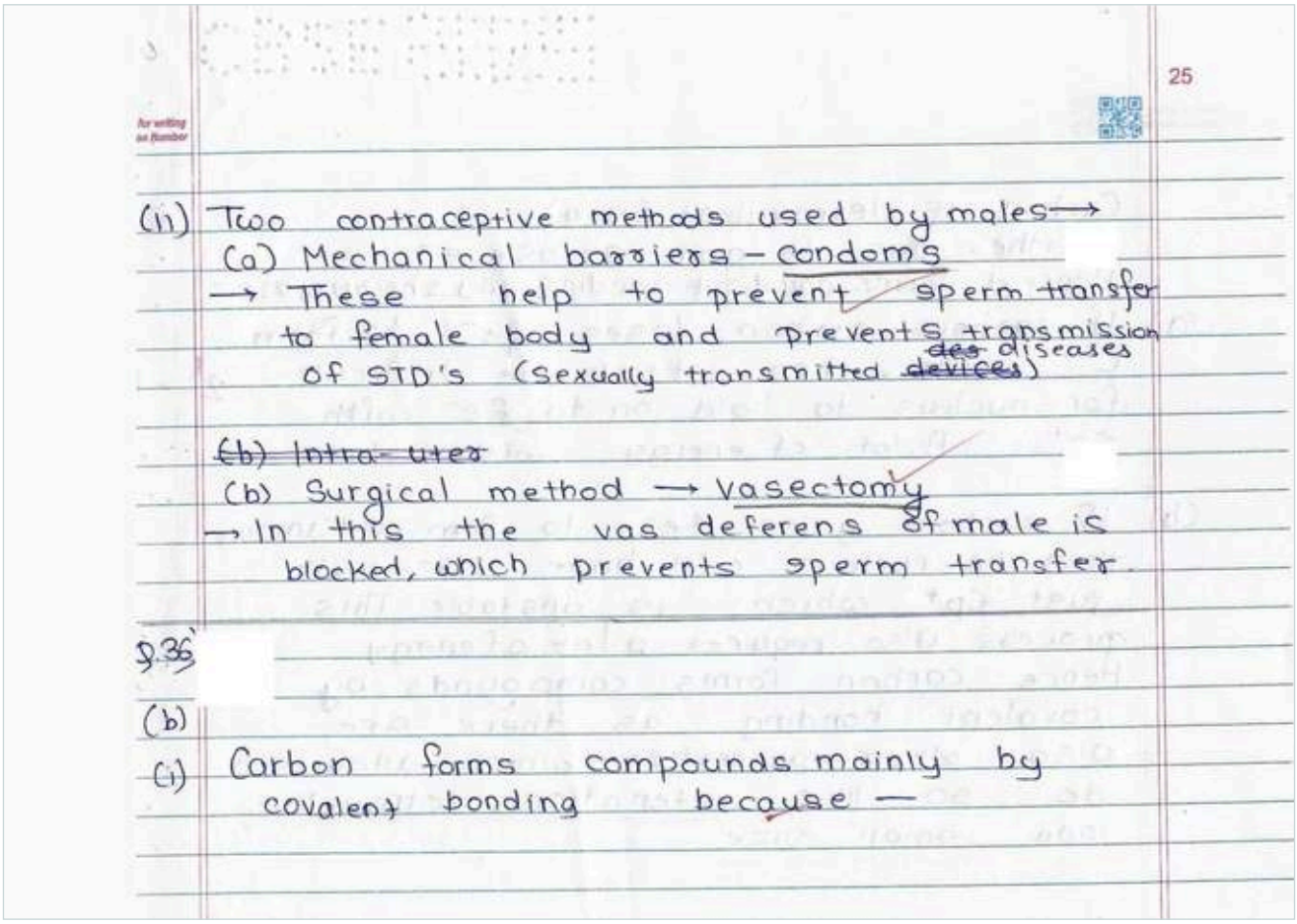
Q.35)

(i) (I) Ovary →
Produces female gametes, eggs which take part in the the process of fertilisation.

(II) Fallopian tube →
It is the site of fertilisation of female gamete (egg) and male gamete(sperm).

(III) Uterus →
Provides nutrients for nourishment of foetus. It has thick and spongy lining which provides necessary conditions if egg is fertilised. During pregnancy, Placenta develops which helps to provide nutrients from mother's body and removal of waste from body of foetus.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.16 and Q.25 of this paper (reproductive health; covalent carbon compounds)



25

(ii) Two contraceptive methods used by males →
(a) Mechanical barriers – Condoms
→ These help to prevent sperm transfer to female body and prevents transmission of STD's (Sexually transmitted diseases)

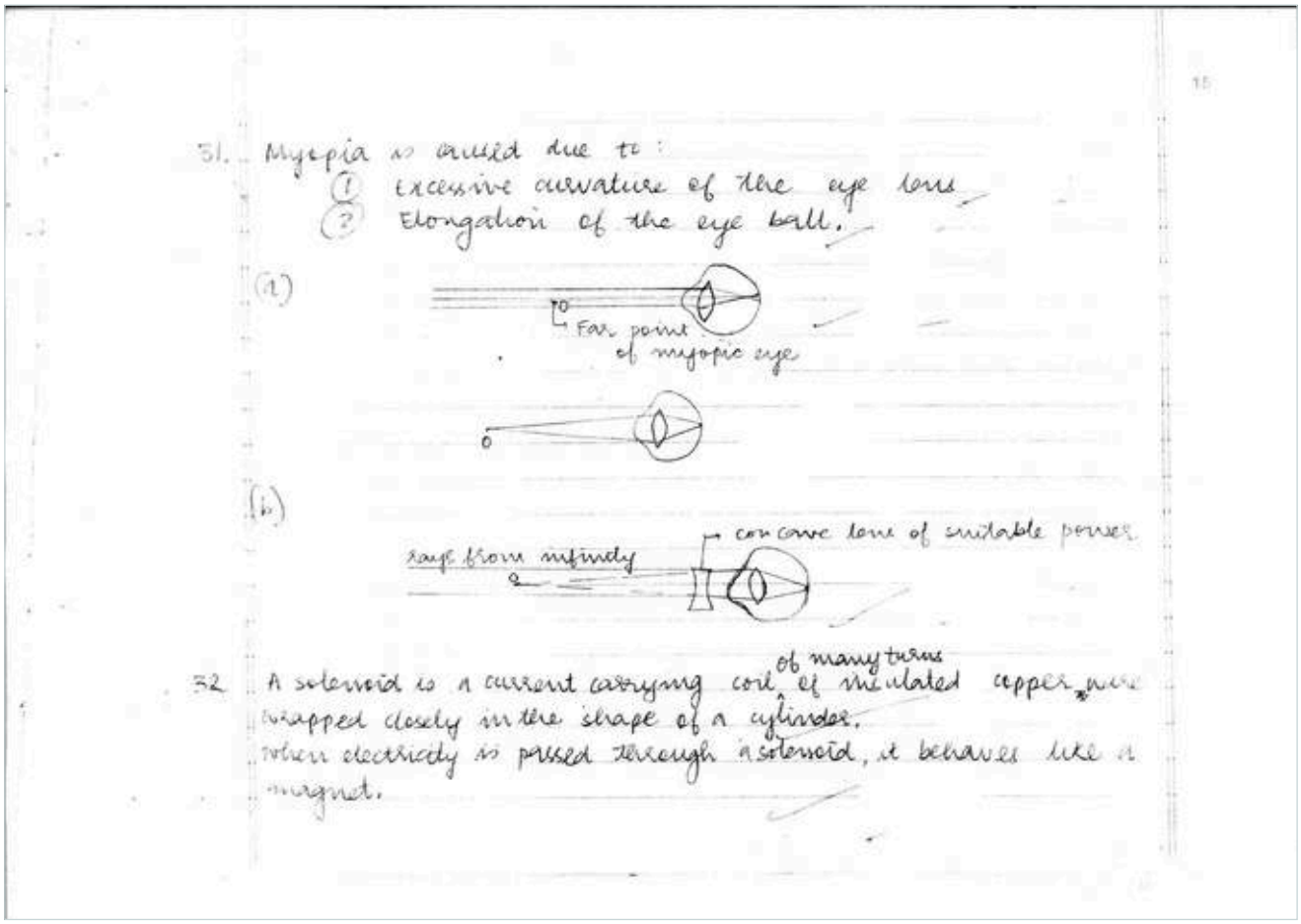
~~(b) Intra-uter~~
(b) Surgical method → Vasectomy
→ In this the vas deferens of male is blocked, which prevents sperm transfer.

Q.36

(b)

(i) Carbon forms compounds mainly by covalent bonding because —

Real CBSE topper's answer · Board Exam 2023 · similar to Q.36 of this paper (myopia, its causes and correction)



What to learn from this topper's answer

1. The myopia answer names the defect, gives both causes (elongated eyeball, excess curvature of the eye lens) and the correcting concave lens.
2. Two labelled ray diagrams, the defective eye and the corrected eye, collect the diagram marks; the far point is clearly marked.
3. The saliva answer links the observation to the reason in one line: starch is digested by salivary amylase, so there is no blue-black colour.
4. Functions of the reproductive organs are written one organ per line, so each part can be ticked separately.

SAMPLE PAPER 06

SCIENCE · CODE NO. 086

Sample Question Paper 6

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 (1)

Manjeet's doctor finds that the gastric glands in his stomach secrete very little hydrochloric acid. Which of the following processes will be most directly affected?

A. Digestion of proteins, because pepsin acts only in an acidic medium

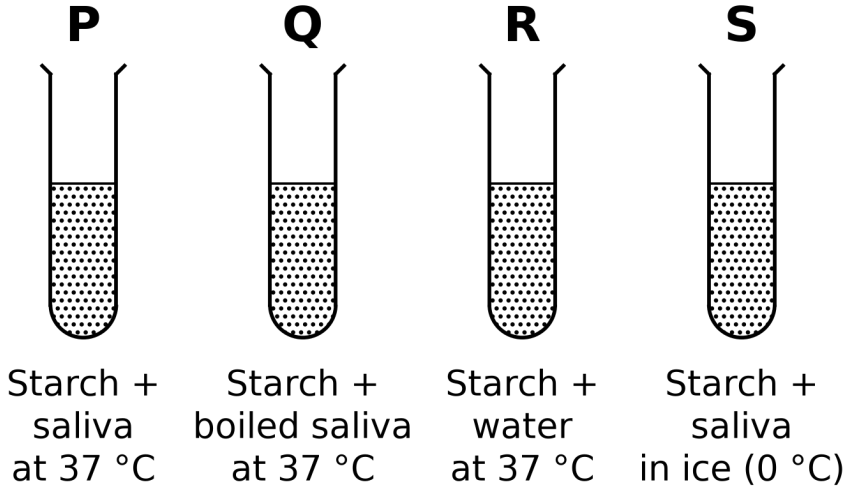
B. Emulsification of fats, because bile is not released

C. Digestion of starch in the mouth, because salivary amylase is not activated

D. Absorption of digested food, because the villi stop working

2 (1)

Neha set up four test tubes containing equal amounts of starch solution, as shown below, and kept them for 30 minutes. She then added two drops of dilute iodine solution to each test tube.



In which test tube(s) will the blue-black colour **not** appear?

	A. P only **B.** P and S **C.** Q and R **D.** P, Q and S **For visually impaired students** Neha took four test tubes containing starch solution. To P she added saliva and kept it at 37 °C; to Q she added boiled saliva and kept it at 37 °C; to R she added water and kept it at 37 °C; to S she added saliva and kept it in ice. After 30 minutes she added iodine solution to each. In which test tube(s) will the blue-black colour not appear? **A.** P only **B.** P and S **C.** Q and R **D.** P, Q and S	
3	Several people in a hilly village have a swelling in the neck. The health officer advises the villagers to use iodised salt. Iodine is needed by the body to make the hormone **A.** insulin **B.** thyroxine **C.** adrenaline **D.** growth hormone	**1**
4	A gardener places a fleshy leaf of *Bryophyllum* on moist soil. After a few days, small plants develop from the buds present in the notches along the leaf margin. This method of reproduction is **A.** fragmentation **B.** budding **C.** spore formation **D.** vegetative propagation	**1**
5	A doctor advises a woman to use a contraceptive device that is placed in the uterus to prevent pregnancy. Which of the following correctly identifies the device and its limitation? **A.** Condom – it does not prevent pregnancy **B.** Oral pill – it changes the hormonal balance of the body **C.** Copper-T – it does not protect against sexually transmitted diseases **D.** Vasectomy – it blocks the vas deferens permanently	**1**
6	Ozone at the higher levels of the atmosphere is formed when **A.** oxygen combines with nitrogen during lightning **B.** high-energy UV radiation splits O_2 molecules into free oxygen atoms, which then combine with O_2 molecules **C.** chlorofluorocarbons react with oxygen in the presence of sunlight **D.** carbon dioxide released by vehicles breaks down into oxygen	**1**
7	During a 400 m race, Ravi's leg muscles get cramps. Under these conditions, pyruvate in his muscle cells is converted into **A.** ethanol and carbon dioxide **B.** carbon dioxide and water **C.** glucose **D.** lactic acid	**1**

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** Arteries have thick and elastic walls. **Reason (R):** Blood in the arteries flows under high pressure as it is pumped out of the heart.	**1**
9	**Assertion (A):** Offspring produced by asexual reproduction show very little variation. **Reason (R):** DNA copying during asexual reproduction is absolutely accurate and never	**1**

produces any change.

10 A potted seedling kept on a windowsill is seen bending towards the window after a few days. **2**

A. Name the plant hormone responsible for this bending and explain how it causes the shoot to bend towards light.

B. Name the plant hormone that inhibits growth and is responsible for the wilting of leaves.

11 ***Students to attempt either option A or B.*** **2**

A. List any four ways by which plants get rid of their excretory products.

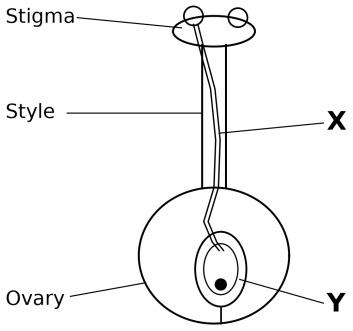
OR

B. A patient whose kidneys have stopped working is put on an artificial kidney (haemodialysis).

I. State the principle on which the artificial kidney works.

II. Mention one way in which its working differs from that of a nephron.

12 The figure shows a pistil of a flower after pollination. **2**



A. Identify the parts labelled X and Y.

B. What happens after X reaches Y? Name the structure into which Y develops after this event.

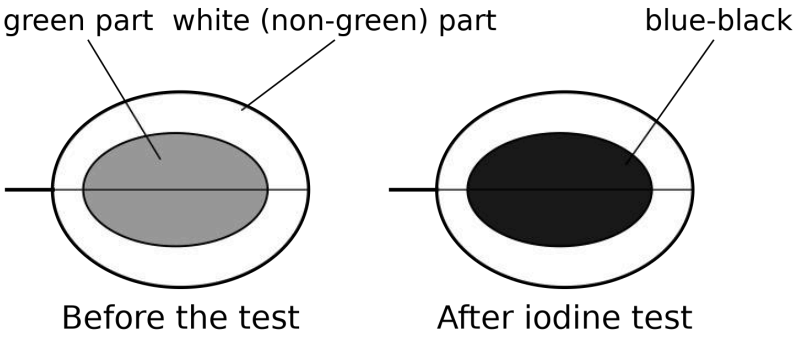
For visually impaired students

After pollination, a pollen grain on the stigma grows a tube that travels through the style and reaches a structure inside the ovary.

A. Name the tube and the structure in the ovary that it reaches.

B. What happens after the tube reaches this structure? Name the structure into which it develops after this event.

13 Aryan kept a potted plant with variegated leaves (leaves with green and white patches) in a dark room for three days. He then placed it in bright sunlight for six hours, plucked a leaf, boiled it in alcohol and tested it with iodine solution. His observations are shown below. **3**



A. Why was the plant kept in the dark for three days before the experiment?

B. Why was the leaf boiled in alcohol before adding iodine?

C. Only the green part of the leaf turned blue-black. What conclusion can be drawn from this observation?

For visually impaired students

Aryan kept a potted plant with variegated leaves (leaves with green and white patches) in the dark for three days, then placed it in sunlight for six hours. He boiled a leaf in alcohol and tested it with iodine. Only the part that was green turned blue-black.

A. Why was the plant kept in the dark for three days?

B. Why was the leaf boiled in alcohol before adding iodine?

C. What conclusion can be drawn from the observation?

14 A pesticide was sprayed regularly on farms near a lake. A study of the lake ecosystem gave the following data: **3**

Part of the lake ecosystem	Pesticide concentration (ppm)
Water	0.02
Phytoplankton	0.5
Zooplankton	4
Small fish	25
Kingfisher (bird)	120

A. Name the phenomenon shown by the data.

B. Explain why the concentration of the pesticide increases at each trophic level.

C. If the phytoplankton trap 1,00,000 J of solar energy, how much energy will be available to the kingfisher in the food chain phytoplankton → zooplankton → small fish → kingfisher?

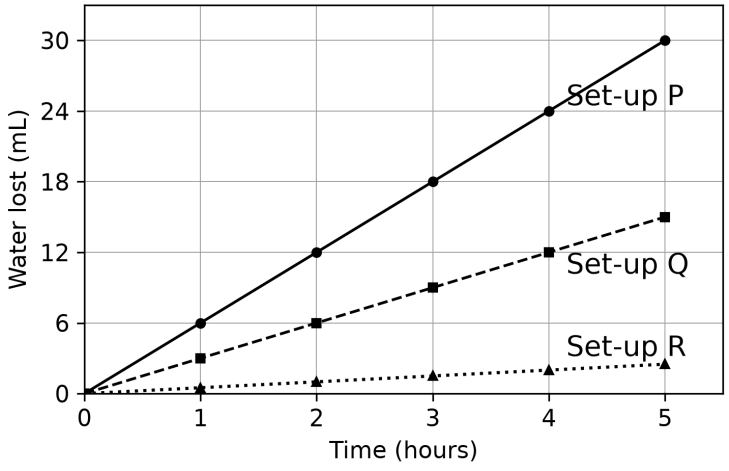
15 Ishita set up three similar potted plants of the same size to study the loss of water from plants. She watered them equally, covered the soil of each pot with a plastic sheet and kept them in sunlight. **4**

Set-up P: plant kept in front of a table fan (moving air)

Set-up Q: plant kept in still air

Set-up R: plant kept in still air with both surfaces of all its leaves coated with petroleum jelly

She measured the water lost by each plant every hour and plotted the graph shown below.



A. Which set-up shows the maximum loss of water? Calculate its rate of water loss per hour.

B. Why did set-up R lose the least amount of water?

Attempt either sub-part C or D.

C. Why was the soil in each pot covered with a plastic sheet? State two functions of the process studied in this experiment that are useful for the plant.

OR

D. Explain how the loss of water from the leaves helps in the upward movement of water in the xylem. What causes the movement of water in the xylem at night, when the stomata are closed?

For visually impaired students

Ishita took three similar potted plants and covered the soil of each pot with a plastic sheet. Plant P was kept in front of a fan, plant Q in still air, and plant R in still air with both surfaces of its leaves coated with petroleum jelly. In 5 hours, P lost 30 mL of water, Q lost 15 mL and R lost only 2.5 mL.

A. Which plant lost water at the maximum rate? Calculate its rate of water loss per hour.

B. Why did plant R lose the least amount of water?

Attempt either sub-part C or D.

C. Why was the soil in each pot covered with a plastic sheet? State two functions of the process studied that are useful for the plant.

OR

D. Explain how the loss of water from the leaves helps in the upward movement of water in the xylem. What causes the movement of water in the xylem at night?

16 ***Attempt either option A or B.*** **5**

A. I. Mendel crossed a pure-breeding pea plant having round and yellow seeds (RRYY) with a pure-breeding pea plant having wrinkled and green seeds (rryy).
(a) What will be the appearance of the seeds of the F_1 plants? Give the genotype of the F_1 plants.
(b) When the F_1 plants are self-pollinated, which four types of seeds appear in the F_2 generation, and in what ratio?
(c) What does the appearance of round-green and wrinkled-yellow seeds in the F_2 generation indicate?

II. How do genes control the expression of traits? Explain with the example of tallness in pea plants.

OR

B. I. With the help of a flow chart, explain how the sex of a child is determined in human beings.

II. Sunita's in-laws blame her for giving birth to two daughters and no son. Are they justified? Give a reason. Name one animal in which the sex of the offspring is decided by an environmental factor, and name the factor.

SECTION B Chemistry

17 Silver chloride turns grey when it is kept in sunlight. This property is used in black and white photography. The reaction taking place is **1**

A. thermal decomposition **B.** displacement
C. combination **D.** photolytic decomposition

18 Ayaan eats chocolates often and does not rinse his mouth afterwards. Bacteria in his mouth break down the sugar into acids. Tooth enamel starts getting corroded when the pH in the mouth falls below **1**

A. 7.0 **B.** 6.5
C. 5.5 **D.** 8.5

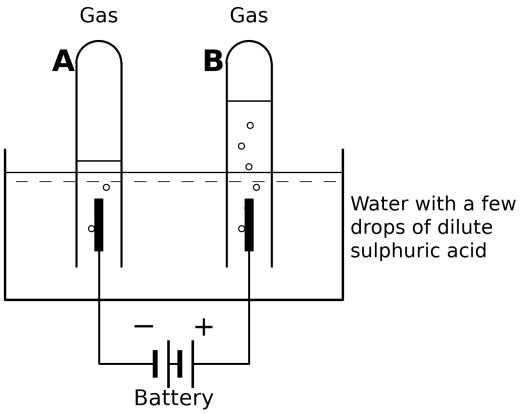
19 A metal X reacts so vigorously with cold water that the hydrogen evolved catches fire. It is stored under kerosene in the laboratory. The metal X is **1**

A. sodium **B.** calcium
C. magnesium **D.** aluminium

20 Which of the following is **not** a member of the same homologous series as methane? **1**

A. C_2H_6 **B.** C_3H_8
C. C_5H_{12} **D.** C_4H_8

21 The set-up for the electrolysis of water is shown below. **1**



Which gas is collected in test tube A, and why is its volume about double the volume of the gas in test tube B?

A. Oxygen, because a water molecule contains two oxygen atoms
B. Hydrogen, because water contains hydrogen and oxygen in the ratio 2 : 1 by number of atoms
C. Hydrogen, because hydrogen is lighter than oxygen
D. Oxygen, because oxygen is collected at the negative electrode

For visually impaired students

During the electrolysis of water, the gas collected at the negative electrode (cathode) and the reason for its volume being about double that of the gas at the positive electrode is:

A. Oxygen, because a water molecule contains two oxygen atoms
B. Hydrogen, because water contains hydrogen and oxygen in the ratio 2 : 1 by number of atoms
C. Hydrogen, because hydrogen is lighter than oxygen
D. Oxygen, because oxygen is collected at the negative electrode

22 Riya notices that soap forms a white curdy precipitate (scum) with the tap water in her town and lathers poorly. This is because **1**

A. the water contains dissolved sodium chloride
B. calcium and magnesium salts present in the water react with soap to form insoluble substances
C. soap molecules are non-ionic in nature
D. the tap water is strongly basic

23 Vegetable oils are converted into vanaspati ghee by heating them with hydrogen in the presence of nickel. Which of the following compounds can undergo a similar reaction? **1**

A. C_3H_6 **B.** C_2H_6
C. CH_4 **D.** C_3H_8

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Bags of potato chips are flushed with nitrogen gas before sealing. **Reason (R):** Nitrogen is an unreactive gas that prevents the oxidation of fats and oils present in the chips. **1**

25 Sana dipped a clean iron nail in a blue solution of copper sulphate for about 20 minutes. **2**

A. State two changes that she will observe.

B. Write the balanced chemical equation for the reaction and name the type of reaction.

26 ***Students to attempt either option A or B.*** **3**

A. **I.** Draw the electron dot structure of ethene (C_2H_4).

II. Why does carbon form compounds mainly by covalent bonding and not by forming ions?

III. LPG burns with a clean blue flame, whereas a candle often gives a sooty yellow flame. Give a reason.

OR

B. **I.** Draw the structural formulae of the two isomers of butane (C_4H_{10}).

II. Identify the functional group present in (a) CH_3CH_2CHO and (b) CH_3COCH_3.

III. What is meant by a homologous series? Write the molecular formula of the next higher homologue of C_2H_5OH.

27 **A.** Show the formation of magnesium chloride ($MgCl_2$) by the transfer of electrons. (Atomic numbers: Mg = 12, Cl = 17) **3**

B. Why do ionic compounds have high melting points?

C. Why does solid sodium chloride not conduct electricity, while its aqueous solution does?

28 Kabir and Meera performed an experiment to compare the reactivity of four metals W, X, Y and Z. They put a clean strip of each metal into separate test tubes containing solutions of zinc sulphate, iron(II) sulphate, copper sulphate and silver nitrate. Their observations are recorded below. **4**

Metal	$ZnSO_4$	$FeSO_4$	$CuSO_4$	$AgNO_3$
W	No change	No change	No change	No change
X	No change	No change	Deposit formed	Deposit formed
Y	No change	Deposit formed	Deposit formed	Deposit formed
Z	Deposit formed	Deposit formed	Deposit formed	Deposit formed

A. Which is the most reactive metal among W, X, Y and Z?

B. Can copper sulphate solution be stored in a container made of metal X? Give a reason.

Attempt either sub-part C or D.

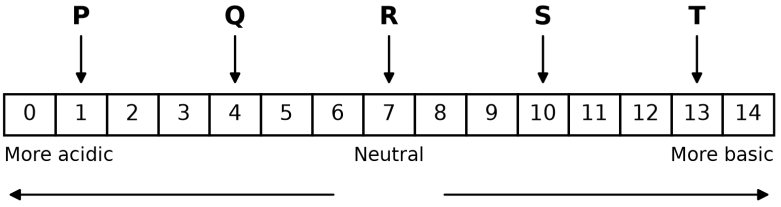
C. Arrange the metals W, X, Y and Z in increasing order of reactivity. Justify your answer on the basis of the observations.

OR

D. Aluminium is more reactive than zinc. In which of the four solutions will a strip of aluminium form a deposit? Write the balanced chemical equation for the reaction of aluminium with copper sulphate solution.

29 ***Attempt either option A or B.*** **5**

A. I. Five colourless solutions P, Q, R, S and T have pH values 1, 4, 7, 10 and 13 respectively, as marked on the pH scale below.



(a) Which of these solutions could be gastric juice, and which could be milk of magnesia?
(b) Arrange the solutions in increasing order of hydrogen ion concentration.
(c) What will happen to the pH of solution P when solution T is added to it gradually? Name the type of reaction that takes place.

II. Chlorine obtained from brine is used to manufacture bleaching powder. Write the chemical equation for its preparation and state two uses of bleaching powder.

OR

B. I. Rohan's mother adds baking powder to the cake batter to make the cake soft and spongy.
(a) Write the chemical equation for the preparation of baking soda from sodium chloride.
(b) Write the equation for the reaction that takes place when baking soda is heated, and explain how it makes the cake spongy.
(c) Why is baking powder (baking soda mixed with tartaric acid) preferred over baking soda alone?

II. Classify the aqueous solutions of the following salts as acidic, basic or neutral: NaCl, Na_2CO_3, NH_4Cl and K_2SO_4.

SECTION C Physics

30 A dentist wants to see an enlarged and erect image of a patient's tooth. Which mirror should she use, and where should the tooth be placed? **1**

A. Plane mirror, at any distance from the mirror
B. Convex mirror, very close to the mirror
C. Concave mirror, between the pole and the principal focus
D. Concave mirror, beyond the centre of curvature

31 A wire of resistance R is replaced by another wire of the same material having double the length and half the area of cross-section. The resistance of the new wire is **1**

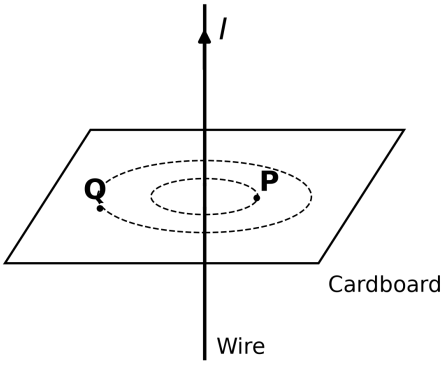
A. R/4 **B.** R
C. 2R **D.** 4R

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Magnetic field lines are closed curves. **1**
Reason (R): Outside a bar magnet, magnetic field lines emerge from the south pole and merge at the north pole.

33 A straight vertical wire passes through a horizontal sheet of cardboard and carries a current *I* upwards, as shown. P and Q are two points on the cardboard. **2**



A. What is the direction of the magnetic field lines on the cardboard, as seen from above? Name the rule used.

B. At which of the two points, P or Q, is the magnetic field stronger? Give a reason.

For visually impaired students

A straight vertical wire passes through a horizontal sheet of cardboard and carries a current upwards. Point P is 2 cm from the wire and point Q is 4 cm from the wire.

A. What is the direction of the magnetic field lines on the cardboard, as seen from above? Name the rule used.

B. At which of the two points, P or Q, is the magnetic field stronger? Give a reason.

34 ***Students to attempt either option A or B.*** **2**

A. **I.** What does the pattern of parallel straight magnetic field lines inside a long current-carrying solenoid indicate?

II. What is formed when a soft iron rod is placed inside a current-carrying solenoid? State one use of it.

OR

B. **I.** Why is it necessary to earth the metallic body of an electric iron?

II. What is meant by short-circuiting? How does a fuse protect a circuit in such a case?

35 An object 4 cm tall is placed 15 cm in front of a convex lens of focal length 10 cm. **3**

A. Find the position of the image.

B. Find the magnification and the height of the image.

C. State the nature of the image.

36 ***Students to attempt either option A or B.*** **3**

A. A concave mirror of focal length 12 cm forms a real image of an object that is three times the size of the object. Find the distance of the object and of the image from the mirror. Where should a screen be placed to obtain this image?

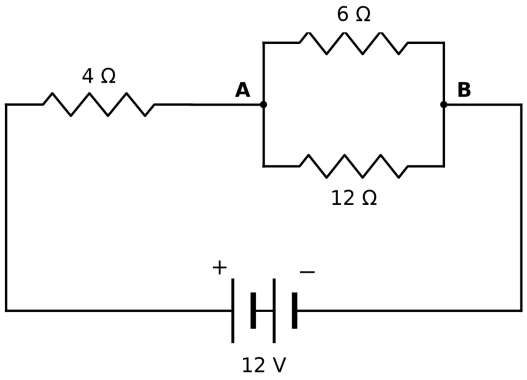
OR

B. **I.** The refractive index of glass is 1.5. Calculate the speed of light in glass. (Speed of light in air = 3×10^8 m s^{-1})

II. A ray of light enters a rectangular glass slab obliquely. Does it bend towards or away from the normal? Give a reason.

III. Why is the emergent ray from the slab parallel to the incident ray?

37 Study the circuit shown below. **3**



A. Calculate the equivalent resistance of the circuit.

B. Calculate the current drawn from the battery.

C. Find the potential difference between points A and B and the current through the 12 Ω resistor.

For visually impaired students

A 4 Ω resistor is connected in series with a parallel combination of a 6 Ω and a 12 Ω resistor across a 12 V battery.

A. Calculate the equivalent resistance of the circuit.

B. Calculate the current drawn from the battery.

C. Find the potential difference across the parallel combination and the current through the 12 Ω resistor.

38 Aditi's family wants to reduce their electricity bill. She notes the daily use of the main appliances in their home, all of which work at 220 V. **4**

Appliance	Power rating	Use per day
Bulbs (4 in number)	15 W each	5 hours
Ceiling fan	75 W	8 hours
Electric iron	1000 W	24 minutes
Electric geyser	2200 W	1 hour

A. Calculate the electrical energy consumed by the ceiling fan in one day, in kWh.

B. Which single appliance in the table has the highest resistance? Give a reason.

Attempt either sub-part C or D.

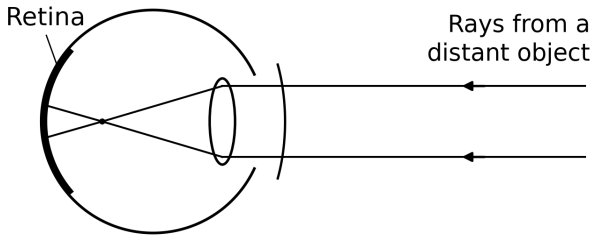
C. Calculate the total energy consumed by these appliances in 30 days, and the cost of this energy at ₹ 6 per kWh.

OR

D. Calculate the current drawn by the geyser. Would it be safe to connect the geyser to a circuit with a 5 A fuse? Give a reason.

39 ***Attempt either option A or B.*** **5**

A. Rahul can read his book clearly but cannot see the writing on the classroom board clearly. The figure shows how parallel rays from a distant object are focused in his eye.



I. Name the defect of vision Rahul is suffering from and state two possible causes of this defect.

II. Rahul's far point is 2.5 m. Find the nature, focal length and power of the lens he needs to see distant objects clearly.

III. Draw a labelled ray diagram to show how this lens corrects the defect.

OR

B. **I.** What is meant by dispersion of light? Why does white light split into its component colours when it passes through a glass prism?

II. Draw a labelled diagram to show the dispersion of white light by a glass prism, and mark the colour that deviates the least.

III. Explain why the clear sky appears blue, while clouds appear white.

For visually impaired students

A. I. Name the defect of vision in which a person can see nearby objects clearly but cannot see distant objects clearly. State two possible causes of this defect.
II. A person's far point is 2.5 m. Find the nature, focal length and power of the lens needed to see distant objects clearly.
III. Explain how this lens helps the person to see distant objects clearly.

OR

B. I. What is meant by dispersion of light? Why does white light split into its component colours when it passes through a glass prism?
II. Which colour deviates the least and which colour deviates the most in the spectrum formed by a prism?
III. Explain why the clear sky appears blue, while clouds appear white.

END OF QUESTION PAPER

Quick revision: The Human Eye

1. Near point of a normal eye: 25 cm. Far point: infinity.
2. Myopia is corrected by a concave lens of focal length equal to the far point (negative).
3. Hypermetropia is corrected by a convex lens.
4. Power of accommodation: the eye lens changes its focal length using the ciliary muscles.

SOLUTION PAPER 06

SCIENCE · CODE NO. 086

Sample Paper 6: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

A. Digestion of proteins, because pepsin acts only in an acidic medium (1)

(HCl makes the medium acidic, which is needed for the action of the enzyme pepsin.)

2 Ans

A. P only (1)

(Salivary amylase in P digests starch at body temperature; it is destroyed by boiling in Q, absent in R and inactive at 0 °C in S.)

3 Ans

B. thyroxine (1)

4 Ans

D. vegetative propagation (1)

5 Ans

C. Copper-T – it does not protect against sexually transmitted diseases (1)

6 Ans

B. high-energy UV radiation splits O_2 molecules into free oxygen atoms, which then combine with O_2 molecules (1)

($O_2 \rightarrow O + O$; $O + O_2 \rightarrow O_3$)

7 Ans

D. lactic acid (1)

8 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

9 Ans

C. A is true but R is false. (1)

(DNA copying is not absolutely accurate; small errors produce slight variations.)

10 Ans

A. Auxin.

When light falls on one side, auxin diffuses towards the shaded side of the shoot. Auxin stimulates the cells on the shaded side to grow longer, so the shoot bends towards light. (½ + ½)

B. Abscisic acid. (1)

Total 2 marks

11 Ans

A.

(i) Oxygen produced in photosynthesis and CO_2 (at night) are removed through stomata; (ii) excess water is removed by transpiration; (iii) wastes are stored in leaves and bark that fall off; (iv) wastes are stored as resins and gums in old xylem; (v) some wastes are excreted into the soil around the roots. (any four) (½ × 4)

OR

B.

I. Dialysis: blood is passed through long tubes with semi-permeable lining suspended in a dialysing fluid; nitrogenous wastes diffuse out of the blood into the fluid (by diffusion across a concentration gradient). (1)

II. In an artificial kidney there is no reabsorption of useful substances (glucose, salts, water), whereas a nephron selectively reabsorbs them. (1)

Total 2 marks

12 **Ans**

A. X – Pollen tube; Y – Ovule (containing the female germ cell / egg) (½ + ½)

B. The male germ cell travels through the pollen tube and fuses with the female germ cell in the ovule (fertilisation) to form a zygote. (½)

After fertilisation, the ovule develops into a seed. (½)

For visually impaired students: same value points.

Total 2 marks

13 **Ans**

A. To destarch the leaves, i.e., to use up the starch already present, so that any starch found later is made during the experiment. (1)

B. Boiling in alcohol removes the chlorophyll (decolourises the leaf), so that the colour change with iodine can be seen clearly. (1)

C. Chlorophyll is necessary for photosynthesis; starch was formed only in the green parts. (1)

Total 3 marks

14 **Ans**

A. Biological magnification (1)

B. The pesticide is non-biodegradable; it is neither broken down nor excreted, so it accumulates in the body. Each organism eats a large number of organisms of the lower level, so the concentration increases at every trophic level. (1)

C. According to the 10% law, energy at each next level = 10% of the previous level: 1,00,000 J → 10,000 J → 1,000 J → **100 J** available to the kingfisher. (1)

Total 3 marks

15 **Ans**

A. Set-up P (plant in moving air). (½)

Rate = 30 mL ÷ 5 h = 6 mL per hour. (½)

B. Petroleum jelly blocks the stomata on both surfaces of the leaves, so very little water is lost by transpiration. (1)

C.

To prevent the evaporation of water from the soil, so that only the water lost by the plant (transpiration) is measured. (1)

Functions (any two): (i) creates a suction (transpiration pull) that helps in the absorption and upward movement of water and dissolved minerals; (ii) helps in cooling the leaf / regulates temperature. (½ + ½)

OR

D.

Evaporation of water from the leaf cells through stomata creates a suction (transpiration pull) that pulls water upward in a continuous column through the xylem from the roots. (1)

At night, root pressure (created by the active uptake of ions by root cells, so that water moves into the root xylem) pushes water up the xylem. (1)

For visually impaired students: same value points (A: plant P, 30 mL ÷ 5 h = 6 mL per hour).

Total 4 marks

16 **Ans**

A. **I.** (a)

All F_1 seeds are round and yellow; genotype RrYy. **(½ + ½)**

(b) F_1 gametes: RY, Ry, rY, ry. F_2 (Punnett square):

♀ \ ♂	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

Round yellow : Round green : Wrinkled yellow : Wrinkled green = 9 : 3 : 3 : 1 **(1)**

(c)

New combinations show that the two pairs of traits (seed shape and seed colour) are inherited independently of each other (law of independent assortment). **(1)**

II. Genes carry information for making proteins (enzymes/hormones). The trait depends on how much of the protein is made. **(1)**

In pea plants, height depends on the amount of a plant growth hormone made by an enzyme. If the gene makes an efficient enzyme, more hormone is formed and the plant is tall; if the gene is altered and the enzyme is less efficient, less hormone is formed and the plant is short. **(1)**

OR

B. **I.** Women have a perfect pair of sex chromosomes (XX); men have a mismatched pair (XY). **(½)**

Parents	Mother (XX)	×	Father (XY)
Gametes	X (all eggs)		X sperm or Y sperm
Offspring	X egg + X sperm → XX (Girl)		X egg + Y sperm → XY (Boy)

All eggs carry an X chromosome; half the sperms carry X and half carry Y. **(1)**

If an X-carrying sperm fertilises the egg, the child is a girl (XX); if a Y-carrying sperm fertilises it, the child is a boy (XY). Chances of a boy or a girl are 50 : 50. **(1)**

(Correct flow chart) **(½)**

II. No. The mother always contributes an X chromosome; the sex of the child is decided by whether the sperm from the father carries X or Y. **(1)**

In some reptiles (e.g., turtles), the temperature at which the fertilised eggs are incubated decides the sex of the offspring. (Snails can change sex – also accepted.) **(½ + ½)**

Total 5 marks

SECTION B Chemistry

17 **Ans**

D. photolytic decomposition **(1)**

($2AgCl \rightarrow 2Ag + Cl_2$ in sunlight)

18 **Ans**

C. 5.5 **(1)**

19

A. sodium **(1)**

Ans

20 **Ans**

D. C_4H_8 **(1)**

(Methane belongs to the alkane series C_nH_{2n+2}; C_4H_8 is C_nH_{2n}.)

21 **Ans**

B. Hydrogen, because water contains hydrogen and oxygen in the ratio 2 : 1 by number of atoms **(1)**

(Tube A is connected to the negative terminal.)

22 **Ans**

B. calcium and magnesium salts present in the water react with soap to form insoluble substances **(1)**

23 **Ans**

A. C_3H_6 **(1)**

(Only an unsaturated hydrocarbon undergoes addition of hydrogen.)

24 **Ans**

A. Both A and R are true, and R is the correct explanation of A. **(1)**

25 **Ans**

A. The blue colour of the solution fades and becomes light green. **(½)**

A reddish-brown coating (of copper) is deposited on the iron nail. **(½)**

B. $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$ **(½)**

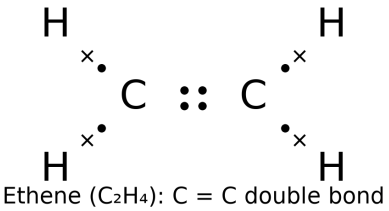
Displacement reaction **(½)**

Total 2 marks

26 **Ans**

A.

I.



Ethene (C_2H_4): C = C double bond

(Correct structure with a double bond between the two carbon atoms) **(1)**

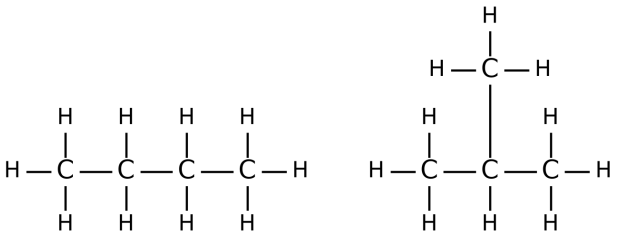
II. Carbon has four valence electrons. Losing or gaining four electrons needs a very large amount of energy (C^{4-} / C^{4+} would be unstable), so carbon shares its electrons. **(1)**

III. LPG is a saturated hydrocarbon and burns completely in air, giving a blue flame; candle wax gives a sooty flame because of incomplete combustion when the supply of air is limited. **(1)**

OR

B.

I.



n-Butane Isobutane (2-methylpropane)

n-butane and isobutane **(½ + ½)**

II. (a) Aldehyde (–CHO); (b) Ketone (>C=O) (½ + ½)

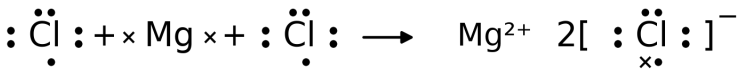
III. A series of compounds with the same functional group and similar chemical properties, in which successive members differ by a $–CH_2–$ unit. (½)

C_3H_7OH (propanol) (½)

Total 3 marks

27 **Ans**

A. Mg (2, 8, 2) → Mg^{2+} (2, 8) + $2e^-$; Cl (2, 7) + e^- → Cl^- (2, 8, 8) (½)



Mg loses two electrons, one to each chlorine atom, forming Mg^{2+} and $2Cl^-$ ions held together as $MgCl_2$. (½)

B. Strong electrostatic forces of attraction exist between oppositely charged ions; a large amount of energy is needed to break them. (1)

C. In the solid, the ions are held in fixed positions and cannot move. In aqueous solution the ions are free to move and carry charge. (1)

Total 3 marks

28 **Ans**

A. Z (1)

B. No. X displaces copper from copper sulphate solution (X is more reactive than copper), so the container would get corroded. (½ + ½)

C.

W < X < Y < Z (1)

Z displaces all four metals; Y displaces Fe, Cu, Ag but not Zn; X displaces only Cu and Ag; W displaces none. A metal that displaces more metals is more reactive. (1)

OR

D.

Aluminium will form a deposit in all four solutions ($ZnSO_4$, $FeSO_4$, $CuSO_4$ and $AgNO_3$). (1)

$2Al(s) + 3CuSO_4(aq) → Al_2(SO_4)_3(aq) + 3Cu(s)$ (1)

Total 4 marks

29 **Ans**

A. **I.** (a) Gastric juice – P (pH 1); milk of magnesia – S (pH 10) (½ + ½)

(b) T < S < R < Q < P (1)

(c) The pH of P increases (towards 7, and beyond 7 if excess T is added); neutralisation reaction. (½ + ½)

II. $Ca(OH)_2 + Cl_2 → CaOCl_2 + H_2O$ (1)

Uses (any two): bleaching cotton and linen / wood pulp; disinfecting drinking water; oxidising agent in industries; bleaching washed clothes in laundry. (½ + ½)

OR

B. **I.** (a) $NaCl + H_2O + CO_2 + NH_3 → NH_4Cl + NaHCO_3$ (1)

(b) $2NaHCO_3$ –(heat)→ $Na_2CO_3 + H_2O + CO_2$. The CO_2 gas produced makes the batter rise, so the cake becomes soft and spongy. (½ + ½)

(c) Tartaric acid neutralises the sodium carbonate formed, which would otherwise give the cake a bitter taste. (1)

II. NaCl – neutral; Na_2CO_3 – basic; NH_4Cl – acidic; K_2SO_4 – neutral (½ × 4)

Total 5 marks

SECTION C Physics

30 Ans

C. Concave mirror, between the pole and the principal focus (1)

31 Ans

D. 4R (1)

($R' = \rho(2l)/(A/2) = 4\rho l/A = 4R$)

32 Ans

C. A is true but R is false. (1)

(Outside a magnet, field lines emerge from the north pole and merge at the south pole.)

33 Ans

A. Concentric circles in the anticlockwise direction (as seen from above). (½)

Right-hand thumb rule (½)

B. At P. (½)

The magnetic field produced by a straight current-carrying wire becomes weaker as the distance from the wire increases; P is nearer to the wire. (½)

Total 2 marks

34 Ans

A.

I. The magnetic field is uniform (same strength and direction) at all points inside the solenoid. (1)

II. An electromagnet; used in electric cranes, electric bells, door locks, etc. (any one) (½ + ½)

OR

B.

I. If the live wire touches the metal body, the current flows to the earth through the earth wire (low resistance path), keeping the potential of the body equal to the earth; so the user does not get a severe shock. (1)

II. Short-circuiting occurs when the live wire and neutral wire come in direct contact, so the resistance becomes very small and the current increases abruptly. (½)

The fuse wire melts due to the heating effect of the large current and breaks the circuit. (½)

Total 2 marks

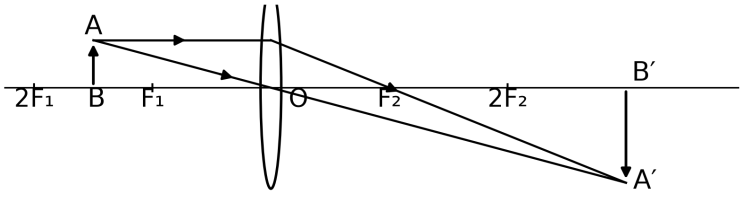
35 Ans

u = −15 cm, f = +10 cm, h = +4 cm

A. $1/v - 1/u = 1/f \Rightarrow 1/v = 1/10 + 1/(-15) = 1/30 \Rightarrow v = +30$ cm (on the other side of the lens) (1)

B. $m = v/u = 30/(-15) = -2$; $h' = m \times h = -2 \times 4 = -8$ cm (½ + ½)

C. Real, inverted and magnified (twice the size of the object). (1)



(Ray diagram for reference)

Total 3 marks

36 Ans

A.

Real image ⇒ inverted, so m = −3 = −v/u ⇒ v = 3u (1)

1/v + 1/u = 1/f ⇒ 1/(3u) + 1/u = 1/(−12) ⇒ 4/(3u) = −1/12 ⇒ u = −16 cm (1)

v = 3u = −48 cm. The object is 16 cm and the image is 48 cm in front of the mirror; the screen should be placed 48 cm in front of the mirror (on the same side as the object). (1)

OR

B.

I. $n = c/v \Rightarrow v = c/n = (3 \times 10^8)/1.5 = 2 \times 10^8$ m s^{-1} (1)

II. Towards the normal, because light travels from a rarer medium (air) to a denser medium (glass), where its speed decreases. (½ + ½)

III. The extent of bending at the two opposite parallel faces is equal and opposite (refraction at the second face reverses the first), so the emergent ray is parallel to the incident ray. (1)

Total 3 marks

37 Ans

A. $R_p = (6 \times 12)/(6 + 12) = 4\ \Omega$; $R_{eq} = 4 + 4 = 8\ \Omega$ (½ + ½)

B. $I = V/R = 12/8 = 1.5$ A (1)

C. $V_{AB} = I \times R_p = 1.5 \times 4 = 6$ V (½)

$I_{12} = 6/12 = 0.5$ A (½)

For visually impaired students: same value points.

Total 3 marks

38 Ans

A. $E = P \times t = 0.075$ kW × 8 h = 0.6 kWh (1)

B. A bulb (15 W). $R = V^2/P$; at the same voltage, the appliance with the least power has the highest resistance ($R = 220^2/15 \approx 3227\ \Omega$). (½ + ½)

C.

Energy per day = bulbs 4 × 15 W × 5 h = 0.3 kWh; fan 0.6 kWh; iron 1 kW × 0.4 h = 0.4 kWh; geyser 2.2 kW × 1 h = 2.2 kWh; total = 3.5 kWh (1)

In 30 days = 3.5 × 30 = 105 kWh; cost = 105 × ₹ 6 = ₹ 630 (½ + ½)

OR

D.

$I = P/V = 2200/220 = 10$ A (1)

No. The current (10 A) is more than the fuse rating (5 A), so the fuse would melt every time the geyser is switched on; it must be connected to a power circuit with a 15 A fuse. (1)

Total 4 marks

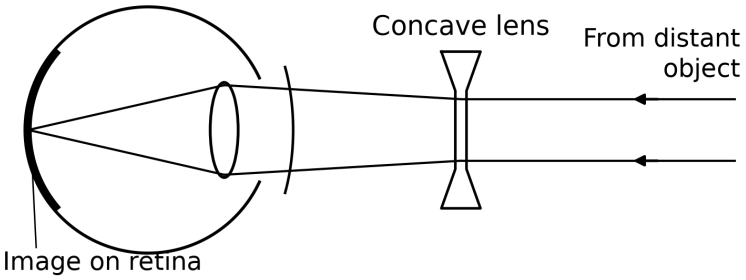
39 Ans

A. **I.** Myopia (near-sightedness) (1)

Causes: (i) excessive curvature of the eye lens; (ii) elongation of the eyeball. (½ + ½)

II. Concave (diverging) lens; f = −(far point) = −2.5 m; P = 1/f = 1/(−2.5 m) = −0.4 D (½ + ½)

III.



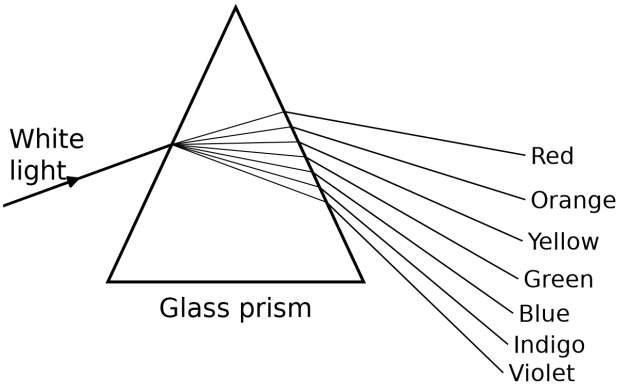
A concave lens of suitable power diverges the parallel rays so that they appear to come from the far point and are focused on the retina. (Correct labelled diagram) **(2)**

OR

B. I. The splitting of white light into its component colours is called dispersion. **(1)**

Different colours of light bend through different angles with respect to the incident ray when passing through the prism (red bends the least, violet the most). **(1)**

II.



Labelled diagram showing the spectrum VIBGYOR; red deviates the least. **(1)**

III. The fine particles (molecules) of air scatter light of shorter wavelength (blue) much more strongly than red; this scattered blue light enters our eyes. **(1)**

Clouds contain larger water droplets that scatter all colours (wavelengths) almost equally, so clouds appear white. **(1)**

For visually impaired students: A. I and II – same as above; III. A concave lens of suitable power diverges the rays from a distant object so that they appear to come from the far point; the eye lens then focuses them on the retina. (2)

B. I and III – same as above; II. Red deviates the least, violet deviates the most. (1)

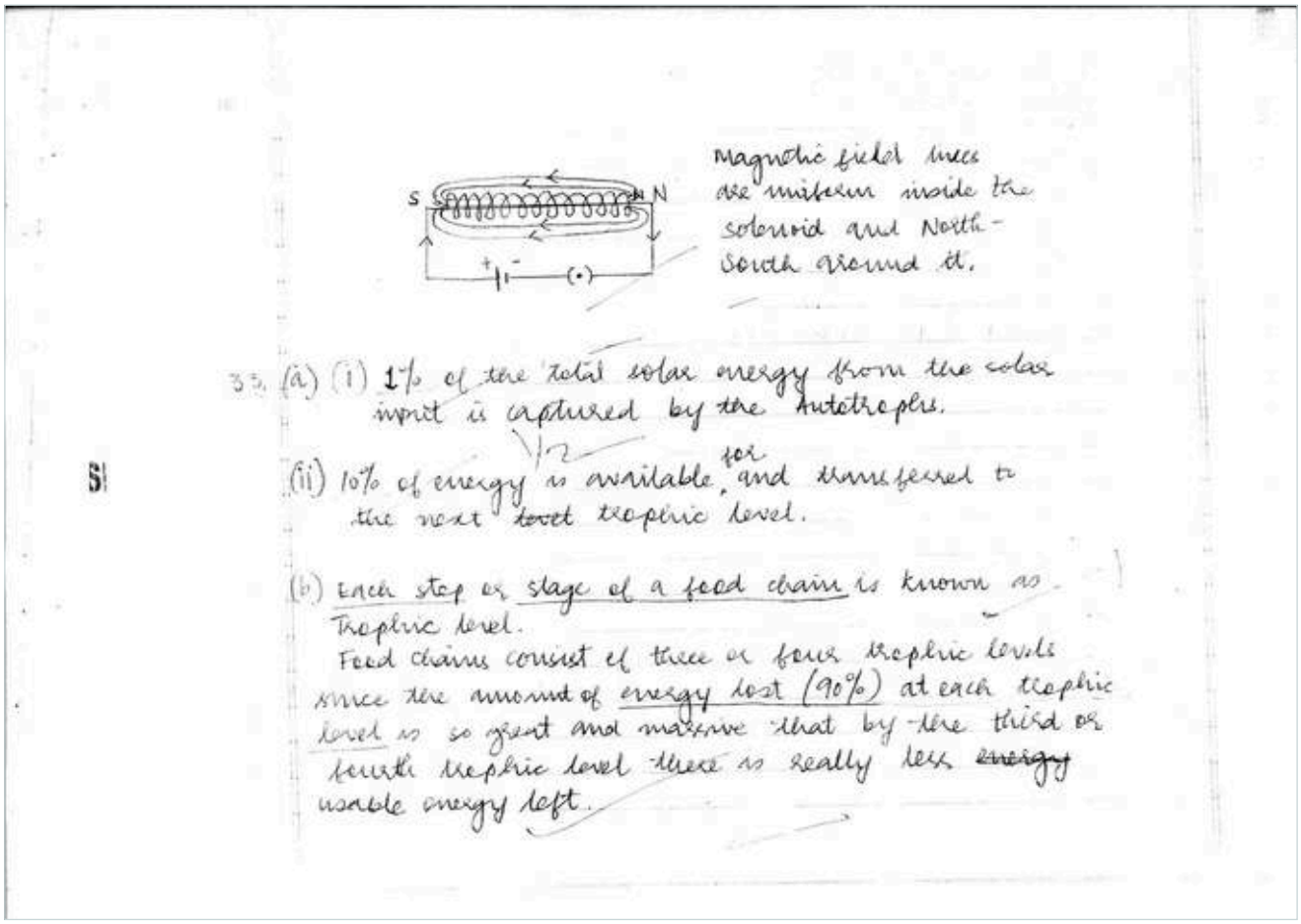
Total 5 marks

END OF SOLUTIONS

Remember while checking

1. Each step in the solution carries a mark; write the formula even if the final value goes wrong.
2. Labelled diagrams score faster than long paragraphs.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.14 of this paper (food chains and flow of energy)



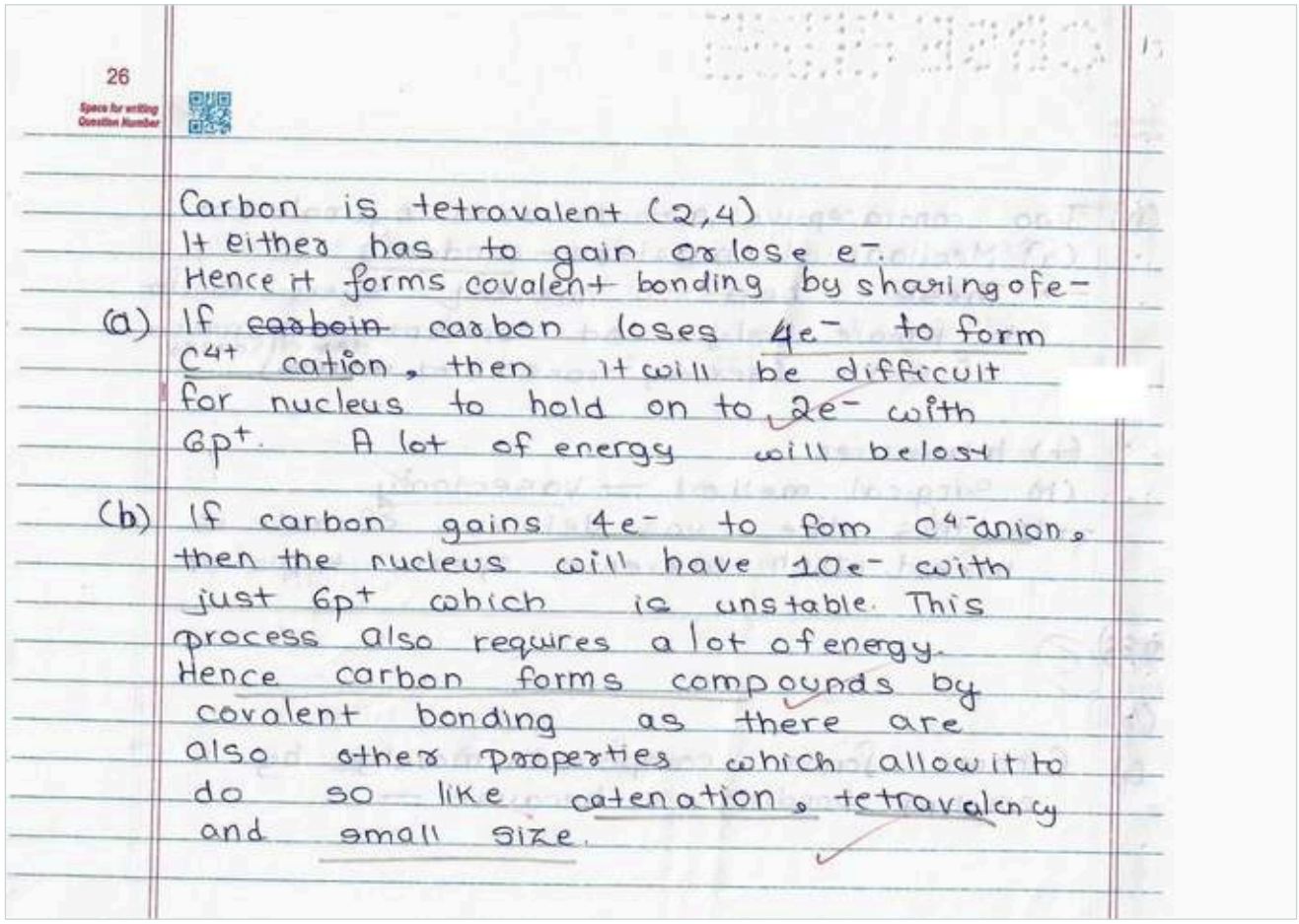
Magnetic field lines are uniform inside the solenoid and North-South around it.

35. (a) (i) 1% of the total solar energy from the solar input is captured by the autotrophs.

(ii) 10% of energy is available for and transferred to the next level trophic level.

(b) Each step or stage of a food chain is known as Trophic level.
Food chains consist of three or four trophic levels since the amount of energy lost (90%) at each trophic level is so great and massive that by the third or fourth trophic level there is really less ~~energy~~ usable energy left.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.26 of this paper (covalent bonding in carbon)



26

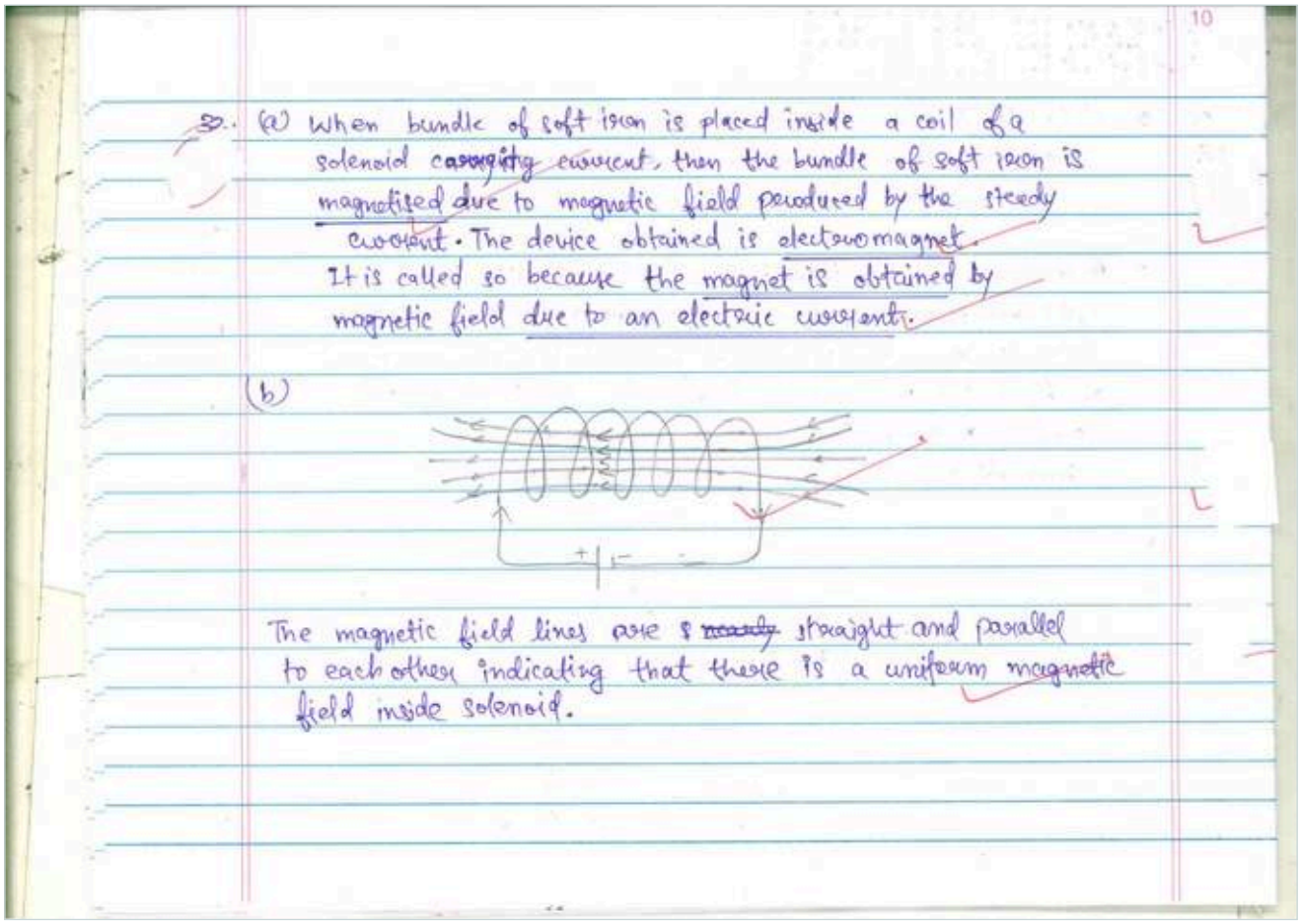
Space for writing Question Number

Carbon is tetravalent (2,4)
It either has to gain or lose e^-.
Hence it forms covalent bonding by sharing of e^-

(a) If ~~carbon~~ carbon loses $4e^-$ to form C^{4+} cation, then it will be difficult for nucleus to hold on to $2e^-$ with $6p^+$. A lot of energy will be lost

(b) If carbon gains $4e^-$ to form C^{4-} anion, then the nucleus will have $10e^-$ with just $6p^+$ which is unstable. This process also requires a lot of energy.
Hence carbon forms compounds by covalent bonding as there are also other properties which allow it to do so like catenation, tetravalency and small size.

Real CBSE topper's answer · Board Exam 2024 · similar to Q.34 of this paper (magnetic field lines inside a solenoid)

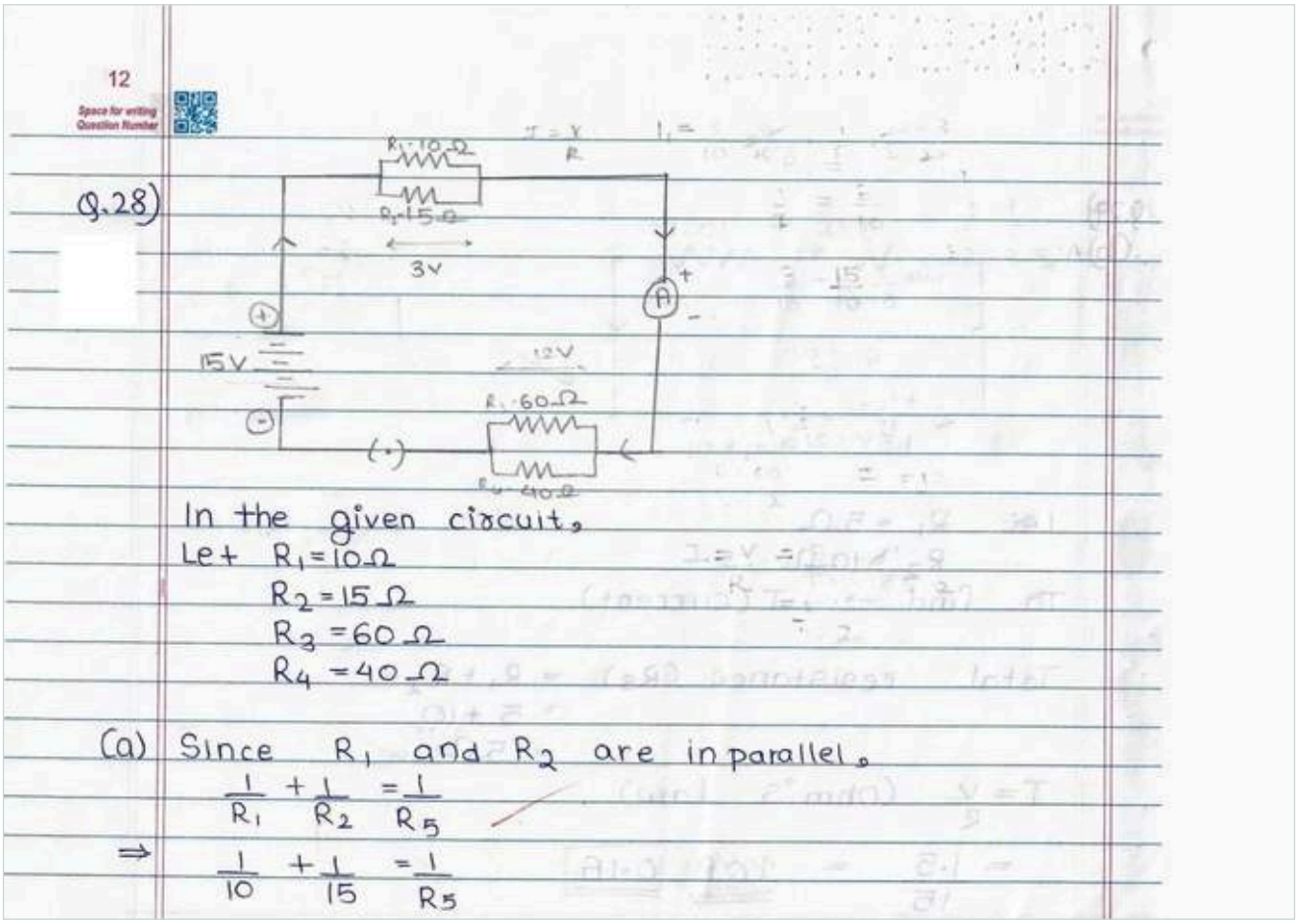


32. (a) When bundle of soft iron is placed inside a coil of a solenoid carrying current, then the bundle of soft iron is magnetised due to magnetic field produced by the steady current. The device obtained is electromagnet. It is called so because the magnet is obtained by magnetic field due to an electric current.

(b)

The magnetic field lines are ~~nearly~~ straight and parallel to each other indicating that there is a uniform magnetic field inside solenoid.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.37 of this paper (circuit diagram, resistors in parallel)



Q.28)

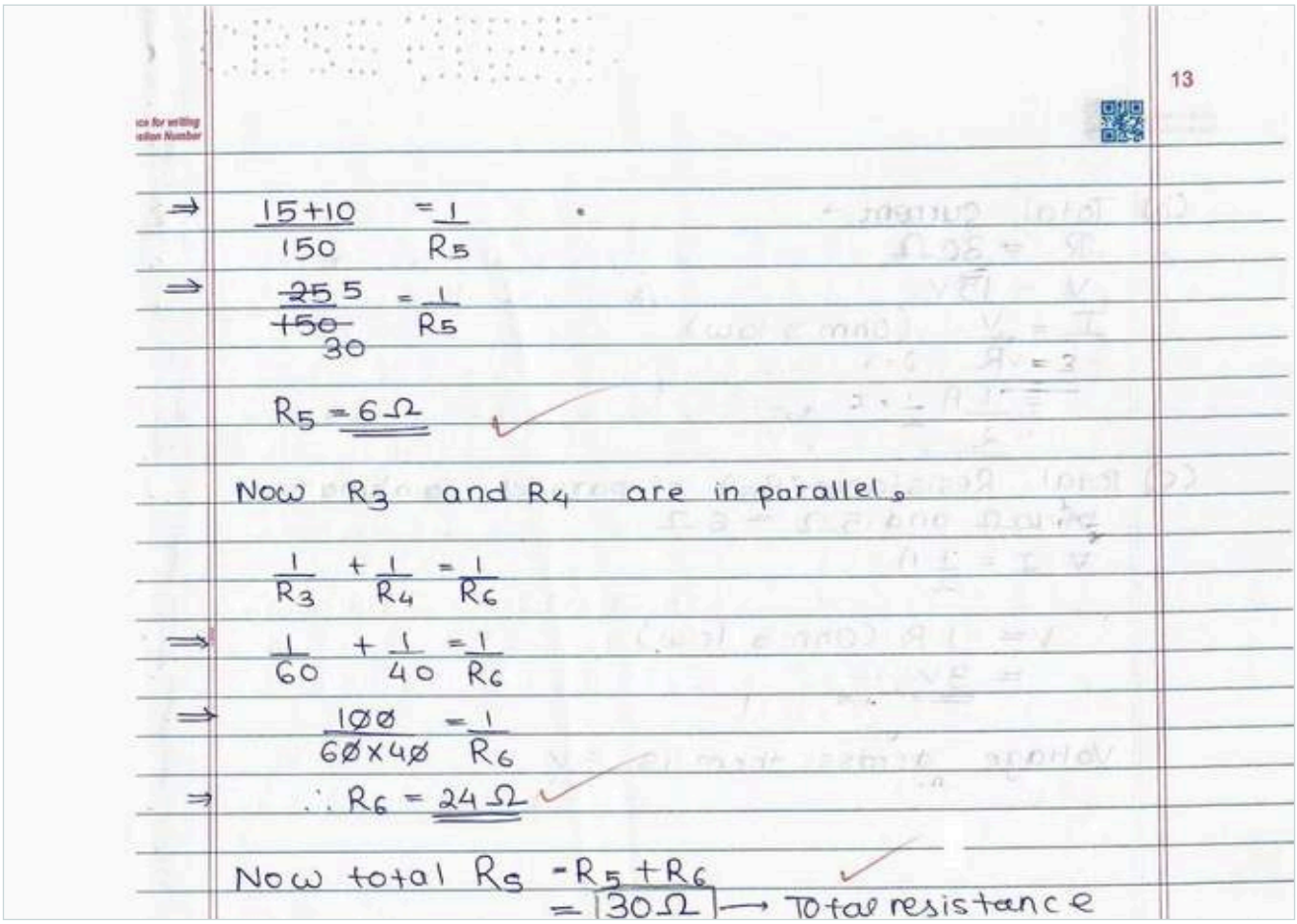
In the given circuit,
Let $R_1 = 10\,\Omega$
$R_2 = 15\,\Omega$
$R_3 = 60\,\Omega$
$R_4 = 40\,\Omega$

(a) Since R_1 and R_2 are in parallel,

$$\frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{R_S}$$

$$\Rightarrow \frac{1}{10} + \frac{1}{15} = \frac{1}{R_S}$$

Real CBSE topper's answer · Board Exam 2025 · similar to Q.37 of this paper (equivalent resistance of the whole circuit)



13

$\Rightarrow \frac{15+10}{150} = \frac{1}{R_5}$

$\Rightarrow \frac{25}{150} = \frac{1}{R_5}$

30

$R_5 = 6\,\Omega$

Now R_3 and R_4 are in parallel.

$\frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{R_6}$

$\Rightarrow \frac{1}{60} + \frac{1}{40} = \frac{1}{R_6}$

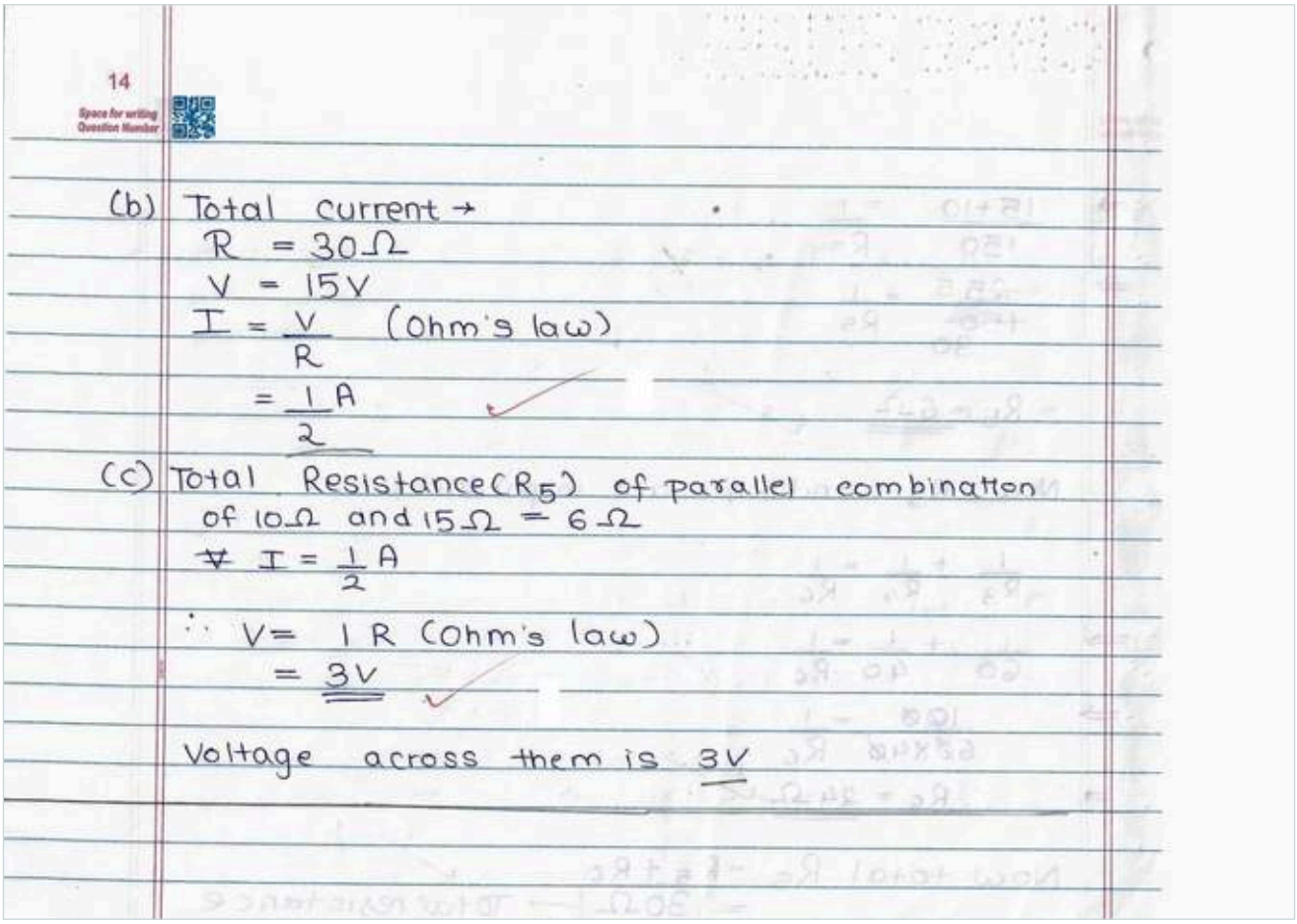
$\Rightarrow \frac{100}{60 \times 40} = \frac{1}{R_6}$

$\Rightarrow \therefore R_6 = 24\,\Omega$

Now total $R_S = R_5 + R_6$

$= 130\,\Omega$ → Total resistance

Real CBSE topper's answer · Board Exam 2025 · similar to Q.37 of this paper (current and potential difference by Ohm's law)



14

(b) Total current →

$R = 30\,\Omega$

$V = 15\,V$

$I = \frac{V}{R}$ (Ohm's law)

$= \frac{1}{2}\,A$

(c) Total Resistance(R_5) of parallel combination of $10\,\Omega$ and $15\,\Omega = 6\,\Omega$

$\therefore I = \frac{1}{2}\,A$

$\therefore V = IR$ (Ohm's law)

$= 3\,V$

Voltage across them is 3V

SAMPLE PAPER 07

SCIENCE · CODE NO. 086

Sample Question Paper 7

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 (1)

Arjun chews a piece of plain bread slowly for about two minutes and notices that it begins to taste sweet. Which of the following best explains this observation?

A. Pepsin present in saliva breaks down proteins of the bread into sugar.

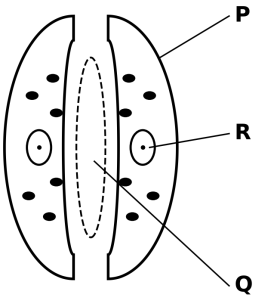
B. Salivary amylase breaks down the starch of the bread into sugar.

C. Bile salts in the mouth break the bread into smaller particles.

D. Hydrochloric acid in saliva converts starch into glucose.

2 (1)

The figure shows a stoma observed on the lower surface of a leaf.



Which of the following correctly describes how the pore Q opens?

A. Cells P lose water, shrink and pull the pore open.

B. Cells P take in carbon dioxide and expand.

C. Water flows into cells P, they swell and the pore Q opens.

D. Nucleus R divides and pushes the two cells apart.

For visually impaired students

The opening and closing of the stomatal pore is a function of:

A. epidermal cells losing water.

B. guard cells taking in carbon dioxide.
C. guard cells swelling when water flows into them.
D. the nucleus of the guard cells dividing.

3 A laboratory technician examines the blood sample of a malaria patient and observes that a single parasite cell divides to form many daughter cells at the same time. This mode of reproduction is: **1**

A. budding **B.** binary fission
C. fragmentation **D.** multiple fission

4 A gardener crosses a pure tall pea plant (TT) with a pure dwarf pea plant (tt). The F_1 plants are self-pollinated and 800 plants are raised in the F_2 generation. How many of these F_2 plants are expected to be tall but **not** pure-breeding? **1**

A. 400 **B.** 200
C. 600 **D.** 800

5 A couple wants to use a method of contraception that will also protect them from sexually transmitted infections such as HIV-AIDS. Which method should the doctor recommend? **1**

A. Copper-T **B.** Condom
C. Oral contraceptive pills **D.** Vasectomy

6 Ozone at the higher levels of the atmosphere is formed when: **1**

A. CFCs released from refrigerators react with oxygen.
B. carbon dioxide breaks down in the presence of sunlight.
C. nitrogen and oxygen combine during lightning.
D. UV radiation splits O_2 molecules into free oxygen atoms, which then combine with O_2 molecules.

7 Study the food chain given below. **1**

If the grass traps 20,000 J of solar energy as food energy, how much energy will be available to the snake?

A. 2000 J **B.** 200 J
C. 20 J **D.** 2 J

For visually impaired students

In a food chain grass → grasshopper → frog → snake → eagle, the grass traps 20,000 J of solar energy as food energy. How much energy will be available to the snake?

A. 2000 J **B.** 200 J
C. 20 J **D.** 2 J

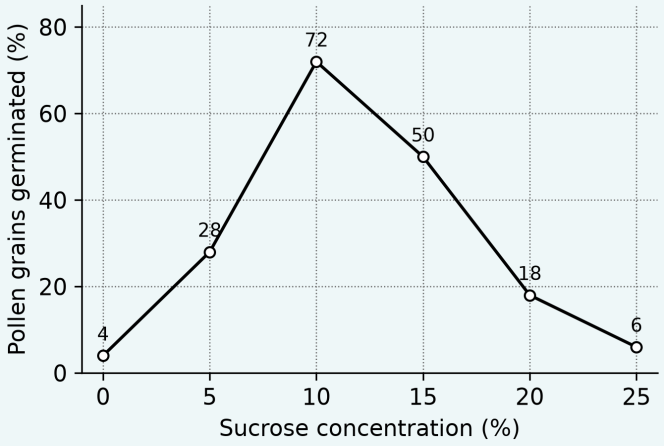
The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** Veins have valves along their length. **Reason (R):** Blood in the veins flows under very high pressure.	**1**
9	**Assertion (A):** Decomposers are essential for maintaining the balance of an ecosystem. **Reason (R):** Decomposers break down the complex organic substances present in dead remains into simple inorganic substances that go back into the soil.	**1**
10	Kabir runs a 400 m race at full speed and soon feels painful cramps in his leg muscles. **A.** Write the pathway of the breakdown of glucose taking place in his leg muscles at that time. **B.** Why do the cramps disappear after he rests for some time?	**2**
11	***Students to attempt either option A or B.*** **A.** Hot water from a factory is released into a pond and the temperature of the pond rises sharply. Most of the bacteria living in the pond die, but a few survive. **I.** What could be the reason for the survival of these few bacteria? **II.** How does this observation show the importance of variation for a species? **OR** **B.** Distinguish between regeneration and fragmentation as modes of reproduction. Give one example of an organism showing each.	**2**
12	Meera's family collects the following household waste in a single bin: vegetable peels, polythene bags, used tea leaves, glass bottles, old cotton cloth and ball-point pen refills. **A.** Sort these items into biodegradable and non-biodegradable wastes. **B.** State one harmful effect of non-biodegradable waste on the environment.	**2**
13	In pea plants, yellow seed colour (Y) is dominant over green seed colour (y). Ritika crosses a pure-breeding yellow-seeded plant with a pure-breeding green-seeded plant. **A.** What will be the seed colour and the genotype of the F_1 plants? **B.** The F_1 plants are self-pollinated. Show the F_2 generation with the help of a checkerboard and write the phenotypic ratio. **C.** If 1600 seeds are obtained in the F_2 generation, how many of them are expected to be green?	**3**
14	***Students to attempt either option A or B.*** **A.** Sana took two identical leafy shoots and placed them in two measuring cylinders, each filled with water up to the 50 mL mark. She covered the water surface in both cylinders with a thin layer of oil. Cylinder X was kept in sunlight and cylinder Y in a shady room. After 6 hours, the water level was 38 mL in X and 46 mL in Y. **I.** Why was a layer of oil poured over the water? **II.** Why is the loss of water greater in cylinder X? **III.** State two ways in which this loss of water is useful to the plant. **OR** **B.** **I.** State any two ways by which plants get rid of their excretory products. **II.** Name the structural and functional unit of the human kidney and the part of this unit where blood is filtered. **III.** On a cold winter day, a person passes more urine than on a hot summer day. Give reason.	**3**

15 **4**

Members of a school botany club placed pollen grains of a garden flower on glass slides containing sucrose solutions of different concentrations. After two hours, they observed the slides under a microscope and counted the pollen grains that had developed pollen tubes. Their results are shown in the graph.



A. At which sucrose concentration was the germination of pollen grains maximum? Which part of a flower provides a similar sugary medium to pollen grains in nature?

B. What is the role of the pollen tube in sexual reproduction of flowering plants?

Attempt either sub-part C or D.

C. List any four changes that take place in a flower after fertilisation.

OR

D. Rahul removes all the stamens from a flower bud before its anthers mature. He says that this flower can never produce seeds. Do you agree with him? Justify your answer.

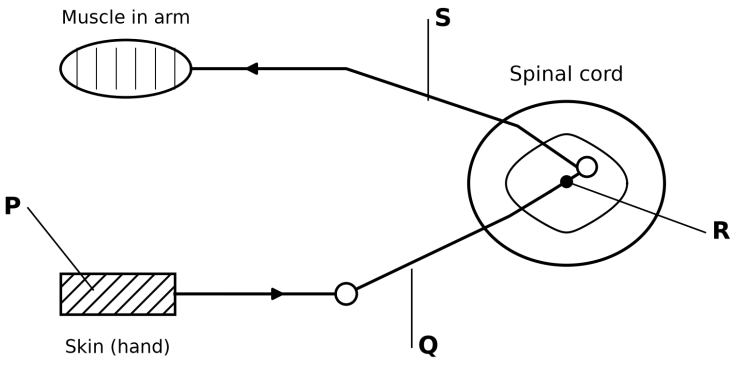
For visually impaired students

Members of a school botany club placed pollen grains of a garden flower in sucrose solutions of 0%, 5%, 10%, 15%, 20% and 25% concentration. After two hours, the percentage of pollen grains that had developed pollen tubes was 4%, 28%, 72%, 50%, 18% and 6% respectively. Answer sub-parts A, B and C or D given above on the basis of these results.

16 **5**

Students to attempt either option A or B.

A. **I.** Rohan accidentally touches a hot electric iron and pulls his hand away at once. The figure shows the pathway followed in this response.



(a) Identify the parts labelled P, Q, R and S. (2)

(b) Why is it an advantage that this response is completed at the level of the spinal cord instead of waiting for the brain to think? (1)

II. Name the part of the human brain that: (a) maintains the posture and balance of the body while walking along a narrow wall; (b) controls involuntary actions such as

salivation and blood pressure. (2)

OR

B. **I.** Ananya's grandfather has been diagnosed with diabetes and the doctor has prescribed insulin injections.
(a) Name the gland that secretes insulin and state its function. (1)
(b) Explain how the timing and amount of insulin released in the body are regulated. (2)

II. (a) People living in some hilly regions are advised to use iodised salt in their food. Why? (1)
(b) Name the hormone released when a person faces an emergency, such as being chased by a dog, and state any one effect it has on the body. (1)

For visually impaired students

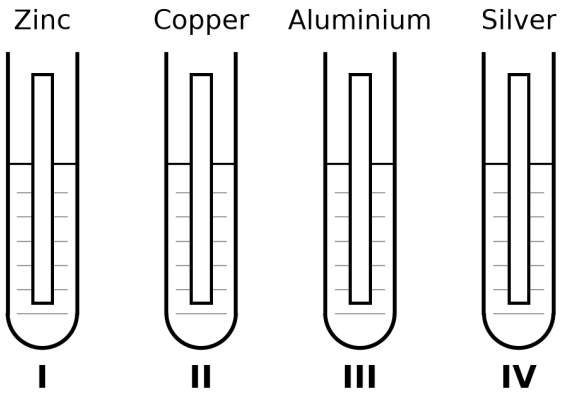
A. I. Rohan accidentally touches a hot electric iron and pulls his hand away at once.
(a) Name the receptor that senses the heat, the neurons that carry the message from the skin to the spinal cord and from the spinal cord to the arm, and the effector in this response. (2)
(b) Why is it an advantage that this response is completed at the level of the spinal cord instead of waiting for the brain to think? (1)

SECTION B Chemistry

17 Rahul's dentist tells him that tooth decay starts when the pH in the mouth falls below about 5.5 and advises him to brush his teeth after meals. Brushing with toothpaste helps because toothpaste is: **1**

A. basic and neutralises the excess acid formed in the mouth.
B. acidic and dissolves the food particles left in the mouth.
C. neutral and simply washes away the bacteria.
D. acidic and hardens the tooth enamel.

18 Strips of four metals are placed in iron(II) sulphate solution in four test tubes as shown. **1**



Each test tube contains iron(II) sulphate solution

In which test tubes will the pale green colour of the solution fade and a dark deposit form on the metal strip?

A. I and II **B.** I and III
C. II and IV **D.** III and IV

For visually impaired students

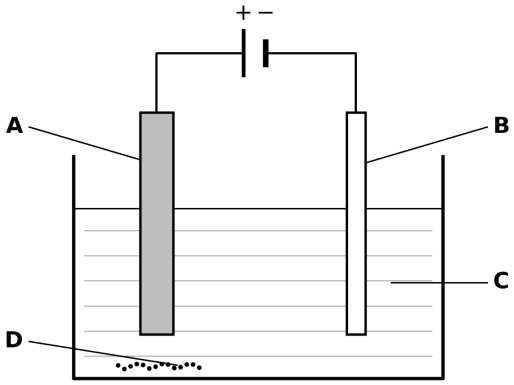
Strips of zinc, copper, aluminium and silver are placed separately in iron(II) sulphate solution. In which cases will the pale green colour of the solution fade?

	A. Zinc and copper **B.** Zinc and aluminium **C.** Copper and silver **D.** Aluminium and silver	
19	The number of single covalent bonds in a molecule of propane (C_3H_8) is: **A.** 8 **B.** 9 **C.** 11 **D.** 10	1
20	When ethanol is heated with alkaline potassium permanganate solution, it is converted into ethanoic acid. In this reaction, alkaline potassium permanganate acts as: **A.** a dehydrating agent **B.** a reducing agent **C.** an oxidising agent **D.** a catalyst	1
21	Aman adds zinc granules to sodium hydroxide solution and warms the mixture. A gas is evolved that burns with a pop sound. The products formed in this reaction are: **A.** sodium zincate and hydrogen **B.** zinc hydroxide and oxygen **C.** sodium zincate and oxygen **D.** zinc oxide and sodium	1
22	Consider the reaction of iron with steam: a Fe + b H_2O → c Fe_3O_4 + d H_2. The values of a, b, c and d in the balanced equation are respectively: **A.** 3, 4, 1, 3 **B.** 3, 4, 1, 4 **C.** 1, 4, 3, 4 **D.** 2, 3, 1, 3	1
23	Which one of the following non-metals is a liquid at room temperature? **A.** Iodine **B.** Mercury **C.** Chlorine **D.** Bromine	1

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24	**Assertion (A):** Copper vessels develop a green coating when left in moist air for a long time. **Reason (R):** Copper is placed below hydrogen in the reactivity series.	1
25	A farmer finds that the pH of the soil in his field is 4.5 and his crops are not growing well. **A.** Is the soil acidic or basic? **B.** Name a substance he should add to the soil and explain how it helps.	2
26	***Students to attempt either option A or B.*** **A.** An element X forms an oxide X_2O_3 which reacts with hydrochloric acid as well as with sodium hydroxide solution to form salt and water. **I.** Identify X and state the nature of its oxide. **II.** Write balanced chemical equations for the reactions of X_2O_3 with hydrochloric acid and with sodium hydroxide. **OR** **B.** The figure shows the set-up used for the electrolytic refining of copper.	3



I. Identify A, B, C and D.

II. What happens at A and at B when electric current is passed?

For visually impaired students

(I) In the electrolytic refining of copper, name the anode, the cathode and the electrolyte, and state where the insoluble impurities collect.

(II) What happens at the anode and at the cathode when electric current is passed?

27 Give reasons for the following: **3**

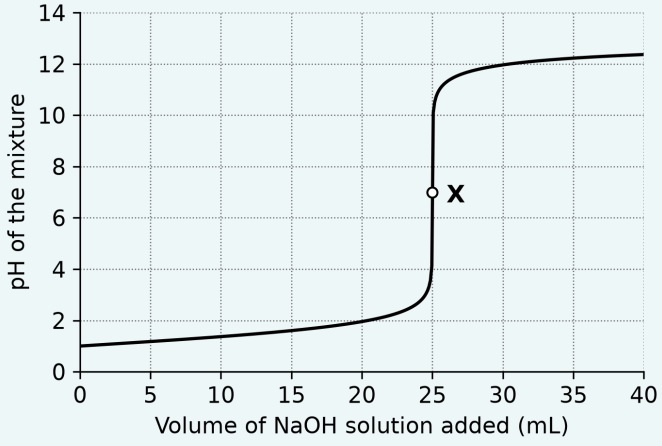
A. Most carbon compounds are poor conductors of electricity.

B. The bottoms of cooking vessels get blackened when the air holes of a gas stove get blocked.

C. Pure ethanoic acid is called glacial acetic acid.

28 **4**

Tanvi takes 25 mL of dilute hydrochloric acid in a beaker and adds sodium hydroxide solution of the same concentration from a burette, a little at a time, stirring continuously. She measures the pH of the mixture after each addition using a pH meter. Her results are shown in the graph.



A. What is the pH of the acid before any sodium hydroxide is added? What does this value tell us about the acid?

B. What does the point X on the graph represent?

Attempt either sub-part C or D.

C. Write the balanced chemical equation for the reaction and name the salt formed. What would be the colour of phenolphthalein in the beaker after 15 mL and after 35 mL of the base has been added?

OR

D. (i) If Tanvi had used ethanoic acid of the same concentration instead of hydrochloric acid, would the initial pH be lower, higher or the same? Justify.
(ii) Her friend says that the solution at point X is acidic because a salt has been formed. Do you agree? Give reason.

For visually impaired students

Tanvi takes 25 mL of dilute hydrochloric acid in a beaker and adds sodium hydroxide solution of the same concentration, a little at a time. The pH of the acid is 1 at the start; it rises slowly, becomes exactly 7 when 25 mL of the base has been added (point X), and then rises to about 12 when 40 mL of the base has been added. Answer sub-parts A, B and C or D given above on the basis of these results.

29 ***Students to attempt either option A or B.*** **5**

A. **I.** Classify each of the following reactions as combination, decomposition, displacement or double displacement. Write the balanced chemical equation and one observation for each.
(i) Lead nitrate crystals are heated strongly in a boiling tube.
(ii) A clean copper wire is dipped in silver nitrate solution.
(iii) Barium chloride solution is mixed with sodium sulphate solution. (3)

II. Copper(II) oxide is heated and hydrogen gas is passed over it: $CuO + H_2 \rightarrow Cu + H_2O$.
Identify the substance oxidised and the substance reduced. Justify your answer. (2)

OR

B. **I.** Riya observes that (a) the iron gate of her house in a coastal town is covered with a reddish-brown flaky coating, and (b) potato chips left in an opened packet for some days taste and smell bad. Name the process involved in each case and suggest two ways to prevent each. (3)

II. Silver chloride kept in a china dish turns grey when exposed to sunlight. Name the type of reaction, write the balanced chemical equation and state one use of this reaction in daily life. (2)

SECTION C Physics

30 The power of a lens is −2.0 D. The lens and its focal length are: **1**

A. concave, −50 cm **B.** convex, +50 cm
C. concave, −0.5 cm **D.** convex, +2 m

31 Shreya has a nichrome wire of resistance 10 Ω. She stretches it uniformly to twice its original length. The new resistance of the wire is: **1**

A. 5 Ω **B.** 10 Ω
C. 20 Ω **D.** 40 Ω

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Red colour is used for danger signal lights. **1**
Reason (R): Red light is scattered the most by fog and smoke.

33 Mr. Sharma, aged 62, can read the newspaper only when he holds it far from his eyes. He also finds it difficult to see distant objects clearly. **2**

A. Name the defect of vision related to reading and state its cause.

B. Suggest one type of spectacle lens that can correct both his problems and describe its construction.

34 ***Students to attempt either option A or B.*** **2**

A. **I.** Calculate the resistance of a copper wire 1 km long and 1 mm^2 in cross-sectional area. (Resistivity of copper = 1.7×10^{-8} Ω m)

II. What will be the resistance of another copper wire of the same length but double the diameter?

OR

B. **I.** Why do two magnetic field lines never intersect each other?

II. How does the strength of the magnetic field at the centre of a circular current-carrying coil change when (a) the number of turns is increased, (b) the radius of the coil is increased?

35 The table gives the absolute refractive indices of some media. **3**

Medium	Air	Water	Kerosene	Crown glass	Diamond
Refractive index	1.0003	1.33	1.44	1.50	2.42

A. In which of these media does light travel the fastest? Give reason.

B. Calculate the speed of light in crown glass. (Speed of light in vacuum = 3×10^8 m/s)

C. A ray of light travels obliquely from water into crown glass. Will it bend towards or away from the normal? Justify.

36 ***Students to attempt either option A or B.*** **3**

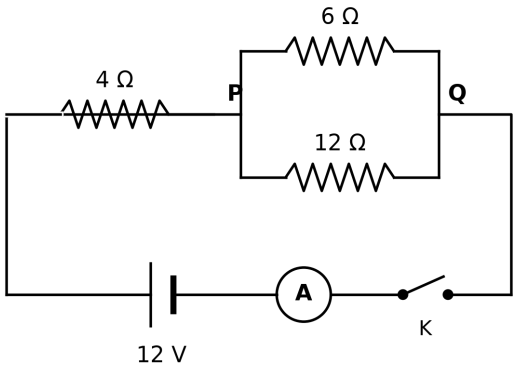
A. **I.** Explain why stars twinkle on a clear night.

II. Why do planets not twinkle?

OR

B. The Sun is visible to us about two minutes before the actual sunrise and about two minutes after the actual sunset. Explain this observation with the help of a labelled diagram.

37 Study the circuit given below. **3**



A. Find the equivalent resistance of the circuit.

B. Find the reading of the ammeter.

C. Calculate the potential difference across PQ and the current through the 12 Ω resistor.

For visually impaired students

A 4 Ω resistor is connected in series with a parallel combination of a 6 Ω and a 12 Ω resistor. The combination is connected to a 12 V battery through an ammeter and a key.
A. Find the equivalent resistance of the circuit.
B. Find the reading of the ammeter.
C. Calculate the potential difference across the parallel combination and the current through the 12 Ω resistor.

38 **4**

Mr. Gupta's kitchen circuit runs on a 220 V supply and is protected by a 10 A fuse. One evening he switched on a room heater (1100 W), a microwave oven (880 W) and an electric iron (660 W) together on this circuit. Within a few seconds the fuse blew and all the appliances stopped working. The electrician who came to repair it noticed that the electric iron had a three-pin plug whose longest pin was connected to a wire with green insulation.

A. Calculate the total current drawn from the supply when all three appliances work together.

B. Why did the fuse blow? Name this condition in the circuit.

Attempt either sub-part C or D.

C. What is the function of the green wire connected to the longest pin of the plug? Why is it important for an electric iron?

OR

D. Why is the fuse always connected in the live wire and in series with the appliances? What is the usual colour of the insulation of the live wire?

39 **5**

Students to attempt either option A or B.

A. **I.** An object 4 cm tall is placed 15 cm in front of a concave mirror of focal length 10 cm. Using the mirror formula, find the position, nature and size of the image. (3)

II. Draw a labelled ray diagram to show the formation of the image in this case. (2)

OR

B. **I.** A convex mirror used as a rear-view mirror in a bus has a radius of curvature of 4 m. A car is 6 m behind the bus. Find the position of the image of the car and the magnification. State the nature of the image. (3)

II. Why is a convex mirror preferred as a rear-view mirror in vehicles? Draw a ray diagram to show the formation of the image by a convex mirror. (2)

For visually impaired students

A. II. State the position and nature of the image formed by a concave mirror when the object is placed (a) at the focus, (b) between the pole and the focus. (2)
B. II. Give two reasons why a convex mirror is preferred as a rear-view mirror in vehicles. (2)

END OF QUESTION PAPER

Exam tip

1. In section E, read all three sub-parts before you start; part (iii) often has an internal choice.
2. Spend about 8 minutes on each case-based question.

SOLUTION PAPER 07

SCIENCE · CODE NO. 086

Sample Paper 7: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

B. Salivary amylase breaks down the starch of the bread into sugar. (1)

2 Ans

C. Water flows into cells P, they swell and the pore Q opens. (1)

For visually impaired students: C. guard cells swelling when water flows into them.

3 Ans

D. multiple fission (1)

4 Ans

A. 400 (1)

(F_2 genotypes TT : Tt : tt = 1 : 2 : 1; Tt = ½ × 800 = 400)

5 Ans

B. Condom (1)

6 Ans

D. UV radiation splits O_2 molecules into free oxygen atoms, which then combine with O_2 molecules. (1)

7 Ans

C. 20 J (1)

(10% law: grass 20,000 J → grasshopper 2000 J → frog 200 J → snake 20 J)

8 Ans

C. A is true but R is false. (1)

(Blood in veins flows under low pressure; valves prevent its backflow.)

9 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

10 Ans

A.

Glucose (6-carbon) —(in cytoplasm)→ Pyruvate (3-carbon) + energy (½)

Pyruvate —(lack of oxygen, in muscle cells)→ Lactic acid (3-carbon) + energy (½)

B.

The cramps are caused by the build-up of lactic acid in the muscles due to anaerobic respiration. (½)

During rest, the oxygen supply to the muscles is restored and the lactic acid is broken down / removed, so the cramps disappear. (½)

Total 2 marks

11 Ans

A.

I. These few bacteria had variations (heat-resistance) that arose during reproduction, which enabled them to survive the high temperature. (1)

II. If all bacteria were identical, the whole population would have been wiped out. Variations help a species survive sudden changes in its environment / niche. (1)

OR

B.

Regeneration: an organism is cut into many pieces and each piece grows into a complete organism using specialised cells; e.g. Planaria / Hydra. (1)

Fragmentation: a simple multicellular organism simply breaks up into pieces on maturing and each piece grows into a new individual; e.g. Spirogyra. (1)

Total 2 marks

12

Ans

A.

Biodegradable: vegetable peels, used tea leaves, old cotton cloth (½)

Non-biodegradable: polythene bags, glass bottles, ball-point pen refills (½)

B.

Any one: they persist in the environment for a long time and pollute soil and water / block drains / harm animals that swallow them / harmful chemicals may enter food chains. (1)

Total 2 marks

13

Ans

A.

All F_1 seeds are yellow; genotype Yy. **(½ + ½)** (1)

B.

F_1 (Yy) × F_1 (Yy); gametes Y and y (½)

♂ \ ♀	Y	y
Y	YY (yellow)	Yy (yellow)
y	Yy (yellow)	yy (green)

Phenotypic ratio: yellow : green = 3 : 1 (½)

C.

Green seeds = ¼ of the F_2 seeds = ¼ × 1600 = 400 (1)

Total 3 marks

14

Ans

A.

I. To prevent loss of water by evaporation from the surface of the water, so that the fall in level is only due to the shoot. (1)

II. Higher temperature and light in sunlight make the stomata open wider and increase evaporation, so the rate of transpiration is higher in X (12 mL lost) than in Y (4 mL lost). (1)

III. (Any two) It creates a suction pull that helps in the absorption and upward movement of water and minerals from roots to leaves; it helps in cooling the plant / regulating its temperature. **(½ + ½)** (1)

OR

B.

I. (Any two) Oxygen and carbon dioxide diffuse out through stomata; excess water is lost by transpiration; wastes are stored in old leaves that fall off; wastes stored as resins and gums in old xylem; some wastes are excreted into the soil. **(½ + ½)** (1)

II. Nephron; blood is filtered in the glomerulus (inside Bowman's capsule). **(½ + ½)** (1)

III. On a cold day, very little water is lost as sweat, so less water is reabsorbed in the tubules of the nephrons and a larger amount of urine is produced. (The amount of water reabsorbed depends on how much excess water there is in the body.) (1)

Total 3 marks

15

Ans

A.

Maximum germination (72%) at 10% sucrose. (½)

The stigma provides a sticky, sugary medium for pollen to germinate. (½)

B.

The pollen tube grows out of the pollen grain, travels through the style and carries the male germ cell to the female germ cell (egg) in the ovule. (1)

C.

(Any four) The zygote divides to form the embryo; the ovule develops a tough coat and becomes the seed; the ovary grows rapidly and ripens to form the fruit; petals, sepals, stamens, style and stigma shrivel and fall off. **(½ × 4)** (2)

OR

D.

No, I do not agree. (½)

Removing the stamens only prevents self-pollination. The carpel (pistil) is still present, so pollen from another flower can reach its stigma by cross-pollination (through insects, wind, etc.), fertilisation can occur and seeds can be formed. (1½)

Total 4 marks

16 **Ans**

A.

I. (a) P – receptor in the skin; Q – sensory neuron; R – relay neuron in the spinal cord; S – motor neuron. **(½ × 4)** (2)

(b) Thinking in the brain takes more time. The spinal cord connects the input nerve directly to the output nerve, so the response is very quick and protects the hand from serious burns. (1)

II. (a) Cerebellum (part of the hind-brain) (1)

(b) Medulla (in the hind-brain) (1)

For visually impaired students (A. I.):

(a) Receptor: heat/pain receptors in the skin; sensory neuron carries the message from skin to spinal cord; motor neuron carries it from spinal cord to arm; effector: muscle of the arm. **(½ × 4)** (2)

(b) As above. (1)

OR

B.

I. (a) Pancreas; insulin regulates the blood sugar level (helps cells take up glucose and lowers excess sugar in blood). **(½ + ½)** (1)

(b) The timing and amount of hormone released are regulated by a feedback mechanism. When the sugar level in blood rises, it is detected by the cells of the pancreas, which respond by producing more insulin. As the blood sugar level falls, insulin secretion is reduced. (2)

II. (a) Iodine is essential for the thyroid gland to make thyroxin. Soil and water in some hilly regions are deficient in iodine; iodine deficiency may cause goitre (swollen neck). (1)

(b) Adrenaline (from the adrenal glands). Any one effect: heart beats faster supplying more oxygen to muscles; blood to digestive system and skin is reduced and diverted to skeletal muscles; breathing rate increases. **(½ + ½)** (1)

Total 5 marks

SECTION B Chemistry

17 **Ans**

A. basic and neutralises the excess acid formed in the mouth. (1)

18 **Ans**

B. I and III (1)

(Zinc and aluminium are more reactive than iron and displace it; copper and silver are less reactive.)

For visually impaired students: B. Zinc and aluminium

19 **Ans**

D. 10
(2 C–C bonds + 8 C–H bonds) (1)

20 **Ans**

C. an oxidising agent (1)

21 **Ans**

A. sodium zincate and hydrogen
($2NaOH + Zn \rightarrow Na_2ZnO_2 + H_2$) (1)

22 **Ans**

B. 3, 4, 1, 4
($3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$) (1)

23 **Ans**

D. Bromine (1)

24 **Ans**

B. Both A and R are true, and R is not the correct explanation of A. (1)
(The green coating is basic copper carbonate formed by the reaction of copper with moist carbon dioxide in air.)

25 **Ans**

A.

The soil is acidic (pH less than 7). (½)

B.

Quicklime (calcium oxide) / slaked lime (calcium hydroxide) / chalk (calcium carbonate). (½)

It is basic in nature and neutralises the excess acid in the soil, raising its pH towards neutral, which is suitable for the growth of most crops. (1)

Total 2 marks

26 **Ans**

A.

I. X is aluminium (Al); Al_2O_3 is an amphoteric oxide. **(½ + ½)** (1)

II. $Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ (1)

$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ (sodium aluminate) (1)

OR

B.

I. A – impure copper (anode); B – thin strip of pure copper (cathode); C – acidified copper sulphate solution (electrolyte); D – anode mud (insoluble impurities). **(½ × 4)** (2)

II. At A, impure copper dissolves into the electrolyte as Cu^{2+} ions; at B, an equivalent amount of pure copper from the electrolyte is deposited. **(½ + ½)** (1)

For visually impaired students:

I. Anode – impure copper; cathode – pure copper strip; electrolyte – acidified copper sulphate solution; insoluble impurities settle below the anode as anode mud. **(½ × 4)** (2)

II. As above. (1)

Total 3 marks

27 **Ans**

A.

Carbon forms covalent bonds by sharing electrons; these compounds do not form ions, hence they are poor conductors of electricity. (1)

B.

When the air holes are blocked, the supply of oxygen is insufficient and the fuel undergoes incomplete combustion; the unburnt carbon (soot) deposits on the vessels. (1)

C.

The melting point of pure ethanoic acid is 290 K (about 17 °C), so it often freezes in cold climates and looks like ice (glacier). (1)

Total 3 marks

28 **Ans**

A.

pH = 1; it is a strong acid (high concentration of H^+ ions). **(½ + ½)** (1)

B.

Point X is the neutralisation point: 25 mL of NaOH has completely neutralised the acid and the pH is 7. (1)

C.

$HCl + NaOH \rightarrow NaCl + H_2O$; salt: sodium chloride **(½ + ½)** (1)

After 15 mL: colourless (solution still acidic); after 35 mL: pink (solution is basic). **(½ + ½)** (1)

OR

D.

(i) Higher. Ethanoic acid is a weak acid; it ionises only partially and gives fewer H^+ ions than HCl of the same concentration. **(½ + ½)** (1)

(ii) No. Sodium chloride is a salt of a strong acid and a strong base, so its solution is neutral (pH 7, as shown by point X). **(½ + ½)** (1)

Total 4 marks

29 **Ans**

A.

I. (i) Decomposition (thermal): $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + O_2$; brown fumes of nitrogen dioxide are seen. **(½ + ½)** (1)

(ii) Displacement: $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$; solution turns blue and a shiny greyish-white deposit of silver forms on the wire. **(½ + ½)** (1)

(iii) Double displacement (precipitation): $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4\downarrow + 2NaCl$; a white precipitate of barium sulphate forms. **(½ + ½)** (1)

II. CuO is reduced to Cu as it loses oxygen. (1)

H_2 is oxidised to H_2O as it gains oxygen. (It is a redox reaction.) (1)

OR

B.

I. (a) Corrosion (rusting) of iron. (½)

Prevention (any two): painting, oiling/greasing, galvanisation (coating with zinc), chrome plating, alloying. **(½ + ½)** (1)

(b) Rancidity (oxidation of fats and oils). (½)

Prevention (any two): adding antioxidants; packing in nitrogen gas; keeping in airtight containers; refrigeration. **(½ + ½)** (1)

II. Photolytic decomposition (decomposition by light). (½)

$2AgCl(s) \xrightarrow{\text{sunlight}} 2Ag(s) + Cl_2(g)$ (1)

Use: in black and white photography. (½)

Total 5 marks

SECTION C Physics

30 **Ans**

A. concave, −50 cm (1)

(f = 1/P = 1/(−2.0) m = −0.5 m = −50 cm)

31 **Ans**

D. 40 Ω (1)

(Length doubles and area halves, so R = ρl/A becomes 4 times.)

32 Ans

C. A is true but R is false. (1)

(Red light has the longest wavelength and is scattered the least by fog and smoke, so it can be seen from a distance.)

33 Ans

A.

Presbyopia. (½)

Cause: gradual weakening of the ciliary muscles and diminishing flexibility of the eye lens with age, so the near point recedes. (½)

B.

Bifocal lens. (½)

Its upper portion is a concave lens for distant vision and the lower portion is a convex lens for near vision (reading). (½)

Total 2 marks

34 Ans

A.

I. $R = \rho l/A = (1.7 \times 10^{-8} \times 1000)/(1 \times 10^{-6}) = 17\ \Omega$ (1)

II. Doubling the diameter makes the area 4 times, so $R' = R/4 = 17/4 = 4.25\ \Omega$ (1)

OR

B.

I. If two field lines intersected, the compass needle at that point would point in two directions at once, which is not possible. (1)

II. (a) The field strength increases (it is proportional to the number of turns). (½)

(b) The field strength decreases (it is inversely related to the radius). (½)

Total 2 marks

35 Ans

A.

Air, because it has the lowest refractive index (n = c/v; lower n means higher speed). **(½ + ½)** (1)

B.

$v = c/n = (3 \times 10^8)/1.50$ (½)

$= 2 \times 10^8$ m/s (½)

C.

Towards the normal, (½)

because light travels from an optically rarer medium (water, 1.33) to an optically denser medium (crown glass, 1.50), where its speed decreases. (½)

Total 3 marks

36 Ans

A.

I. Starlight entering the Earth's atmosphere undergoes continuous refraction before reaching the Earth. Since the physical conditions (temperature, density) of the atmosphere keep changing, the apparent position of the star fluctuates slightly. (1)

Stars are very far away and act as point sources, so the amount of starlight entering the eye also flickers – the star appears sometimes brighter and sometimes fainter, i.e. it twinkles. (1)

II. Planets are much closer and behave as extended sources (a collection of many point sources). The variations in light from all these points average out to zero, so planets do not twinkle. (1)

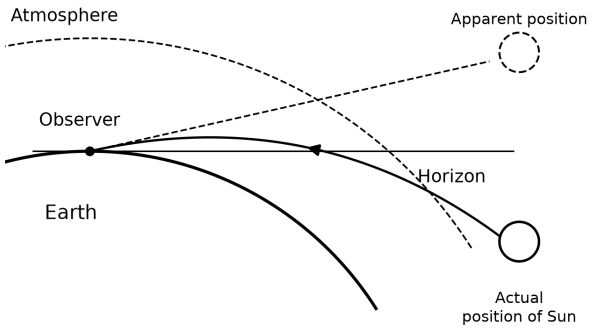
OR

B.

Due to atmospheric refraction, light from the Sun when it is slightly below the horizon bends towards the normal as it passes through layers of the atmosphere of increasing density and reaches the observer. (1)

The Sun therefore appears raised above its actual position; so it is seen about 2 minutes before actual sunrise and about 2 minutes after actual sunset (the time difference between actual and apparent sunset/sunrise is about 2 minutes). (1)

Labelled diagram: (1)



Total 3 marks

37 Ans

A.

Parallel combination: $1/R_p = 1/6 + 1/12 = 3/12$, so $R_p = 4\ \Omega$ (½)

Equivalent resistance $R = 4\ \Omega + 4\ \Omega = 8\ \Omega$ (½)

B.

$I = V/R = 12/8 = 1.5$ A (ammeter reading) (1)

C.

$V_{PQ} = I \times R_p = 1.5 \times 4 = 6$ V (½)

Current through 12 Ω = 6/12 = 0.5 A (½)

Total 3 marks

38 Ans

A.

Total power P = 1100 + 880 + 660 = 2640 W; I = P/V = 2640/220 = 12 A (1)

B.

The total current (12 A) exceeded the rating of the fuse (10 A), so the fuse wire heated up and melted. This condition is called overloading. **(½ + ½)** (1)

C.

It is the earth wire; it provides a low-resistance conducting path for current to the earth. (1)

The iron has a metallic body. If current leaks to the body, it flows to the earth instead of through the user, keeping the body at the potential of the earth and preventing a severe electric shock. (1)

OR

D.

In series, the whole current through the appliances passes through the fuse, so it melts whenever the current exceeds its rating. (½)

In the live wire, so that when it blows the appliances are disconnected from the high potential and are safe to touch. (1)

Live wire: red insulation (brown in the new convention). (½)

Total 4 marks

39

A.

Ans

I. h = +4 cm, u = −15 cm, f = −10 cm. Mirror formula: 1/v + 1/u = 1/f (½)

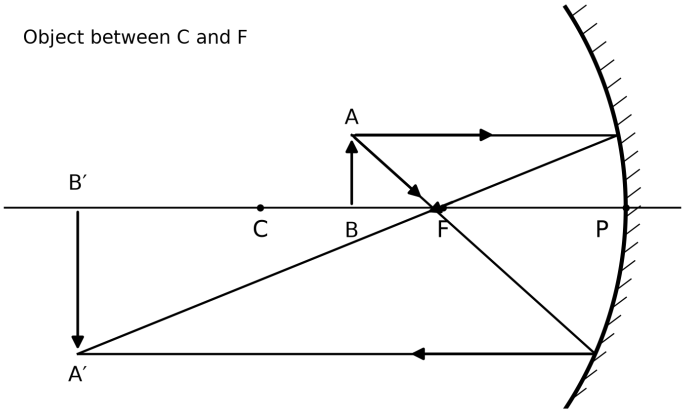
1/v = 1/f − 1/u = −1/10 + 1/15 = −1/30, so v = −30 cm (1)

m = −v/u = −(−30)/(−15) = −2; h′ = m × h = −2 × 4 = −8 cm (1)

The image is formed 30 cm in front of the mirror; it is real, inverted and magnified (8 cm tall). (½)

II. Correct ray diagram with arrows (1)

Correct labelling (C, F, P, object and image) (1)



For visually impaired students (A. II.):

(a) At the focus: image at infinity, highly enlarged, real and inverted. (1)

(b) Between P and F: image behind the mirror, virtual, erect and enlarged. (1)

OR

B.

I. R = +4 m, so f = R/2 = +2 m; u = −6 m (½)

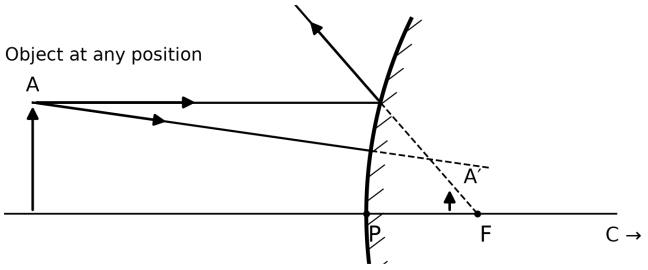
1/v = 1/f − 1/u = 1/2 + 1/6 = 4/6, so v = +1.5 m (1)

m = −v/u = −(1.5)/(−6) = +0.25 (1)

The image is 1.5 m behind the mirror; it is virtual, erect and diminished. (½)

II. A convex mirror always gives an erect, diminished image and has a wider field of view, so the driver can see a large area of traffic behind. (1)

Ray diagram: (1)



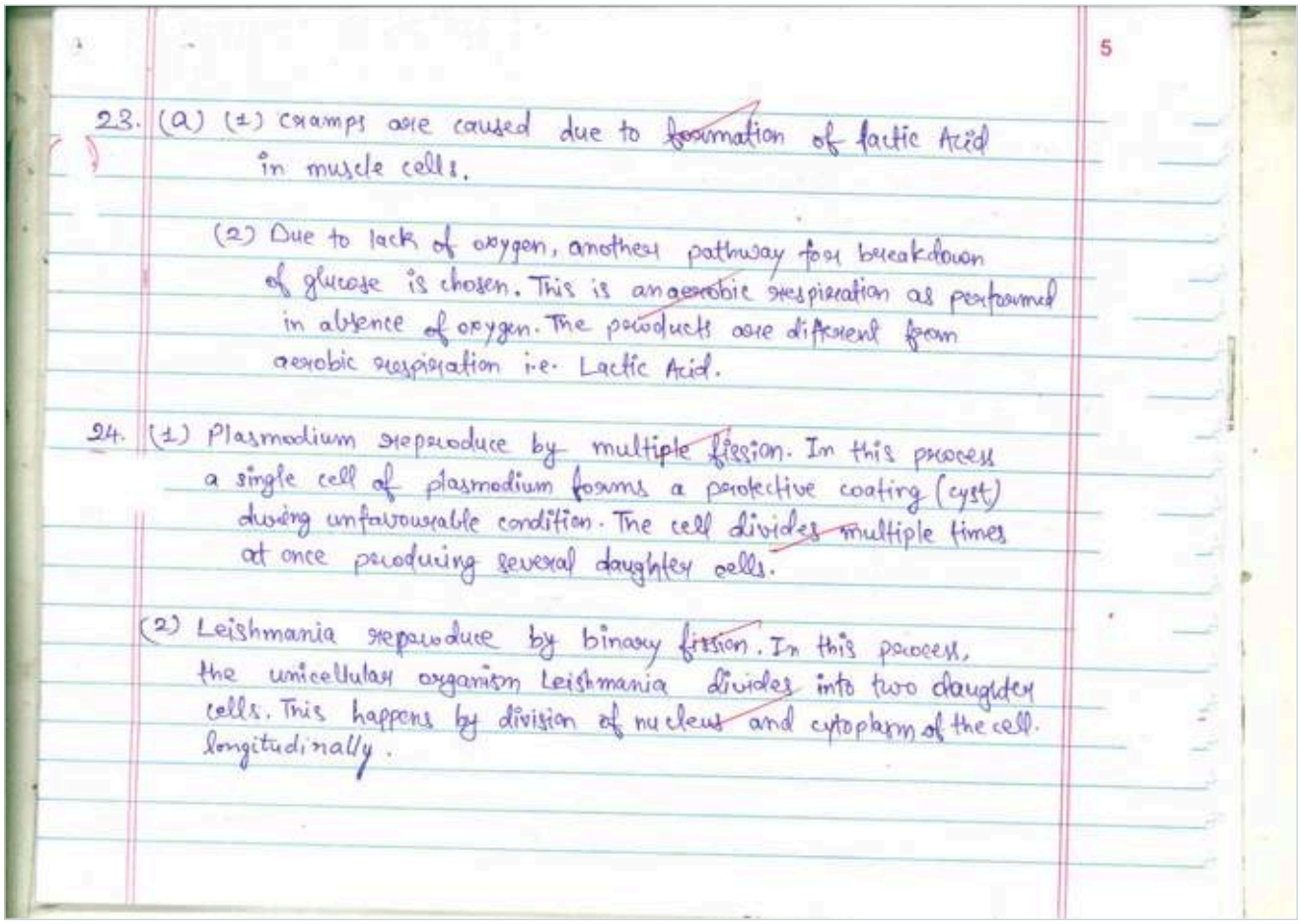
For visually impaired students (B. II.):

It always forms an erect image; the image is diminished so it gives a wider field of view (any two). **(1 + 1)** (2)

Total 5 marks

END OF SOLUTIONS

Real CBSE topper's answer · Board Exam 2024 · similar to Q.10 of this paper (muscle cramps and anaerobic respiration)



5

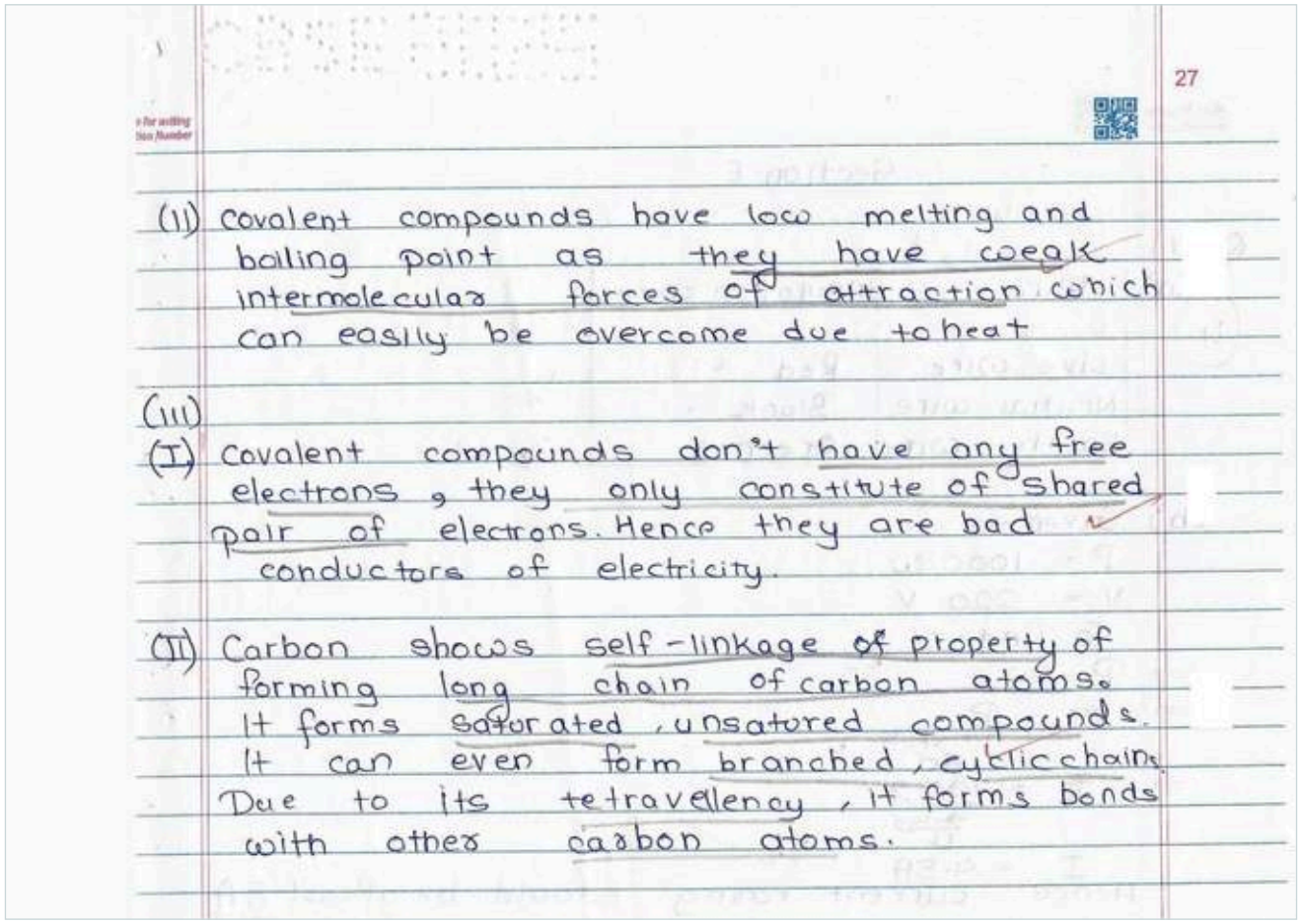
23. (a) (1) cramps are caused due to formation of lactic Acid in muscle cells.

(2) Due to lack of oxygen, another pathway for breakdown of glucose is chosen. This is anaerobic respiration as performed in absence of oxygen. The products are different from aerobic respiration i.e. Lactic Acid.

24. (1) Plasmodium reproduce by multiple fission. In this process a single cell of plasmodium forms a protective coating (cyst) during unfavourable condition. The cell divides multiple times at once producing several daughter cells.

(2) Leishmania reproduce by binary fission. In this process, the unicellular organism Leishmania divides into two daughter cells. This happens by division of nucleus and cytoplasm of the cell. longitudinally.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.27 of this paper (why carbon compounds are poor conductors)



27

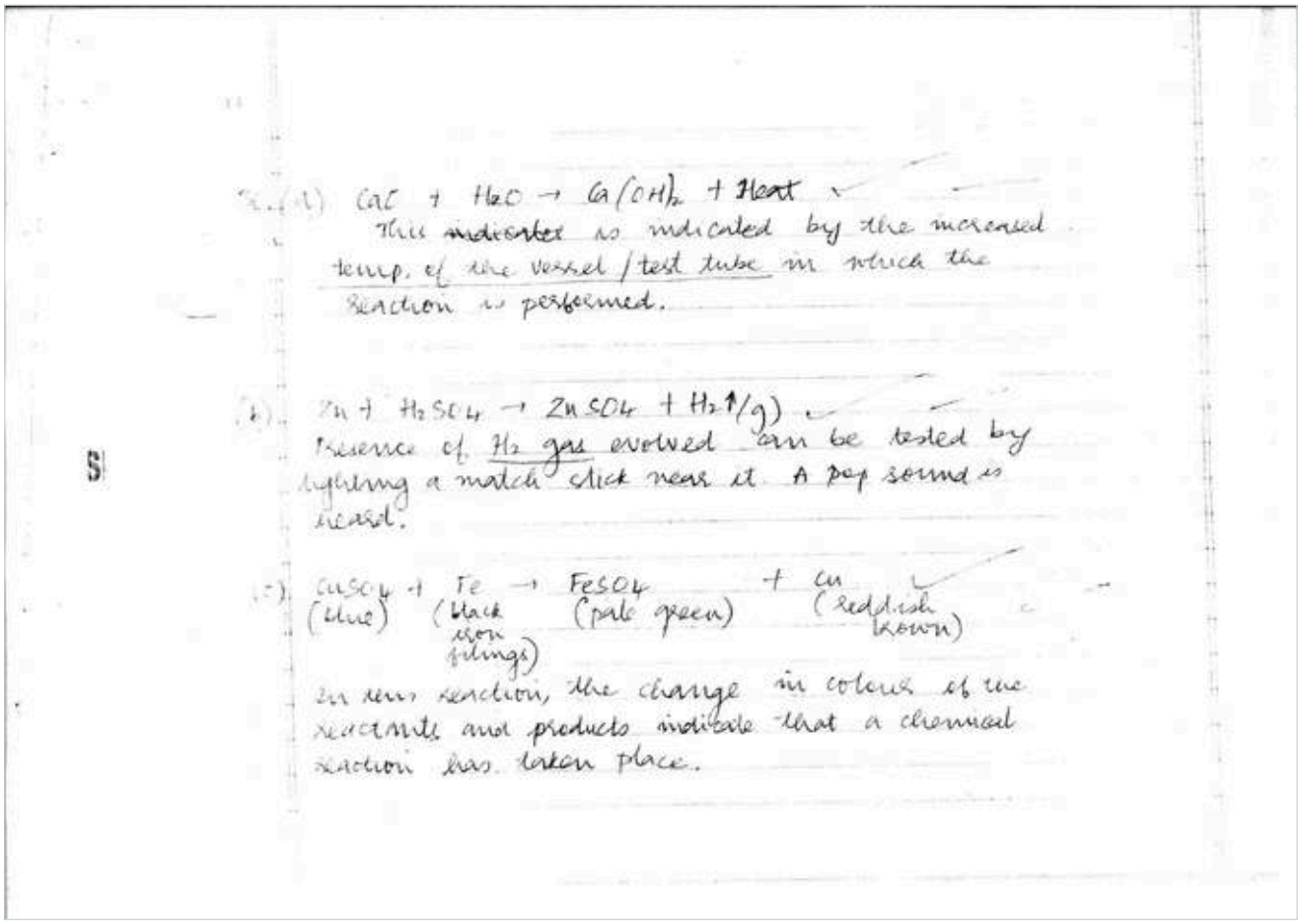
(II) Covalent compounds have low melting and boiling point as they have weak intermolecular forces of attraction which can easily be overcome due to heat

(III)

(I) Covalent compounds don't have any free electrons, they only constitute of shared pair of electrons. Hence they are bad conductors of electricity.

(II) Carbon shows self-linkage of property of forming long chain of carbon atoms. It forms saturated, unsatured compounds. It can even form branched, cyclic chains. Due to its tetravellency, it forms bonds with other carbon atoms.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.29 of this paper (classifying chemical reactions)



(a) $CaC + H_2O \rightarrow Ca(OH)_2 + Heat$

This reaction is indicated by the increased temp. of the vessel / test tube in which the reaction is performed.

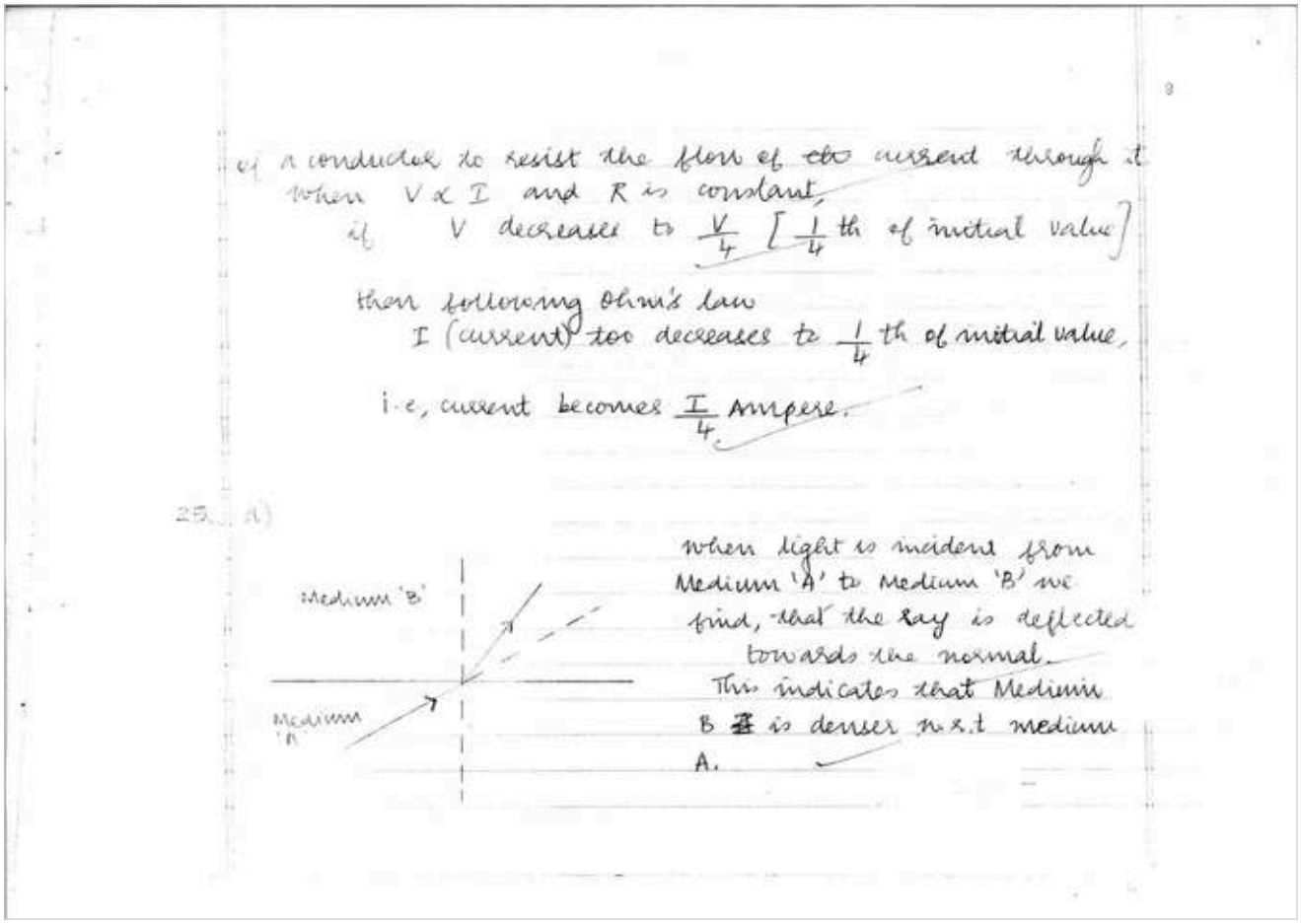
(b) $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2\uparrow(g)$

Presence of H_2 gas evolved can be tested by lighting a match stick near it. A pop sound is heard.

(c) $CuSO_4$ (blue) + Fe (black iron filings) → $FeSO_4$ (pale green) + Cu (reddish brown)

In this reaction, the change in colour of the reactants and products indicate that a chemical reaction has taken place.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.35 of this paper (refractive index of media)



of a conductor to resist the flow of the current through it when $V \propto I$ and R is constant,

if V decreases to $\frac{V}{4}$ [$\frac{1}{4}$ th of initial value]

then following Ohm's law

I (current) too decreases to $\frac{1}{4}$ th of initial value,

i.e, current becomes $\frac{I}{4}$ Ampere.

25. A)

When light is incident from Medium 'A' to Medium 'B' we find, that the ray is deflected towards the normal.

This indicates that Medium B is denser w.r.t medium A.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.35 of this paper (optical density and refractive index)

b) Speed of light in Medium A → v_a [v_1]
Speed of light in Medium B = v_b [v_2]

$$n_{21} = \frac{v_1}{v_2} = \frac{v_a}{v_b} \quad [\text{Ans}]$$

using formula
Refractive index of B w.r.t. A → $\frac{\text{speed of light in A}}{\text{speed of light in B}}$

5

26. a)

Right Hand Thumb Rule ~~of Max~~ right
if the direction of our ~~hand~~ thumb on a conductor indicates the direction of current, then the direction in which our fingers of right hand wrap around the conductor indicates the direction of magnetic field around the straight current carrying wire.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.38 of this paper (domestic circuits, fuse and current rating)

28

Section-E

Q.37)

(a)

Wire	Insulation cover
Live wire	Red
Neutral wire	Black
Earth wire	Green

(b) Given →
P = 1000 ω
V = 220 V
To find → I
⇒ P = VI
⇒ $I = \frac{P}{V}$
⇒ $I = \frac{1000}{220}$ (4.5, 50)
I = 4.5A
Hence current rating should be atleast 5A.

SAMPLE PAPER 08

SCIENCE · CODE NO. 086

Sample Question Paper 8

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 (1)

Due to an infection, the pancreatic duct of a patient gets blocked and the pancreatic juice cannot reach the small intestine. Which of the following will be affected the most?

A. Only the digestion of starch in the mouth.

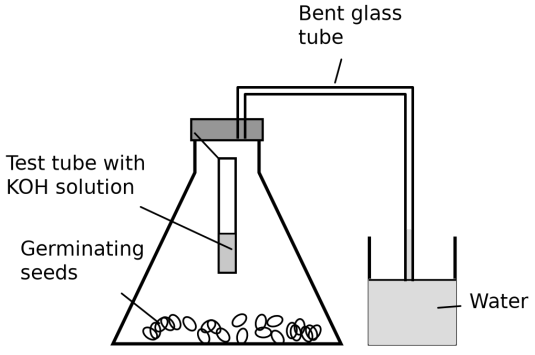
B. Only the absorption of water in the large intestine.

C. The digestion of proteins, fats and carbohydrates in the small intestine.

D. The secretion of hydrochloric acid in the stomach.

2 (1)

Rahul set up the apparatus shown below to study respiration in germinating seeds. After two hours, the water rose in the bent glass tube.



Which of the following is the most appropriate explanation of this observation?

A. The seeds released oxygen, which dissolved in the KOH solution.

B. The seeds used up oxygen and the carbon dioxide released by them was absorbed by KOH, lowering the air pressure inside the flask.

C. KOH absorbed the oxygen present in the flask.

D. The seeds absorbed water vapour from the flask and became heavier.

For visually impaired students

Which of the following is kept in a small test tube inside a flask of germinating seeds to absorb the carbon dioxide released during respiration?

A. Potassium hydroxide solution **B.** Sodium chloride solution
C. Distilled water **D.** Dilute hydrochloric acid

3 The tendril of a pea plant touches a bamboo stick and coils around it within a few days. Which of the following statements correctly explains this movement? **1**

A. The cells of the tendril in contact with the stick grow faster than the cells on the other side.
B. Gibberellins collect on the side of the tendril in contact with the stick and stop its growth.
C. It is a growth-independent movement caused by a sudden loss of water, as seen in the sensitive plant.
D. The cells of the tendril on the side away from the stick grow faster than the cells in contact with it.

4 Which of the following is a correct pair of an organism and its mode of asexual reproduction? **1**

A. *Plasmodium* – Multiple fission **B.** *Leishmania* – Budding
C. *Hydra* – Fragmentation **D.** *Spirogyra* – Budding

5 Mr and Mrs Sharma have two sons. They are hoping that their third child will be a daughter. What is the probability of the third child being a girl? **1**

A. 100%, because two boys have already been born to the couple.
B. 25%, because the chance of a girl reduces with every boy born.
C. 0%, because the mother contributes a Y chromosome to boys.
D. 50%, because half of the sperms produced by the father carry an X chromosome.

6 In a dihybrid cross, a pure-breeding pea plant with round, yellow seeds (RRYY) is crossed with a plant having wrinkled, green seeds (rryy). The F_1 plants are self-pollinated and 1600 seeds are obtained in the F_2 generation. How many of these seeds are expected to be round and green? **1**

A. 100 **B.** 600
C. 300 **D.** 900

7 The ozone layer in the upper atmosphere shields the Earth from harmful ultraviolet radiation. Ozone at this level is formed when **1**

A. high-energy UV radiation splits oxygen molecules (O_2) into free oxygen atoms, which then combine with O_2 molecules.
B. chlorofluorocarbons react with oxygen in the presence of sunlight.
C. lightning combines atmospheric nitrogen with oxygen.
D. green plants release ozone as a by-product of photosynthesis.

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8 **Assertion (A):** A person whose cerebellum is injured in an accident may find it difficult to walk in a straight line. **1**

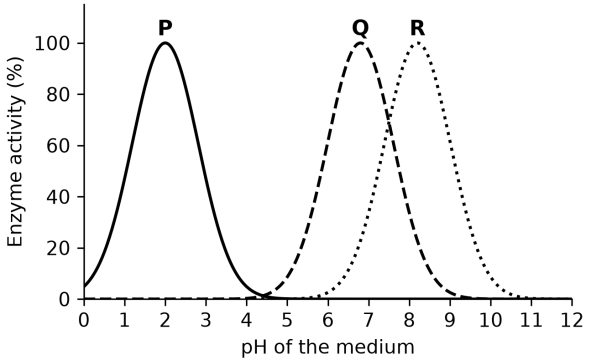
	Reason (R): The cerebellum is responsible for maintaining the posture and balance of the body.	
9	**Assertion (A):** Food chains in nature generally consist of only three or four trophic levels. **Reason (R):** The flow of energy in an ecosystem is bidirectional.	**1**
10	Tanvi kept a potted croton plant having variegated leaves (green and white patches) in a dark room for three days. She then placed it in bright sunlight for six hours, plucked a leaf, boiled it in alcohol and added iodine solution to it. **A.** Which regions of the leaf will turn blue-black? What does this observation show? **B.** Why was the plant kept in the dark room for three days before the experiment?	**2**
11	***Students to attempt either option A or B.*** **A.** Rohan noticed that a slice of bread left in a warm, moist corner of the kitchen developed a white cottony growth with tiny black dots after a few days. Name the organism and its mode of reproduction. What do the black dots contain, and why does the growth not appear on dry bread? **OR** **B.** How is regeneration in *Planaria* different from fragmentation in *Spirogyra*? Why can complex multicellular organisms not reproduce by fragmentation?	**2**
12	A potted seedling kept near a window bends towards the window after a few days. Name the plant hormone responsible for this bending and explain how it brings about the movement.	**2**
13	Riya's grandmother often blames Riya's mother for having given birth to three daughters. Using a flow chart of sex determination in human beings, explain why this belief is scientifically incorrect. Also state the chance of the next child being a boy.	**3**
14	***Students to attempt either option A or B.*** **A.** Study the pond food chain given below, which shows the concentration of the pesticide DDT in each organism. **Phytoplankton** DDT: 0.04 ppm → **Zooplankton** DDT: 0.23 ppm → **Small fish** DDT: 2.07 ppm → **Kingfisher** DDT: 25.0 ppm **I.** Name the phenomenon shown by the increasing concentration of DDT along the food chain. **II.** Why is the concentration of DDT highest in the kingfisher? **III.** If the phytoplankton trap 20,000 J of energy, how much energy will be available to the kingfisher? **For visually impaired students** In a pond, the concentration of the pesticide DDT was found to be 0.04 ppm in phytoplankton, 0.23 ppm in zooplankton, 2.07 ppm in small fish and 25.0 ppm in kingfishers that feed on the fish. **I.** Name the phenomenon shown by the increasing concentration of DDT along this food chain. **II.** Why is the concentration of DDT highest in the kingfisher? **III.** If the phytoplankton trap 20,000 J of energy, how much energy will be available to the kingfisher? **OR** **B.** The following waste items were collected from a housing society in one day: vegetable peels, plastic bottles, newspapers, glass jars, dry leaves and aluminium foil.	**3**

I. Classify these items into biodegradable and non-biodegradable wastes.

II. State two harmful effects of the accumulation of non-biodegradable wastes.

III. Suggest one practice the residents can adopt to reduce the waste they produce.

15 Manjeet studied the effect of pH on the activity of three digestive enzymes P, Q and R found in the human alimentary canal. His results are shown in the graph below. **4**



A. Which enzyme is most likely to be pepsin? Give a reason. (1)

B. Enzyme Q is secreted by the salivary glands. Name the food component it acts upon and the product formed. (1)

Attempt either sub-part C or D.

C. What will happen to the activity of enzyme Q when the food reaches the stomach? Name the substance responsible for this change. Why does pepsin not digest the inner lining of the stomach? (2)

OR

D. Enzyme R acts in the small intestine. Name enzyme R and the gland that secretes it. How is the acidic food coming from the stomach made suitable for the action of R? (2)

For visually impaired students

Manjeet studied three digestive enzymes P, Q and R. He found that P works best at pH 2, Q works best at about pH 6.8 and R works best at about pH 8.

A. Which enzyme is most likely to be pepsin? Give a reason.

B. Enzyme Q is secreted by the salivary glands. Name the food component it acts upon and the product formed.

Attempt either sub-part C or D.

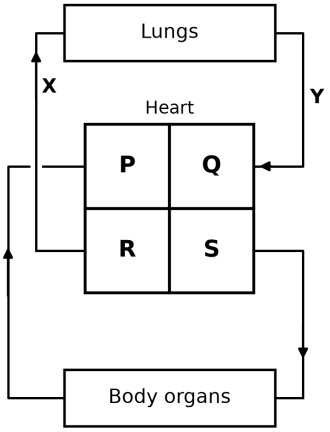
C. What will happen to the activity of enzyme Q when the food reaches the stomach? Name the substance responsible for this change. Why does pepsin not digest the inner lining of the stomach?

OR

D. Enzyme R acts in the small intestine. Name enzyme R and the gland that secretes it. How is the acidic food coming from the stomach made suitable for the action of R?

16 ***Attempt either option A or B.*** **5**

A. I. The diagram below shows the path of blood through the human heart. P, Q, R and S are the four chambers of the heart and X and Y are blood vessels.



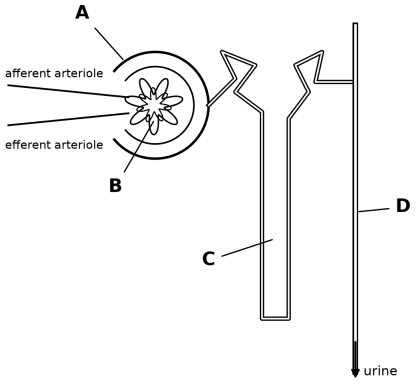
a. Name the chambers P and S. Which two chambers of the heart contain oxygenated blood?

b. Name the blood vessels X and Y. Why is the complete separation of oxygenated and deoxygenated blood important for birds and mammals?

II. Water absorbed by the roots reaches the top of a 30 m tall tree. Explain the role of transpiration in this upward movement. How is the transport of food in the phloem different from the transport of water in the xylem?

OR

B. I. The diagram below shows the structure of a nephron, the basic filtration unit of the human kidney.



a. Identify the parts labelled A and B.

b. What happens in part B, and which substances are reabsorbed as the filtrate passes along the tubule C?

c. How is the amount of urine produced regulated by the body?

II. What is the principle of working of an artificial kidney (haemodialysis)? Name two ways by which plants get rid of their excretory products.

For visually impaired students

A. I. (a) Name the chambers of the human heart that receive blood from the body and pump blood to the body. Which two chambers contain oxygenated blood? (b) Name the blood vessels that carry blood from the heart to the lungs and from the lungs to the heart. Why is the complete separation of oxygenated and deoxygenated blood important for birds and mammals?

II. Water absorbed by the roots reaches the top of a 30 m tall tree. Explain the role of transpiration in this upward movement. How is the transport of food in the phloem different from the transport of water in the xylem?

OR

B. I. (a) Name the cup-shaped structure of a nephron and the cluster of capillaries inside it. (b) What happens in the cluster of capillaries, and which substances are reabsorbed as the filtrate passes along the tubule? (c) How is the amount of urine produced regulated by the body?
II. What is the principle of working of an artificial kidney (haemodialysis)? Name two ways by which plants get rid of their excretory products.

SECTION B Chemistry

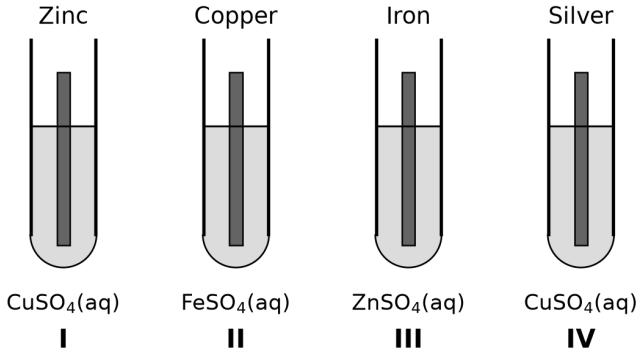
17 When a few drops of potassium iodide solution are added to a test tube containing colourless lead nitrate solution, a bright yellow solid settles at the bottom. This reaction is an example of **1**

A. a combination reaction.
B. a decomposition reaction.
C. a displacement reaction.
D. a double displacement (precipitation) reaction.

18 Sodium hydrogencarbonate is an ingredient of many antacids because **1**

A. it is a strong alkali that removes all the acid from the stomach.
B. it is a mild, non-corrosive base that neutralises the excess acid in the stomach.
C. it is an acidic salt that lowers the pH of the stomach.
D. it releases chlorine, which kills the bacteria causing acidity.

19 A student placed strips of four metals in four different salt solutions as shown below. **1**



In which test tube(s) will a change in the colour of the solution be observed after some time?

A. I only **B.** I and III
C. II and IV **D.** I and II

For visually impaired students

A student placed a zinc strip in copper sulphate solution, a copper strip in iron sulphate solution, an iron strip in zinc sulphate solution and a silver strip in copper sulphate solution. In which case will a displacement reaction take place?

A. Zinc in copper sulphate solution
B. Copper in iron sulphate solution
C. Iron in zinc sulphate solution
D. Silver in copper sulphate solution

20 Liquefied petroleum gas contains propane (C_3H_8). How many covalent bonds are present in one molecule of propane? **1**

A. 8 **B.** 10
C. 11 **D.** 13

21 Black-and-white photographic film works because silver bromide changes as follows when light falls on it: **1**

$2AgBr(s) \rightarrow 2Ag(s) + Br_2(g)$

This reaction is an example of

A. thermal decomposition. **B.** electrolytic decomposition.

C. photolytic decomposition. **D.** displacement reaction.

22 A doctor advises that Plaster of Paris should be stored in a moisture-proof container. This is because Plaster of Paris **1**

A. combines with water to form gypsum, which sets into a hard mass.

B. absorbs carbon dioxide from the air and turns into chalk.

C. decomposes into quicklime in moist air.

D. becomes strongly acidic when it absorbs moisture.

23 Which of the following pairs consists of amphoteric oxides only? **1**

A. Na_2O and MgO **B.** CO_2 and SO_2

C. Al_2O_3 and ZnO **D.** CaO and K_2O

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Hydrogen gas is not evolved when most metals react with dilute nitric acid. **Reason (R):** Nitric acid is a strong oxidising agent; it oxidises the hydrogen produced to water. **1**

25 Aman brought dry blue litmus paper in contact with dry hydrogen chloride gas but saw no change in its colour. When he moistened the litmus paper, it turned red. **2**

A. Explain these observations.

B. While diluting concentrated sulphuric acid, why must the acid be added slowly to water and not water to the acid?

26 ***Students to attempt either option A or B.*** **3**

A. An element X is stored under kerosene. It burns with a golden-yellow flame and reacts with chlorine to form a white solid Y that dissolves in water.

I. Identify X and Y.

II. Show the formation of Y by the transfer of electrons, using electron-dot structures.

III. Y does not conduct electricity in the solid state but conducts it in the molten state. Why?

OR

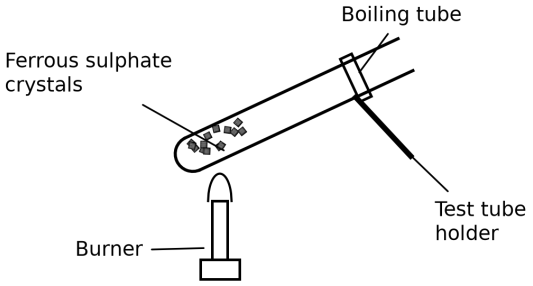
B. Railway workers join broken rails by igniting a mixture of iron(III) oxide and aluminium powder; molten iron flows into the gap.

I. Write the balanced chemical equation for this reaction.

II. Identify the substance oxidised and the substance reduced.

III. Why can aluminium displace iron from its oxide, and why is the reaction suitable for welding?

27 Study the experimental set-up shown below, in which a few crystals of ferrous sulphate are heated in a dry boiling tube. **3**



A. What change in the colour of the crystals is observed on (i) gentle heating and (ii) strong heating?

B. What smell is noticed during strong heating? Write the balanced chemical equation for the reaction.

C. Name the type of reaction and state whether it is exothermic or endothermic.

For visually impaired students

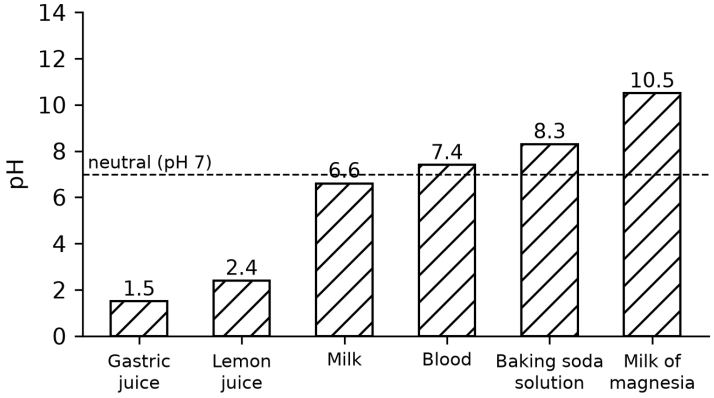
A few green crystals of ferrous sulphate are heated in a dry boiling tube, first gently and then strongly.

A. What change in the colour of the crystals is observed on (i) gentle heating and (ii) strong heating?

B. What smell is noticed during strong heating? Write the balanced chemical equation for the reaction.

C. Name the type of reaction and state whether it is exothermic or endothermic.

28 Neha measured the pH of some substances found in the human body and at home with a pH meter. Her readings are shown in the bar graph. **4**



A. Which sample has the highest concentration of hydrogen ions? (1)

B. Which of these samples – lemon juice, blood, baking soda solution – will turn red litmus blue? (1)

Attempt either sub-part C or D.

C. A person suffering from acidity is given milk of magnesia. Name the acid present in gastric juice and explain how milk of magnesia gives relief. Write the balanced chemical equation for the reaction. (2)

OR

D. Fresh milk (pH 6.6) is left in a warm place overnight and turns into curd. Predict the change in its pH and give a reason. Why does a milkman add a very small amount of baking soda to fresh milk? (2)

For visually impaired students

Neha measured the pH of some substances: gastric juice 1.5, lemon juice 2.4, milk 6.6, blood 7.4, baking soda solution 8.3 and milk of magnesia 10.5.

A. Which sample has the highest concentration of hydrogen ions?

B. Which of these samples – lemon juice, blood, baking soda solution – will turn red litmus blue?

Attempt either sub-part C or D.

C. A person suffering from acidity is given milk of magnesia. Name the acid present in gastric juice and explain how milk of magnesia gives relief. Write the balanced chemical equation for the reaction.

OR

D. Fresh milk (pH 6.6) is left in a warm place overnight and turns into curd. Predict the change in its pH and give a reason. Why does a milkman add a very small amount of baking soda to fresh milk?

29 ***Attempt either option A or B.*** **5**

A. I. Study the following compounds: CH_4, CH_3OH, C_2H_4, C_3H_8, C_2H_5OH, C_3H_7OH

(a) Select two compounds that belong to the same homologous series as CH_3OH.

(b) State two characteristics of a homologous series, including the difference in molecular mass between two successive members.

II. (a) What are structural isomers?

(b) Draw the structures of the two isomers of butane (C_4H_{10}).

(c) Why does propane (C_3H_8) have no isomers?

OR

B. I. Simran washed a greasy shirt with soap solution and the dirt came off easily.

(a) Explain, with the help of a labelled diagram, how soap removes oily dirt from the cloth.

(b) When she used the same soap with water from a hand pump, very little lather formed and a white curdy solid appeared. Why?

II. (a) Why are detergents preferred over soaps for washing clothes in hard water?

(b) Give one reason why the use of some detergents is a concern for the environment.

SECTION C Physics

30 A beam of light from a torch, held in the air, falls obliquely on the surface of the water in a swimming pool. As the light enters the water, it **1**

A. bends away from the normal and its speed increases.

B. bends towards the normal and its speed increases.

C. bends towards the normal and its speed decreases.

D. goes undeviated and its speed decreases.

31 A wire of resistance R is stretched uniformly so that its length becomes double, its volume remaining the same. The resistance of the stretched wire will be **1**

A. R/2 **B.** R

C. 2R **D.** 4R

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Danger signal lights on towers and vehicles are red in colour. **1**
Reason (R): Red light has the shortest wavelength and is scattered the most by fog and smoke.

33 Kabir, who sits in the last row of his classroom, cannot read the words on the blackboard clearly, but he can read his textbook easily. **2**

A. Name the defect of vision Kabir is suffering from and state two possible causes of it.

B. Name the type of lens that will correct his vision and state how it helps.

34 ***Students to attempt either option A or B.*** **2**

A. **I.** Why is tungsten used almost exclusively for the filaments of electric bulbs?
II. Give two reasons why an alloy such as nichrome is used for the heating elements of electric irons and toasters.

OR

B. **I.** Draw a schematic diagram of a circuit to verify Ohm's law, consisting of a battery, a plug key, a resistor, an ammeter, a voltmeter and a rheostat.
II. What will be the shape of the V–I graph obtained for the resistor?

35 ***Students to attempt either option A or B.*** **3**

A. A dentist uses a small concave mirror of focal length 4 cm and holds it 2 cm away from a tooth.

I. Why does the dentist use a concave mirror and not a convex mirror?

II. Find the position of the image formed.

III. Find the magnification and state the nature of the image.

OR

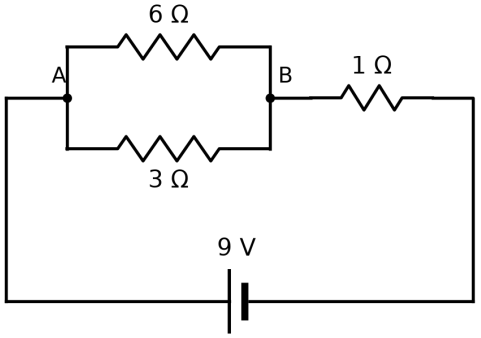
B. An object 2 cm high is placed 15 cm in front of a convex lens of focal length 10 cm.

I. Find the position of the image.

II. Find the magnification and the height of the image.

III. State the nature of the image.

36 Study the circuit given below. **3**



A. Find the equivalent resistance of the circuit.

B. Find the current drawn from the battery and the potential difference between A and B.

C. Find the current through the 6 Ω resistor.

For visually impaired students

A 6 Ω resistor and a 3 Ω resistor are connected in parallel, and this combination is connected in series with a 1 Ω resistor across a 9 V battery.

A. Find the equivalent resistance of the circuit.

B. Find the current drawn from the battery and the potential difference across the parallel combination.

C. Find the current through the 6 Ω resistor.

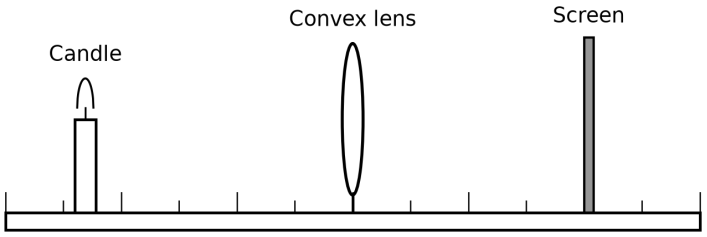
37 An electric iron is rated 1100 W, 220 V. It is used for 2 hours every day. **3**

A. Find the current drawn by the iron and the resistance of its heating element.

B. Find the electrical energy consumed by the iron in 30 days in kWh.

C. Find the cost of this energy if one unit costs ₹ 6.

38 Riya performed an experiment with a convex lens on an optical bench, as shown below. She placed a lighted candle at different distances from the lens and moved the screen each time to get a sharp image. Her observations are recorded in the table. **4**



Observation	1	2	3	4
Object distance u (cm)	−60	−40	−30	−25
Image distance v (cm)	+30	+40	+60	+100

A. Using observation 2, find the focal length of the lens. (1)

B. In which observation is the image of the same size as the object? (1)

Attempt either sub-part C or D.

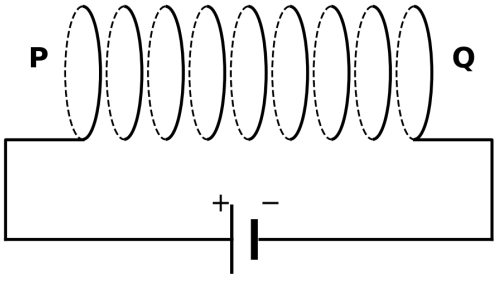
C. Calculate the power of the lens and state the nature of the lens from the sign of its power. (2)

OR

D. Find the magnification in observation 4 and state the nature of the image formed. (2)

39 ***Attempt either option A or B.*** **5**

A. I. The figure shows a solenoid PQ connected to a battery. When viewed from end P, the current in the turns appears to flow anticlockwise.



(a) Draw the pattern of magnetic field lines inside and around the solenoid, and mark the polarity of ends P and Q.

(b) State two ways of increasing the strength of the magnetic field inside the solenoid.

II. (a) State Fleming's left-hand rule.

(b) A straight horizontal wire carries current from south to north in a magnetic field directed vertically upwards. In which direction will the wire experience a force?

OR

B. I. (a) Give two reasons why electrical appliances in a house are connected in parallel and not in series.
(b) What is the function of the earth wire? Why is it necessary for appliances with a metallic body?
(c) Distinguish between short-circuiting and overloading.
II. An electric kettle of 880 W and two bulbs of 110 W each are switched on together in a 220 V domestic circuit.
(a) Find the total current drawn from the mains.
(b) Would a fuse of rating 3 A be suitable for this circuit? Give a reason.

For visually impaired students

A. I. A solenoid PQ is connected to a battery. When viewed from end P, the current in the turns appears to flow anticlockwise. (a) Describe the pattern of the magnetic field lines inside and outside the solenoid and state the polarity of ends P and Q. (b) State two ways of increasing the strength of the magnetic field inside the solenoid.
II. (a) State Fleming's left-hand rule. (b) A straight horizontal wire carries current from south to north in a magnetic field directed vertically upwards. In which direction will the wire experience a force?

END OF QUESTION PAPER

Quick revision: diagrams asked most often

1. Human digestive system, with the liver, pancreas and small intestine labelled.
2. Structure of a nephron, with Bowman's capsule and glomerulus.
3. Human heart showing the four chambers and the direction of blood flow.
4. Stomata (open and closed) and the guard cells.
5. Neuron and the reflex arc.
6. Longitudinal section of a flower; germination of pollen on the stigma.
7. Ray diagrams for a convex lens and a concave mirror for every object position.
8. Magnetic field lines around a bar magnet and a solenoid.
9. Electrolytic refining of copper and the activity series experiment.

Nitesh Bhaiya ki tip: diagram pencil se banao, labels seedhi line se aur right side me.

SOLUTION PAPER 08

SCIENCE · CODE NO. 086

Sample Paper 8: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 **Ans**

C. The digestion of proteins, fats and carbohydrates in the small intestine. (1)
(Pancreatic juice contains trypsin for proteins, lipase for fats and amylase for starch.)

2 **Ans**

B. The seeds used up oxygen and the carbon dioxide released by them was absorbed by KOH, lowering the air pressure inside the flask. (1)
For visually impaired students: A. Potassium hydroxide solution

3 **Ans**

D. The cells of the tendril on the side away from the stick grow faster than the cells in contact with it. (1)

4 **Ans**

A. *Plasmodium* – Multiple fission (1)

5 **Ans**

D. 50%, because half of the sperms produced by the father carry an X chromosome. (1)
(Each birth is an independent event.)

6 **Ans**

C. 300 (1)
(Round green = 3/16 of F_2; 3/16 × 1600 = 300)

7 **Ans**

A. High-energy UV radiation splits oxygen molecules (O_2) into free oxygen atoms, which then combine with O_2 molecules. (1)

8 **Ans**

A. Both A and R are true, and R is the correct explanation of A. (1)

9 **Ans**

C. A is true but R is false. (1)
(Energy flow is unidirectional; only about 10% is passed on at each level, so little energy is left after 3–4 levels.)

10 **Ans**

A. Only the green regions turn blue-black; the white regions do not. This shows that chlorophyll is necessary for photosynthesis (starch is formed only where chlorophyll is present). (½+½)

B. To destarch the leaves, i.e., to use up the starch already present, so that any starch detected is formed only during the experiment. (1)

Total 2 marks

11 **Ans**

A.

Organism: *Rhizopus* (bread mould); mode: spore formation. (½+½)

The black dots are sporangia, which contain spores (that can grow into new *Rhizopus*). (½)

Spores need moisture (and nutrition) to germinate and grow, so the mould does not grow on dry bread. (½)

OR

B.

Regeneration: *Planaria* cut into pieces – each piece grows into a complete organism using specialised cells that divide and differentiate. Fragmentation: the filament of *Spirogyra* simply breaks into pieces, each of which grows into a new filament. (1)

Complex organisms have cells organised into tissues, organs and organ systems; a fragment cannot carry out all functions or form all the specialised parts, so they cannot reproduce by fragmentation. (1)

Total 2 marks

12

Ans

Hormone: auxin. (½)

Auxin is synthesised at the shoot tip. When light falls from one side, auxin diffuses towards the shaded side of the shoot. (½)

The higher concentration of auxin on the shaded side makes those cells grow (elongate) faster than the cells on the illuminated side. (½)

As a result, the shoot bends towards light (positive phototropism). (½)

Total 2 marks

13

Ans

Women have a perfect pair of sex chromosomes (XX); men have a mismatched pair (XY). (½)

All eggs carry an X chromosome; half of the sperms carry X and half carry Y. (½)

Flow chart: Parents XX × XY → Gametes: X (egg) and X, Y (sperm) → X + X = XX (girl); X + Y = XY (boy). (1)

The sex of the child depends on whether the sperm (from the father) carrying X or Y fertilises the egg; the mother always contributes X, so she cannot be held responsible. (½)

Chance of the next child being a boy: 50%. (½)

Total 3 marks

14

Ans

A.

I. Biological magnification. (1)

II. DDT is non-biodegradable and is not excreted; it accumulates in the body and gets concentrated at each trophic level. The kingfisher occupies the highest trophic level, so it has the highest concentration. (1)

III. 10% law: 20,000 J → 2000 J → 200 J → **20 J** available to the kingfisher. (1)

OR

B.

I. Biodegradable: vegetable peels, newspapers, dry leaves. Non-biodegradable: plastic bottles, glass jars, aluminium foil. (1)

II. Any two: they persist for long and pile up; harm soil fertility; block drains and water bodies; harm animals that swallow them; may enter food chains and get biomagnified. (½+½)

III. Any one: segregate wet and dry waste and compost the wet waste; use cloth bags instead of plastic bags; reuse jars and bottles; send newspapers, glass and foil for recycling. (1)

Total 3 marks

15

Ans

A. P is pepsin, because it shows maximum activity at about pH 2, i.e., in the acidic medium of the stomach (due to HCl). (1)

B. Starch is broken down into a simple sugar (maltose) by enzyme Q (salivary amylase). (½+½)

C.

The activity of Q decreases and stops, because the medium in the stomach is strongly acidic. (½)

Substance responsible: hydrochloric acid secreted by the gastric glands. (½)

Mucus secreted by the gastric glands protects the inner lining of the stomach from the action of the acid and pepsin. (1)

OR

D.

R is trypsin, secreted by the pancreas. (½+½)

Bile juice from the liver makes the acidic food coming from the stomach alkaline, so that pancreatic enzymes like trypsin can act (bile salts also emulsify fats). (1)

Total 4 marks

16 Ans

A. I.

(a) P – right atrium; S – left ventricle. (½+½)

Q (left atrium) and S (left ventricle) contain oxygenated blood. (½)

(b) X – pulmonary artery; Y – pulmonary vein. (½+½)

Separation prevents mixing of oxygenated and deoxygenated blood and ensures a highly efficient supply of oxygen, needed by birds and mammals, which have high energy needs to maintain a constant body temperature. (½)

II.

Loss of water vapour through stomata (transpiration) creates a suction (transpiration pull) that pulls water up through the continuous column of xylem from the roots to the leaves. (1)

Food (sugars) moves in the phloem in both directions (upward and downward) and needs energy from ATP (translocation); water moves in the xylem only upwards, mainly by physical forces (root pressure and transpiration pull). (1)

OR

B. I.

(a) A – Bowman's capsule; B – glomerulus. (½+½)

(b) In the glomerulus, blood is filtered (under pressure) and the filtrate collects in Bowman's capsule. (½)

Useful substances such as glucose, amino acids, salts and a major amount of water are selectively reabsorbed in the tubule. (½)

(c) The amount of water reabsorbed depends on how much excess water there is in the body and how much dissolved waste has to be excreted; this regulates the volume of urine. (1)

II.

Haemodialysis: blood is passed through long tubes with a semi-permeable lining suspended in dialysing fluid of the same osmotic pressure as blood (without nitrogenous wastes); the wastes diffuse from the blood into the fluid and purified blood is returned to the patient. (No reabsorption takes place.) (1)

Plants: (any two) excess water lost by transpiration; wastes stored in old leaves that fall off; stored as resins and gums in old xylem; some excreted into the soil around the roots. (½+½)

For visually impaired students: the same value points apply (right atrium / left ventricle; pulmonary artery / pulmonary vein; Bowman's capsule / glomerulus).

Total 5 marks

SECTION B Chemistry

17 Ans

D. a double displacement (precipitation) reaction. (1)

$Pb(NO_3)_2 + 2KI \rightarrow PbI_2$ *(yellow ppt)* $+ 2KNO_3$

18 Ans

B. it is a mild, non-corrosive base that neutralises the excess acid in the stomach. (1)

19 Ans

A. I only (1)

(Only zinc is more reactive than the metal in its salt solution; the blue solution fades.)

For visually impaired students: A. Zinc in copper sulphate solution

20 **Ans**

B. 10
(2 C–C bonds + 8 C–H bonds) (1)

21 **Ans**

C. photolytic decomposition. (1)

22 **Ans**

A. combines with water to form gypsum, which sets into a hard mass. (1)
$CaSO_4 \cdot ½H_2O + 1½H_2O \rightarrow CaSO_4 \cdot 2H_2O$

23 **Ans**

C. Al_2O_3 and ZnO (1)

24 **Ans**

A. Both A and R are true, and R is the correct explanation of A. (1)

25 **Ans**

A. HCl gas does not give H^+ ions in the absence of water, so dry HCl shows no acidic behaviour. In the presence of water, HCl dissociates to give H^+ (H_3O^+) ions, which turn blue litmus red. (½+½)

B. Mixing an acid with water is highly exothermic. If water is added to the acid, the heat generated may cause the mixture to splash out and cause burns, and the glass container may break. Adding acid slowly to water spreads the heat safely. (1)

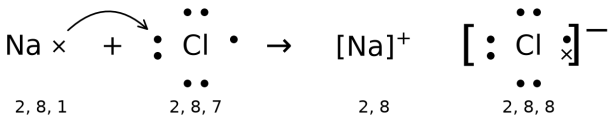
Total 2 marks

26 **Ans**

A.

I. X – sodium (Na); Y – sodium chloride (NaCl). (½+½)

II. Sodium (2, 8, 1) loses its one valence electron to chlorine (2, 8, 7); Na^+ and Cl^- ions are formed and held together by electrostatic attraction: (1)



III. In the solid state, the ions are held in fixed positions by strong electrostatic forces and cannot move; in the molten state, the ions become free to move and carry charge. (1)

OR

B.

I. $Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 + \text{heat}$ (1)

II. Substance oxidised: aluminium (Al); substance reduced: iron(III) oxide (Fe_2O_3). (½+½)

III. Aluminium is more reactive than iron, so it displaces iron from its oxide. The reaction is highly exothermic, so iron is obtained in the molten state and fills the gap (thermit reaction). (½+½)

Total 3 marks

27 **Ans**

A. (i) On gentle heating, the green crystals turn white (they lose their water of crystallisation). (ii) On strong heating, a reddish-brown solid (Fe_2O_3) is left. (½+½)

B. Smell of burning sulphur (due to SO_2 and SO_3). (½)

$2FeSO_4(s) \rightarrow Fe_2O_3(s) + SO_2(g) + SO_3(g)$ (½)

C. Thermal decomposition reaction; it is endothermic (heat is absorbed). (½+½)

Total 3 marks

28

A. Gastric juice (lowest pH, 1.5). (1)

Ans

B. Baking soda solution (pH 8.3, basic). **(1)**

C.

Hydrochloric acid (HCl). **(½)**

Milk of magnesia, $Mg(OH)_2$, is a mild base; it neutralises the excess acid in the stomach and relieves the pain. **(½)**

$Mg(OH)_2 + 2HCl \rightarrow MgCl_2 + 2H_2O$ **(1)**

OR

D.

The pH decreases (falls below 6.6), because lactic acid is formed when milk turns into curd, making it more acidic. **(1)**

Baking soda makes the milk slightly alkaline (pH a little above 6.6), so it does not turn sour quickly and can be kept for a longer time (it also takes longer to set as curd). **(1)**

Total 4 marks

29

Ans

A. I.

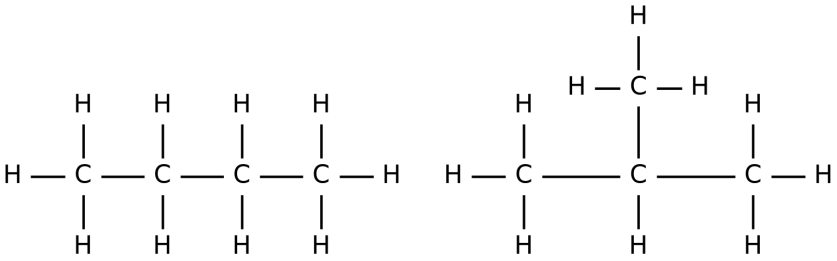
(a) C_2H_5OH and C_3H_7OH (alcohols). **(½+½)**

(b) Any two: all members have the same functional group / general formula; successive members differ by a $–CH_2$ unit and by 14 u in molecular mass; similar chemical properties; gradation in physical properties such as boiling point. **(½+½)**

II.

(a) Compounds having the same molecular formula but different structures are called structural isomers. **(1)**

(b) Structures of n-butane and isobutane (2-methylpropane): **(½+½)**



n-Butane (butane) Isobutane (2-methylpropane)

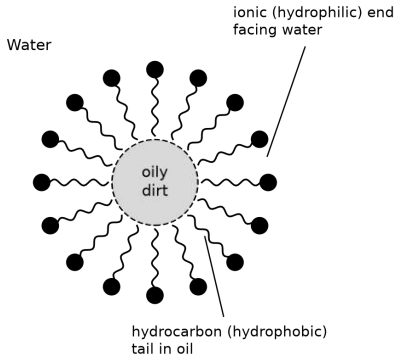
(c) The three carbon atoms of propane can be arranged only in one way (a straight chain); no branched structure is possible. **(1)**

OR

B. I.

(a) A soap molecule has an ionic (hydrophilic) end and a long hydrocarbon (hydrophobic) tail. In water, the tails attach to the oily dirt and the ionic ends face the water, forming clusters called micelles with the dirt trapped at the centre. The micelles stay suspended in water and are washed away on rinsing. **(1)**

Labelled diagram of a micelle: (1)



(b) Hand-pump water is hard; it contains calcium and magnesium salts which react with soap to form an insoluble precipitate (scum), so little lather forms. (1)

II.

(a) The charged ends of detergents do not form insoluble precipitates with calcium and magnesium ions in hard water, so they remain effective. (1)

(b) Some detergents (with branched hydrocarbon chains) are non-biodegradable; they cause foaming and pollute rivers and lakes, harming aquatic life. (1)

Total 5 marks

SECTION C Physics

30 **Ans** C. bends towards the normal and its speed decreases. (1)

31 **Ans** D. 4R (1)
(l → 2l, A → A/2, so R = ρl/A becomes 4 times.)

32 **Ans** C. A is true but R is false. (1)
(Red light has the longest wavelength and is scattered the least, so it can be seen from a distance.)

33 **Ans**

A. Myopia (near-sightedness). (½)

Causes: (i) excessive curvature of the eye lens; (ii) elongation of the eyeball. (½)

B. Concave (diverging) lens of suitable power. (½)

It diverges the rays from a distant object so that they appear to come from his far point, and the image is formed on the retina. (½)

Total 2 marks

34 **Ans**

A.

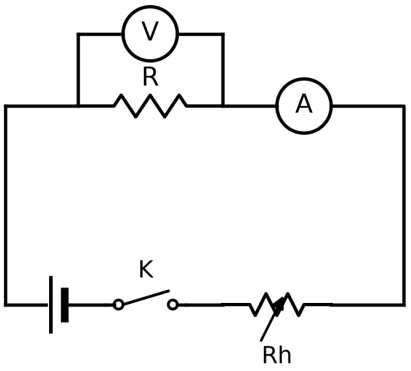
I. Tungsten has a very high melting point (about 3380 °C) and does not melt at the high temperature at which it glows and emits light. (1)

II. Any two: an alloy has a higher resistivity than its constituent metals; it does not oxidise (burn) readily at high temperatures; it has a high melting point. (½+½)

OR

B.

I. Correct circuit: resistor, ammeter, rheostat and key in series with the battery, voltmeter in parallel with the resistor. (1)



II. A straight line passing through the origin ($V \propto I$). (1)

Total 2 marks

35 Ans

A.

I. A concave mirror forms an erect and magnified image when the object is between the pole and the focus, so the dentist sees a larger image of the tooth; a convex mirror always forms a diminished image. (1)

II. $f = -4$ cm, $u = -2$ cm (1)

$1/v = 1/f - 1/u = -1/4 + 1/2 = 1/4 \Rightarrow$ **v = +4 cm** (4 cm behind the mirror)

III. $m = -v/u = -(4)/(-2) =$ **+2**; the image is virtual, erect and magnified (twice the size). (½+½)

OR

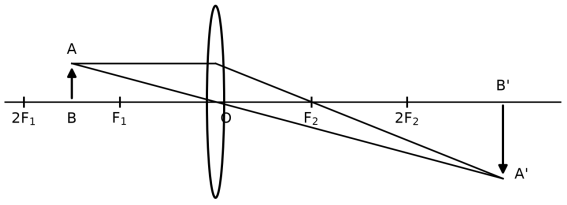
B.

I. $f = +10$ cm, $u = -15$ cm (1)

$1/v = 1/f + 1/u = 1/10 - 1/15 = 1/30 \Rightarrow$ **v = +30 cm** (on the other side of the lens)

II. $m = v/u = 30/(-15) =$ **−2**; $h' = m \times h = -2 \times 2 =$ **−4 cm** (½+½)

III. Real, inverted and magnified. (1)



Total 3 marks

36 Ans

A. 6 Ω and 3 Ω in parallel: $1/R_p = 1/6 + 1/3 = 1/2 \Rightarrow R_p = 2\ \Omega$; $R = 2\ \Omega + 1\ \Omega =$ **3 Ω** (1)

B. $I = V/R = 9/3 =$ **3 A**; $V_{AB} = I \times R_p = 3 \times 2 =$ **6 V** (½+½)

C. $I_{6\Omega} = V_{AB}/6 = 6/6 =$ **1 A** (1)

Total 3 marks

37 Ans

A. $I = P/V = 1100/220 =$ **5 A**; $R = V/I = 220/5 =$ **44 Ω** (½+½)

B. $E = P \times t = 1.1$ kW $\times 2$ h $\times 30 =$ **66 kWh** (1)

C. Cost = 66 × ₹ 6 = **₹ 396** (1)

Total 3 marks

38

Ans

A. In observation 2, u = −40 cm and v = +40 cm, i.e., the object is at 2F. So 2f = 40 cm ⇒ **f = 20 cm**. (1)

(Check: 1/f = 1/v − 1/u = 1/40 + 1/40 = 1/20)

B. Observation 2 (object and image both at 2F; m = −1). (1)

C.

P = 1/f (in m) = 1/0.20 m = **+5 D** (1)

The power is positive, so the lens is converging (convex). (1)

OR

D.

m = v/u = (+100)/(−25) = **−4** (1)

The image is real, inverted and magnified (four times the size of the object). (1)

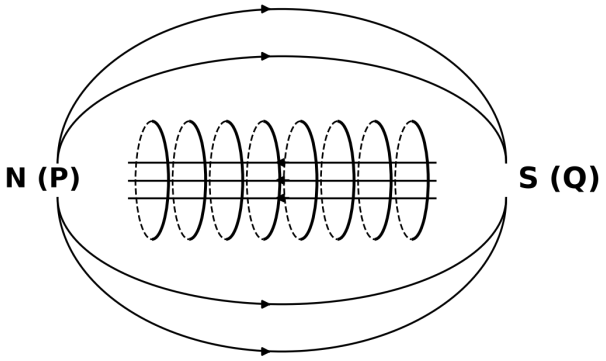
Total 4 marks

39

Ans

A. I.

(a) Field lines: parallel straight lines inside the solenoid (uniform field), and closed curves outside going from one end to the other, like those of a bar magnet. (1)



Polarity: end P – North pole (current anticlockwise as seen from P); end Q – South pole. (½+½)

(b) Any two: increase the current; increase the number of turns; insert a soft iron core inside the solenoid. (½+½)

II.

(a) Stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of current, the thumb gives the direction of the force (motion) on the conductor. (1)

(b) Towards the east. (1)

OR

B. I.

(a) In parallel, each appliance gets the full mains voltage (220 V), and each can be switched on or off independently; if one fails, the others keep working (and each draws the current it needs). (½+½)

(b) The earth wire provides a low-resistance path to the earth for any current leaking to the metallic body, keeping the body at earth potential. This protects the user from a severe electric shock. (½+½)

(c) Short-circuiting: the live and neutral wires come in direct contact, the resistance becomes very small and the current rises suddenly. Overloading: too many appliances are connected to one socket, or the supply voltage rises, so the circuit draws more current than it is designed for. (1)

II.

(a) Total power P = 880 + 2 × 110 = 1100 W; I = P/V = 1100/220 = **5 A** (1)

(b) No. The circuit draws 5 A, which is more than 3 A, so the fuse wire would melt and break the circuit as soon as all the appliances are switched on. A fuse of slightly higher rating than 5 A should be used. **(1)**

Total 5 marks

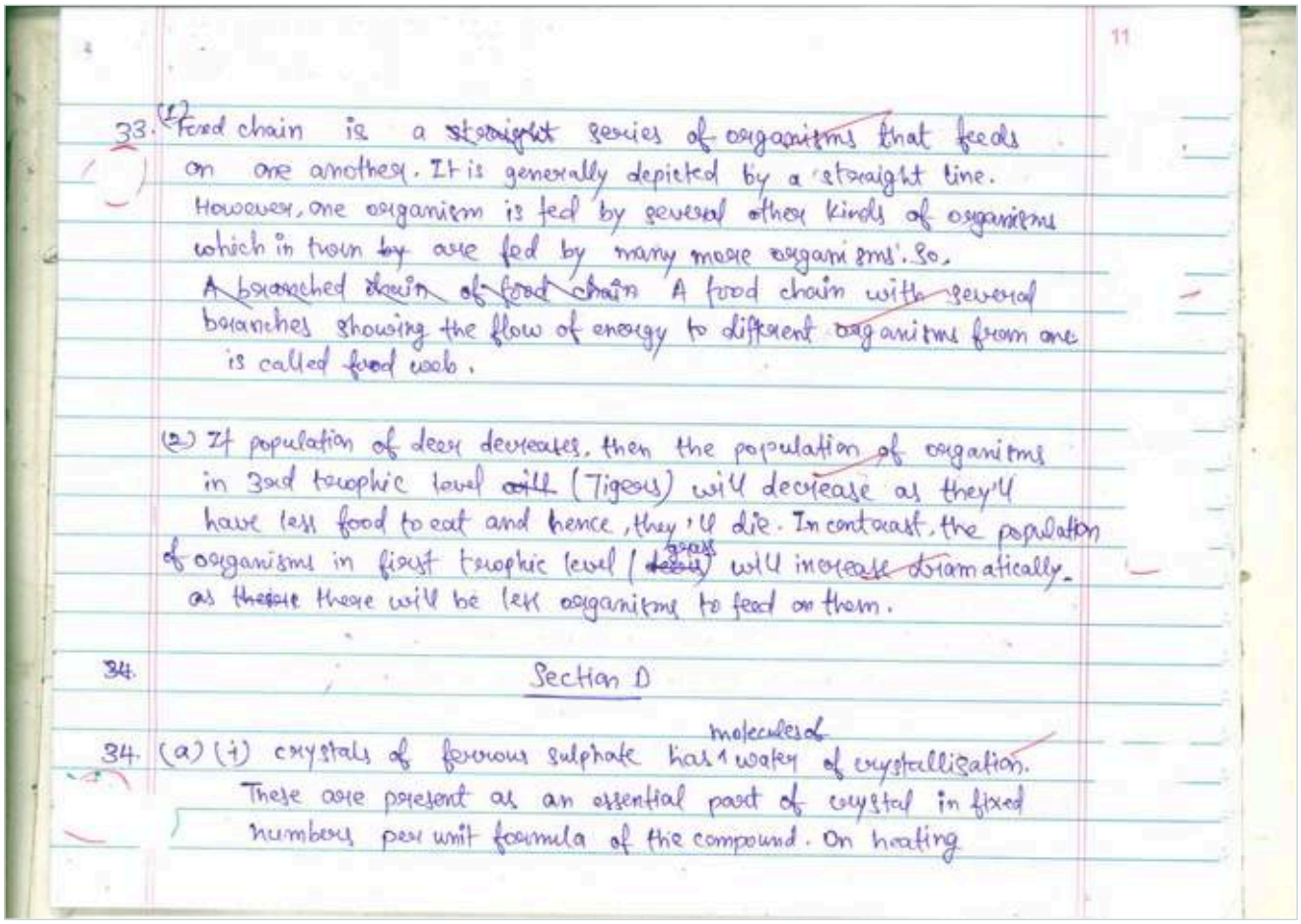
END OF SOLUTIONS

Common mistakes: Biology answers

1. Writing 'oxygen is released in respiration': it is used; CO_2 is released.
2. Mixing up arteries (carry blood away from the heart) and veins (carry blood to the heart).
3. Saying plants do not respire: they respire all the time, day and night.
4. Calling the nephron a kidney: the nephron is the unit of the kidney.
5. Writing auxin and gibberellin functions the wrong way round.
6. Missing the reason in heredity: show the cross with gametes and the ratio 3:1 or 9:3:3:1.
7. Calling an asexual method 'budding' for every organism; name the right one: fission, fragmentation, regeneration.
8. Food chains drawn with arrows pointing from the eater to the eaten: arrows show energy flow, from food to eater.

Nitesh Bhaiya ki tip: biology me keyword ka spelling sahi likho, jaise 'alveoli', 'villi', 'glomerulus'.

Real CBSE topper's answer · Board Exam 2024 · similar to Q.14 and Q.27 of this paper (food chains; heating ferrous sulphate)



11

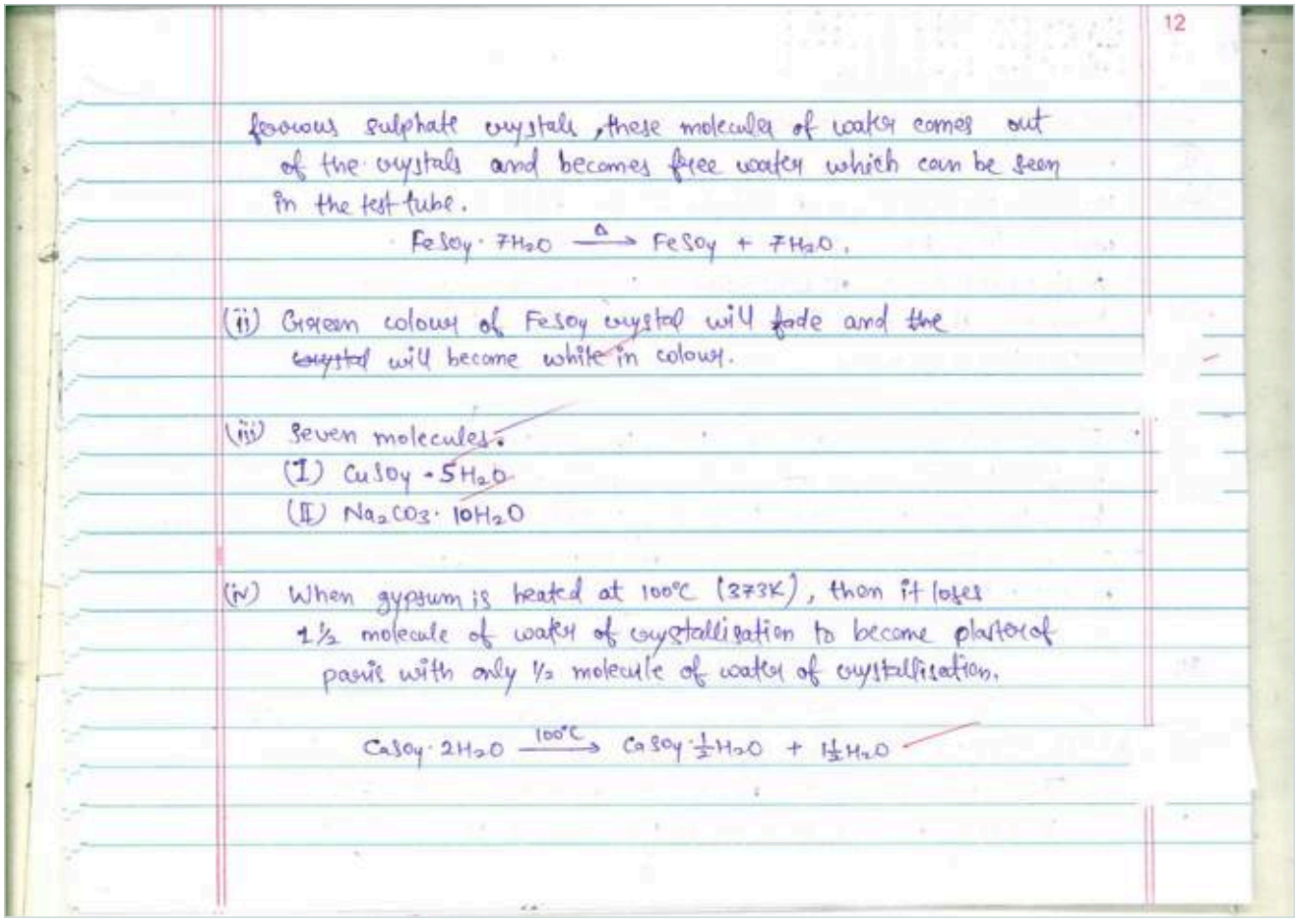
33. (i) Food chain is a ~~straight~~ series of organisms that feeds on one another. It is generally depicted by a straight line. However, one organism is fed by several other kinds of organisms which in turn by are fed by many more organisms. So, ~~A branched chain of food chain~~ A food chain with several branches showing the flow of energy to different organisms from one is called food web.

(ii) If population of deer decreases, then the population of organisms in 3rd trophic level ~~will~~ (Tigers) will decrease as they'll have less food to eat and hence, they'll die. In contrast, the population of organisms in first trophic level (grass) will increase dramatically as there there will be less organisms to feed on them.

34. Section D

34. (a) (i) crystals of ferrous sulphate has 7 molecules of water of crystallisation. These are present as an essential part of crystal in fixed numbers per unit formula of the compound. On heating

Real CBSE topper's answer · Board Exam 2024 · similar to Q.27 of this paper (heating ferrous sulphate crystals, water of crystallisation)



12

ferrous sulphate crystals, these molecules of water comes out of the crystals and becomes free water which can be seen in the test tube.

$$FeSO_4 \cdot 7H_2O \xrightarrow{\Delta} FeSO_4 + 7H_2O$$

(ii) Green colour of $FeSO_4$ crystal will fade and the crystal will become white in colour.

(iii) Seven molecules.
(I) $CuSO_4 \cdot 5H_2O$
(II) $Na_2CO_3 \cdot 10H_2O$

(iv) When gypsum is heated at 100°C (373K), then it loses 1½ molecule of water of crystallisation to become plaster of paris with only ½ molecule of water of crystallisation.

$$CaSO_4 \cdot 2H_2O \xrightarrow{100°C} CaSO_4 \cdot \tfrac{1}{2}H_2O + 1\tfrac{1}{2}H_2O$$

Real CBSE topper's answer · Board Exam 2023 · similar to Q.16 of this paper (path of blood through the heart)

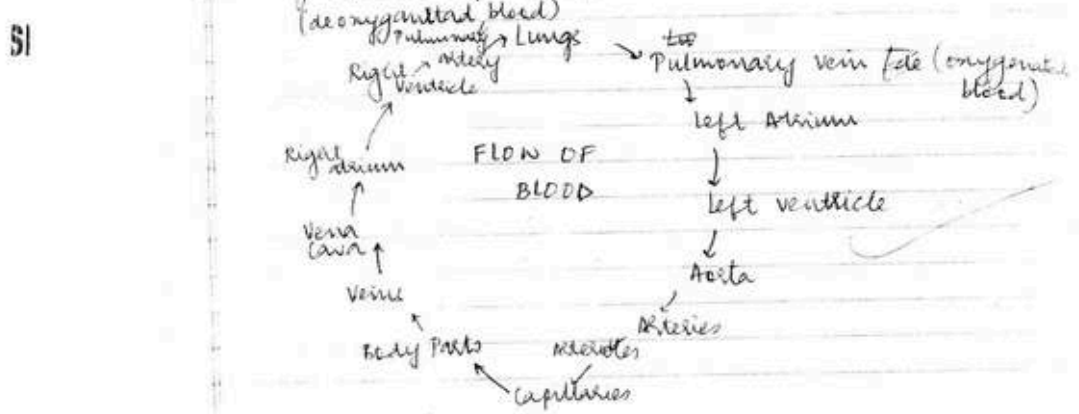
b) Fleming's Left Hand Rule
If we place our forefinger, middle finger and thumb in such a way that they are mutually perpendicular, and the direction of our middle finger gives the direction of current, direction of forefinger indicates direction of magnetic field, then the thumb indicates the direction of force experienced by the conductor.

SECTION C

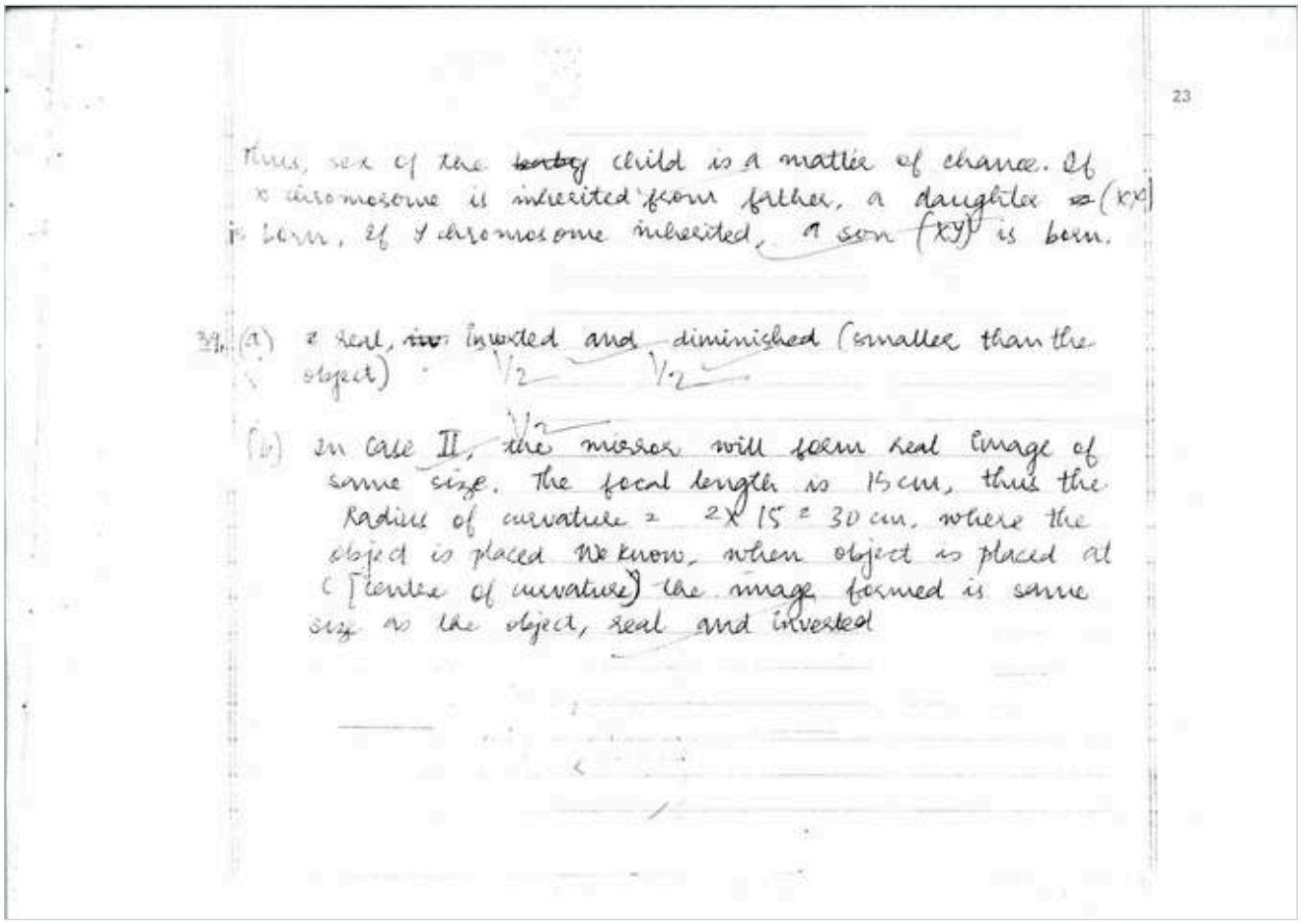
24 The oxygenated blood from the lungs is transported to the heart for pumping through the pulmonary vein. The pulmonary vein transports blood to the left Atrium which relaxes as it receives it then contracts to transport blood to the left ventricle which relaxes. When the left ventricle contracts, blood is pumped to all organs

Real CBSE topper's answer · Board Exam 2023 · similar to Q.16 of this paper (double circulation in humans)

through the thick walled Aorta.
The deoxygenated blood from the body is transported to the heart by vena cava from upper and lower body. The ~~left~~ Right Atrium relaxes as it receives blood. when it contracts, the blood is transferred to Right ventricle which relaxes Finally the ~~left~~ Right Ventricle contracts to pump blood to Lungs ~~&~~ ~~to~~, for oxygenation through pulmonary artery.



Real CBSE topper's answer · Board Exam 2023 · similar to Q.35 of this paper (uses of a concave mirror)



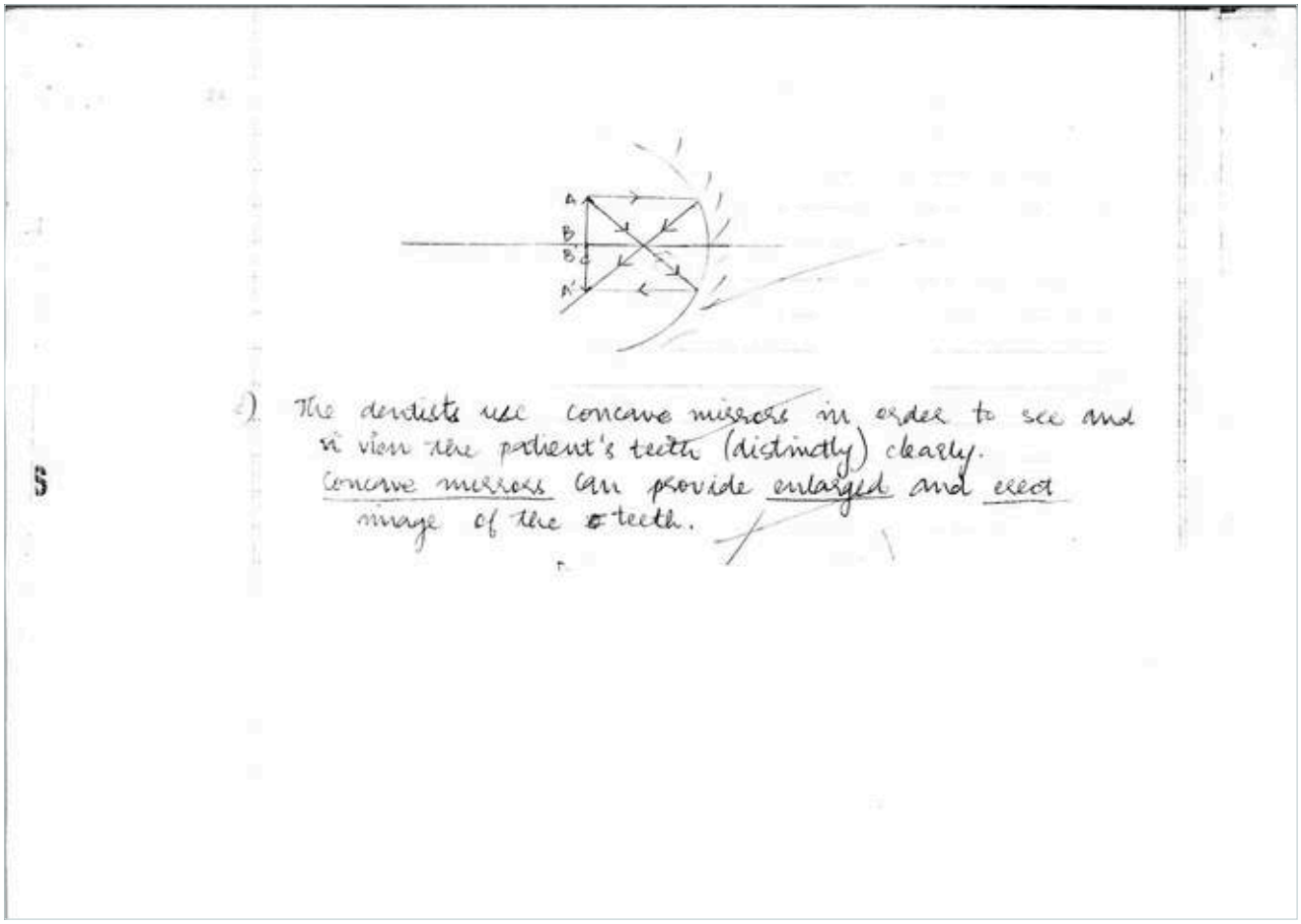
23

Thus, sex of the ~~baby~~ child is a matter of chance. If X chromosome is inherited from father, a daughter ≡(XX) is born, if Y chromosome inherited, a son (XY) is born.

39. (a) a real, ~~inv~~ inverted and diminished (smaller than the object) ½ ½

(b) In case II, the mirror will form real image of same size. The focal length is 15 cm, thus the Radius of curvature = 2 × 15 = 30 cm, where the object is placed We know, when object is placed at C [Centre of curvature] the image formed is same size as the object, real and inverted

Real CBSE topper's answer · Board Exam 2023 · similar to Q.35 of this paper (dentist's concave mirror, ray diagram)



c) The dentists use concave mirrors in order to see and ~~s~~ view the patient's teeth (distinctly) clearly.
Concave mirrors can provide enlarged and erect image of the ~~o~~ teeth.

5

SAMPLE PAPER 09

SCIENCE · CODE NO. 086

Sample Question Paper 9

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

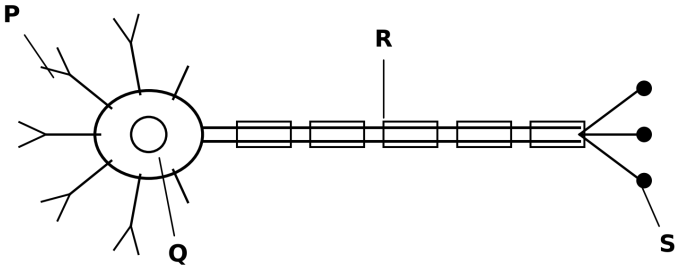
1 (1)

Riya's doctor told her that the finger-like projections lining her small intestine had been damaged by an infection. Which process is most likely to be affected first?

A. Secretion of hydrochloric acid
B. Absorption of digested food into the blood
C. Emulsification of fats by bile
D. Breakdown of starch by salivary amylase

2 (1)

The figure below shows a nerve cell with four parts marked P, Q, R and S.



In which part is the electrical impulse converted into a chemical signal that passes on to the next cell?

A. P **B.** Q
C. R **D.** S

For visually impaired students

In which part of a neuron is the electrical impulse converted into a chemical signal that passes on to the next cell?

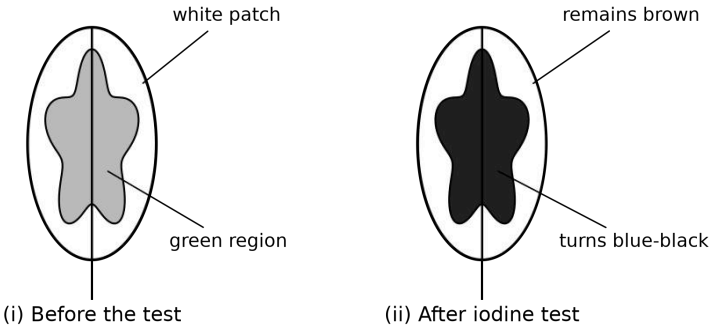
A. Dendrite
B. Cell body

	C. Axon **D.** Nerve ending at the end of the axon	
3	In a hilly village, many people developed a swelling in the neck. A health worker advised all families to use iodised salt. The deficiency of which hormone, secreted by which gland, was the most likely cause? **A.** Insulin – pancreas **B.** Growth hormone – pituitary **C.** Thyroxin – thyroid **D.** Adrenaline – adrenal	**1**
4	Papaya plants bear either only male flowers or only female flowers. Which statement about fruit formation in papaya is correct? **A.** A single papaya plant can produce fruits by self-pollination. **B.** Fruits form only on plants with female flowers, after pollen reaches them from a plant with male flowers. **C.** Fruits develop from the anthers of the male flowers. **D.** Both male and female plants produce fruits with seeds.	**1**
5	A couple already has three daughters. What is the probability that their fourth child will be a son? **A.** 0% **B.** 25% **C.** 50% **D.** 100%	**1**
6	A pure tall pea plant (TT) was crossed with a pure dwarf pea plant (tt). The F_1 plants were self-pollinated and 1200 plants were obtained in the F_2 generation. How many of these are expected to be dwarf? **A.** 0 **B.** 300 **C.** 600 **D.** 900	**1**
7	A housing society wants to reduce its contribution to the depletion of the ozone layer. Which step will help the most? **A.** Replacing old refrigerators and air conditioners that use CFCs with CFC-free models **B.** Segregating wet and dry waste at source **C.** Using cloth bags instead of plastic bags **D.** Planting more trees in the society park	**1**

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** Pancreatic juice contains enzymes that digest proteins and fats. **Reason (R):** Pancreatic juice is poured directly into the stomach.	**1**
9	**Assertion (A):** A potted plant kept near a window shows its shoot bending towards the light. **Reason (R):** Auxin is synthesised at the tip of the shoot.	**1**
10	Raghav took a potted croton plant with variegated (green and white) leaves and kept it in a dark room for three days. He then placed it in bright sunlight for six hours, plucked a leaf and performed the iodine test on it. The leaf before and after the test is shown below.	**2**



A. What conclusion can be drawn from the result of the test?

B. Why was the plant kept in the dark for three days before the experiment?

For visually impaired students

Raghav kept a potted plant with green and white (variegated) leaves in the dark for three days and then in sunlight for six hours. On testing a leaf with iodine solution, only the green parts turned blue-black.

A. What conclusion can be drawn from this result?

B. Why was the plant kept in the dark for three days before the experiment?

11 ***Students to attempt either option A or B.*** **2**

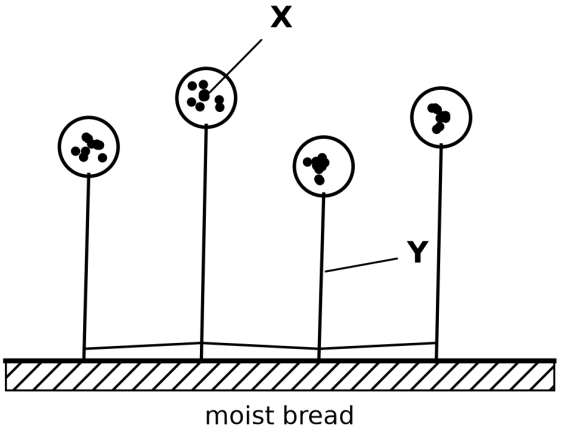
A. A gardener noticed that the tendrils of a pea plant had coiled around a bamboo stick placed near it. Name this response and explain how the tendril is able to coil around the support.

OR

B. Rohan accidentally touched a hot iron and pulled his hand back immediately, even before he felt the pain. Name this type of action and draw a flow chart to show the pathway of the nerve impulse involved in it.

12 A. State two functions of the testes in human males. **2**
B. Why are the testes located outside the abdominal cavity, in the scrotum?

13 Rani noticed a white, cotton-like growth with tiny black dots on a slice of bread left in a moist, warm kitchen for a few days. The growth as seen under a magnifying glass is shown below. **3**



A. Name the organism and identify the structures X and Y.

B. Name the mode of asexual reproduction shown by this organism and describe how it helps the organism spread.

C. Why does a slice of bread kept in a refrigerator not show such growth for many days?

For visually impaired students

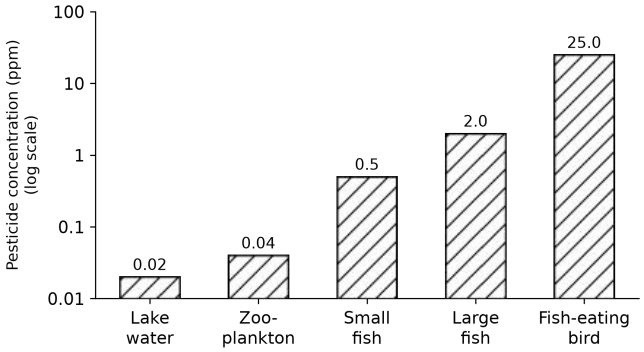
Rani noticed a white, cotton-like growth with tiny black dots on a slice of bread left in a moist, warm kitchen for a few days.

A. Name the organism and state what the thread-like structures and the black dots are.

B. Name the mode of asexual reproduction shown by this organism and describe how it helps the organism spread.

C. Why does a slice of bread kept in a refrigerator not show such growth for many days?

14 Scientists measured the concentration of a pesticide in a lake and in the organisms of a food chain living in and around it. Their results are shown in the bar graph below. **3**



A. Name the phenomenon shown by the data.

B. Which organism of the food chain will be harmed the most? Give a reason based on the data.

C. The pesticide was not sprayed in the lake. Suggest how it entered the lake and state one way to prevent this.

For visually impaired students

In a lake, the concentration of a pesticide was found to be 0.02 ppm in water, 0.04 ppm in zooplankton, 0.5 ppm in small fish, 2 ppm in large fish and 25 ppm in fish-eating birds.

A. Name the phenomenon shown by the data.

B. Which organism will be harmed the most? Give a reason based on the data.

C. The pesticide was not sprayed in the lake. Suggest how it entered the lake and state one way to prevent this.

15 **4**

Mr. Verma, aged 55, was advised regular blood tests because of diabetes and high blood pressure. His doctor explained how the kidneys filter about 180 litres of fluid from the blood every day but produce only about 1.5 litres of urine. The table below compares the composition of blood plasma, the filtrate formed in the Bowman's capsule and urine in a healthy person.

Substance (g per 100 mL)	Blood plasma	Filtrate in Bowman's capsule	Urine
Proteins	7.0	0	0
Glucose	0.1	0.1	0
Urea	0.03	0.03	2.0
Salts	0.72	0.72	1.5

A. Why are proteins present in the blood plasma but absent in the filtrate? (1)

B. Glucose is present in the filtrate but absent in the urine of a healthy person. Explain why. (1)

Attempt either sub-part C or D.

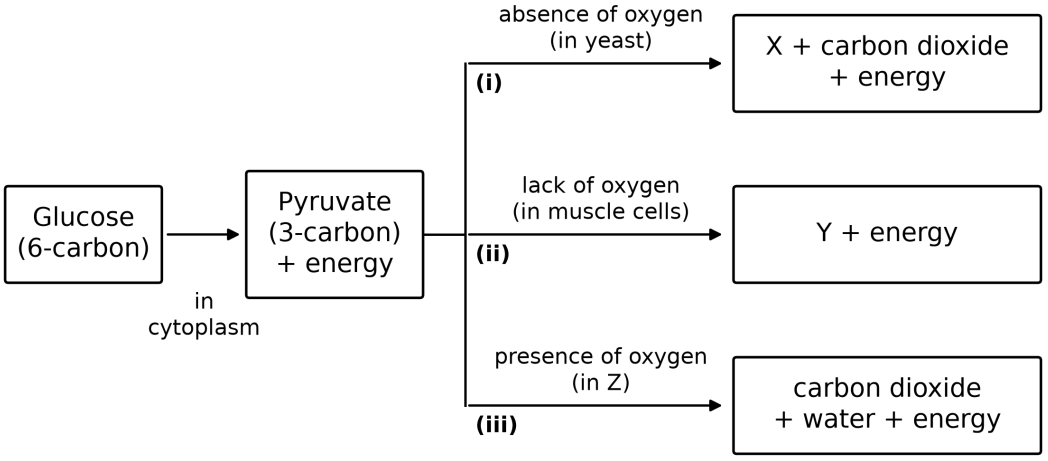
C. The concentration of urea in urine is about 66 times its concentration in the filtrate. Explain how this happens and name the part of the nephron responsible for it. (2)

OR

D. How will the volume of urine produced by Mr. Verma change on a hot summer day when he sweats a lot? Explain why. Name the organ where urine is stored before it is passed out. (2)

16 ***Attempt either option A or B.*** **5**

A. The flow chart below shows the breakdown of glucose by different pathways in living organisms.



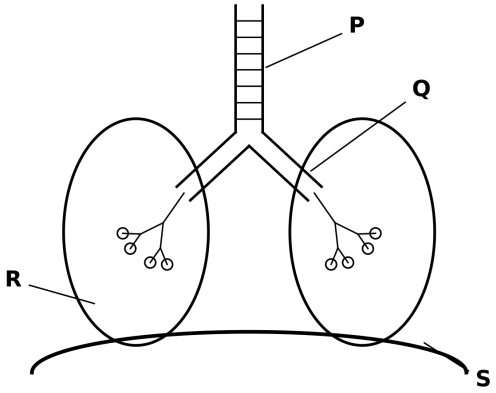
I. Identify X, Y and Z. Name the type of respiration shown by pathway (i). (2)

II. During a 100 m race, a sprinter gets painful cramps in her leg muscles. Using the flow chart, explain why this happens. Why does pathway (iii) release much more energy than pathway (ii)? (2)

III. Why is ATP called the energy currency of the cell? (1)

OR

B. The figure below shows the human respiratory system.



I. Identify P and Q. Why do the walls of P not collapse even when there is less air in it? (2)

II. Describe the role of S and the rib muscles during exhalation. (2)

III. In R, the smallest tubes end in balloon-like structures. Name these structures and state how they are suited for the exchange of gases. (1)

For visually impaired students

A. Glucose is first broken down into a three-carbon compound, pyruvate, in the cytoplasm.

I. Name the products formed from pyruvate (a) in yeast in the absence of oxygen and (b) in muscle cells during a lack of oxygen. Name the cell organelle where pyruvate is broken down in the presence of oxygen. (2)

II. During a 100 m race, a sprinter gets painful cramps in her leg muscles. Explain why this happens. Why does aerobic respiration release much more energy than anaerobic respiration? (2)

III. Why is ATP called the energy currency of the cell? (1)

OR

B.

I. Name the tube that carries air from the throat to the lungs and the tubes into which it divides. Why do the walls of this tube not collapse even when there is less air in it? (2)

II. Describe the role of the diaphragm and the rib muscles during exhalation. (2)

III. Name the balloon-like structures at the end of the smallest tubes in the lungs and state how they are suited for the exchange of gases. (1)

SECTION B Chemistry

17 Neel dipped a clean copper wire in a colourless silver nitrate solution. After some time, the solution turned blue and a shiny grey deposit formed on the wire. This is an example of a: **1**

A. combination reaction
B. decomposition reaction
C. displacement reaction
D. double displacement reaction

18 When steam is passed over red-hot iron, iron(II, III) oxide and hydrogen are formed. Which of the following is the correctly balanced equation for this reaction? **1**

A. $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
B. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
C. $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$
D. $2Fe + 3H_2O \rightarrow Fe_3O_4 + 3H_2$

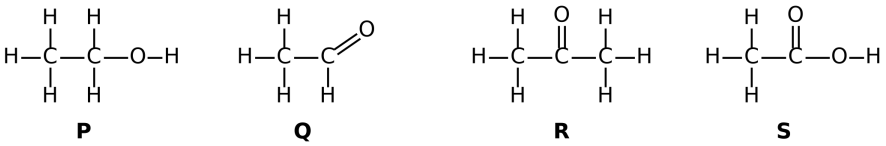
19 A honeybee sting injects an acidic liquid into the skin. Four solutions with different pH values are available in a first-aid box. Which one is the most suitable to apply on the sting? **1**

A. Solution with pH 2
B. Solution with pH 5
C. Solution with pH 7
D. Solution with pH 9

20 An orthopaedic doctor uses a white powder which, on mixing with water, sets into a hard mass to support a fractured bone. The powder and the hard mass are respectively: **1**

A. $CaSO_4 \cdot \frac{1}{2}H_2O$ and $CaSO_4 \cdot 2H_2O$
B. $CaSO_4 \cdot 2H_2O$ and $CaSO_4 \cdot \frac{1}{2}H_2O$
C. $CaOCl_2$ and $CaCO_3$
D. $CaCO_3$ and $Ca(OH)_2$

21 The structures of four carbon compounds P, Q, R and S are shown below. **1**



P **Q** **R** **S**

Which compound will produce brisk effervescence when added to a solution of sodium hydrogencarbonate?

A. P **B.** Q
C. R **D.** S

For visually impaired students

Which of the following compounds will produce brisk effervescence when added to a solution of sodium hydrogencarbonate?

A. Ethanol **B.** Ethanal
C. Propanone **D.** Ethanoic acid

22 The oxide of a metal M has the formula M_2O_3. This oxide reacts with dilute hydrochloric acid as well as with sodium hydroxide solution to form a salt and water. The metal M is most likely: **1**

A. sodium **B.** magnesium
C. aluminium **D.** copper

23 How many covalent bonds are present in a molecule of propane (C_3H_8)? **1**

A. 8 **B.** 10
C. 11 **D.** 13

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** When an iron nail is kept in copper sulphate solution, the blue colour of the solution fades and a brown coating forms on the nail.
Reason (R): Iron is more reactive than copper. **1**

25 Silver chloride is a white solid. When a watch glass containing silver chloride is kept in sunlight for some time, it turns grey. **2**

A. Write the balanced chemical equation for the reaction and name the type of reaction.

B. State one use of this reaction in daily life.

26 ***Students to attempt either option A or B.*** **3**

A. A cook noticed that the bottom of a cooking vessel was getting blackened and the flame of the gas stove was yellow instead of blue.

I. What do these observations indicate about the supply of air to the burner? Give a reason.

II. Saturated hydrocarbons generally burn with a clean flame, whereas unsaturated hydrocarbons give a sooty flame. Explain.

III. Write the chemical equation for the complete combustion of methane.

OR

B. **I.** Draw the electron dot structure of ethene (C_2H_4).

II. Ethene undergoes addition reactions but ethane does not. Give a reason.

III. Write the chemical equation for the reaction of methane with chlorine in the presence of sunlight and name the type of reaction.

27 Give reasons for the following: **3**

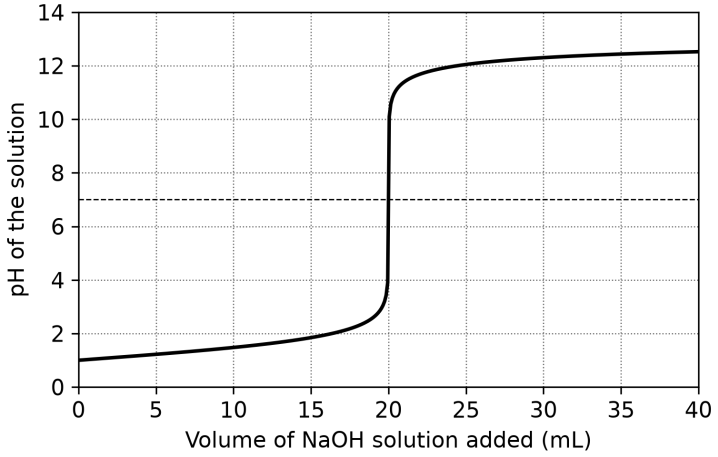
A. Dry hydrogen chloride gas does not change the colour of dry blue litmus paper, but it turns moist blue litmus paper red.

B. Dentists advise brushing the teeth after every meal using a toothpaste, which is generally basic.

C. While diluting a concentrated acid, the acid must be added slowly to water and not water to the acid.

28 **4**

Kabir took 20 mL of dilute hydrochloric acid in a beaker and added a few drops of universal indicator to it. He then added sodium hydroxide solution of the same concentration from a burette, a little at a time, and measured the pH after each addition with a pH meter. He also noticed that the beaker became slightly warm. His readings are plotted in the graph below.



A. What was the pH of the solution in the beaker before any sodium hydroxide was added, and what colour would the universal indicator show at that point? (1)

B. At what volume of sodium hydroxide solution added was the acid exactly neutralised? Name the salt formed. (1)

Attempt either sub-part C or D.

C. Write the balanced chemical equation for the reaction. Why did the beaker become warm? (2)

OR

D. If a few drops of phenolphthalein were added to the beaker instead, what colour would the solution show after 10 mL and after 30 mL of sodium hydroxide solution have been added? Justify using the graph. (2)

For visually impaired students

Kabir added sodium hydroxide solution, a little at a time, to 20 mL of dilute hydrochloric acid of the same concentration. The pH was 1 at the start, rose slowly to about 2.6 by 19 mL, became 7 at 20 mL and was about 12 at 30 mL. The beaker became slightly warm.

A. What was the pH of the solution before any sodium hydroxide was added, and what colour would the universal indicator show?

B. At what volume was the acid exactly neutralised? Name the salt formed.

C. Write the balanced chemical equation for the reaction. Why did the beaker become warm?

OR

D. What colour would phenolphthalein show after 10 mL and after 30 mL of sodium hydroxide solution have been added? Justify.

29 ***Attempt either option A or B.*** **5**

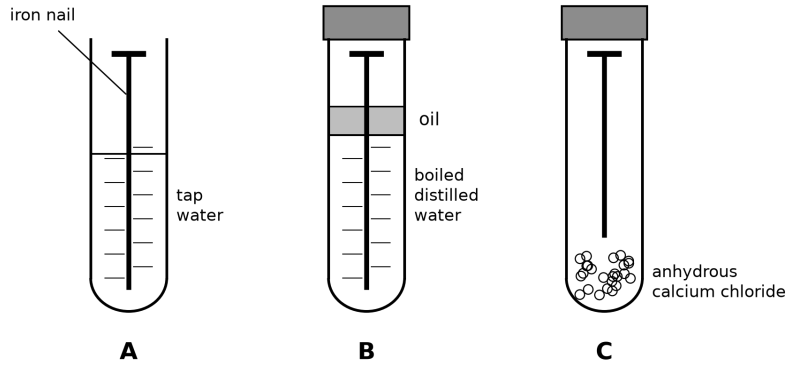
A. **I.** Element X has atomic number 12 and element Y has atomic number 17. Show the formation of the compound between X and Y by the transfer of electrons. Write the formula of the compound and name the type of bond formed. (2)

II. Give reasons: (a) the compound formed in (I) has a high melting point; (b) it conducts electricity in the molten state but not in the solid state. (2)

III. Why is this compound soluble in water but not in kerosene? (1)

OR

B. Meera set up three test tubes A, B and C with clean iron nails as shown below and left them for a week.



I. In which test tube(s) will the nails rust? Explain the purpose of the oil layer in B and of the anhydrous calcium chloride in C. (2)

II. Iron sheets used for roofing are coated with zinc. Name this process. Why does the iron remain protected even if the zinc coating gets scratched? (2)

III. Name the substances formed on the surface of silver articles and copper vessels when they are exposed to air for a long time. (1)

For visually impaired students

B. Meera kept three clean iron nails: A in a test tube with tap water and air, B in boiled distilled water with a layer of oil on top, and C in a corked test tube with anhydrous calcium chloride (dry air).

I. In which test tube(s) will the nails rust? Explain the purpose of the oil layer in B and of the anhydrous calcium chloride in C. (2)

II. Iron sheets used for roofing are coated with zinc. Name this process. Why does the iron remain protected even if the zinc coating gets scratched? (2)

III. Name the substances formed on the surface of silver articles and copper vessels when they are exposed to air for a long time. (1)

SECTION C Physics

30 Two-wheeler manufacturers fit a particular type of mirror as the rear-view mirror. Which option correctly names the mirror and the reason for its use? **1**

A. Concave mirror, as it forms an enlarged image of vehicles behind

B. Convex mirror, as it gives an erect, diminished image over a wider field of view

C. Plane mirror, as it gives an image of the same size

D. Concave mirror, as it always forms a real image

31 A long solenoid carries a steady current. Which statement correctly describes the magnetic field inside the solenoid? **1**

A. It is zero at all points.
B. It is in the form of concentric circles.
C. It is uniform and is shown by parallel straight lines.
D. It is strongest near the ends and weakest at the centre.

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** A person suffering from myopia can see distant objects clearly.
Reason (R): In a myopic eye, the image of a distant object is formed in front of the retina. **1**

33 On a clear morning, the Sun can be seen about two minutes before it actually rises above the horizon, and it remains visible for about two minutes after it has actually set. Explain this observation with the help of a labelled diagram. **2**

For visually impaired students

On a clear morning, the Sun can be seen about two minutes before it actually rises above the horizon, and it remains visible for about two minutes after it has actually set. Name the phenomenon responsible and explain how it happens.

34 ***Students to attempt either option A or B.*** **2**

A. In a house, the metal body of a washing machine is connected to the earth wire. What is the purpose of this connection? What is meant by a short circuit?

OR

B. State any two properties of magnetic field lines. What does the closeness of the field lines in a region indicate?

35 A dentist holds a concave mirror of focal length 4 cm at a distance of 3 cm from a tooth. **3**

A. Find the position of the image of the tooth.
B. Find the magnification and state the nature of the image.
C. Draw a ray diagram to show the formation of the image.

For visually impaired students

A dentist holds a concave mirror of focal length 4 cm at a distance of 3 cm from a tooth.

A. Find the position of the image of the tooth.
B. Find the magnification and state the nature of the image.
C. Why is a concave mirror, and not a convex mirror, used by dentists?

36 ***Students to attempt either option A or B.*** **3**

A. A convex lens of focal length 20 cm forms a real image of an object on a screen placed 30 cm from the lens. Find (i) the distance of the object from the lens, (ii) the magnification produced, and (iii) the power of the lens.

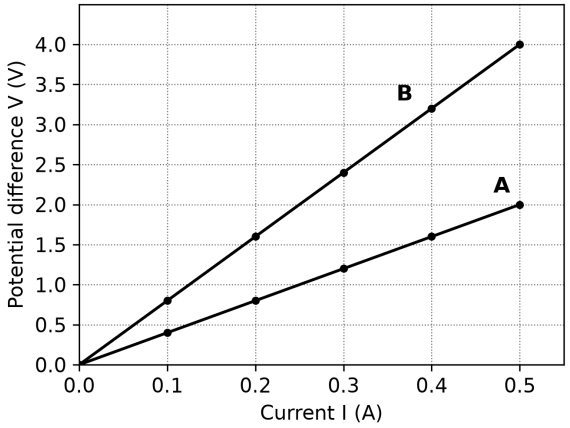
OR

B. Draw a labelled ray diagram to show the path of a ray of light passing obliquely through a rectangular glass slab. Mark the angle of incidence, angle of refraction, angle of emergence and the lateral displacement. How is the emergent ray related to the incident ray?

For visually impaired students

A ray of light passes obliquely through a rectangular glass slab. State the relation between the angle of incidence and the angle of emergence. How is the emergent ray related to the incident ray? Name two factors on which the lateral displacement depends.

37 Ananya performed an experiment with two wires A and B of the same material and the same length. She recorded the potential difference across each wire for different currents and plotted the graph shown below. **3**



A. Which law is verified by the graph? State the law.

B. Calculate the resistance of each wire from the graph.

C. Which of the two wires is thicker? Justify your answer.

For visually impaired students

Ananya measured the current through two wires A and B of the same material and the same length. With 2 V across each, the current was 0.5 A in A and 0.25 A in B. The ratio V/I stayed constant for each wire.

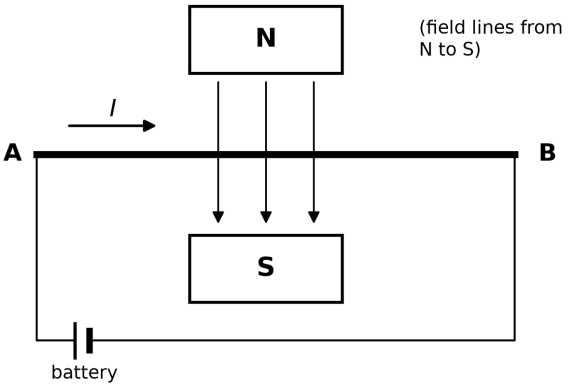
A. Which law is verified? State the law.

B. Calculate the resistance of each wire.

C. Which of the two wires is thicker? Justify your answer.

38 **4**

In a physics laboratory, Arjun suspended a light aluminium rod AB horizontally between the poles of a strong magnet, as shown in the figure. The magnetic field is directed vertically downwards from the north pole to the south pole. When he switched on the current from A to B, the rod was displaced.



A. In which direction will the rod be displaced? (1)

B. What will happen to the displacement of the rod if the direction of the current is reversed? (1)

Attempt either sub-part C or D.

C. State the rule used to find the direction of the force on the rod. Give two ways of increasing the displacement of the rod. (2)

OR

D. If the rod is turned so that it lies along the direction of the magnetic field (vertical), what will be the force on it? Explain. When is the force on a current-carrying conductor in a magnetic field maximum? (2)

For visually impaired students

Arjun suspended a light aluminium rod horizontally between the poles of a strong magnet. The magnetic field is directed vertically downwards. When a current was passed through the rod from left to right (as seen by him), the rod was displaced.

A. In which direction will the rod be displaced?

B. What will happen to the displacement if the current is reversed?

Attempt either sub-part C or D.

C. State the rule used to find the direction of the force. Give two ways of increasing the displacement.

OR

D. If the rod lies along the magnetic field, what will be the force on it? When is the force maximum?

39 ***Attempt either option A or B.*** **5**

A. I. State Joule's law of heating and write its mathematical expression. (1)

II. A room heater is rated 1100 W, 220 V. Calculate (a) the current drawn by it, (b) its resistance, and (c) the heat produced by it in 10 minutes. (3)

III. Why is nichrome used for the heating element of the heater, while tungsten is used for the filament of an electric bulb? (1)

OR

B. I. A family uses the following appliances every day:

Appliance	Number	Power of each	Use per day
Tube light	4	40 W	5 h
Ceiling fan	2	75 W	8 h
Water heater (geyser)	1	2000 W	1 h

Calculate the electrical energy consumed in 30 days in kilowatt-hour, and the cost of this energy at ₹6 per kWh. (3)

II. Two identical resistors, each of resistance R, are connected first in series and then in parallel across the same source of potential difference V. Find the ratio of the heat produced per second in the parallel combination to that in the series combination. (2)

END OF QUESTION PAPER

Exam tip

Keep the last 10 minutes to read through your answers and check units, signs and labels.

SOLUTION PAPER 09

SCIENCE · CODE NO. 086

Sample Paper 9: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

B. Absorption of digested food into the blood (1)

2 Ans

D. S (1)

For visually impaired students: D. Nerve ending at the end of the axon

3 Ans

C. Thyroxin – thyroid (1)

(Iodine is needed by the thyroid gland to make thyroxin; its deficiency causes goitre.)

4 Ans

B. Fruits form only on plants with female flowers, after pollen reaches them from a plant with male flowers. (1)

5 Ans

C. 50% (1)

(Each child gets an X from the mother and either X or Y from the father with equal chance; earlier births do not affect it.)

6 Ans

B. 300 (1)

(F_2 ratio tall : dwarf = 3 : 1; ¼ × 1200 = 300.)

7 Ans

A. Replacing old refrigerators and air conditioners that use CFCs with CFC-free models (1)

8 Ans

C. A is true but R is false. (1)

(Pancreatic juice contains trypsin and lipase but is poured into the duodenum, i.e. the small intestine.)

9 Ans

B. Both A and R are true, and R is not the correct explanation of A. (1)

(The bending is due to auxin diffusing to the shaded side and causing the cells there to elongate more.)

10 Ans

A. Starch is formed only in the green parts of the leaf; hence chlorophyll is necessary for photosynthesis. (1)

B. To destarch the leaves, i.e. to use up the starch already present in the leaves, so that any starch detected is formed only during the experiment. (1)

Total 2 marks

11 Ans

A.

Thigmotropism / a tropic movement in response to touch (contact). (½)

The part of the tendril in contact with the support grows less, while the part away from the support grows faster (cells elongate more), so the tendril coils around the support. (1½)

OR

B.

Reflex action. (½)

Receptor (skin) → sensory neuron → relay neuron in spinal cord → motor neuron → effector (muscles of the arm), which pulls the hand back. (1½)

Total 2 marks

12 Ans

A. Production of sperms (male germ cells). (½)
Secretion of the hormone testosterone. (½)

B. Sperm formation requires a temperature lower than the normal body temperature; the scrotum keeps the testes about 1–3 °C cooler. (1)

Total 2 marks

13 Ans

A. Rhizopus (bread mould). (½)
X – sporangium; Y – hypha. (½)

B. Spore formation. (½)
Each sporangium contains a large number of spores with thick walls that protect them; when the sporangium bursts, the light spores are dispersed by air and germinate into new mould on any moist food surface they land on. (1)

C. Low temperature slows down the growth of the mould and the germination of its spores, so no visible growth appears for many days. (½)

Total 3 marks

14 Ans

A. Biological magnification (biomagnification). (1)

B. Fish-eating bird. It is at the highest trophic level of this food chain, and the pesticide, which is not degraded, accumulates at each step; the bird has the highest concentration (25 ppm). (1)

C. Pesticides sprayed on fields are washed into the lake by rainwater (run-off) / seep in through the soil. (½)
Prevention: use of biological pest control / organic farming / limited and careful use of pesticides away from water bodies (any one). (½)

Total 3 marks

15 Ans

A. Proteins are large molecules that cannot pass through the walls of the glomerular capillaries during filtration, so they stay in the blood. (1)

B. All the glucose in the filtrate is selectively reabsorbed into the blood by the cells of the tubule as the filtrate passes along it. (1)

C.

As the filtrate flows along the tubule, a large amount of water is reabsorbed into the blood (about 178.5 L of the 180 L), while urea is not reabsorbed; hence urea becomes concentrated in urine. (1½)
Part: the tubular part of the nephron (renal tubule). (½)

OR

D.

The volume of urine will decrease (urine will be more concentrated). (½)
Much water is lost as sweat, so the body has less excess water; therefore more water is reabsorbed in the tubules of the nephrons. (1)
Urine is stored in the urinary bladder. (½)

Total 4 marks

16 Ans

A.

I. X – ethanol; Y – lactic acid; Z – mitochondria. (½+½+½)

Pathway (i) is anaerobic respiration (fermentation). (½)

II. During the sprint, the muscles need energy faster than oxygen can be supplied, so pyruvate is converted into lactic acid (pathway ii). Accumulation of lactic acid causes cramps. (1)

In pathway (iii), pyruvate is completely broken down into carbon dioxide and water using oxygen in the mitochondria, releasing much more energy; in pathway (ii) the breakdown is incomplete. (1)

III. The energy released during respiration is stored in ATP, which is used (broken down) to supply energy for all activities of the cell, just as currency is used for transactions. (1)

OR

B.

I. P – trachea (windpipe); Q – bronchus. (½+½)

The walls of the trachea are supported by rings of cartilage, which keep it open. (1)

II. During exhalation, the diaphragm (S) relaxes and becomes dome-shaped (moves up), and the rib muscles relax so that the ribs move down and inwards. (1)

This decreases the volume of the chest cavity, so air is pushed out of the lungs. (1)

III. Alveoli. They provide a very large surface area, have very thin walls and are richly supplied with blood capillaries for efficient exchange of gases. (½+½)

For visually impaired students

A. I. (a) Ethanol and carbon dioxide (b) lactic acid; mitochondria. II and III as above. B. I. Trachea, which divides into bronchi; rings of cartilage keep it open. II and III as above.

Total 5 marks

SECTION B Chemistry

17 Ans C. displacement reaction (1)
($Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$)

18 Ans B. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ (1)

19 Ans D. Solution with pH 9 (1)
(A mild base such as baking soda solution neutralises the acid.)

20 Ans A. $CaSO_4 \cdot ½H_2O$ and $CaSO_4 \cdot 2H_2O$ (1)
(Plaster of Paris sets into gypsum.)

21 Ans D. S (1)
For visually impaired students: D. Ethanoic acid

22 Ans C. aluminium (1)
(Al_2O_3 is an amphoteric oxide.)

23 Ans B. 10 (1)
(2 C–C bonds + 8 C–H bonds)

24 Ans A. Both A and R are true, and R is the correct explanation of A. (1)

25 Ans **A.** $2AgCl(s) \xrightarrow{\text{Sunlight}} 2Ag(s) + Cl_2(g)$ (1)

Photolytic decomposition (decomposition reaction). (½)

B. Used in black and white photography. (½)

Total 2 marks

26 **Ans**

A.

I. The air holes of the burner are blocked / supply of air (oxygen) is insufficient, so the fuel is not burning completely. (1)

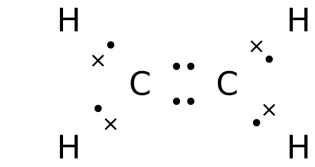
II. Unsaturated hydrocarbons have a higher proportion of carbon, which does not burn completely, producing soot (unburnt carbon); saturated hydrocarbons burn completely in sufficient air. (1)

III. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light (1)

OR

B.

I. Electron dot structure of ethene (two shared pairs between the carbon atoms form the double bond): (1)



• electron of carbon, × electron of hydrogen

II. Ethene is an unsaturated compound with a C=C double bond, to which atoms can add; ethane is saturated (only single bonds), so no addition is possible. (1)

III. $CH_4 + Cl_2 \xrightarrow{\text{Sunlight}} CH_3Cl + HCl$; substitution reaction. (½+½)

Total 3 marks

27 **Ans**

A. Acids show acidic behaviour only in the presence of water, which allows the formation of H^+ (H_3O^+) ions. Dry HCl gas does not form H^+ ions, so it does not affect dry litmus. (1)

B. Bacteria in the mouth produce acids by degrading sugar and food particles left after eating; when the pH falls below 5.5, tooth enamel corrodes. The basic toothpaste neutralises the excess acid and prevents tooth decay. (1)

C. Mixing an acid with water is highly exothermic. If water is added to the acid, the heat generated may cause the mixture to splash out and cause burns, or the glass container may break. (1)

Total 3 marks

28 **Ans**

A. pH ≈ 1 (strongly acidic); universal indicator shows red colour. (½+½)

B. 20 mL; sodium chloride (NaCl). (½+½)

C.

$NaOH(aq) + HCl(aq) \rightarrow NaCl(aq) + H_2O(l)$ (1)

The neutralisation reaction is exothermic (releases heat). (1)

OR

D.

After 10 mL: colourless, because the pH is about 1.5, i.e. the solution is still acidic. (1)

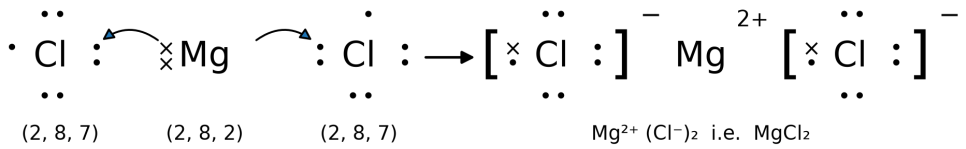
After 30 mL: pink, because the pH is about 12, i.e. excess NaOH makes the solution basic. (1)

Total 4 marks

29 Ans

A.

I. X (2, 8, 2) is magnesium and Y (2, 8, 7) is chlorine. Mg loses its two valence electrons, one to each of two Cl atoms: (1)



Formula: $MgCl_2$; ionic (electrovalent) bond. (½+½)

II. (a) A strong force of attraction exists between the oppositely charged ions, so a large amount of energy is needed to break it. (1)

(b) In the solid state the ions are held in fixed positions and cannot move; in the molten state the ions become free to move and carry charge. (1)

III. Water is a polar solvent and separates the ions; kerosene (non-polar, organic solvent) cannot do so. (1)

OR

B.

I. Rusting takes place only in test tube A, where both air and water are present. (1)

The oil layer in B prevents air from dissolving in the boiled water (boiling removes dissolved air); anhydrous calcium chloride in C absorbs moisture and keeps the air dry. (½+½)

II. Galvanisation. (1)

Zinc is more reactive than iron, so it reacts with air and moisture first (and protects the iron) even when the coating is broken. (1)

III. Silver – silver sulphide (black coating); copper – basic copper carbonate (green coating). (½+½)

Total 5 marks

SECTION C Physics

30 Ans

B. Convex mirror, as it gives an erect, diminished image over a wider field of view (1)

31 Ans

C. It is uniform and is shown by parallel straight lines. (1)

32 Ans

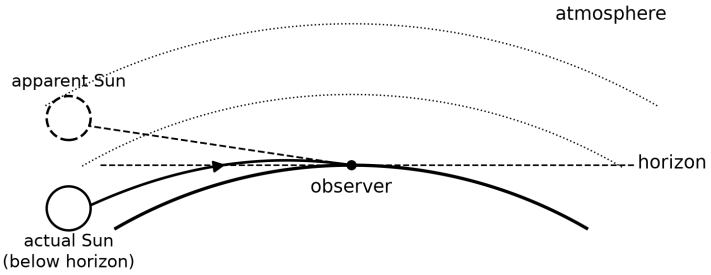
D. A is false but R is true. (1)

(A myopic person can see nearby objects clearly but not distant objects.)

33 Ans

This is due to atmospheric refraction. When the Sun is slightly below the horizon, its light passes from rarer to denser layers of the atmosphere and bends towards the normal. (1)

Hence the Sun appears to be raised above the horizon and is seen about 2 minutes before actual sunrise and about 2 minutes after actual sunset. (½)



Labelled diagram (½)

Total 2 marks

34 Ans

A.

The earth wire provides a low-resistance path for current, so if live wire touches the metal body due to leakage, the current flows to the earth and the user does not get a severe electric shock. (1)

Short circuit: when the live wire and the neutral wire come into direct contact, the resistance of the circuit becomes very small and the current increases abruptly. (1)

OR

B.

Any two: field lines emerge from the north pole and merge at the south pole outside a magnet; they are closed curves; no two field lines cross each other. (½+½)

Closeness of field lines indicates a stronger magnetic field in that region. (1)

Total 2 marks

35 Ans

A. u = −3 cm, f = −4 cm; 1/v + 1/u = 1/f ⇒ 1/v = 1/f − 1/u = −1/4 + 1/3 = 1/12 (½)

v = +12 cm, i.e. the image is 12 cm behind the mirror. (½)

B. m = −v/u = −(12)/(−3) = +4 (½)

The image is virtual, erect and magnified (4 times). (½)

C. Ray diagram: (1)

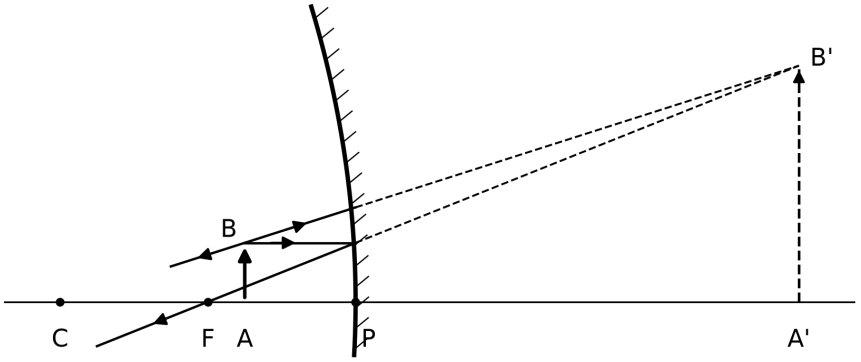


Image A'B': behind the mirror, virtual, erect and magnified

For visually impaired students: C. A concave mirror forms an enlarged, erect image of a tooth held between its pole and focus, so the dentist can see the tooth clearly; a convex mirror always forms a diminished image. (1)

Total 3 marks

36 Ans

A.

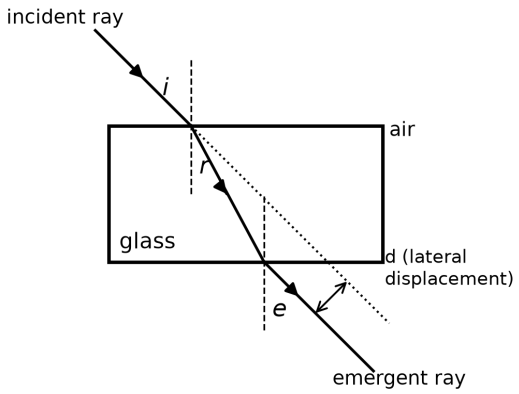
f = +20 cm, v = +30 cm; 1/v − 1/u = 1/f ⇒ 1/u = 1/30 − 1/20 = −1/60 ⇒ u = −60 cm (object is 60 cm in front of the lens) (1)

m = v/u = 30/(−60) = −0.5 (real, inverted, half the size) (1)

P = 1/f (in m) = 1/0.20 m = +5 D (1)

OR

B.



Correct labelled ray diagram showing the bending towards the normal on entering the slab and away from the normal on leaving it. (1)

Marking of angles i, r, e and the lateral displacement d. (1)

The emergent ray is parallel to the direction of the incident ray but laterally displaced (∠i = ∠e). (1)

For visually impaired students: ∠i = ∠e (1); the emergent ray is parallel to the incident ray but laterally displaced (1); lateral displacement depends on the thickness of the slab, the angle of incidence and the refractive index of the glass (any two) (½+½).

Total 3 marks

37 **Ans**

A. Ohm's law: the potential difference across the ends of a metallic wire is directly proportional to the current through it, provided its temperature remains the same (V = IR). (1)

B. R_A = V/I = 2 V/0.5 A = 4 Ω (½)

R_B = V/I = 2 V/0.25 A = 8 Ω (½)

C. Wire A is thicker. For the same material and length, R = ρl/A, so resistance is inversely proportional to the area of cross-section; A has the lower resistance, hence the larger area of cross-section. (1)

Total 3 marks

38 **Ans**

A. The rod will be displaced into the plane of the paper (away from the observer), perpendicular to both the current and the field. (1)

B. The rod will be displaced in the opposite direction (out of the plane of the paper, towards the observer). (1)

C.

Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular; if the forefinger points along the field and the middle finger along the current, the thumb gives the direction of the force. (1)

Increase the current through the rod; use a stronger magnet (any two ways, e.g. also a longer rod in the field). (½+½)

OR

D.

The force will be zero, because no force acts on a current-carrying conductor placed parallel (or antiparallel) to the magnetic field. (1)

The force is maximum when the direction of the current is perpendicular to the direction of the magnetic field. (1)

For visually impaired students: A. Horizontally, away from Arjun (perpendicular to both the current and the field). B. In the opposite direction. C and D as above.

Total 4 marks

39 **Ans**

A.

I. The heat produced in a resistor is directly proportional to the square of the current, to the resistance and to the time for which the current flows: $H = I^2Rt$. (½+½)

II. (a) $I = P/V = 1100\ W/220\ V = 5\ A$ (1)

(b) $R = V/I = 220\ V/5\ A = 44\ \Omega$ (1)

(c) $H = Pt = 1100\ W \times (10 \times 60)\ s = 6.6 \times 10^5\ J$ (1)

III. Nichrome has high resistivity and does not oxidise (burn) readily at high temperature; tungsten has a very high melting point (3380 °C), so it can glow without melting. (½+½)

OR

B.

I. Energy per day: tube lights = 4 × 40 W × 5 h = 800 Wh; fans = 2 × 75 W × 8 h = 1200 Wh; geyser = 2000 W × 1 h = 2000 Wh (1)

Total per day = 4000 Wh = 4 kWh; in 30 days = 4 × 30 = 120 kWh (1)

Cost = 120 × ₹6 = ₹720 (1)

II. $R_{series} = 2R$, $R_{parallel} = R/2$; heat per second (power) $P = V^2/R$ (1)

$P_{parallel} : P_{series} = (V^2/(R/2)) : (V^2/2R) = 2 : ½ = 4 : 1$ (1)

Total 5 marks

END OF SOLUTIONS

Quick revision: Chemical Reactions and Equations

1. Combination: A + B → AB. Decomposition: AB → A + B (by heat, light or electricity).
2. Displacement: a more reactive element replaces a less reactive one.
3. Double displacement: ions are exchanged; a precipitate often forms.
4. Redox: oxidation and reduction happen together.
5. Rancidity is prevented by airtight packing, antioxidants and nitrogen flushing.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.13 of this paper (modes of reproduction in organisms)

35. (a) Cellular DNA is the information source of making proteins in the cell. DNA provides the blueprint for all structures and functions of cell. Inheritence of ~~DNA~~ different proteins will thus lead to altered body designs. Proteins determine the function of hormones and enzymes in the body. Altered body design will lead to difference in efficiency of hormones, enzymes and body functions, and variation.

(b) Pollination is the transfer of pollen grains from anther of a flower to stigma ~~of flower~~ of a flower. If pollination does not occur, the pollen grains will not be able to reach the egg cell for fusion of male and female gametes (fertilization)

(c) It is unsuitable to depend on being cut accidentantly for mode of reproduction. The organisms cannot rely on being cut and divided at any point on body.

(d) Multicellular organisms have specialised cells for different functions. Their cells are divided and differentiated and specific. Thus they have seperate gamete cells for reproduction. Gametes may on differentiate and divide to form tissues for different functions.

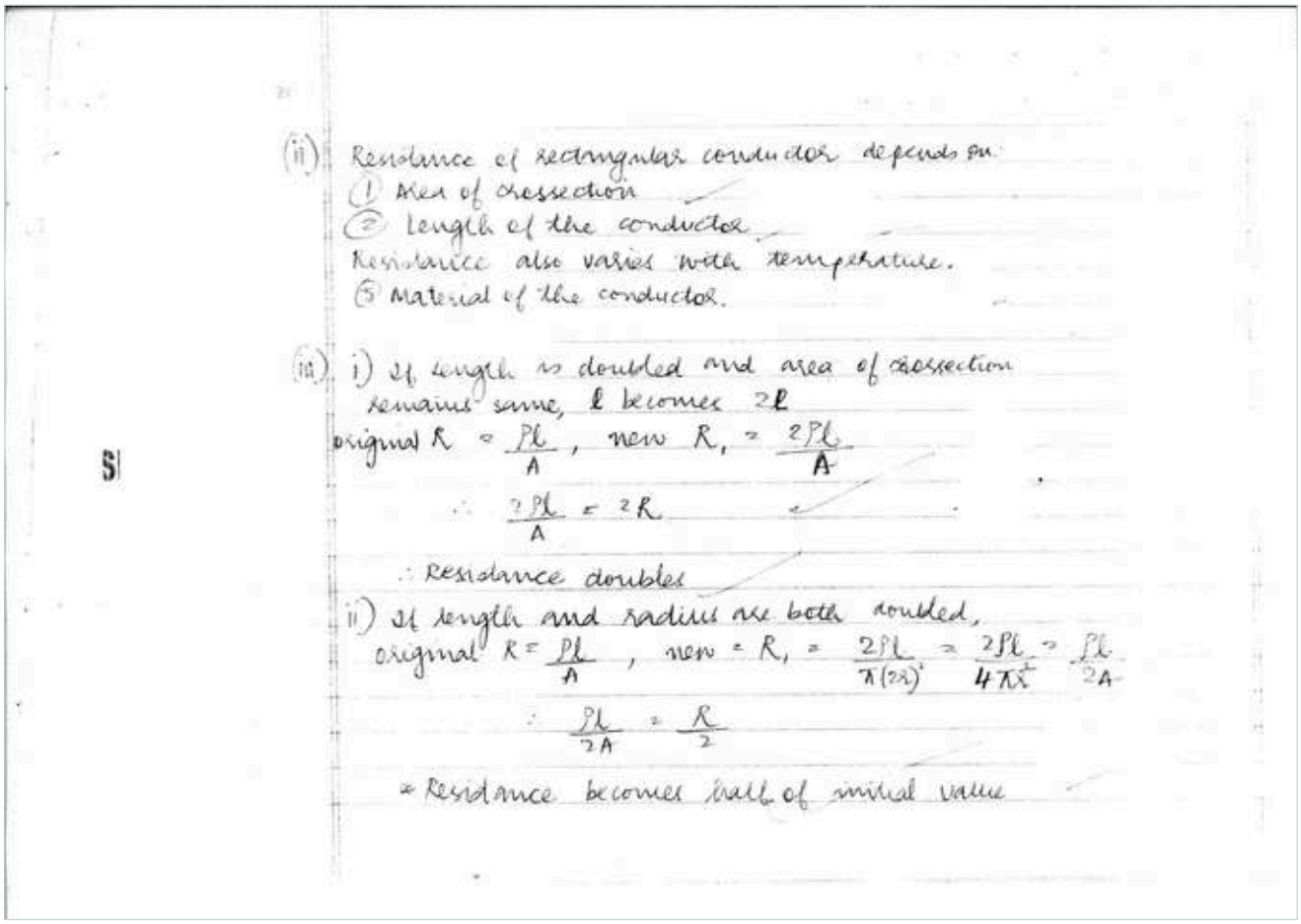
Real CBSE topper's answer · Board Exam 2023 · similar to Q.13 and Q.37 of this paper (vegetative propagation; resistance of a wire)

c) Vegetative propegation is used for plants which have lost ~~its~~ ability to produce viable seeds for reproduction. Moreover, vegetative propegation provides ~~seed~~ little variations between parent and offsprings. Variations are sometimes desirable.

e) The answer is Law of Segregation, which states that during gamete formation, the alleles of genes seperate and only one allele for each gene is taken up by the gamete. Thus, gametes are haploid and contain half the no. of chromosomes. When the male and female gamete fertilise, they restore the total no of chromosomes in the offspring.

3) (i) Resistance of a conductor is the property of the conductor to resist / retard the flow of current through it. ~~It~~ 'Ohm' is the unit of resistance. 1 ohm is the resistance when 1 volt of potential difference is applied across the conductor and 1 Ampere of current flows through it.

Real CBSE topper's answer · Board Exam 2023 · similar to Q.37 of this paper (resistance, length and area of a wire)



(ii) Resistance of rectangular conductor depends on
(1) Area of cross section
(2) Length of the conductor
Resistance also varies with temperature.
(3) Material of the conductor.

(iii) i) If length is doubled and area of cross section remains same, l becomes $2l$

original $R = \frac{\rho l}{A}$, new $R_1 = \frac{2\rho l}{A}$

$\therefore \frac{2\rho l}{A} = 2R$

$\therefore$ Resistance doubles

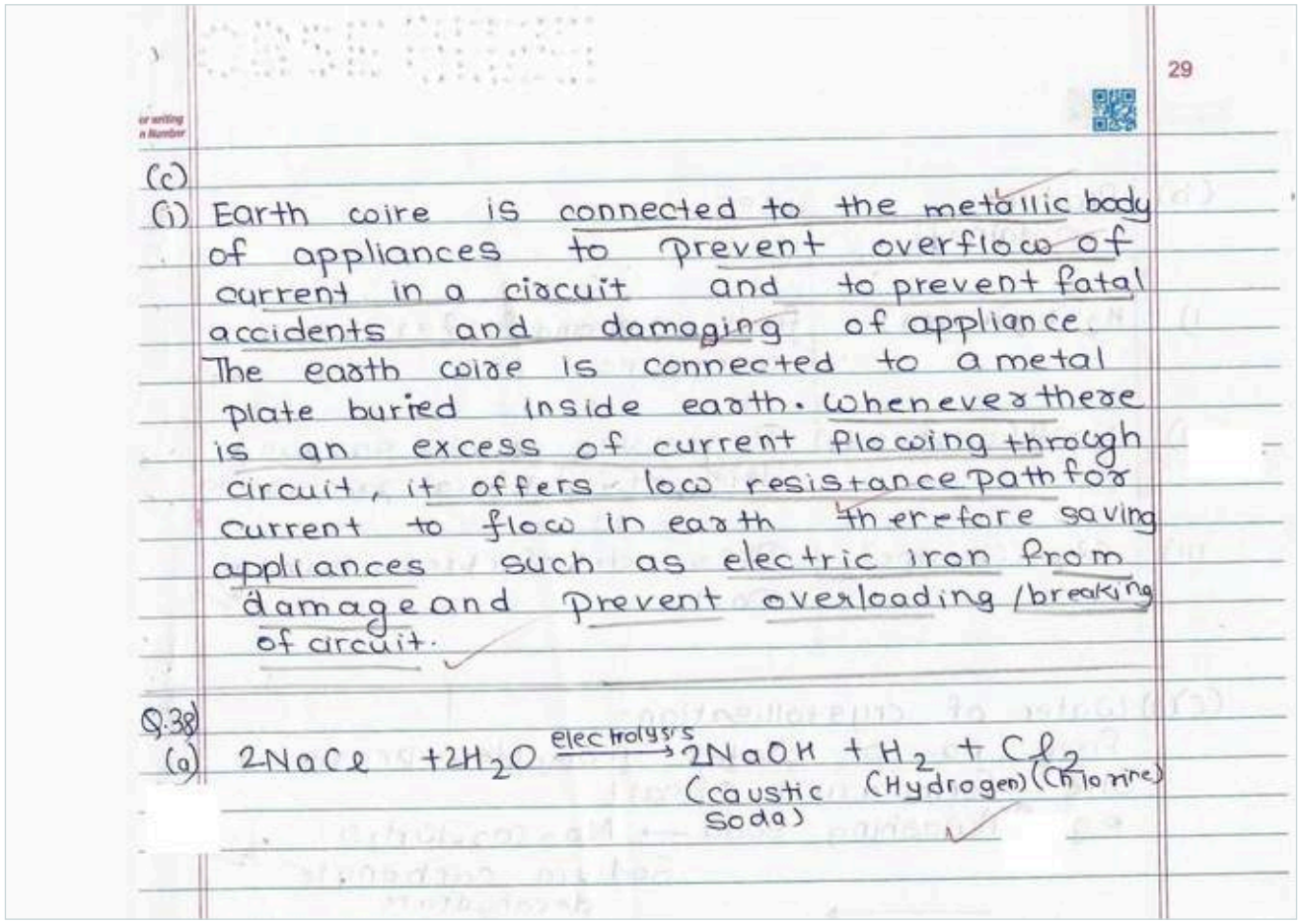
ii) If length and radius are both doubled,

original $R = \frac{\rho l}{A}$, new $= R_1 = \frac{2\rho l}{\pi(2r)^2} = \frac{2\rho l}{4\pi r^2} = \frac{\rho l}{2A}$

$\therefore \frac{\rho l}{2A} = \frac{R}{2}$

$\therefore$ Resistance becomes half of initial value

Real CBSE topper's answer · Board Exam 2025 · similar to Q.34 of this paper (earthing of metal-bodied appliances)

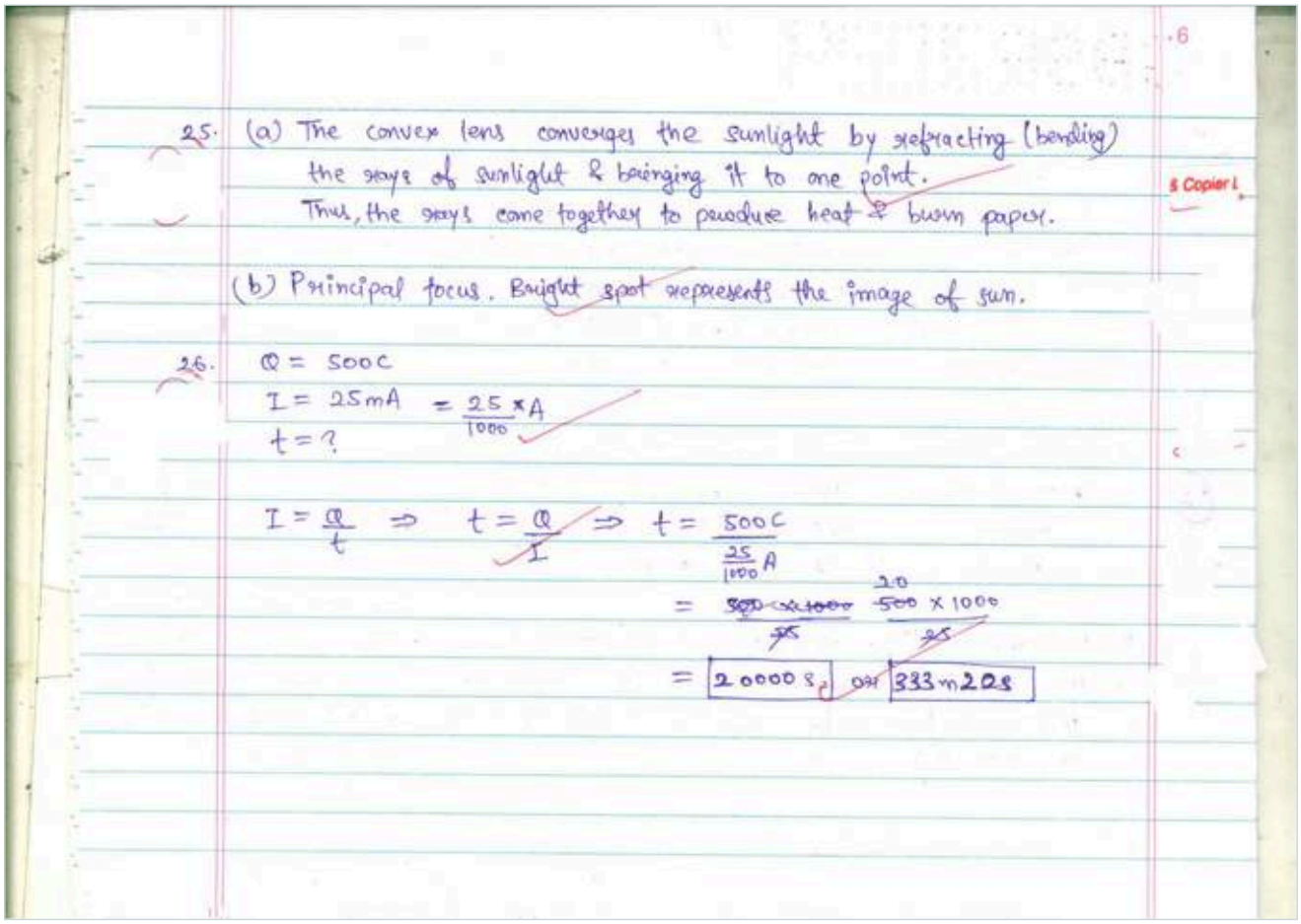


29

(c)
(i) Earth coire is connected to the metallic body of appliances to prevent overfloco of current in a circuit and to prevent fatal accidents and damaging of appliance. The earth coire is connected to a metal plate buried inside earth. Whenever there is an excess of current flocoing through circuit, it offers loco resistance path for current to floco in earth therefore saving appliances such as electric iron from damage and prevent overloading /breaking of arcuit.

Q.38
(a) $2NaCl + 2H_2O \xrightarrow{\text{electrolysis}} 2NaOH + H_2 + Cl_2$
(Caustic Soda) (Hydrogen) (Chlorine)

Real CBSE topper's answer · Board Exam 2024 · similar to Q.36 of this paper (convex lens and its focus)



25. (a) The convex lens converges the sunlight by refracting (bending) the rays of sunlight & bringing it to one point. Thus, the rays come together to produce heat & burn paper.

(b) Principal focus. Bright spot represents the image of sun.

26. Q = 500C
I = 25mA = $\frac{25}{1000}$ A
t = ?

$I = \frac{Q}{t} \Rightarrow t = \frac{Q}{I} \Rightarrow t = \frac{500C}{\frac{25}{1000}A}$

$= \frac{500 \times 1000}{25} = \frac{20}{500} \times \frac{1000}{25}$

= 20000 s or 333 min 20 s

What to learn from this topper's answer

1. The charge question is solved in clear steps: formula Q = It, 25 mA converted to 0.025 A, then t = Q/I with the unit in the answer.
2. For the convex lens, the topper states the idea (parallel rays of sunlight meet at the focus) before giving the distance.
3. The resistance answer uses R = ρl/A to explain the effect of length and area instead of only stating the result.
4. The earthing answer gives both the reason (leakage current flows to the earth) and the benefit (the user is protected from a severe shock).

SAMPLE PAPER 10

SCIENCE · CODE NO. 086

Sample Question Paper 10

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

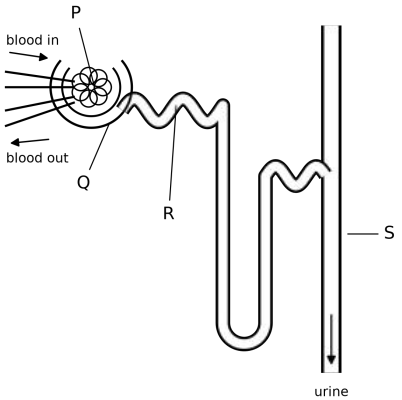
1 Reena's doctor found that her stomach secretes very little hydrochloric acid. Which of the following processes in her stomach will be directly affected? **1**

A. Emulsification of fats
B. Activation of pepsin for the digestion of proteins
C. Absorption of digested food
D. Breakdown of starch into sugars

2 Seeds of a desert plant remain dormant during long dry spells, and the stomata of its leaves close when water is scarce. Which plant hormone is mainly responsible for these responses? **1**

A. Auxin **B.** Gibberellin
C. Cytokinin **D.** Abscisic acid

3 The figure shows a nephron, the structural and functional unit of the human kidney. In which labelled part does selective reabsorption of glucose, amino acids, salts and water take place? **1**



	A. P **B.** Q **C.** R **D.** S **For visually impaired students** In a nephron, selective reabsorption of glucose, amino acids, salts and water takes place in the: **A.** glomerulus **B.** Bowman's capsule **C.** tubular part of the nephron **D.** ureter	
4	Seema crossed a pea plant having round seeds (genotype unknown) with a pea plant having wrinkled seeds. Of the 200 seeds obtained, 102 were round and 98 were wrinkled. If R (round) is dominant over r (wrinkled), the genotype of the round-seeded parent is: **A.** RR **B.** Rr **C.** rr **D.** either RR or Rr	**1**
5	Both the oviducts (fallopian tubes) of a woman are blocked. Which of the following will NOT be possible for her naturally? **A.** Release of eggs from the ovary **B.** Fertilisation of the egg by a sperm **C.** Monthly menstruation **D.** Secretion of oestrogen	**1**
6	Which of the following groups contains only biodegradable substances? **A.** Vegetable peels, dry leaves, cotton cloth **B.** Paper, polythene bag, dry leaves **C.** Glass bottle, wood, vegetable peels **D.** Aluminium foil, paper, cotton cloth	**1**
7	In a grassland, the grass traps 20,000 J of solar energy. In the food chain Grass → Grasshopper → Frog → Snake, the energy available to the snake, according to the 10 percent law, will be: **A.** 2000 J **B.** 200 J **C.** 20 J **D.** 2 J	**1**
The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true.		
8	**Assertion (A):** The walls of arteries are thick and elastic. **Reason (R):** Blood in the arteries flows under high pressure.	**1**
9	**Assertion (A):** Pancreatic juice contains the enzyme trypsin. **Reason (R):** Trypsin digests emulsified fats in the small intestine.	**1**
10	Anil kept a potted plant near a window. After a few days, he noticed that the shoot had bent towards the window. Name this response and explain how the plant hormone auxin brings about this bending.	**2**
11	***Students to attempt either option A or B.***	**2**

A. A couple has two daughters, and some relatives blame the mother for not having a son. With the help of a flow chart, explain why this belief is scientifically incorrect.

OR

B. How do genes control the expression of a trait? Explain using the example of the height of a plant.

12 (2)

A pure-bred pea plant with red flowers (RR) is crossed with a pure-bred pea plant with white flowers (rr). All the F_1 plants have red flowers. The F_1 plants are self-pollinated and 240 plants are obtained in the F_2 generation.

A. Which trait is dominant?

B. Show the F_2 cross and find how many of the 240 plants are expected to have white flowers.

13 (3)

Answer the following:

A. Doctors advise people living in hilly regions far from the sea to use iodised salt. Why?

B. A ten-year-old child is unusually short for his age. Deficiency of which hormone, secreted by which gland, could be the cause?

C. Rahul feels his heart beating fast and breathes faster just before going on stage. Name the hormone responsible and the gland that secretes it, and state how the faster heartbeat helps his body.

14 (3)

Students to attempt either option A or B.

A. Two identical potted plants are watered equally. Plant X is kept in a dry, sunny and windy place, and plant Y is kept in a closed glass chamber with humid air.

I. In which plant will water move up through the xylem faster?

II. Explain how transpiration helps in the upward movement of water in plants.

III. State one other function of transpiration.

OR

B. Answer the following about lymph:

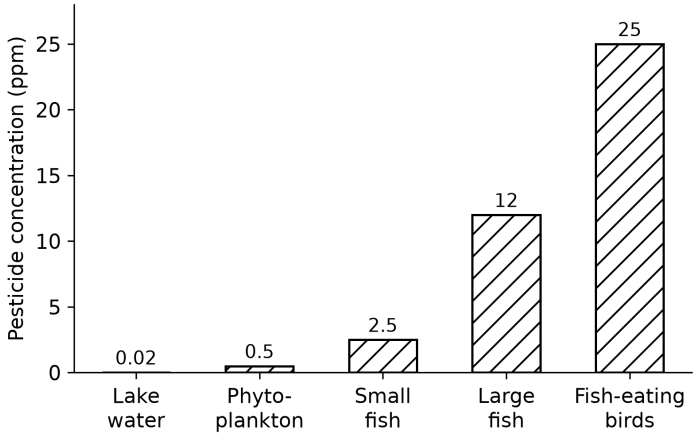
I. What is lymph and how is it formed?

II. State two differences between lymph and blood.

III. State two functions of lymph.

15 (4)

A pesticide was sprayed on crops grown near a lake. Rainwater carried the pesticide into the lake. Scientists measured the pesticide concentration in the lake water and in the organisms of the food chain Phytoplankton → Small fish → Large fish → Fish-eating birds. Their results are shown in the graph.



A. Name the phenomenon shown by the graph.

B. How many times greater is the pesticide concentration in the fish-eating birds than in the lake water?

Attempt either sub-part C or D.

C. Why does the concentration of the pesticide increase at each trophic level instead of being broken down or excreted?

OR

D. Mohan says that vegetables sprayed with pesticides become completely safe once they are washed well. Do you agree? Justify your answer and suggest one way to reduce the entry of such chemicals into food chains.

For visually impaired students

A pesticide was sprayed on crops near a lake. The pesticide concentration was measured as: lake water 0.02 ppm, phytoplankton 0.5 ppm, small fish 2.5 ppm, large fish 12 ppm and fish-eating birds 25 ppm.

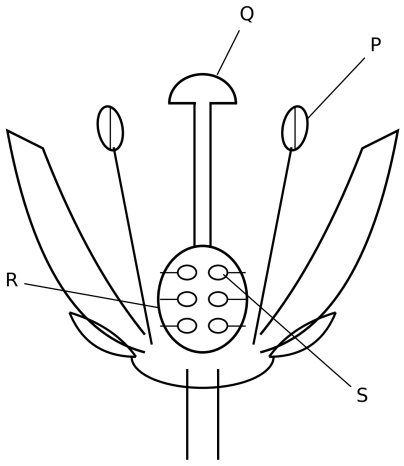
A. Name the phenomenon shown by this data.

B. How many times greater is the pesticide concentration in the fish-eating birds than in the lake water?

Sub-parts C and D: same as above.

16 ***Attempt either option A or B.*** **5**

A. The figure shows the longitudinal section of a flower.



I. (a) Name the parts labelled P, Q, R and S.
(b) Which of these parts develops into the seed and which into the fruit after fertilisation?

II. A honeybee visiting this flower leaves a compatible pollen grain on part Q. Describe the events that follow, up to the formation of the zygote.

OR

B. **I.** Priya makes three observations:
(a) A slice of moist bread left in a warm place gets covered with a black, cotton-like mass in a few days.
(b) A Planaria cut into three pieces grows into three complete organisms.
(c) Small plants grow from the notches along the margin of a Bryophyllum leaf.
Name the mode of asexual reproduction in each case, and explain how the organism in (a) spreads so fast.

II. State two advantages of vegetative propagation to a farmer growing bananas or roses.

For visually impaired students

A. **I.** (a) Name the parts of a flower in which pollen grains are produced and on which they are received.
(b) Which part of the flower develops into the seed and which into the fruit after fertilisation?

II. A honeybee leaves a compatible pollen grain on the stigma of a flower. Describe the events that follow, up to the formation of the zygote.

B. Same as option B above.

SECTION B Chemistry

17 Raghav kept some white silver chloride in a watch glass in sunlight. After some time, it turned grey. This change is an example of a: **1**

A. combination reaction **B.** displacement reaction
C. decomposition reaction **D.** double displacement reaction

18 Zinc granules are warmed with sodium hydroxide solution. A gas is evolved that burns with a pop sound. The gas evolved and the salt formed are respectively: **1**

A. oxygen and sodium zincate **B.** hydrogen and sodium zincate
C. hydrogen and zinc hydroxide **D.** chlorine and zinc chloride

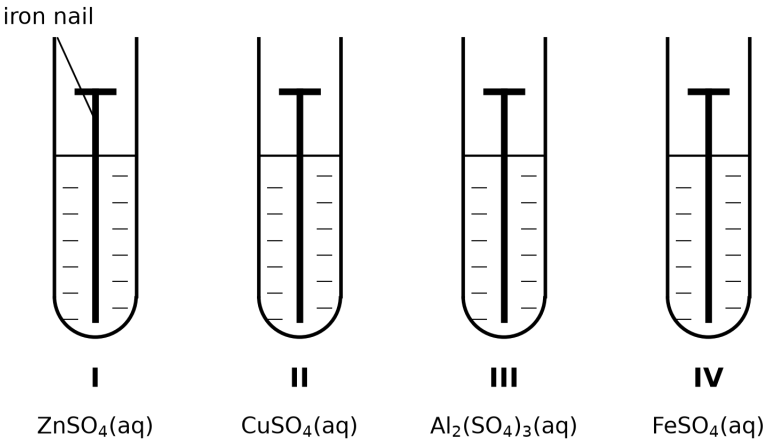
19 A metal X reacts so violently with cold water that the hydrogen evolved catches fire. To prevent accidental fires, X is stored under kerosene. X is most likely: **1**

A. magnesium **B.** sodium
C. aluminium **D.** zinc

20 Which of the following pairs of compounds belongs to the same homologous series? **1**

A. CH_4 and C_2H_4 **B.** C_2H_6 and C_3H_8
C. C_2H_2 and C_2H_4 **D.** C_3H_6 and C_4H_{10}

21 Clean iron nails were placed in four test tubes containing different solutions, as shown. After 30 minutes, in which test tube will the colour of the solution change and a brownish coating appear on the nail? **1**



A. I **B.** II
C. III **D.** IV

For visually impaired students

Clean iron nails are placed separately in solutions of zinc sulphate, copper sulphate, aluminium sulphate and iron sulphate. In which solution will the colour change and a brownish coating appear on the nail?

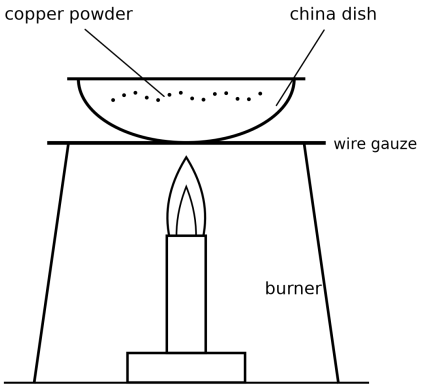
	A. Zinc sulphate **B.** Copper sulphate **C.** Aluminium sulphate **D.** Iron sulphate	
22	Bacteria present in the mouth break down sugar and food particles left on the teeth and produce acids. Tooth enamel starts getting corroded when the pH of the mouth falls below about: **A.** 7.5 **B.** 6.5 **C.** 5.5 **D.** 8.5	**1**
23	The IUPAC name of the compound $CH_3–CH_2–CHO$ is: **A.** propanone **B.** propanal **C.** propanol **D.** propanoic acid	**1**

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24	**Assertion (A):** Aluminium is more reactive than iron, yet aluminium articles do not corrode easily. **Reason (R):** Aluminium reacts with oxygen to form a thin protective layer of aluminium oxide on its surface.	**1**
25	Neha mixed colourless solutions of lead(II) nitrate and potassium iodide in a test tube. A yellow solid settled at the bottom of the test tube. **A.** Write the balanced chemical equation for the reaction, with state symbols. **B.** Name two types of reactions that this reaction represents.	**2**
26	***Students to attempt either option A or B.*** **A.** **I.** Why does carbon form compounds mainly by covalent bonding rather than by forming C^{4+} or C^{4-} ions? **II.** Draw the electron dot structure of ethyne (C_2H_2). **III.** Why are most carbon compounds poor conductors of electricity? **OR** **B.** Kunal burnt two hydrocarbons, X (C_2H_6) and Y (C_2H_4), in air. One of them burnt with a yellow, sooty flame. **I.** Which hydrocarbon gave the sooty flame? Give a reason. **II.** Name the reaction by which Y can be converted into X. Write the chemical equation, mentioning the condition required. **III.** State one industrial application of this reaction.	**3**
27	Element X has atomic number 12 and element Y has atomic number 17. **A.** Write the electronic configurations of X and Y and state which of them is a metal. **B.** Show the formation of the compound between X and Y by the transfer of electrons. **C.** Why does this compound have a high melting point? Why does it conduct electricity in the molten state but not in the solid state?	**3**
28	Tanvi took some brown copper powder in a china dish and heated it, as shown. The surface of the powder became coated with a black substance. She then passed hydrogen gas over	**4**

the heated black substance and it turned brown again.



A. Name the black substance formed on heating copper powder in air.

B. When hydrogen gas is passed over the heated black substance, which substance gets oxidised?

Attempt either sub-part C or D.

C. Write the balanced chemical equation for the reaction in B and identify the oxidising agent and the reducing agent.

OR

D. Tanvi notices that (i) her mother's copper vessel slowly develops a dull green coating, and (ii) an open packet of fried snacks starts to smell and taste bad after a few days. Name the phenomenon in each case and suggest one method to prevent each.

29 ***Attempt either option A or B.*** **5**

A. **I.** Sonal has three unlabelled test tubes containing dilute hydrochloric acid, sodium hydroxide solution and distilled water. She has only red litmus paper. Explain how she can identify the contents of each test tube.

II. Write balanced chemical equations for the reactions of dilute hydrochloric acid with (a) zinc granules and (b) sodium carbonate. How will you test the gas evolved in each case?

OR

B. **I.** Mohan took 10 mL of dilute sodium hydroxide solution in a conical flask, added two drops of phenolphthalein, and then added dilute hydrochloric acid drop by drop while shaking.
(a) What colour change will he observe, and at what stage?
(b) Name the reaction taking place and write its balanced chemical equation.
(c) Will the flask feel warm or cold? Give a reason.

II. (a) A student uses finely chopped onion as an indicator. What will she notice when it is added separately to sodium hydroxide solution and to dilute hydrochloric acid?
(b) A person suffering from acidity is advised to take milk of magnesia. Why?

SECTION C Physics

30 A student cannot see objects placed beyond 2 m clearly. The power of the lens required to correct this defect is: **1**

A. +0.5 D **B.** −0.5 D
C. +2.0 D **D.** −2.0 D

31 Riya has three identical resistors of 6 Ω each. Which arrangement will give an equivalent resistance of 9 Ω? **1**

A. All three in series
B. All three in parallel
C. Two in parallel, with this combination in series with the third
D. Two in series, with this combination in parallel with the third

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

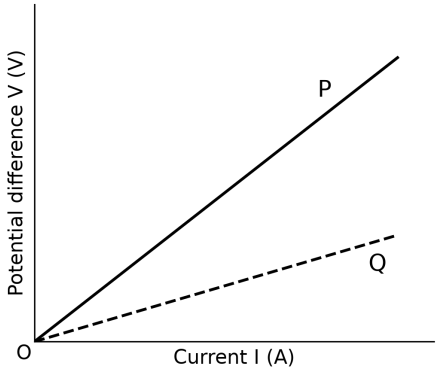
A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Two magnetic field lines never intersect each other.
Reason (R): At any point, the magnetic field has only one direction. **1**

33 **A.** Why is an alloy such as nichrome used for the heating element of an electric iron instead of a pure metal such as copper?
B. Give two reasons why household appliances are connected in parallel and not in series. **2**

34 ***Students to attempt either option A or B.*** **2**

A. The V–I graphs of two wires P and Q, made of the same material and having the same length, are shown.



I. Which wire has the higher resistance? Give a reason.
II. Which wire is thicker? Justify your answer.

For visually impaired students

Two wires P and Q of the same material and the same length have resistances 8 Ω and 3 Ω respectively.

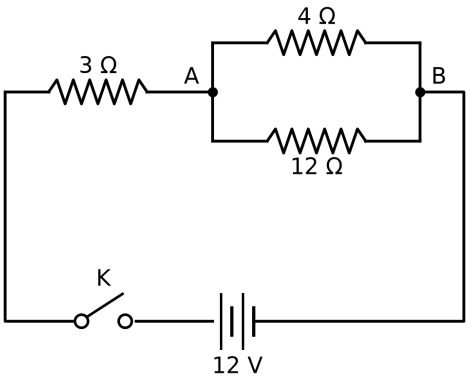
I. Which wire has the higher resistance?
II. Which wire is thicker? Justify your answer.

OR

B. A wire of resistance 10 Ω is stretched uniformly so that its length becomes double its original length.

I. Calculate its new resistance.
II. Will the resistivity of the wire change? Justify your answer.

35 Study the circuit shown and calculate: **3**



A. the equivalent resistance of the circuit,

B. the current drawn from the battery,

C. the current through the 12 Ω resistor.

For visually impaired students

A 3 Ω resistor is connected in series with a parallel combination of 4 Ω and 12 Ω resistors across a 12 V battery. Calculate A, B and C as above.

36 Give reasons for the following: **3**

A. Stars twinkle, but planets do not.

B. The sky appears blue on a clear day.

C. The path of a beam of sunlight becomes visible when it enters a smoke-filled room through a small hole.

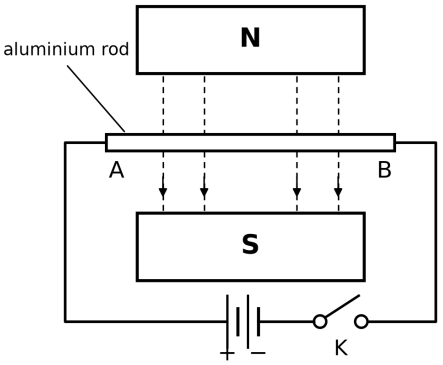
37 A candle flame 2 cm tall is placed 15 cm in front of a concave mirror of focal length 10 cm. **3**

A. At what distance from the mirror should a screen be placed to obtain a sharp image of the flame?

B. Find the size of the image.

C. State the nature of the image.

38 Ishaan suspended a light aluminium rod AB horizontally between the poles of a strong horseshoe magnet so that it lay perpendicular to the magnetic field (N pole above, S pole below), as shown. When he closed the key K, the rod was displaced. **4**



A. State the rule used to find the direction of the force acting on the rod.

B. In which direction will the rod AB be displaced?

Attempt either sub-part C or D.

C. What will happen to the displacement of the rod if (i) the direction of the current is reversed, and (ii) the rod is placed parallel to the magnetic field?

OR

D. Suggest two ways to increase the displacement of the rod. How will the direction of the force change if the directions of both the current and the magnetic field are reversed?

For visually impaired students

A light aluminium rod is placed between the poles of a strong horseshoe magnet, perpendicular to the magnetic field, and a current is passed through it. The rod is displaced.

A. State the rule used to find the direction of the force acting on the rod.

B. What will happen to the rod if it is placed parallel to the magnetic field and the current is switched on?

Attempt either sub-part C or D.

C. What will happen to the displacement of the rod if the direction of the current is reversed? What if the current is increased?

OR

D. Suggest two ways to increase the displacement of the rod. How will the direction of the force change if the directions of both the current and the magnetic field are reversed?

39 ***Attempt either option A or B.*** **5**

A. **I.** State the two laws of refraction of light.

II. Rohan traced the path of a ray of light through a rectangular glass slab for an angle of incidence of 45°. Draw a labelled ray diagram showing the incident ray, refracted ray, emergent ray, the angles i, r and e, and the lateral displacement. How is the angle of emergence related to the angle of incidence?

III. The refractive index of the glass is 1.5. Find the speed of light in the glass. (Speed of light in air = 3×10^8 m/s)

OR

B. **I.** Define the power of a lens. Write its SI unit.

II. A convex lens of focal length 25 cm and a concave lens of focal length 20 cm are placed in contact. Find the power and the focal length of the combination. Will the combination converge or diverge light?

III. The absolute refractive indices of some media are given below.

Medium	Refractive index
Water	1.33
Kerosene	1.44
Crown glass	1.52
Diamond	2.42

(a) In which of these media does light travel the fastest?

(b) A ray of light passes obliquely from kerosene into water. Will it bend towards or away from the normal?

(c) Find the refractive index of diamond with respect to water.

For visually impaired students

In place of A. II: Why is the emergent ray from a rectangular glass slab parallel to the incident ray? What is meant by lateral displacement? (All other parts remain the same.)

END OF QUESTION PAPER

SOLUTION PAPER 10

SCIENCE · CODE NO. 086

Sample Paper 10: Solutions

STEP-WISE MARKING AS PER CBSE · CLASS X · 2026-27

Time Allowed 3 Hours

Maximum Marks 80

SECTION A Biology

1 Ans

B. Activation of pepsin for the digestion of proteins (1)

(HCl creates the acidic medium in which pepsin acts.)

Total 1 mark

2 Ans

D. Abscisic acid (1)

Total 1 mark

3 Ans

C. R (tubular part of the nephron) (1)

For visually impaired students:

C. tubular part of the nephron (1)

Total 1 mark

4 Ans

B. Rr (1)

(Rr × rr gives round : wrinkled in a 1 : 1 ratio; 102 : 98 is nearly 1 : 1.)

Total 1 mark

5 Ans

B. Fertilisation of the egg by a sperm (1)

Total 1 mark

6 Ans

A. Vegetable peels, dry leaves, cotton cloth (1)

Total 1 mark

7 Ans

C. 20 J (1)

(20,000 J → 2000 J → 200 J → 20 J)

Total 1 mark

8 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

Total 1 mark

9 Ans

C. A is true but R is false. *(Trypsin digests proteins; lipase digests emulsified fats.)* (1)

Total 1 mark

10 Ans

Positive phototropism (bending of the shoot towards light). (½)

Auxin is synthesised at the shoot tip. (½)

When light falls from one side, auxin diffuses towards the shaded side of the shoot. (½)

The higher concentration of auxin makes the cells on the shaded side grow longer, so the shoot bends towards light. (½)

Total 2 marks

11 Ans

A.

Mother (XX) produces only one type of egg: X. Father (XY) produces two types of sperms: X and Y, in equal numbers. (½)

Flow chart: (1)

Egg (X) + Sperm (X) → Zygote XX → Girl

Egg (X) + Sperm (Y) → Zygote XY → Boy

Since the egg always carries X, the sex of the child depends on which sperm (X or Y) fertilises the egg, i.e. on the father; the mother cannot be blamed. (½)

OR

B.

Genes carry the information for making proteins (enzymes). (½)

Plant height depends on the amount of a growth hormone, whose formation is controlled by an enzyme. (½)

If the gene makes an efficient enzyme, more hormone is made and the plant is tall. (½)

If the gene is altered and the enzyme is less efficient, less hormone is made and the plant is short. (½)

Total 2 marks

12

Ans

A.

Red flower colour is dominant. (½)

B.

F_1: Rr × Rr (1)

	R	r
R	RR (red)	Rr (red)
r	Rr (red)	rr (white)

F_2 ratio: red : white = 3 : 1

White-flowered plants = ¼ × 240 = 60 (½)

Total 2 marks

13

Ans

A.

Iodine is needed by the thyroid gland to make thyroxin, which regulates carbohydrate, protein and fat metabolism. Food in such regions is low in iodine, and its deficiency causes goitre (swollen neck), so iodised salt is advised. (1)

B.

Growth hormone, secreted by the pituitary gland. (1)

C.

Adrenaline, secreted by the adrenal glands. (½)

Faster heartbeat supplies more oxygen to the muscles, preparing the body to deal with the situation. (½)

Total 3 marks

14

Ans

A.

I. Plant X. (1)

II. Water evaporates from the leaves through the stomata (transpiration). This creates a suction (pull) that draws water up through the xylem from the roots to the leaves. (1)

III. It helps in the absorption and upward movement of minerals dissolved in water / it helps in temperature regulation (cooling) of the plant. (any one) (1)

OR

B.

I. Lymph (tissue fluid) is a colourless fluid. It is formed when some plasma, proteins and blood cells escape through the pores in the walls of capillaries into the intercellular spaces. (1)

II. Lymph is colourless while blood is red; lymph has less protein than blood; lymph has no red blood cells. (any two) (1)

III. It carries digested and absorbed fat from the small intestine; it drains excess fluid from the intercellular spaces back into the blood. (any two, ½ each) (1)

Total 3 marks

15 Ans

A.

Biological magnification (1)

B.

25 ÷ 0.02 = 1250 times (1)

C.

The pesticide is non-biodegradable, so it cannot be broken down or excreted and gets stored in the body tissues of organisms. (1)

Each organism feeds on many organisms of the lower trophic level, so the pesticide gets accumulated progressively at each level. (1)

OR

D.

No. Washing removes only the pesticide on the surface; the pesticide absorbed by the plant from the soil and water along with nutrients remains inside the vegetable. (1)

Measures (any one): use of biopesticides / organic farming / minimal and careful use of chemical pesticides. (1)

Total 4 marks

16 Ans

A.

I. (a) P – Anther, Q – Stigma, R – Ovary, S – Ovule (2)

(b) The ovule (S) develops into the seed and the ovary (R) develops into the fruit. (1)

II. The pollen grain germinates on the stigma and a pollen tube grows out of it. (½)

The pollen tube grows through the style and reaches the ovary. (½)

It enters the ovule and releases the male germ cell. (½)

The male germ cell fuses with the female germ cell (egg) to form the zygote (fertilisation). (½)

For visually impaired students:

I. (a) Pollen grains are produced in the anther and received on the stigma. (b) Ovule → seed; ovary → fruit. II. As above.

OR

B.

I. (a) Spore formation (Rhizopus / bread mould)
(b) Regeneration
(c) Vegetative propagation (by leaves) (1½)

Bread mould produces sporangia at the tips of its hyphae; each sporangium contains a large number of spores. (½)

Spores are light and covered by thick walls, which protect them until they come in contact with a moist surface. (½)

Spores are dispersed easily by air and germinate on moist food to form new hyphae, so the mould spreads quickly. (½)

II. (any two) Plants that have lost the capacity to produce seeds (banana, rose) can be propagated; plants bear flowers and fruits earlier than those grown from seeds; all new plants are genetically similar to the parent, so desirable traits are retained. (2)

Total 5 marks

SECTION B Chemistry

17 **Ans**

C. decomposition reaction *(2AgCl → 2Ag + Cl_2 in sunlight)* (1)

Total 1 mark

18 **Ans**

B. hydrogen and sodium zincate *(2NaOH + Zn → Na_2ZnO_2 + H_2)* (1)

Total 1 mark

19 **Ans**

B. sodium (1)

Total 1 mark

20 **Ans**

B. C_2H_6 and C_3H_8 *(both alkanes, differ by –CH_2–)* (1)

Total 1 mark

21 **Ans**

B. II *(Fe is more reactive than Cu: Fe + $CuSO_4$ → $FeSO_4$ + Cu)* (1)

For visually impaired students:

B. Copper sulphate (1)

Total 1 mark

22 **Ans**

C. 5.5 (1)

Total 1 mark

23 **Ans**

B. propanal (1)

Total 1 mark

24 **Ans**

A. Both A and R are true, and R is the correct explanation of A. (1)

Total 1 mark

25 **Ans**

A.

$Pb(NO_3)_2(aq) + 2KI(aq) \rightarrow PbI_2(s) + 2KNO_3(aq)$ (1)

B.

Double displacement reaction; precipitation reaction (1)

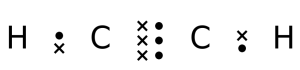
Total 2 marks

26 **Ans**

A.

I. Carbon (2, 4) cannot easily lose 4 electrons (very high energy needed) or gain 4 electrons (6 protons cannot hold 10 electrons), so it shares electrons. (1)

II. Electron dot structure of ethyne (H–C≡C–H): three shared pairs between the two carbon atoms and one shared pair between each C and H. (1)



(dots and crosses represent electrons of the two different atoms)

III. Carbon compounds are covalent; they do not form ions, so they cannot conduct electricity. (1)

OR

B.

I. Y (C_2H_4, ethene); it is unsaturated (C=C double bond) and does not burn completely, giving a sooty flame. (1)

II. Hydrogenation (addition reaction): (1)

$CH_2=CH_2 + H_2 \rightarrow CH_3–CH_3$ (in the presence of nickel catalyst)

III. Hydrogenation of vegetable oils to make vanaspati ghee (saturated fat). (1)

Total 3 marks

27 Ans

A.

X (Mg): 2, 8, 2; Y (Cl): 2, 8, 7. X (magnesium) is a metal. (1)

B.

$Mg \rightarrow Mg^{2+} + 2e^{-}$ (2, 8, 2 → 2, 8) (1)

$2Cl + 2e^{-} \rightarrow 2Cl^{-}$ (2, 8, 7 → 2, 8, 8)

$Mg^{2+} + 2Cl^{-} \rightarrow MgCl_2$ [$Mg^{2+}(Cl^{-})_2$]

C.

Strong electrostatic forces of attraction between the oppositely charged ions need a lot of energy to break, so the melting point is high. (½)

In the solid state the ions are held in fixed positions and cannot move; in the molten state the ions are free to move and carry current. (½)

Total 3 marks

28 Ans

A.

Copper(II) oxide, CuO (1)

B.

Hydrogen (it gains oxygen to form water) (1)

C.

$CuO + H_2 \rightarrow Cu + H_2O$ (on heating) (1)

Oxidising agent: CuO; reducing agent: H_2 (1)

OR

D.

(i) Corrosion of copper (copper reacts with moist CO_2 in air to form a green basic copper carbonate). Prevention: cleaning regularly / coating (tinning or lacquering) the vessel. (1)

(ii) Rancidity (oxidation of fats and oils). Prevention: keeping in airtight containers / flushing the packet with nitrogen / adding antioxidants. (1)

Total 4 marks

29 Ans

A.

I. Dip red litmus paper in each test tube. The solution that turns it blue is sodium hydroxide. (1)

Dip this blue litmus paper in the other two test tubes. The solution that turns it red is dilute hydrochloric acid; the one that brings no change in either litmus paper is distilled water. (1)

II. (a) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$ (½)

Test: bring a burning splinter near the gas; it burns with a pop sound (hydrogen). (½)

(b) $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$ (1)

Test: pass the gas through lime water; it turns milky (carbon dioxide). (1)

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

OR

B.

I. (a) The pink colour of the solution disappears (becomes colourless) when all the sodium hydroxide has been neutralised by the acid. (1)

(b) Neutralisation reaction: $NaOH + HCl \rightarrow NaCl + H_2O$ (1)

(c) Warm, because neutralisation is an exothermic reaction (heat is released). (1)

II. (a) The smell of onion is not detected in sodium hydroxide solution (base), but it is retained in dilute hydrochloric acid. Onion acts as an olfactory indicator. (1)

(b) Milk of magnesia (magnesium hydroxide) is a mild base; it neutralises the excess hydrochloric acid in the stomach and gives relief. (1)

Total 5 marks

SECTION C Physics

30 Ans

B. −0.5 D *(myopia; concave lens with f = −2 m, P = 1/f = −0.5 D)* (1)

Total 1 mark

31 Ans

C. Two in parallel, with this combination in series with the third *(3 Ω + 6 Ω = 9 Ω)* (1)

Total 1 mark

32 Ans

A. Both A and R are true, and R is the correct explanation of A. (1)

Total 1 mark

33 Ans

A.

Alloys have higher resistivity than pure metals and do not oxidise (burn) readily at high temperatures. (1)

B.

(any two) Each appliance gets the full supply voltage; each appliance can be switched on or off independently; failure of one appliance does not affect the others. (1)

Total 2 marks

34 Ans

A.

I. Wire P; the slope of the V–I graph (V/I) gives the resistance, and the slope of P is greater. (1)

II. Wire Q; for the same material and length, R = ρl/A, so resistance is inversely proportional to the area of cross-section. Q has lower resistance, so it is thicker. (1)

For visually impaired students:

I. P (8 Ω). II. Q, as R ∝ 1/A for the same material and length.

OR

B.

I. On stretching, the volume stays the same: l′ = 2l, A′ = A/2. (1)

R′ = ρ(2l)/(A/2) = 4ρl/A = 4R = 4 × 10 = 40 Ω

II. No. Resistivity depends only on the nature of the material (and temperature), not on its dimensions. (1)

Total 2 marks

35 Ans

A.

4 Ω and 12 Ω in parallel: $1/R_p = 1/4 + 1/12 = 4/12$, so $R_p = 3\ \Omega$ (1)

$R_{eq} = 3\ \Omega + 3\ \Omega = 6\ \Omega$

B.

$I = V/R_{eq} = 12\ V / 6\ \Omega = 2\ A$ (1)

C.

Potential difference across the parallel part = $I \times R_p = 2 \times 3 = 6\ V$ (1)

Current through 12 Ω = 6 V / 12 Ω = 0.5 A

Total 3 marks

36 Ans

A.

Stars are very far away and act as point sources of light. Due to atmospheric refraction in layers of continuously changing refractive index, the apparent position and brightness of a star keep changing, so it twinkles. (½)

Planets are much closer and act as extended sources (a collection of point sources); the variations average out, so they do not twinkle. (½)

B.

The fine particles and molecules in the atmosphere scatter light of shorter wavelength (blue) much more strongly than light of longer wavelength (red). The scattered blue light enters our eyes, so the sky appears blue. (1)

C.

The smoke particles scatter the light of the beam (Tyndall effect); the scattered light reaches our eyes, making the path visible. (1)

Total 3 marks

37 **Ans**

Given: h = +2 cm, u = −15 cm, f = −10 cm

A.

$1/v + 1/u = 1/f \Longrightarrow 1/v = 1/f - 1/u = -1/10 + 1/15 = -1/30$ (1)

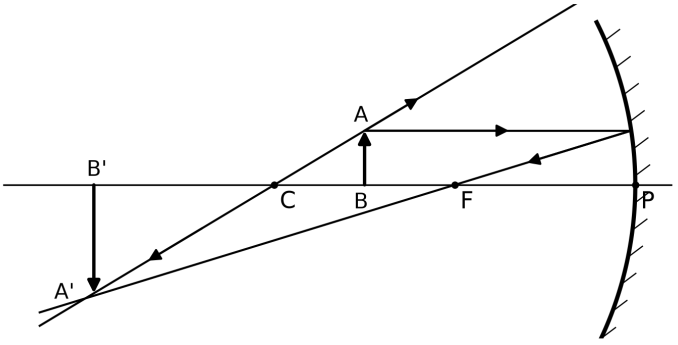
v = −30 cm; the screen should be placed 30 cm in front of the mirror.

B.

m = −v/u = −(−30)/(−15) = −2; h′ = m × h = −2 × 2 = −4 cm. The image is 4 cm tall. (1)

C.

Real, inverted and magnified (twice the size of the object). (1)



Ray diagram (for reference): object between C and F, image beyond C

Total 3 marks

38 **Ans**

A.

Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the left hand so that they are mutually perpendicular. If the forefinger points in the direction of the magnetic field and the middle finger in the direction of the current, the thumb points in the direction of the force (motion) on the conductor. (1)

B.

The current flows from A to B and the field is from N (above) to S (below); so the rod is displaced perpendicular to the plane of the paper, into the paper. (1)

C.

(i) The rod will be displaced in the opposite direction (out of the paper). (1)

(ii) No force acts on the rod, so it will not be displaced. (1)

OR

D.

Increase the current through the rod; use a stronger magnet (or a longer length of the rod in the field). (any two) (1)

If both the current and the field are reversed, the direction of the force remains the same. (1)

For visually impaired students:

A. Fleming's left-hand rule (as above). B. No force acts; the rod is not displaced. C. Displacement is reversed; it increases with current. D. As above.

Total 4 marks

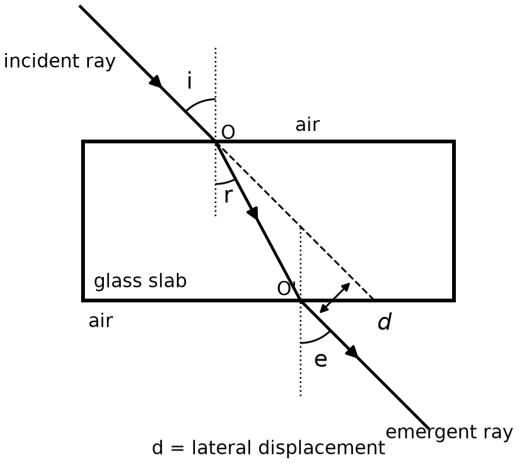
39 Ans

A.

I. (i) The incident ray, the refracted ray and the normal to the interface at the point of incidence all lie in the same plane. **(1)**

(ii) For a given pair of media and a given colour of light, sin i / sin r = constant (Snell's law); the constant is the refractive index of the second medium with respect to the first. **(1)**

II. Labelled ray diagram: **(1½)**



Angle of emergence = angle of incidence (∠e = ∠i = 45°); the emergent ray is parallel to the incident ray. **(½)**

III. $v = c/n = (3 \times 10^8) / 1.5 = 2 \times 10^8$ m/s **(1)**

For visually impaired students (A II):

The bending at the two parallel faces is equal and opposite, so the emergent ray is parallel to the incident ray. Lateral displacement is the perpendicular distance between the incident ray (produced) and the emergent ray.

OR

B.

I. The power of a lens is the reciprocal of its focal length expressed in metres (P = 1/f). SI unit: dioptre (D). **(1)**

II. $P_1 = 1/(+0.25\ m) = +4$ D; $P_2 = 1/(-0.20\ m) = -5$ D **(½)**

$P = P_1 + P_2 = +4\ D - 5\ D = -1$ D **(½)**

$f = 1/P = -1\ m = -100$ cm; since the power is negative, the combination diverges light (acts as a concave lens). **(1)**

III. (a) Water (lowest refractive index) **(½)**

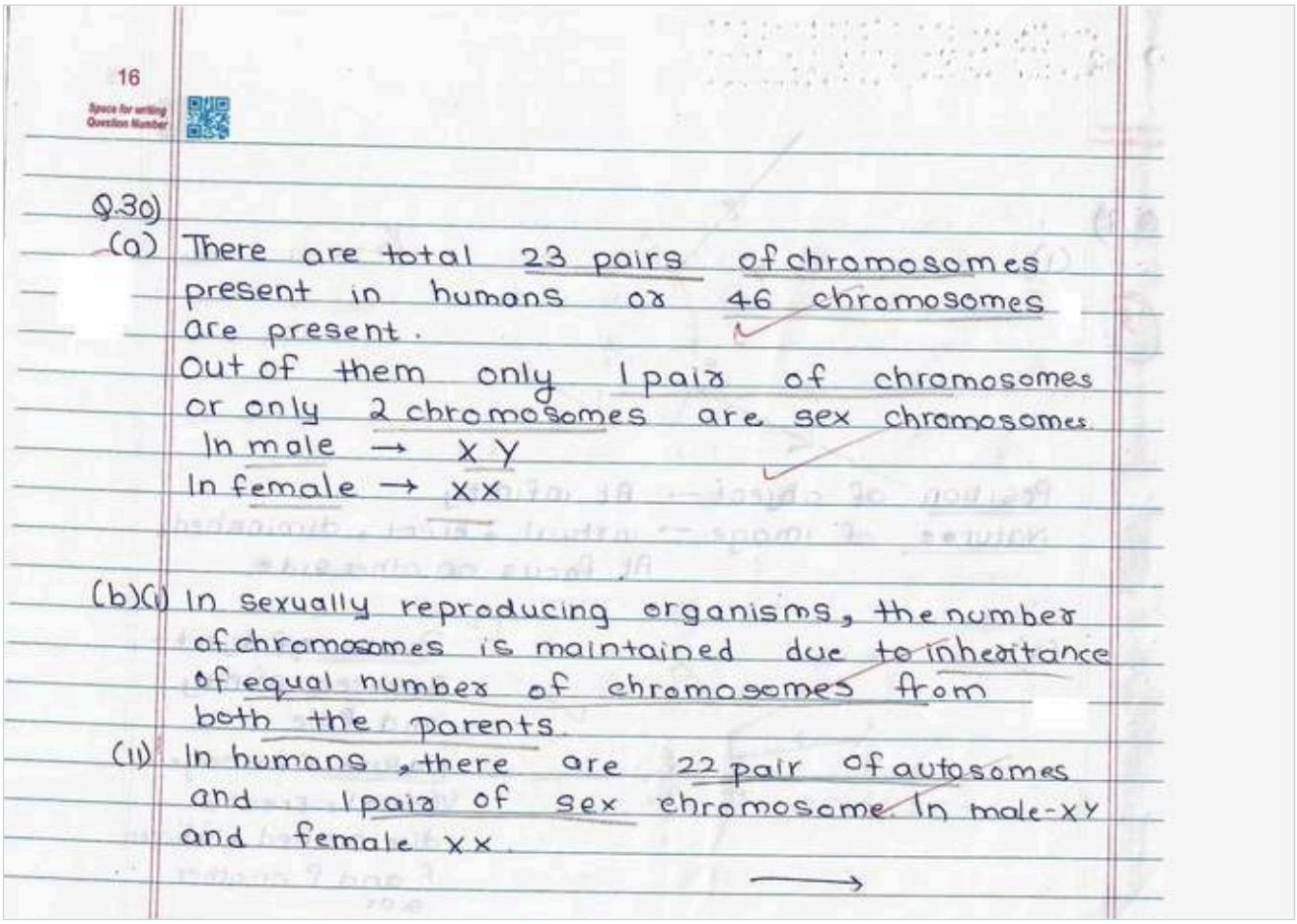
(b) Away from the normal (light goes from an optically denser to an optically rarer medium). **(½)**

(c) $n_{dw} = n_d/n_w = 2.42/1.33 \approx 1.82$ **(1)**

Total 5 marks

END OF SOLUTIONS

Real CBSE topper's answer · Board Exam 2025 · similar to Q.11 of this paper (sex determination in human beings)



16

Q.30)

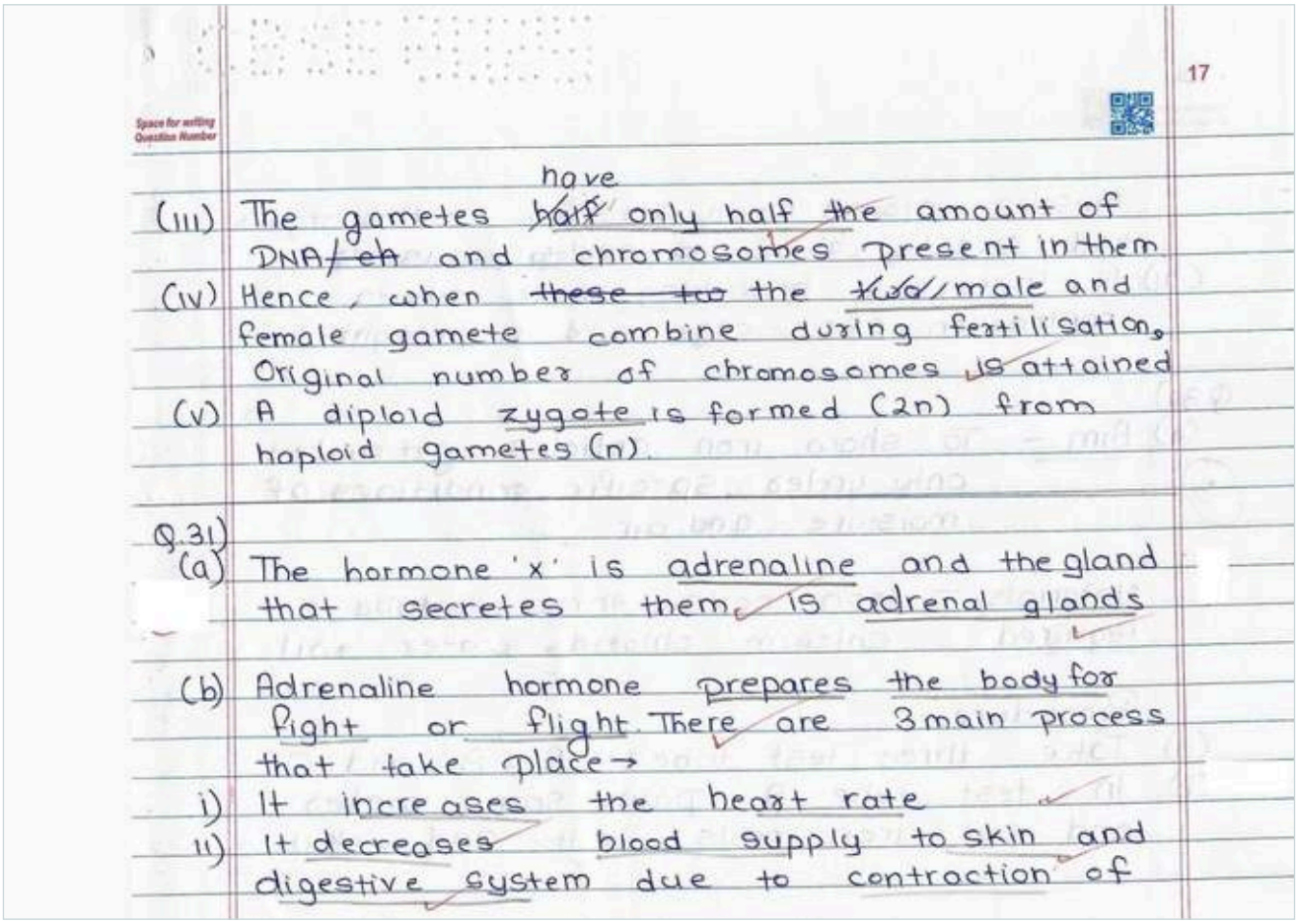
(a) There are total 23 pairs of chromosomes present in humans or 46 chromosomes are present.
Out of them only 1 pair of chromosomes or only 2 chromosomes are sex chromosomes.
In male → XY
In female → XX

(b)(i) In sexually reproducing organisms, the number of chromosomes is maintained due to inheritance of equal number of chromosomes from both the parents.

(ii) In humans, there are 22 pair of autosomes and 1 pair of sex chromosome. In male-XY and female XX.

→

Real CBSE topper's answer · Board Exam 2025 · similar to Q.11 and Q.13 of this paper (sex determination; hormones and glands)



17

(iii) The gametes ~~have~~ only half ~~the~~ amount of DNA/~~ch~~ and chromosomes present in them.

(iv) Hence, when ~~these too~~ the ~~two~~/male and female gamete combine during fertilisation, original number of chromosomes is attained

(v) A diploid zygote is formed (2n) from haploid gametes (n).

Q.31)

(a) The hormone 'x' is adrenaline and the gland that secretes them, is adrenal glands

(b) Adrenaline hormone prepares the body for fight or flight. There are 3 main process that take place →

i) It increases the heart rate

ii) It decreases blood supply to skin and digestive system due to contraction of

Real CBSE topper's answer · Board Exam 2024 · similar to Q.28 of this paper (heating copper powder in air)

18. (D) A is false but R is true

19. (B) Both Assertion (A) and Reason (R) is correct, but Reason (R) is NOT correct explanation of Assertion (A).

20. (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is NOT correct explanation of Assertion (A).

Section B

21. (a)
* Copper (II) oxide is formed. Its colour is Black.

* $2Cu + O_2 \xrightarrow{Heat} 2CuO$

22. (1) The intermolecular forces of attraction between carbon compounds is weak. Due to this, they have low melting & boiling point.

(2) The carbon compounds do not contain ions and they form covalent bonds. As they don't conduct electricity.

Real CBSE topper's answer · Board Exam 2025 · similar to Q.39 of this paper (refraction through a glass slab)

22

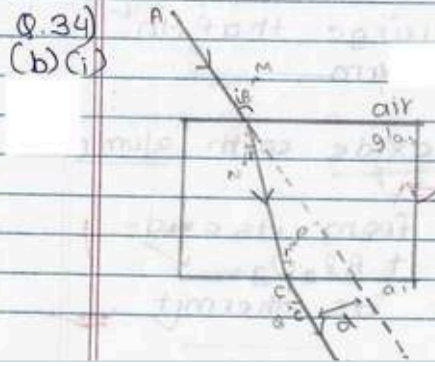
(b) Metals high up in the reactivity series cannot be obtained by reduction of ~~metal~~ oxides by carbon as –

(i) These metals (K, Na, Ca) are highly reactive

(ii) They have more affinity for oxygen rather than carbon.

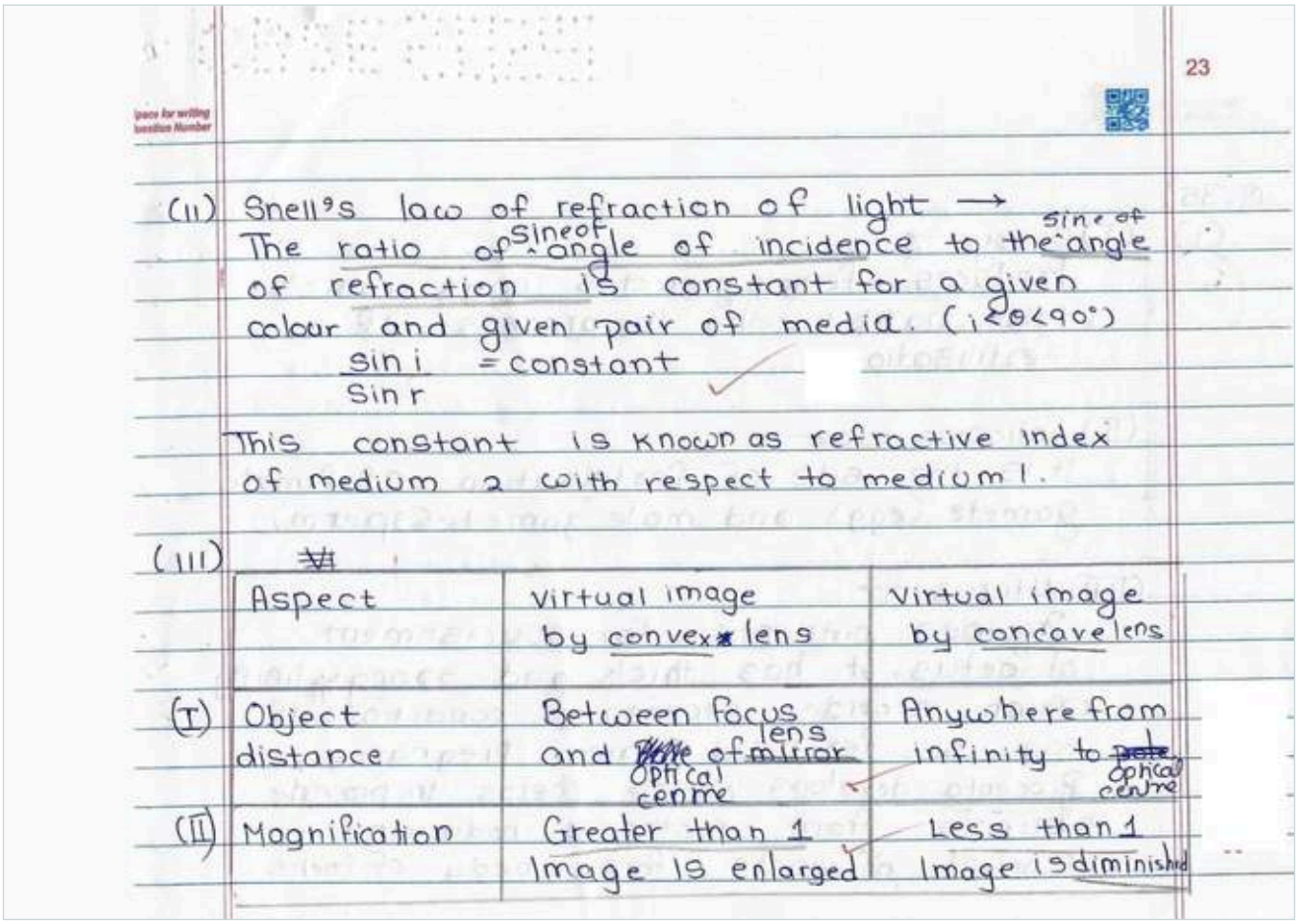
Section - D

Q.34
(b)(i)



Here,
AB – ~~re~~ incident ray
∠ABM → angle of incidence
∠NBC → angle of refraction
BC → refracted ray
∠QCP → angle of emergence
CP → emergent ray
d → lateral displacement

Real CBSE topper's answer · Board Exam 2025 · similar to Q.39 of this paper (the two laws of refraction)



23

(II) Snell's law of refraction of light → The ratio of sine of angle of incidence to the sine of angle of refraction is constant for a given colour and given pair of media. ($i \geq 0 < 90°$)

$$\frac{\sin i}{\sin r} = \text{constant}$$

This constant is known as refractive index of medium 2 with respect to medium 1.

(III)

Aspect	Virtual image by convex lens	Virtual image by concave lens
(I) Object distance	Between focus and lens / optical centre	Anywhere from infinity to optical centre
(II) Magnification	Greater than 1 Image is enlarged	Less than 1 Image is diminished

What to learn from this topper's answer

1. Snell's law is written as a ratio, sin i / sin r = constant, and the constant is named: the refractive index of the second medium with respect to the first.
2. The glass-slab diagram carries a key for every ray and angle: incident ray, angle of incidence, refracted ray, emergent ray and lateral displacement.
3. The convex and concave lens comparison is set out as a neat table, aspect by aspect, which is quick to read and easy to mark.
4. The sex-determination answer gives the numbers first (23 pairs, one pair of sex chromosomes) and then the XX and XY combinations.

SECTION

PRACTICE PAPERS

Sample Papers 11 to 15

Five full papers for timed practice in exam conditions. Answers to all MCQ and Assertion-Reason questions are given in the Answer Key at the end of the book, along with a score tracker for all 15 papers.

Unsolved, latest CBSE pattern: write each paper in 3 hours

SAMPLE PAPER 11

SCIENCE · CODE NO. 086

Sample Question Paper 11

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 During a 400 m race, Ishaan felt a painful cramp in his leg muscles. Which substance had accumulated in his muscle cells, and how was it formed? [1]

A. Ethanol – breakdown of pyruvate in the absence of oxygen
B. Lactic acid – breakdown of pyruvate when oxygen is in short supply
C. Carbon dioxide – breakdown of pyruvate in the mitochondria
D. Glycogen – conversion of excess glucose in the muscles

2 A gardener sprays a dilute solution of a plant hormone on a dwarf variety of a plant and notices that its stems become noticeably longer within a few weeks. The hormone sprayed is most likely [1]

A. abscisic acid **B.** cytokinin
C. gibberellin **D.** adrenaline

3 Which of the following organisms is correctly matched with its mode of reproduction? [1]

A. *Planaria* – budding
B. *Plasmodium* – binary fission
C. *Rhizopus* – fragmentation
D. *Bryophyllum* – vegetative propagation through buds on leaf margins

4 Pure tall pea plants (TT) were crossed with pure dwarf pea plants (tt). The F_1 plants were self-pollinated and 800 plants were obtained in the F_2 generation. How many of the F_2 plants are expected to be tall but heterozygous? [1]

A. 200 **B.** 400
C. 600 **D.** 800

5 Study the food chain given below. [1]

If the grass in this food chain traps 20,000 J of solar energy as food, how much energy will be available to the snake?

A. 2000 J **B.** 200 J
C. 20 J **D.** 2 J

For visually impaired students

In the food chain Grass → Grasshopper → Frog → Snake → Hawk, if the grass traps 20,000 J of solar energy as food, how much energy will be available to the snake?

A. 2000 J **B.** 200 J
C. 20 J **D.** 2 J

6 Which of the following shows the correct path of blood in the human body? **1**

A. Right atrium → right ventricle → pulmonary artery → lungs
B. Left atrium → left ventricle → pulmonary vein → lungs
C. Right ventricle → right atrium → aorta → body organs
D. Lungs → pulmonary artery → left atrium → left ventricle

7 A news report states that the ozone hole over Antarctica has been slowly shrinking over the last few decades. This improvement is mainly the result of **1**

A. an increase in forest cover all over the world
B. restrictions on the manufacture of chlorofluorocarbons (CFCs)
C. a reduction in the use of plastic carry bags
D. the use of biodegradable packaging material

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8 **Assertion (A):** The growth of the pollen tube towards the ovule is an example of chemotropism. **1**
Reason (R): Plant parts can show directional movement in response to chemical stimuli.

9 **Assertion (A):** The use of condoms can help prevent the transmission of sexually transmitted diseases. **1**
Reason (R): Condoms are a hormonal method of contraception that change the hormonal balance of the body.

10 Mrs. Sharma has two daughters. Some of her relatives blame her for not having a son. **2**

A. Explain, with the help of the sex chromosomes involved, why this blame is not justified.
B. What is the probability that her next child will be a son? Give reason.

11 ***Students to attempt either option A or B.*** **2**

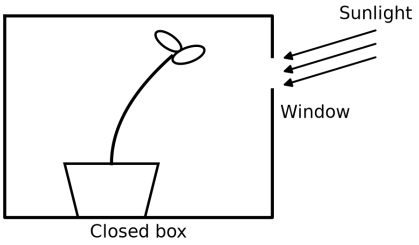
A. The heart of birds and mammals has four chambers, whereas the heart of amphibians has three chambers. Explain why the complete separation of the right side and the left side of the heart is useful to birds and mammals.

OR

B. **I.** Name the two kinds of conducting elements of xylem that transport water in plants.

II. How does transpiration help in the upward movement of water from the roots to the leaves?

12 A potted seedling is kept inside a closed box that has a small window on one side through which sunlight enters (see figure). After a week, the stem is seen bending towards the window. **2**



A. Name this response and the plant hormone responsible for it.

B. Explain how this hormone makes the stem bend towards light.

For visually impaired students

A potted seedling is kept inside a closed box with a small window on one side through which sunlight enters. After a week the stem bends towards the window.

A. Name this response and the plant hormone responsible for it.

B. Explain how this hormone makes the stem bend towards light.

13 Riya placed a few pollen grains of a flower in a drop of sugar solution on a glass slide. After some time, she observed tube-like outgrowths from the pollen grains under a microscope. **3**

A. What does the pollen tube carry, and up to which part of the flower does it grow?

B. After fertilisation, which parts of the flower develop into the seed and the fruit?

C. Distinguish between self-pollination and cross-pollination.

14 ***Students to attempt either option A or B.*** **3**

A. The table shows the amount of some substances (in g per 100 mL) in the blood plasma entering a nephron, in the filtrate inside Bowman's capsule and in the urine of a healthy person.

Substance	Blood plasma	Filtrate in Bowman's capsule	Urine
Proteins	8.0	0.0	0.0
Glucose	0.10	0.10	0.0
Urea	0.03	0.03	2.0
Salts	0.72	0.72	1.5

I. Which substance is present in the blood plasma but absent in the filtrate? Why?

II. Glucose is present in the filtrate but absent in the urine. Explain.

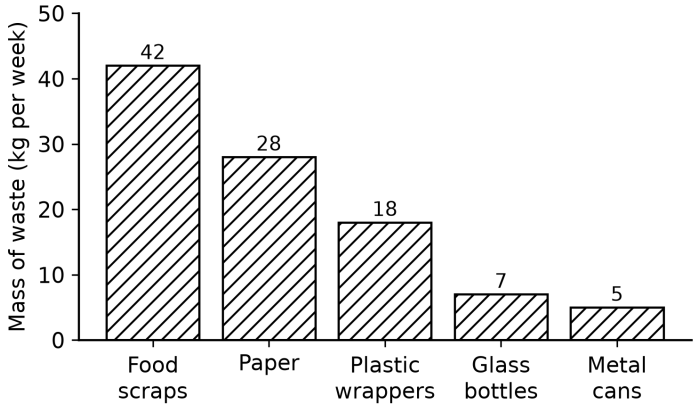
III. Why is the concentration of urea much higher in the urine than in the filtrate?

OR

B. Plants do not have a special excretory system like animals. Describe three different ways in which plants get rid of their excretory products.

15 **4**

The eco-club of a school weighed the waste collected from the school canteen and classrooms over one week. Their results are shown in the bar graph.



A. Which two categories of waste shown in the graph are biodegradable?

B. Why do plastic wrappers remain in the environment for a very long time?

Attempt either sub-part C or D.

C. The eco-club wants to halve the amount of waste sent from the school to the landfill. Suggest two practical steps they could take and explain how each step helps.

OR

D. Some of the food scraps came from vegetables sprayed with a non-biodegradable pesticide. Explain how this pesticide can enter the human body and why its concentration is the highest in human beings.

For visually impaired students

The eco-club of a school weighed the waste collected from the school over one week: food scraps 42 kg, paper 28 kg, plastic wrappers 18 kg, glass bottles 7 kg and metal cans 5 kg.

A. Which two of these categories of waste are biodegradable?

B. Why do plastic wrappers remain in the environment for a very long time?

Attempt either sub-part C or D.

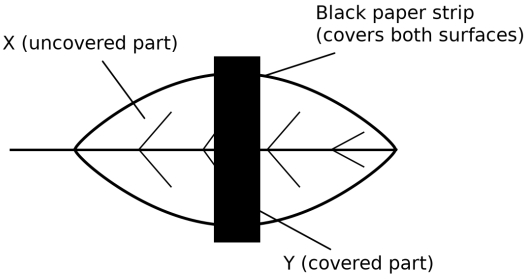
C. The eco-club wants to halve the amount of waste sent to the landfill. Suggest two practical steps they could take and explain how each step helps.

OR

D. Some of the food scraps came from vegetables sprayed with a non-biodegradable pesticide. Explain how this pesticide can enter the human body and why its concentration is the highest in human beings.

16 ***Students to attempt either option A or B.*** **5**

A. **I.** Aman kept a potted plant in a dark room for 48 hours. He then covered a part of one leaf with a strip of black paper on both surfaces (see figure) and placed the plant in sunlight for six hours. Finally, he plucked the leaf and tested it for starch with iodine solution.



(a) Why was the plant kept in the dark for 48 hours before the experiment?

(b) What colour will the parts X and Y show after the iodine test? What do you conclude from this?

(c) Write a balanced chemical equation for photosynthesis.

For visually impaired students

For (b): A part of a leaf of a destarched plant was covered with black paper and the plant was kept in sunlight. What colour will the uncovered part and the covered part show after the iodine test? What do you conclude from this?

II. **(a)** Desert plants keep their stomata closed during the day. How are they still able to carry out photosynthesis?

(b) State one advantage to desert plants of keeping their stomata closed during the day.

OR

B. **I.** Trace the changes that a bite of chapati smeared with ghee (containing starch, protein and fat) undergoes as it passes from the mouth to the small intestine. Name the digestive juice or enzyme acting at each stage.

II. **(a)** Why is the small intestine of a herbivore such as a cow longer than that of a carnivore such as a tiger?

(b) How are the villi of the small intestine suited for the absorption of digested food?

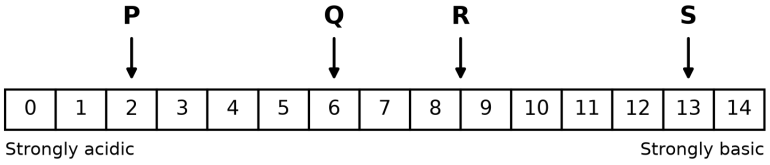
SECTION B Chemistry

17 Dhruv dipped a clean copper wire in a colourless solution of silver nitrate. After some time, the solution turned bluish and a shiny greyish-white deposit appeared on the wire. This reaction is an example of a **1**

A. combination reaction **B.** decomposition reaction

C. displacement reaction **D.** double displacement reaction

18 Four samples P, Q, R and S were tested with universal indicator. Their positions on the pH scale are shown below. **1**



A honey-bee sting injects an acidic liquid into the skin. Which sample is the most suitable to apply on the sting to get relief?

A. P **B.** Q

C. R **D.** S

For visually impaired students

Four samples P, Q, R and S have pH values 2, 6, 8.5 and 13 respectively. A honey-bee sting injects an acidic liquid into the skin. Which sample is the most suitable to apply on the sting to get relief?

A. P **B.** Q

C. R **D.** S

19 An element X forms an oxide with the formula X_2O_3. This oxide reacts with dilute hydrochloric acid as well as with sodium hydroxide solution to form a salt and water. The element X is **1**

A. aluminium **B.** sodium

C. magnesium **D.** sulphur

20 The correct IUPAC name of the compound $CH_3–CH_2–CHO$ is **1**

A. propanone **B.** propanol
C. propanoic acid **D.** propanal

21 Which of the following is an endothermic process? **1**

A. Burning of LPG in a kitchen stove
B. Decomposition of calcium carbonate into calcium oxide and carbon dioxide
C. Reaction of quicklime with water
D. Respiration in the cells of the body

22 Plaster of Paris is always stored in an airtight, moisture-proof container because it **1**

A. absorbs carbon dioxide from air and changes into chalk
B. reacts with moisture and sets into a hard mass of gypsum
C. loses all its water of crystallisation and changes into quicklime
D. turns acidic on exposure to air

23 Which of the following properties is NOT shown by ionic compounds? **1**

A. High melting and boiling points
B. Conduction of electricity in the molten state
C. Hard and brittle nature
D. Solubility in kerosene and petrol

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Soaps form a scum (curdy white precipitate) when used with hard water. **Reason (R):** The calcium and magnesium salts of the long-chain carboxylic acids present in soap are insoluble in water. **1**

25 A compound X, with molecular formula C_2H_6O, is the main ingredient of many hand sanitisers. When X is heated with alkaline potassium permanganate solution, it is converted into a compound Y with molecular formula $C_2H_4O_2$. **2**

A. Identify X and Y and write the chemical equation for the conversion.
B. Why is alkaline potassium permanganate called an oxidising agent in this reaction?

26 ***Students to attempt either option A or B.*** **3**

A. A white powder X is used as an antacid and in soda-acid fire extinguishers. On heating, X gives a solid Y along with carbon dioxide and water. Y is used in the glass and paper industries. When Y is recrystallised from water, it forms Z, which is used for removing permanent hardness of water.

I. Identify X, Y and Z by writing their chemical names and formulae.
II. Write the balanced chemical equation for the action of heat on X.

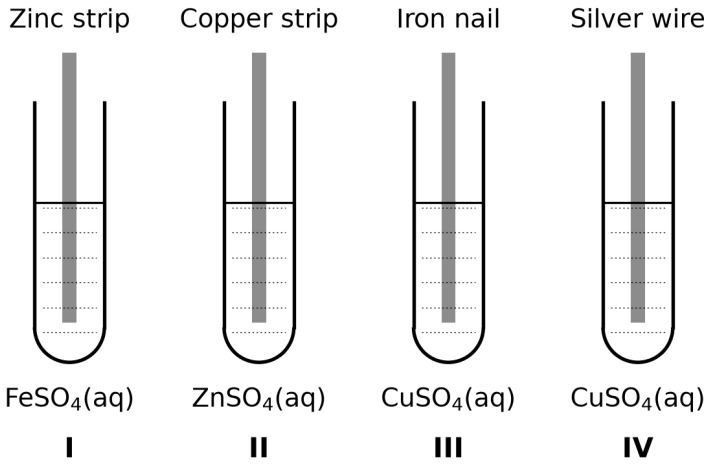
OR

B. When electricity is passed through an aqueous solution of sodium chloride (brine), it decomposes to form three useful products.

I. Why is this process called the chlor-alkali process?
II. Write the balanced chemical equation for the process.

III. Name the gas produced at the anode and give one use of it.

27 Ananya set up four test tubes, I to IV, each containing a clean metal strip in a salt solution, as shown in the figure. **3**



A. In which test tubes will a chemical change be observed?

B. Write the balanced chemical equation for the reaction in test tube III and state the change in colour of the solution.

C. On the basis of this experiment, arrange zinc, iron, copper and silver in decreasing order of reactivity.

For visually impaired students

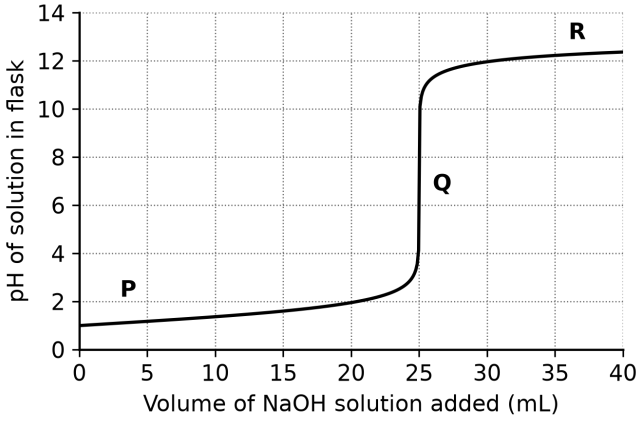
Ananya set up four test tubes: I – zinc strip in iron(II) sulphate solution; II – copper strip in zinc sulphate solution; III – iron nail in copper sulphate solution; IV – silver wire in copper sulphate solution.

A. In which test tubes will a chemical change be observed?

B. Write the balanced chemical equation for the reaction in test tube III and state the change in colour of the solution.

C. On the basis of this experiment, arrange zinc, iron, copper and silver in decreasing order of reactivity.

28 Kavya took 25 mL of dilute hydrochloric acid in a conical flask and added a few drops of phenolphthalein. She then added sodium hydroxide solution from a burette in small amounts and recorded the pH of the solution after each addition using a pH meter. Her results are shown in the graph. **4**



A. What is the pH of the solution at point P? What does it tell you about the solution?

B. Name the salt present in the flask at point Q and state the pH of its solution.

Attempt either sub-part C or D.

C. Phenolphthalein is colourless in acidic and neutral solutions and pink in basic solutions. State the colour of the solution in the flask at P, Q and R, and write the balanced chemical equation for the reaction taking place.

OR

D. If Kavya repeats the experiment with 50 mL of the same acid, what volume of the same sodium hydroxide solution will be needed to make the solution neutral? Justify your answer. Also state whether the reaction is exothermic or endothermic.

For visually impaired students

Kavya took 25 mL of dilute hydrochloric acid in a flask with a few drops of phenolphthalein and added sodium hydroxide solution in small amounts. The pH was 1 at the start (P), became exactly 7 when 25 mL of the base had been added (Q), and was about 12 when 40 mL had been added (R).

A. What does the pH at the start (P) tell you about the solution?

B. Name the salt present in the flask at point Q and state the pH of its solution.

Attempt either sub-part C or D.

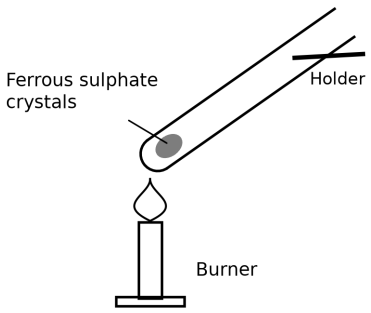
C. Phenolphthalein is colourless in acidic and neutral solutions and pink in basic solutions. State the colour of the solution at P, Q and R, and write the balanced chemical equation for the reaction.

OR

D. If Kavya repeats the experiment with 50 mL of the same acid, what volume of the same base will be needed to make the solution neutral? Justify. Also state whether the reaction is exothermic or endothermic.

29 ***Students to attempt either option A or B.*** **5**

A. **I.** Shalini heated a few crystals of ferrous sulphate in a dry boiling tube as shown in the figure.



(a) State two observations she would make.

(b) Write the balanced chemical equation for the reaction and name the type of reaction.

(c) Is this reaction exothermic or endothermic? Give reason.

For visually impaired students

Shalini heated a few green crystals of ferrous sulphate in a dry boiling tube. (a) State two observations she would make. (b) Write the balanced chemical equation and name the type of reaction. (c) Is this reaction exothermic or endothermic? Give reason.

II. For the reaction $ZnO + C \rightarrow Zn + CO$, a student says, “Zinc oxide is oxidised in this reaction.” Do you agree? Justify your answer by identifying the substance oxidised,

the substance reduced, the oxidising agent and the reducing agent.

OR

B. **I.** Neha mixed the following pairs of aqueous solutions in three test tubes:
Test tube 1: lead nitrate + potassium iodide
Test tube 2: barium chloride + sodium sulphate
Test tube 3: sodium hydroxide + hydrochloric acid

(a) In which test tubes will a precipitate be formed? Name each precipitate and state its colour.

(b) Write the balanced chemical equation for the reaction in test tube 2.

(c) Why are all three reactions classified as double displacement reactions?

II. White silver chloride kept in a china dish turns grey when left in sunlight.

(a) Write the balanced chemical equation for this change and name the type of reaction.

(b) State one practical application of this reaction.

SECTION C Physics

30 A dentist uses a small mirror that gives an enlarged and erect image of a patient's tooth. Which mirror is used, and where is the tooth placed? **1**

A. Convex mirror, with the tooth placed anywhere in front of it
B. Concave mirror, with the tooth placed between its pole and principal focus
C. Concave mirror, with the tooth placed at its centre of curvature
D. Plane mirror, with the tooth placed very close to it

31 Three resistors of 6 Ω each are available. The maximum and the minimum resistance that can be obtained by combining all three of them are respectively **1**

A. 18 Ω and 3 Ω **B.** 12 Ω and 4 Ω
C. 18 Ω and 2 Ω **D.** 9 Ω and 2 Ω

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** A concave lens always forms a virtual image of a real object.
Reason (R): Rays from the object diverge after refraction through a concave lens and only appear to meet on the same side of the lens as the object. **1**

33 Vikram allowed a narrow beam of white light to pass through a glass prism and obtained a band of colours on a white screen. **2**

A. Why do the different colours of white light bend through different angles in the prism?

B. He now places a second identical prism in an inverted position next to the first prism. What will he observe on the screen? Give reason.

34 ***Students to attempt either option A or B.*** **2**

A. An optician's prescription for a student reads "−2.5 D" for both eyes.

I. What type of lens is prescribed? Calculate its focal length.

II. Name the defect of vision that this lens corrects.

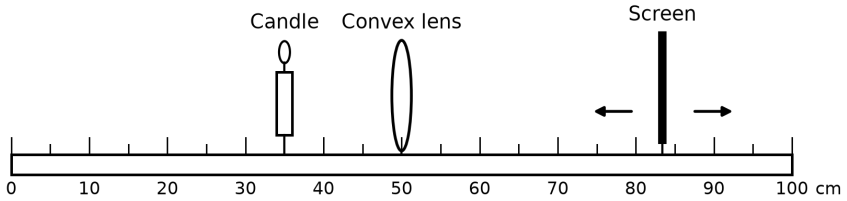
OR

B. The refractive index of a type of glass is 1.5.

I. Calculate the speed of light in this glass. (Speed of light in vacuum = 3×10^8 m s^{-1})

II. A ray of light passes from this glass into water (refractive index 1.33). Will it bend towards or away from the normal? Give reason.

35 A student set up a convex lens of focal length 10 cm on an optical bench as shown. A candle flame 4 cm tall is placed at the 35 cm mark and the lens at the 50 cm mark. The screen can be moved along the bench. **3**



A. At which mark on the bench should the screen be placed to obtain a sharp image of the flame?

B. Find the size of the image and state its nature.

C. The candle is now moved to the 40 cm mark. Will the student be able to obtain the image on the screen? Give reason.

For visually impaired students

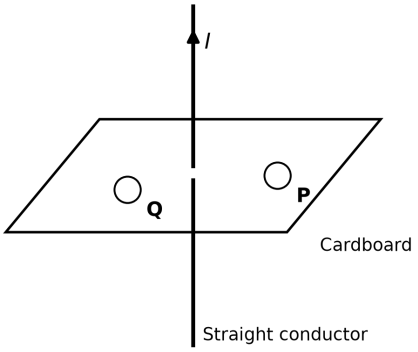
A candle flame 4 cm tall is placed 15 cm in front of a convex lens of focal length 10 cm.

A. At what distance from the lens should a screen be placed to obtain a sharp image?

B. Find the size of the image and state its nature.

C. If the candle is moved to 10 cm from the lens, will an image be obtained on the screen? Give reason.

36 A straight conductor passes vertically through a horizontal sheet of cardboard and a current *I* flows upward through it, as shown. Two small compasses are placed on the cardboard at P (2 cm from the wire) and Q (5 cm from the wire). **3**



A. Describe the pattern of the magnetic field lines on the cardboard and state their direction as seen from above.

B. At which point, P or Q, will the magnetic field be stronger? Give reason.

C. State two changes that will be observed in the compass needles if the current is reversed and made stronger.

For visually impaired students

A straight conductor passes vertically through a horizontal cardboard and a current flows upward through it. Two small compasses are placed at P (2 cm from the wire) and Q (5 cm from the wire).

A. Describe the pattern of the magnetic field lines on the cardboard and state their direction as seen from above.

B. At which point, P or Q, will the magnetic field be stronger? Give reason.

C. State two changes that will be observed in the compass needles if the current is reversed and made stronger.

37 ***Students to attempt either option A or B.*** **3**

A. Convex mirrors are fitted as rear-view mirrors in vehicles. A bus is 6 m behind a car whose rear-view mirror has a radius of curvature of 4 m.

I. Why is a convex mirror preferred as a rear-view mirror?

II. Find the position of the image of the bus.

III. Find the magnification and state the nature of the image.

OR

B. A student obtains a sharp image of a distant window on a screen placed 12 cm in front of a concave mirror. She then places a candle 18 cm in front of the same mirror.

I. What is the focal length of the mirror?

II. Find the position of the image of the candle.

III. Find the magnification and state the nature of the image.

38 **4**

> Mr. Verma's house has a 220 V electricity supply. The kitchen circuit is protected by a 10 A fuse. He uses an electric kettle rated 1540 W and a toaster rated 880 W on this circuit. Both appliances have metal bodies and three-pin plugs.

A. Why is the metal body of the kettle connected to the earth wire?

B. In which wire of the circuit (live, neutral or earth) is the fuse connected? Give reason.

Attempt either sub-part C or D.

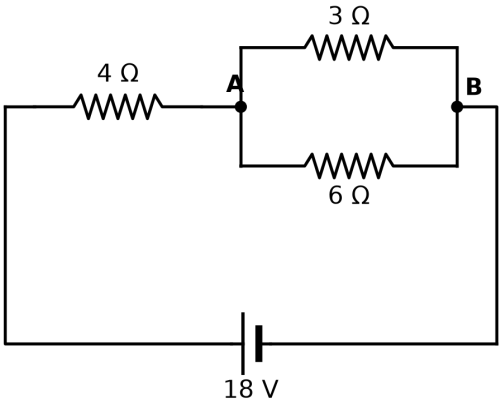
C. Calculate the total current drawn when both appliances are switched on together. Will the fuse blow? Justify your answer.

OR

D. Distinguish between short-circuiting and overloading. Give one cause of each.

39 ***Students to attempt either option A or B.*** **5**

A. **I.** Study the circuit shown in the figure.



(a) Calculate the effective resistance of the circuit.

(b) Calculate the current drawn from the battery.

(c) Find the potential difference between A and B and the current through the 6 Ω resistor.

For visually impaired students

A 4 Ω resistor is connected in series with a parallel combination of 3 Ω and 6 Ω resistors across an 18 V battery. (a) Calculate the effective resistance. (b) Calculate the current drawn from the battery. (c) Find the potential difference across the parallel combination and the current through the 6 Ω resistor.

II. Give two reasons why household appliances are connected in parallel and not in series.

OR

B. I. A wire of length 2 m and area of cross-section 0.55 mm^2 is made of an alloy of resistivity 1.1×10^{-6} Ω m.

(a) Calculate the resistance of the wire.

(b) The wire is stretched uniformly to twice its original length. Calculate its new resistance.

(c) Will the resistivity of the wire change on stretching? Justify your answer.

II. (a) Why are alloys, rather than pure metals, used in the heating elements of electric irons and toasters?

(b) Why is tungsten used for making the filaments of electric bulbs?

END OF QUESTION PAPER

Quick revision: Acids, Bases and Salts

1. Acids turn blue litmus red; bases turn red litmus blue.
2. Acid + base → salt + water (neutralisation).
3. Acid + metal carbonate → salt + water + CO_2; CO_2 turns lime water milky.
4. pH scale runs from 0 to 14; our body works between pH 7.0 and 7.8.
5. Tooth decay starts below pH 5.5.
6. Bleaching powder: $CaOCl_2$. Common salt is the raw material for NaOH, bleaching powder, baking soda and washing soda.
7. Water of crystallisation: blue $CuSO_4 \cdot 5H_2O$ turns white on heating.

Nitesh Bhaiya ki tip: practice paper khud time lagake karo, phir answer key se check karo.

SAMPLE PAPER 12

SCIENCE · CODE NO. 086

Sample Question Paper 12

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed	3 Hours
Maximum Marks	80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

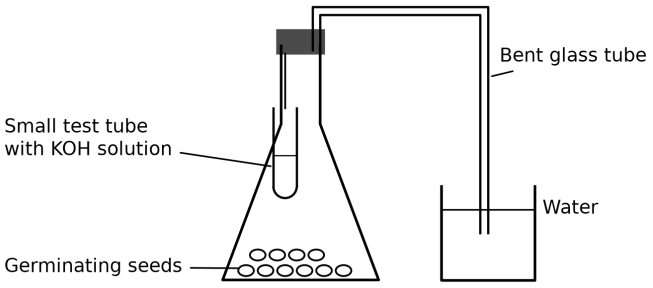
Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 Bile juice secreted by the liver helps in the digestion of fats by — **1**

A. converting fats into fatty acids and glycerol
B. breaking large fat globules into smaller droplets
C. absorbing digested fats into the blood
D. making the food acidic for the action of pancreatic enzymes

2 Arjun set up the apparatus shown in the figure. After about an hour, he noticed that water had risen in the bent glass tube. — **1**



Which of the following explains this observation?

A. The germinating seeds released oxygen, which dissolved in the water.
B. The seeds absorbed water from the bent glass tube.
C. The KOH solution released water vapour, which condensed in the flask.
D. The KOH solution absorbed the carbon dioxide released by the seeds, lowering the pressure in the flask.

For visually impaired students

Germinating seeds are kept in a closed flask along with a small test tube of potassium hydroxide (KOH) solution. A bent glass tube connects the flask to a beaker of water. After an

hour, water rises in the bent tube. Which of the following explains this observation?

A. The germinating seeds released oxygen, which dissolved in the water.
B. The seeds absorbed water from the bent glass tube.
C. The KOH solution released water vapour, which condensed in the flask.
D. The KOH solution absorbed the carbon dioxide released by the seeds, lowering the pressure in the flask.

3 Which of the following actions is controlled by the medulla in the hind-brain? **1**

A. Vomiting **B.** Riding a bicycle
C. Writing a letter **D.** Solving a puzzle

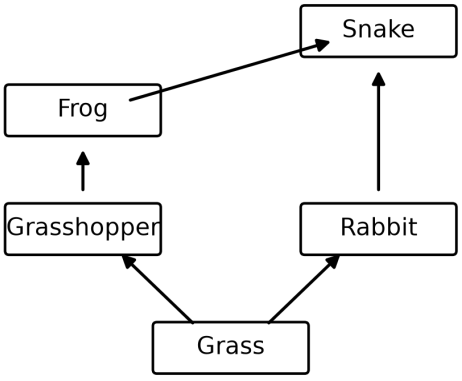
4 A doctor advises a person with a swollen neck to use iodised salt in food. Iodine is essential for the synthesis of the hormone **1**

A. insulin **B.** adrenaline
C. thyroxin **D.** growth hormone

5 In human females, the fusion of the egg and the sperm normally takes place in the **1**

A. uterus **B.** oviduct (fallopian tube)
C. ovary **D.** cervix

6 Study the food web shown below. **1**



Which organism in this food web feeds at more than one trophic level?

A. Grasshopper **B.** Rabbit
C. Frog **D.** Snake

For visually impaired students

In a food web, grasshoppers and rabbits feed on grass, frogs feed on grasshoppers, and snakes feed on both frogs and rabbits. Which organism feeds at more than one trophic level?

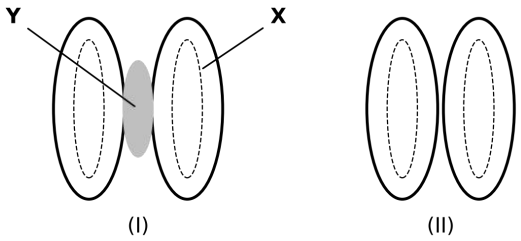
A. Grasshopper **B.** Rabbit
C. Frog **D.** Snake

7 Which of the following correctly describes how an artificial kidney (haemodialysis) differs from a natural kidney? **1**

A. No reabsorption of useful substances takes place in it.
B. It removes only excess water and not urea.
C. It filters blood proteins into the dialysing fluid.
D. It produces urea from excess amino acids.

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** The walls of the ventricles of the human heart are thicker than the walls of the atria. **Reason (R):** The ventricles have to pump blood out of the heart to various organs under high pressure.	**1**
9	**Assertion (A):** Decomposers are essential for the replenishment of nutrients in the soil. **Reason (R):** Decomposers convert simple inorganic substances into complex organic substances.	**1**
10	**A.** Why is the flow of energy in an ecosystem said to be unidirectional? **B.** Why do food chains in nature generally consist of only three or four trophic levels?	**2**
11	***Students to attempt either option A or B.*** **A.** The leaves of the touch-me-not plant fold up when touched, while the tendrils of a pea plant coil around a support. Compare these two movements with respect to (I) the stimulus involved and (II) whether growth is involved. **OR** **B.** **I.** What is a synapse? **II.** Explain how a nerve impulse is passed from one neuron to the next across a synapse.	**2**
12	The figure shows two states, (I) and (II), of a stomatal pore in a leaf.  **A.** Name the cells labelled X and the structure labelled Y. **B.** Which state, (I) or (II), is seen when the cells X take in water? Explain how this happens. **For visually impaired students** **A.** Name the cells that surround a stomatal pore and control its opening and closing. **B.** What happens to the stomatal pore when these cells take in water? Explain how this happens.	**2**
13	At a health camp, a doctor explains some methods of family planning: condoms, copper-T, oral contraceptive pills and surgical methods. **A.** Which of these methods also protects against sexually transmitted infections such as HIV-AIDS? How? **B.** How do oral contraceptive pills prevent pregnancy? **C.** Prenatal sex determination is prohibited by law in India. Give one reason for this.	**3**
14	***Students to attempt either option A or B.*** **A.** Rohit's grandfather grows banana plants in his farm without using any seeds, and the new plants are exactly like the parent plants.	**3**

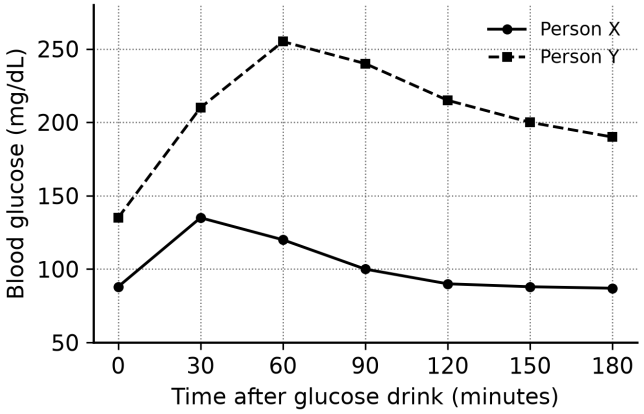
I. Name the method by which banana plants are grown.
II. Give two advantages of this method to the farmer.
III. State one disadvantage of growing all plants of a crop by this method.

OR

B. I. Name the tissue that transports food from the leaves to other parts of a plant.
II. Explain how food is moved through this tissue. Is energy used in this process?
III. How does the direction of movement of food differ from that of water in a plant?

15 **4**

A doctor gave two persons, X and Y, a glucose drink after an overnight fast and measured their blood glucose level every 30 minutes. The results are shown in the graph.



A. Name the hormone that lowers the blood glucose level and the gland that secretes it.
B. Which person, X or Y, is likely to be suffering from diabetes? Give one reason from the graph.

Attempt either sub-part C or D.

C. Explain how the timing and the amount of this hormone released are regulated in person X after the glucose drink.

OR

D. Give two differences between the ways in which the nervous system and the hormonal system bring about coordination in the body.

For visually impaired students

A doctor gave two persons, X and Y, a glucose drink after an overnight fast. In person X, blood glucose rose from 88 mg/dL to 135 mg/dL in 30 minutes and returned to 90 mg/dL within 2 hours. In person Y, it rose from 135 mg/dL to 255 mg/dL at 60 minutes and was still 190 mg/dL after 3 hours.

A. Name the hormone that lowers the blood glucose level and the gland that secretes it.
B. Which person, X or Y, is likely to be suffering from diabetes? Give one reason.

Attempt either sub-part C or D.

C. Explain how the timing and the amount of this hormone released are regulated in person X after the glucose drink.

OR

D. Give two differences between the ways in which the nervous system and the hormonal system bring about coordination in the body.

16 **5**

Students to attempt either option A or B.

A.

I. Mendel crossed a pea plant having round, yellow seeds (RRYY) with a pea plant having wrinkled, green seeds (rryy).

(a) What will be the appearance of the seeds of the F_1 plants?

(b) Write all the types of gametes that the F_1 plants can produce.

(c) The F_1 plants are self-pollinated and 1600 seeds are obtained in the F_2 generation. How many of these seeds are expected to be round and green? Show your working.

II. **(a)** The F_2 generation contains some new combinations (round-green and wrinkled-yellow) not seen in the parents. What does this show about the inheritance of seed shape and seed colour?

(b) Give one reason why Mendel chose garden pea plants for his experiments.

OR

B. I. Ravi says, "The sex of a child is decided by the mother." Do you agree? With the help of a flow chart showing the sex chromosomes of the parents, their gametes and the offspring, explain how the sex of a child is determined in human beings and why there is an equal chance of a boy or a girl being born.

II. **(a)** What is a gene?

(b) Explain, with the example of the height of a plant, how a gene controls a characteristic.

SECTION B Chemistry

17 Steam is passed over red-hot iron to form iron(II, III) oxide and hydrogen gas. Which of the following equations is correctly balanced? **1**

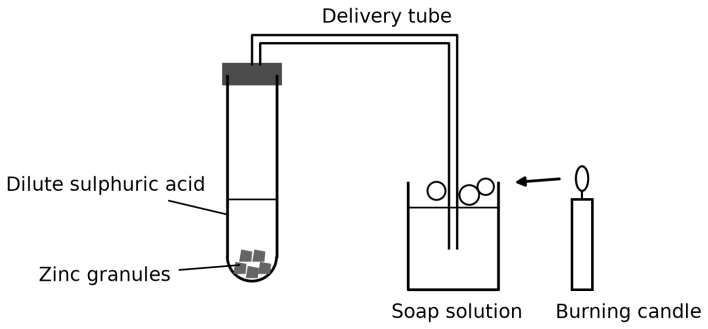
A. $Fe + H_2O \rightarrow Fe_3O_4 + H_2$

B. $3Fe + 2H_2O \rightarrow Fe_3O_4 + 2H_2$

C. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

D. $2Fe + 3H_2O \rightarrow Fe_3O_4 + 3H_2$

18 Sneha set up the apparatus shown below and passed the gas evolved into soap solution. **1**



What will she observe when the burning candle is brought near a gas-filled soap bubble, and which gas is it?

A. The candle is extinguished; carbon dioxide

B. The gas burns with a pop sound; hydrogen

C. The candle burns more brightly; oxygen

D. Nothing happens; water vapour

For visually impaired students

Sneha added zinc granules to dilute sulphuric acid in a test tube and passed the gas evolved into soap solution. What will she observe when a burning candle is brought near a gas-filled soap bubble, and which gas is it?

	A. The candle is extinguished; carbon dioxide **B.** The gas burns with a pop sound; hydrogen **C.** The candle burns more brightly; oxygen **D.** Nothing happens; water vapour	
19	Mercury is obtained from its sulphide ore, cinnabar (HgS), by **A.** electrolysis of the molten ore **B.** reduction of the ore with carbon **C.** roasting the ore in air and then heating the oxide formed **D.** heating the ore with aluminium powder	**1**
20	The total number of covalent bonds in a molecule of propane (C_3H_8) is **A.** 10 **B.** 8 **C.** 11 **D.** 13	**1**
21	Equal amounts of four salts are dissolved separately in distilled water. Which solution will have a pH greater than 7? **A.** Sodium chloride **B.** Ammonium chloride **C.** Sodium sulphate **D.** Sodium carbonate	**1**
22	Silver anklets gradually turn black when exposed to air. The black coating is mainly **A.** silver oxide **B.** silver sulphide **C.** silver nitrate **D.** silver chloride	**1**
23	Equal-sized magnesium ribbons are added to dilute hydrochloric acid and to dilute ethanoic acid of the same concentration. Fizzing is much more vigorous in hydrochloric acid because **A.** hydrochloric acid is a weak acid **B.** ethanoic acid does not contain any hydrogen **C.** hydrochloric acid produces a higher concentration of H^+ ions in water **D.** magnesium does not react with ethanoic acid	**1**

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24	**Assertion (A):** Aluminium is more reactive than iron, yet aluminium articles do not corrode easily in air. **Reason (R):** Aluminium gets coated with a thin, protective layer of aluminium oxide that prevents further reaction.	**1**
25	**A.** Write the balanced chemical equation for the burning of a magnesium ribbon in air and name the type of reaction. **B.** Why is the magnesium ribbon cleaned with sandpaper before it is burnt?	**2**
26	***Students to attempt either option A or B.*** **A.** Bleaching powder is used for disinfecting drinking water. **I.** How is bleaching powder manufactured? Write the balanced chemical equation. **II.** Give two uses of bleaching powder other than disinfecting water. **OR**	**3**

B. Explain the following on the basis of pH:

I. Using toothpaste helps prevent tooth decay.

II. Aquatic animals find it difficult to survive when acid rain flows into a river.

III. Milk of magnesia gives relief from indigestion caused by excess acid in the stomach.

27 3

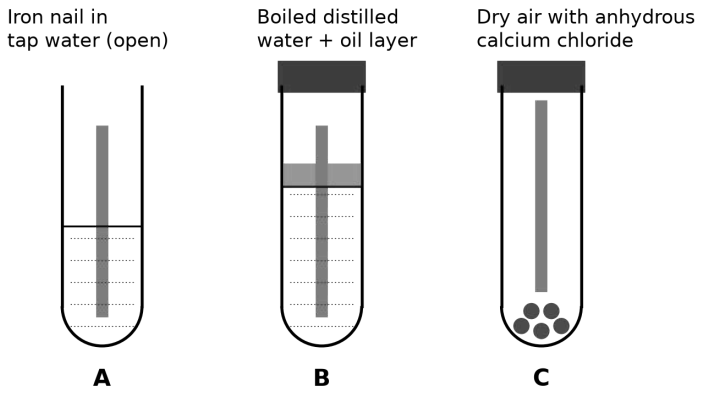
A. Show the formation of sodium oxide (Na_2O) by the transfer of electrons. (Atomic numbers: Na = 11, O = 8)

B. Name the ions present in sodium oxide.

C. Why does solid sodium oxide not conduct electricity, although it conducts electricity in the molten state?

28 4

Meera set up three test tubes A, B and C, each containing a clean iron nail, as shown in the figure, and observed them after one week.



A. In which test tube will the iron nail rust?

B. Why is the water in test tube B boiled, and why is a layer of oil poured over it?

Attempt either sub-part C or D.

C. What is galvanisation? Explain why a galvanised iron article remains protected from rusting even if its zinc coating is broken at a few places.

OR

D. Stainless steel is preferred over pure iron for making kitchen utensils. Name the metals mixed with iron to make stainless steel and give one reason why it is preferred.

For visually impaired students

Meera kept three clean iron nails: A – in tap water in an open test tube; B – in boiled distilled water covered with a layer of oil, in a corked tube; C – in dry air with anhydrous calcium chloride, in a corked tube. She observed them after one week.

A. In which test tube will the iron nail rust?

B. Why is the water in test tube B boiled, and why is a layer of oil poured over it?

Attempt either sub-part C or D.

C. What is galvanisation? Explain why a galvanised iron article remains protected from rusting even if its zinc coating is broken at a few places.

OR

D. Stainless steel is preferred over pure iron for making kitchen utensils. Name the metals mixed with iron to make stainless steel and give one reason why it is preferred.

29 5

Students to attempt either option A or B.

A. **I.** Study the following open-chain hydrocarbons: CH_4, C_2H_4, C_3H_6, C_3H_8, C_4H_8, C_4H_{10}

(a) Which of these belong to the same homologous series as ethene (C_2H_4)?

(b) By how much do the molecular masses of two successive members of a homologous series differ? (Atomic masses: C = 12 u, H = 1 u)

(c) Draw the structural formulae of two isomers of butane (C_4H_{10}).

II. Explain why carbon forms its compounds mainly by covalent bonding. Give two reasons why carbon is able to form a very large number of compounds.

OR

B. I. Ritu warmed 1 mL of ethanol with 1 mL of glacial ethanoic acid and a few drops of concentrated sulphuric acid in a water bath. After five minutes, she poured the mixture into a beaker of water.

(a) What does she observe about the smell of the product? Name the class of compounds to which the product belongs.

(b) Write the chemical equation for the reaction.

(c) What happens when this product is heated with sodium hydroxide solution? Name this reaction.

II. With the help of a labelled diagram of a micelle, explain how soap removes oily dirt from clothes.

SECTION C Physics

30 Which type of mirror is used as a rear-view mirror in vehicles, and why? **1**

A. Concave mirror, because it forms erect and enlarged images

B. Plane mirror, because it forms images of the same size

C. Convex mirror, because it forms erect, diminished images over a wider field of view

D. Concave mirror, because it forms real images

31 An electric heater draws a current of 4 A from a 220 V supply. The electrical energy it consumes in 2 hours is **1**

A. 0.88 kWh **B.** 1.76 kWh

C. 3.52 kWh **D.** 1760 kWh

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** A person suffering from myopia is advised to wear spectacles with concave lenses. **1**
Reason (R): A concave lens of suitable power diverges the incoming rays so that the image of a distant object is formed on the retina.

33 Aditi sees a rainbow in the sky just after a shower in the evening. **2**

A. Name the phenomena of light that together produce a rainbow.

B. In which direction with respect to the Sun should she look to see the rainbow? What role do the raindrops play?

34 ***Students to attempt either option A or B.*** **2**

A. Two bulbs are rated 100 W, 220 V and 60 W, 220 V.

I. Which bulb has the higher resistance? Calculate the resistance of the 100 W bulb.

II. Which bulb will glow brighter when both are connected in parallel to a 220 V supply?

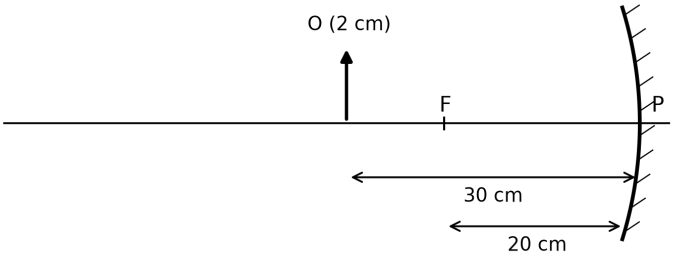
OR

B. An electric iron draws a current of 5 A from a 220 V supply.

I. Calculate the resistance of its heating element.

II. Calculate the heat produced by it in 30 s.

35 An object 2 cm tall is placed 30 cm in front of a concave mirror of focal length 20 cm, as shown in the figure. **3**



A. Find the position of the image.

B. Find the size of the image.

C. State the nature of the image.

For visually impaired students

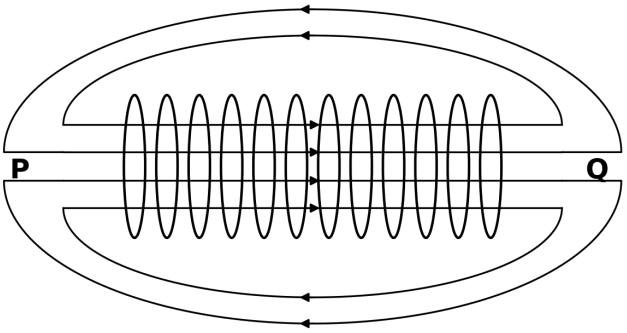
An object 2 cm tall is placed 30 cm in front of a concave mirror of focal length 20 cm.

A. Find the position of the image.

B. Find the size of the image.

C. State the nature of the image.

36 The figure shows the pattern of magnetic field lines of a current-carrying solenoid. **3**



A. What does the pattern of the field lines inside the solenoid indicate?

B. Which end of the solenoid, P or Q, behaves like the north pole of a magnet? Give reason.

C. How can this solenoid be used to make an electromagnet?

For visually impaired students

A long solenoid carries a steady current. Inside it, the magnetic field lines are parallel straight lines directed from end P towards end Q, and outside they form closed loops from Q back to P.

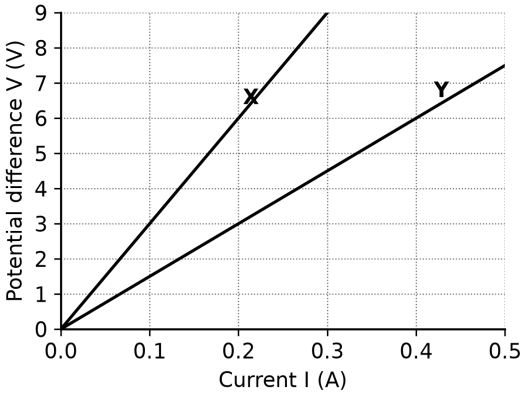
A. What does the pattern of the field lines inside the solenoid indicate?

B. Which end of the solenoid, P or Q, behaves like the north pole of a magnet? Give reason.

C. How can this solenoid be used to make an electromagnet?

37 ***Students to attempt either option A or B.*** **3**

A. The graph shows the variation of potential difference V with current I for two wires X and Y.



I. Find the resistance of X and of Y.

II. If X and Y are made of the same material and have the same length, which wire is thicker? Give reason.

III. Calculate the effective resistance when X and Y are connected in parallel.

For visually impaired students

For a wire X, a potential difference of 6 V drives a current of 0.2 A; for a wire Y, 6 V drives 0.4 A. (I) Find the resistance of X and of Y. (II) If X and Y are made of the same material and have the same length, which wire is thicker? Give reason. (III) Calculate their effective resistance when connected in parallel.

OR

B. A copper wire has a resistance of 10 Ω. It is cut into two equal halves, and the two halves are connected in parallel.

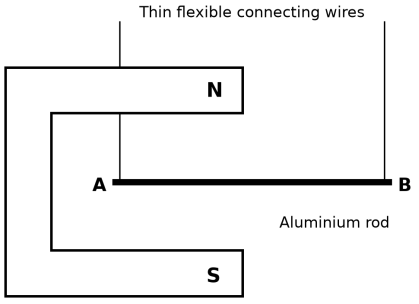
I. Calculate the resistance of the combination.

II. Does the resistivity of copper change when the wire is cut? Justify your answer.

III. Why is copper used for making connecting wires in household circuits?

38 **4**

Neel suspended a light aluminium rod AB horizontally between the poles of a horseshoe magnet using thin flexible wires, and connected the ends of the rod to a battery through a key (figure). When the key was closed and current flowed from A to B, the rod was displaced.



A. Name the rule that gives the direction of the force acting on the rod.

B. What will happen to the displacement of the rod if the direction of the current is reversed?

Attempt either sub-part C or D.

C. State two ways of increasing the displacement of the rod. In which orientation of the rod with respect to the magnetic field will the force on it be zero?

OR

D. Neel's battery supplies direct current. Distinguish between direct current and alternating current. What is the frequency of the AC supply in India? State one advantage of AC over DC.

For visually impaired students

Neel suspended a light aluminium rod horizontally between the poles of a horseshoe magnet and passed a current through it. The rod was displaced.

A. Name the rule that gives the direction of the force acting on the rod.

B. What will happen to the displacement of the rod if the direction of the current is reversed?

Attempt either sub-part C or D.

C. State two ways of increasing the displacement of the rod. In which orientation of the rod with respect to the magnetic field will the force on it be zero?

OR

D. Distinguish between direct current and alternating current. What is the frequency of the AC supply in India? State one advantage of AC over DC.

39 ***Students to attempt either option A or B.*** **5**

A. I. A watchmaker holds a convex lens of focal length 15 cm at a distance of 10 cm from a tiny part of a watch, 3 mm in size.

(a) Find the position of the image.

(b) Find the magnification and the size of the image.

(c) State the nature of the image.

II. Draw a labelled ray diagram to show the formation of the image in this case.

For visually impaired students

Instead of II: State the position in which an object should be placed in front of a convex lens so that the lens works as a magnifying glass, and give the nature of the image formed.

OR

B. I. An object 5 cm tall is placed 30 cm in front of a concave lens of focal length 20 cm.

(a) Find the position of the image.

(b) Find the magnification and the size of the image.

(c) State the nature of the image.

II. A convex lens of power +5 D is placed in contact with a concave lens of power −3 D. Find the power and the focal length of the combination. Will the combination behave as a converging or a diverging lens?

END OF QUESTION PAPER

Exam tip

Write the question number exactly as printed, including the internal choice you attempted.

SAMPLE PAPER 13

SCIENCE · CODE NO. 086

Sample Question Paper 13

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 Rakesh had his gall bladder surgically removed. His doctor advised him to avoid heavy, fatty meals. Which of the following best explains this advice? **1**

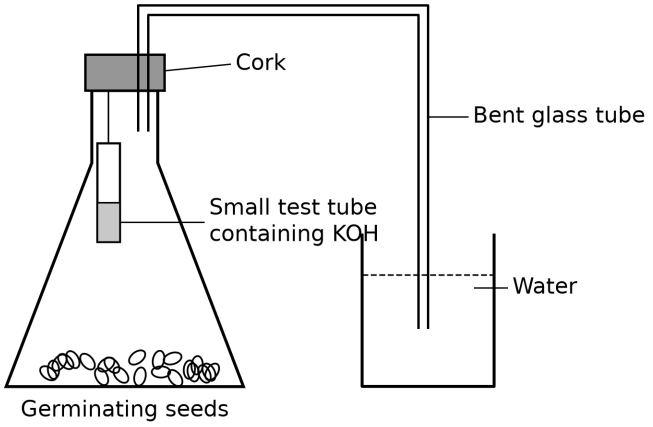
A. The gall bladder secretes lipase, which is the only enzyme that digests fats.

B. Bile, which emulsifies fats, can no longer be stored and released in a large amount when a fatty meal reaches the small intestine.

C. The gall bladder absorbs digested fats and passes them into the blood.

D. Bile makes the food in the stomach acidic so that fats can be digested there.

2 Shreya set up the apparatus shown below using germinating seeds and left it undisturbed for two hours. **1**



What will she observe, and why?

A. The water level in the bent tube falls, because the carbon dioxide released by the seeds pushes the water out.

B. The water level in the bent tube rises, because the carbon dioxide released by the seeds is absorbed by KOH, lowering the pressure inside the flask.
C. The water level remains unchanged, because germinating seeds do not respire.
D. The water level rises, because the seeds release oxygen, which dissolves in KOH.

For visually impaired students

In an experiment to show that germinating seeds release carbon dioxide during respiration, a small test tube containing KOH solution is kept inside the flask with the seeds. What is the role of KOH in this experiment?

A. It provides oxygen to the seeds.
B. It absorbs the carbon dioxide released by the seeds.
C. It supplies water for germination.
D. It absorbs the oxygen present in the flask.

3 A plant physiologist detected a high concentration of a particular plant hormone in developing fruits and seeds, regions where cells divide rapidly. The hormone is most likely: **1**

A. Abscisic acid **B.** Auxin
C. Cytokinin **D.** Gibberellin

4 Meena placed a fleshy leaf of *Bryophyllum* on moist soil. After two weeks, small plantlets appeared along the notches of the leaf margin. This is an example of: **1**

A. regeneration **B.** budding
C. vegetative propagation **D.** fragmentation

5 Study the following pairs of structures and their functions in the human reproductive system: **1**
(i) Testes – produce sperms and the hormone testosterone
(ii) Seminal vesicle – site of sperm formation
(iii) Placenta – exchange of glucose, oxygen and wastes between the mother and the embryo
(iv) Oviduct – site of implantation of the embryo
Which of the pairs are correctly matched?

A. (i) and (ii) **B.** (i) and (iii)
C. (ii) and (iv) **D.** (iii) and (iv)

6 In pea plants, round seed shape (R) is dominant over wrinkled seed shape (r). A plant with genotype Rr is crossed with a plant with genotype rr, and 400 seeds are obtained. How many of these seeds are expected to be wrinkled? **1**

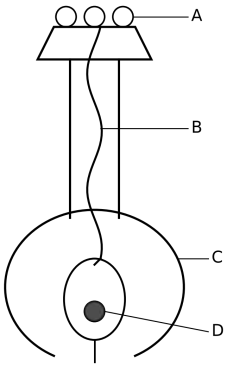
A. 100 **B.** 300
C. 200 **D.** 400

7 Which of the following practices will directly reduce the amount of non-biodegradable waste reaching a landfill? **1**

A. Carrying cloth bags instead of polythene bags while shopping.
B. Composting vegetable peels in a backyard pit.
C. Burning dry leaves in the garden.
D. Feeding leftover food to stray animals.

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** The wall of the right ventricle of the human heart is thicker than that of the left ventricle. **Reason (R):** The left ventricle pumps blood to all parts of the body.	**1**
9	**Assertion (A):** The concentration of a pesticide such as DDT is highest in organisms at the top trophic level of a food chain. **Reason (R):** Pesticides are biodegradable and are broken down at each trophic level.	**1**
10	Sana kept a potted plant having variegated leaves (leaves with green and white patches) in a dark room for 48 hours. She then placed it in bright sunlight for 6 hours, plucked one leaf, boiled it in alcohol and tested it with iodine solution. **A.** Which region of the leaf turned blue-black? What conclusion can be drawn from this observation? **B.** Why was the plant kept in the dark room for 48 hours before the experiment?	**2**
11	***Students to attempt either option A or B.*** **A.** Sunita accidentally touched a hot pan and pulled her hand back even before she felt the pain. Name this type of response. Trace the path followed by the message, from the receptor to the effector, in the form of a flow chart. **OR** **B.** The tendrils of a pea plant coil around a support that they touch. Name this type of movement and explain how the coiling is brought about at the level of cell growth.	**2**
12	A couple has two daughters, and some relatives blame the mother for not having a son. Using your understanding of sex determination in human beings, explain why this view is incorrect.	**2**
13	The figure given below shows the germination of pollen on the stigma of a flower.  **A.** Identify the parts labelled B and D. **B.** What happens when the structure B reaches D? **C.** Name the parts of the flower that develop into the seed and the fruit after fertilisation. **For visually impaired students** **A.** Name the tube that grows out of a pollen grain on the stigma and carries the male germ cell to the ovule. **B.** What happens when this tube reaches the ovule?	**3**

C. Name the parts of the flower that develop into the seed and the fruit after fertilisation.

14 ***Students to attempt either option A or B.*** **3**

A. The daily waste collected from a housing society is given below.

Type of waste	Mass (kg)
Vegetable and fruit peels	40
Dry leaves from the garden	10
Plastic bottles	8
Newspapers and cardboard	12
Glass jars	5
Polythene and foil wrappers	6

(i) Calculate the total mass of biodegradable waste produced by the society each day.

(ii) Give two reasons why the plastic and foil waste is a serious environmental problem.

(iii) Suggest one way in which the society can deal with its biodegradable waste usefully.

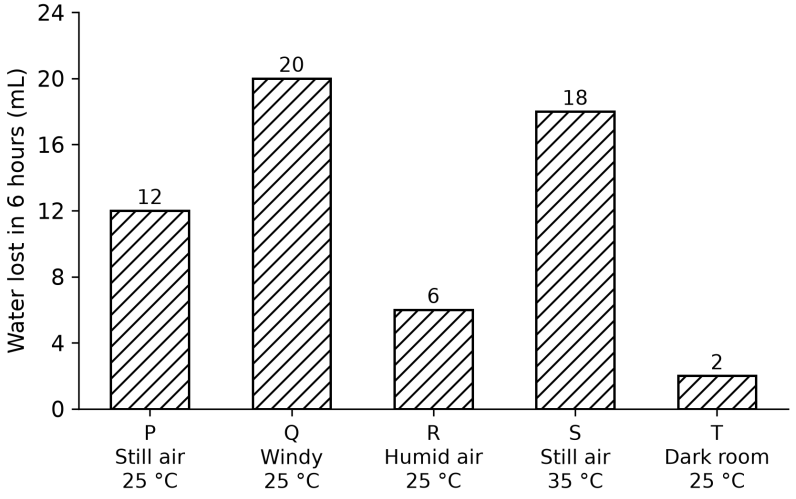
OR

B. (i) How is ozone formed in the higher levels of the atmosphere? Write the equations involved.

(ii) Why is the depletion of the ozone layer a cause of concern for living beings?

(iii) Name the group of chemicals mainly responsible for ozone depletion and state the step taken by the United Nations Environment Programme (UNEP) in 1987 to deal with it.

15 Arjun measured the volume of water lost by five identical potted plants, P to T, kept under different conditions for 6 hours. The pots were sealed with plastic sheets so that water could be lost only through the plants. His results are shown in the bar graph. **4**



A. Under which condition was the rate of transpiration the highest? Name the factor responsible for it.

B. Why did the plant kept in the dark room (T) lose the least amount of water?

Attempt either sub-part C or D.

C. Explain how transpiration helps in the upward movement of water and dissolved minerals from the roots to the leaves of a tall tree.

OR

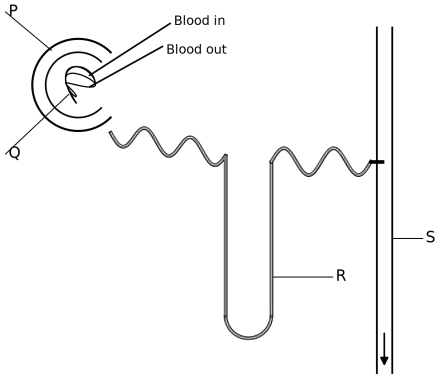
D. Explain why plant R lost less water than plant P. State any two ways in which transpiration is useful to a plant.

For visually impaired students

Arjun measured the water lost in 6 hours by five identical potted plants: P (still air, 25 °C) 12 mL; Q (windy, 25 °C) 20 mL; R (humid air, 25 °C) 6 mL; S (still air, 35 °C) 18 mL; T (dark room, 25 °C) 2 mL. Answer sub-parts A and B, and either C or D, given above.

16 ***Students to attempt either option A or B.*** **5**

A. Study the diagram of a nephron given below and answer the questions that follow.



I. Identify the parts labelled P and Q. What happens to the blood in the part labelled Q?

II. Name two substances that are selectively reabsorbed from the filtrate as it flows along the tubule R. On what factors does the amount of water reabsorbed depend?

III. Trace the path of urine from the tube S until it is passed out of the body.

For visually impaired students

I. Name the cup-shaped structure and the cluster of capillaries present at the beginning of a nephron. What happens to the blood in the cluster of capillaries?

II. Name two substances that are selectively reabsorbed from the filtrate as it flows along the tubule of a nephron. On what factors does the amount of water reabsorbed depend?

III. Trace the path of urine from the collecting duct until it is passed out of the body.

OR

B. **I.** Plants do not have specialised excretory organs. Describe any three ways in which plants get rid of their excretory products.

II. Ramesh, whose kidneys have stopped working, is put on haemodialysis. State the principle on which an artificial kidney works. Mention one way in which it differs from the working of a nephron.

SECTION B Chemistry

17 Riya kept white silver chloride in a watch glass in sunlight. After some time, it turned grey. Which of the following applications is based on this type of reaction? **1**

A. Black and white photography **B.** Galvanisation of iron

C. Preparation of antacids **D.** Electroplating of jewellery

18 The pH values of four samples, P, Q, R and S, measured using a pH meter, are given below. **1**

Sample	P	Q	R	S
pH	2.2	7.0	10.5	13.8

Which sample is most suitable to be taken as an antacid to relieve acidity in the stomach?

A. P **B.** Q
C. R **D.** S

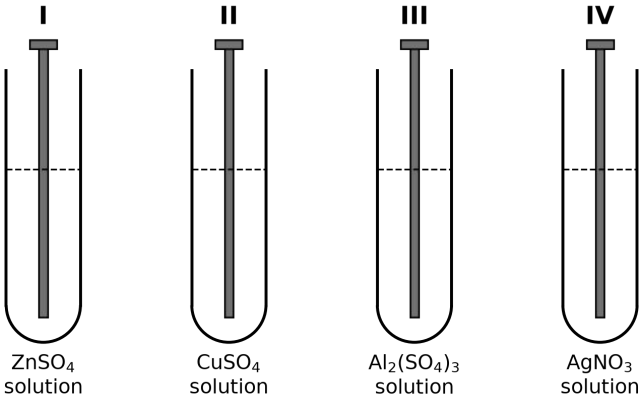
19 An element X forms an oxide with the formula X_2O_3. This oxide reacts with dilute hydrochloric acid as well as with sodium hydroxide solution to form a salt and water. The element X is most likely: **1**

A. sodium **B.** magnesium
C. aluminium **D.** copper

20 Which of the following pairs of compounds belongs to the same homologous series? **1**

A. CH_4 and C_2H_4 **B.** C_2H_4 and C_3H_8
C. C_2H_2 and C_2H_6 **D.** C_2H_6 and C_3H_8

21 Clean iron nails were placed in four test tubes containing the solutions shown below. **1**



In which test tubes will a chemical reaction take place?

A. I and II **B.** II and IV
C. I and III **D.** III and IV

For visually impaired students

Clean iron nails were placed in four test tubes containing solutions of (I) zinc sulphate, (II) copper sulphate, (III) aluminium sulphate and (IV) silver nitrate. In which test tubes will a chemical reaction take place?

A. I and II **B.** II and IV
C. I and III **D.** III and IV

22 Kunal noticed that a soap forms a white scum with water from a hand pump, but a detergent does not. Which of the following explains this observation? **1**

A. The calcium and magnesium salts of detergents are soluble in water, whereas those of soaps are insoluble.
B. Detergents are sodium salts of long-chain carboxylic acids.
C. Hand-pump water contains chloride ions which react only with soap.
D. Soaps work only in acidic water, while detergents work in all types of water.

23 Plaster of Paris should be stored in a moisture-proof container because it: **1**

A. reacts with carbon dioxide of air to form calcium carbonate.
B. loses its water of crystallisation and turns into a powder.
C. absorbs water and sets into a hard mass of gypsum.
D. absorbs moisture and turns into a solution.

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Packets of potato chips are flushed with nitrogen gas by manufacturers. **Reason (R):** Nitrogen is a highly reactive gas which removes oxygen by reacting with it. **1**

25 Before burning a magnesium ribbon in air, Rahul rubbed it with sandpaper. **2**

A. Why did he rub the ribbon with sandpaper?

B. Write the balanced chemical equation for the burning of magnesium in air and name the type of reaction.

26 ***Students to attempt either option A or B.*** **3**

A. Rohan's mother adds a pinch of baking soda to the batter while making dhokla, which makes it soft and spongy.

(i) Write the chemical name and formula of baking soda.

(ii) Write the chemical equation for the reaction that takes place when baking soda is heated during cooking.

(iii) Why is a mild edible acid such as tartaric acid added to baking soda to make baking powder?

OR

B. When electricity is passed through an aqueous solution of sodium chloride (brine), it decomposes to form three useful products.

(i) Name this process and the three products formed.

(ii) Write the chemical equation for the preparation of bleaching powder from one of these products.

(iii) State two uses of bleaching powder.

27 **3**

A. Write the IUPAC names of the compounds CH_3CH_2CHO and $CH_3COCH_2CH_3$.

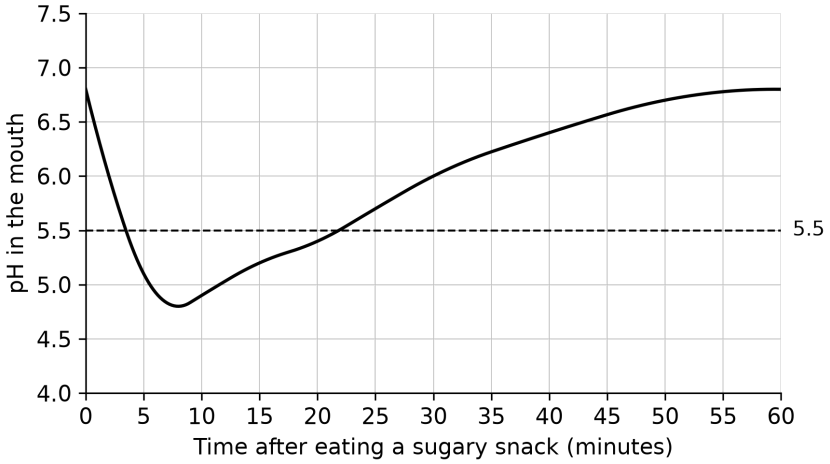
B. Draw the electron dot structure of ethyne (C_2H_2). How many shared pairs of electrons are present between the two carbon atoms?

C. Butane (C_4H_{10}) has a low boiling point and does not conduct electricity. Explain these properties on the basis of the bonding in butane.

28 Bacteria in the mouth break down sugars and food particles left in the mouth and produce acids. Tooth enamel, made of calcium hydroxyapatite (a form of calcium phosphate), is the hardest substance in the body, but it starts to corrode when the pH in the mouth falls below 5.5. Kavya recorded the pH in the mouth of a volunteer after he ate a sugary snack. Her results are shown in the graph. **4**

Exam tip

For assertion-reason, test A and R separately as true or false first; only then decide if R explains A.



A. What was the lowest pH reached in the mouth, and about how many minutes after eating the snack?

B. For approximately how many minutes did the pH stay below 5.5?

Attempt either sub-part C or D.

C. Toothpastes are generally basic. Explain how brushing the teeth after eating sweets helps to prevent tooth decay. Name the type of reaction involved.

OR

D. Kavya says, "Eating a sweet five times a day is more harmful for the teeth than eating the same amount of sweets at one time." Do you agree? Justify your answer using the graph.

For visually impaired students

Kavya recorded the pH in the mouth of a volunteer after he ate a sugary snack: 0 min – 6.8; 5 min – 5.1; 8 min – 4.8; 15 min – 5.2; 20 min – 5.4; 25 min – 5.7; 40 min – 6.4; 60 min – 6.8. Tooth enamel starts to corrode below pH 5.5. Answer sub-parts A and B, and either C or D, given above.

29 ***Students to attempt either option A or B.*** **5**

A. **I.** Four metals P, Q, R and S were added separately to solutions of four salts. The observations are recorded below.

Metal	$FeSO_4$ solution	$CuSO_4$ solution	$ZnSO_4$ solution	$AgNO_3$ solution
P	Reaction	Reaction	No reaction	Reaction
Q	No reaction	No reaction	No reaction	Reaction
R	Reaction	Reaction	Reaction	Reaction
S	No reaction	Reaction	No reaction	Reaction

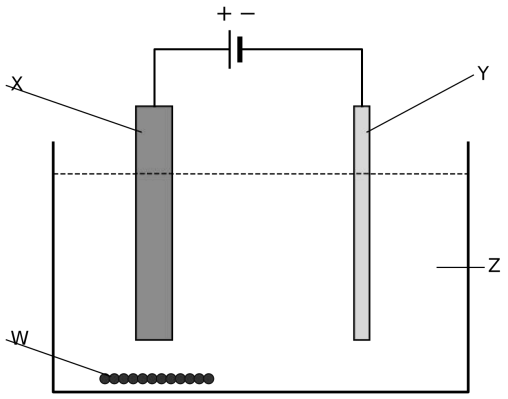
(a) Arrange the metals P, Q, R and S in decreasing order of reactivity.
(b) Which of these metals can be used to make a container for storing copper sulphate solution? Give a reason.

II. Zinc is extracted from its sulphide ore, zinc blende. Describe, with balanced chemical equations, the two steps by which the ore is converted into zinc metal. Why is a sulphide ore converted into an oxide before reduction?

OR

B. **I.** Give reasons for the following:
(a) Sodium is stored under kerosene.
(b) Aluminium, though a highly reactive metal, is used to make cooking utensils.
(c) Gold and platinum are found in the free state in nature.

II. The set-up given below is used for the electrolytic refining of copper.



(a) Identify X, Y, Z and W.
(b) Write the reaction that takes place at Y.

For visually impaired students

In the electrolytic refining of copper, name the substances used as the anode, the cathode and the electrolyte. Write the reaction that takes place at the cathode.

SECTION C Physics

30 A dentist wants to see an erect and enlarged image of a patient's tooth. Which type of mirror should she use, and where should the tooth be with respect to the mirror? **1**

A. Convex mirror; at any distance in front of the mirror
B. Concave mirror; between the pole and the principal focus
C. Concave mirror; between the principal focus and the centre of curvature
D. Plane mirror; very close to the tooth

31 Two resistors of 6 Ω and 3 Ω are connected in parallel across a 6 V battery. The current drawn from the battery is: **1**

A. 0.67 A **B.** 1 A
C. 2 A **D.** 3 A

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Two magnetic field lines never intersect each other.
Reason (R): If two field lines intersected, the compass needle placed at the point of intersection would point in two directions at the same time, which is not possible. **1**

33 A ray of light travels from air into a glass block of refractive index 1.5. **2**

A. Calculate the speed of light in the glass. (Speed of light in air = 3×10^8 m/s)
B. Will the ray bend towards or away from the normal on entering the glass? Give a reason.

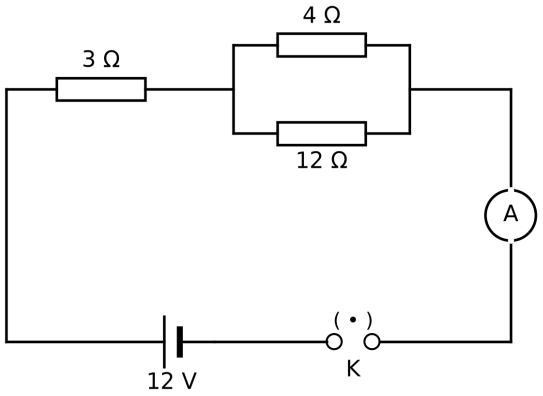
34 ***Students to attempt either option A or B.*** **2**

A. (i) Why is a fuse always connected in series with the live wire of a domestic circuit?
(ii) What is the function of the earth wire in an appliance with a metallic body, such as an electric iron?

OR

B. State Fleming's left-hand rule. What will be the force on a current-carrying conductor placed parallel to the direction of a magnetic field?

35 Study the circuit shown below. **3**



Calculate:

A. the total resistance of the circuit,

B. the reading of the ammeter when the key is closed,

C. the current through the 4 Ω resistor.

For visually impaired students

A 3 Ω resistor is connected in series with a parallel combination of a 4 Ω and a 12 Ω resistor, across a 12 V battery. Calculate (A) the total resistance of the circuit, (B) the current drawn from the battery, and (C) the current through the 4 Ω resistor.

36 A 4 cm tall candle is placed 30 cm in front of a concave mirror of focal length 20 cm. Using the mirror formula, find the position of the image. State the nature of the image and calculate its size. **3**

37 ***Students to attempt either option A or B.*** **3**

A. An electric iron is rated 1100 W, 220 V. It is used for 2 hours every day.

(i) Calculate the current drawn by the iron.

(ii) Calculate the resistance of its heating element.

(iii) Calculate the cost of electrical energy consumed by the iron in 30 days at the rate of ₹6 per kWh.

OR

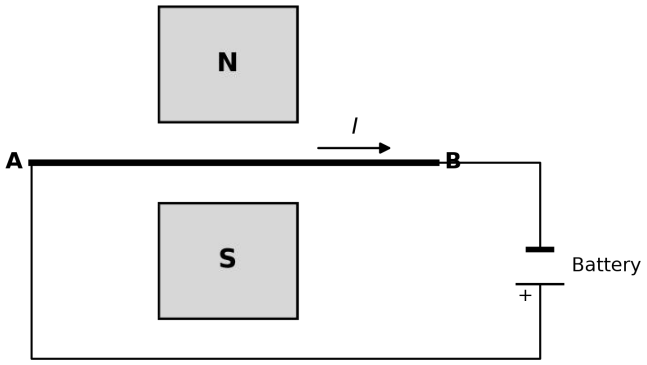
B. A nichrome wire is 2 m long and has an area of cross-section of 1.0×10^{-6} m^2. The resistivity of nichrome is 1.1×10^{-6} Ω m.

(i) Calculate the resistance of the wire.

(ii) The wire is stretched uniformly to twice its original length. Calculate its new resistance.

(iii) Why is nichrome, and not copper, used for the heating elements of electric heaters?

38 Tanvi suspended a light aluminium rod AB horizontally between the poles of a strong magnet, with the north pole above the rod and the south pole below it, and connected it to a battery as shown. When the current was switched on, the rod was displaced. **4**



A. Name the rule used to find the direction of the force acting on the rod.

B. In which direction (into the plane of the paper or out of it) will the rod be displaced?

Attempt either sub-part C or D.

C. What will happen to the displacement of the rod if (i) only the direction of current is reversed, and (ii) the direction of current and the magnetic field are both reversed?

OR

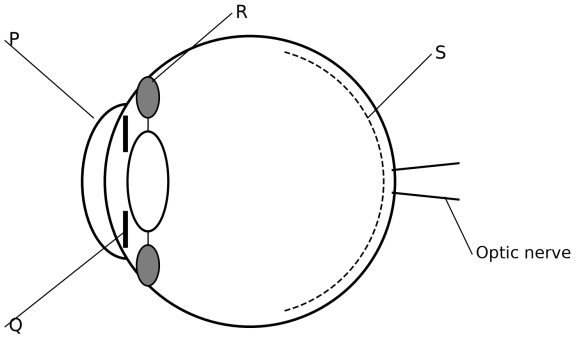
D. Why is no displacement observed if the rod is placed along the direction of the magnetic field? State one way to increase the displacement of the rod.

For visually impaired students

Tanvi suspended a light aluminium rod horizontally between the poles of a strong magnet, with the north pole above the rod and the south pole below it. A current was passed through the rod from its left end A to its right end B, and the rod was displaced. Answer sub-parts A and B, and either C or D, given above.

39 ***Students to attempt either option A or B.*** **5**

A. Study the diagram of the human eye given below.



I. Identify the parts labelled Q and R and state one function of each.

II. The near point of a hypermetropic eye is 50 cm. Calculate the power of the lens required to enable this person to read a book held at 25 cm from the eye. Name the type of lens.

III. Why is a normal eye unable to see clearly an object held 10 cm from it?

For visually impaired students

I. Name the part of the human eye that controls the size of the pupil and the part that changes the focal length of the eye lens.

II. The near point of a hypermetropic eye is 50 cm. Calculate the power of the lens required to enable this person to read a book held at 25 cm from the eye. Name the type of lens.

III. Why is a normal eye unable to see clearly an object held 10 cm from it?

OR

B. I. Explain why stars twinkle.

II. Why do planets not twinkle?

III. A narrow beam of white light is passed through a glass prism. (a) Name the phenomenon observed. (b) Which colour bends the most, and why?

END OF QUESTION PAPER

Quick revision: Carbon and its Compounds

1. Carbon forms covalent bonds by sharing electrons; it shows catenation and tetravalency.

2. Saturated: single bonds (alkanes, C_nH_{2n+2}). Unsaturated: double or triple bonds (alkenes C_nH_{2n}, alkynes C_nH_{2n-2}).

3. A homologous series differs by $-CH_2-$ and 14 u in mass.

4. Functional groups: alcohol −OH, aldehyde −CHO, ketone −CO−, carboxylic acid −COOH.

5. Ethanol + ethanoic acid → ester (sweet smell) + water, with conc. H_2SO_4.

6. Soap molecules form micelles: the ionic end faces water, the carbon chain traps oil.

7. Hard water wastes soap (scum); detergents work in hard water.

8. Addition of hydrogen to vegetable oils uses a nickel catalyst.

Nitesh Bhaiya ki tip: IUPAC naming me pehle carbon count karo, phir functional group ka suffix lagao.

SAMPLE PAPER 14

SCIENCE · CODE NO. 086

Sample Question Paper 14

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

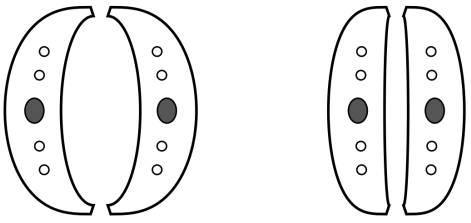
Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 Ravi has been taking antacid tablets regularly for a long time. The activity of which of the following enzymes in his stomach is most likely to be reduced? **1**

A. Salivary amylase
B. Pepsin
C. Trypsin
D. Pancreatic lipase

2 The figure given below shows two stomata, X and Y, as observed in a leaf peel under a microscope. **1**



Which of the following statements is correct?

A. X is open because its guard cells have swollen by absorbing water.
B. Y is open because its guard cells have lost water.
C. X is closed because its guard cells have become flaccid.
D. Both X and Y are open because guard cells contain chloroplasts.

For visually impaired students

The pore of a stoma opens when:

A. its guard cells absorb water and swell.
B. its guard cells lose water and shrink.

	C. the surrounding epidermal cells absorb water. **D.** the chloroplasts in its guard cells absorb carbon dioxide.	
3	A doctor recorded a patient's blood pressure as 120/80 mm of Hg. The value 120 mm of Hg represents the pressure of blood: **A.** in the veins when the heart relaxes. **B.** in the arteries when the ventricles relax. **C.** in the capillaries of the lungs. **D.** in the arteries when the ventricles contract.	**1**
4	A farmer noticed that some papaya plants in his orchard never bear fruits, even though they flower regularly. The most likely reason is that these plants: **A.** bear only staminate (male) flowers. **B.** bear bisexual flowers that are self-pollinated. **C.** bear flowers that have no sepals. **D.** bear flowers that contain both stamens and pistil.	**1**
5	During an awareness camp, a health worker explains different methods of contraception. Which of the following methods also protects against sexually transmitted infections such as HIV-AIDS? **A.** Copper-T **B.** Oral pills **C.** Condom **D.** Vasectomy	**1**
6	Pure-breeding pea plants with round yellow seeds (RRYY) were crossed with plants having wrinkled green seeds (rryy). The F_1 plants were self-pollinated and 1600 F_2 seeds were obtained. How many of these seeds are expected to be round and green? **A.** 100 **B.** 300 **C.** 600 **D.** 900	**1**
7	In the food chain: grass → grasshopper → frog, the frog obtains 50 J of energy. According to the ten per cent law, the energy trapped by the grass is: **A.** 5 J **B.** 500 J **C.** 5000 J **D.** 50,000 J	**1**

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8	**Assertion (A):** The inner lining of the small intestine has numerous finger-like projections called villi. **Reason (R):** The small intestine is the site of complete digestion of carbohydrates, proteins and fats.	**1**
9	**Assertion (A):** Doctors advise the use of iodised salt in our diet. **Reason (R):** Iodine is essential for the thyroid gland to make the hormone thyroxin.	**1**
10	After running a 400 m race, Aman felt painful cramps in his leg muscles. **A.** Explain the cause of the cramps by writing the pathway of the breakdown of glucose in his muscle cells.	**2**

B. Why is ATP called the energy currency of the cell?

11 ***Students to attempt either option A or B.*** **2**

A. When a *Planaria* is accidentally cut into three pieces, each piece grows into a complete organism. Name this process. Why can complex multicellular organisms not reproduce by this method?

OR

B. State two advantages of vegetative propagation to a gardener. Name one plant that is commonly grown by this method.

12 Mendel crossed pure tall pea plants with pure short pea plants. All the F_1 plants were tall, but short plants reappeared in the F_2 generation. Explain why the short trait disappeared in F_1 and reappeared in F_2. **2**

13 **3**

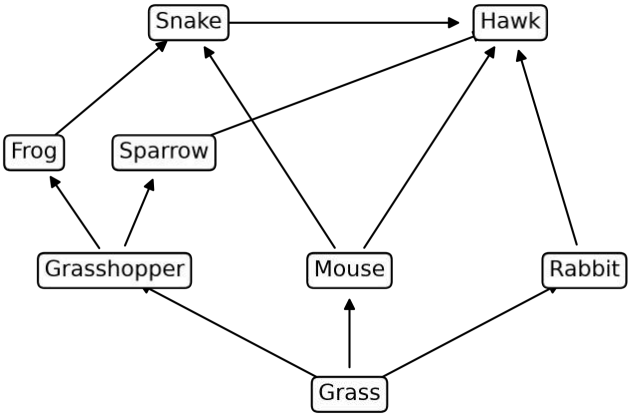
A. Desert plants keep their stomata closed during the day and open them only at night. How do they carry out photosynthesis during the day?

B. List the three events that occur during the process of photosynthesis.

C. Write the balanced chemical equation for photosynthesis.

14 ***Students to attempt either option A or B.*** **3**

A. Study the food web given below.



(i) Write the longest food chain in this food web.

(ii) Name one organism that occupies more than one trophic level. Mention the trophic levels it occupies.

(iii) If all the frogs were removed from this food web, predict the effect on the population of grasshoppers. Give a reason.

For visually impaired students

In a grassland, grass is eaten by grasshoppers, mice and rabbits; grasshoppers are eaten by frogs and sparrows; frogs and mice are eaten by snakes; and mice, rabbits, sparrows and snakes are eaten by hawks. (i) Write the longest food chain. (ii) Name one organism that occupies more than one trophic level and mention the levels. (iii) Predict the effect of removing all the frogs on the population of grasshoppers.

OR

B. (i) What would happen if all the decomposers in an ecosystem were destroyed?

(ii) Why are food chains generally limited to three or four trophic levels?

(iii) Why is a crop field considered an artificial ecosystem while a pond is a natural ecosystem?

15 Priya, a Class X student, recorded her heart rate and breathing rate after different activities in her physical education class. **4**

Activity	Heart rate (beats per minute)	Breathing rate (breaths per minute)
At rest	72	16
After brisk walking for 10 minutes	100	24
After running for 10 minutes	140	38

A. Why do both the heart rate and the breathing rate increase after running?

B. Name the blood vessel that brings oxygenated blood from the lungs to the heart.

Attempt either sub-part C or D.

C. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?

OR

D. Why is the circulation of blood in human beings called double circulation? Explain.

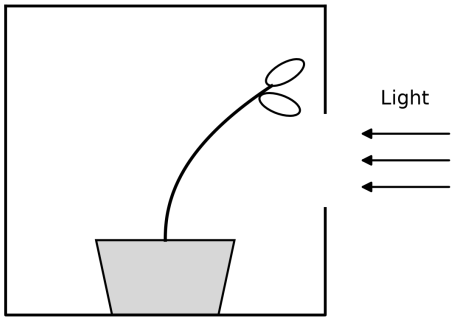
16 ***Students to attempt either option A or B.*** **5**

A. **I.** Name the part of the human brain responsible for each of the following:
(a) maintaining posture and balance of the body while riding a bicycle
(b) involuntary actions such as blood pressure, salivation and vomiting
(c) thinking, memory and voluntary actions

II. In each of the following cases, name the hormone involved and the gland that secretes it:
(a) a child shows very slow growth, which may lead to dwarfism
(b) a person develops a swollen neck due to lack of iodine in the diet
(c) a person has a very high level of sugar in the blood
(d) a boy develops facial hair and a deeper voice at puberty

OR

B. **I.** A potted seedling was kept for a few days in a box having an opening on one side only. The figure shows how the seedling looked after a few days.



(a) Name the response shown by the shoot.
(b) Explain the role of auxin in bringing about the bending of the shoot.
(c) How would the roots of this seedling respond to light?

For visually impaired students

A potted seedling was kept for a few days in a box having an opening on one side only, through which light entered. The shoot bent towards the opening. (a) Name the response shown by the shoot. (b) Explain the role of auxin in bringing about this bending. (c) How would the roots of this seedling respond to light?

II. Why do animals need chemical communication by hormones in addition to communication through nerves? Give two reasons.

SECTION B Chemistry

17 When a solution of lead nitrate is mixed with a solution of potassium iodide, a yellow substance settles at the bottom. This reaction is an example of: **1**

A. displacement and precipitation reaction
B. combination and redox reaction
C. double displacement and precipitation reaction
D. decomposition and redox reaction

18 Shivam passed dry hydrogen chloride gas over a dry blue litmus paper and saw no change in colour. When he moistened the litmus paper, it turned red. This shows that: **1**

A. hydrogen chloride is a weak acid.
B. dry litmus paper behaves as a base.
C. hydrogen chloride gas reacts with litmus only in sunlight.
D. an acid shows acidic behaviour only in the presence of water, which allows it to produce H^+ ions.

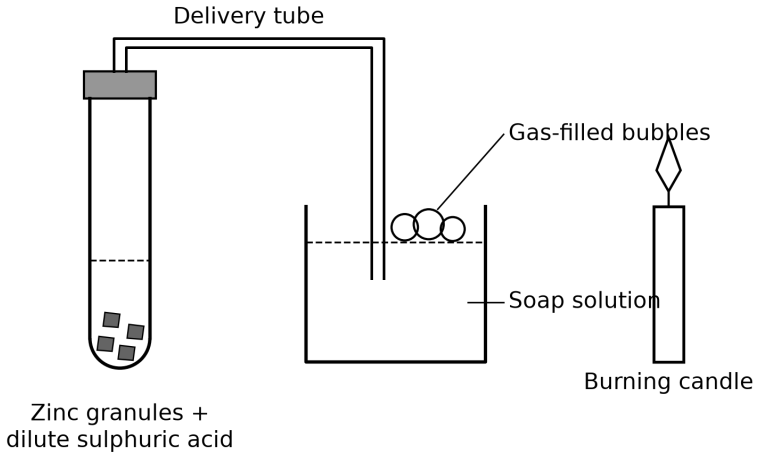
19 A metal M does not react with cold or hot water, but reacts with steam to form a metal oxide and hydrogen gas. M could be: **1**

A. sodium **B.** potassium
C. iron **D.** copper

20 How many covalent bonds are present in a molecule of propane (C_3H_8)? **1**

A. 8 **B.** 11
C. 10 **D.** 13

21 Ananya set up the apparatus shown below. Zinc granules react with dilute sulphuric acid and the gas formed is passed through soap solution. A burning candle is brought near a gas-filled soap bubble. **1**



What will she observe?

A. The candle is extinguished, as the gas is carbon dioxide.
B. The bubble burns with a pop sound, as the gas is hydrogen.
C. The candle burns more brightly, as the gas is oxygen.
D. Nothing happens, as the gas is nitrogen.

For visually impaired students

Zinc granules react with dilute sulphuric acid and the gas formed is passed through soap solution. What will be observed when a burning candle is brought near a gas-filled soap bubble?

A. The candle is extinguished, as the gas is carbon dioxide.
B. The bubble burns with a pop sound, as the gas is hydrogen.
C. The candle burns more brightly, as the gas is oxygen.
D. Nothing happens, as the gas is nitrogen.

22 When ethanol is heated with alkaline potassium permanganate solution, it is converted into ethanoic acid. In this reaction, alkaline potassium permanganate acts as: **1**

A. an oxidising agent **B.** a reducing agent
C. a catalyst **D.** a dehydrating agent

23 The water supplied to a hostel is permanently hard. Which of the following salts can be used to remove its hardness? **1**

A. Baking soda **B.** Bleaching powder
C. Plaster of Paris **D.** Washing soda

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24 **Assertion (A):** Silver articles become black after some time when exposed to air. **1**
Reason (R): Silver reacts with oxygen present in the air to form black silver oxide.

25 ***Students to attempt either option A or B.*** **2**

A. Two unlabelled test tubes contain ethanol and ethanoic acid. Describe a chemical test, with the chemical equation, to identify the test tube containing ethanoic acid.

OR

B. (i) Why do saturated hydrocarbons generally burn with a clean blue flame, while unsaturated hydrocarbons burn with a yellow, sooty flame?
(ii) Why does the bottom of a cooking vessel get blackened when the air holes of a gas stove are blocked?

26 Shreya heated a few crystals of ferrous sulphate in a dry boiling tube. **3**

(i) State two observations she would make.
(ii) Write the balanced chemical equation for the reaction.
(iii) Name the type of reaction. Why does this reaction need energy to be supplied continuously?

27 ***Students to attempt either option A or B.*** **3**

A. (i) Show the formation of magnesium oxide by the transfer of electrons. (Atomic numbers: Mg = 12, O = 8)
(ii) Name the ions present in the compound formed.
(iii) State two properties of this compound.

OR

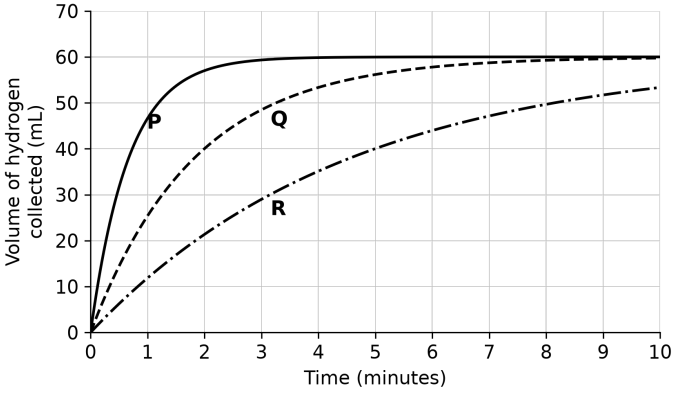
B. Iron nails were placed in three test tubes: tube I with tap water (exposed to air), tube II with boiled distilled water covered with a layer of oil, and tube III with dry air and anhydrous calcium chloride.

(i) In which tube will the nails rust after a few days?

(ii) Why is oil added in tube II and anhydrous calcium chloride in tube III?

(iii) State two methods of preventing the rusting of iron.

28 Ishaan added excess pieces of three metals (magnesium, zinc and iron) of the same size, separately, to equal volumes of the same dilute hydrochloric acid. He measured the volume of hydrogen gas collected with time and plotted the graph shown below. **4**



A. Which curve, P, Q or R, represents magnesium? Give a reason.

B. Why do all the three curves finally level off at the same volume of hydrogen?

Attempt either sub-part C or D.

C. Write the balanced chemical equation for the reaction of zinc with dilute hydrochloric acid. How can the gas evolved be tested?

OR

D. If Ishaan repeated the experiment with copper pieces, what would the graph look like? Explain.

For visually impaired students

Ishaan added excess pieces of magnesium, zinc and iron of the same size, separately, to equal volumes of the same dilute hydrochloric acid and measured the volume of hydrogen collected with time. The final volume collected was 60 mL in each case, but it was reached in about 3 minutes with one metal, about 8 minutes with the second, and was still not reached after 10 minutes with the third. Answer: A. Which metal reacted the fastest? Give a reason. B. Why was the same final volume of hydrogen collected in all three cases? Attempt either C or D given above.

29 ***Students to attempt either option A or B.*** **5**

A. **I.** Gypsum, when heated at 373 K, loses some of its water of crystallisation and forms a substance X that is used by doctors for supporting fractured bones.
(a) Identify X and write its chemical formula.
(b) Write the chemical equation for the formation of X from gypsum.
(c) What happens when X is mixed with water? Write the equation.

II. A student is asked to dilute concentrated sulphuric acid.
(a) Should she add the acid to water or water to the acid? Give a reason.
(b) How does dilution affect the concentration of hydronium ions (H_3O^+) per unit volume?

OR

B. **I.** Four solutions P, Q, R and S have pH values 1, 5, 9 and 13 respectively.
(a) Which solution is a strong acid and which one is a strong base?
(b) Arrange the solutions in increasing order of hydrogen ion concentration.
(c) Which solutions will turn red litmus blue?

II. Zinc granules were warmed separately with dilute hydrochloric acid and with sodium hydroxide solution.
(a) Name the gas evolved in both the cases and state how it can be tested.
(b) Write the balanced chemical equations for both reactions.

SECTION C Physics

30 On an optical bench, Riya placed a candle at a distance of $2f$ from a convex lens, where f is the focal length of the lens. The image formed will be: **1**

A. at infinity, highly enlarged.
B. between F_2 and $2F_2$, real, inverted and diminished.
C. at $2F_2$, real, inverted and of the same size.
D. on the same side as the candle, virtual, erect and enlarged.

31 A wire of resistance 4 Ω is cut into two equal halves, and the two halves are connected in parallel. The equivalent resistance of the combination is: **1**

A. 1 Ω **B.** 2 Ω
C. 4 Ω **D.** 8 Ω

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** Stars appear to twinkle when seen on a clear night.
Reason (R): Starlight undergoes continuous atmospheric refraction because the physical conditions of the atmosphere keep changing. **1**

33 An optician prescribes a lens of power −2.5 D. **2**

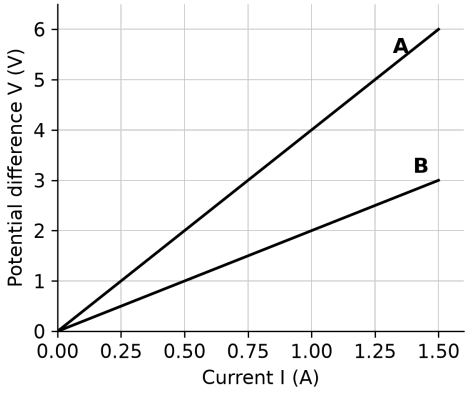
A. Find the focal length of the lens and state its nature.
B. If this lens is placed in contact with a lens of power +1.5 D, find the power of the combination. Will the combination converge or diverge light?

34 ***Students to attempt either option A or B.*** **2**

A. Give two reasons why electrical appliances in a house are connected in parallel and not in series.

OR

B. The V–I graphs of two wires, A and B, are shown below.



(i) Using the graph, find the resistance of each wire.
(ii) If the two wires are made of the same material and have the same length, which wire is thicker? Give a reason.

For visually impaired students

For wire A, a current of 1 A flows when a potential difference of 4 V is applied; for wire B, a current of 1 A flows when 2 V is applied. (i) Find the resistance of each wire. (ii) If the two wires are made of the same material and have the same length, which wire is thicker? Give a reason.

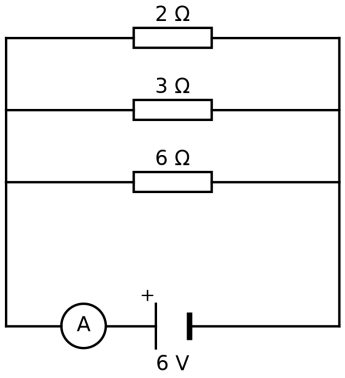
35 Neel uses a convex lens of focal length 15 cm as a magnifying glass to read the fine print on a medicine bottle held 10 cm from the lens. **3**

A. Find the position of the image.
B. Calculate the magnification and state the nature of the image.
C. Draw a ray diagram to show the image formation.

For visually impaired students

C. State the position of the object with respect to the lens for which a convex lens acts as a magnifying glass.

36 Study the circuit given below. **3**



A. Calculate the equivalent resistance of the circuit.
B. What will be the reading of the ammeter?
C. Which resistor carries the maximum current? Calculate this current.

For visually impaired students

Three resistors of 2 Ω, 3 Ω and 6 Ω are connected in parallel across a 6 V battery, with an ammeter in the main circuit. Calculate (A) the equivalent resistance, (B) the reading of the ammeter, and (C) the current through the resistor that carries the maximum current.

37 ***Students to attempt either option A or B.*** **3**

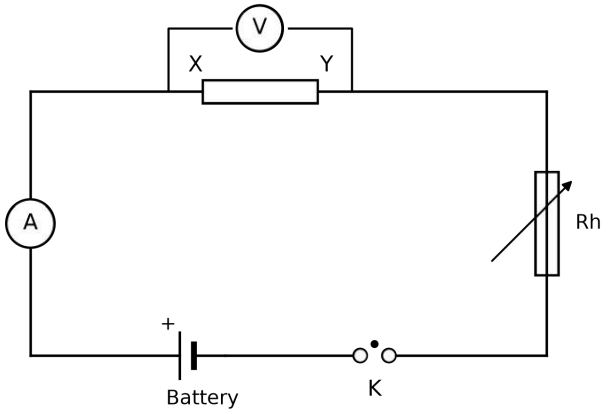
A. A concave mirror forms a real image that is three times the size of the object, when the object is placed 10 cm in front of the mirror.
(i) Find the position of the image.
(ii) Calculate the focal length of the mirror.
(iii) Is the image erect or inverted? Justify using the sign of magnification.

OR

B. A student sitting in the last row of the classroom cannot read the writing on the blackboard clearly, but can read his book easily.
(i) Name the defect of vision he is suffering from and state one cause of it.
(ii) His far point is 2.5 m. Calculate the power of the lens required to correct his vision.

(iii) Name the type of lens used for the correction.

38 In a school laboratory, Diya set up the circuit shown below to verify Ohm's law for a resistance wire XY. She recorded the readings given in the table. **4**



Potential difference V (V)	0.5	1.0	1.5	2.0	2.5
Current I (A)	0.1	0.2	0.3	0.4	0.5

A. How are the ammeter and the voltmeter connected in this circuit?

B. Using the table, find the resistance of the wire XY.

Attempt either sub-part C or D.

C. The wire XY is replaced by another wire of the same material, twice the length and half the area of cross-section. Find its resistance, and the current through it when the potential difference across it is 2 V.

OR

D. State the nature of the V–I graph for the wire and what its slope represents. Why should the current not be passed through the wire for a long time during the experiment?

For visually impaired students

In a school laboratory, Diya connected a resistance wire XY to a battery, a key, an ammeter, a rheostat and a voltmeter to verify Ohm's law. Her readings were: 0.5 V – 0.1 A; 1.0 V – 0.2 A; 1.5 V – 0.3 A; 2.0 V – 0.4 A; 2.5 V – 0.5 A. Answer sub-parts A and B, and either C or D, given above.

39 ***Students to attempt either option A or B.*** **5**

A. **I.** Draw the pattern of magnetic field lines of a current-carrying solenoid. What does the pattern of field lines inside the solenoid indicate?

II. A beam of electrons moving towards the east enters a uniform magnetic field directed vertically downwards. Using Fleming's left-hand rule, find the direction of the force on the beam. Explain your answer.

III. State one advantage of alternating current over direct current. What is the frequency of the alternating current supplied in India?

OR

B. **I.** (a) Name the three wires used in domestic circuits and state the colour of the insulation of each.
(b) Distinguish between short-circuiting and overloading.
(c) Why are separate circuits of 15 A and 5 A current ratings used in a house?

II. In a room, five bulbs of 100 W each and a room heater of 1000 W are used together on a 220 V supply. Calculate the total current drawn. Would a 5 A fuse be suitable for this circuit? Give a reason.

SAMPLE PAPER 15

SCIENCE · CODE NO. 086

Sample Question Paper 15

CLASS X · SESSION 2026-27 · LATEST CBSE PATTERN

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions

i. This question paper consists of 39 questions in 3 sections. Section A is Biology, Section B is Chemistry and Section C is Physics.

ii. All questions are compulsory. However, an internal choice is provided in some questions. A student is expected to attempt only one of the alternatives in these questions.

Marks Distribution

Section	Subject	1 mark	2 marks	3 marks	4 marks (case)	5 marks	Questions	Marks
A	Biology	9 × 1	3 × 2	2 × 3	1 × 4	1 × 5	16	30
B	Chemistry	8 × 1	1 × 2	2 × 3	1 × 4	1 × 5	13	25
C	Physics	3 × 1	2 × 2	3 × 3	1 × 4	1 × 5	10	25
Total							39	80

Unit	Unit name	Marks
I	Chemical Substances	25
II	World of Living	25
III	Natural Phenomena	12
IV	Effects of Current	13
V	Natural Resources	5
Total		80

Unit-wise weightage as per the CBSE Science curriculum 2026-27 (theory 80 marks).

SECTION A Biology

1 (1)

Rehan kept a potted plant in the dark for three days. He then enclosed one of its leaves in a transparent polythene bag containing a watch-glass of potassium hydroxide, and another leaf of the same plant in an identical bag without potassium hydroxide. The plant was kept in bright sunlight for six hours. On testing with iodine solution, only the leaf in the bag without potassium hydroxide turned blue-black. This experiment shows that

A. sunlight is necessary for photosynthesis.

B. chlorophyll is necessary for photosynthesis.

C. carbon dioxide is necessary for photosynthesis.

D. oxygen is released during photosynthesis.

2 (1)

During the last 100 m of a 400 m race, an athlete experiences painful cramps in the leg muscles. Which of the following correctly shows the pathway of glucose breakdown in his muscle cells at that time?

A. Glucose —(in cytoplasm)→ Pyruvate —(in mitochondria)→ Lactic acid + Energy

B. Glucose —(in mitochondria)→ Pyruvate —(in cytoplasm)→ Ethanol + Carbon dioxide + Energy

C. Glucose —(in cytoplasm)→ Pyruvate —(in mitochondria)→ Carbon dioxide + Water + Energy

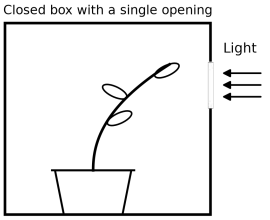
D. Glucose —(in cytoplasm)→ Pyruvate —(in cytoplasm, lack of oxygen)→ Lactic acid + Energy

3 (1)

A seedling grown in a closed box with a single opening on one side was observed after a week, as shown in the figure. The bending of the shoot towards the opening is best explained by:

A. auxin diffusing towards the shaded side of the shoot, where it makes the cells elongate more.

B. auxin being destroyed on the shaded side, so that the cells on the lit side grow faster.
C. abscisic acid accumulating on the lit side and promoting cell division there.
D. gibberellin moving towards the lit side and inhibiting growth on that side.



For visually impaired students

A seedling is grown in a closed box with a single opening on one side through which light enters. After a week its shoot is found bent towards the opening. This bending is best explained by: (options as above)

4 Consider the following statements about sexual reproduction in flowering plants: [1]
(i) The pollen tube grows through the style and carries the male germ-cell to the ovule.
(ii) After fertilisation, the ovary develops into the seed.
(iii) The zygote divides several times to form an embryo within the ovule.
(iv) Self-pollination produces more variation in the offspring than cross-pollination.
Which of the above statements are correct?

A. (i) and (ii) **B.** (i) and (iii)
C. (ii) and (iv) **D.** (iii) and (iv)

5 A pea plant with round, yellow seeds (RrYy) was crossed with a pea plant with wrinkled, green seeds (rryy). Out of 800 seeds obtained from this cross, the expected number of round, green seeds is [1]

A. 50 **B.** 100
C. 450 **D.** 200

6 Which of the following correctly describes the formation of ozone in the upper atmosphere? [1]

A. Ultraviolet radiation splits O_2 molecules into free oxygen atoms, which then combine with O_2 molecules to form O_3.
B. Ultraviolet radiation directly joins three O_2 molecules to form two O_3 molecules without splitting them.
C. Chlorofluorocarbons react with oxygen in the presence of sunlight to form O_3.
D. Visible light splits O_3 into O_2, which then recombines to form a thicker ozone layer.

7 At night, when the stomata of a herbaceous plant are closed and transpiration is negligible, water still continues to move up through its xylem. The main force responsible for this movement is [1]

A. transpiration pull created by the leaves.
B. evaporation of water from the leaf surface.
C. root pressure created by the active uptake of ions by root cells.
D. translocation of water through phloem using energy from ATP.

The following two questions consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

8 **Assertion (A):** Glucose is present in the glomerular filtrate but is normally absent in the urine of a healthy person. [1]

Reason (R): Glucose is selectively reabsorbed into the blood by the tubular part of the nephron.

9 **Assertion (A):** In a reflex action, such as pulling the hand away from a hot object, the response is generated by the brain. **Reason (R):** The spinal cord acts as the processing centre for many reflex actions. **1**

10 Rohit's grandfather underwent surgery in which a large part of his small intestine was removed. Doctors advised him to eat small, easily digestible meals, yet he kept losing weight. **2**

A. Which function of the small intestine has been most affected by the surgery? Name the structures that make the small intestine efficient at this function.

B. Why would the digestion of fats be affected more than the digestion of starch after this surgery?

11 ***Students to attempt either option A or B.*** **2**

A. The leaves of the touch-me-not (sensitive) plant fold up when touched, and the tendrils of a pea plant coil around a support that they touch. Explain how these two responses to touch differ from each other in their mechanism.

OR

B. How are the timing and amount of hormone released in our body regulated? Explain with the example of insulin and blood sugar level.

12 A slice of bread kept in a warm, humid kitchen developed a cottony mass with black dots in three days, while an identical slice kept in a refrigerator did not. **2**

A. Name the organism and the method by which it reproduces. What are the black dots?

B. Why does the cottony mass not appear in the refrigerator? How do the reproductive structures of this organism survive unfavourable conditions?

13 The table below shows the approximate amounts of some substances in the glomerular filtrate formed and in the urine passed by a healthy adult in one day. **3**

Substance	Glomerular filtrate (per day)	Urine (per day)
Water	180 L	1.5 L
Glucose	180 g	0 g
Urea	54 g	30 g
Proteins	0 g	0 g

A. Proteins are present in blood but absent in the filtrate. Give a reason.

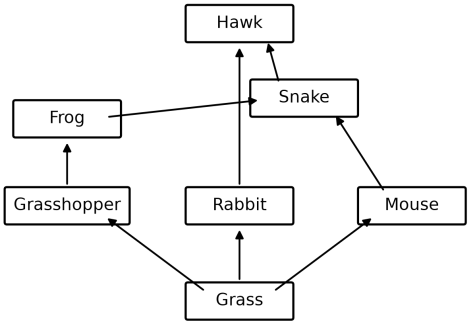
B. Calculate the percentage of water that is reabsorbed. Name the part of the nephron where this occurs.

C. Calculate the concentration of urea (in g/L) in the filtrate and in the urine. Why is the urine much more concentrated in urea than the filtrate?

14 Study the grassland food web given below. **3**

Exam tip

In a 3-mark answer, write three separate points or show three clear steps.



A. Name the organism that occupies three different trophic levels in this food web and state those levels.

B. If the grass traps 20,000 kJ of energy, how much energy will be available to the hawk through the shortest food chain? Show the chain.

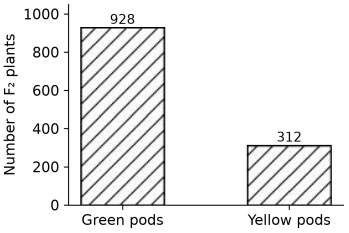
C. If all the frogs are removed from this grassland, predict one effect each on the population of grasshoppers and on the grass.

For visually impaired students

In a grassland, grass is eaten by grasshoppers, rabbits and mice. Frogs eat grasshoppers. Snakes eat frogs and mice. Hawks eat snakes and rabbits. (Sub-parts A to C as above.)

15 **4**

Riya crossed a pure-breeding pea plant having green pods (GG) with a pea plant having yellow pods (gg). All the F_1 plants had green pods. When the F_1 plants were self-pollinated, she obtained 1240 F_2 plants, as shown in the bar graph.



A. Calculate the ratio of green-podded to yellow-podded plants in F_2. Which trait is dominant?

B. What fraction of the green-podded F_2 plants are expected to be pure-breeding? Give a reason.

Attempt either sub-part C or D.

C. Riya picked one green-podded plant from F_2 and wants to know whether it is GG or Gg. Suggest a cross she should perform and state the results expected in each case.

OR

D. In another experiment, Riya crossed two plants both heterozygous for pod colour and plant height (GgTt × GgTt). Out of 1600 plants obtained, how many are expected to have yellow pods and tall height? Show your working.

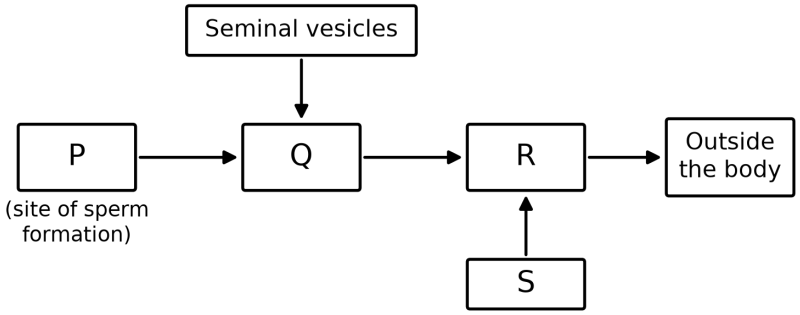
For visually impaired students

In Riya's cross between pure green-podded (GG) and yellow-podded (gg) plants, all F_1 plants had green pods. On selfing the F_1, 928 green-podded and 312 yellow-podded plants were obtained in F_2. (Sub-parts A to D as above.)

16 **5**

Students to attempt either option A or B.

A. **I.** The flow chart below shows the path taken by sperms in the human male reproductive system.



Identify P, Q, R and S. Why is P located in the scrotum outside the abdominal cavity? What is the role of the secretions added by the seminal vesicles and S? (3)

II. Name the hormone produced by P and state two changes it brings about in boys at puberty. (2)

For visually impaired students

I. Trace the path of sperms in the human male from their site of formation to the outside of the body, naming the glands that add their secretions on the way. Why are the testes located outside the abdominal cavity? What is the role of these secretions? (3)

II. Name the hormone produced by the testes and state two changes it brings about in boys at puberty. (2)

OR

B. **I.** The ovary of a woman releases one egg every month. Describe what happens in the uterus if the released egg (a) is fertilised, and (b) is not fertilised. How does the developing embryo get nutrition from the mother? (3)

II. A copper-T is a widely used contraceptive device. How does it prevent pregnancy? Why can it not protect a woman against HIV-AIDS, and which method of contraception can? (2)

SECTION B Chemistry

17 When lead nitrate is heated strongly, it decomposes as: 1

$x\ Pb(NO_3)_2(s) \rightarrow y\ PbO(s) + z\ NO_2(g) + O_2(g)$

The values of x, y and z in the balanced equation are respectively

A. 1, 1, 2 **B.** 2, 2, 4

C. 2, 2, 2 **D.** 4, 4, 2

18 Sneha heated a small amount of a white powder used in cooking. The gas evolved turned lime water milky, and an aqueous solution of the residue left behind turned red litmus blue. The residue is 1

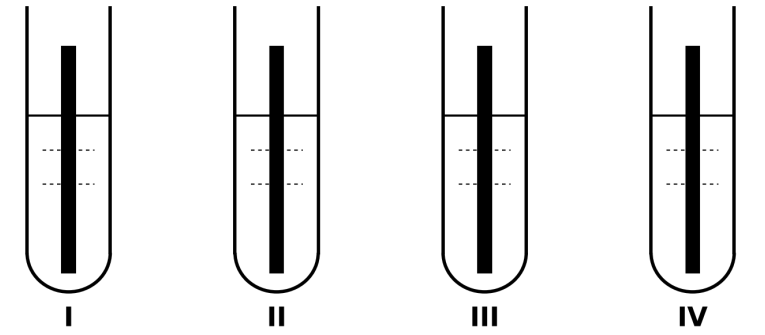
A. sodium hydroxide **B.** sodium oxide

C. sodium carbonate **D.** sodium hydrogencarbonate

19 Four test tubes I to IV were set up as shown and left for some time. 1

Exam tip

Underline the final answer of every numerical and write its unit.



In which test tubes will a change in the colour of the solution be observed?

A. I and III **B.** II and IV
C. I, II and IV **D.** I and II

For visually impaired students

Four test tubes contain: I – iron nail in copper sulphate solution; II – zinc granule in iron sulphate solution; III – copper strip in zinc sulphate solution; IV – silver wire in copper sulphate solution. In which test tubes will a change in colour of the solution be observed? (options as above)

20 An ore, on heating strongly in excess of air, gives off a gas with the choking smell of burning sulphur. The type of ore and the process are — **1**

A. sulphide ore, roasting **B.** carbonate ore, calcination
C. sulphide ore, calcination **D.** oxide ore, reduction

21 Which of the following compounds have the molecular formula C_3H_6O? — **1**
(i) Propanal (ii) Propanone (iii) Propanol (iv) Propanoic acid

A. (ii) and (iii) **B.** (iii) and (iv)
C. (i) and (ii) **D.** (i) and (iv)

22 In the reaction $ZnO + C \rightarrow Zn + CO$, the substance oxidised and the oxidising agent are respectively — **1**

A. Zn and C **B.** C and ZnO
C. ZnO and C **D.** CO and Zn

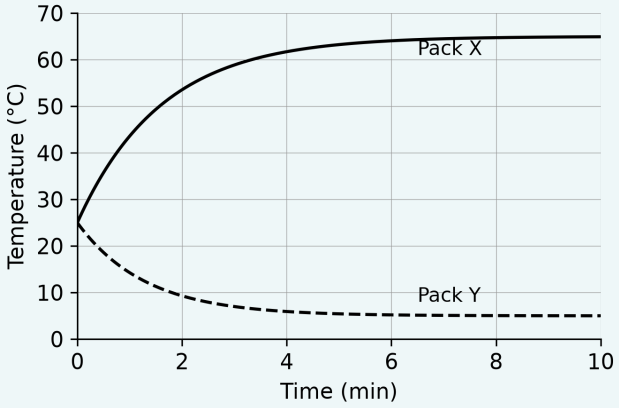
23 A doctor uses a white powder mixed with water to make a cast for a fractured bone; the paste sets into a hard solid mass. Which equation correctly represents this setting? — **1**

A. $CaSO_4 \cdot ½H_2O + 1½H_2O \rightarrow CaSO_4 \cdot 2H_2O$
B. $CaSO_4 \cdot 2H_2O \rightarrow CaSO_4 \cdot ½H_2O + 1½H_2O$
C. $CaO + H_2O \rightarrow Ca(OH)_2$
D. $CaSO_4 + 2H_2O \rightarrow CaSO_4 \cdot 2H_2O$

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

24	**Assertion (A):** Aluminium oxide reacts with both hydrochloric acid and sodium hydroxide to form salt and water. **Reason (R):** Aluminium oxide is an ionic compound, and all ionic oxides react with both acids and bases.	**1**
25	Tanya set up three test tubes, each containing a clean iron nail: tube A – nail in ordinary tap water; tube B – nail in boiled distilled water with a layer of oil on top; tube C – nail in dry air over anhydrous calcium chloride, with the tube corked. After a few days, the nail in only one tube had rusted. **A.** In which tube did the nail rust? Why did the nails in the other two tubes not rust? **B.** Suggest two ways in which the iron grill of a house near the sea can be protected from rusting.	**2**
26	Meena took 20 mL of sodium hydroxide solution in a conical flask, added two drops of phenolphthalein and then added dilute hydrochloric acid from a burette, drop by drop. The pink colour just disappeared when 16 mL of the acid had been added. **A.** Write the balanced chemical equation for the reaction and name the type of reaction. **B.** What volume of the same acid will be needed to just decolourise 30 mL of the same sodium hydroxide solution? **C.** If Meena adds only 20 mL of the acid to 30 mL of the same base, what will be the colour of the solution? Will its pH be more or less than 7? Justify.	**3**
27	***Students to attempt either option A or B.*** **A.** Magnesium (atomic number 12) reacts with chlorine (atomic number 17) to form magnesium chloride. (i) Show the formation of magnesium chloride by the transfer of electrons. (ii) Why does magnesium chloride have a high melting point? (iii) Why does it conduct electricity in the molten state but not in the solid state? **OR** **B.** Cinnabar (HgS) and zinc blende (ZnS) are both sulphide ores. Both are first roasted, but the metal oxides obtained are converted into the metals in different ways. (i) Write the balanced equations for the conversion of cinnabar into mercury. (ii) Write the balanced equations for the conversion of zinc blende into zinc. (iii) Explain why the two oxides are reduced differently.	**3**
28	Ready-to-eat meal packs for soldiers posted at high altitudes contain a self-heating pouch (Pack X). When water is released into it, quicklime in the pouch reacts with the water and heats the food. First-aid kits carry an instant cold pack (Pack Y) for sports injuries, in which a salt dissolves in water and the pack becomes very cold. The graph shows how the temperature of the two packs changes after activation.	**4**



A. Write the balanced chemical equation for the reaction in Pack X and name the product formed.

B. Classify the change in Pack X and the change in Pack Y as exothermic or endothermic, using the graph.

Attempt either sub-part C or D.

C. A solution of the product of Pack X is used for white-washing walls. Explain, with a balanced equation, why the walls acquire a shiny white finish two to three days after white-washing.

OR

D. Decomposition reactions generally need energy. Give one example, with a balanced equation, each of a decomposition reaction brought about by (i) light and (ii) electricity.

For visually impaired students

In the passage, Pack X warms up from 25 °C to about 65 °C, while Pack Y cools from 25 °C to about 5 °C within ten minutes. (Sub-parts A to D as above.)

29 ***Students to attempt either option A or B.*** **5**

A. **I.** An organic compound P (C_2H_6O) is used in cough syrups. P reacts with sodium metal to liberate a gas that burns with a pop sound. On heating with alkaline potassium permanganate, P forms Q ($C_2H_4O_2$). P and Q react in the presence of concentrated sulphuric acid to form a sweet-smelling compound R. P, on heating with excess concentrated sulphuric acid at 443 K, forms S, which decolourises bromine water. Identify P, Q, R and S and write the chemical equations for the formation of Q, R and S.

II. (a) What is the role of concentrated sulphuric acid in the formation of S? (b) How would you distinguish between P and Q using sodium hydrogencarbonate? (3) (2)

OR

B. **I.** (a) Draw the electron dot structure of ethyne (C_2H_2). (b) Why are most carbon compounds poor conductors of electricity and have low melting points? (2)

II. Consider the hydrocarbons: C_2H_6, C_3H_6, C_4H_{10}, C_3H_4, C_5H_{10}.

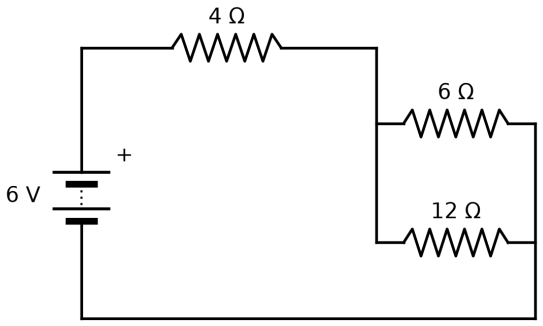
(a) Which of these belong to the same homologous series as ethene? Write the formula of the next higher homologue of C_3H_4.

(b) Draw the structures of two isomers of C_4H_{10} and name them.

(c) Vegetable oils are converted into vanaspati ghee by heating with hydrogen in the presence of nickel. Name this type of reaction and state why animal fats (saturated) are considered less healthy than vegetable oils. (3)

SECTION C Physics

30 In the circuit shown below, the current through the 12 Ω resistor is **1**



A. 0.75 A **B.** 0.5 A
C. 1.0 A **D.** 0.25 A

For visually impaired students

A 4 Ω resistor is connected in series with a parallel combination of 6 Ω and 12 Ω resistors across a 6 V battery. The current through the 12 Ω resistor is: (options as above)

31 Which of the following phenomena are caused by atmospheric refraction? **1**
(i) Twinkling of stars (ii) Blue colour of the clear sky
(iii) The Sun becoming visible about two minutes before the actual sunrise (iv) Formation of a rainbow

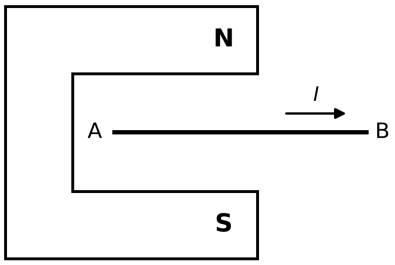
A. (i) and (ii) **B.** (ii) and (iv)
C. (i) and (iii) **D.** (iii) and (iv)

The following question consists of two statements – Assertion (A) and Reason (R). Answer the question by selecting the appropriate option given below:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, and R is not the correct explanation of A.
C. A is true but R is false.
D. A is false but R is true.

32 **Assertion (A):** A ray of light incident obliquely on one face of a rectangular glass slab emerges from the opposite face parallel to the incident ray. **1**
Reason (R): The bending of the ray at the two opposite parallel faces of the slab is equal in extent and opposite in direction.

33 A straight conductor AB is placed horizontally between the poles of a horseshoe magnet, as shown. A current *I* flows from A to B. **2**



A. State the rule used to find the direction of the force on AB, and find the direction of the force.
B. How will the direction of the force change if the current through AB and the polarity of the magnet are both reversed at the same time? Justify.

For visually impaired students

A horizontal conductor AB lies in a vertical magnetic field directed downwards (from the N pole above to the S pole below). A current flows from A to B (towards the right). A. State the rule used to find the direction of the force on AB and find the direction. B. How will the direction of the force change if the current and the magnetic field are both reversed at the same time? Justify.

34 ***Students to attempt either option A or B.*** **2**

A. The absolute refractive indices of diamond and crown glass are 2.42 and 1.50 respectively. (i) Calculate the speed of light in diamond. (ii) Find the refractive index of diamond with respect to crown glass. (Speed of light in air = 3×10^8 m/s)

OR

B. The far point of a myopic person is 2 m in front of the eye. Find the nature, focal length and power of the lens required to correct the defect.

35 The heating element of an electric room heater has a resistance of 44 Ω and is connected to a 220 V supply. **3**

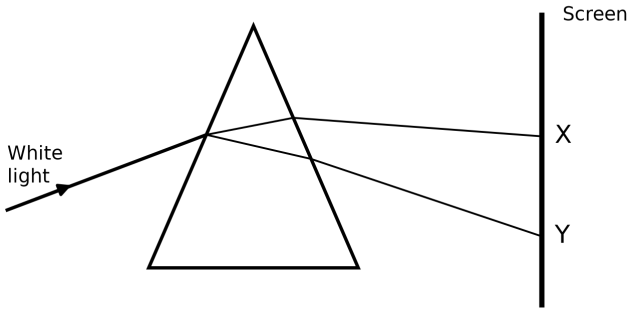
A. Calculate the current drawn by the heater.

B. Calculate the heat produced in 10 minutes.

C. The element is cut into two equal halves and the two halves are connected in parallel across the same supply. Calculate the new power and compare it with the original power.

36 ***Students to attempt either option A or B.*** **3**

A. A narrow beam of white light falls on a glass prism and forms a band of colours on a screen, as shown.



(i) Name the colours at the ends X and Y of the band. (ii) Why does white light split into colours on passing through the prism? (iii) What will be observed on the screen if a second identical prism is placed in an inverted position in the path of the emergent light?

For visually impaired students

(i) Name the colours that deviate the least and the most when white light passes through a glass prism. (ii) Why does white light split into colours on passing through a prism? (iii) What happens if a second identical prism is placed in an inverted position in the path of the emergent light?

OR

B. A hypermetropic person has a near point of 75 cm. (i) Calculate the power of the lens needed so that he can read a book held at 25 cm. (ii) State two possible causes of this defect.

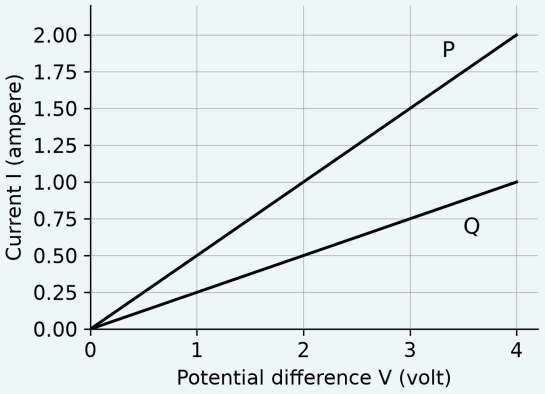
37 A house has a 220 V supply with a 10 A fuse in the circuit. The following appliances are connected in parallel: an electric iron (1000 W), a room heater (1200 W), two bulbs (60 W each) and a fan (80 W). **3**

A. Can all these appliances be used at the same time without the fuse blowing? Show your calculation.

B. Explain how the earth wire protects a person if the metal body of the electric iron accidentally becomes live.

38 **4**

Arjun studied two nichrome wires P and Q of the same thickness (area of cross-section 0.55 mm^2). Wire P is 1.0 m long. He measured the current through each wire for different potential differences and plotted the graph shown below.



A. Which wire has greater resistance? Find its resistance from the graph.

B. How is the length of wire Q related to that of wire P? Give a reason.

Attempt either sub-part C or D.

C. Calculate the resistivity of nichrome using the data for wire P. Will the resistivity change if wire P is stretched to double its length? Justify.

OR

D. Wires P and Q are connected in parallel across a 4 V battery. Calculate the total current drawn from the battery and find which wire produces more heat per second.

For visually impaired students

For wire P a potential difference of 4 V drives a current of 2.0 A; for wire Q the same potential difference drives a current of 1.0 A. Both wires are nichrome, of the same thickness (0.55 mm^2), and wire P is 1.0 m long. (Sub-parts A to D as above.)

39 **5**

Students to attempt either option A or B.

A. **I.** Riya uses a concave shaving mirror of focal length 24 cm and holds her face 16 cm in front of it. Find the position, nature and magnification of the image. (2)

II. At what distance from the same mirror should she hold her face to see an erect image magnified two times? (2)

III. Draw a ray diagram to show the image formation in part I. (1)

OR

B. **I.** The rear-view mirror of a car is a convex mirror of radius of curvature 3.0 m. A bus 3.0 m high is 6.0 m behind the car. Find the position, size and nature of the image of the bus.

II. Draw a ray diagram to show the formation of the image of an object placed in front(3) of a convex mirror, and give two reasons why convex mirrors are preferred as rear-view mirrors. (2)

END OF QUESTION PAPER

Answer Key

Sample Papers 11 to 15

MCQ and Assertion-Reason questions

Paper	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q17	Q18	Q19	Q20	Q21	Q22	Q23	Q24	Q30	Q31	Q32
SP 11	B	C	D	B	C	A	B	A	C	C	C	A	D	B	B	D	A	B	C	A
SP 12	B	D	A	C	B	D	A	A	C	C	B	C	A	D	B	C	A	C	B	A
SP 13	B	B	C	C	B	C	A	D	C	A	C	C	D	B	A	C	C	B	D	A
SP 14	B	A	D	A	C	B	C	B	A	C	D	C	C	B	A	D	C	C	A	A
SP 15	C	D	A	B	D	A	C	A	D	B	C	D	A	C	B	A	C	D	C	A

Section A: Q1 to Q9 · Section B: Q17 to Q24 · Section C: Q30 to Q32. For written answers, check your work against the notes and the solved papers 4 to 10.

My Score Tracker (fill after every paper)

Paper	Date	Time taken	Biology /30	Chemistry /25	Physics /25	Total /80	Weak topics to revise
SP 1							
SP 2							
SP 3							
SP 4							
SP 5							
SP 6							
SP 7							
SP 8							
SP 9							
SP 10							
SP 11							
SP 12							
SP 13							
SP 14							
SP 15							

#All the best! 15 papers, 1200 marks of practice. Board exam is just paper number 16. Keep practising, keep smiling. **Nitesh Bhaiya, Eduspark**